

2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22)

Data File Documentation

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Executive Summary

The 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22), conducted by the National Center for Education Statistics (NCES) at the U.S. Department of Education, is the first follow-up data collection of students who began postsecondary education in the 2019–20 academic year. BPS:20/22 draws from the 2019–20 National Postsecondary Student Aid Study (NPSAS:20) to create a nationally representative sample of first-time beginning undergraduate students (FTBs). The data collected in BPS:20/22 are intended to provide researchers and policymakers with key information on persistence and attainment through the third academic year after students' initial enrollment in postsecondary education. This data file documentation details the methods used to collect, process, and analyze these data.

Sampling Design

The BPS:20/22 cohort was subsampled from the NPSAS:20 sample. NPSAS:20 was a nationally representative survey of students attending Title IV eligible postsecondary institutions in the 50 states, the District of Columbia, and Puerto Rico. NPSAS:20 sample members were enrolled in postsecondary education at all levels, excluding those currently enrolled in high school or exclusively enrolled in a high school equivalency program. Among the group of students eligible for NPSAS:20, students were also identified as potentially eligible for the BPS:20/22 cohort if their records indicated they began their postsecondary education for the first time during the 2019–20 academic year. These potential FTBs were identified during the NPSAS:20 sampling and matching process using institution enrollment list data, and NPSAS:20 purposefully oversampled them to provide a sufficient pool of study members from which to draw the BPS:20/22 cohort.

The BPS:20/22 sample consisted of 37,330 sample members. For the first follow-up (BPS:20/22), the sample consisted of three groups according to their NPSAS:20 sample status along with their administrative and survey response statuses. The three groups were (1) NPSAS:20 study members who completed the student interview and were confirmed to be FTBs ($n = 26,470$), (2) a subsample of potential FTBs identified during NPSAS:20 sampling who were NPSAS:20 study respondents but student interview nonrespondents ($n = 5,510$), and (3) a subsample of potential FTBs who only had administrative data collected and were

NPSAS:20 study respondents ($n = 5,350$). During BPS:20/22 data collection, approximately 3,090 individuals were identified as ineligible or deceased. The BPS:20/22 sample targeted some states for state-level representativeness: California, Florida, Georgia, New York, North Carolina, Pennsylvania, and Texas.

Survey Design, Data Collection, Outcomes, and Evaluation

The purpose of BPS is to contribute to a better understanding of key postsecondary outcomes: persistence, degree attainment, and employment. The BPS:20/22 survey included many long-standing BPS data elements including postsecondary enrollment history, financial aid and borrowing, employment, and career expectations. Additionally, in response to the coronavirus pandemic, several items capturing respondent experiences due to related societal changes were added to the survey. A facsimile of the final survey instrument is included as an appendix to this report.

The BPS:20/22 survey was available by both web-based instrument and telephone interview. In addition to the full survey, an abbreviated version was developed for nonresponse conversion. The abbreviated survey (averaged 18 minutes) consisted of a subset of items from the full survey, including high-priority enrollment, education experiences, financial aid, employment, income and expenses, background, and COVID-19 pandemic information.

To assess respondent burden and instrument performance, survey items were evaluated based on how quickly they were completed, the BPS:20/22 respondents who did not respond to the item, and consistency between the distribution of responses for telephone respondents and web respondents. Coder forms—survey items in which respondents started typing a response and then selected an option from a predicted list—were evaluated based on the proportion of respondents who identified a response from the predicted list.

The BPS:20/22 survey data collection began in March 2022 and continued through January 2023. Overall, 29,260 BPS:20/22 sample members were located, and 23,320 sample members (65 percent of the BPS:20/22 eligible fielded sample) were deemed BPS:20/22 respondents. Sample members were considered a BPS:20/22 respondent if they completed either a full or abbreviated survey. Partial survey completers were considered BPS:20/22 respondents if they completed at least the enrollment section of the survey.

Among BPS:20/22 respondents, 95 percent completed the survey on the web (this includes smartphones and other mobile devices), and 5 percent completed it by telephone. The average completion time for the full survey was 35 minutes for web nonmobile respondents, 32 minutes for web mobile respondents, and 60 minutes for telephone respondents, with substantially shorter completion times the abbreviated survey.

Administrative Records Matching Overview and Outcomes

The BPS:20/22 team collected administrative records to supplement the survey data. The BPS:20/22 cohort was matched to the U.S. Department of Education's Central Processing System (CPS) and the National Student Loan Data System (NSLDS). Additional data sources included the National Student Clearinghouse (NSC) and the Veterans Benefits Administration. CPS provided demographic and enrollment information for 24,430 sample members (70 percent) who applied for federal student aid in the 2020–21 academic year, 19,490 sample members (55 percent) who applied in the 2021–22 academic year, and 15,870 sample members (45 percent) who applied in the 2022–23 academic year. NSLDS provided historical information about disbursement of federal student loans and grants as well as debt and repayment outcomes for all sample members. This match yielded student loan data for 17,200 sample members (49 percent), and Pell Grant data for 19,790 sample members (56 percent). The NSC StudentTracker service provided information on institutions attended, enrollment dates, and degree completions. This match yielded information for 33,430 sample members (90 percent). Finally, the Veterans Benefits Administration data system provided information regarding 1,320 sample members' service status (4 percent).

Data File Processing and Preparation

The BPS:20/22 restricted-use dataset includes derived variables along with source data from the BPS:20/22 survey and administrative data sources. The public can generate tables of estimates and simple regressions based upon restricted-use data using the NCES data analysis tools in the DataLab website (<https://nces.ed.gov/datalab>).

To protect the confidentiality of study member information and to minimize disclosure risks, BPS:20/22 data were subject to data swapping, an Institute of Education Sciences Disclosure Review Board-approved perturbation procedures including the swapping of select records. Perturbation was carried out under

specific, targeted, but undisclosed, swap rates. This process preserved the central tendency estimates but may increase nonsampling error slightly. An extensive data quality check was carried out to assess and limit the impact of swapping. To construct the analysis dataset, data from the BPS:20/22 survey and administrative data sources were edited, upcoded, and combined to make analysis variables. The resulting variables were extensively reviewed for quality and accuracy. Details about variable construction and sources are available on the restricted-use files and on the DataLab tools.

Weighting and Variance Estimation

Because the BPS:20/22 cohort is a subset of the NPSAS:20 sample, statisticians derived the weights for analyzing the BPS:20/22 data from the NPSAS:20 student design weights and follow-up data collection design weights. These design weights were adjusted to account for subsampling, unknown student eligibility, and nonresponse. They were also calibrated to weighted estimates obtained from NPSAS:20 and population estimates.

Missing values in the analysis dataset were imputed for most variables using a weighted sequential hot deck (stochastic) process that replaced missing values with valid values from other respondents (Cox 1980; Iannacchione 1982).

Acknowledgments

The authors acknowledge and thank the respondents who participated in the first follow-up of the 2022/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22). This study would not have been possible without their invaluable contributions.

We would also like to express sincere gratitude to the members of the Technical Review Panel. Thank you for contributing your time and expertise to help develop this study.

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Chapter 1. Overview of BPS:20/22

This report describes the design, methods, and results of the 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) and serves as a resource for users of the study's data file. It is the first follow-up with a panel of first-time beginning students (FTBs) originally identified during the 2019–20 National Postsecondary Student Aid Study (NPSAS:20). The following legislation authorizes the Beginning Postsecondary Students Longitudinal Study (BPS):

- the Higher Education Act of 1965, as amended by the Higher Education Opportunity Act of 2008, 20 U.S.C. § 1015(a) (2012), and
- the Education Sciences Reform Act of 2002, 20 U.S.C. §§ 9541 to 9548 (2012).

Chapter 1 of this report provides an overview of the background and purpose of BPS, as well as the study design, schedule, and products. Chapter 2 presents details on the institution and student samples for NPSAS:20 and the subsequent sample for the BPS:20/22 cohort. Chapter 3 describes the development of the survey and the data collection processes and results and provides analysis of survey data quality. Chapter 4 contains information on administrative data collection processes and outcomes. Chapter 5 contains a description of BPS:20/22 study files, along with post-data-collection data file processing and editing. Chapter 6 provides information on weighting, imputation, bias analysis, and variance estimation. Appendix A provides a list of acronyms and abbreviations used throughout the report.

Tables and figures throughout this report present relevant analyses from the full-scale study. Unless otherwise indicated, a probability level of .05 was used for all tests of statistical significance conducted for BPS:20/22 evaluations. Because of rounding, row and column entries in tables may not sum to their respective totals and reported percentages may differ somewhat from those that would result from the rounded numbers. Rounding is used to ensure the confidentiality of study respondents.

1.1 Background and Purpose of BPS

The Beginning Postsecondary Students Longitudinal Study (BPS) is one of several NCES studies developed to address the need for nationally representative data on important postsecondary education issues. BPS cohorts include FTBs in postsecondary institutions. Follow-up data collections capture the academic progress and persistence in postsecondary education of a BPS cohort after its initial entry into a postsecondary institution.

NPSAS studies traditionally serve as the base-year data collection for one of two longitudinal studies: the current study, BPS, or the Baccalaureate and Beyond Longitudinal Study (B&B). NPSAS:20 is the base-year data collection for a cohort of FTBs —BPS:20/22 in 2022 and the 2020/25 Beginning Postsecondary Students Longitudinal Study (BPS:20/25), planned for 2025. Consequently, the NPSAS:20 student survey included modules that collect information about the experiences of BPS-eligible students in their first year of postsecondary education, including student debt, education experiences, majors, persistence, and completion. Users should refer to the NPSAS:20 Full-Scale Data File Documentation (DFD) for more information about the base-year data collection (nces.ed.gov/pubs2024/2024482.pdf).

1.2 Overview of Study Design

Data collection for BPS:20/22 included a survey and administrative record matching. The respondent universe for the full-scale study included all individuals who began postsecondary education during the 2019–20 academic year. Data from sample members were collected via a web-based multimodal survey designed for web, mobile, and telephone administration.

In addition to survey data, the sample was matched with several administrative data sources, including: the Central Processing System (CPS), which houses and processes data contained in the Free Application for Federal Student Aid (FAFSA) forms; the National Student Clearinghouse (NSC), which provides enrollment and degree verification; the National Student Loan Data System (NSLDS), containing federal loan and grant files; and the Veterans Benefits Administration (VBA), which identifies veterans education benefit amounts. A full description of administrative records matching processes and outcomes is provided in Chapter 4.

1.2.1 Study Management Systems and Security

The design of BPS:20/22 required the collection and management of personally identifiable information (PII) at multiple points throughout the course of the study. PII was required when updating student enrollment status at the schools, tracing sample members, and collecting data. These data were also managed during data processing and data product preparation activities. For more information on the study management systems used to securely collect, transmit, and process BPS:20/22 data, which are comparable to those used in NPSAS:20, refer to sections 4.1.2 and 5.1 of the 2019–20 National Postsecondary Student Aid Study DFD (Wine, Siegel, and Haynes 2024) for student data collection systems and administrative data matching, respectively. For additional information on study management systems and security—including the Enhanced Security Network, Secure Sockets Layer (SSL)-certified servers, and the Integrated Management System (IMS)—refer to section 1.2.1 of the NPSAS:20 DFD.

1.3 Schedule and Products

Table 1 shows the schedule for the major activities of the full-scale study.

Table 1. Schedule of major activities for the BPS:20/22 full-scale study: 2022

BPS:20/22 activity	Start date	End date
Contact information updates	Oct. 21, 2021	Mar. 6, 2022
Web-based survey data collection	Mar. 7, 2022	Dec. 12, 2022
Outbound telephone interviewing	Mar. 14, 2022	Dec. 12, 2022
Nonresponse conversion efforts	Sep. 28, 2022	Dec. 12, 2022
Administrative data matching	Jul. 27, 2022	Feb. 14, 2023
Process data, construct data files	Mar. 8, 2022	Jan. 12, 2026
Prepare/update reports	Mar. 8, 2022	Jan. 12, 2026

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

In addition to this DFD, a report titled *Persistence and Attainment of 2019–20 First-Time Postsecondary Students After 3 Years*, will be available on the NCES website at <https://nces.ed.gov/surveys/bps/>.

BPS microlevel data files, associated codebooks, and DFD are available to researchers who have obtained a restricted-use data license from NCES. To apply for a restricted-use data license, visit the NCES website at <https://nces.ed.gov/statprog/instruct.asp>. Further information on the process for

obtaining a restricted-use data license is available in the NCES *Restricted-Use Data Procedures Manual* at <https://nces.ed.gov/statprog/rudman/>.

The public may use NCES data analysis tools in the DataLab application, found at <https://nces.ed.gov/datalab>, to review and analyze the derived data file without a restricted-use license. Within DataLab, PowerStats can produce summary statistics and complex tables, as well as estimate regression models. It permits analysis without disclosing microlevel data to the user and suppresses or flags any estimates that fail to meet NCES reporting standards. DataLab also contains the Tables Library, which houses thousands of published analysis tables sortable by topic, publication, and source.

Chapter 2. Sampling Design

This chapter describes the student universe and sampling methods used to produce a nationally representative sample of FTBs for BPS:20/22. It also refers to the sampling universe and the two-stage sampling design used for the base-year study, NPSAS:20, which produced this BPS cohort. That design began with the selection of postsecondary institutions and was followed by selection of students from these institutions. Pertinent to this sampling design was the identification of FTBs.

The student universe for BPS:20/22 consisted of all students who began their postsecondary education for the first time during the 2019–20 academic year at any postsecondary institution in the United States that was eligible for inclusion in NPSAS:20. The BPS:20/22 sample was created from the set of confirmed and potential FTBs identified in NPSAS:20.

2.1 NPSAS:20 Institution and Student Universe

The target population includes all students enrolled in Title IV eligible postsecondary institutions during the 2019–20 academic year in each of the 50 states, the District of Columbia, and Puerto Rico. A sample was generated from this target population with the goal to be nationally representative as well as representative at the state level for undergraduate students. The institution eligibility conditions for NPSAS:20 were consistent with those in the most recent iterations of NPSAS. Additional details for the institution and student universe are covered in section 2.1 of the NPSAS:20 DFD (Wine, Seigel, and Haynes 2024).

2.2 NPSAS:20 Institution and Student Samples

NPSAS:20 used a two-stage sampling design. In the first stage of sampling, a total of 3,110 institutions were selected from three strata: public 2-year institutions, public 4-year institutions, and an “all other sectors” stratum (i.e., public less-than-2-year institutions, private nonprofit institutions, and private for-profit institutions). The 3,110 sample institutions included all public 2-year ($n = 960$) and all public 4-year ($n = 780$) institutions as well as a sample of 1,370 institutions from the “all other sectors” stratum.

For the second stage of sampling for the NPSAS:20 student sample, 380,100 students were sampled overall, of which 352,740 were undergraduate students. After initial undergraduate student sampling, within each student strata and institution, undergraduates were randomly divided into two groups. One group was asked to complete the student survey, while the other group was not offered the student survey. Only administrative data were collected for the second group. From this random selection, 146,150 of the sampled undergraduate students were selected for the student survey. Additionally, because NPSAS:20 served as the base year for BPS:20/22, potential FTBs were oversampled. For NPSAS:20, 52,620 of the students sampled were identified as potential FTBs. More information on the sample design and results samples can be found in sections 2.2, 2.3, and 2.4 of the NPSAS:20 DFD (Wine, Seigel, and Haynes 2024).

Respondents were defined as follows:

- *Survey respondents:* A survey respondent was defined as any undergraduate or graduate sample member who was determined to be eligible for the study and completed at least the enrollment and FAFSA sections of the student survey.
- *Administrative student respondents:* An administrative student respondent was defined as any undergraduate sample member who was determined to be eligible for the study, was enrolled for at least 1 month (based on student records or administrative sources), and at a minimum, had institution-provided valid key data in student records (i.e., FTB indicator, degree program, class level, high school graduation date/completion program completion date, enrollment in high school/completion program, and date of birth). Administrative student respondents include both students who were sampled for the survey and those who were not invited to participate in the survey.
- *Study respondents:* A study respondent was defined as any undergraduate sample member who was either an administrative student respondent or a survey respondent, or both. Study respondents who were survey respondents, but not administrative student respondents, had their administrative data imputed.

This NPSAS:20 study respondent definition was a departure from prior traditional NPSAS collections, which had a study member definition. For more information on the prior study member definition see the NPSAS:16 DFD (Wine, Siegel, and Stollberg 2018).

2.3 BPS:20/22 First Follow-up Sampling Design

This section describes the sampling design and methods for BPS:20/22. Section 2.3.1 describes the eligibility requirements for the BPS cohort and the procedures used in the identification of the BPS-eligible sample members in NPSAS:20. Section 2.3.2 details the process followed to create the BPS:20/22 sample and decisions for subsampling and oversampling of certain groups.

2.3.1 Identification of BPS:20/22 Cohort Eligibility

NPSAS:20 consisted of an administrative sample of 380,100 students. The NPSAS:20 survey sample was a subset of approximately 173,360 students from the total student sample. The remaining individuals who were part of the administrative sample but were not selected for the survey sample were referred to as administrative-only sample or students. For the BPS:20/22 cohort, the sampling frame was based on the full NPSAS:20 student sample. This frame contained survey-confirmed FTBs as well as survey nonrespondent and administrative-only potential FTBs. Some previous BPS administrations did not require that the NPSAS institution be the first institution a student attended after completing high school for the student to be eligible for the study. Therefore, the first institution a respondent attended was not necessarily the NPSAS institution. In BPS:20/22, analysts excluded students from the sample who reported in the eligibility section that the NPSAS institution was not the first postsecondary institution they attended after completing high school.

The BPS:20/22 frame only included confirmed and potential FTBs who were defined as study respondents in NPSAS:20 and who were not found to be deceased during NPSAS:20 data collection. A study respondent was any individual who is a survey or administrative student respondent. It follows that all confirmed FTBs were also NPSAS:20 study respondents. The BPS:20/22 sample consisted of three different groups based on their NPSAS:20 sample status along with their administrative and survey response statuses. The groups were (1) NPSAS:20 survey respondents who were confirmed FTBs, (2) NPSAS:20 survey nonrespondents who were potential FTBs and NPSAS:20 study respondents, and (3) NPSAS:20 administrative-only students who were potential FTBs and NPSAS:20 study respondents. As stated earlier, all three groups were considered study respondents. Only a random portion, or subsample, of the second and third groups were included in the final BPS:20/22 sample. Table 2 shows the distribution of the BPS:20/22 frame.

Table 2. BPS:20/22 FTB sample members, by NPSAS:20 sample outcome and inclusion in BPS:20/22 sample: 2022

NPSAS:20 full-scale outcome	Count	Included in BPS:20/22 Sample
Total confirmed and potential FTBs	60,960	
Confirmed FTBs		
NPSAS survey respondents	26,470	Yes
Potential FTBs		
NPSAS study respondents		
Survey nonrespondents	14,710	Yes (Subsampled) ²
Administrative-only students ¹	10,730	Yes (Subsampled) ²
NPSAS study nonrespondents		
Survey nonrespondents	7,620	No
Administrative-only students	1,430	No

¹ Excludes 370 administrative-only students sampled from institutions without NPSAS survey students.

² Randomly subsampled for inclusion in the final BPS:20/22 sample

NOTE: FTB = first-time beginning student. Details may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Some NPSAS:20 administrative-only potential FTBs (approximately 370) who were considered administrative student respondents were sampled from institutions that were unable to provide accurate enrollment list information necessary for student survey contacting. These students were therefore not able to be included in the student survey portion of the NPSAS:20 sample and were not included in the BPS:20/22 sampling frame. These institutions were not found to be statistically significantly different, based on size and control or level of institution, from the remaining institutions or administrative-only potential FTBs and thus should not impact sample representativeness. The decision to include administrative-only FTBs was based on two primary factors: (1) field test data showed statistically significantly higher response rates for the administrative-only group compared to survey nonrespondents and (2) simulation results indicated national financial aid and state subgroup precision goals would be met, which could not be achieved without including these students.

2.3.2 BPS:20/22 Sample

The BPS:20/22 sample consisted of approximately 37,330 total students including approximately 26,470 confirmed FTB NPSAS:20 survey respondents and a subsample of approximately 10,860 potential FTBs who did not complete the survey but who were NPSAS:20 study respondents. The potential FTBs were split nearly evenly between NPSAS:20 survey nonrespondents and administrative-only students.

The base NPSAS:20 administrative sample that BPS:20/22 follows was designed to be state representative, which allowed for the BPS:20/22 sample to target some

states for state-level representativeness. The state samples were designed to be representative of the state as a whole and not representative of control or level of institution within the state. The distribution of confirmed and potential FTBs from NPSAS:20 was reviewed by state to arrive at the final decision to target state representativeness for California, Florida, Georgia, New York, North Carolina, Pennsylvania, and Texas in BPS:20/22. The minimum desired sample sizes for each state subgroup were based on the ability to measure a relative change of 20 percent in proportions across rounds and an assumed design effect of 3. To meet this goal, approximately 740 survey respondents per desired state were necessary by the end of BPS:20/25 data collection.

An oversample of approximately 330 students was required to achieve the desired sample size for Georgia while oversampling was not required for the other six targeted states. The decision to oversample students for Georgia was made as Georgia was close to the target sample size, had enough additional administrative student respondents to be oversampled and was considered an important state to target.

All NPSAS:20 survey respondents who were confirmed FTBs were sampled with certainty. The approximately 25,440 potential FTBs who did not complete a survey, but who were study respondents, were explicitly stratified by “survey nonrespondent” and “administrative-only student” as well as control and level of institution. Additionally, the oversample of approximately 330 students for Georgia was allocated by sampling all potential FTBs within Georgia. Within the explicit strata, a simple random sample of students was selected. Table 3 displays the final BPS:20/22 sample by control and level of institution including the oversample of approximately 330 students within Georgia, while Table 4 details the sample by state.

Table 3. BPS:20/22 sample by NPSAS:20 sample outcome and control and level of base-year institution: 2022

Control and level of base-year institution	Total	NPSAS:20 Sample Outcome		
		NPSAS:20 survey respondents	NPSAS:20 survey nonrespondents	NPSAS:20 administrative-only respondents
Total	37,330	26,470	5,510	5,350
Control and level of base-year institution				
Public				
Less-than-2-year	400	270	120	#
2-year	12,370	9,000	2,210	1,160
4-year, non-doctorate-granting, primarily sub-baccalaureate	2,610	2,010	550	50
4-year, non-doctorate-granting, primarily baccalaureate	2,290	1,850	260	180
4-year, doctorate-granting	7,840	4,750	790	2,300
Private nonprofit				
Less-than-4-year	290	190	100	#
4-year, non-doctorate-granting	2,940	2,000	220	720
4-year, doctorate-granting	3,600	2,400	330	880
Private for-profit				
Less-than-2-year	980	780	150	50
2-year	1,600	1,280	300	10
4-year	2,430	1,940	480	#

Rounds to zero.

NOTE: FTB = first-time beginning student. Potential FTBs are individuals who did not complete a NPSAS:20 student survey and NPSAS:20 student records or administrative data (National Student Loan Data System, Central Processing System, etc.) indicated first postsecondary enrollment in the 2019–20 academic year. Sector reflects institution categorization as determined from the 2018–19 Integrated Postsecondary Education Data System (IPEDS) files, which served as the basis for the sampling frame. Details may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table 4. BPS:20/22 sample by NPSAS:20 sample outcome and key state subsamples: 2022

State	Total	NPSAS:20 sample outcome		
		NPSAS:20 survey respondents	NPSAS:20 survey nonrespondents	NPSAS:20 administrative-only respondents
Total	37,330	26,470	5,510	5,350
California	2,750	2,230	500	20
Florida	1,970	1,580	350	40
Georgia	1,410	860	390	170
North Carolina	1,370	970	230	180
New York	2,030	1,660	280	90
Pennsylvania	1,340	940	170	230
Texas	2,340	1,810	410	120
All other states	24,120	16,410	3,190	4,510

NOTE: FTB = first-time beginning student. Potential FTBs are individuals who did not complete a NPSAS:20 student survey and NPSAS:20 student records or administrative data (National Student Loan Data System, Central Processing System, etc.) indicated first postsecondary enrollment in the 2019–20 academic year. Details may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Chapter 3. Survey Design, Data Collection, Outcomes, and Evaluation

This chapter provides an overview of the BPS:20/22 survey design, data collection, outcomes, and evaluations.

3.1 Survey Design and Systems

The BPS:20/22 full-scale survey collected information on students who were enrolled in postsecondary education for the first time during the 2019–20 academic year. This section provides an overview of the BPS:20/22 full-scale survey design process, an introduction to survey features and functionalities, descriptions of the data elements included in each key content area of the survey and ends with a summary of the survey and data collection systems.

3.1.1 *Survey Design*

The data elements included in the BPS:20/22 full-scale survey were determined with consideration of past BPS studies, while also incorporating new and modified data elements. The full-scale survey included data elements from prior BPS studies, such as enrollment and employment history, as well as data elements pertaining to student experiences, such as persistence, attainment, and social and academic activities. New data elements were identified and developed with Technical Review Panel (TRP) member input. For a list of TRP members, see appendix B. The TRP supported recommendations to carry forward new FTB-only data elements from NPSAS:20 to enhance analysis of student choices, persistence, and attainment. These data elements included measures of teacher effectiveness, as well as frequency and types of communication and interactions with family and friends, faculty members, and other students. Additionally, to capture the continued impact of the coronavirus pandemic (COVID-19) on students, several items related to academic and employment disruptions and personal experiences were added to the survey.

The survey consisted of seven key sections focused on specific content areas: enrollment, education experiences, financial aid, employment, income and expenses, background, and COVID-19 experiences. A 15-minute abbreviated version of the survey included a subset of content from each key section as a nonresponse conversion technique (see section 3.2.4), and for those wishing to take the survey in Spanish. A brief overview of each section follows. In addition to these content areas, a locating and incentives section collected information for future contacting purposes but contains no substantive data for public use.

Enrollment confirmed eligibility to participate in the BPS:20/22 full-scale survey for NPSAS:20 survey nonrespondents. The survey collected additional information on continued enrollment at the sampled institution (referred to hereafter as the NPSAS institution) as well as any additional postsecondary institutions attended between July 1, 2020, and June 30, 2022. This section also determined the primary institution for each respondent. The primary institution was the current or most recent institution the respondent attended between July 1, 2020, and June 30, 2022. Questions throughout the remainder of the survey were anchored to experiences at respondents' primary institution.

Education experiences collected information on remedial education, grade point average, and online coursetaking at the primary institution. This section also captured information regarding the general academic and social experiences of college students such as questions related to interactions with family and friends, faculty, and other students; participation in social and academic groups and clubs; and services used at primary institution and the importance of those services. Finally, this section collected information about the respondent's expected occupation and wages after completing all their postsecondary education.

Financial aid captured information about sources of aid used in the 2021–22 academic year, such as federal and private loans, grants or scholarships, employer assistance, and veterans' benefits. Information about cumulative private loan amounts and repayment plans and the out-of-pocket costs for academic materials were also collected. Additionally, awareness and use of federal income-driven repayment plans and loan forgiveness programs, and information about the primary institution's emergency aid options were included.

For the first time in BPS, this section collected information about receipt of state financial aid programs for select states based on state representativeness in NPSAS:20 and other state-level characteristics. These data will be used to inform how state aid may affect retention, persistence, and attainment. Respondents whose NPSAS institution was in a select state that identified as having a major

financial aid program (e.g., Helping Outstanding Pupils Educationally [HOPE] scholarship in Georgia), were administered questions about receipt of the applicable state financial aid program in the 2019–20, 2020–21, and 2021–22 academic years.

Employment items collected information about all work for pay during the 2020–21 and 2021–22 academic years. This included dates employed, hours worked while enrolled and while not enrolled, amount earned, and whether the job was on or off campus, for all full- and part-time employment, paid internships, and self-employment. The employment section also collected more detailed information about the current or most recent employer such as job title, whether the job was related to respondents’ major, and whether respondents had looked for work while not employed.

The **income and expenses** section of the survey collected information including respondent income, military status, information about dependent children and other dependents, marital status, parental income, financial support for education and living expenses from family and friends, number of credit cards and balances, residence while attending school, mortgage and rent amount for those who did not live on campus, broad financial concept knowledge, and untaxed federal benefits receipt. For respondents with a spouse, information on spousal income and education attainment was collected.

Background items obtained information about student demographic characteristics such as U.S. citizenship, immigration status, race and ethnicity, sex assigned at birth, gender identity, sexual orientation, and voting behavior. Disability status, housing security, risk of homelessness, and items from the U.S. Department of Agriculture 10-item food security module were also included in the background section.¹

The **COVID-19 experiences** section of the survey included items that pertained to student experiences during the COVID-19 in the 2020–21 academic year. This section collected information on changes due to the COVID-19 in enrollment, student loan borrowing, academic engagement, access to support resources, and mental health. Finally, this section collected information about respondents’ experiences with online coursetaking, including appropriate hardware and software requirements and residential access to high-speed internet.

¹ For more information about the U.S. Department of Agriculture food security module, see <https://www.ers.usda.gov/topics/food-nutrition-assistance/food-security-in-the-us/survey-tools>.

Finally, the **locating** section collected the respondent's contacting information for the final follow-up, including primary address e-mail and phone numbers, along with contact information for parents/guardians.

3.1.2 Survey Mode of Administration

The BPS:20/22 survey instrument was a multimodal survey designed for web and telephone administration, with both modes offered for the entirety of data collection (see 3.2.4 for more information about contacting methods). Self-administered surveys could be completed on a mobile device (e.g., smartphone or tablet, etc.), referred to hereafter as “web mobile” mode, or on a nonmobile device, referred to hereafter as “web nonmobile” mode.² Respondents advanced through the survey according to skip logic based on information collected from the NPSAS:20 data collection or responses earlier in the BPS:20/22 survey. The survey consisted of forms, which are survey screens that typically contained a single question, but sometimes multiple questions, organized by content area; form-specific help text; the response options to each question; and navigation buttons.

To minimize mode effects, specific methodological features were programmed to provide web nonmobile and mobile respondents with the assistance normally provided by a trained data collection interviewer (DCI), such as

- Help text on every form to define key terms and clarify question intent;
- validation pop-up messages prompting participants to correct responses that were out of range or in an incorrect format;
- conversion text pop-up messages to encourage responses to critical questions when left unanswered; and
- conversion text pop-up messages prompting a response when three consecutive forms were left blank.

Help text was provided at the form level and was written specifically for the question(s) on-screen. Section 3.5.2 provides analysis of help text usage in the survey. To reduce nonresponse on critical items (e.g., annual income), conversion text appeared when a respondent did not provide a response. The conversion text was written to communicate the importance of the form and encouraged the respondent to reconsider providing a response.

² Two elements of instrument paradata identified web mobile mode: the rendering of the instrument (e.g., how the instrument displayed on a browser for a respondent) and parsing the browser agent string to obtain information about device type, browser, and touchscreen capability.

Given the higher timing burden observed for the full Spanish survey in NPSAS:20 and the number of respondents participating in the Spanish survey, only the abbreviated BPS:20/22 survey was translated into Spanish.³ NPSAS:20 survey nonrespondents and those who completed the base-year survey in Spanish were asked in which language they would prefer to complete the BPS:20/22 survey. If Spanish was selected, the abbreviated survey launched and allowed respondents or trained bilingual DCIs to progress through the translated survey. If English was selected, the full survey was administered.⁴ Those who completed the NPSAS:20 survey in English were not offered the opportunity to complete the BPS:20/22 abbreviated Spanish survey.

3.1.3 Survey Response Coding Systems

The BPS:20/22 survey used coding systems, or “coders,” to provide standardized codes for certain text string responses. Predictive text string coders were used for the following survey items: postsecondary institutions attended, major/field of study, and occupations. For each coder, respondents entered their response as a text string into a text box. As respondents typed, a keyword search of the underlying database returned a list of possible matches for selection, displayed in a drop-down menu. If a suitable match was not returned, respondents were prompted to provide additional information so that expert coding staff could attempt to assign a code.

See section 5.2 for an explanation of how codes were assigned to text strings that were not initially coded in the survey. The following are brief descriptions of the individual coding systems, sometimes used on multiple forms, and the underlying databases for each coding system:

- **The postsecondary institution coder** was linked to the complete set of institutions contained in the 2017–18 IPEDS, developed by NCES (<https://nces.ed.gov/ipeds>). Any postsecondary institutions, other than the NPSAS institution, attended in the 2020–21 or 2021–22 academic years were coded using this coder. For institutions not listed in the database, the coder saved the entered institution name and prompted respondents to provide the control (e.g., public, private nonprofit, or private for-profit) and level (e.g., 4-year or 2-year) of the institution, as well as the city and state in which the institution was located.

³ The Spanish BPS:20/22 abbreviated survey is available upon request by contacting bps@ed.gov.

⁴ The English abbreviated survey was still offered as a nonresponse conversion technique later in data collection; see section 3.4.6 for more information about outcomes across phases.

- **The major/field of study coder** used the 2020 Classification of Instructional Programs (CIP) taxonomy, developed by NCES (<https://nces.ed.gov/ipeds/cipcode>). For any major or field of study not found in the CIP database, the coder saved the entered major and asked respondents to select a general area of study and a specific discipline within that area.
- **The occupation coder** linked respondents' occupation titles to Version 26.1 of the Occupational Information Network (O*NET) database (<https://onetonline.org>), which uses the 2018 Standard Occupational Classification taxonomy (<https://www.bls.gov/soc/home.htm>). For any occupations not listed in the database, the coder saved the entered occupation and asked respondents to provide a general area, specific area, and finally a detailed classification for the occupations.

3.1.4 Data Collection Systems

Several data collection systems were used to develop the survey instrument, manage sample member cases, contact sample members (including phone, e-mail, mail, and text reminders), report data collection progress, and evaluate DCI performance. These systems and software, mostly proprietary in nature and tailored to meet the specific needs of the project staff, included a survey development system (Hatteras), computer-assisted telephone interviewing case management system (CATI-CMS), an application to coordinate e-mail and text message reminders (Harmony), and a quality evaluation system (QUEST).

The BPS:20/22 survey instrument was developed using a web-based system called Hatteras. Hatteras allows multiple users to interact with the survey during specification, review, and testing. All instrument design specifications were stored in a Structured Query Language (SQL) database via the survey editor interface. When published to the web server, the survey forms were dynamically rendered, reflecting the specified content of each form, question routing, valid ranges and, if applicable, any previous responses entered by the respondent. The survey's appearance was automatically adjusted to fit the screen size of the respondent's computer, mobile phone, or other device. Both self-administered web surveys (nonmobile and mobile) and telephone interviews used the same Hatteras survey instrument to collect data.

The case management system used by DCIs, CATI-CMS, managed all sample member locating information and all activity related to outbound and inbound calls. Any contact updates, including new telephone numbers, were added to CATI-CMS as they were identified via DCI contacts, batch tracing services, or

intensive tracing methods. The CATI-CMS, in concert with the tracing system, dictated the appropriate next step for each case, whether additional outbound calls were required, or the case needed additional tracing efforts (see section 3.2.3 for more information on tracing operations).

The CATI-CMS was used to assign cases to DCIs by prior contact status (e.g., cases that had been recently contacted or had never been contacted), best day and time to call, and previously scheduled appointments. Sample members who had previously refused to participate were placed into a separate queue to be contacted by DCIs specifically trained in refusal conversion techniques designed to encourage sample members to complete the survey after a refusal. Additionally, BPS:20/22 project staff members prioritized other cases, as needed, that could be accessed separately by a subset of telephone DCIs. For example, sample members who were eligible for the shortened version of the survey could be assigned to this priority dialing queue shortly after becoming eligible for the shorter survey to maximize our contact efforts and encourage participation.

Project staff utilized a suite of data collection applications called Harmony to coordinate e-mail and text message reminders to sample members. The system allowed staff to deliver personalized reminders by incorporating the latest contacting and survey progress updates. Data collection activities were monitored via real-time reports as well as daily reporting of survey completion, response, timing, and trend analysis. Project staff utilized a proprietary Adaptive Total Design dashboard to check quality indicators from a variety of data sources during survey data collection. This included response rates, case dispositions, completes by wave and mode, and DCI performance reports, allowing for prompt decisionmaking during data collection. Additionally, staff used Tableau, a visual analytics platform to create need-specific dashboards, charts, and worksheets for analyzing specific data collection performance, such as response by contacting mode (mailing, e-mail, text, and phone) and tracing activities.

QUEST facilitated DCI performance monitoring. Protocols for evaluating DCI performance were used in real time while an interview was being performed or through recordings after the survey had been administered. The quality evaluation system supported all phases of DCI quality monitoring including selecting interviews, observing DCIs' work, evaluating DCI performance, providing feedback, and analyzing performance data across DCIs to identify cross-cutting instrument or performance issues.

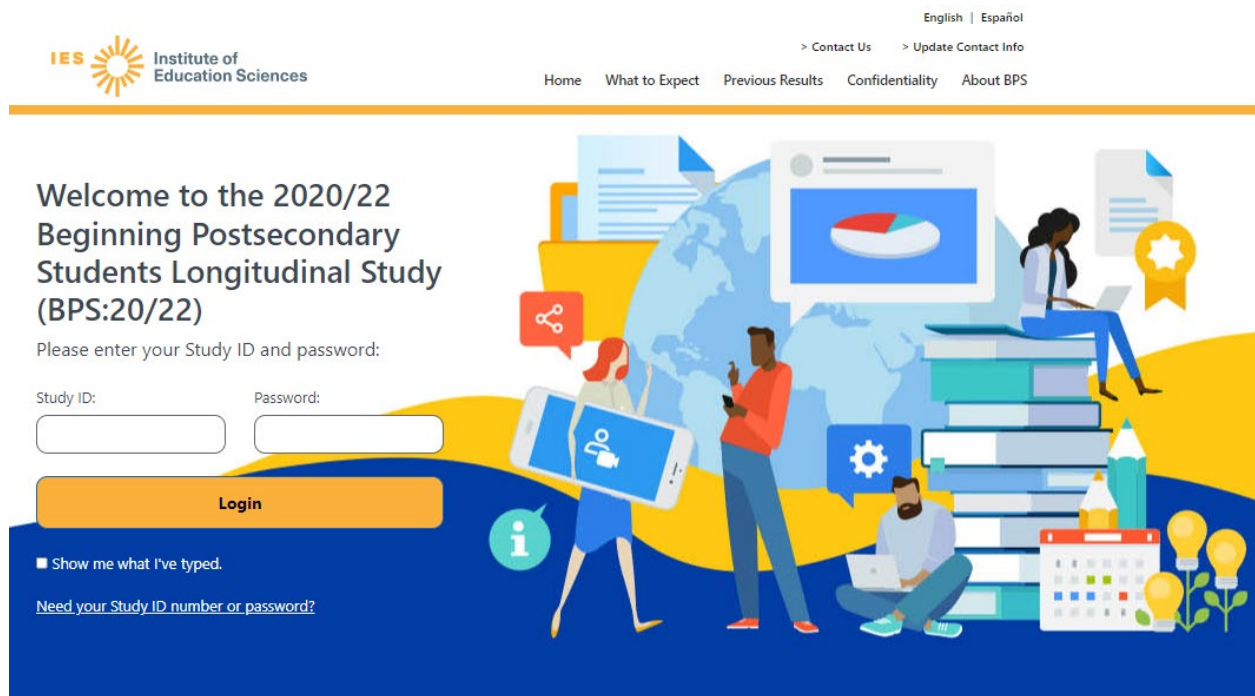
3.2 Survey Data Collection

The primary mode of BPS:20/22 survey data collection was a web survey available through the study website. Sample members also had the option of completing the survey with a DCI trained in computer-assisted telephone interviewing (CATI) methods. The BPS:20/22 data collection study website and help desk offered information and support to sample members. Prior to and during data collection, sample members were traced utilizing batch and intensive tracing processes. DCIs completed extensive training on interviewing processes and protocols. Additionally, data collection staff were trained on locating, tracing, and contacting procedures to ensure efficiency and consistency.

3.2.1 Study Website and Help Desk

Communications with sample members included a link to the BPS:20/22 website, which provided general information about the study, details about who is conducting the study, how the data can be used, answers to frequently asked questions (FAQs), information security, and selected findings from previous BPS studies. The website also included contact information for the study help desk and information for the RTI Project Director and NCES Study Director. Sample members could log in to a secure section of the website to update their contact information and complete the survey. Figure 1 shows a screenshot of the home page for the BPS:20/22 website.

Figure 1. Home page for BPS:20/22 study website: 2022



SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

The study website, designed according to NCES web policies, protects all survey data using secure logins composed of a study ID and password unique to each sample member, and SSL technology for data encryption during transmission and at rest.

BPS:20/22 DCIs also served as help-desk agents and responded to sample members' questions related to technical issues or completing the web survey. For each call received, data collection staff confirmed contact information for the sample member for security purposes and recorded a description of the issue and resolution in a shared database.

The two most common reasons for contacting the help desk were for sample members to retrieve login credentials and to request assistance in completing the survey over the telephone. To minimize the need for telephone assistance, a link on the website allowed sample members to submit a request for login information. Sample members that successfully submitted several key pieces of identifying information were provided login credentials via e-mail.

3.2.2 *Training of Data Collection Staff*

Members of the BPS:20/22 data collection team included DCIs, call center supervisory staff, and intensive tracing staff. All data collection staff, regardless of role, completed a comprehensive training program before beginning work on the study (see appendix E for training content and training schedule). Additionally, all data collection staff completed a general training program on topics such as proper interviewing techniques, confidentiality procedures, and sample member rights.

In total, 29 data collection staff were trained over three training sessions held between March and July 2022. Additionally, four intensive tracers were trained for the project. Training sessions took place virtually on Zoom videoconferencing software. During BPS:20/22 there was a 3-night training session for new DCIs, and two 2-night training sessions for DCIs with experience working on recent NCES studies. Both trainings included direct instruction and hands-on practice, and the additional date of training for new DCIs allowed for more direct instruction and hands-on practice.

DCIs. DCIs were the primary point of contact for sample members, conducting telephone interviews and providing reminders and information used to encourage sample members' participation in BPS:20/22. DCIs also served as help-desk agents to respond to sample member concerns and questions, e-mail login instructions, and address incentive receipt inquiries and issues. DCIs familiarized themselves with the survey instrument and received training specific to each survey question. They developed proficiency with the survey through mock interviews, hands-on practice with case management systems, and instruction on conversational interviewing techniques. Training materials included a DCI manual and associated materials addressing survey administration and conversational interviewing. DCIs completed a project certification after they conducted a mock interview and provided appropriate and accurate responses to BPS:20/22 FAQs. Periodic Quality Circle (QC) meetings of DCIs and call center supervisory staff were held to review proper administration of the survey and ad hoc topics related to BPS:20/22 or general interview protocol.

Call Center Supervisory Staff. Call center supervisory staff supervised DCIs, performed general day-to-day monitoring responsibilities, provided constructive feedback and coaching to DCIs after monitoring live or recorded BPS:20/22 interviews, and helped troubleshoot issues that arose. Supervisory staff attended DCI training to learn survey basics and interviewing conventions specific to the BPS:20/22 study.

Intensive tracing staff. Intensive tracing staff were experienced with general tracing procedures and only required 2 hours of project-specific training, including the tracing techniques most appropriate for tracing BPS:20/22 sample members.

3.2.3 Locating, Tracing, and Contacting Sample Members

To achieve a high rate of response, project staff implemented procedures to identify sample members' updated contact information (i.e., tracing) and to confirm the contact information was accurate (i.e., locating). Prior to data collection, an e-mail was sent to sample members requesting that they provide up-to-date contact information. Batch tracing services and intensive tracing were also used to update contact information.

Tracing efforts were considered successful if a match produced contact information for the sample member that was not previously known to data collection staff. A sample member was then deemed *located* if at any point during data collection, contact information was confirmed to be accurate for the individual. Thus, a sample member was only considered *not located* if no contact information was ever verified as accurate for the individual. Descriptions of tracing, locating, and contacting efforts are described below. The results of these efforts are presented in section 3.4.1.

Contact updates. Approximately one month before the beginning of data collection, project staff utilized all known e-mail addresses and the last known address to request that sample members update their contact information. An e-mail and letter were sent with a link to the student website where information could be updated. Sample members could submit this information through the end of data collection.

Batch tracing. Also prior to data collection, known contact information for sample members was matched to the U.S. Postal Service (USPS) National Change of Address (NCOA) database, LexisNexis's Single Best Address search, the CPS, and the NSLDS. The NCOA database contains change-of-address records submitted to USPS over the last 4 years, and Single Best Address can search multiple data sources using progressive search logic to match to the most current address available. Matched records were compared with last known addresses, and any new or updated addresses for sample members were loaded into the database. To update telephone numbers, all cases were matched to LexisNexis's Phone Append prior to beginning CATI data collection.

CATI locating. If a sample member logged in to the survey via the web and verified their identity, they were considered located. For cases not completed via the web, DCIs attempted to conduct an interview over the telephone. They called the number with the best likelihood of reaching the sample member as determined by CATI-CMS. If the DCI could not reach the sample member at that number, they attempted to gather locating information from the contact who answered the call. If this approach was unsuccessful, the DCI used other telephone numbers available for the sample member. Once all telephone numbers proved inaccurate for the sample member, production tracing was initiated.

Production tracing. If all telephone locating methods were exhausted, and no other telephone number was available, further telephone tracing was initiated. As a first step prior to intensive tracing, project staff sent cases through LexisNexis Single Best Phone and Premium Phone searches. If neither of these searches returned new telephone information, these cases were initiated to intensive tracing.

As part of intensive tracing, tracers used an approach that included public domain and proprietary databases to identify updated contact information. Tracers identified sample members in consumer databases (e.g., LexisNexis, Experian, and Accurant) using Social Security number (SSN) and/or date of birth (DOB). If this search resulted in an updated telephone number, tracers designated the case for DCI follow-up. This approach minimized the effort required to locate cases through intensive tracing and the time that cases were unavailable to DCIs.

Data collection mailings, e-mails, and text messages. Using the best-known mailing address information, project staff sent mailings to the two best-known addresses. Mailing addresses were prioritized based on the most recent source report date and an analysis of rates of undeliverable addresses for each source. All mail correspondence was sent via USPS mail. The first mailing included a lead letter and a study brochure. The lead letter notified sample members of the start of data collection and the incentive they were eligible to receive for completing the survey. The letter also included unique login information for the web survey instrument and encouraged participation during the early response period. The brochure provided information about the purpose of the study, confidentiality and security concerns, and study contact information. Project staff sent additional mailings such as postcards and letters, periodically as reminders to complete the survey.

Project staff sent e-mail and Short Message Service (SMS) text communications to all sample members periodically throughout data collection. Each e-mail

included a click-here link as well as unique identification and password credentials that could be used to complete the survey. Additionally, each SMS text message provided a unique login link to access the survey. See appendix D for examples of the mailing, e-mail, and text message contact materials sent to sample members.

3.2.4 Survey Data Collection Phases and Contacting Methods

Data collection for BPS:20/22 consisted of four phases: the early completion phase, production phase I, production phase II, and the nonresponse conversion phase. Sample members had access to both the web and telephone versions of the survey throughout data collection. The Web and telephone versions of the survey were identical except that the telephone option included administration instructions for the DCI.

Data collection began in mid-March 2022, when e-mails were sent inviting BPS:20/22 sample members to take the survey and were followed up with a data collection announcement letter on April 5, 2022. The two data collection groups were defined by prior response status as follows:

- Group 1: NPSAS:20 survey respondents. Sample members who responded to NPSAS:20 and self-identified that they began their postsecondary education between July 1, 2019, and June 30, 2020, received the default data collection protocol.
- Group 2: NPSAS:20 survey nonrespondents and administrative-only cases. This group includes NPSAS:20 survey nonrespondents and NPSAS:20 administrative-only sample who were potential 2019–20 academic year FTBs based on administrative data. The additional data collection efforts used for Group 2 (discussed below) were needed to maximize participation of NPSAS:20 survey nonrespondents and sample members who had never been contacted (i.e., NPSAS:20 administrative-only cases).

Early completion phase. Both groups received panel maintenance invitations in October 2021 to gather updated contact information for the sample prior to the start of data collection. Data collection started on March 7, 2022, for Group 1 and on April 7, 2022, for Group 2. The beginning of data collection, known as the early completion phase, included mailings, e-mails, and text messages encouraging all sample members to complete the survey, as well as the beginning of outbound calling efforts for targeted cases in both groups. Group 1 was offered

a \$30 base incentive, and Group 2 was sent a \$2 prepaid incentive in addition to a base incentive offer of \$45.

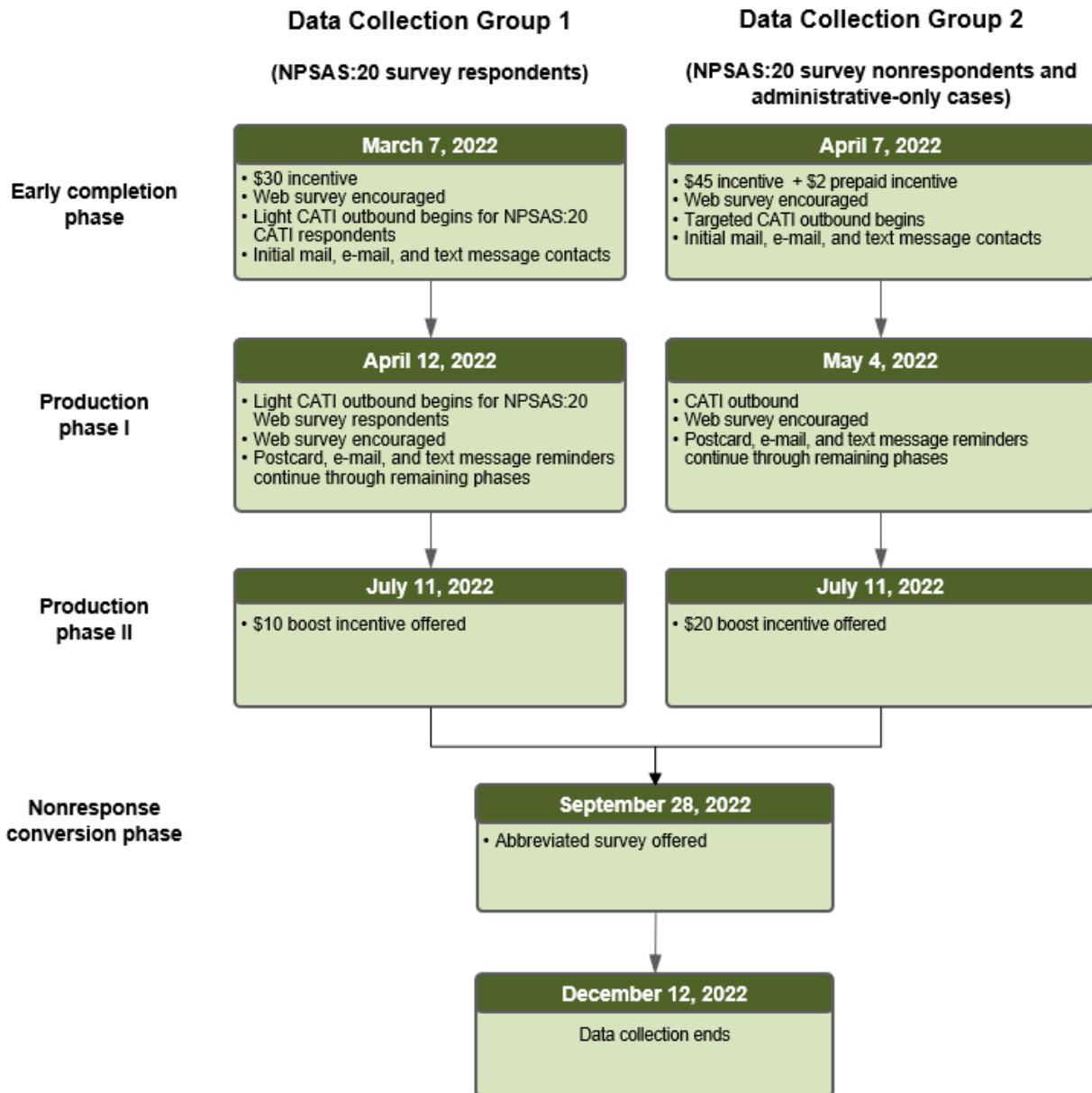
Production phase I. During this phase, both groups were encouraged to complete the survey either online or by telephone, and the data collection staff was available to answer any questions. Light outbound calling began for all remaining sample members in Group 1, and more frequent outbound calling began for all remaining sample members in Group 2.

Production phase II. This phase began on July 10, 2022. Group 1 was offered a permanent incentive boost of \$10 for a total of \$40, and Group 2 was offered a permanent incentive boost of \$20 for a total of \$65.

Nonresponse conversion phase. This phase began in September 2022 and an abbreviated, 15-minute- survey was offered to both groups.

Figure 2 displays the activities conducted in each phase for Groups 1 and 2.

Figure 2. Timeline of survey data collection activities by data collection group: 2022



SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Contacting methods. The availability of the web survey was announced to sample members through mail and e-mail, which included the URL and the sample members' login credentials. E-mails and text messages included case specific direct links to the survey (and mailings included QR codes) providing sample members access to their unique survey, bypassing the need to manually enter their login credentials. The web survey was available 24 hours a day, 7 days a week throughout the entire data collection. Throughout data collection, e-mail,

letters, and text messages continued to encourage sample members to complete the web survey.

Once outbound calling began, an automated call scheduler assigned cases for DCIs to call. The scheduler organized cases into queues based on a variety of factors, including prior contact status (e.g., cases that had been recently contacted, or had never been contacted), refusal status, and appointments set during a prior contact attempt. The scheduler also automatically ordered numbers to call by prioritizing those most likely to reach the sample member. New numbers were added continuously, based on telephone contacts and tracing efforts, as well as updates received through mailings, e-mails, or telephone calls to the help desk. The call scheduler reprioritized telephone numbers based on new information as it came in. As DCIs successfully reached sample members, they answered questions about the study, encouraged sample members' cooperation, and conducted interviews. For sample members who preferred to participate on the web, the DCI could e-mail secure survey login credentials to the sample member.

3.3 Survey Data Collection Quality Assurance

This section summarizes the survey quality assurance procedures employed throughout the course of data collection.

3.3.1 *Call Monitoring*

Telephone interviews and other calls resulting in telephone contact were regularly monitored during BPS:20/22 data collection to meet the following data quality objectives:

- identify difficult items in the survey;
- reduce the number of DCI errors;
- improve DCI performance through reinforcement of effective strategies such as active listening and probing; and
- assess the quality of the data collected.

As call center supervisory staff monitored DCI interactions with sample members, they recorded feedback on standardized forms that covered such topics as DCI professionalism, question administration, conversational interviewing, and familiarity with the survey instrument. DCIs received feedback from monitoring

sessions, and QC meetings frequently incorporated issues identified during monitoring to improve the overall quality of telephone interviews.

3.3.2 Quality Circle Meetings

QC meetings served as a tool for communication among project staff, call center supervisory staff, and DCIs. Project staff reinforced concepts from interview training sessions in biweekly QC meetings, reminding DCIs of proper administration of the survey and other topics as needed. Project staff encouraged attendees to ask questions, which helped identify topics for subsequent meetings. During BPS:20/22, some of the topics covered during QC meetings included:

- use of help text within the survey;
- clarification of questions and item responses from the survey instrument;
- proper administration of specific survey questions;
- reinforcement of successful interviewing and refusal conversion techniques;
- guidelines for providing detailed sample member-level comments in the CATI-CMS;
- strategies for gaining cooperation from sample members and other contacts;
- data security protocols; and
- study progress and outline of upcoming interventions.

After each QC meeting, summaries were disseminated to data collection staff collectively via Microsoft Teams. The notes provided guidance on the topics discussed at each meeting and were posted in a cumulative format, so that data collection staff had an updated and searchable document containing all QC notes compiled over the course of the project.

3.3.3 Debriefings

At both the midpoint and the end of data collection, debriefings were conducted with data collection staff regarding their experiences during the study. Feedback was collected from DCIs and call center supervisory staff through an anonymous online survey and in-person meetings. Topics of the debriefing discussions

included DCI training, support and monitoring, systems for locating and contacting sample members, gaining sample member cooperation, and BPS:20/22 survey design and administration. Feedback from DCIs and call center supervisory staff from the midpoint debriefing informed QC meeting content and procedures for the last half of data collection; feedback from both debriefings will be used to inform the planning and implementation of future BPS studies.

Debriefing feedback. With regard to data collection staff training, DCIs were appreciative of “hands-on” training which included practicing interviews. Based on suggestions raised by DCIs during prior study debriefings, BPS:20/22 trainings included additional CATI-CMS practice scenarios that allowed DCIs to gain experience navigating and using the system, including features for incident reports (used to communicate any issues sample members are having such as with the survey or incentives). DCIs appreciated this additional practice and specifically noted it was helpful in preparing them for creating reports and updating contact information.

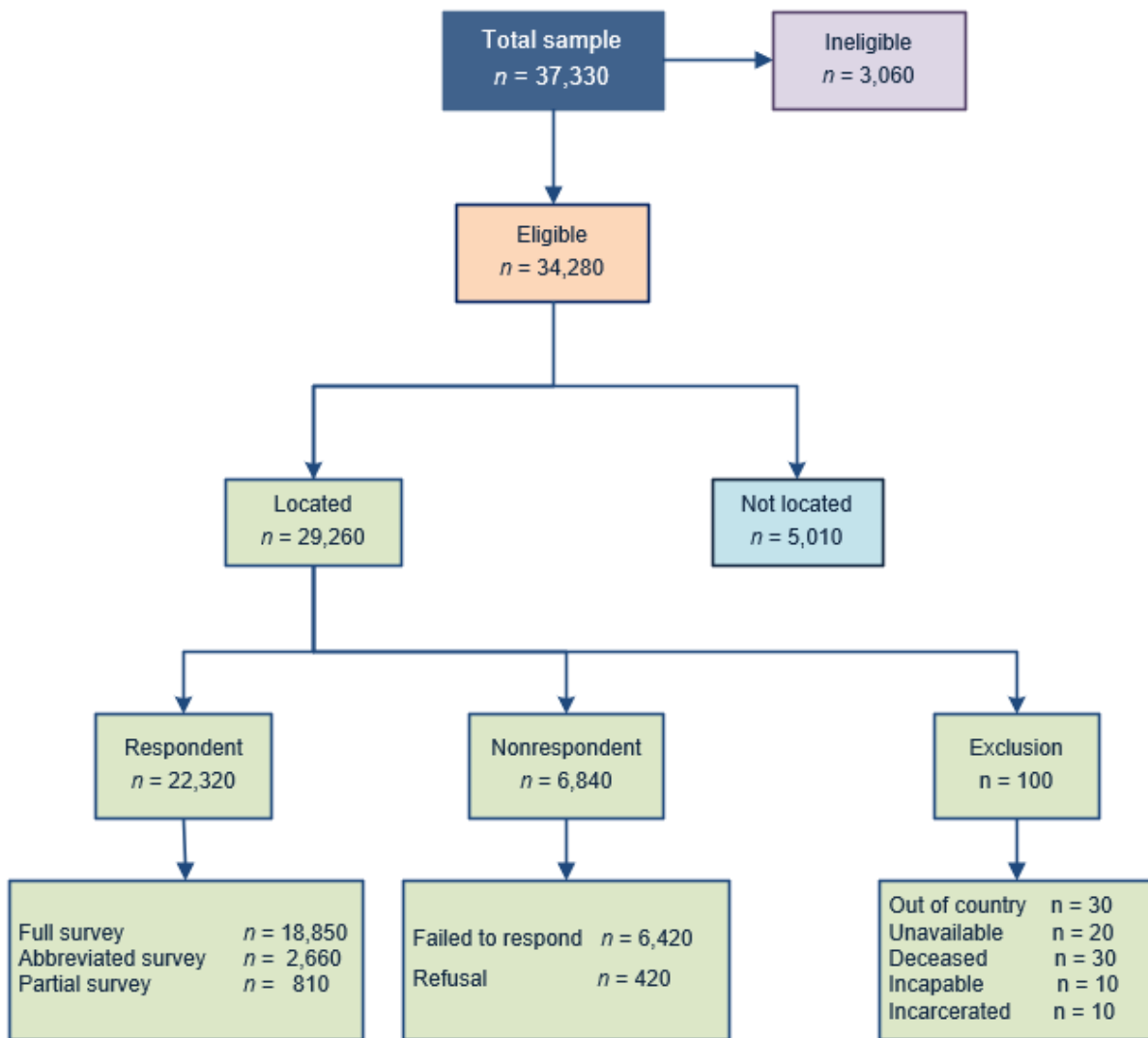
DCIs thought that the ease of question administration, flow of the survey, and help text were the best characteristics of the BPS:20/22 survey. In addition, DCIs reported that the resources provided in the survey, such as help text and conversion text, were helpful in successfully administering the survey by telephone. Finally, DCIs provided suggestions for improving future trainings. Suggestions included more time for additional coder practice and providing tips on how to mitigate sample member confusion during survey administration.

3.4 Survey Data Collection Outcomes

This section summarizes the results of the BPS:20/22 data collection in terms of the success rates and administrative burden of reaching and surveying sample members. Some analyses in this section include only eligible sample members, while others include all sample members. Similarly, some analyses include partial surveys, and other analyses exclude partial surveys where it does not make sense to include them. Sample members were identified as ineligible either through their survey responses or through information available from administrative student records. Partial survey respondents are defined as sample members who completed at least the enrollment section of the survey but did not complete the entire survey. Footnotes on each table indicate any excluded cases. For reporting response rates, data users should use the response rate that includes eligible sample members and partial survey completes.

To assess student survey data collection outcomes, project staff determined the number of sample members located and surveyed, the survey timing burden, the average number of contacts sent to sample members, and the rate of conversion for survey refusals. Overall, project staff located approximately 29,260 (85 percent) of the 34,280 eligible BPS:20/22 sample members. Of the sample members located, approximately 76 percent ($n = 22,320$) responded, including partial survey respondents, resulting in an overall response rate of 65 percent of eligible sample members ($n = 34,280$). See figure 3 below for overall locating and surveying results. Section 3.4.2 provides more detail on student survey response rates.

Figure 3. BPS:20/22 survey locating and response results: 2022



NOTE: Sample sizes rounded to the nearest 10. Detail may not sum to totals because of rounding. "Ineligible" cases include sample members determined ineligible from the survey or from administrative student records.
SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

3.4.1 Locating Results

Project staff employed a multistage locating approach to identify updated contact information and locate sample members. See section 3.2.3 for details about locating and tracing activities.

Contact update results. A total of 5,650 BPS:20/22 sample members (15 percent) responded to a request for updated contact information prior to the start of data collection.⁵ Of those who provided updated contact information, 85 percent responded to the BPS:20/22 survey, while those who did not provide updated contact information responded to the BPS:20/22 survey at a rate of 69 percent.

Number of sample members located, eligible, and survey respondents. Overall, 85 percent of eligible sample members were located during BPS:20/22. The locate rate for sample members who completed the NPSAS:20 survey was 90 percent, compared to 62 percent for sample members who were NPSAS:20 nonrespondents. Of the 29,260 eligible sample members who were located, 22,320 (76 percent) completed the BPS:20/22 survey. The number and percentage of total eligible sample members who were located and responded, by prior-round response status and control and level of institution, are presented in Table 5.

⁵ A total of 37,110 sample members with valid e-mail and mailing addresses were sent requests to provide updated contact information.

Table 5. BPS:20/22 located and respondent status, by prior-round response status and control and level of institution: 2022

Prior-round response status and control and level of base-year institution	Eligible sample ¹	Located ²		Responded ³		
		Number	Percent of eligible	Number	Percent of eligible	Percent of located
Total	34,280	29,260	85.4	22,320	65.1	76.3
NPSAS:20 (prior-round) response status						
Survey respondents	25,250	22,760	90.1	18,900	74.8	83.0
Survey nonrespondents	4,480	2,800	62.4	980	21.9	35.1
Administrative-only respondents	4,550	3,710	81.5	2,450	53.8	66.0
Control of base-year institution						
Public	23,360	19,830	84.9	14,930	63.9	75.3
Private nonprofit	6,340	5,630	88.7	4,470	70.4	79.4
Private for-profit	4,580	3,810	83.3	2,930	64.0	76.8
Level of base-year institution						
Less-than-2-year	1,220	1,010	83.0	760	62.0	74.8
2-year	12,980	10,780	83.1	7,900	60.8	73.2
4-year, non-doctorate-granting	9,030	7,840	86.8	6,100	67.6	77.9
4-year, doctorate-granting	11,050	9,630	87.2	7,570	68.4	78.5
Control and level of base-year institution						
Public less-than-2-year	330	270	81.3	180	54.6	67.2
Public 2-year	11,310	9,420	83.3	6,890	60.9	73.1
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	2,390	1,990	83.2	1,450	60.6	72.9
Public 4-year, non-doctorate-granting, primarily baccalaureate	2,130	1,890	88.9	1,500	70.6	79.4
Public 4-year, doctorate-granting	7,210	6,260	86.8	4,910	68.2	78.5
Private nonprofit 2-year or less	220	190	84.3	130	59.6	70.7
Private nonprofit 4-year, non-doctorate-granting	2,750	2,450	89.2	1,970	71.6	80.3
Private nonprofit 4-year, doctorate-granting	3,370	2,980	88.6	2,360	70.2	79.2
Private for-profit less-than-2-year	880	730	83.4	570	64.8	77.7
Private for-profit 2-year	1,470	1,190	81.0	890	60.7	75.0
Private for-profit 4-year	2,230	1,890	84.7	1,470	65.8	77.7

¹ Table excludes 3,060 sample members determined to be ineligible for the study using data from survey and administrative sources.

² Sample members were considered located if they were ever located at some point during data collection.

³ Respondents count includes sample members who met the criteria for qualification as a survey respondent, which required completing at least a partial survey.

NOTE: Sector reflects institution categorization as determined from the 2018–19 Integrated Postsecondary Education Data System (IPEDS) files, which served as the basis for the sampling frame. Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Batch tracing results. Prior to and during data collection, known contact information was sent to several batch tracing sources (NCOA, Phone Append, Premium Phone, Single Best Phone, Single Best Address, CPS, and NSLDS) to collect updated locating information. Match rates varied by batch tracing source and are presented in table 6.

Table 6. Number of sample member cases sent, and number and percentage matched to each batch tracing source: 2022

Batch tracing source	Number sent	Number matched	Percent matched
National Change of Address (NCOA)	37,010	7,060	19.1
Phone Append	37,010	20,000	54.0
Premium Phone	710	230	32.9
Single Best Phone Search	1,800	1,410	78.7
Single Best Address Search	35,270	32,140	91.1
Central Processing System (CPS)	35,400	20,570	58.1
National Student Loan Data System (NSLDS)	37,330	25,990	69.6

NOTE: Number of sample member cases sent to each source varies based on the timing of the matching procedure, available information for matching, and the need for new contact information. Percent matched includes instances when a sample member's contact information was confirmed or when new information was provided. For NCOA and Premium Phone, percent matched includes only instances when new information was provided.

Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Intensive tracing results. After exhausting all known telephone information and batch tracing services, 4.3 percent of sample members required intensive tracing efforts once CATI efforts began. Table 7 shows the sample members requiring intensive tracing by prior-round response status control and level of institution.

Table 7. Number and percentage of BPS:20/22 sample members who required intensive tracing, by prior-round response status, and control and level of institution: 2022

Prior-round response status and control and level of base-year institution	Total sample	Required intensive tracing ¹	
		Number	Percent of total
Total	37,330	1,590	4.3
NPSAS:20 (prior-round) response status			
Survey respondents	26,470	610	2.3
Survey nonrespondents	5,510	640	11.6
Administrative-only respondents	5,350	340	6.4
Control of base-year institution			
Public	25,510	1,050	4.1
Private nonprofit	6,830	310	4.6
Private for-profit	5,000	230	4.5
Level of base-year institution			
Less-than-2-year	1,400	70	4.9
2-year	14,230	680	4.8
4-year, non-doctorate-granting	9,740	350	3.6
4-year, doctorate-granting	11,970	490	4.1
Control and level of base-year institution			
Public less-than-2-year	400	20	5.3
Public 2-year	12,370	580	4.7
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	2,610	100	4.0
Public 4-year, non-doctorate-granting, primarily baccalaureate	2,290	50	2.2
Public 4-year, doctorate-granting	7,840	290	3.7
Private nonprofit 2-year or less	290	20	8.1
Private nonprofit 4-year, non-doctorate-granting	2,940	120	4.0
Private nonprofit 4-year, doctorate-granting	3,600	170	4.8
Private for-profit less-than-2-year	980	50	4.8
Private for-profit 2-year	1,600	80	4.7
Private for-profit 4-year	2,430	110	4.3

¹ Required intensive tracing count excludes cases initiated to tracing that were not traced.

NOTE: Sample sizes rounded to the nearest 10. Sector reflects institution categorization as determined from the 2018–19 Integrated Postsecondary Education Data System (IPEDS) files, which served as the basis for the sampling frame. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

3.4.2 Survey Response Rates

This subsection reports and analyzes unit response rates for the survey, by prior-round response status, data collection phase, survey type, and mode of administration. Overall, 65 percent (22,320) of the eligible sample responded to the survey. Partial survey completers were considered respondents if they completed at least the enrollment section of the survey. Excluding partial survey completers, 63 percent (21,510) of the eligible sample of 34,280 completed the BPS:20/22 full or abbreviated survey (see table 8).

Completion by prior response status and data collection phase. BPS:20/22 survey completion rates differed by prior response status. Seventy-two percent of NPSAS:20 survey respondents completed the survey, compared to 20 percent of

NPSAS:20 survey nonrespondents. Of NPSAS:20 administrative-only respondents, 51 percent completed the survey. The data collection design consisted of four operational phases: the early response phase, production phase I, production phase II, and a nonresponse conversion phase. Of all completed surveys, 42 percent were completed during the early completion phase. Twenty-nine percent were completed during production phase I, and 17 percent were completed during production phase II. Twelve percent were completed during the nonresponse conversion phase. These results are displayed in table 8. To be counted as a completed survey during a phase of data collection, the survey could not be a partial complete; therefore, partial survey completes are excluded from table 8. Response rates by survey completion type are displayed in table 9.

Table 8. BPS:20/22 completion, by data collection phase, prior-round response status, and control and level of institution: 2022

Prior-round response status and control and level of base-year institution	Eligible sample ¹	Data collection phase									
		Total completes ²		Early response		Production phase I		Production phase II		Nonresponse conversion ³	
		Number	Percent of eligible	Number	Percent of completes	Number	Percent of completes	Number	Percent of completes	Number	Percent of completes
Total	34,280	21,510	62.8	8,940	41.6	6,280	29.2	3,690	17.1	2,600	12.1
NPSAS:20 (prior-round) response status											
Survey respondents	25,250	18,290	72.4	7,960	43.5	5,350	29.3	2,990	16.3	1,990	10.9
Survey nonrespondents	4,480	890	19.9	160	18.3	240	27.4	250	28.2	230	26.1
Administrative-only respondents	4,550	2,330	51.3	820	35.1	690	29.5	450	19.2	380	16.3
Control of base-year institution											
Public	23,360	14,380	61.6	5,930	41.2	4,190	29.1	2,490	17.3	1,780	12.4
Private nonprofit	6,340	4,340	68.4	1,820	41.9	1,290	29.7	740	17.0	500	11.5
Private for-profit	4,580	2,800	61.1	1,200	42.9	800	28.8	470	16.6	330	11.7
Level of base-year institution											
Less-than-2-year	1,220	710	58.4	280	39.6	200	28.2	130	18.8	100	13.5
2-year	12,980	7,550	58.2	3,080	40.8	2,210	29.2	1,330	17.6	940	12.4
4-year, non-doctorate-granting	9,030	5,910	65.5	2,500	42.3	1,680	28.3	1,020	17.2	720	12.2
4-year, doctorate-granting	11,050	7,330	66.3	3,080	42.0	2,200	30.0	1,210	16.5	850	11.5
Control and level of base-year institution											
Public less-than-2-year	330	170	51.2	50	31.1	40	24.6	50	28.7	30	15.6
Public 2-year	11,310	6,590	58.3	2,700	40.9	1,920	29.2	1,150	17.5	820	12.4
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	2,390	1,400	58.5	570	40.9	410	29.3	240	17.2	180	12.7
Public 4-year, non-doctorate-granting, primarily baccalaureate	2,130	1,460	68.6	600	41.0	410	28.0	260	17.5	200	13.5
Public 4-year, doctorate-granting	7,210	4,770	66.1	2,010	42.2	1,410	29.5	790	16.6	560	11.7
Private nonprofit 2-year or less	220	130	57.8	50	35.7	30	22.5	30	23.3	20	18.6
Private nonprofit 4-year, non-doctorate-granting	2,750	1,910	69.5	820	42.7	540	28.3	340	17.7	220	11.3
Private nonprofit 4-year, doctorate-granting	3,370	2,290	68.1	950	41.5	720	31.2	370	16.0	260	11.2
Private for-profit less-than-2-year	880	530	61.0	220	41.8	160	30.0	80	15.7	70	12.5
Private for-profit 2-year	1,470	850	57.7	340	40.7	260	30.1	150	17.5	100	11.7
Private for-profit 4-year	2,230	1,420	63.4	630	44.6	390	27.5	230	16.5	160	11.4

¹ Excludes 3,060 sample members determined to be ineligible for the study using data from survey and administrative sources.

² Total completes count excludes 810 respondents who met the criteria as a survey respondent but were partial completes. Partial survey respondents are defined as sample members who completed at least the enrollment section of the survey but did not complete the entire survey.

³ Nonresponse conversion count includes 258 cases that were a partially-completed survey at the time the abbreviated survey was offered; these cases were excluded from that offer.

NOTE: Sample sizes rounded to the nearest 10. Sector reflects institution categorization as determined from the 2018–19 Integrated Postsecondary Education Data System (IPEDS) files, which served as the basis for the sampling frame. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table 9. BPS:20/22 survey completion type, by prior-round response status, and control and level of institution: 2022

Prior-round response status and control and level of base-year institution	Total respondents ¹	Survey completion type					
		Full survey		Abbreviated survey ²		Partial survey ³	
		Number	Percent of total	Number	Percent of total	Number	Percent of total
Total	22,320	18,850	84.4	2,660	11.9	810	3.6
NPSAS:20 (prior-round) response status							
Survey respondents	18,900	16,230	85.9	2,070	10.9	610	3.2
Survey nonrespondents	980	660	66.7	240	23.9	90	9.4
Administrative-only sample	2,450	1,970	80.5	360	14.8	110	4.7
Control of base-year institution							
Public	14,930	12,680	84.9	1,700	11.4	550	3.7
Private nonprofit	4,470	3,820	85.4	520	11.6	130	2.9
Private for-profit	2,930	2,360	80.5	440	15.0	130	4.5
Level of base-year institution							
Less-than-2-year	760	610	80.6	100	13.6	40	5.8
2-year	7,900	6,640	84.0	920	11.6	340	4.4
4-year, non-doctorate-granting	6,100	5,090	83.5	820	13.4	190	3.1
4-year, doctorate-granting	7,570	6,510	86.1	820	10.9	230	3.1
Control and level of base-year institution							
Public less-than-2-year	180	140	78.7	30	15.2	10	6.2
Public 2-year	6,890	5,830	84.7	760	11.0	300	4.3
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	1,450	1,230	84.6	170	11.9	50	3.5
Public 4-year, non-doctorate-granting, primarily baccalaureate	1,500	1,260	83.8	200	13.3	40	2.9
Public 4-year, doctorate-granting	4,910	4,220	85.9	540	11.1	150	3.0
Private nonprofit 2-year or less	130	100	75.9	30	21.1	#	3.0
Private nonprofit 4-year, non-doctorate-granting	1,970	1,670	84.9	240	12.1	60	2.9
Private nonprofit 4-year, doctorate-granting	2,360	2,040	86.4	250	10.7	70	2.9
Private for-profit less-than-2-year	570	460	81.0	80	13.2	30	5.8
Private for-profit 2-year	890	720	80.3	130	14.7	40	4.9
Private for-profit 4-year	1,470	1,180	80.4	230	15.9	50	3.7

= rounds to zero.

¹ Total respondents count includes sample members who met the criteria for qualification as a survey respondent, which required completing at least a partial survey.

² Abbreviated survey count includes both English and Spanish abbreviated survey completes.

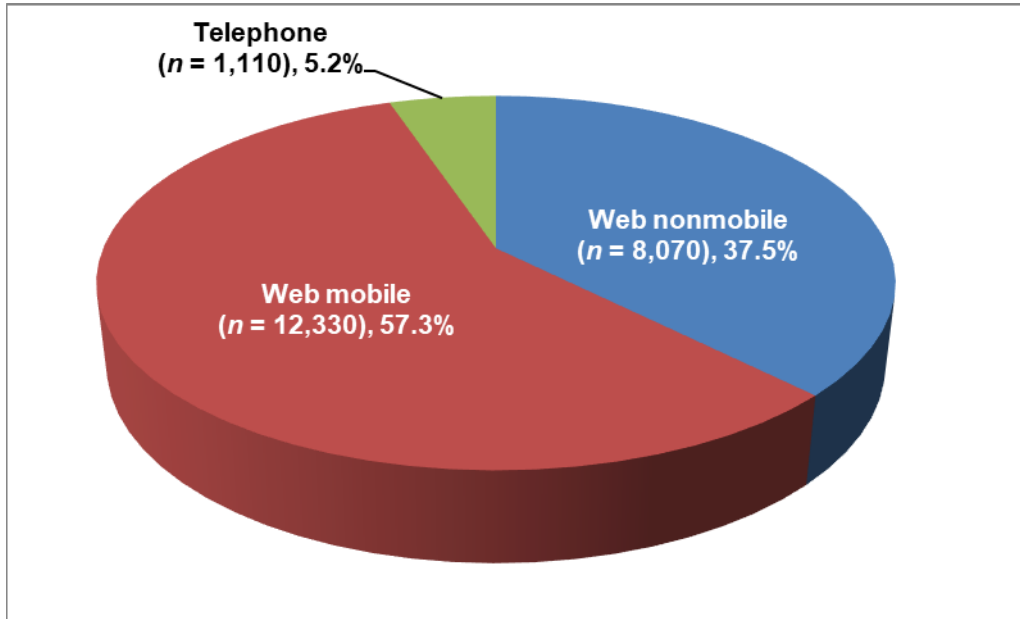
³ Partial survey respondents are defined as sample members who completed at least the enrollment section of the survey, but did not complete the entire survey.

NOTE: Sample sizes rounded to the nearest 10. Sector reflects institution categorization as determined from the 2018–19 Integrated Postsecondary Education Data System (IPEDS) files, which served as the basis for the sampling frame. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Completion by mode of administration. Respondents completed the BPS:20/22 survey by web or by telephone. The majority of surveys (57 percent) were completed by web on a mobile device⁶ such as a tablet or smartphone. Web completion on a nonmobile device such as a desktop computer accounted for 38 percent of completed surveys, while telephone interviews accounted for 5 percent of all completed surveys. Figure 4 displays survey completions by mode of administration.

Figure 4. Number and percentage of BPS:20/22 respondents, by mode of completion: 2022



NOTE: Web survey completers were analyzed separately by device type: nonmobile (e.g., desktop or laptop) or mobile (e.g., smartphone or tablet). The mode of completion is the mode associated with the final session. Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.
SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

⁶ Two elements of instrument paradata identified web mobile mode: the rendering of the instrument (i.e., how the instrument displayed on a browser for a respondent) and parsing the browser agent string to obtain information about device type, browser, and touchscreen capability.

3.4.3 Survey Timing Burden

This section describes the timing burden associated with completing the 2020/22 Beginning Postsecondary Student Longitudinal Study (BPS:20/22) full-scale survey. Analyses include the average times required for respondents to complete the survey across several characteristics, including mode of administration, survey type (i.e., full, or abbreviated survey), prior response status, and survey language.

To calculate the time required to complete the survey, the survey instrument recorded the elapsed time respondents took to complete each form, or web screen. Individual form-level timing values were summed to calculate times for discrete survey sections (i.e., enrollment, employment, etc.) and total survey times. Respondents were able to complete the survey in multiple sessions. When beginning a new session to continue the survey, respondents began on the form they last saw in their prior session. Among sample members that completed in multiple sessions, the timing for the last form the respondent saw before exiting the survey was not able to be measured, therefore, it was imputed to the median time other respondents spent on the same form.

Of the 22,320 full and abbreviated full-scale completed or partial surveys, 19,620 (88 percent) are included in the timing report (table 10). About 2,700 cases (12 percent) were excluded from analyses, including partial cases, cases who completed in multiple sessions (i.e., cases with more than two forms requiring mean time imputation⁷), and total time outliers.⁸

⁷ To avoid introducing excessive imputation and uncertainty into the timing estimates, cases that required more than two form imputations were excluded.

⁸ To detect total time outliers, the distribution of all survey times (highly right-skewed) was first normalized using a Box-Cox power transformation (Box and Cox 1964). Cases were then removed from the overall total using an interquartile range formula adopted from Tukey (1977) with a multiplier of 1.5. Cases were excluded as outliers if total time > 75th percentile + (1.5 * interquartile range) or if total time < 25th percentile – (1.5 * interquartile range).

Table 10. Number and percentage of BPS:20/22 full-scale surveys included and excluded from timing report: 2022

	Number of cases	Percent
Total surveys	22,320	100
Completed surveys included in timing report	19,620	87.9
Full	17,040	76.3
Abbreviated	2,590	11.6
Surveys excluded from timing report	2,700	12.1
Completed in multiple sessions ¹	1,360	6.1
Total time outliers ²	530	2.4
Partial surveys	810	3.6

¹ Interviews completed in more than one session were included in the timing analysis when possible. However, if a respondent completed the survey in more than two sessions, typically more than two forms would be missing timing information; thus, the case would be excluded from analysis.

² Outliers were calculated by normalizing the data and excluding extreme values identified by Tukey's formula (1977), which is not sensitive to distributional assumptions. Timing cases outside of the interquartile range (IQR = third quartile - first quartile) multiplied by 1.5 were identified as total time outliers.

NOTE: Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

3.4.3.1 Timing by survey type

Overall, BPS:20/22 full surveys in web nonmobile mode took an average of 34.5 minutes to complete, while web mobile surveys took statistically significantly less time with an average of 31.7 minutes ($p < .0001$).⁹ Full surveys in telephone mode took an average of 59.9 minutes to complete, statistically significantly the highest timing burden of all modes of administration, compared with both web nonmobile ($p < .0001$) and web mobile ($p < .0001$). As anticipated, NPSAS:20 nonrespondents took statistically significantly longer than NPSAS:20 respondents to complete the BPS:20/22 full survey, an average of 37.5 minutes, compared to 33.6 minutes ($p < .0001$). NPSAS:20 nonrespondents took statistically significantly longer in telephone mode (61.7 minutes) than in both web nonmobile mode (36.8 minutes) ($p < .0001$) and in web mobile mode (34.4 minutes) ($p < .0001$). The full survey was also statistically significantly longer in web nonmobile mode than web mobile mode for NPSAS:20 nonrespondents ($p < .001$). DCIs read survey questions aloud to respondents for telephone mode, therefore, surveys administered by telephone are often associated with longer completion times.

The BPS:20/22 survey contained eight sections: (1) enrollment, (2) education experiences, (3) financial aid, (4) employment, (5) income and expenses, (6) background, (7) COVID-19 experiences, and (8) locating and incentives. Table 11 provides a breakdown of the average time to complete the full survey by

⁹ Results use Satterthwaite (1946) approximation in difference-of-means tests with unequal variances.

prior response status and mode of administration. NPSAS:20 nonrespondents received more survey questions than NPSAS:20 respondents to collect eligibility data elements that would have been collected in the base-year survey.

Table 11. Average time, in minutes, to complete the full BPS:20/22 survey, by mode of administration and prior response status: 2022

Prior response status	Full survey respondents			Mode of administration ¹								
				Web nonmobile			Web mobile			Telephone		
	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time
Overall	17,040	34.13	30.79	6,710	34.48	31.44	9,520	31.7	29.17	810	59.94	60.75
NPSAS:20 nonrespondent	2,400	37.47	34.06	1,000	36.80	34.22	1,220	34.38	31.54	180	61.67	61.45
NPSAS:20 respondent	14,640	33.58	30.26	5,710	34.07	30.99	8,300	31.31	28.86	620	59.43	60.14

¹ Mode of administration is the mode in which the respondent completed the student survey; this mode may be different than starting mode for respondents who completed the survey in more than one session.
NOTE: NPSAS:20 = 2019–20 National Postsecondary Student Aid Study. Analysis includes only completed full surveys that have two or fewer forms of imputed timing data. Respondents who completed the abbreviated survey, respondents who completed the survey in multiple sessions, total time outliers, and respondents who did not complete the entire survey (partial surveys) were excluded. Sample sizes rounded to the nearest 10. Detail may not sum to totals because of rounding.
SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

The abbreviated survey included a subset of questions from the full survey, collecting high-priority enrollment, education experiences, financial aid, employment, income and expenses, background, and COVID-19 information. The abbreviated version of the survey was also translated into Spanish. Combining both English and Spanish language surveys, the abbreviated survey took an average of 18.3 minutes to complete. Separately, the abbreviated English took on average 17.6 minutes to complete, statistically significantly faster than the 23.1 minutes to complete the Spanish abbreviated survey ($p < .0001$).

3.4.3.2 Timing by survey section

Table 12 provides the average timing for the full survey, by survey section and mode of administration. Overall, the education experiences section took respondents the longest time to complete: 7 minutes on average, representing 21 percent of the entire full survey burden. Comparatively, financial aid was the shortest section, taking just 1.5 minutes on average for full survey respondents. This distribution of burden aligns with BPS:20/22 design, since BPS follows students' experiences during, and transitions through, postsecondary education and into the labor force, whereas financial aid was of greater focus for the base-year survey.

Table 12. Average time, in minutes, to complete the full BPS:20/22 survey, by mode of administration and survey section: 2022

Survey section	Mode of administration ¹											
	Full survey respondents			Web nonmobile			Web mobile			Telephone		
	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time
Total	17,040	34.13	30.79	6,710	34.48	31.44	9,520	31.70	29.17	810	59.94	60.75
Enrollment	17,040	5.06	4.19	6,710	5.20	4.30	9,520	4.70	3.97	810	8.20	7.80
Education experiences	17,040	7.06	6.29	6,710	7.50	6.60	9,520	6.39	5.90	810	11.36	12.99
Financial aid	17,040	1.50	1.18	6,710	1.67	1.30	9,520	1.29	1.04	810	2.48	2.42
Employment	17,040	4.58	3.85	6,710	4.46	3.80	9,520	4.35	3.70	810	8.37	8.10
Income and expenses	17,040	5.10	4.27	6,710	5.19	4.38	9,520	4.75	4.02	810	8.35	8.01
Background	17,040	2.99	2.47	6,710	2.78	2.37	9,520	2.85	2.45	810	6.35	6.21
COVID	17,040	3.72	3.20	6,710	3.85	3.39	9,520	3.30	2.87	810	7.57	8.25
Locating and incentives	17,040	4.12	3.47	6,710	3.83	3.26	9,520	4.06	3.45	810	7.27	6.89

¹ Mode of administration is the mode in which the respondent completed the student survey; this mode may be different than starting mode for respondents who completed the survey in more than one session.

NOTE: Analysis includes only completed cases that have two or fewer forms of imputed timing data. Respondents who completed the abbreviated survey, respondents who completed the survey in multiple sessions, total time outliers, and respondents who did not complete the entire survey (partial surveys) were excluded. Sample sizes rounded to the nearest 10. Detail may not sum to totals because of rounding.
SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Student Longitudinal Study (BPS:20/22).

3.4.3.3 Timing of individual forms

This section details the average time to administer each individual substantive form within the survey—excluding the front end (which confirmed the correct person was being surveyed and collected consent to proceed with the survey), locating, and incentive sections. The form-level timing analysis separated coders from other form types. Coder forms use an underlying database that provides standardized classifications to open-ended responses and represent some of the longest overall form times in the survey. Coders required respondents or the DCI to interact with the predictive text search functionality of the form, which required more time than simply selecting a response option on noncoder forms (i.e., forms not linked to an underlying database), for more information about coders, refer to section 3.5.1. Table 13 provides the average form-level timing for all coders and the top 10 longest noncoder forms.

Four of the nine coders took an average of over 40 seconds. The *Expected occupation coder* (B22BEXOCC) had the longest average form time at 51.8 seconds (table 5). This form asked respondents to code their own expected occupation using the Occupational Information Network Online database (O*NET-SOC v 24.0). The next longest coder in the survey also used the O*NET-SOC database: the *Reference employer coder* (B22DOCC) took an average of 45.5 seconds for respondents to code their current occupation. The shorter time to report the reference employer occupation is likely due to the

reporting of actual events, for example, providing a current or most recent occupation is less burdensome than providing an expected future occupation.

The last two coders that took more than 40 seconds on average were the *Major 1 coder: base-year NPSAS degree/certificate* (B22AMAJ1), with an average administration time of 42.8 seconds, and *Major 1 coder: other attendance* (B22AOTMAJ101), with an average administration time of 42.5 seconds. These forms asked respondents to provide and code their major or field of study using the 2020 CIP codes.

The noncoder forms that took the longest to answer represent some of the most cognitively burdensome questions in the survey, either due to asking the respondent to recall detailed information or due to interaction with the form type. The top four noncoder forms with the highest timing burden were calendar forms, which required respondents to select specific months within a time range in which they attended or worked. The calendar form with the highest timing burden, *NPSAS attendance: months full-time in years 2/3* (B22ANENRLFT), took respondents an average of 51.9 seconds to select the months they attended their NPSAS institution between July 1, 2020, and June 30, 2022. Five of the longest noncoder forms were in a grid format, which displayed several items and scaled response options in a grid. *Primary institution: participation in academic activities* (B22BACDPART) displayed a list of activities in a grid and took respondents an average of 41 seconds to answer.

Table 13. Forms with the highest average administration time, in seconds, by form and mode of administration: 2022

			Mode of administration ¹											
Form Name	Form Description	Type	All respondents			Web nonmobile			Web mobile			Telephone		
			Number of cases	Mean	Median	Number of cases	Mean	Median	Number of cases	Mean	Median	Number of cases	Mean	Median
Coders														
B22BEXOCC	Expected occupation coder	Coder	16,080	51.75	35.02	6,370	48.95	33.11	8,950	53.01	35.02	760	60.33	48.44
B22DOCC	Reference employer coder	Coder	4,110	45.54	30.06	860	44.92	27.27	2,960	44.98	30.06	290	53.11	41.72
B22AMAJ1	Major 1 coder: base-year NPSAS degree/certificate	Coder	5,280	42.83	27.21	2,430	43.54	26.76	2,620	41.13	27.21	230	54.5	38.8
B22AOTMAJ101 ²	Major 1 coder: other attendance	Coder	4,770	42.49	25.73	1,690	37.47	22.76	2,810	43.84	25.73	260	60.31	38.27
B22AMAJ2	Major 2 coder: base-year NPSAS degree/certificate	Coder	790	28.99	17.65	510	25.16	16.13	260	34.04	19.68	20	58.07	47.67
B22AOTSCH01 ²	School coder: other attendance	Coder	4,390	28.9	18.31	1,680	25.01	16.28	2,480	29.06	18.31	240	54.7	36.22
B22AOMJ1A	Original major coder: base-year NPSAS degree/certificate	Coder	3,590	26.43	17.9	1,660	23.47	16.38	1,790	28.08	17.9	130	41.8	29.93
B22AOTMAJ201 ²	Major 2 coder: other attendance	Coder	180	26.32	16.95	90	23.68	14.99	80	27.99	16.95	10	39.24	42.05
B22AOTOMAJ01 ²	Original major coder: other attendance	Coder	950	23.16	15.81	360	21.18	14.36	540	23.44	15.81	50	34.86	25.93
Noncoder forms														
B22ANENRLFT	NPSAS attendance: months full-time in years 2/3	Calendar	10,520	51.93	39.53	5,040	53.48	39.86	5,080	47.73	38.04	410	85.28	70.61
B22DWRKMON01 ²	Employer 1: months worked for pay in years 2/3	Calendar	14,400	43.87	32.41	5,730	42.04	31.69	7,970	42.47	31.31	700	75.05	61.63
B22AOTENRLFT01 ²	Other attendance: months full-time in years 2/3	Calendar	4,400	42.09	31.04	1,750	40.87	29.02	2,450	40.71	30.67	200	69.13	58.27
B22ANENRLPT	NPSAS attendance: months part-time in years 2/3	Calendar	2,650	41.25	25.97	900	41.89	25.91	1,630	38	24.74	120	79.79	60.9
B22BACDPART	Primary institution: participation in academic activities	Grid	16,350	41.08	32.17	6,860	41.25	31.85	8,730	38.41	30.62	760	69.91	64.11
B22ALAST	NPSAS attendance: last date	Drop-down	30	40.35	21.48	10	28.04	21.11	20	45.39	22.68	#	#	#
B22BFAMCOMM	Frequency of communication with family and friends	Grid	14,300	36.01	28.52	6,240	34.97	27.81	7,450	34.1	28.09	610	70.15	68.19
B22BSRVUSE	Primary institution: used school services	Grid	14,300	35.5	27.03	6,240	35.94	28.17	7,460	31.41	25.34	610	81.28	80.08
B22FCVTHINK	Primary institution: COVID social/academic experiences	Grid	13,050	35.32	27.66	5,820	33.41	26.67	6,650	33.3	27.3	570	78.2	73.96
B22DEDBENFTS	Importance of nonmonetary benefits compared to salary	Grid	16,950	33.74	25.42	6,680	32.14	24.48	9,460	31.9	24.84	800	68.97	64.42

Rounds to zero.

¹ Mode of administration is the mode in which the respondent completed the student survey; this mode may be different than starting mode for respondents who completed the survey in more than one session.

² This form may be administered multiple times to each respondent, so each respondent's time value is the average of all administrations. That is, it is the number of seconds to complete all iterations of the form divided by the number of iterations.

NOTE: Analysis includes only completed cases that have two or fewer forms of imputed timing data. Respondents who completed the survey in multiple sessions, total time outliers, and respondents who did not complete the entire survey (partial surveys) were excluded. For individual form time calculations, forms from the front end and locating and incentive sections and forms administered to fewer than 25 respondents were excluded from the analysis. Sample sizes rounded to the nearest 10.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Longitudinal Study (BPS:20/22).

3.4.4 *Contacts to Sample Members*

BPS:20/22 sample members received data collection contacts of various forms, including mailings, e-mails, and SMS text messages. In addition, nonrespondents began receiving telephone call reminders starting 1 to 3 weeks after initial mailing contacts in addition to periodic reminder e-mails, mailings, and SMS text messages. On average, sample members received 33 contacts during the 9-month data collection period. Sample members received, on average, six calls and around five text messages during the course of data collection. Table 14 shows the average number of contacts by response status and mode.

Table 14. Average number of contacts, by response status and mode of administration: 2022

Response status and mode	Eligible sample ¹	Contact mode									
		Total contacts ²		Total telephone calls		Total mailings		Total e-mails		Total text messages	
		Number	Average per case	Number	Average per case	Number	Average per case	Number	Average per case	Number	Average per case
Total	34,280	1,113,170	32.5	202,240	5.9	244,580	7.1	488,520	14.3	177,820	5.2
NPSAS:20 (prior-round) response status											
Survey respondents	25,250	731,180	29.0	126,800	5.0	164,310	6.5	322,120	12.8	117,960	4.7
Survey nonrespondents	4,480	211,450	47.2	43,530	9.7	43,720	9.8	91,730	20.5	32,470	7.2
Administrative-only respondents	4,550	170,540	37.5	31,910	7.0	36,560	8.0	74,680	16.4	27,390	6.0
BPS:20/22 response status											
Completed full survey	18,850	289,310	15.3	28,400	1.5	79,770	4.2	144,650	7.7	36,490	1.9
Completed abbreviated survey ³	2,660	124,840	46.9	26,320	9.9	26,920	10.1	50,400	18.9	21,210	8.0
Partial survey respondent	810	46,390	57.2	10,230	12.6	8,690	10.7	18,790	23.2	8,680	10.7
Nonrespondent or exclusion ⁴	11,950	652,620	54.6	137,300	11.5	129,200	10.8	274,690	23.0	111,440	9.3
BPS:20/22 mode of administration ⁵											
Web survey (all)	20,400	380,130	18.6	48,380	2.4	98,660	4.8	180,670	8.9	52,430	2.6
Web survey (excluding early response cases)	11,490	315,470	27.5	48,290	4.2	76,230	6.6	141,240	12.3	49,700	4.3
Telephone survey	1,110	34,020	30.6	6,340	5.7	8,030	7.2	14,370	12.9	5,280	4.7

¹ Excludes 3,060 cases determined to be ineligible for the study using data obtained from one or more sources.

² Total contacts includes all telephone calls, mailings, e-mails, and text messages sent (at a case level).

³ Completed abbreviated survey count includes both English and Spanish survey completes.

⁴ Exclusion cases are sample members who are out of the country, unavailable for the duration of study, deceased, institutionalized or incarcerated, or incapacitated.

⁵ Mode of administration includes only completed cases in the mode in which the respondent completed the survey; this mode may be different than starting mode for respondents who completed the survey in more than one session. Excludes the 810 partial survey respondents.

NOTE: NPSAS:20 = 2019–20 National Postsecondary Student Aid Study. Sample sizes rounded to the nearest 10. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

3.4.5 Refusal Conversion

For the purposes of data collection, a “refusal” occurred when a sample member refused to participate or when someone else refused on a sample member’s behalf (i.e., a “gatekeeper”). Overall, 4 percent of sample members had a refusal; of those, 18 percent subsequently completed the survey (table 15).

Sample members provided a variety of reasons for refusing to complete the survey. The most common reasons provided were general phrases such as “not interested,” “too busy,” or “take me off your calling list.” Other less common reasons for refusing

were not wanting the government to have their information, survey legitimacy/ confidentiality concerns, and concerns about the overall length of the survey.

Table 15. Refusal and refusal conversion rates, by prior-round response status and control and level of institution: 2022

Prior response status and control and level of base-year institution	Eligible sample ¹	Ever any refusal ¹		Surveyed after refusal		
		Number	Percent of eligible	Number	Percent of refusals	Percent of eligible
Total	34,280	1,430	4.2	260	17.8	0.74
NPSAS:20 (prior-round) response status						
Survey respondents	25,250	840	3.3	200	23.3	0.77
Survey nonrespondents	4,480	310	7.0	20	6.7	0.47
Administrative-only respondents	4,550	280	6.1	40	14.0	0.86
Control of base-year institution						
Public	23,360	1,010	5.5	180	17.5	0.75
Private nonprofit	6,340	270	4.4	60	20.9	0.88
Private for-profit	4,580	160	3.6	20	14.8	0.50
Level of base-year institution						
Less-than-2-year	1,220	40	4.0	10	16.7	0.49
2-year	12,980	560	4.1	90	16.5	0.71
4-year, non-doctorate-granting	9,030	370	4.3	80	21.3	0.86
4-year, doctorate-granting	11,050	470	2.1	80	16.8	0.71
Control and level of base-year institution						
Public less-than-2-year	330	20	5.5	#	16.7	0.92
Public 2-year	11,310	500	4.4	90	17.5	0.77
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	2,390	90	3.6	20	17.4	0.63
Public 4-year, non-doctorate-granting, primarily baccalaureate	2,130	100	4.5	20	22.1	0.99
Public 4-year, doctorate-granting	7,210	310	4.3	50	16.1	0.69
Private nonprofit 2-year or less	220	10	4.0	#	11.1	0.45
Private nonprofit 4-year, non-doctorate-granting	2,750	110	4.1	30	23.0	0.95
Private nonprofit 4-year, doctorate-granting	3,370	150	4.3	30	19.9	0.86
Private for-profit less-than-2-year	880	20	2.1	#	16.7	0.34
Private for-profit 2-year	1,470	50	3.5	#	7.8	0.27
Private for-profit 4-year	2,230	90	3.8	20	18.6	0.72

¹ Refusals include eligible sample members who ever refused on BPS:20/22 or had a gatekeeper (parent or other contact) refuse on their behalf.

² Excludes 3,060 cases determined to be ineligible for the study using data obtained from one or more sources.

NOTE: NPSAS:20 = 2019–20 National Postsecondary Student Aid Study. Sample sizes rounded to the nearest 10. Sector reflects institution categorization as determined from the 2018–19 Integrated Postsecondary Education Data System (IPEDS) files, which served as the basis for the sampling frame. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

3.4.6 Contact Tailoring by Phase

As discussed in section 3.2.4, the BPS:20/22 data collection used different data collection protocols, group 1 (the default protocol) and group 2 (the aggressive protocol), based on the success of these protocols in previous studies.

To summarize, the two data collection groups were defined as follows:

- Group 1: NPSAS:20 survey respondents. Sample members who responded to NPSAS:20 and self-identified that they began their postsecondary

education between July 1, 2019, and June 30, 2020 received the default data collection protocol.

- Group 2: NPSAS:20 survey nonrespondents and administrative-only cases. This group includes NPSAS:20 survey nonrespondents and NPSAS:20 administrative-only sample who were potential 2019–20 academic year FTBs based on administrative data. The additional data collection efforts used for group 2 (discussed below) were needed to maximize participation of NPSAS:20 survey nonrespondents and sample members who had never been contacted (i.e., NPSAS:20 administrative-only cases).

These data collection groups received a targeted data collection protocol that varied in intensity to increase response rates and decrease the potential for nonresponse bias. Specifically, the treatments included (1) a post-paid incentive (both data collection groups); (2) a prepaid incentive (group 2 only); (3) mode tailoring, that is, contacting sample members in their NPSAS:20 completion mode; (4) telephone prompting; (5) an incentive boost; and (6) an abbreviated survey. Both groups received an initial e-mail, text, and letter, and reminder e-mails, texts, and postcards to complete the survey throughout data collection. An overview of the differences in the data collection protocol are shown in table 16.

Table 16. Data collection group assignments, by protocol and data collection phase: 2022

	Group 1 (default protocol)	Group 2 (aggressive protocol)
Data collection phase	NPSAS:20 survey respondents (<i>n</i> = 26,470)	NPSAS:20 nonrespondents (<i>n</i> = 5,510) Administrative-only (<i>n</i> = 5,350)
Early completion phase	Data collection announcement mail, text, and e-mail \$30 post-paid incentive Mode tailoring using NPSAS:20 completion mode	Data collection announcement mail, text, and e-mail \$2 prepaid incentive + \$45 post-paid incentive Outbound calling started 2 weeks after mail-outs and continued through all phases
Production phase I	Targeted light CATI outbound begins Postcard, e-mail, and text message reminders began and continued throughout all phases	Postcard, e-mail, and text message reminders began and continued throughout all phases
Production phase II	\$10 boost incentive offered	\$20 boost incentive offered
Nonresponse conversion phase	Abbreviated survey offered	Abbreviated survey offered

NOTE: CATI = computer-assisted telephone interviewing; NPSAS:20 = 2019–20 National Postsecondary Student Aid Study. Sample sizes correspond to assignment prior to data collection. Sample sizes rounded to the nearest 10. The abbreviated survey was offered as a survey nonresponse conversion effort to cases that had not yet started the survey. The Spanish survey was only available in abbreviated form and therefore was not a part of the nonresponse conversation phase.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table 17 displays the completion rates for all eligible fielded cases at the conclusion of each data collection phase. Despite the increased data collection efforts for group 2, completion rates for this group are consistently lower in all data collection phases compared to group 1. For example, 32 percent of eligible sample members in group 1 responded by the end of the early completion phase, and 53 percent responded by the end of production phase I. Comparatively, for group 2, 11 percent of sample members responded by the end of the early completion phase, and 21 percent responded by the end of production phase I. Overall, 72 percent of sample members in group 1 responded to the BPS:20/22 survey, while 36 percent of sample members in group 2 responded.

Table 17. Completion rates at the end of each data collection phase, by data collection protocol group: 2022

Data collection phase	Data collection protocol group ¹					
	Group 1 (Default)			Group 2 (Aggressive)		
	Eligible sample ²	Number ³	Percent	Eligible sample ²	Number ³	Percent
Early completion phase	25,250	7,960	31.5%	9,030	980	10.9%
Production phase I	25,250	13,310	52.7%	9,030	1,910	21.2%
Production phase II	25,250	16,300	64.6%	9,030	2,610	28.9%
Nonresponse conversion phase	25,250	18,290	72.4%	9,030	3,220	35.7%
Overall	25,250	18,290	72.4%	9,030	3,220	35.7%

¹ Data collection activities were determined by the sample member's assigned data collection protocol group. Group 1 consisted of 2019–20 National Postsecondary Student Aid Study (NPSAS:20) survey respondents. Group 2 consisted of NPSAS:20 survey nonrespondents and administrative-only cases.

² Excludes 3,060 sample members determined to be ineligible for the study using data from one or more sources.

³ Number of completes excludes 810 respondents who met the criteria as a survey respondent but were partial completes. Partial survey respondents are defined as sample members who completed at least the enrollment section of the survey but did not complete the entire survey.

NOTE: Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

As indicated in table 16, both data collection groups were offered an abbreviated survey during the nonresponse conversion phase. Overall, 14,380 sample members were offered the abbreviated survey and 16 percent completed it.

Table 18 presents the results for the abbreviated survey.

Table 18. Number and percentage of sample members who were offered and completed the abbreviated survey, by data collection protocol group

Data collection protocol group ¹	Eligible sample ²	Offered abbreviated survey ³		Completed the abbreviated survey ⁴	
		Number	Percent of eligible sample	Number	Percent of offered abbreviated survey
Total	34,280	14,380	41.9%	2,340	16.3%
Group 1	25,250	12,030	47.7%	1,780	14.8%
Group 2	9,030	2,340	26.0%	560	24.1%

¹ Data collection activities were determined by the sample member's assigned data collection protocol group. Group 1 consisted of 2019–20 National Postsecondary Student Aid Study (NPSAS:20) survey respondents. Group 2 consisted of NPSAS:20 survey nonrespondents and administrative-only cases.

² Excludes 3,060 cases determined to be ineligible for the study using data obtained from one or more sources.

³ Includes only cases offered the abbreviated survey as part of survey nonresponse conversion efforts.

⁴ Excludes 319 Spanish abbreviated survey cases that completed the survey prior to the nonresponse conversion phase in which the abbreviated survey was offered.

NOTE: Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

3.5 Evaluation of Survey Items

This section evaluates the survey items, including analyses of the performance of the instrument coders, the use of help text to assist sample members, and item-level nonresponse.

3.5.1 Coder System Usability

Coder usability rates (e.g., successful rate of coding strings within the survey) were reported by coder type. Success rates for each coding system were calculated by dividing the number of strings with a complete code selected, by either the respondent or a DCI in the survey, by the total number of strings collected. Analysis of successful coding in the survey was limited to strings from cases with a final complete or final partial complete status. Overall, 92.5 percent of all text strings were successfully coded (table 19).

The major coder had the highest overall success rate, with 98 of the entered strings being coded in the survey, followed by the postsecondary institution coder with an overall success rate of 96 percent, and finally the occupation coder with a success rate of 85 percent. Overall coding success rates by mode of administration ranged from 95 percent for web nonmobile mode, 92 percent for telephone mode, and 91 percent for web mobile mode. Table 19 shows a summary of coding success rate, by mode of administration and coding system.

Table 19. Percentage of strings coded in survey, by mode of administration and coding system: 2022

Coding system	Mode of administration ¹			
	Overall	Web nonmobile	Web mobile	Telephone
Total	92.5	95.2	90.7	92.3
Major ²	97.6	98.7	96.8	96.7
Postsecondary institution ²	95.6	97.6	94.8	91.4
Occupation	85.2	89.2	82.6	88.5

¹ Mode of administration is the mode in which the respondent completed the survey; this mode may be different than starting mode for respondents who completed the survey in more than one session.

² When a form is administered multiple times to a respondent (e.g., the postsecondary institution coder is administered for every additional school attended), the coder system success rate is calculated using the total number of strings across respondents (i.e., the number of successful coding attempts of the form divided by the number of strings collected).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

3.5.2 Help Text

Respondents and DCIs were able to click a help button provided on each form to obtain form-specific help text. Additionally, some questions provided embedded hyperlinks within question wording or response option text to prompt respondents to access the help text on the form. Whether accessed through the help button or through an embedded hyperlink, the help text provided definitions of key terms and phrases used on the form and any explanations thought to help standardize interpretation for respondents.

Overall, respondents and DCIs accessed help text less than 1 percent of the time. Overall help text access rates were calculated by dividing the number of times help text was accessed by the number of times forms were administered. The rate of help text access was also analyzed at the form level, overall and by mode of administration, to identify specific forms that may have been difficult for users. Form-level help text access rates were calculated by dividing the number of times help text was accessed on each form by the number of times that form was administered. Table 20 shows the 10 forms administered to at least 10 percent of respondents for which help text was accessed at a rate of 1 percent or greater, by mode of administration.

Aware of emergency aid program at primary institution (B22CEAIDAWARE) was administered to 17,520 respondents and had the highest rate of help text access at 6 percent. The term “emergency aid program” was displayed as an embedded help text hyperlink that defined these types of programs. The lower familiarity with these programs and the embedded hyperlink likely contributed to the higher help

text access rate.¹⁰ *Employer 1: work-study job* (B22DWRKSTD01) had the next highest level of help text access at 4.5 percent. This question asked respondents in each employer loop whether the employment was a work-study job. The phrase “work-study” was displayed with an embedded help text hyperlink that defined the concept and requirements of work-study positions.

DCIs are trained to use survey tools such as help text, which can often lead to higher help text access rates in telephone mode than in self-administered web modes. For example, the help text access rate for DCIs was 13 percent for B22CEAIDAWARE, statistically significantly higher than the rate of web nonmobile and web mobile modes (9 percent and 4 percent, respectively, $p < .0001$). This pattern of higher help text access rates for DCIs holds for the majority of forms reported in table 20, although not all differences between telephone and other modes are statistically significant.

¹⁰ Results from BPS:20/22 cognitive testing indicated that students typically have little awareness regarding emergency aid programs (see appendix J for details on cognitive testing). This question also had the highest help text access rate in the BPS:20/22 field test and NPSAS:20, this cohort’s base-year survey.

Table 20. Forms with at least 1 percent help text access rate, by mode of administration: 2022

Form ¹	Form description	Overall		Web nonmobile		Web mobile		Telephone	
		Analysis cases	Percent of help text access	Analysis cases	Percent of help text access	Analysis cases	Percent of help text access	Analysis cases	Percent of help text access
B22CEAIDAWARE	Aware of emergency aid at primary institution	17,520	6.4	7,430	9.0	9,300	3.9	800	13.2
B22DWRKSTD01	Employer 1: work-study job	2,950	4.5	1,470	5.2	1,350	3.6	120	6.5
B22ECARRYBAL	Credit card balance carried over each month	12,260	4.1	4,590	5.7	7,020	3.0	650	4.6
B22ASTDABR	Participated in a study-abroad program at NPSAS in years 2 and 3	12,650	2.4	5,860	2.4	6,310	2.4	480	3.1
B22ADBLMAJ	Declared major for base-year NPSAS degree/certificate	14,320	1.8	6,350	2.0	7,370	1.6	600	2.8
B22DEDBENFTS	Importance of nonmonetary characteristics compared to salary	18,850	1.5	7,370	1.2	10,610	1.6	870	2.3
B22DEMPLOY01	Employer 1: employer name	29,060	1.4	11,480	1.2	15,960	1.7	1,620	1.0
B22AOTDBLM01	Other attendance 1: declared major for degree/certificate	6,640	1.3	2,220	1.5	4,060	1.1	360	1.7
B22FHOUSEC	Housing security in the last 12 months	18,850	1.1	7,370	1.8	10,610	0.5	870	1.2
B22AMJCHGNUM	Number of times formally changed major for base-year NPSAS degree/certificate	11,720	1.0	5,560	1.7	5,730	0.4	430	0.9

¹ Results are based on respondent behavior on the survey forms and may not align with variables found in source files on the restricted-use file.

NOTE: Mode of administration is the mode in which the respondent completed the survey; this mode may be different than starting mode for respondents who completed the survey in more than one session. This table only includes those items that were administered to at least 10 percent of respondents. Due to exclusions resulting from data processing, the analysis cases may differ from the total cases administered the form. Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals due to rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

3.5.3 Item-Level Nonresponse

An item in the survey is a single response option (e.g., check box) or set of response options (e.g., radio button list) that will make a variable in data processing. A form in the survey can include more than one item. The rate of nonresponse for individual items in the survey is used to identify potentially burdensome survey questions and to better understand the experiences of respondents completing the survey. Item-level nonresponse rates in the BPS:20/22 survey were calculated for all items administered to at least 25 respondents, and items with a nonresponse rate of 10 percent or more are reported. Of the 510 items administered to more than 25 respondents, only 1 percent (six items) had a nonresponse rate of 10 percent or more. Table 21 shows item nonresponse rates for these six items, overall and by mode of administration, ordered by item nonresponse rate.

Table 21. Item nonresponse for items with 10 percent or more of data missing, by mode of administration: 2022

Item ¹	Item Description	Overall		Web nonmobile		Web mobile		Telephone	
		Analysis cases	Percent missing	Analysis cases	Percent missing	Analysis cases	Percent missing	Analysis cases	Percent missing
B22FLGBTQ_DK	Sexual orientation: don't know: specify	490	65.4	200	64.7	270	66.7	10	44.4
B22FGENDERQR_OTHER	Gender identity: genderqueer or gender nonconforming: specify	550	23.6	230	18.9	300	26.3	20	43.8
B22AOTMJSPE105	Other attendance 5: major 1: specific code	30	16.1	10	33.3	20	10.0	#	#
B22AOTMJGEN105	Other attendance 5: major 1: general code	30	16.1	10	33.3	20	10.0	#	#
B22FNOTSURE_OTHER	Gender identity: a different identity: specify	120	16.1	40	7.3	70	21.1	10	16.7
B22AOTCMPDGN04	Other attendance 4: completed degree or certificate	30	14.3	10	22.2	20	13.3	#	#

Rounds to zero.

¹ Results are based on respondent behavior on the survey forms and may not align with variables found in source files on the restricted-use file.

NOTE: Mode of administration is the mode in which the respondent completed the survey; this mode may be different than starting mode for respondents who completed the survey in more than one session. This table only includes those items that were administered to at least 25 respondents. Due to exclusions resulting from data processing, the analysis cases may differ from the total cases administered the form. Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Chapter 4. Administrative Records

Matching Overview and Outcomes

Administrative data are available for the BPS:20/22 cohort to supplement the survey data and reduce respondent burden. Table 22 identifies the administrative sources available for the BPS:20/22 cohort across all rounds of data collection and indicates whether the data were new, refreshed to include updated data, or carried forward from the prior round.

Table 22. Selected administrative data sources for NPSAS:20 and BPS:20/22

Data source	NPSAS:20	BPS:2022
SAT	New	Carried Over
ACT	New	Carried Over
Student Records	New	Carried Over
Central Processing System (CPS)	New	Refreshed
National Student Loan Data System (NSLDS)	New	Refreshed
National Student Clearinghouse (NSC)	New	Refreshed
Veterans Benefits Administration (VBA)	New	Refreshed

NOTE: Carried over = data carried over from previous round and not refreshed; New = new data source; Refreshed = data carried over from previous round and refreshed.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2019–20 National Postsecondary Student Aid Study (NPSAS:20), and 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

4.1 Administrative Records Matching Process and Outcomes

As indicated in table 22, in addition to the survey, sample member data for BPS:20/22 came from selected administrative databases, including two databases from the U.S. Department of Education’s Federal Student Aid (FSA) Office: CPS and NSLDS. Other data sources included the NSC and VBA. This chapter provides details on the processes used to match these administrative data and the outcomes associated with each process.

4.1.1 Central Processing System (CPS)

Federal financial aid application data were obtained from CPS. As part of the process of applying for federal student financial aid, students enter information about themselves and their family into the FAFSA form. CPS then processes the FAFSA information and provides it to postsecondary institutions as part of the process for determining student eligibility for federal financial aid. CPS data were collected for the BPS:20/22 sample for the 2020–21, 2021–22, and 2022–23 financial aid years. CPS ID—the student’s SSN concatenated with the first two letters of the sample member’s last name—was used to match BPS:20/22 and CPS records. Any SSNs that had been obtained since the previous match were included in later matches. Records without sample members’ SSNs were not submitted for matching to CPS. Data were transmitted to FSA using their SSL-encrypted website, and from FSA to project staff using EdConnect, a software program provided by the Department of Education to securely transmit data. Table 23 summarizes the results of CPS matching for academic years 2020–21, 2021–22, and 2022–23 with the number of cases sent and matched. The match rates were approximately 70 percent, 55 percent, and 45 percent, respectively. The decline in CPS match rates over the three academic years is most likely to be primarily driven by reductions in postsecondary enrollment over time (e.g., due to degree/certificate completion, stopout, or dropout) among BPS sample members. Since the CPS match rate is not conditional on enrollment (the denominator is based on all BPS sample members for whom an SSN was available), the CPS match rate is expected to decline over time if fewer sample members are enrolled in postsecondary education in later years, as is typical for BPS cohorts. Fellowship and assistantship recipients who did not otherwise receive federal aid were not included because these aid sources are not usually need based and therefore do not require the completion of a federal financial aid application.

Table 23. Results of Central Processing System (CPS) matching, by academic year: 2020–21, 2021–22, and 2022–23

Academic year	Total sample	Sent to CPS ¹		Matched to CPS	
	Number	Number	Percent of eligible	Number	Percent of sent
2020–21	37,330	35,130	94.1	24,430	69.6
2021–22	37,330	35,390	94.8	19,490	55.1
2022–23	37,330	35,440	94.9	15,870	44.8

¹ Records for sample members without Social Security numbers (SSNs) were not sent to CPS.

NOTE: The number of SSNs sent for matching may vary slightly by data source because different agencies have different requirements for matching identifiers and because SSNs, names, and dates of birth are updated for accuracy and completeness throughout the administrative records matching process. Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

4.1.2 National Student Loan Data System (NSLDS)

Project staff obtained student-level data on Pell Grants and federal student loans by matching sample members to the NSLDS database. In a cooperative effort, project staff and the U.S. Department of Education conducted a match between BPS:20/22 records and NSLDS once during the data collection period. As with CPS, sample members missing SSNs were not part of the match. A BPS:20/22 sample member had to have at least one valid grant or loan record in the NSLDS database for a successful match. The NSLDS Pell Grant and loan files included a complete Pell Grant and loan history for each student. All NSLDS data transfers used a password-protected NCES system, transmitting over an SSL-encrypted connection. NSLDS matching only returned records of sample members who, at some point in time during their postsecondary education enrollment, had received Pell Grant or federal student loan funding. The NSLDS match yielded loan records for 17,200 sample members, 49 percent of sample members. The match yielded Pell Grant records for 19,790 sample members, or 56 percent of sample members. Table 24 shows the overall NSLDS match rates for sample members.

Table 24. Results of National Student Loan Data System (NSLDS) loan and Pell Grant matching: 2023

Sample members	Number	Percent
Total Sample	37,330	100.0
Sent to NSLDS	35,440	94.9
Not sent to NSLDS ¹	1,900	5.1
NSLDS loans		
Matched	17,200	48.5
Did not match	18,230	51.5
Pell Grants		
Matched	19,790	55.8
Did not match	15,660	44.2

¹ Records for sample members without Social Security numbers (SSNs) were not sent to NSLDS.

NOTE: Matching was completed on historical files that include awards made in 2022–23 and prior years. Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding. The number of Social Security numbers (SSNs) sent for matching may vary slightly by data source because different agencies have different requirements for matching identifiers and because SSNs, names, and dates of birth are updated for accuracy and completeness throughout the administrative records matching process.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

4.1.3 National Student Clearinghouse (NSC)

Enrollment and attainment data were obtained for the sample from the NSC StudentTracker service. This administrative record match provided information on institutions attended, enrollment dates, and degree completions. An individual student record would match with NSC only if the student’s institution was an NSC participant. All files were encrypted and transmitted over encrypted Secure

File Transfer Protocol connections. Personally identifying data used for the match included sample member name, SSN, and DOB. NSC matches for sample members included the institution in which they were sampled for NPSAS and any other participating institutions they had attended as of September 2022. Of the total sample members, 33,430 or 90 percent, matched to NSC (table 25). Reasons for a nonmatch include unavailable personally identifying data and school nonparticipation.

Table 25. Results of National Student Clearinghouse (NSC) matching: 2022

Sampled Cases	Number	Percent
Total Sample	37,330	100.0
Matched	33,430	89.5
Did Not Match	3,910	10.5

NOTE: All sampled cases were provided for the match. Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

4.1.4 Veterans Benefits Administration (VBA)

A file match with VBA was performed to identify veterans, amounts of federal veterans' education benefits, and any associated enrollment information. During data collection, project staff provided a file containing SSN, name, and DOB to VBA for data matching. The match used SSN as the primary identifier, with the other information used to ensure a match to the proper person. As with NSLDS file matching, all data transmissions used an NCES secure file transfer system. Federal veterans' education benefits information was obtained for approximately 1,320 (4 percent) of the sample members sent for matching (table 26).

Table 26. Results of Veterans Benefits Administration (VBA) matching: 2022

Sampled Cases	Number	Percent
Total Sample	37,330	100.0
Sent	35,250	94.4
Matched	1,320	3.7
Did Not Match	33,930	96.3
Not Sent	2,080	5.6

NOTE: All cases with a valid Social Security number (SSN) were provided for the match. Sample sizes rounded to the nearest 10.

Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding. The number of SSNs sent for matching may vary slightly by data source because different agencies have different requirements for matching identifiers and because SSNs, names, and dates of birth are updated for accuracy and completeness throughout the administrative records matching process.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

4.2 Administrative Records Matching Quality Control

Several checks were performed to verify the quality of the administrative data received. File layouts and code to read in the files were compared to ensure that the code was current and accurate. File completeness was also checked; for example, with NSLDS, we compared the variables in the files against a data dictionary to ensure all variables were present. If a source provided PII from their database, names and dates of birth were compared to what was in the study database to make sure the data were for the correct person. If the names and dates of birth did not match, the case was excluded from the administrative file. Project staff examined individual data files by running basic summary statistics such as number of records and value ranges (e.g., dates and amounts) to check for potential outliers or abnormalities. Then the files were checked to confirm they were related to one another as expected. For example, student veterans who received veterans' education benefit payments were expected to have at least one military service record.

Chapter 5. Data File Processing and Preparation

BPS:20/22 unit-level data were compiled from surveys and matches to government and other administrative databases. These files are fully documented and available to researchers as a set of restricted-use, microlevel data files. The public may generate tables of estimates and simple linear and logistic regressions based upon restricted-use data via PowerStats and other publicly facing web tools available on the NCES website (<https://nces.ed.gov/datalab>). This chapter provides details on the contents of the restricted-use BPS:20/22 files, post-data-collection editing of the survey data files, and creation of analysis variables.

5.1 Overview of the BPS:20/22 Study Files

Source and derived data for BPS:20/22 are contained in restricted-use files and documented in detail in the associated codebooks. Source files are the individual data files collected as part of the study (e.g., the student interview data, administrative data from NSLDS). The derived data file combines information from these individual source files into one data set that is designed for research use. While researchers will primarily use the derived data file, the source data files are included in the restricted-use files as well, as they may be of interest for particular research projects. The restricted-use files listed in table 27 are available to researchers who have applied for and received authorization from NCES to access those files. Researchers may obtain authorization by contacting the Institute of Education Sciences Data Security Office (see <https://nces.ed.gov/statprog/rudman>).

The primary analysis file (or derived file) for the BPS:20/22 cohort contains data for approximately 22,300 cohort members who responded to NPSAS:20 or BPS:20/22 and more than 950 variables, 550 of which were created using data collected for BPS:20/22. NSC and VBA data were also used to create analysis variables, although the source files are not available as restricted-use data files.

Table 27. BPS:20/22 restricted-use files, description, and file path: 2022

Restricted-use file	Description	File path
BPS:20/22 analysis (derived)	Contains the analytic variables derived from all data sources as well as selected student survey variables for the 22,320 student study respondents to the BPS:20/22 student survey.	\\DATA\DERIVED\BPS22DERIVED_DATAFILE.CSV
CPS 2019–20 data	Contains data from CPS and NSLDS for the study respondents whose records matched to a 2019–20 financial aid application (FAFSA). There is a separate record for each BPS:20/22 survey respondent.	\\DATA\SOURCE\BPS22CPS20\BPS22CPS20_DATAFILE.CSV
CPS 2020–21 data	Contains data from CPS and NSLDS for the BPS:20/22 survey respondents whose records matched to a 2020–21 financial aid application (FAFSA). There is a separate record for each BPS:20/22 survey respondent.	\\DATA\SOURCE\BPS22CPS21\BPS22CPS21_DATAFILE.CSV
CPS 2021–22 data	Contains data from CPS and NSLDS for the BPS:20/22 survey respondents whose records matched to a 2021–22 financial aid application (FAFSA). There is a separate record for each BPS:20/22 survey respondent.	\\DATA\SOURCE\BPS22CPS22\BPS22CPS22_DATAFILE.CSV
CPS 2022–23 data	Contains data from CPS and NSLDS for the BPS:20/22 survey respondents whose records matched to a 2022–23 financial aid application (FAFSA). There is a separate record for each BPS:20/22 survey respondent.	\\DATA\SOURCE\BPS22CPS23\BPS22CPS23_DATAFILE.CSV
CPS 2023–24 data	Contains data from CPS and NSLDS for the BPS:20/22 survey respondents whose records matched to a 2023–24 financial aid application (FAFSA). There is a separate record for each BPS:20/22 survey respondent.	\\DATA\SOURCE\BPS22CPS24\BPS22CPS24_DATAFILE.CSV
FEMA data	Contains data from FEMA for the institutions BPS:20/22 survey respondents attended. There is a separate record for each institution.	\\DATA\SOURCE\BPS22FEMA\BPS22FEMA_DATAFILE.CSV
Imputation flag	Contains a flag indicating a variable's imputation status. There is a separate record for each BPS:20/22 survey respondent.	\\DATA\SOURCE\BPS22FLAG\BPS22FLAG_DATAFILE.CSV
NPSAS:20 institution data	Contains institution-level data for the BPS:20/22 sample members' NPSAS institution collected during 2019–20.	\\DATA\SOURCE\BPS22N20INSTITUTION\BPS22N20INSTITUTION_DATAFILE.CSV
NPSAS:20 student records	Contains base-year data included in the NPSAS:20 data file collected for the BPS:20/22 survey respondents from institutional records. There is a separate record for each BPS:20/22 survey respondent.	\\DATA\SOURCE\BPS22N20STUDRECS\BPS22N20STUDRECS_DATAFILE.CSV
NPSAS:20 student survey	Contains base-year data included in the NPSAS:20 data file collected for the BPS:20/22 survey respondents from the student survey. There is a separate record for each BPS:20/22 survey respondent.	\\DATA\SOURCE\BPS22N20SURVEY\BPS22N20SURVEY_DATAFILE.CSV
NSLDS award origin	Contains raw award-level data extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file of loan awards with separate records for each transaction in the award file.	\\DATA\SOURCE\BPS22NSLDSCOD_LOAN\BPS22NSLDSCOD_LOAN_DATAFILE.CSV
NSLDS enrollment status	Contains raw enrollment-level data extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file of loan enrollments with separate records for each transaction in the enrollment file.	\\DATA\SOURCE\BPS22NSLDSENRL_PGM_STATUS_CERT\BPS22NSLDSENRL_PGM_STATUS_CERT_DATAFILE.CSV
NSLDS FAFSA history	Contains raw data at the FAFSA level extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file with separate records for each transaction in the FAFSA file.	\\DATA\SOURCE\BPS22NSLDSFAFSA\BPS22NSLDSFAFSA_DATAFILE.CSV
NSLDS loan deferment	Contains raw deferment-level data extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file of loan deferrals with separate records for each transaction in the loan file.	\\DATA\SOURCE\BPS22NSLDSLOAN_DEFER\BPS22NSLDSLOAN_DEFER_DATAFILE.CSV

See notes at end of table.

Table 27. BPS:20/22 restricted-use files, description, and file path: 2022—Continued

Restricted-use file	Description	File path
NSLDS loan delinquency	Contains raw delinquency-level data extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file of loan delinquency status with separate records for each transaction in the loan file.	\\DATA\\SOURCE\\BPS22NSLDSLOAN_DELINQ\\BPS22NSLDSLOAN_DELINQ_DATAFILE.CSV
NSLDS loan disbursement	Contains raw loan-disbursement-level data extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file with separate records for each transaction in the loan distribution file.	\\DATA\\SOURCE\\BPS22NSLDSLOAN_DISB\\BPS22NSLDSLOAN_DISB_DATAFILE.CSV
NSLDS loan	Contains raw loan-level data extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file with separate records for each transaction in the loan file.	\\DATA\\SOURCE\\BPS22NSLDSLOAN_DTL\\BPS22NSLDSLOAN_DTL_DATAFILE.CSV
NSLDS loan forbearance	Contains raw forbearance-level data extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file of forbearance loans with separate records for each transaction in the loan file.	\\DATA\\SOURCE\\BPS22NSLDSLOAN_FORBEAR\\BPS22NSLDSLOAN_FORBEAR_DATAFILE.CSV
NSLDS loan repayment history	Contains raw loan-repayment-history-level data extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file of repayments with separate records for each transaction in the loan distribution file.	\\DATA\\SOURCE\\BPS22NSLDSLOAN_LAST_PYMT_HIST\\BPS22NSLDSLOAN_LAST_PYMT_HIST_DATAFILE.CSV
NSLDS loan maturation	Contains raw loan-level maturation data received from NSLDS for respondents who received federal loans during 2021–22 or prior years. This is a history file with separate records for each transaction in the loan files.	\\DATA\\SOURCE\\BPS22NSLDSLOAN_MATURE_HIST\\BPS22NSLDSLOAN_MATURE_HIST_DATAFILE.CSV
NSLDS outstanding interest balance	Contains raw outstanding-interest-balance-level data extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file with separate records for each transaction in the loan distribution file.	\\DATA\\SOURCE\\BPS22NSLDSLOAN_OIB\\BPS22NSLDSLOAN_OIB_DATAFILE.CSV
NSLDS outstanding principal balance	Contains raw outstanding-principal-balance-level data extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file with separate records for each transaction in the loan distribution file.	\\DATA\\SOURCE\\BPS22NSLDSLOAN_OPB\\BPS22NSLDSLOAN_OPB_DATAFILE.CSV
NSLDS loan repayment plan	Contains raw loan-repayment-plan-level data extracted from NSLDS for the BPS:20/22 survey respondents who received federal loans during the BPS:20/22 year or prior years. This is a history file of repayment plans with separate records for each transaction in the loan distribution file.	\\DATA\\SOURCE\\BPS22NSLDSLOAN_REPYMT_PLAN\\BPS22NSLDSLOAN_REPYMT_PLAN_DATAFILE.CSV
NSLDS Pell Grant data	Contains raw grant-level data extracted from NSLDS for the BPS:20/22 survey respondents who received Pell Grants during the BPS:20/22 year or prior years. This is a history file with separate records for each transaction in the Pell Grant files	\\DATA\\SOURCE\\BPS22NSLDSPELL_IASG_GRANT\\BPS22NSLDSPELL_IASG_GRANT_DATAFILE.CSV
BPS:20/22 program information	Contains a separate record for each degree program students reported (up to eight programs).	\\DATA\\SOURCE\\BPS22PROGINF\\BPS22PROGINF_DATAFILE.CSV
BPS:20/22 school information	Contains a separate record for each postsecondary institution students reported in the survey as somewhere they had attended during each of the study years (up to five institutions each study year).	\\DATA\\SOURCE\\BPS22SCHINFO\\BPS22SCHINFO_DATAFILE.CSV
BPS:20/22 student survey data	Contains raw data collected from 20/22 student surveys for the BPS:20/22 survey respondents.	\\DATA\\SOURCE\\BPS22SURVEY\\BPS22SURVEY_DATAFILE.CSV
BPS:20/22 weights	Contains all intermediate weight adjustment factors and all weight and variance estimation variables for BPS:20/22. There is a separate record for each BPS:20/22 survey respondent.	\\DATA\\SOURCE\\BPS22WEIGHTS\\BPS22WEIGHTS_DATAFILE.CSV

NOTE: CPS = Central Processing System; FAFSA = Free Application for Federal Student Aid; FEMA = Federal Emergency Management Agency; NSLDS = National Student Loan Data System.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

5.2 Post-Data-Collection Survey Data Editing

During data collection, quality control checks were performed on all survey items to ensure the quality and accuracy of data. Survey item responses were processed into a dataset to be delivered as a restricted-use file (bps22survey_datafile.csv). The survey data file, which serves as a principal source file for derived variables, includes data as reported by respondents. Inconsistencies in self-reported data are neither edited nor documented in source files. During derived variable construction, inconsistencies that are identified may be remedied at that point. Outlier data in continuous variables are not bottom coded, top coded, or blanked out but remain in the source file for reference during derived variable construction. Documentation for the survey variables includes question wording, response options, logical imputations, and administration descriptions (see the full-scale survey instrument in appendix C). Preparing this survey data file was a multifaceted process described in the steps below.

Assigning missing data codes. All missing data from the survey were assigned missing data codes to indicate why data were missing. Project staff defined gate-nest question relationships, in which “gate” questions must first be answered before dependent “nest” questions. Some values were missing due to appropriate question routing (e.g., a respondent with no dependents would not be administered a question about a dependent’s age). These values were assigned a missing data code of -3, “item does not apply.” If a value was missing because the respondent completed the abbreviated survey and the item was excluded from those surveys, the value was assigned a -7, “not included in abbreviated survey.” If enough information was available to determine that a variable not included in the abbreviated survey would still not apply had the respondent been administered the full survey, then a -3 is applied in place of -7. Therefore, all variables not included in the abbreviated survey will not have the same number of -7s. Sometimes an item was not administered when, due to prior missing survey data, it was not possible to determine whether the item applied to the respondent; these items were assigned a value of -4, “Missing—unable to determine applicability.”

Some items received a missing data code when they were administered, but the response could be inferred from other responses. Specifically, if a form displayed multiple items, each with yes/no checkboxes and the respondent checked “yes” for at least one item but left all other items missing, it was assumed that the respondent intended for the missing items to be “no.” A value of -5, to indicate an “implied no/zero,” was assigned.

Assigning these codes during data collection served as a quality control check for the instrument operation, final data file quality, and documentation accuracy. For example, if an investigation revealed survey routing was not operating properly, an update was deployed to the survey and the item was assigned a -8 value (“Missing - survey error”) signifying that the item was missing due to an instrument error. Any remaining missing data were determined to be missing because the respondent did not provide an answer and were assigned a code of -9 (“Missing - response not provided”). See survey missing data codes and descriptions in table 28.

Table 28. BPS:20/22 survey missing data codes and descriptions: 2022

Survey missing data code	Data label
-3	Skipped - not applicable
-4	Missing - unable to determine applicability
-5	Missing - implied no/zero
-7	Missing - not included in abbreviated survey
-8	Missing - survey error
-9	Missing - response not provided

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Applying logical recodes. Logical recodes of values were performed when the value of missing items could be definitively determined (as opposed to implied) from answers to previous survey questions. For example, if a respondent answered “yes” to *Did you vote in the November 2020 presidential election* (B22FVTNEL), then the item, *Have you ever voted in a national, state, or local election* (B22FEVRVT) was skipped and logically recoded to “yes.”

Sanitizing. All open-ended responses collected in the survey were systematically reviewed. Any PII was “sanitized,” or removed from the text string. Any other text was left unchanged. All open-ended text strings released on the restricted-use datasets were sanitized.

Coding. Predictive coding systems, or coder forms, were used to assign codes to standardized data elements such as postsecondary institutions, majors, occupations, and addresses. For each coder form, respondents entered their answer as a text string. As respondents typed, a keyword search of an underlying database returned a list of possible matches that were displayed in a drop-down menu for respondents to select. See section 3.1.3 for detailed coder form descriptions and examples, and see section 3.5.1 for respondent coding rates.

When an item on a coder form was not coded in the survey but an open-ended response was provided, the responses were reviewed to assign a valid code. First, the open-ended responses were automatically compared against a list of established text/code combinations. If the response matched, the corresponding code was automatically assigned for that response. The remaining uncoded responses were loaded into an application where staff searched the coder database and assigned a code when possible. For example, if the respondent provided “Education-math” as their major but did not select a CIP code, the text string would first be compared to the list of established text/code combinations. If the text string “Education-math” existed in the list of previously coded established text/code combinations, a corresponding CIP code would be assigned (e.g., 13.1311 Mathematics Teacher Education). If not, the text string would be loaded into an application for staff review. Then, upon review, staff could assign “Education-math” to CIP code 13.1311 based on the text’s similarity to “Mathematics Teacher Education.” Results of post-data-collection edits to coder responses (i.e., upcoding) are provided below.

5.2.1 Upcoding

Upcoding is a process by which expert coding staff attempt to assign a code to any text string that a respondent or DCI did not select a code for in the survey. Text strings from each coding system went through the upcoding process. Upcode rates were calculated by dividing the number of nonmissing coder values because of expert staff upcoding, by the total number of nonmissing coder values in the survey data. Overall, all coders had upcode rates of 6 percent or less. The *occupation* coder had an upcode rate of 12 percent and the *major* coder had an upcode rate of 2 percent. Table 29 shows the upcode rates for the coding systems, overall and by mode of administration.

Table 29. Summary of upcoding rates, by mode of administration and coding system: 2022

Coding system	Overall	Mode of administration ¹		
		Web nonmobile	Web mobile	Telephone
Total	5.9	3.7	7.3	5.9
Postsecondary institution	2.3	1.3	2.7	4.6
Occupation	12.2	8.7	14.4	9.9
Major	1.9	1.0	2.6	2.4

¹ Mode of administration is the mode in which the respondent completed the survey; this mode may be different than starting mode for respondents who completed the survey in more than one session.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

5.3 Derived Variable Construction

Analysts derived the analytic variables by examining data available from the various data sources, prioritizing the data sources on an item-by-item basis,¹¹ and reconciling discrepancies within and between sources. In some cases, staff created derived variables by assigning a value from the available source with the highest priority or if there was only one source (e.g., variables whose only source is the survey). If there was no available data from any source, the derived variable was imputed (see section 6.5 for more information about imputation procedures). The Derived file provides a refined aggregation of data, informed by logical decision rules, subject matter expertise, and policy considerations. It is considered the main analysis file. As BPS is a fully integrated data product, incorporating multiple sources is a critical step in increasing variable accuracy and reducing missingness. Further detail on variable derivation is available in PowerStats on the “Get more info” tab for each variable, and in the restricted-use file codebooks.

For derived variables, a value of -3 was applied when the variable does not apply to the study respondent (e.g., a respondent with no dependents would not have a valid value derived for number of dependents). Table 30 provides a listing of the reserve codes that are found on the Derivable Variable file. Note that additional reserve codes are used on source files (e.g., see section 5.2 for reserve codes for the survey data file).

For a complete listing of the analysis variables derived for BPS:20/22, see appendix F.

Table 30. BPS:20/22 analysis variable missing data codes and descriptions: 2022

Missing data code	Item source	Description(s)
-3	Any	Item does not apply, i.e., the item was “skipped” or a “legitimate skip”

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

¹¹ Analysts prioritized data sources according to factors such as their relevance, completeness, timing, and intended purpose. As a result, administrative data often received priority over self-reported data (e.g., FAFSA data took precedence over survey responses for financial aid variables). However, in some cases self-reported data offered better insight into the constructs that the derived variables sought to measure and therefore received priority over some administrative sources (e.g., survey responses about race/ethnicity received priority over all other sources).

Chapter 6. Weighting and Variance Estimation

This chapter provides an overview of the statistical analysis weights for the BPS:20/22 sample. Variance estimation, response rates, unit-level and item-level nonresponse bias analysis, imputation for missing data, and disclosure procedures to protect confidentiality are also described.

6.1 Analysis Weight

A single statistical analysis weight (WTA000) was developed for analyzing the BPS:20/22 sample respondents. Because the BPS:20/22 sample is a subset of the NPSAS:20 undergraduate study respondent sample, the weight for analyzing the BPS:20/22 data was derived from the NPSAS:20 undergraduate study respondent weights and can be used to analyze respondents' data from both collections. For more information on the NPSAS:20 weights, please see section 7.1 of the NPSAS:20 DFD (Wine, Seigel, and Haynes 2024).

As described in chapter 2, the BPS:20/22 full-scale sample consists of approximately 37,330 sample members who were NPSAS:20 study respondents and confirmed or potential FTBs. The statistical team designed the BPS:20/22 sample to be nationally representative of individuals who began their postsecondary education for the first time during the 2019–20 academic year at any Title IV eligible postsecondary institution in the United States and who were not concurrently enrolled in high school.

The team also designed the sample to be representative of this population for a selected number of states. The determination of whether the respondent sample was state representative within states was informed by sample response rates, nonresponse bias analysis results, and substantive knowledge about the states. Generally, the following informed the determination:

- the response counts needed to meet thresholds based on power calculations;
- the percentage of characteristics with statistically significant bias and effect size needed to be reasonable relative to other states;

- key estimates needed to be reasonable compared to external estimates; and
- most of the essential institutions (i.e., institutions with greater student enrollment and institutions within large systems) and unique institutions (based on multiple characteristics) needed to have individuals that participated in the BPS:20/22 data collection.

The steps in developing the weight components are described in sections 6.1.1 through 6.1.3 and include the following:

- creating the BPS:20/22 base weight;
- performing nonresponse weight adjustments; and
- performing poststratification adjustments.

Table 31 lists the components used to construct the BPS:20/22 analysis weight. Each weight component represents a probability of selection, a weight adjustment, or an interim weight.

Table 31. Components of final analysis weight: 2022

Weight component	Purpose
NPSAS:20 adjustments	
WTB_WT1	Institution sampling weight
WTB_adj1	Institution nonresponse adjustment
WTB_adj2	Institution poststratification adjustment
WTB_INSTWT	Institution weight; $WTB_INSTWT = WTB_WT1 * WTB_adj1 * WTB_adj2$
WTB_WT4	Student within-institution sampling weight
WTB_WT5	Student sampling weight; $WTB_WT5 = WTB_INSTWT * WTB_WT4$
WTB_adj3	Student multiplicity adjustment
WTB_adj4	Student unknown eligibility status adjustment
WTB_adj5	Student nonresponse adjustment
BPS:20/22 adjustments	
BPS22WTA_adj1	Student subsampling adjustment
BPS22WTA_adj2	Student not located nonresponse adjustment
BPS22WTA_adj3	Student refusal nonresponse adjustment
BPS22WTA_adj4	Student other nonresponse adjustment
BPS22WTA_adj5	Student poststratification adjustment

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2019–20 National Postsecondary Student Aid Study (NPSAS:20) and 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

All nonresponse and poststratification adjustments were computed using the WTADJUST procedure in SUDAAN. SUDAAN is a statistical software package developed by RTI to analyze complex survey data. The WTADJUST procedure used a constrained logistic model to predict response. A key feature of the WTADJUST procedure is that the weight adjustments and weight trimming and smoothing are all accomplished in one step. Bounds can be specified on the weight going into the WTADJUST procedure, and bounds can be specified on the

weight adjustment factors that are output by the WTADJUST procedure. The exact formula for the weight adjustment factors calculated by the SUDAAN WTADJUST procedure is in the *SUDAAN Language Manual, Release 11.0* (RTI International 2012).

During model refinement, which involved collapsing categories of candidate predictor variables and/or excluding candidate predictor variables so that models would converge, no upper or lower bounds were specified. Once model convergence was achieved, the weight adjustment bounds were tightened to reduce the occurrence of extreme weights and the variance inflation caused by unequal weighting (referred to as the unequal weighting effect, or UWE).

The UWE is a measure of unequal weighting on the variance. When there is no variation in weights (as with a simple random sample, or SRS, design), the UWE equals one. The UWE can be inflated by weight adjustments in addition to unequal probabilities of selection during sampling. The UWE is defined as follows:

$$UWE = n \frac{\sum w^2}{(\sum w)^2}$$

where n = the sample size and w = the adjusted weight.

UWEs at each weight adjustment stage were examined, overall and by stratum, before and after the adjustment to evaluate the effect of each weighting adjustment on the precision of survey estimates. If an adjustment resulted in a much larger UWE compared to the UWE before the adjustment, weight adjustment bounds were tightened to reduce the loss of precision.

BPS statisticians loosened the upper and lower bounds for the adjustment factors, as necessary, and dropped variables from the models to ensure that the models still converged while maintaining reasonable values for the variances, the UWEs, and the minimum and maximum adjustment factors.

6.1.1 Base Weight for BPS:20/22

As discussed above, the BPS:20/22 base weight was derived from the NPSAS:20 weights, and the weight components from NPSAS:20 are listed in table 31. Statisticians multiplied the product of the NPSAS:20 weight components by the weight associated with subsampling from the BPS frame to create the BPS base weight. Specifically, for each sample member, the BPS:20/22 base weight was computed as

$$\text{BPS:20/22 base weight} = \text{WTB_WT1} \times \text{WTB_adj1} \times \text{WTB_adj2} \times \\ \text{WTB_WT4} \times \text{WTB_adj3} \times \text{WTB_adj4} \times \text{WTB_adj5} \times \text{BPS22WTA_adj1}.$$

The BPS:20/22 base weight differed from prior BPS cohorts in that the NPSAS base-year student study nonresponse adjustment was included. The decision to base the BPS:20/22 sample on NPSAS study respondents was made to protect the quality of imputed base-year data. The NPSAS study respondent definition included individuals with adequate key information for imputation purposes. NPSAS study nonrespondents who were potential FTBs were eligible for BPS, but their base-year data could not be confidently imputed were not included in the sample to be fielded.

Despite this, NPSAS study nonrespondent potential FTBs needed to be included in the BPS analytic weights and overall response rate calculations so that the BPS sample was representative of the entire FTB population. This was done by using the NPSAS base-year student study nonresponse-adjusted weights as the BPS base weights and reporting overall response rates as the product of the NPSAS study response rates and the BPS:20/22 data collection response rates.

6.1.2 Nonresponse Adjustment

Statisticians adjusted the BPS:20/22 analysis weight for nonresponse in three stages. These stages included an inability to locate the sample member, survey refusal, and other nonresponse (sample member located, did not refuse, did not respond) because the predictors of response propensity were different for each of these nonresponse outcomes. Adjusting for different types of nonresponse separately can help reduce nonresponse bias when the predictors of response propensity are different for the nonresponse outcomes. Individuals who were found to be ineligible¹² for BPS:20/22 or deceased as of the BPS:20/22 survey were not included in the nonresponse adjustments.

The models for each nonresponse adjustment were developed in the following three stages:

1. identification of model predictors;
2. chi-square automatic interaction detection (CHAID) analysis to determine statistically significant interactions between model predictors; and
3. WTADJUST procedure to calculate nonresponse adjustment.

¹² For details on the number of ineligible individuals, see Figure 3 in section 3.4.

The statistical team selected candidate predictor variables for the nonresponse adjustment models because they were thought to predict response status and were nonmissing for most respondents and nonrespondents. If needed, missing data were recoded into an “unknown” category when there was a small percentage of missing data. The following variables were identified for inclusion:

- control and level of institution (categorical, from NPSAS:20);
- Bureau of Economic Analysis region (categorical, from NPSAS:20);
- composite variable of seven-state sample is representative of or Census region of remaining states (categorical, from BPS:20/22);
- institution full-year total undergraduate enrollment from IPEDS 2020–21 (categorical, from NPSAS:20);
- age group as of December 31, 2019 (categorical, from NPSAS:20);
- veteran status (yes/no, from NPSAS:20);
- race/ethnicity (categorical, from NPSAS:20);
- gender (male/female/unknown, from NPSAS:20);
- total amount of Direct Loans disbursed to undergraduate students in 2019–20 (categorical, from NPSAS:20);
- institution aid status in 2019–20 (received/did not receive /unknown, from NPSAS:20);
- state aid status in 2019–20 (received/did not receive /unknown, from NPSAS:20);
- federal aid status in 2019–20 (received/did not receive /unknown, from NPSAS:20);
- any aid status excluding private loans in 2019–20 (received/did not receive/unknown, from NPSAS:20);
- total amount of PLUS Loans disbursed to undergraduate students in 2019–20 (categorical, from NPSAS:20);
- total amount of federal Pell Grant awarded to undergraduate students in 2019–20 (categorical, from NPSAS:20);
- parents’ highest education (categorical, from CPS data from NPSAS:20 [CPS:20 data]);
- has dependents (yes/no/unknown/not matched into CPS, from CPS:20 data); and

- marital status (categorical, from CPS:20 data).

In addition to the candidate predictor variables, statisticians included two-way and three-way interactions between predictor variables in the nonresponse adjustment models. To identify these interactions, the CHAID algorithm (Kass 1980) was used. CHAID is a hierarchical clustering algorithm that successively partitions individuals according to categorical predictors for a categorical dependent variable. The algorithm begins with all records and cycles over each predictor, finding for each predictor an optimal partition of the records according to its levels. The most statistically significant optimal partition was retained, and the CHAID algorithm was applied to the members of that partition to find further partitions using the remaining predictors. The algorithm was stopped after a specified number of partitioning steps or if the algorithm failed to find statistical significance among any of the partitions at a given step.

Statisticians retained all variables in the nonresponse models unless the models failed to converge. In the event a model failed to converge, the model was then reduced sequentially in a backward selection method until a converging model was attained. Categories of candidate predictor variables that impeded the creation of a convergent model were first collapsed to attempt to achieve model convergence. If the model still failed to converge after collapsing categories, the variable was dropped from the final model.

Before SUDAAN's WTADJUST procedure was implemented, an initial lower bound for the nonresponse adjustment factors was set at 1, but no upper limit was set. Once convergence of the model was achieved, the weight adjustment bounds were tightened to reduce the magnitude of the weight adjustment factors and the UWEs. The results of the nonresponse adjustment models follow.

Not located nonresponse adjustment (BPS22WTA_adj2). Table 32 shows the final predictor variables used in the model to determine the not located nonresponse adjustments and the average weight adjustment factors results from these variables. Summary statistics of the weight adjustment factors follow:

- minimum: 1.00;
- median: 1.18; and
- maximum: 2.60.

To achieve model convergence, the final lower bound was 1.0 and the final upper bound was 3.0.

Table 32. Weighted response rate and average weight adjustment factors for nonresponse due to sample member not located adjustment: 2022

Model predictor variable	Number of students who were located	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj2)
Total	29,230	82.2	1.22
Control and level of institution			
Public less-than-2-year	300	80.8	1.24
Public 2-year	9,160	79.0	1.26
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	2,110	77.8	1.27
Public 4-year, non-doctorate-granting, primarily baccalaureate	1,970	86.6	1.15
Public 4-year, doctorate-granting	6,270	83.9	1.19
Private nonprofit 2-year or less	190	81.9	1.23
Private nonprofit 4-year, non-doctorate-granting	2,360	87.5	1.14
Private nonprofit 4-year, doctorate-granting	3,070	84.5	1.17
Private for-profit less-than-2-year	760	78.4	1.25
Private for-profit 2-year	1,170	79.8	1.24
Private for-profit 4-year	1,870	79.7	1.25
Bureau of Economic Analysis region ²			
New England	2,150	83.3	1.20
Mideast	4,340	84.0	1.19
Great Lakes	3,220	82.8	1.20
Plains	2,310	83.6	1.19
Southeast	8,200	82.4	1.22
Southwest	3,230	78.4	1.28
Rocky Mountains	1,700	85.8	1.18
Far West	3,570	80.4	1.23
Outlying Areas	520	84.0	1.18
Representative state or Census region			
California	2,060	80.7	1.22
Florida	1,520	79.6	1.26
Georgia	1,080	84.9	1.20
New York	1,670	83.4	1.20
North Carolina	1,120	87.5	1.16
Pennsylvania	1,070	83.0	1.20
Texas	1,820	78.3	1.28
Northeast and Puerto Rico	3,510	83.2	1.20
Midwest	5,520	83.1	1.20
South	5,720	82.1	1.23
West	4,140	82.2	1.22
Institution total undergraduate enrollment ³			
1–2,983	7,350	85.5	1.19
2,984–8,499	7,270	81.7	1.23
8,500–19,798	7,300	81.6	1.23
19,799 or more	7,310	81.6	1.22
Age as of December 31, 2019			
15–23	26,780	82.5	1.21
24–29	1,220	78.4	1.25
30 or more	1,230	78.6	1.25
Unknown	0	46.4	1.77
Veteran status			
Yes	320	85.0	1.16
No	28,910	82.2	1.22

See notes at end of table.

Table 32. Weighted response rate and average weight adjustment factors for nonresponse due to sample member not located adjustment: 2022—Continued

Model predictor variable	Number of students who were located	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj2)
Race/ethnicity			
American Indian or Alaska Native, not Hispanic or Latino	210	64.7	1.59
Asian, not Hispanic or Latino	1,850	80.7	1.24
Black, not Hispanic or Latino	4,000	83.6	1.20
Hispanic or Latino, of any race	6,710	80.4	1.24
Native Hawaiian or Other Pacific Islander, not Hispanic or Latino	100	73.7	1.49
White, not Hispanic or Latino	14,460	85.1	1.18
Two or more races, not Hispanic or Latino	1,390	81.8	1.23
Unknown	520	55.3	1.75
Gender			
Male	12,700	81.5	1.22
Female	16,170	82.8	1.21
Unknown	370	83.3	1.19
Total amount of Direct Loans disbursed to undergraduate students ³			
\$0	17,830	80.0	1.25
\$1–\$4,568	2,760	84.1	1.21
\$4,569–\$5,499	640	85.4	1.19
\$5,500	5,480	88.6	1.14
\$5,501 or more	2,510	84.3	1.20
Institutional aid status			
Received	10,730	86.0	1.17
Did not receive	14,870	77.8	1.28
Unknown	3,630	90.2	1.12
State aid status			
Received	7,520	84.2	1.20
Did not receive	17,860	80.0	1.24
Unknown	3,850	89.1	1.13
Federal aid status			
Received	19,380	85.0	1.19
Did not receive	8,380	76.5	1.30
Unknown	1,470	87.7	1.15
Any aid status excluding private loans			
Received	24,270	84.4	1.19
Did not receive	3,840	71.6	1.38
Unknown	1,120	87.4	1.15
Total amount of PLUS Loans disbursed to undergraduate students ³			
\$0	26,970	81.7	1.22
\$1–\$7,450	550	85.5	1.19
\$7,451–\$12,904	560	87.1	1.16
\$12,905–\$20,628	580	88.7	1.14
\$20,629 or more	580	90.9	1.10
Total amount of federal Pell Grants awarded to undergraduate students ⁴			
\$0	15,460	81.5	1.22
\$1–\$3,097	3,290	79.9	1.26
\$3,098–\$6,194	5,140	83.3	1.21
\$6,195 or more	5,350	85.4	1.18

See notes at end of table.

Table 32. Weighted response rate and average weight adjustment factors for nonresponse due to sample member not located adjustment: 2022—Continued

Model predictor variable	Number of students who were located	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj2)
Central Processing System (CPS) - Parents' highest education			
Middle school/junior high	1,020	81.0	1.24
High school	6,490	81.9	1.23
College or beyond	12,560	87.0	1.16
Other	1,140	81.2	1.23
Unknown	230	81.0	1.23
Not matched in CPS:20 data	7,800	75.7	1.30
CPS - Has dependents			
Yes	1,740	82.6	1.21
No	19,690	85.0	1.18
Not matched in CPS:20 data	7,800	75.7	1.30
CPS - Marital status			
Single	20,450	84.9	1.19
Married/remarried	740	85.3	1.17
Separated	110	75.0	1.36
Divorced/widowed	130	82.6	1.22
Not matched in CPS:20 data	7,800	75.7	1.30
CHAID segments			
Undergraduate students with Asian, not Hispanic or Latino, Hispanic or Latino, of any race, or Two or more races, not Hispanic or Latino race/ethnicity whose marital status in CPS:20 data was Single, Married/remarried, or Separated and did not receive institutional aid	2,590	79.9	1.25
Undergraduate students with Asian, not Hispanic or Latino, Hispanic or Latino, of any race, or Two or more races, not Hispanic or Latino race/ethnicity whose marital status in CPS:20 data was Single, Married/remarried, or Separated and did receive institutional aid	1,740	87.3	1.15
Undergraduate students with Asian, not Hispanic or Latino, Hispanic or Latino, of any race, or Two or more races, not Hispanic or Latino race/ethnicity whose marital status in CPS:20 data was Single, Married/remarried, or Separated and it is unknown whether they received institutional aid	1,080	80.4	1.24
Undergraduate students with Asian, not Hispanic or Latino, Hispanic or Latino, of any race, or Two or more races, not Hispanic or Latino race/ethnicity whose marital status in CPS:20 data was Divorced/widowed or did not match into CPS:20 data and received federal aid or it is unknown whether they received federal aid	1,740	87.9	1.14
Undergraduate students with Asian, not Hispanic or Latino, Hispanic or Latino, of any race, or Two or more races, not Hispanic or Latino race/ethnicity whose marital status in CPS:20 data was Divorced/widowed or did not match into CPS:20 data and did not receive federal aid	1,430	67.5	1.46
Undergraduate students with White, not Hispanic or Latino race/ethnicity who did not receive institutional aid and their parents' highest education from CPS:20 data was Middle school/junior high, Other/unknown, or did not match into CPS:20 data	790	76.5	1.30
Undergraduate students with White, not Hispanic or Latino race/ethnicity who did not receive institutional aid and their parents' highest education from CPS:20 data was High school	580	87.2	1.15
Undergraduate students with White, not Hispanic or Latino race/ethnicity who did not receive institutional aid and their parents' highest education from CPS:20 data was College or beyond or Unknown	1,020	77.2	1.30

See notes at end of table.

Table 32. Weighted response rate and average weight adjustment factors for nonresponse due to sample member not located adjustment: 2022—Continued

Model predictor variable	Number of students who were located	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj2)
Undergraduate students with White, not Hispanic or Latino race/ethnicity who received institutional aid and their parents' highest education from CPS:20 data was High school, Other, Unknown, or did not match into CPS:20 data	2,240	82.3	1.21
Undergraduate students with White, not Hispanic or Latino race/ethnicity who received institutional aid and their parents' highest education from CPS:20 data was Middle school/junior high or College or beyond	1,490	87.0	1.15
Undergraduate students with White, not Hispanic or Latino race/ethnicity for whom it is unknown if they received institutional aid and the institution's control and level was Public less-than-2-year, Public 4-year, non-doctorate-granting, primarily baccalaureate, Public 4-year, doctorate-granting, Private nonprofit 4-year, non-doctorate-granting, or Private nonprofit 4-year, doctorate-granting	2,060	76.5	1.30
Undergraduate students with White, not Hispanic or Latino race/ethnicity for whom it is unknown if they received institutional aid and the institution's control and level was Public 2-year, Public 4-year, non-doctorate-granting, primarily subbaccalaureate, Private for-profit 2-year, or Private for-profit 4-year	2,230	84.4	1.18
Undergraduate students with White, not Hispanic or Latino race/ethnicity for whom it is unknown if they received institutional aid and the institution's control and level was Private for-profit less-than-2-year	3,790	90.5	1.10
Undergraduate students with American Indian or Alaska Native, not Hispanic or Latino or Native Hawaiian or Other Pacific Islander, not Hispanic or Latino race/ethnicity whose institution's control and level was Public less-than-2-year, Public 2-year, Public 4-year, non-doctorate-granting, primarily baccalaureate, Public 4-year, doctorate-granting, Private nonprofit 2-year or less, or Private for-profit 2-year and institution's Bureau of Economic Analysis region was New England, Southwest, or Rocky Mountains	260	84.2	1.19
Undergraduate students with American Indian or Alaska Native, not Hispanic or Latino or Native Hawaiian or Other Pacific Islander, not Hispanic or Latino race/ethnicity whose institution's control and level was Public less-than-2-year, Public 2-year, Public 4-year, non-doctorate-granting, primarily baccalaureate, Public 4-year, doctorate-granting, Private nonprofit 2-year or less, or Private for-profit 2-year and institution's Bureau of Economic Analysis region was Mideast, Great Lakes, Plains, Southeast, or Far West	680	91.5	1.08
Undergraduate students with American Indian or Alaska Native, not Hispanic or Latino or Native Hawaiian or Other Pacific Islander, not Hispanic or Latino race/ethnicity whose institution's control and level was Public 4-year, non-doctorate-granting, primarily subbaccalaureate, Private nonprofit 4-year, non-doctorate-granting, Private nonprofit 4-year, doctorate-granting, Private for-profit less-than-2-year, or Private for-profit 4-year	690	94.7	1.06
Undergraduate students with Black, not Hispanic or Latino race/ethnicity whose institution's control and level was Public less-than-2-year, Public 2-year, Public 4-year, non-doctorate-granting, primarily subbaccalaureate, Private for-profit less-than-2-year, or Private for-profit 2-year and their marital status in CPS:20 data was Single or Divorced/widowed	80	59.5	1.66
Undergraduate students with Black, not Hispanic or Latino race/ethnicity whose institution's control and level was Public less-than-2-year, Public 2-year, Public 4-year, non-doctorate-granting, primarily subbaccalaureate, Private for-profit less-than-2-year, or Private for-profit 2-year and they did not match into CPS:20 data	140	85.9	1.16

See notes at end of table.

Table 32. Weighted response rate and average weight adjustment factors for nonresponse due to sample member not located adjustment: 2022—Continued

Model predictor variable	Number of students who were located	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj2)
Undergraduate students with Black, not Hispanic or Latino race/ethnicity whose institution's control and level was Public less-than-2-year, Public 2-year, Public 4-year, non-doctorate-granting, primarily subbaccalaureate, Private for-profit less-than-2-year, or Private for-profit 2-year and their marital status in CPS:20 data was Married/ remarried or Separated	90	48.3	2.08
Undergraduate students with Black, not Hispanic or Latino race/ethnicity whose institution's control and level was Public 4-year, non-doctorate-granting, primarily baccalaureate, Private nonprofit 2-year or less, Private nonprofit 4-year, doctorate-granting, or Private for-profit 4-year and their institution's Bureau of Economic Analysis region was Great Lakes or Southwest	1,550	80.4	1.24
Undergraduate students with Black, not Hispanic or Latino race/ethnicity whose institution's control and level was Public 4-year, non-doctorate-granting, primarily baccalaureate, Private nonprofit 2-year or less, Private nonprofit 4-year, doctorate-granting, or Private for-profit 4-year and their institution's Bureau of Economic Analysis region was Mideast, Rocky Mountains, or Far West	390	74.7	1.33
Undergraduate students with Black, not Hispanic or Latino race/ethnicity whose institution's control and level was Public 4-year, non-doctorate-granting, primarily baccalaureate, Private nonprofit 2-year or less, Private nonprofit 4-year, doctorate-granting, or Private for-profit 4-year and their institution's Bureau of Economic Analysis region was New England, Plains, or Southeast	80	96.9	1.03
Undergraduate students with Black, not Hispanic or Latino race/ethnicity whose institution's control and level was Public 4-year, doctorate-granting or Private nonprofit 4-year, non-doctorate-granting and they did or did not receive institutional aid	100	100.0	1.00
Undergraduate students with Black, not Hispanic or Latino race/ethnicity whose institution's control and level was Public 4-year, doctorate-granting or Private nonprofit 4-year, non-doctorate-granting and it is unknown whether they received institutional aid	260	94.4	1.07
Undergraduate students with unknown race/ethnicity who received federal aid or it is unknown whether they received federal aid and who did not receive state aid or it is unknown if they received state aid	540	87.3	1.16
Undergraduate students with unknown race/ethnicity who received federal aid or it is unknown whether they received federal aid and who received state aid	930	84.4	1.18
Undergraduate students with unknown race/ethnicity who did not receive federal aid	150	94.5	1.06

¹ The weighted response rates were calculated using the BPS:20/22 base weight.

² New England = Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont; Mid East = Delaware, District of Columbia, Maryland, New Jersey, New York, Pennsylvania; Great Lakes = Illinois, Indiana, Michigan, Ohio, Wisconsin; Plains = Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, South Dakota; Southeast = Alabama, Arkansas, Florida, Georgia, Kentucky, Mississippi, North Carolina, South Carolina, Tennessee, Virginia, West Virginia; Southwest = Arizona, New Mexico, Oklahoma, Texas; Rocky Mountains = Colorado, Idaho, Montana, Utah, Wyoming; Far West = Alaska, California, Hawaii, Nevada, Oregon, Washington; Outlying Areas = Puerto Rico.

³ Categories were defined by quartiles. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of "None" or "Unknown" were excluded from quartile calculations.

⁴ For Total amount of federal Pell Grants awarded to undergraduate students, the second and third categories were defined by the median of Pell amounts greater than \$0 and less than \$6,195. Categories were defined by quartiles. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of "None" or "Unknown" were excluded from quartile calculations.

NOTE: CHAID = chi-square automatic interaction detection; CPS:20 data = Central Processing System data from the 2019–20 National Postsecondary Student Aid Study. Number of respondents rounded to the nearest 10. Detail may not sum to totals because of rounding. SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Survey refusal nonresponse adjustment (BPS22WTA_adj3). Table 33 shows the final predictor variables used in the model to determine the survey refusal nonresponse adjustments and the average weight adjustment factors results from these variables. Summary statistics of the weight adjustment factors follow:

- minimum: 1.00;
- median: 1.04; and
- maximum: 1.82.

To achieve model convergence, the final lower bound was 1.0 and the final upper bound was 7.0.

Table 33. Weighted response rate and average weight adjustment factors for nonresponse due to sample member refusal adjustment: 2022

Model predictor variable	Number of students who were located and did not refuse to complete the survey	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj3)
Total	28,060	95.1	1.05
Control and level of institution			
Public less-than-2-year	280	95.5	1.06
Public 2-year	8,770	94.4	1.06
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	2,020	95.3	1.05
Public 4-year, non-doctorate-granting, primarily baccalaureate	1,900	95.7	1.04
Public 4-year, doctorate-granting	6,010	95.1	1.05
Private nonprofit 2-year or less	180	97.1	1.04
Private nonprofit 4-year, non-doctorate-granting	2,280	96.0	1.04
Private nonprofit 4-year, doctorate-granting	2,950	95.7	1.04
Private for-profit less-than-2-year	740	97.0	1.03
Private for-profit 2-year	1,130	95.6	1.04
Private for-profit 4-year	1,800	94.6	1.05
Bureau of Economic Analysis region ²			
New England	2,050	95.0	1.05
Mideast	4,160	95.3	1.05
Great Lakes	3,100	95.3	1.05
Plains	2,190	94.2	1.06
Southeast	7,870	95.2	1.05
Southwest	3,120	94.6	1.05
Rocky Mountains	1,620	96.3	1.04
Far West	3,450	95.1	1.05
Outlying Areas	510	96.3	1.04

See notes at end of table.

Table 33. Weighted response rate and average weight adjustment factors for nonresponse due to sample member refusal adjustment: 2022—Continued

Model predictor variable	Number of students who were located and did not refuse to complete the survey	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj3)
Representative state or Census region			
California	1,990	94.9	1.05
Florida	1,460	95.4	1.05
Georgia	1,040	96.5	1.04
New York	1,600	95.0	1.05
North Carolina	1,060	94.2	1.06
Pennsylvania	1,030	95.7	1.04
Texas	1,760	94.6	1.05
Northeast and Puerto Rico	3,360	95.2	1.05
Midwest	5,290	95.0	1.05
South	5,490	95.1	1.05
West	3,970	95.5	1.04
Institution total undergraduate enrollment ³			
1–2,983	7,080	96.1	1.04
2,984–8,499	6,970	95.2	1.05
8,500–19,798	7,000	94.8	1.05
19,799 or more	7,010	94.8	1.05
Age as of December 31, 2019			
15–23 or Unknown	25,740	95.3	1.05
24–29	1,160	93.0	1.07
30 or more	1,170	90.9	1.07
Veteran status			
Yes	290	88.7	1.12
No	27,770	95.2	1.05
Race/ethnicity			
American Indian or Alaska Native, not Hispanic or Latino	200	93.9	1.06
Asian, not Hispanic or Latino	1,790	95.3	1.05
Black, not Hispanic or Latino	3,870	96.0	1.04
Hispanic or Latino, of any race	6,500	95.9	1.04
Native Hawaiian or Other Pacific Islander, not Hispanic or Latino	90	97.2	1.03
White, not Hispanic or Latino	13,780	94.6	1.05
Two or more races, not Hispanic or Latino	1,340	96.1	1.04
Unknown	490	92.0	1.08
Gender			
Male	12,060	94.0	1.06
Female	15,640	96.0	1.04
Unknown	360	99.6	1.00
Total amount of Direct Loans disbursed to undergraduate students ³			
\$0	17,030	94.5	1.06
\$1–\$4,568	2,680	97.1	1.03
\$4,569–\$5,499	630	97.9	1.02
\$5,500	5,280	95.8	1.04
\$5,501 or more	2,430	97.0	1.03

See notes at end of table.

Table 33. Weighted response rate and average weight adjustment factors for nonresponse due to sample member refusal adjustment: 2022—Continued

Model predictor variable	Number of students who were located and did not refuse to complete the survey	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj3)
Institutional aid status			
Received	10,310	95.7	1.04
Did not receive	14,220	94.1	1.06
Unknown	3,530	97.8	1.02
State aid status			
Received	7,230	95.9	1.04
Did not receive	17,090	94.3	1.06
Unknown	3,740	97.5	1.03
Federal aid status			
Received	18,710	96.0	1.04
Did not receive	7,920	93.2	1.07
Unknown	1,430	97.6	1.03
Any aid status excluding private loans			
Received	23,370	95.7	1.04
Did not receive	3,600	91.7	1.09
Unknown	1,090	98.2	1.02
Total amount of PLUS Loans disbursed to undergraduate students ³			
\$0	25,890	95.0	1.05
\$1–\$7,450	530	97.3	1.03
\$7,451–\$12,904	530	94.0	1.06
\$12,905–\$20,628	560	96.7	1.03
\$20,629 or more	550	96.4	1.04
Total amount of federal Pell Grants awarded to undergraduate students ⁴			
\$0	14,730	94.4	1.06
\$1–\$3,097	3,160	95.8	1.04
\$3,098–\$6,194	4,970	95.9	1.04
\$6,195 or more	5,200	96.7	1.03
Central Processing System (CPS) - Parents' highest education			
Middle school/junior high	990	96.7	1.03
High school	6,260	96.0	1.04
College or beyond	12,050	95.3	1.05
Other	1,090	92.5	1.08
Unknown	220	95.0	1.05
Not matched in CPS:20 data	7,450	94.3	1.05
CPS - Has dependents			
Yes	1,700	95.5	1.04
No	18,910	95.4	1.05
Not matched in CPS:20 data	7,450	94.3	1.05
CPS - Marital status			
Single	19,660	95.5	1.05
Married/remarried	710	94.8	1.05
Separated	110	98.2	1.01
Divorced/widowed	130	88.6	1.10
Not matched in CPS:20 data	7,450	94.3	1.05

See notes at end of table.

Table 33. Weighted response rate and average weight adjustment factors for nonresponse due to sample member refusal adjustment: 2022—Continued

Model predictor variable	Number of students who were located and did not refuse to complete the survey	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj3)
CHAID segments			
Undergraduate students who received any aid excluding private loans whose gender is female and their institution's control and level was Private nonprofit 4-year, non-doctorate-granting, Private nonprofit 4-year, doctorate-granting, Private for-profit less-than-2-year, or Private for-profit 2-year	3,900	97.3	1.03
Undergraduate students who received any aid excluding private loans whose gender is female and their institution's control and level was Public 2-year, Public 4-year, non-doctorate-granting, primarily subbaccalaureate, Public 4-year, non-doctorate-granting, primarily baccalaureate, Public 4-year, doctorate-granting, or Private for-profit 4-year	9,390	96.1	1.04
Undergraduate students who received any aid excluding private loans whose gender is female and their institution's control and level was Private nonprofit 2-year or less	70	100.0	1.00
Undergraduate students who received any aid excluding private loans whose gender is female and their institution's control and level was Public less-than-2-year	70	99.3	1.01
Undergraduate students who received any aid excluding private loans whose gender is male whose race/ethnicity was American Indian or Alaska Native, not Hispanic or Latino, Asian, not Hispanic or Latino, Black, not Hispanic or Latino, Hispanic or Latino, of any race, Native Hawaiian or Other Pacific Islander, not Hispanic or Latino, or Two or more races, not Hispanic or Latino	4,710	96.0	1.04
Undergraduate students who received any aid excluding private loans whose gender is male whose race/ethnicity was White, not Hispanic or Latino or Unknown	4,950	93.6	1.07
Undergraduate students who received any aid excluding private loans whose gender is unknown and parents' highest education from CPS:20 data was Middle school/junior high, College or beyond, Other, Unknown, or did not match into CPS:20 data	230	100.0	1.00
Undergraduate students who received any aid excluding private loans whose gender is unknown and parents' highest education from CPS:20 data was High school	50	97.9	1.02
Undergraduate students for whom it is unknown if they received any aid excluding private loans whose institution's control and level was Public less-than-2-year, Private nonprofit 2-year or less, or Private nonprofit 4-year, non-doctorate-granting	100	100.0	1.00
Undergraduate students for whom it is unknown if they received any aid excluding private loans whose institution's control and level was Public 2-year, Public 4-year, non-doctorate-granting, primarily subbaccalaureate, Public 4-year, non-doctorate-granting, primarily baccalaureate, or Private for-profit less-than-2-year and institution's Bureau of Economic Analysis region was Plains or Southeast	190	100.0	1.00
Undergraduate students for whom it is unknown if they received any aid excluding private loans whose institution's control and level was Public 2-year, Public 4-year, non-doctorate-granting, primarily subbaccalaureate, Public 4-year, non-doctorate-granting, primarily baccalaureate, or Private for-profit less-than-2-year and institution's Bureau of Economic Analysis region was Mideast or Rocky Mountains	190	99.2	1.01

See notes at end of table.

Table 33. Weighted response rate and average weight adjustment factors for nonresponse due to sample member refusal adjustment: 2022—Continued

Model predictor variable	Number of students who were located and did not refuse to complete the survey	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj3)
Undergraduate students for whom it is unknown if they received any aid excluding private loans whose institution's control and level was Public 2-year, Public 4-year, non-doctorate-granting, primarily subbaccalaureate, Public 4-year, non-doctorate-granting, primarily baccalaureate, or Private for-profit less-than-2-year and institution's Bureau of Economic Analysis region was New England, Great Lakes, Southwest, or Far West	380	96.3	1.04
Undergraduate students for whom it is unknown if they received any aid excluding private loans whose institution's control and level was Public 4-year, doctorate-granting or Private nonprofit 4-year, doctorate-granting and institution's state or Census region was South	50	99.3	1.01
Undergraduate students for whom it is unknown if they received any aid excluding private loans whose institution's control and level was Public 4-year, doctorate-granting or Private nonprofit 4-year, doctorate-granting and institution's state or Census region was California, Florida, Georgia, New York, North Caroline, Pennsylvania, Texas, Northeast & Puerto Rico, Midwest, or West	130	100.0	1.00
Undergraduate students for whom it is unknown if they received any aid excluding private loans whose institution's control and level was Private for-profit 2-year or Private for-profit 4-year	50	91.9	1.11
Undergraduate students who did not receive any aid excluding private loans whose age group was 15–23, 24–29, or Unknown and whose gender was Female	1,560	93.5	1.07
Undergraduate students who did not receive any aid excluding private loans whose age group was 15–23, 24–29, or Unknown and whose gender was Male	1,790	91.2	1.10
Undergraduate students who did not receive any aid excluding private loans whose age group was 15–23, 24–29, or Unknown and whose gender was Unknown	60	99.2	1.01
Undergraduate students who did not receive any aid excluding private loans whose age group was 30 or more and institution's total undergraduate enrollment was 1–2,983 or 2,984–8,499	90	94.9	1.05
Undergraduate students who did not receive any aid excluding private loans whose age group was 30 or more and institution's total undergraduate enrollment was 8,500–19,798 or 19,799 or more	90	74.7	1.33

Rounds to zero.

¹ The weighted response rates were calculated using the BPS:20/22 base weight adjusted for not located nonresponse.

² New England = Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont; Mid East = Delaware, District of Columbia, Maryland, New Jersey, New York, Pennsylvania; Great Lakes = Illinois, Indiana, Michigan, Ohio, Wisconsin; Plains = Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, South Dakota; Southeast = Alabama, Arkansas, Florida, Georgia, Kentucky, Mississippi, North Carolina, South Carolina, Tennessee, Virginia, West Virginia; Southwest = Arizona, New Mexico, Oklahoma, Texas; Rocky Mountains = Colorado, Idaho, Montana, Utah, Wyoming; Far West = Alaska, California, Hawaii, Nevada, Oregon, Washington; Outlying Areas = Puerto Rico.

³ Categories were defined by quartiles. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of "None" or "Unknown" were excluded from quartile calculations.

⁴ For Total amount of federal Pell Grants awarded to undergraduate students, the second and third categories were defined by the median of Pell amounts greater than \$0 and less than \$6,195. Categories were defined by quartiles. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of "None" or "Unknown" were excluded from quartile calculations.

NOTE: CHAID = chi-square automatic interaction detection; CPS:20 data = Central Processing System data from the 2019–20 National Postsecondary Student Aid Study. Number of respondents rounded to the nearest 10. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Other nonresponse adjustment (BPS22WTA_adj4). Table 34 shows the final predictor variables used in the model to determine the not located nonresponse adjustments and the average weight adjustment factor results from these variables. Summary statistics of the weight adjustment factors follow:

- minimum: 1.02;
- median: 1.28; and
- maximum: 2.72.

To achieve model convergence, the final lower bound was 1.0 and the final upper bound was 3.0.

Table 34. Weighted response rate and average adjustment factors for nonresponse due to sample member other nonresponse adjustment: 2022

Model predictor variable	Number of respondents	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj4)
Total	22,320	75.0	1.33
Control and level of institution			
Public less-than-2-year	180	57.8	1.73
Public 2-year	6,730	71.3	1.39
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	1,530	67.9	1.45
Public 4-year, non-doctorate-granting, primarily baccalaureate	1,560	78.5	1.27
Public 4-year, doctorate-granting	4,930	78.3	1.26
Private nonprofit 2-year or less	130	72.9	1.39
Private nonprofit 4-year, non-doctorate-granting	1,900	78.7	1.27
Private nonprofit 4-year, doctorate-granting	2,440	77.7	1.27
Private for-profit less-than-2-year	580	75.9	1.32
Private for-profit 2-year	880	74.8	1.34
Private for-profit 4-year	1,460	73.6	1.34
Bureau of Economic Analysis region ²			
New England	1,580	70.9	1.40
Mideast	3,410	78.7	1.27
Great Lakes	2,490	75.1	1.32
Plains	1,740	76.8	1.29
Southeast	6,200	74.5	1.35
Southwest	2,450	72.8	1.37
Rocky Mountains	1,290	77.5	1.31
Far West	2,760	74.0	1.34
Outlying Areas	410	75.3	1.30

See notes at end of table.

Table 34. Weighted response rate and average adjustment factors for nonresponse due to sample member other nonresponse adjustment: 2022—Continued

Model predictor variable	Number of respondents	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj4)
Representative state or Census region			
California	1,610	76.0	1.29
Florida	1,150	73.7	1.36
Georgia	830	79.5	1.27
New York	1,320	79.0	1.26
North Carolina	800	70.6	1.45
Pennsylvania	840	78.7	1.26
Texas	1,390	72.8	1.36
Northeast & Puerto Rico	2,630	71.9	1.37
Midwest	4,230	75.7	1.31
South	4,390	75.8	1.32
West	3,120	72.8	1.37
Institution total undergraduate enrollment ³			
1–2,983	5,690	77.9	1.29
2,984–8,499	5,470	73.2	1.37
8,500–19,798	5,510	73.5	1.36
19,799 or more	5,650	76.2	1.31
Age as of December 31, 2019			
15–23 or Unknown	20,540	75.7	1.31
24–29	890	66.6	1.45
30 or more	890	62.6	1.56
Veteran status			
Yes	210	65.5	1.50
No	22,110	75.1	1.33
Race/ethnicity			
American Indian or Alaska Native, not Hispanic or Latino	150	75.7	1.30
Asian, not Hispanic or Latino	1,520	82.2	1.22
Black, not Hispanic or Latino	3,100	74.4	1.35
Hispanic or Latino, of any race	5,180	74.7	1.34
Native Hawaiian or Other Pacific Islander, not Hispanic or Latino	80	72.9	1.42
White, not Hispanic or Latino	10,960	75.5	1.33
Two or more races, not Hispanic or Latino	1,080	76.5	1.31
Unknown	270	54.1	1.82
Gender			
Male	9,100	70.2	1.42
Female	12,910	79.1	1.27
Unknown	320	83.0	1.19
Total amount of Direct Loans disbursed to undergraduate students ³			
\$0	13,320	73.3	1.36
\$1–\$4,568	2,160	77.9	1.30
\$4,569–\$5,499	530	81.1	1.24
\$5,500	4,310	78.5	1.28
\$5,501 or more	2,020	78.9	1.29
Institutional aid status			
Received	8,440	78.9	1.28
Did not receive	10,840	70.0	1.42
Unknown	3,040	86.0	1.16

See notes at end of table.

Table 34. Weighted response rate and average adjustment factors for nonresponse due to sample member other nonresponse adjustment: 2022—Continued

Model predictor variable	Number of respondents	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj4)
State aid status			
Received	5,840	78.1	1.29
Did not receive	13,280	71.9	1.38
Unknown	3,200	84.5	1.18
Federal aid status			
Received	15,130	77.3	1.30
Did not receive	6,000	70.1	1.42
Unknown	1,200	82.5	1.21
Any aid status excluding private loans			
Received	18,850	77.3	1.30
Did not receive	2,560	63.7	1.57
Unknown	920	82.0	1.22
Total amount of PLUS Loans disbursed to undergraduate students ³			
\$0	20,550	74.8	1.33
\$1–\$7,450	420	74.9	1.33
\$7,451–\$12,904	430	80.4	1.24
\$12,905–\$20,628	460	79.3	1.26
\$20,629 or more	460	80.3	1.25
Total amount of federal Pell Grants awarded to undergraduate students ⁴			
\$0	11,590	74.0	1.34
\$1–\$3,097	2,450	72.3	1.39
\$3,098–\$6,194	4,010	76.4	1.32
\$6,195 or more	4,280	79.6	1.27
Central Processing System (CPS) - Parents' highest education			
Middle school/junior high	790	74.1	1.36
High school	4,930	75.0	1.34
College or beyond	9,740	78.1	1.29
Other	860	74.7	1.35
Unknown	170	65.8	1.56
Not matched in CPS:20 data	5,840	71.0	1.38
CPS - Has dependents			
Yes	1,350	71.9	1.38
No	15,140	77.0	1.31
Not matched in CPS:20 data	5,840	71.0	1.38
CPS - Marital status			
Single	15,720	76.9	1.31
Married/remarried	580	74.5	1.33
Separated	80	59.5	1.67
Divorced/widowed	100	67.9	1.45
Not matched in CPS:20 data	5,840	71.0	1.38

See notes at end of table.

Table 34. Weighted response rate and average adjustment factors for nonresponse due to sample member other nonresponse adjustment: 2022—Continued

Model predictor variable	Number of respondents	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj4)
CHAID segments			
Undergraduate students for whom it is unknown if they received institutional aid whose race/ethnicity was American Indian or Alaska Native, not Hispanic or Latino, Asian, not Hispanic or Latino, or Black, not Hispanic or Latino and received federal Pell Grant aid	410	89.1	1.13
Undergraduate students for whom it is unknown if they received institutional aid whose race/ethnicity was American Indian or Alaska Native, not Hispanic or Latino, Asian, not Hispanic or Latino, or Black, not Hispanic or Latino and did not receive federal Pell Grant aid	260	93.1	1.08
Undergraduate students for whom it is unknown if they received institutional aid whose race/ethnicity was Hispanic or Latino, of any race, White, not Hispanic or Latino, or Two or more races, not Hispanic or Latino and for whom the total amount of federal Pell Grants awarded was \$0, \$1-\$3,097, or \$3,098-\$6,194	1,880	84.9	1.18
Undergraduate students for whom it is unknown if they received institutional aid whose race/ethnicity was Hispanic or Latino, of any race, White, not Hispanic or Latino, or Two or more races, not Hispanic or Latino and for whom the total amount of federal Pell Grants awarded was \$6,195 or more	410	91.4	1.09
Undergraduate students for whom it is unknown if they received institutional aid whose race/ethnicity was Native Hawaiian or Other Pacific Islander, not Hispanic or Latino or Unknown	70	64.8	1.51
Undergraduate students who did not receive institutional aid whose gender was Female or Unknown and received any aid excluding private loans	5,070	77.3	1.30
Undergraduate students who did not receive institutional aid whose gender was Female or Unknown and who did not receive any aid excluding private loans or for whom it is unknown if they received any aid excluding private loans	1,350	69.5	1.45
Undergraduate students who did not receive institutional aid whose gender was Male and whose parents' highest education from CPS:20 data was College or beyond	1,910	71.3	1.41
Undergraduate students who did not receive institutional aid whose gender was Male and whose parents' highest education from CPS:20 data was Middle school/junior high, Unknown, or they did not match into CPS:20 data	1,350	58.4	1.71
Undergraduate students who did not receive institutional aid whose gender was Male and whose parents' highest education from CPS:20 data was High school or Other	1,160	64.4	1.57
Undergraduate students who received institutional aid whose gender was Female and whose race/ethnicity was American Indian or Alaska Native, not Hispanic or Latino, Black, not Hispanic or Latino, Native Hawaiian or Other Pacific Islander, not Hispanic or Latino, White, not Hispanic or Latino, or Two or more races, not Hispanic or Latino	3,580	82.2	1.22
Undergraduate students who received institutional aid whose gender was Female and whose race/ethnicity was Asian, not Hispanic or Latino or Hispanic or Latino, of any race	1,260	85.5	1.18
Undergraduate students who received institutional aid whose gender was Female and whose race/ethnicity was Unknown	60	57.0	1.78
Undergraduate students who received institutional aid whose gender was Male and whose race/ethnicity was Black, not Hispanic or Latino, White, not Hispanic or Latino, or Two or more races, not Hispanic or Latino	2,500	75.0	1.35

See notes at end of table.

Table 34. Weighted response rate and average adjustment factors for nonresponse due to sample member other nonresponse adjustment: 2022—Continued

Model predictor variable	Number of respondents	Weighted response rate ¹	Average weight adjustment factor (BPS22WTA_adj4)
Undergraduate students who received institutional aid whose gender was Male and whose race/ethnicity was Hispanic or Latino, of any race	570	68.4	1.48
Undergraduate students who received institutional aid whose gender was Male and whose race/ethnicity was American Indian or Alaska Native, not Hispanic or Latino, Asian, not Hispanic or Latino, or Native Hawaiian or Other Pacific Islander, not Hispanic or Latino	330	85.2	1.19
Undergraduate students who received institutional aid whose gender was Male and whose race/ethnicity was Unknown	30	46.6	2.18
Undergraduate students who received institutional aid whose gender was Unknown	110	90.3	1.11

Rounds to zero.

¹ The weighted response rates were calculated using the BPS:20/22 base weight adjusted for not located and refusal nonresponse.

² New England = Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont; Mid East = Delaware, District of Columbia, Maryland, New Jersey, New York, Pennsylvania; Great Lakes = Illinois, Indiana, Michigan, Ohio, Wisconsin; Plains = Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, South Dakota; Southeast = Alabama, Arkansas, Florida, Georgia, Kentucky, Mississippi, North Carolina, South Carolina, Tennessee, Virginia, West Virginia; Southwest = Arizona, New Mexico, Oklahoma, Texas; Rocky Mountains = Colorado, Idaho, Montana, Utah, Wyoming; Far West = Alaska, California, Hawaii, Nevada, Oregon, Washington; Outlying Areas = Puerto Rico.

³ Categories were defined by quartiles. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of "None" or "Unknown" were excluded from quartile calculations.

⁴ For Total amount of federal Pell Grants awarded to undergraduate students, the second and third categories were defined by the median of Pell amounts greater than \$0 and less than \$6,195. Categories were defined by quartiles. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of "None" or "Unknown" were excluded from quartile calculations.

NOTE: CHAID = chi-square automatic interaction detection; CPS:20 data = Central Processing System data from the 2019–20 National Postsecondary Student Aid Study. Number of respondents rounded to the nearest 10. Detail may not sum to totals because of rounding. SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

6.1.3 Poststratification Adjustment

To ensure that the weighted sample adequately represents individuals who began their postsecondary education for the first time during the 2019–20 academic year at any Title IV eligible postsecondary institution in the United States and who were not concurrently enrolled in high school, statisticians further adjusted the nonresponse-adjusted weight so that weighted sums would equal known population control totals for key characteristics. This adjustment also helped increase the precision of the estimates for these key characteristics and any related characteristics. Part of this poststratification adjustment included trimming adjustments. Specifically, bounds on trimming were set as the median nonresponse-adjusted weight plus or minus three times the interquartile range, where the median and interquartile range were defined by control and level of institution. This adjustment helped increase the precision of characteristics related to the study outcomes.

The statistical team derived control totals by calculating the weighted sums of NPSAS:20 study respondents who were confirmed or potential FTBs for the following variables:

- full-year undergraduate student enrollment for 2019–20, by race/ethnicity, control and level of the institution, and selected states;
- fall undergraduate student enrollment for 2019–20, by control and level of the institution and selected states;
- number of Pell Grant undergraduate recipients in 2019–20, by control and level of the institution and selected states;
- total amount of Pell Grants awarded to undergraduate students in 2019–20, by control and level of the institution and selected states;
- total amount of Parent PLUS Loans disbursed to undergraduate students in 2019–20, by control and level of the institution and selected states;
- number of subsidized Direct Loan undergraduate student recipients in 2019–20, by control and level of the institution and selected states;
- total amount of subsidized Direct Loans disbursed to undergraduate students in 2019–20, by control and level of the institution and selected states;
- number of unsubsidized Direct Loan undergraduate student recipients in 2019–20, by control and level of the institution and selected states;
- total amount of unsubsidized Direct Loans disbursed to undergraduate students in 2019–20, by control and level of the institution and selected states;
- number of state aid undergraduate in 2019–20, by control and level of the institution and selected states; and
- total amount of state aid awarded to undergraduate students in 2019–20, by control and level of the institution and selected states.

If convergence problems with the poststratification model were encountered, variables were selected to be collapsed or dropped from the model. The control totals for variables dropped from the poststratification model were used as benchmark totals in the quality control of the weights.

The WTADJUST procedure was executed with no lower or upper limit on the adjustment factors. Results of the poststratification adjustment models follow.

Table 35 shows the control total and the average poststratification adjustment factor (defined as the product of the trim adjustment factor and model adjustment factor) result from each poststratification category for analysis WTA000 (BPS:20/22 respondents). The poststratification adjustment factor has the following characteristics:

- minimum: 0.01;
- median: 1.36; and
- maximum: 24.19.

Table 35 also provides the weighted sums for each poststratification category for the final, eligible BPS:20/22 sample.

Table 35. Control totals, average poststratification weight adjustment factor, and weighted sums for BPS:20/22 respondents: 2022

Poststratification category	Control total ¹	Average weight adjustment factor (BPS22WTA_adj5)	Weighted sum of eligible respondents ²
Total, full-year first-time beginning undergraduate student enrollment	3,256,967	1.73	3,251,341
Full-year first-time beginning undergraduate student enrollment, by race, within control and level of institution			
Black, not Hispanic or Latino ³			
Public 2-year	159,973	1.67	158,299
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	31,080	1.44	31,080
Public 4-year, non-doctorate-granting, primarily baccalaureate	17,445	1.19	17,445
Public 4-year, doctorate-granting	85,190	1.41	85,190
Private nonprofit 2-year or less	3,324	2.99	3,324
Private nonprofit 4-year, non-doctorate-granting	23,851	1.33	23,529
Private nonprofit 4-year, doctorate-granting	33,143	1.46	33,143
Private for-profit less-than-2-year	15,241	2.99	15,241
Private for-profit 2-year	11,562	3.92	11,169
Private for-profit 4-year	17,316	3.15	17,316
Hispanic or Latino, of any race			
Public less-than-2-year	2,851	2.38	2,851
Public 2-year	300,613	1.57	299,763
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	83,363	1.46	83,363
Public 4-year, non-doctorate-granting, primarily baccalaureate	30,712	1.34	30,712
Public 4-year, doctorate-granting	132,859	1.24	132,859
Private nonprofit 2-year or less	1,776	1.99	1,776
Private nonprofit 4-year, non-doctorate-granting	27,833	1.28	27,833
Private nonprofit 4-year, doctorate-granting	49,555	1.27	49,555
Private for-profit less-than-2-year	23,539	2.08	23,371
Private for-profit 2-year	15,211	2.35	15,011
Private for-profit 4-year	15,262	1.46	15,210

See notes at end of table.

Table 35. Control totals, average poststratification weight adjustment factor, and weighted sums for BPS:20/22 respondents: 2022—Continued

Poststratification category	Control total ¹	Average weight adjustment factor (BPS22WTA_adj5)	Weighted sum of eligible respondents ²
White, not Hispanic or Latino			
Public less-than-2-year	11,850	2.64	11,850
Public 2-year	506,056	1.82	504,881
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	107,189	1.69	107,008
Public 4-year, non-doctorate-granting, primarily baccalaureate	78,274	1.28	78,212
Public 4-year, doctorate-granting	426,053	1.28	425,708
Private nonprofit 2-year or less	3,601	1.70	3,601
Private nonprofit 4-year, non-doctorate-granting	103,744	1.09	103,744
Private nonprofit 4-year, doctorate-granting	180,944	1.27	180,770
Private for-profit less-than-2-year	24,605	2.61	24,605
Private for-profit 2-year	19,118	3.73	19,087
Private for-profit 4-year	33,818	2.16	33,818
Other, not Hispanic or Latino			
Public less-than-2-year ³	2,291	1.45	2,291
Public 2-year	195,164	2.31	195,164
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	53,251	2.60	53,251
Public 4-year, non-doctorate-granting, primarily baccalaureate	29,342	1.91	29,342
Public 4-year, doctorate-granting	203,635	1.82	203,635
Private nonprofit 2-year or less	1,219	3.91	1,219
Private nonprofit 4-year, non-doctorate-granting	45,127	1.85	45,127
Private nonprofit 4-year, doctorate-granting	102,150	2.24	102,150
Private for-profit less-than-2-year	9,330	2.95	9,330
Private for-profit 2-year	6,742	5.08	6,742
Private for-profit 4-year	31,765	7.14	31,765
Fall first-time beginning undergraduate student enrollment, by control and level of institution			
Public less-than-2-year	11,656	2.84	11,656
Public 2-year	782,867	1.50	781,565
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	194,169	1.49	194,104
Public 4-year, non-doctorate-granting, primarily baccalaureate	134,474	1.26	134,412
Public 4-year, doctorate-granting	765,692	1.28	765,347
Private nonprofit 2-year or less	4,672	2.54	4,672
Private nonprofit 4-year, non-doctorate-granting	171,122	1.12	170,800
Private nonprofit 4-year, doctorate-granting	315,345	1.28	315,171
Private for-profit less-than-2-year	37,722	2.42	37,554
Private for-profit 2-year	27,935	3.17	27,356
Private for-profit 4-year	43,779	1.91	43,779
Number of Pell Grant first-time beginning undergraduate recipients, by control and level of institution			
Public less-than-2-year	5,348	2.11	5,348
Public 2-year	534,344	1.65	532,848
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	120,756	1.53	120,691
Public 4-year, non-doctorate-granting, primarily baccalaureate	70,744	1.38	70,744
Public 4-year, doctorate-granting	329,377	1.61	329,377
Private nonprofit 2-year or less	8,743	3.31	8,743
Private nonprofit 4-year, non-doctorate-granting	81,779	1.46	81,457
Private nonprofit 4-year, doctorate-granting	125,064	1.86	124,957
Private for-profit less-than-2-year	62,028	2.77	61,860
Private for-profit 2-year	48,171	3.91	47,547
Private for-profit 4-year	74,158	2.92	74,106

See notes at end of table.

Table 35. Control totals, average poststratification weight adjustment factor, and weighted sums for BPS:20/22 respondents: 2022—Continued

Poststratification category	Control total ¹	Average weight adjustment factor (BPS22WTA_adj5)	Weighted sum of eligible respondents ²
Total amount of Pell Grants awarded to first-time beginning undergraduate students, by control and level of institution			
Public less-than-2-year	17,311,483	2.11	17,311,483
Public 2-year	2,077,943,265	1.65	2070234522
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	472,630,230	1.53	472,428,387
Public 4-year, non-doctorate-granting, primarily baccalaureate	332,461,000	1.38	332,461,000
Public 4-year, doctorate-granting	1,588,418,219	1.61	1588418219
Private nonprofit 2-year or less	36,888,053	3.31	36,888,053
Private nonprofit 4-year, non-doctorate-granting	383,334,598	1.46	381,857,288
Private nonprofit 4-year, doctorate-granting	542,355,801	1.86	541,690,175
Private for-profit less-than-2-year	252,771,124	2.77	252,250,840
Private for-profit 2-year	210,214,529	3.91	208,453,895
Private for-profit 4-year	313,982,727	2.92	313,841,937
Total amount of PLUS loans disbursed to first-time beginning undergraduate student recipients, by control and level of institution			
Public less-than-2-year and Public 2-year	22,808,725	1.13	22,808,725
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	20,197,528	1.16	20,197,528
Public 4-year, non-doctorate-granting, primarily baccalaureate	154,372,834	1.47	154,372,834
Public 4-year, doctorate-granting	1,472,922,061	1.50	1472922061
Private nonprofit 2-year or less and Private nonprofit 4-year, non-doctorate-granting	425,905,057	1.14	424,537,013
Private nonprofit 4-year, doctorate-granting	838,463,129	1.49	837,972,499
Private for-profit less-than-2-year	50,106,049	1.63	50,106,049
Private for-profit 2-year	75,933,400	3.71	75,933,400
Private for-profit 4-year	118,310,783	2.14	118,319,195
Number of subsidized Direct Loan first-time beginning undergraduate student recipients, by control and level of institution			
Public 2-year	108,322	1.63	108,305
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	26,959	1.26	26,959
Public 4-year, non-doctorate-granting, primarily baccalaureate	56,802	1.41	56,740
Public 4-year, doctorate-granting	316,027	1.58	316,027
Private nonprofit 2-year or less	7,596	3.72	7,596
Private nonprofit 4-year, non-doctorate-granting	92,512	1.32	92,190
Private nonprofit 4-year, doctorate-granting	161,070	1.75	160,896
Private for-profit less-than-2-year	51,971	2.35	51,803
Private for-profit 2-year	43,903	3.82	43,310
Private for-profit 4-year	70,961	2.90	70,909
Total amount of subsidized Direct Loans disbursed to first-time beginning undergraduate students, by control and level of institution			
Public 2-year	286,263,217	1.63	286,202,141
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	72,684,954	1.26	72,684,954
Public 4-year, non-doctorate-granting, primarily baccalaureate	178,395,893	1.41	178,179,282
Public 4-year, doctorate-granting	1,037,787,791	1.58	1037787791
Private nonprofit 2-year or less	23,876,470	3.72	23,876,470
Private nonprofit 4-year, non-doctorate-granting	307,325,502	1.32	306,733,337
Private nonprofit 4-year, doctorate-granting	533,255,484	1.75	532,646,012
Private for-profit less-than-2-year	150,904,269	2.35	150,610,371
Private for-profit 2-year	145,068,911	3.82	144,006,079
Private for-profit 4-year	237,661,136	2.90	237,540,983

See notes at end of table.

Table 35. Control totals, average poststratification weight adjustment factor, and weighted sums for BPS:20/22 respondents: 2022—Continued

Poststratification category	Control total ¹	Average weight adjustment factor (BPS22WTA_adj5)	Weighted sum of eligible respondents ²
Number of unsubsidized Direct Loan first-time beginning undergraduate student recipients, by control and level of institution			
Public 2-year	95,011	1.49	93,688
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	23,778	1.40	23,778
Public 4-year, non-doctorate-granting, primarily baccalaureate	59,673	1.39	59,673
Public 4-year, doctorate-granting	338,426	1.51	338,297
Private nonprofit 2-year or less	7,593	3.80	7,593
Private nonprofit 4-year, non-doctorate-granting	98,603	1.28	98,281
Private nonprofit 4-year, doctorate-granting	171,168	1.72	171,061
Private for-profit less-than-2-year	51,593	2.37	51,425
Private for-profit 2-year	43,350	3.86	42,757
Private for-profit 4-year	68,056	2.92	68,004
Total amount of unsubsidized Direct Loans disbursed to first-time beginning undergraduate students, by control and level of institution			
Public 2-year	274,966,059	1.49	268,374,140
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	68,085,717	1.40	68,085,717
Public 4-year, non-doctorate-granting, primarily baccalaureate	187,397,431	1.39	187,397,431
Public 4-year, doctorate-granting	1,050,139,118	1.51	1049732663
Private nonprofit 2-year or less	21,613,917	3.80	21,613,917
Private nonprofit 4-year, non-doctorate-granting	310,318,464	1.28	309,996,460
Private nonprofit 4-year, doctorate-granting	515,103,364	1.72	514,894,217
Private for-profit less-than-2-year	193,554,889	2.37	193,051,064
Private for-profit 2-year	182,294,442	3.86	181,811,226
Private for-profit 4-year	263,235,785	2.92	263,028,434
Number of Direct Loan first-time beginning undergraduate student recipients: Public less-than-2-year institutions	1,238	0.98	1,238
Total amount of Direct Loans disbursed to first-time beginning undergraduate students: Public less-than-2-year institutions	6,947,900	0.98	6,947,900
Full-year first-time beginning undergraduate student enrollment, by race, within selected states			
California			
White, not Hispanic or Latino	161,955	1.64	161,815
Hispanic or Latino, of any race	79,961	1.43	79,899
Other race, not Hispanic or Latino	130,588	2.14	130,588
Florida			
White, not Hispanic or Latino	73,350	1.92	73,319
Hispanic or Latino, of any race	59,976	1.63	59,808
Other race, not Hispanic or Latino	68,192	2.69	68,178
Georgia			
White, not Hispanic or Latino	39,811	1.59	39,695
Black, not Hispanic or Latino	29,201	1.35	29,201
Other race/ethnicity	23,663	1.62	23,663
North Carolina			
White, not Hispanic or Latino	52,159	1.46	52,052
Black, not Hispanic or Latino	19,968	1.59	19,646
Other race/ethnicity	24,300	1.67	24,300

See notes at end of table.

Table 35. Control totals, average poststratification weight adjustment factor, and weighted sums for BPS:20/22 respondents: 2022—Continued

Poststratification category	Control total ¹	Average weight adjustment factor (BPS22WTA_adj5)	Weighted sum of eligible respondents ²
New York			
White, not Hispanic or Latino	79,773	1.33	79,731
Hispanic or Latino, of any race	41,159	1.11	41,159
Other race, not Hispanic or Latino	86,970	1.80	86,970
Pennsylvania			
White, not Hispanic or Latino	67,632	1.54	67,407
Black, not Hispanic or Latino	13,786	1.20	13,786
Other race/ethnicity	34,587	2.02	34,587
Texas			
White, not Hispanic or Latino	120,312	1.67	120,312
Hispanic or Latino, of any race	80,623	1.81	80,558
Other race, not Hispanic or Latino	78,505	1.91	77,460
Fall first-time beginning undergraduate student enrollment, by selected states			
California	269,504	1.38	269,302
Florida	143,492	1.78	143,324
Georgia	71,279	1.37	71,279
North Carolina	78,399	1.37	77,970
New York	166,011	1.29	165,969
Pennsylvania	94,155	1.27	93,930
Texas	216,580	1.55	216,515
Number of Pell Grant first-time beginning undergraduate recipients, by selected states			
California	156,048	1.78	155,908
Florida	107,259	2.36	107,046
Georgia	50,214	1.71	50,214
North Carolina	43,080	1.57	42,651
New York	93,967	1.39	93,967
Pennsylvania	45,356	1.83	45,356
Texas	145,322	1.94	145,257
Total amount of Pell Grants awarded to first-time beginning undergraduate students, by selected states			
California	683,448,040	1.78	682,361,319
Florida	484,194,042	2.36	483,491,642
Georgia	226,643,673	1.71	226,643,673
North Carolina	187,253,160	1.57	185,110,224
New York	434,766,416	1.39	434,766,416
Pennsylvania	185,114,556	1.83	185,114,556
Texas	663,598,461	1.94	663,396,618
Total amount of PLUS loans disbursed to first-time beginning undergraduate student recipients, by selected states			
California	204,636,743	1.82	204,642,834
Florida	134,547,965	1.92	134,547,965
Georgia	104,631,652	1.24	104,631,652
North Carolina	89,972,216	1.87	88,107,451
New York	261,780,526	1.15	261,780,526
Pennsylvania	268,565,104	1.81	268,565,104
Texas	208,539,802	2.26	208,539,802

See notes at end of table.

Table 35. Control totals, average poststratification weight adjustment factor, and weighted sums for BPS:20/22 respondents: 2022—Continued

Poststratification category	Control total ¹	Average weight adjustment factor (BPS22WTA_adj5)	Weighted sum of eligible respondents ²
Number of subsidized Direct Loan first-time beginning undergraduate student recipients, by selected states			
California	53,537	2.13	53,475
Florida	55,253	2.33	55,071
Georgia	31,287	1.62	31,287
North Carolina	25,422	1.66	24,993
New York	57,658	1.35	57,658
Pennsylvania	57,687	1.89	57,687
Texas	74,467	2.05	74,467
Total amount of subsidized Direct Loans disbursed to first-time beginning undergraduate students, by selected states			
California	176,992,670	2.13	176,776,522
Florida	182,510,171	2.33	182,166,962
Georgia	102,178,461	1.62	102,178,461
North Carolina	84,539,296	1.66	83,571,071
New York	180,730,361	1.35	180,730,361
Pennsylvania	185,615,688	1.89	185,615,688
Texas	236,736,922	2.05	236,736,922
Number of unsubsidized Direct Loan first-time beginning undergraduate student recipients, by selected states			
California	44,324	2.12	44,324
Florida	53,989	2.46	53,807
Georgia	33,301	1.73	33,301
North Carolina	27,486	1.66	27,057
New York	60,566	1.30	60,524
Pennsylvania	60,171	1.85	59,946
Texas	70,237	2.01	69,192
Total amount of unsubsidized Direct Loans disbursed to first-time beginning undergraduate students, by selected states			
California	135,280,416	2.12	135,280,681
Florida	198,817,185	2.46	198,228,826
Georgia	101,571,475	1.73	101,571,475
North Carolina	89,860,331	1.66	89,328,915
New York	191,869,863	1.30	191,639,552
Pennsylvania	171,383,118	1.85	170,799,323
Texas	219,446,757	2.01	213,699,723
Number of state aid first-time beginning undergraduate student recipients, by selected states			
California	153,754	1.50	153,692
Florida	59,263	1.71	59,095
Georgia	63,741	1.85	63,625
North Carolina	27,332	1.81	26,903
New York	81,273	1.39	81,273
Pennsylvania	40,545	2.21	40,545
Texas	100,625	3.04	99,580

See notes at end of table.

Table 35. Control totals, average poststratification weight adjustment factor, and weighted sums for BPS:20/22 respondents: 2022—Continued

Poststratification category	Control total ¹	Average weight adjustment factor (BPS22WTA_adj5)	Weighted sum of eligible respondents ²
Total amount of state aid disbursed to first-time beginning undergraduate students, by selected states			
California	565,041,247	1.50	564,685,881
Florida	239,123,310	1.71	238,997,354
Georgia	259,391,151	1.85	258,985,103
North Carolina	83,322,031	1.81	81,508,343
New York	389,144,386	1.39	389,144,386
Pennsylvania	91,274,926	2.21	91,274,926
Texas	364,064,561	3.04	362,482,559

¹ Control totals were derived from the weighted sum of BPS:20/22-eligible 2019–20 National Postsecondary Student Aid Study (NPSAS:20) study respondents (including deceased cases). They were weighted using the NPSAS:20 study base weight adjusted for student multiplicity, unknown eligibility, and nonresponse.

² The weighted sums of eligible respondents are the final BPS:20/22 weighted sums after removing deceased cases.

³ The control total for Black, not Hispanic or Latino students from public less-than-2-year institutions was collapsed with other race, not Hispanic or Latino students from public less-than-2-year institutions.

NOTE: Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

6.1.4 Weighted Adjustment Summary and Evaluation

To determine the effect of weighting adjustments, BPS statisticians analyzed the UWEs and the distribution of weights. Table 36 summarizes the BPS:20/22 survey weight distributions and the variance inflation caused by the UWE, by control and level of institution. The UWE is 2.09 overall and ranges from 1.61 for sample members at public 4-year, non-doctorate-granting, primarily baccalaureate base-year institutions to 6.69 for sample members at private nonprofit less-than-4-year base-year institutions, with most UWEs being below 3.00.

Table 36. Sample member weight distribution and unequal weighting effects for BPS:20/22 respondents, by control and level of base-year institution: 2022

Institution characteristics	Minimum	First quartile	Median	Third quartile	Maximum	Mean	Unequal weighting effect ¹
Total	1.0	55.6	105.6	182.7	2,481.4	145.6	2.09
Public less-than-2-year	1.5	30.8	61.8	121.0	569.0	94.9	2.16
Public 2-year	2.2	75.9	123.6	209.7	1,637.3	172.0	1.86
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	8.7	83.1	139.3	221.8	1,362.5	179.4	1.68
Public 4-year, non-doctorate-granting, primarily baccalaureate	2.5	53.5	83.4	120.7	794.3	99.8	1.61
Public 4-year, doctorate-granting	1.2	84.1	137.1	214.1	2,069.4	172.0	1.70
Private nonprofit less-than-4-year	1.0	11.6	37.1	101.1	1,995.6	74.6	6.69
Private nonprofit 4-year, non-doctorate-granting	1.2	32.3	73.6	131.2	1,616.6	105.6	2.38
Private nonprofit 4-year, doctorate-granting	1.0	48.8	99.3	182.0	2,481.4	150.2	2.66
Private for-profit less-than-2-year	3.8	38.2	91.0	159.5	893.1	124.2	2.06
Private for-profit 2-year	1.8	20.5	36.5	69.2	476.7	58.8	2.27
Private for-profit 4-year	1.0	6.0	17.9	70.5	1,892.5	67.2	5.24

¹ The unequal weighting effect is calculated as $n\sum w^2/(\sum w)^2$, where n is the sample size and w is the final sample member weight.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Additionally, statisticians created a Receiver Operating Characteristic (ROC) curve (Hanley and McNeil 1982) to assess the performance of the nonresponse adjustment models. Specifically, the ROC curve measures how well the model correctly classified individuals with a known response status.¹³ The ROC curve was developed in the following manner. The predicted probabilities of response (c) for the ROC curve associated with the nonresponse are the product of the predicted response probabilities obtained at each of the three nonresponse adjustment steps. For any specified probability of response, c , two proportion were calculated:

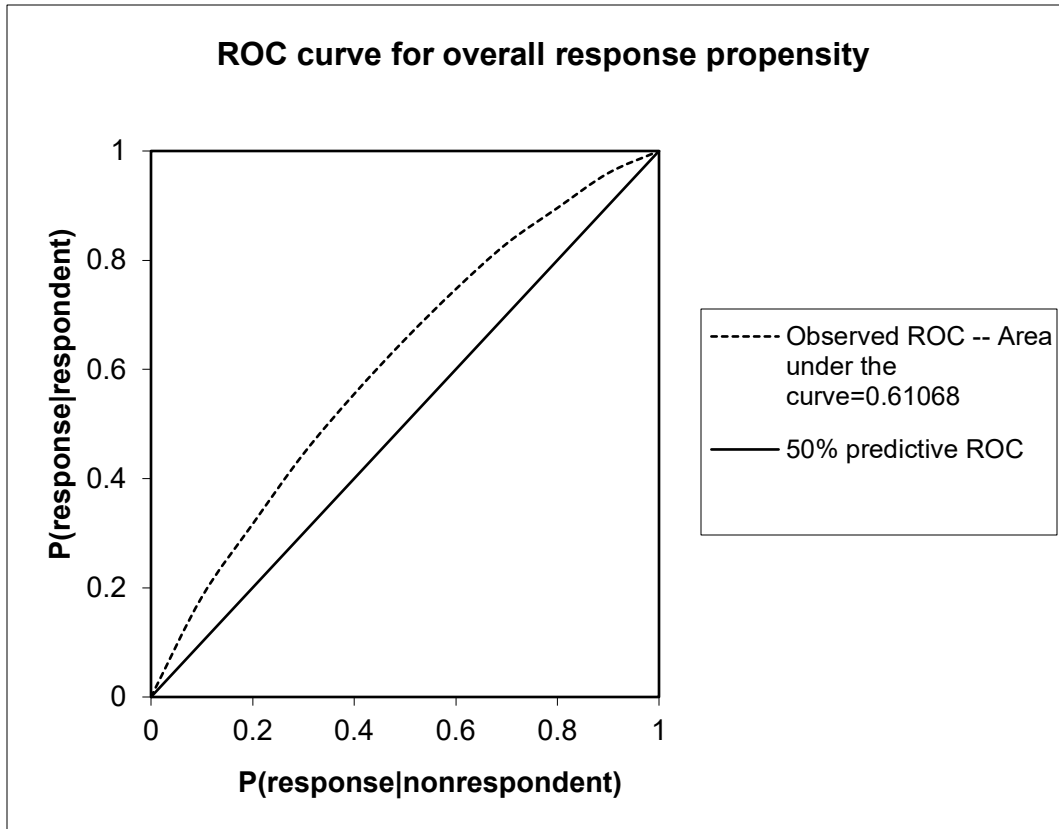
- the proportion of respondents with a predicted probability of response greater than c (the true positive rate); and
- the proportion of nonrespondents with a predicted probability of response greater than c (the false positive rate).

The ROC curve is created by plotting the true positive rate against the false positive rate for all c . The area under the curve represents the probability that the nonresponse adjustment models correctly classify individuals. An area of 0.5 under a ROC curve indicates that a correct classification is made 50 percent of the time. This is equivalent to random assignment and would indicate the model provided no predictive benefit. An area of 1.0 indicates that the model always classified individuals correctly.

Figure 5 shows that the area under the ROC curve for BPS:20/22 sample members is approximately 0.61, so the predicted probabilities give the correct classification about 61 percent of the time. Additionally, the area under the ROC curve can serve as the nonparametric Wilcoxon test, which can determine whether the predicted probability of response is different between respondents and nonrespondents. In this case, the Wilcoxon test rejects the null hypothesis that the nonresponse models have no predictive ability for response status. Thus, the variables used in the model are informative predictors of a sample member's overall response propensity.

¹³ For a more detailed example of the ROC curve used in nonresponse modeling, see Iannacchione (2003).

Figure 5. Receiver Operating Characteristic (ROC) curve for overall sample member response propensity for BPS:20/22: 2022



SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

6.2 Weighted and Unweighted Response Rates

The overall BPS:20/22 response rate is an estimate of the proportion of the study population directly represented by the respondents.

The BPS:20/22 sample only included confirmed and potential FTBs who were defined as study respondents in NPSAS:20. A study respondent was defined as any undergraduate sample member who was an administrative student respondent or survey respondent and is discussed in more detail in the executive summary of the NPSAS:20 DFD (Wine, Seigel, and Haynes 2024). Because of this, the overall BPS:20/22 response rate is the product of the NPSAS:20 study sample response rate times the BPS:20/22 survey response rate. Furthermore, the overall BPS:20/22 response rates can only be estimated directly by institution characteristics, and not student characteristics, given that they are missing for those who do not respond.

Table 37 displays the final unweighted and weighted NPSAS:20 institution and study response rates and BPS:20/22 sample member response rates, by control and level of institution. Section 6.3.1 analyzes the potential bias due to unit nonresponse and the effect the weight adjustments had in reducing the bias.

Table 37. Unweighted and weighted NPSAS:20 institution and administrative study response rates and BPS:20/22 sample member response rates, by control and level of base-year institution: 2022

Control and level of institution	NPSAS:20				BPS:20/22					
	Institution sample		Study sample ¹		Number eligible	Number of respondents	Survey data collection		Overall BPS sample ²	
	Unweighted response rate	Weighted response rate	Unweighted response rate	Weighted response rate			Unweighted response rate	Weighted response rate ³	Unweighted response rate	Weighted response rate ³
Total	71.5	63.9	80.0	83.7	34,240	22,320	65.2	59.6	52.1	49.8
Public										
Less-than-2-year	53.8	68.5	91.6	96.0	380	180	47.7	44.8	43.7	43.1
2-year	75.1	75.2	74.8	75.4	11,000	6,730	61.2	54.3	45.8	40.9
4-year										
Non-doctorate-granting										
Primarily subbaccalaureate	75.4	75.4	82.6	87.9	2,540	1,530	60.4	51.1	49.9	45.0
Primarily baccalaureate	81.2	81.2	77.2	81.2	2,210	1,560	70.6	65.6	54.5	53.2
Doctorate-granting	87.4	87.4	87.0	89.5	7,220	4,930	68.2	62.9	59.3	56.3
Private nonprofit										
Less-than-4-year	43.8	58.8	80.8	83.7	220	130	59.6	59.2	48.2	49.6
4-year										
Non-doctorate-granting	71.7	71.2	81.4	86.4	2,640	1,900	71.9	66.7	58.5	57.6
Doctorate-granting	76.6	78.0	83.5	88.9	3,470	2,440	70.1	63.8	58.5	56.7
Private for-profit										
Less-than-2-year	42.2	38.9	68.6	70.3	920	580	63.5	59.2	43.5	41.7
2-year	49.7	49.7	67.1	72.3	1,440	880	61.6	58.7	41.3	42.4
4-year	51.3	62.3	86.0	87.5	2,210	1,460	65.9	56.8	56.7	49.6

¹ The 2019–20 National Postsecondary Student Aid Study (NPSAS:20) defined a study respondent as any undergraduate sample member who was an administrative student respondent or NPSAS:20 survey respondent.

² Overall BPS:20/22 response rate is the product of the NPSAS:20 study response rate and the BPS:20/22 survey data collection response rate.

³ Weighted rates use the BPS:20/22 base weight WTA000.

NOTE: Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2019–20 National Postsecondary Student Aid Study (NPSAS:20) and 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

6.3 Nonresponse Bias Analysis

The accuracy of survey statistics is affected by both random and nonrandom errors. Random errors reduce the precision of survey estimates; however, precision can be improved with increased sample size. For example, all sample surveys are subject to random error in the form of sampling error (i.e., an error that occurs because the survey is based on a sample of population members rather than the population). Nonrandom errors may result in bias (i.e., estimates that do not converge to the true population parameter as the sample size increases without limit). Nonrandom errors can result in biased estimates for administrative data as well. The BPS:20/22 data collection procedures were thoroughly developed and tested to minimize nonrandom errors because these errors are difficult to quantify.

Sample studies typically encounter two general types of nonresponse: unit nonresponse and item nonresponse. Unit nonresponse occurs when an individual did not respond to the BPS:20/22 study. Item nonresponse occurs when an individual respondent is missing values for one or more items. Errors due to nonresponse are often, but not always, nonrandom; hence, when nonresponse rates are high, an assessment of the potential for study results to be biased due to nonresponse is required under NCES statistical standards. NCES Statistical Standard 4-4-1 states that “Any survey stage of data collection with a unit or item response rate (RRI) less than 85 percent must be evaluated for the potential magnitude of nonresponse bias before the data or any analysis using the data may be released. Estimates of survey characteristics for nonrespondents and respondents are required to assess the potential nonresponse bias” (Seastrom 2014).

Due to weighted unit response rates below 85 percent (section 6.1), unit nonresponse bias analyses were conducted for the sample overall and across institution level and control regardless of whether the unit response rate was below 85 percent. Item nonresponse bias analyses were performed for all survey items that had a weighted response rate below 85 percent, using the analysis weight (WTA000) for individuals overall. The bias in an estimated mean based on respondents $\bar{y}_R$ is the difference between the expected value of this mean and the target parameter, π , the population mean: $B(\bar{y}_R) = E(\bar{y}_R) - \pi$. This bias can be estimated as the difference between the mean for respondents and the mean for nonrespondents, multiplied by the weighted nonresponse rate: $\hat{B}(\bar{y}_R) = \eta(\bar{y}_R - \bar{y}_{NR})$. A general overview of the procedures for unit and item nonresponse bias analyses can be found in section 7.3.1 of the NPSAS:20 DFD (Wine, Seigel, and Haynes 2024).

The unit-level results are summarized in section 6.3.1, and item-level results and response rates are summarized in section 6.3.2. Detailed nonresponse bias tables for all analyses are provided in appendix G.

6.3.1 Nonresponse Bias Analysis: Unit Level

BPS statisticians conducted unit-level bias analysis for the analytic weight (WTA000) overall and within each control and level of institution. The results are summarized below. Nonresponse bias was estimated for variables known for all respondents and nonrespondents. Bias estimates for characteristic categories that did not meet reporting requirements (i.e., they had fewer than 30 nonrespondents) were excluded from calculations of summary statistics. The following variables were used for the nonresponse bias analysis:

- control and level of institution (categorical, from NPSAS:20);
- state of institution (categorical, from NPSAS:20);
- composite variable of selected states and Census region of institution (categorical, from BPS:20/22);
- Bureau of Economic Analysis region of institution (categorical, from NPSAS:20);
- federal aid status in 2019–20 (received/did not receive/missing, from NPSAS:20);
- Direct Loan status in 2019–20 (received/did not receive, from NPSAS:20);
- total Direct Loan amount received in 2019–20 (quartiles, from NPSAS:20);
- age group as of December 31, 2019 (categorical, from NPSAS:20);
- institution aid status in 2019–20 (received/did not receive/missing, from NPSAS:20);
- state aid status in 2019–20 (received/did not receive/missing, from NPSAS:20);
- any aid status in 2019–20 (received/did not receive/missing, from NPSAS:20);
- veteran status (yes/no, from NPSAS:20);
- race/ethnicity (categorical, from NPSAS:20);
- gender (male/female/unknown, from NPSAS:20);

- institution total undergraduate full-year enrollment from IPEDS 2020–21 file (quartiles, from NPSAS:20);
- undergraduate major (categorical, from NPSAS:20);
- undergraduate degree program (from NPSAS:20);
- Pell Grant status in 2019–20 (received/did not receive, from NPSAS:20);
- total Pell Grant amount received in 2019–20 (quartiles, from NPSAS:20);
- total income (quartiles, from CPS:20 data);
- parents' highest education (categorical, from CPS:20 data);
- has dependents (categorical, from CPS:20 data); and
- marital status (categorical, from CPS:20 data).

As seen in table 38, there were many small but statistically significant biases on observable characteristics before the nonresponse weight adjustment, and these were largely diminished after weight adjustment overall and across institution control and level. Before weight adjustment, the median percent relative bias across characteristics was 7.1 overall, ranging from 4.9 to 12.4 across institution control and level. After weight adjustment, the median was nearly zero overall and ranged from 2.4 to 9.4 across institution control and level.

The percentage of characteristics with statistically significant bias before weight adjustment was 53 overall and between 9 and 42 across institution control and level. After weight adjustment, the percentage was 11 overall and between 2 and 14 across institution control and level. The median effect size before weight adjustment was 0.09 overall and between zero and 0.13 across institution control and level, while after weight adjustment, the median effect size was zero overall and 0.01 to 0.11 across institution control and level.

As seen in table 39, the mean absolute difference across the same frame characteristics between respondent means before and after poststratification was 0.8 overall, and mean values ranged from 0.6 to 14.6 by institution control and level. The median absolute difference across characteristics was 0.4 overall, and median values ranged from 0.3 to 12.2 by institution control and level. While poststratification is expected to increase bias for frame characteristics as compared to nonresponse-adjusted weights, these values show relatively small increases in the mean absolute differences. Appendix G, table G-2, details the differences for each level of the characteristics and whether the difference is statistically significant.

The mean absolute difference across characteristics between the full-sample and respondent means after poststratification was 0.8 overall, and mean values ranged from 0.7 to 14.2 by institution and level. The median absolute difference across characteristics was 0.4 overall, and median values ranged from 0.4 to 7.6 by institution and level. These values are very similar to the mean and median absolute differences between the respondent means before and after poststratification. This is expected as table 38 showed that, after nonresponse weight adjustments, very little bias remained, meaning the respondent sample had distributions similar to the full sample.

Table 38. Summary of unit-level nonresponse bias analysis for BPS:20/22 response, by control and level of institution: 2022

Nonresponse bias statistics ¹	Overall	Public less-than- 2-year	Public 2-year	Public 4-year, non- doctorate- granting, primarily subbaccalaureate	Public 4-year, non- doctorate- granting, primarily baccalaureate	Public 4-year, doctorate- granting	Private nonprofit less-than- 4-year	Private nonprofit 4-year, non- doctorate- granting	Private nonprofit 4-year, doctorate- granting	Private for-profit less-than- 2-year	Private for-profit 2-year	Private for-profit 4-year
Before nonresponse weight adjustments²												
Mean percent relative bias across characteristics	9.38	14.81	10.88	9.54	6.33	6.77	7.03	8.40	7.95	9.34	10.82	12.41
Median percent relative bias across characteristics	7.11	12.42	8.14	7.23	5.01	5.26	4.94	6.33	5.64	5.11	7.38	8.81
Percentage of characteristics with significant bias	52.94	14.63	41.91	41.18	32.50	39.67	9.09	41.33	35.87	10.94	14.29	17.78
Median effect size	0.09	0.13	0.08	0.07	0.05	0.05	0.00	0.04	0.08	0.06	0.09	0.08
After nonresponse weight adjustments³												
Mean percent relative bias across characteristics	2.74	11.67	5.60	4.70	4.32	4.18	5.73	5.15	4.25	4.36	7.28	8.88
Median percent relative bias across characteristics	#	9.43	3.25	2.44	3.37	2.66	3.49	3.37	2.79	2.79	4.71	7.25
Percentage of characteristics with significant bias	10.59	2.44	11.03	8.24	8.75	5.79	13.64	6.67	4.35	1.56	6.49	5.56
Median effect size	#	0.11	0.01	0.04	0.03	0.02	0.02	0.03	0.01	0.04	0.04	0.06

Rounds to zero.

¹ Relative bias and effect size are calculated using the weighted differences between respondent and full-sample means. Relative bias is calculated as 100 times the ratio of estimated bias to the weighted full-sample mean. Effect size is calculated as the square root of the sum over categories of the squared differences over full-sample means.² Respondent and full-sample means are weighted using the BPS:20/22 base weight.³ Full-sample means are weighted using the BPS:20/22 base weight, and the respondent means are weighted using the BPS:20/22 base weight adjusted for nonresponse.

NOTE: Characteristics that had fewer than 30 nonrespondents were excluded from calculation of summary statistics.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table 39. Summary of unit-level analysis after poststratification for BPS:20/22 response, by control and level of institution: 2022

Nonresponse bias statistics ¹	Overall	Public less-than- 2-year	Public 2-year	Public 4-year, non- doctorate- granting, primarily subbaccalaureate	Public 4-year, non- doctorate- granting, primarily baccalaureate	Public 4-year, doctorate- granting	Private nonprofit less-than- 4-year	Private nonprofit 4-year, non- doctorate- granting	Private nonprofit 4-year, doctorate- granting	Private for-profit less-than- 2-year	Private for-profit 2-year	Private for-profit 4-year
Difference between means for respondents before and after poststratification adjustment¹												
Mean absolute difference across characteristics	0.78	5.92	0.61	1.23	0.79	1.13	14.63	1.43	1.80	2.84	5.11	3.38
Median absolute difference across characteristics	0.37	5.33	0.33	0.76	0.61	0.42	12.20	1.08	1.02	2.43	2.64	2.02
Difference between means for full sample and respondents after poststratification adjustment²												
Mean absolute difference across characteristics	0.79	4.01	0.71	1.44	0.87	1.08	14.24	1.33	2.00	2.77	5.14	4.01
Median absolute difference across characteristics	0.37	2.83	0.43	0.96	0.58	0.51	7.56	1.14	1.14	2.09	2.81	2.83

¹ Respondent means before poststratification adjustment are weighted using the BPS:20/22 base weight adjusted for nonresponse. Respondent means after poststratification adjustment are weighted using the BPS:20/22 base weight adjusted for nonresponse and poststratification.

² Full-sample means are weighted using the BPS:20/22 base weight, and respondent means are weighted using the BPS:20/22 base weight adjusted for nonresponse and poststratification.

NOTE: Characteristics that had fewer than 30 nonrespondents were excluded from calculation of summary statistics.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

6.3.2 Nonresponse Bias Analysis: Item Level

Item nonresponse bias analyses were conducted overall for each item with a weighted response rate below 85 percent. The variables used are the same as those used in the unit analyses (section 6.3.1). The results are summarized below.

Weighted item response rates were calculated for all items in the derived variable files using the analysis weight (WTA000) and appear in appendix I. In accordance with NCES Statistical Standard 1-3-5 (Seastrom 2014), RRI's were calculated as the ratio of the number of unit respondents for whom in-scope data were obtained (I^x for item x) to the number of eligible respondents, which is calculated as the number of unit respondents (I) minus the number of item respondents with a valid skip for item x (V^x):

$$RRI^x = I^x / (I - V^x).$$

If a unit respondent's eligibility for an item is unknown, it is treated as an eligible item nonrespondent. An example where item-level unknown eligibility can occur would be if an individual chose to not complete a question that informs a skip pattern for subsequent questions in the interview. Without a response to the informing logical gate, missingness for the following questions cannot be attributed to logical skips or eligible missingness.

When an abbreviated questionnaire is used to convert refusals, the eliminated questions are treated as item nonresponse, so items on the abbreviated questionnaire tend to have higher response rates than items not on the questionnaire. A sample member was defined as an item respondent for a variable if that sample member had data for that variable from any source, including logical imputation.¹⁴

Appendix table I-1 contains weighted item response rates for 455 variables in the derived file. Response rates overall ranged from 6 percent for *Private student loans: monthly payment amount* (PRRPY22) to 100 percent for several variables from administrative sources. Another 94 variables had response rates over 95 percent. There were 267 variables with overall response rates below 85 percent.

As displayed in appendix I, tables I-2 and I-3, statistically significant differences between the pre- and postimputation means were found for about 51 percent of

¹⁴ Logical imputation is filling in missing data for cases where values can be deduced with certainty based upon logical or mathematical relationships among observed variables.

the continuous variables, and roughly 60 percent of the categorical variables had at least one level that had a statistically significantly different percent. The differences between the preimputation and postimputation means were statistically significant for these variables, suggesting that the bias was reduced by imputation.

6.4 Variance Estimation

Every estimate calculated from a probability-based sample study, such as a mean, a percentage, or a regression coefficient, has an associated variance. Hypothesis testing, calculation of confidence intervals, and modeling that uses complex sample data all require the calculation of variances using appropriate methods that account for the sampling design. Complex sample designs like those used for NPSAS:20 and BPS:20/22 result in data that violate the assumptions that are normally required to assess the statistical significance of results. The variances of the estimates from complex sample studies may differ from those that would be expected if the sample were a simple random sample, and the observations were independent and identically distributed random variables.

When testing hypotheses (conducting t tests, regression analyses, etc.) using weighted data from a study such as NPSAS that has a complex design, analysts should use methods that properly estimate variances. Two procedures for estimating variances of statistics from complex sample studies are the Taylor series linearization procedure and the bootstrap replication procedure, which are both available for the NPSAS data files. Stratum, primary sampling unit (PSU), and secondary sampling unit (SSU) identifiers are provided in the data file for use with the Taylor series method, and bootstrap weights are provided for bootstrap replicate variance estimation.

The analysis strata and PSUs created for the Taylor series procedure are discussed in section 6.4.1. Section 6.4.2 contains a discussion of the replicate weights created for the bootstrap procedure. Use of software packages for proper variance estimation is discussed in section 6.4.3.

Design effects were also calculated for key variables to assist analysts who do not have access to software that can compute Taylor series or bootstrap variance estimates. The design effect for a statistic is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that sample had been selected). It is often used to measure the effects that sample design features have on the

precision of estimates. For example, stratification tends to decrease the variance, but multistage sampling and unequal sampling rates usually increase the variance. In addition, weight adjustments for nonresponse (performed to reduce nonresponse bias) and poststratification increase the variance by increasing the weight variation. Design effects are discussed in section 6.4.4 and appendix H.

6.4.1 Taylor Series

The Taylor series variance estimation procedure is a well-known technique used to estimate the variances of nonlinear statistics. The procedure takes the first-order Taylor series approximation of a nonlinear statistic and substitutes the linear representation into the appropriate variance formula based on the sample design (Woodruff 1971).

For stratified multistage sample studies, the Taylor series procedure requires variance estimation strata and variance estimation PSUs, defined from the sampling strata and PSUs used in the first stage of sampling (NPSAS:20 for BPS:20/22). The steps used in the construction of the NPSAS:20 strata and PSU variables are described in chapter 7 of the NPSAS:20 DFD (Wine, Seigel, and Haynes 2024). The variance estimation formulas require at least two PSUs in each stratum. When BPS:20/22 strata did not contain two PSUs, statisticians collapsed the NPSAS:20 variance estimation strata or PSUs following three rules:

1. variance estimation strata were combined with others within the original NPSAS:20 sampling strata;
2. schools sampled with certainty were combined with other schools sampled with certainty; and
3. noncertainty schools were combined with other noncertainty schools.

In addition, the sort order that was used for constructing the NPSAS:20 variance estimation strata and PSUs was used for BPS:20/22. The statistical team combined variance estimation strata with the next stratum in the sorted list. If the stratum was the first in the sorted list, then statisticians combined them with the next stratum in the list. The single PSU then became an additional PSU in the new variance estimation strata.

The NPSAS:20 restricted-use data files provided two sets of variables for Taylor series variance estimation, and BPS:20/22 also provides two sets of variables. One set of variables is used in software that assumes that the first-stage sampling units (institutions) were sampled with replacement (or with small selection

probabilities) and does not account for the finite population corrections (FPCs) at the institution level of sampling. The other set of variables is used in software that assumes sampling of institutions without replacement in the calculation of variances and does account for the FPC. Statisticians provide both sets of variables because not all survey data analysis packages have the option to incorporate the FPC in the variance calculations. When the first-stage units are sampled with very small probabilities, the estimated variances using the with-replacement variance formulas and the without-replacement variance formulas are the same.

The set of variables that assume the first-stage units were sampled without replacement and account for the FPC includes the analysis stratum (BPS22FANALSTR), analysis PSU (BPS22FANALPSU), the analysis SSU (BPS22FANALSSU), and the count of PSUs in an analysis stratum (BPS22PSUCOUNT). The set of variables that assume the first-stage units were sampled with replacement includes the analysis stratum (BPS22ANALSTR) and analysis PSU (BPS22ANALPSU). Ultimately, BPS22FANALSTR equals the institutional variance estimation stratum BPS22ANALSTR, and BPS22FANALPSU equals BPS22ANALPSU. BPS22FANALSSU was created by randomly dividing the NPSAS:20 analysis PSUs into two parts. These variables are a by-product of the bootstrap variance estimation weights (described in section 6.4.2), and the justification for using the without-replacement variance formulas follows from the assumptions described by Kott (1988). Some values of the variance estimation strata, PSU variables, and SSU variables were combined to have at least two SSUs in each PSU and at least two PSUs in each stratum. The same stratum and PSU terms, under with-replacement and without-replacement assumptions, were used for analysis with the cross-sectional weight.

6.4.2 Bootstrap Replicate Weights

Statisticians also constructed a set of 200 bootstrap weights separate from those done for NPSAS:20. The analysis weight WTA000, defined in section 6.1, is used for computing estimates such as means, percentages, and regression coefficients, and the vector of replicate weights allows for computation of additional estimates for the sole purpose of estimating variances.

Section 7.3.2 of the NPSAS:20 DFD (Wine, Seigel, and Haynes 2024) provides an overview of the approach to create the bootstrap replicate weights. This includes creations of the replicate analysis strata and PSUs and algorithms to account for FPCs.

The number of replicate weights was set to 200 to ensure stable variance estimates for a variety of estimates. Statisticians applied the nonresponse and poststratification adjustments described in section 6.1 to each replicate to create the 200 replicate weights included on the analysis file (WTA001–WTA200), so that the variances could be estimated to account for these weight adjustments. For some of the replicates, the statistical team had to loosen the bounds on the nonresponse and poststratification adjustment factors or collapse model variables because of model convergence problems (i.e., there was no solution to satisfy all model equations simultaneously). However, the model adjustments were not necessary for many replicates, and when it was necessary, the adjustments were minimal. Therefore, this approach worked well for BPS:20/22 to achieve model convergence for all replicates and to minimize the effect of different models on the variance estimates.¹⁵

6.4.3 Software Use for Variance Estimation

Table 40 summarizes the weight and variance estimation variables and how they are used in selected software packages that allow for Taylor series variance estimation with replacement (SUDAAN, Stata, the SAS survey data analysis procedures, IBM SPSS complex samples, and the R survey package), Taylor series variance estimation without replacement (SUDAAN, Stata, and the R survey package), and bootstrap variance estimation (SUDAAN, Stata, the SAS survey data analysis procedures, WesVar, and the R survey package). The provided code is intended for use within respective program statements or procedures and cannot be used alone as shown in the table. The code may need to be revised to be appropriate for a user's specific data file and coding decisions, and for that reason, the provided code may not work for all users and may require editing before it is implemented.

¹⁵ The sums of the replicate weights vary slightly due to the model adjustments, but replicate weights are only used for variance estimates and not for point estimates.

Table 40. Use of analysis weights, replicate weights, and variance estimation strata, primary sampling unit (PSU), secondary sampling unit (SSU), and PSU count variables available from NPSAS:20 in selected survey data analysis software: 2022

Analysis weight for estimates	WTA000
Taylor series variance estimation (with replacement)	
Variance estimation stratum and PSU variables	BPS22ANALSTR and BPS22ANALPSU
Software: statements, parameters, and keywords for Taylor series variance estimation (with replacement)	
SUDAAN	DESIGN = WR WEIGHT WTA000; NEST BPS22ANALSTR BPS22ANALPSU;
Stata	svyset BPS22ANALPSU [pweight = WTA000], strata (BPS22ANALSTR) vce(LINEARIZED)
SAS survey data analysis procedures	VARMETHOD = TAYLOR WEIGHT WTA000; STRATA BPS22ANALSTR; CLUSTER BPS22ANALPSU;
IBM SPSS complex samples ¹	CSPLAN ANALYSIS /PLAN FILE='myfile.csaplan' /PLANVARS ANALYSISWEIGHT=WTA000 /DESIGN STRATA=BPS22ANALSTR CLUSTER=BPS22ANALPSU /ESTIMATOR TYPE=WR
R survey package ²	mydesign<-svydesign(id=~BPS22ANALPSU, strata=~BPS22ANALSTR, weights=~WTA000, data=mydata)
Taylor series variance estimation (without replacement)	
Variance estimation stratum, PSU, SSU, and count variables	BPS22FANALSTR, BPS22FANALPSU, BPS22FANALSSU, and BPS22PSUCOUNT
Software: statements, parameters, and keywords for Taylor series variance estimation (without replacement)	
SUDAAN	DESIGN = WOR WEIGHT WTA000; NEST BPS22FANALSTR BPS22FANALPSU BPS22FANALSSU; TOTCNT BPS22PSUCOUNT _minus1__zero_;
Stata	svyset BPS22FANALPSU [pw=WTA000], strata(BPS22FANALSTR) fpc(BPS22PSUCOUNT) BPS22FANALSSU, vce(LINEARIZED)
R survey package ²	mydesign<-svydesign(id=~BPS22FANALPSU, strata=~BPS22FANALSTR, weights=~WTA000, fpc=~BPS22PSUCOUNT, data=mydata)

See notes at end of table.

Table 40. Use of analysis weights, replicate weights, and variance estimation strata, primary sampling unit (PSU), secondary sampling unit (SSU), and PSU count variables available from NPSAS:20 in selected survey data analysis software: 2022—Continued

Analysis weight for estimates	WTA000
Bootstrap variance estimation	
Replicate weight variables	WTA001 – WTA200
Software: statements, parameters, and keywords for BRR variance estimation	
SUDAAN	DESIGN = BRR WEIGHT WTA000; REPWGT WTA001 – WTA200;
Stata	svyset [pweight=WTA000], brrweight(WTA001 - WTA200) vce(BRR) mse
SAS survey data analysis procedures	VARMETHOD = BRR WEIGHT WTA000; REPWEIGHTS WTA001 – WTA200;
WesVar	Method: BRR Full sample weight: WTA000 Replicates: WTA001 – WTA200
R survey package ²	mydesign <- svrepdesign(type="BRR", weights=~WTA000, repweights="WTA00[1-200]", mse=TRUE, data=mydata)

¹ The name "myfile" should be replaced with the desired file name.

² For the R survey package (Lumley, 2014), "mydesign" can be renamed to any name for an R object to hold the specification of the survey design, and "mydata" is the name of the current dataset. For the without-replacement design, the R survey package does not account for the second stage of sampling.

NOTE: BRR = balanced repeated replication. The survey data analysis software specifications are given for the following versions of the software packages: SUDAAN 11.0.3, Stata 12 and newer, SAS 9.3 and newer, IBM SPSS complex samples 25, and WesVar 4.3 and newer. SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

6.4.4 Variance Approximation

The design effect (*DEFF*) for a given estimate, $\hat{\theta}$, is defined as

$$DEFF(\hat{\theta}) = \frac{Var_{design}(\hat{\theta})}{Var_{srs}(\hat{\theta})},$$

where $Var_{design}(\hat{\theta})$ = the variance for a given estimate accounting for the study design and

$Var_{srs}(\hat{\theta})$ = the unadjusted variance for a given estimate for an SRS.

The square root of the design effect (*DEFT*) is another measure, which analysts can express as the ratio of the standard errors, or

$$\text{DEFT}(\hat{\theta}) = \frac{SE_{\text{design}}(\hat{\theta})}{SE_{\text{srs}}(\hat{\theta})},$$

where

$SE_{\text{design}}(\hat{\theta})$ = the variance for a given estimate accounting for the study design and

$SE_{\text{srs}}(\hat{\theta})$ = the unadjusted variance for a given estimate for SRS.

Most complex multistage sampling designs like NPSAS:20 and BPS:20/22 result in design effects greater than 1.0; that is, the design-based variance is larger than the SRS variance. Appendix H provides design effect estimates for important domains to summarize the effects of stratification, multistage sampling, unequal probabilities of selection, and the weight adjustments. BPS statisticians estimated design effects using SUDAAN and the bootstrap variance estimation procedure described above.

Large design effects imply large standard errors and relatively poor precision. Small design effects imply small standard errors and good precision. In general terms, a design effect less than 2.0 is low, from 2.0 to 3.0 is moderate, and greater than 3.0 is high. Moderate and high design effects often occur in complex sample studies such as BPS:20/22. Unequal weighting causes large design effects and is often due to nonresponse and poststratification adjustments; however, in BPS:20/22, the unequal weighting is also due to the sample design and different sampling rates among institution strata, as well as to the different sampling rates among NPSAS student-level strata.

As discussed above, Taylor series linearization and replication techniques can be used to compute more precise standard errors for data from complex sample studies. If data users conduct statistical analyses using software packages that assume the data were collected using simple random sampling (i.e., adjustments are not made using the Taylor series or bootstrap replication methods), the standard errors will be calculated under this assumption and will be incorrect. They can be adjusted using the average *DEFT*, although this method is less precise than Taylor series or replication techniques. Those who must perform an analysis of BPS:20/22 data without using one of the software packages for analysis of complex sample study data should begin by computing weighted point

estimates, regression coefficients, etc., using WTA000 and then using the design effect tables in appendix H to make approximate adjustments to the standard errors of statistics computed with the standard software packages that assume simple random sampling designs.

As the first step in the approximation of a standard error, the analyst should normalize the overall sample weights for packages that use the weighted population size (N) in the calculation of standard errors (SPSS but not SAS). The normalized weight will sum to the sample size (n) and is calculated as

$$\text{normalized weight} = \text{weight} * n/N,$$

where n is the sample size (i.e., the number of cases with a valid main sampling weight) and N is the sum of weights.

As the second step in the approximation, the standard errors produced by the statistical software, the test statistics, or the sample weight used in analysis can be adjusted to reflect the actual complex design of the study. To adjust the standard error of an estimate, the analyst should multiply the standard error produced by the statistical software by the square root of the *DEFF*, or the *DEFT*. The *DEFF* and *DEFT* used to make adjustments can be calculated for specific estimates, can be the median *DEFF* and *DEFT* across a number of variables, or can be the median *DEFF* and *DEFT* for a specific subgroup in the population. Adjusted standard errors can then be used in hypothesis testing, for example, when calculating t and F statistics.

A second option is to adjust the t and F statistics produced by statistical software packages using unadjusted (i.e., SRS) standard errors. To do this, first conduct the desired analysis weighted by the normalized weight and then divide a t statistic by the *DEFT* or divide an F statistic by the *DEFF*. A third alternative is to create a new analytic weight variable in the data file by dividing the normalized analytic weight by the *DEFF* and using the adjusted weight in the analyses.

6.5 Imputations

BPS statisticians imputed item-level missing data for most analysis variables included on the primary analysis file (BB20DERIVED_DATAFILE) on the restricted-use file and available in DataLab. Variables in the source data files were not imputed.

Statisticians imputed missing data, with the exception of missing code -3 (see section 6.2), for all variables included in the restricted-use derived files (also used in PowerStats) in accordance with mass imputation procedures described by Krotki, Black, and Creel (2005). After filling in missing data for cases where values could be deduced with certainty based upon logical or mathematical relationships among observed variables (logical imputation),¹⁶ the statistical team used the weighted sequential hot deck (WSHD) method to replace missing data by imputing plausible values from statistically selected donor cases (stochastic imputation) (Cox 1980; Iannacchione 1982). Statisticians then imputed missing data separately for each key content area.

The first stage in the imputation procedure was to determine the pattern and level of missingness and produce an initial set of imputations. Depending on patterns of missing data, some variables that were related substantively and required imputation were grouped into blocks (vectors), and the variables within a block were imputed simultaneously (vector imputation). Then, variables and vectors were prioritized for imputation based upon their level of missing data. Variables and vectors with low levels of missingness were imputed before variables where the rate of missingness was greater. That is, variables with smaller amounts of uncertainty were imputed first, and variables with larger amounts of uncertainty were imputed next. For each variable and vector, imputation classes were identified from which donor cases for the hot deck procedure would be selected. To develop those classes, nonparametric classification or regression trees were used to identify homogeneous subgroups of item respondents (Breiman et al. 1984) using complete response variables and any previously imputed variables as possible predictor variables. Within these classes, WSHD was used to select donors.

In the second stage of imputation, for each variable or vector in the same sequence as in the first stage, the missingness was reintroduced and the missing items were reimputed. In this stage, all complete response variables and imputed variables on the dataset were available to form the imputation classes. To improve imputation quality, the previously described procedure using trees and WSHD was combined and implemented with the cyclic *p*-partition hot deck (Marker, Judkins, and Winglee 2002) technique (cycling), as discussed in Judkins (1997). This imputation approach reinforces existing patterns within the observed data.

¹⁶ An example of logical imputation follows: If a student has valid values for the total number of dependents and the number of dependent children but not the number of other dependents, the third value may be calculated as the difference of the first value minus the second value. Likewise, if a student has zero total dependents, it may be logically inferred that the student has zero dependent children.

This is an iterative process, and typically, the result of cycling is a convergence to plausible values and maintenance of relationships that already exist. For BPS:20/22, there were eight iterations, which improved quality without significantly slowing down the imputation process.

To minimize the potential error due to imputation, statisticians performed quality checks throughout the imputation process. Specifically, the distributions of the observed, imputed, and completed (observed and imputed) data were compared to screen variables for further investigation. RRI and distributions (observed and imputed) are shown in appendix I for 414 imputed variables. Results for 15 imputed variables were not included in the continuous table as they were for date or occupation type variables and results were not interpretable.

6.6 Disclosure Risk Analysis and Avoidance

In preparing data files for release, NCES takes steps, including a formal disclosure risk analysis, to minimize the likelihood that individuals participating in the study can be identified. Every effort is made to protect the confidentiality of information about specific individuals, including perturbation procedures (including the swapping of records) on BPS:20/22 data to minimize disclosure risk. In data swapping, the values of the variables being swapped are exchanged between carefully selected pairs of records: a target record and a donor record.

All cases were eligible for swapping. Perturbation was carried out through specific targeted, but undisclosed, swap rates. Because perturbation of the BPS:20/22 data could have changed the relationships between data items, an extensive data quality check was carried out to assess and limit the impact of swapping on these relationships. For example, a set of correlations for a variety of variables was evaluated pre- and posttreatment to verify that the swapping did not greatly affect the associations.

Therefore, the modifications used to reduce the likelihood that any respondent could be identified in the data generally did not affect the overall data quality. The swapping procedures, which the IES Disclosure Review Board reviewed and approved, preserved central tendency estimates but may have resulted in slight increases in nonsampling errors.

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Appendix A. Acronyms and Abbreviations

Acronym/Abbreviation	Name
AP	Advanced Placement
AY	academic year
B&B	Baccalaureate and Beyond Longitudinal Study
BPS	Beginning Postsecondary Students Longitudinal Study
BPS:20/22	2020/22 Beginning Postsecondary Students Longitudinal Study
BPS:20/25	2020/25 Beginning Postsecondary Students Longitudinal Study
CATI	computer-assisted telephone interviewing
CATI-CMS	computer-assisted telephone interviewing case management system
CHAID	chi-square automatic interaction detection
CIP	Classification of Instructional Programs
COVID-19	coronavirus pandemic
CPS	Central Processing System
CPS:20	Central Processing System data from the 2019–20 National Postsecondary Student Aid Study
DCI	data collection interviewer
<i>DEFF</i>	design effect
<i>DEFT</i>	square root of the design effect
DFD	data file documentation
DOB	date of birth
DOD	Department of Defense
EFC	expected family contribution
FAFSA	Free Application for Federal Student Aid
FAQs	frequently asked questions
FEMA	Federal Emergency Management Agency
FPC	finite population correction
FSA	Federal Student Aid
FTB	first-time beginning student
FWS	federal work-study
HOPE	Helping Outstanding Pupils Educationally
IES	Institute of Education Sciences
IMS	Integrated Management System
IPEDS	Integrated Postsecondary Education Data System
IQR	interquartile range
NCES	National Center for Education Statistics
NCOA	National Change of Address
NPSAS	National Postsecondary Student Aid Study

Acronym/Abbreviation	Name
NPSAS:20	2019–20 National Postsecondary Student Aid Study
NSC	National Student Clearinghouse
NSF	National Science Foundation
NSLDS	National Student Loan Data System
O*NET	Occupational Information Network
PII	personally identifiable information
PR	Puerto Rico
PSU	primary sampling unit
QC	Quality Circle
QUEST	quality evaluation system
ROC	Receiver Operating Characteristic
RRI	item response rate
RTI	Research Triangle Institute
SEOG	Supplemental Educational Opportunity Grant
SMS	Short Message Service
SQL	Structured Query Language
SSL	Secure Sockets Layer
SSN	Social Security number
SSU	secondary sampling unit
STEM	science, technology, engineering, and mathematics
TANF	Temporary Assistance for Needy Families
TRP	Technical Review Panel
USPS	United States Postal Service
UWE	unequal weighting effect
VBA	Veterans Benefits Administration
WIC	Special Supplemental Nutrition Program for Women, Infants, and Children
WSHD	weighted sequential hot deck

Appendix B. Technical Review Panel

Technical Review Panel

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Appendix C. Survey Data Elements and English Instrument

The core data elements used in the 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) full-scale survey consisted of seven key sections focused on specific content areas: enrollment, education experiences, financial aid, employment, income and expenses, background, and COVID-19 pandemic experiences. In addition to the seven content areas, the survey included nonsubstantive forms at the front (security questions to confirm respondent identity and consent) and navigation forms (introductory screens, etc.). Table C-1 provides a list of the data elements by section and topic.

Table C-1. Full-scale survey core data elements, by section and topic: 2022

Section	Topic
Enrollment	Eligibility determination for base-year nonrespondents Additional postsecondary enrollment in 2020–21 and 2021–22 academic years: • Postsecondary institution (if not NPSAS) • Degree program or level of coursework • Enrollment intensity • Dates of attendance • Date degree completed (or expected date of completion) • Year or level Primary institution identification
Education experiences	Remedial education GPA and online coursetaking at primary institution Academic and social experiences at primary institution: • Interactions with family, friends, faculty, and students • Social and academic group involvement • Educational services used Expected occupation and wages after completing education
Financial aid	Receipt of state financial aid in 2019–20, 2020–21, and 2021–22 academic years Applied for any financial aid in 2020–21 and 2021–22 academic years Financial sources used to pay for education in 2020–21 and 2021–22 academic years: • Amount of fellowships, grants, scholarships, and veteran's benefits received • Amount borrowed in private loans • Refunds of scholarships, grants, or loans received Total amount borrowed in private loans for education Other sources of aid and knowledge of aid: • Emergency aid programs at primary institution • Awareness and likelihood of using loan forgiveness programs and repayment plans • Knowledge of federal student loan terms Cost of books in 2020–21 and 2021–22 academic years
Employment	Employment information for each employer from July 1, 2020, through June 30, 2022: • Months worked • Currently employed • Earnings • Hours worked per week • Job was on or off campus • Job was related to major Worked at any employers while also enrolled Primarily student or employee Looked for work while not employed

See notes at end of table.

Table C-1. Full-scale survey core data elements, by section and topic: 2022—Continued

Section	Topic
Income and expenses	Income from all sources for 2021 (including spousal income if applicable)
	Military status
	Dependents, and others financially supported
	Number of dependent children in child care
	Average monthly cost of child care
	Help from family or friends for education or living expenses
	Use of and amount owed on credit cards
	Checking or savings account(s)
	Confidence can come up with \$2,000 or \$500 next month
	Residence during 2020–21 and 2021–22 academic years
	Rent or mortgage amount
	Financial literacy
Background	Citizenship and nativity status
	Ethnicity and race
	Sex and gender identity
	Sexual orientation
	Disability status
	Physical and mental health status
	Frequency of missing school and work due to health in past 30 days
	Felt safe attending primary institution
	Homelessness
	Meal plans in the 2020–21 and 2021–22 academic years
	Food security
COVID-19 pandemic experiences	Institutional communication/information about pandemic to students
	Changes to enrollment and borrowing
	Changes in academic engagement
	Access to support resources
	Access to internet
Locating	Personal contact information
	Parent(s)' contact information
	Other contact information
	Social Security number

NOTE: GPA = grade point average; NPSAS = National Postsecondary Student Aid Study.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Survey questions administered in the BPS:20/22 abbreviated student survey are notated with an asterisk (*). The Spanish BPS:20/22 survey is available upon request by contacting bps@ed.gov.

Enrollment

B22AELIGINTRO *

To get started, [{if USERMODE = WEB} we {else} I] need to collect some information about your enrollment at [NPSAS Institution] [{if USERMODE = CATI} to {else} so that we can] confirm your eligibility to participate in the study.

B22AELIG *

Did you attend [NPSAS Institution] at any time between July 1, 2019, and June 30, 2020?

1 = Yes

2 = Yes, but left [NPSAS Institution] before completing any classes

3 = Yes, but also attended high school between July 1, 2019, and June 30, 2020

0 = No

-1 = Don't know

B22AEVREN *

Have you ever attended [NPSAS Institution]?

1 = Yes

0 = No

B22ALAST *

In what month and year did you **last** attend [NPSAS Institution]?

Month:

January - December

Year:

Before 2019 - 2022

B22ADRPTM *

When you last attended [NPSAS Institution], did you leave at the end of the term, or did you leave before the term ended?

1 = Left at the end of the term

2 = Left before the term ended

B22ADRPRF *

Did you receive a full refund of your tuition when you left [NPSAS Institution]?

(A full tuition refund occurs when all money paid for tuition a semester or term is refunded to you [or your parents].)

1 = Yes

0 = No

B22ADRPCMP *

Did you complete at least one course [{if NPSAS Institution not a continuous calendar system} or term] at [NPSAS Institution] at any time between July 1, 2019, and June 30, 2020?

1 = Yes

0 = No

B22AASENR *

Were you in an associate's degree program at [NPSAS Institution] in the 2019–20 academic year (July 1, 2019–June 30, 2020)?

(An associate's degree usually requires 2 years of full-time, college-level work.)

1 = Yes

0 = No

B22ABACHENR *

Were you in a bachelor's degree program at [NPSAS Institution] in the 2019–20 academic year (July 1, 2019–June 30, 2020)?

(A bachelor's degree is usually awarded by a 4-year college or university and usually requires at least 4 years of full-time, college-level work.)

1 = Yes

0 = No

B22ADEGREE *

What degree or certificate were you working on at [NPSAS Institution] in the 2019–20 academic year (July 1, 2019–June 30, 2020)?

(Please check all that apply.)

- ☐ Associate's degree (usually a 2-year degree)
- ☐ Bachelor's degree (usually a 4-year degree)
- ☐ Master's degree
- ☐ Doctoral degree—Research/scholarship (e.g., PhD, EdD, etc.)

- ☐ Doctoral degree—Professional practice (e.g., chiropractic, dentistry, law, medicine, optometry, pharmacy, podiatry, or veterinary medicine)
- ☐ Doctoral degree—Other
- ☐ Undergraduate certificate or diploma (usually less than 2 years), including those leading to a certification or license (e.g., cosmetology)
- ☐ Postbaccalaureate certificate
- ☐ Post-master's certificate
- ☐ Undergraduate-level classes
- ☐ Graduate-level classes

B22AUGSTVER *

Were you primarily an undergraduate working toward a bachelor's degree at [NPSAS Institution] in the 2019–20 academic year (July 1, 2019–June 30, 2020)?

- 1 = Yes
0 = No

B22ACKHOUR *

Did your certificate or diploma program require at least 3 months or 300 hours of instruction?

- 1 = Yes
0 = No

B22AELCRD *

In the 2019–20 academic year, were you taking at least one course for credit at [NPSAS Institution] that could be applied toward fulfilling the requirements for an academic degree?

- 1 = Yes
0 = No

B22ADIPL *

Which of the following best describes your high school completion?

- 1 = Received a high school diploma in the United States
2 = Received a GED (General Educational Development) certificate or other equivalent credential (such as HiSET or TASC)
3 = Received a high school completion certificate
4 = Attended a foreign high school
6 = Homeschooled
5 = Did not complete high school or a high school equivalency program

B22AHSGRAD *

In what month and year did you complete high school?

Month:

January - December

Year:

Before 1990 - 2022

B22AHSCMP *

Were you completing high school requirements for the entire time you attended [NPSAS Institution] between July 1, 2019, and June 30, 2020?

1 = Yes

0 = No

-1 = Don't know

B22ANFST *

Was [NPSAS Institution] the first college, university, or trade school you attended after [{if B22ADIPL = 5} high school {else} completing your high school requirements]?

1 = Yes

0 = No

B22ASCHSTR *

In which month and year did you first attend [NPSAS Institution] after [{if B22ADIPL = 5} high school {else} completing your high school requirements]?

Month:

January - December

Year:

Before 2019 - 2022

B22APRDG *

After you completed your high school requirements and **before** you attended [NPSAS Institution] in the 2019–20 academic year, did you earn a degree or certificate at any college, university, or trade school?

1 = Yes

0 = No

-1 = Don't know

B22AWHYSM *

Based on your responses, it seems you may not be eligible for this study.

Information provided by [NPSAS Institution] indicates that you first attended [NPSAS Institution] at some time between July 1, 2019, and June 30, 2020.

Please provide any additional information in the textbox below regarding when you **first began** your enrollment at [NPSAS Institution].

B22ABYE *

We will review your responses and we may need to contact you again.

Please provide your e-mail address:

Street Address:

Phone Number:

☐ Please check here if the address is an international address.

B22ANPINTRO *

Last time we contacted you, you provided us with information about your attendance at [NPSAS Institution] in the 2019–20 academic year.

To begin, [{if USERMODE = WEB} we'd {else} I'd] like to collect information about any additional attendance for your [base-year degree] at [NPSAS Institution] between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022].

B22ASAMESCH *

Did you attend [NPSAS Institution] at any time between July 1, 2020, and June 30, 2022?

1 = Yes

0 = No

B22ASAMEDEG *

Between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022], were you continuing at [NPSAS Institution] for your [base-year degree] that you began in the 2019–20 academic year?

1 = Yes

0 = No

B22ACURENR *

Are you currently attending [NPSAS Institution] [{if undergraduate classes from base-year} for the undergraduate classes that are not part of a degree or certificate program {else} for your [base-year degree]]?

1 = Yes

0 = No

B22ACMPDGN

Did you complete all the requirements for the [base-year degree] you began at [NPSAS Institution] in the 2019–20 academic year before July 1, 2022?

(Answer **No** if you transferred schools and completed your [base-year degree] at a different school. [{If USERMODE = WEB} We will {else} I'll] ask you about your enrollment at any other colleges, universities, or trade schools later.)

1 = Yes

0 = No

B22ADGN

In what month and year were you awarded your [base-year degree] from [NPSAS Institution]?

Month:

January - December

Year:

2019 - 2022

☐ Have not yet been awarded [base-year degree]

B22ASTST

Between July 1, 2020, and June 30, 2022, [{if before July 1, 2022} have you been, or do you plan to be, {else} were you] a full-time or part-time student or a mix of both at [NPSAS Institution]?

	1 = Full-time	2 = Part-time	3 = Mix of both full-time and part-time	4 = Not enrolled
2020–21 academic year (July 1, 2020–June 30, 2021)	☐	☐	☐	☐
2021–22 academic year (July 1, 2021–June 30, 2022)	☐	☐	☐	☐

B22ANENRLFT

[{If B22ASTYR2 = 3 or B22ASTYR3 = 3} You just told [{if USERMODE = WEB} us {else} me] you [{if before July 1, 2022 and B22ASTYR3 in (1 3)} have, or plan to have, {else} had] a mix of full-time and part-time attendance at [NPSAS Institution] for your [base-year degree]. First, [{if USERMODE = WEB} we'd {else} I'd] like to collect information about your full-time attendance.]

Between [{if B22ASTYR3 not in (1 3)} July 1, 2020, and June 30, 2021, in which months did you attend {else if B22ASTYR2 not in (1 3)} July 1, 2021, and June 30, 2022, [{if before July 1, 2022 and B22ASTYR3 in (1 3)} have you attended, or do you plan to attend, {else} did you attend] {else} July 1, 2020, and June 30, 2022, [{if before July 1, 2022 and B22ASTYR3 in (1 3)} have you attended, or do you plan to attend, {else} did you attend]] [NPSAS Institution] for your [base-year degree] as a full-time student? Do not include months during which [{if before July 1, 2022 and B22ASTYR3 in (1 3)} you are not taking classes {else} you did not take classes], such as summer break.

([{If USERMODE = WEB} Selected months will be shaded dark gray.] If you attended [{if before July 1, 2022 and B22ASTYR3 in (1 3)}, or plan to attend,] [NPSAS Institution] for your [base-year degree] full time for only a portion of a month, please include that month.)

2020–21

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

2021–22

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

B22ANENRLPT

[{If B22ASTYR2 = 3 or B22ASTYR3 = 3} Now, [{if USERMODE = WEB} we'd {else} I'd] like to collect information about your part-time attendance.]

Between [{if B22ASTYR3 not in (2 3)} July 1, 2020, and June 30, 2021, in which months did you attend {else if B22ASTYR2 not in (2 3)} July 1, 2021, and June 30, 2022, in which months [{before July 1, 2022 and B22ASTYR3 in (2 3)} have you attended, or do you plan to attend, {else} did you attend] {else} July 1, 2020, and June 30, 2022, in which months [{before July 1, 2022 and B22ASTYR3 in (2 3)} have you attended, or do you plan to attend, {else} did you attend]] [NPSAS Institution] for your [base-year degree] as a part-time student? Do not include months during which [{if before July 1, 2022 and B22ASTYR3 in (2 3)} you are not taking classes {else} you did not take classes], such as summer break.

([If USERMODE = WEB} Selected months will be shaded dark gray.] If you attended [if before July 1, 2022 and B22ASTYR3 in (2 3)}, or plan to attend,] [NPSAS Institution] for your [base-year degree] part-time for only a portion of a month, please include that month.)

2020-21

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

2021-22

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

B22ASTDABR

Between July 1, 2020, and [if enrolled in years 2 and 3 at NPSAS Institution} June 30, 2022 {else if enrolled in year 2 at NPSAS Institution} June 30, 2021 {else} June 30, 2022], did you participate in a study-abroad program as part of your education at [NPSAS Institution]?

1 = Yes

0 = No

B22ACLSDGREE

Between July 1, 2020, and [if enrolled in years 2 and 3 at NPSAS Institution} June 30, 2022 {else if enrolled in year 2 at NPSAS Institution} June 30, 2021 {else} June 30, 2022], were you taking these [base-year degree] at [NPSAS Institution] primarily to fulfill a degree requirement or transfer course credit to a degree or certificate program?

1 = Yes

0 = No

B22AREASON

Which of these reasons best describes why you were enrolled in classes at [NPSAS Institution] between July 1, 2020, and [if enrolled in years 2 and 3 at NPSAS Institution} June 30, 2022 {else if enrolled in year 2 at NPSAS Institution} June 30, 2021 {else} June 30, 2022]?

- 1 = To prepare to earn a degree later
 2 = To prepare for a job certification or license
 3 = To gain job or occupational skills
 4 = To take courses solely for recreation, self-improvement, or personal interest

B22ADBLMAJ *

Had you already declared a major [{if base-year degree is certificate or diploma or associate's degree} or field of study] for your [base-year degree] when you last attended [NPSAS Institution]?

- 1 = [{If B22ACMPDGN = 1} Declared {else} Yes, declared] a single major [{if base-year degree is certificate or diploma or associate's degree} or field of study]
 2 = [{If B22ACMPDGN = 1} Declared {else} Yes, declared] a double major [{if base-year degree is certificate or diploma or associate's degree} or field of study]
 3 = No

B22ADECIDMAJ *

Even though you did not formally declare your major, did you decide on a major for your [base-year degree] at [NPSAS Institution]?

(Answer **Yes** if you have left [NPSAS Institution] and do not plan to attend again to earn your [base-year degree] there but you had decided what your major would be prior to leaving.)

- 1 = Yes
 0 = No

B22AVERN2MJ *

Last time we contacted you, you told us you were majoring in [base-year major 1].

[{If currently enrolled at NPSAS Institution} Are {else} When you last attended [NPSAS Institution] for your [base-year degree] were] you still majoring in...

	1 = Yes	0 = No
[Base-year major 1]?	☐	☐
[Base-year major 2]?	☐	☐

B22AMAJ1

What [{if currently enrolled at NPSAS Institution} is {else} was] your [{if B22ADBLMAJ = 3} intended] major [{if base-year degree is certificate or diploma or associate's degree} or field of study] at [NPSAS Institution]?

(Please provide your [{if B22ADBLMAJ = 3} intended] major in the textbox. If you are unable to find a match in the results, please select the “Major not listed” option located at the bottom of the listed results.)

B22AMAJ2

[{If B22ANPMJ1STG = 1 and B22ACMPDGN = 1} You just told [{if USERMODE = CATI} me {else} us] one of your final majors was [base-year major 1]. {else if B22ANPMJ1STG = 1} You just told [{if USERMODE = CATI} me {else} us] you [{if currently enrolled at NPSAS Institution} are {else} were] majoring in [base-year major 1].]

What [{if currently enrolled at NPSAS Institution} is {else} was] your second major [{if base-year degree is certificate or diploma or associate’s degree} or field of study] at [NPSAS Institution]?

(Please provide your second major in the textbox. If you are unable to find a match in the results, please select the “Major not listed” option located at the bottom of the listed results.)

B22AMJCHGNUM

[{If on or after July 1, 2022} Before July 1, 2022, how {else} How] many times did you formally change your major [{if base-year degree is certificate or diploma or associate’s degree} or field of study] at [NPSAS Institution] for your [base-year degree]?

- 0 = Never
- 1 = Once
- 2 = More than once

B22AOMJ1A

What was your original declared major [{if base-year degree is certificate or diploma or associate’s degree} or field of study] at [NPSAS Institution]?

(Please provide your original major [{if base-year degree is certificate or diploma or associate’s degree} or field of study] in the textbox. If you are unable to find a match in the results, please select the “Major not listed” option located at the bottom of the listed results.)

B22ANPOTHENR *

[{If Y_N20CMPDGN = 0} You just provided [{if USERMODE = CATI} me {else} us] with information about your [{if B22ACMPDGN = 1} completed] [base-year degree] at [NPSAS Institution] that you began in the 2019–20 academic year.]

Aside from [{if undergraduate classes from base-year} these {else} this] [{if completed base-year degree at NPSAS Institution or B22ACMPDGN = 1} completed] [base-year degree], [{if before July 1, 2022} have you attended {else} did you attend] [NPSAS Institution] for any [{if not undergraduate classes from base-year} other] degree or certificate programs between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022]?

[{If not undergraduate classes from base-year} (Answer Yes if you attended [NPSAS Institution] for any nondegree coursework that was not part of your [base-year degree].)]

1 = Yes

0 = No

B22AOTSCHINT *

Thank you for providing your attendance information at [NPSAS Institution].

Next, we will be asking you about any **other** colleges, universities, or trade schools you may have attended for any reason.

B22AOTSCHENR *

Other than [NPSAS Institution], have you attended another college, university, or trade school at any time between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022]?

1 = Yes

0 = No

B22APREVSCH01 *

Last time we contacted you, you told us you attended the following [{if Y_N20SCHNUM = 1} school {else} schools] in the 2019–20 academic year (July 1, 2019–June 30, 2020).

Have you attended [{if Y_N20SCHNUM = 1} this school {else} any of these schools] at any time between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022]?

[{If iteration = 1 and Y_N20SCHNUM > 1} If you have attended more than one other school between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022], tell [{if USERMODE= WEB} us {else} me] about the most recent school first.]

1 = [base-year other postsecondary institution 1]

2 = [base-year other postsecondary institution 2]

3 = [base-year other postsecondary institution 3]

4 = [base-year other postsecondary institution 4]

5 = [base-year other postsecondary institution 5]

99 = [{If Y_N20SCHNUM = 1} No, did not attend this school {else} No, did not attend any of these schools]

B22AOTSCH01 *

What is the name of [{if not enrolled in years 2 or 3 at NPSAS Institution and iteration = 1 } a {else} another] school you attended between July 1, 2020, and June 30, 2022?

(Please provide the name of [{if not enrolled in years 2 or 3 at NPSAS Institution and iteration = 1 } a {else} another] school you attended in the textbox. [{If iteration = 1 } If you attended more than one other school between July 1, 2020, and June 30, 2022, tell [{if USERMODE= WEB} us {else} me] about the most recent school first.] If you are unable to find a match in the results, please select the “School not listed” option at the bottom of the listed results.)

B22AOTDEGREE01 *

[{If iteration 1 and B22ASAMESCH = 1 and B22ASAMEDEG = 0 } You just provided [{if USERMODE = CATI} me {else} us] with information about your [{if B22ACMPDGN = 1 } completed] [base-year degree] at [NPSAS Institution] that you began in the 2019–20 academic year.]

What [{if (Other Institution is NPSAS Institution) or (iteration > 1 and B22AOTDG01 = 1 from the prior iteration) other] degree or certificate were you working on at [Other Institution] [{if iteration 1 and B22ASAMESCH = 1 and B22ASAMEDEG = 0 } between July 1, 2020, and June 30, 2022]?

(If you have more than one enrollment at [Other Institution] or if you were in a joint degree program, tell [{if USERMODE = WEB} us {else} me] about only one of these now. You will have the opportunity to tell [{if USERMODE = WEB} us {else} me] about all enrollment at [Other Institution].)

- 1 = Undergraduate-level classes
- 2 = Undergraduate certificate or diploma (usually less than 2 years), including those leading to a license (e.g., cosmetology, etc.)
- 3 = Associate’s degree (usually a 2-year degree)
- 4 = Bachelor’s degree (usually a 4-year degree)
- 5 = Graduate-level classes
- 6 = Postbaccalaureate certificate
- 7 = Master’s degree
- 8 = Post-master’s certificate
- 9 = Doctoral degree—Research/scholarship (e.g., PhD, EdD, etc.)
- 10 = Doctoral degree—Professional practice (e.g., chiropractic, dentistry, law, medicine, optometry, pharmacy, podiatry, or veterinary medicine, etc.)
- 11 = Doctoral degree—Other

B22AOTCURENR01 *

Are you currently attending [Other Institution] for your [Other Degree]?

- 1 = Yes
- 0 = No

B22AOTCMPDGN01

Did you complete all of the requirements for your [Other Degree] at [Other Institution] before July 1, 2022?

(Answer **No** if you transferred schools and completed your [Other Degree] at a different school. [{If USERMODE = WEB} We will {else} I'll] ask you about your enrollment at other schools later.)

1 = Yes

0 = No

B22AOTDGN01

In what month and year were you awarded your [Other Degree] from [Other Institution]?

Month:

January - December

Year:

2019 - 2022

☐ Have not yet been awarded [Other Degree]

B22AOTSTST01

Between July 1, 2020, and June 30, 2022, [{if before July 1, 2022} have you been, or do you plan to be, {else} were you] a full-time or part-time student or a mix of both for your [Other Degree] at [Other Institution]?

	1 = Full-time	2 = Part-time	3 = Mix of both full-time and part-time	4 = Not enrolled
2020–21 academic year (July 1, 2020–June 30, 2021)	☐	☐	☐	☐
2021–22 academic year (July 1, 2021–June 30, 2022)	☐	☐	☐	☐

B22AOTENRLFT01

[{If B22AOTST3YR01 = 3 or B22AOTST2YR01 = 3} You just told [{if USERMODE = WEB} us {else} me] you [{if before July 1, 2022 and B22AOTST3YR01 in (1 3)} have, or plan to have, {else} had] a mix of full-time and part-time attendance at [Other Institution] for your [Other Degree]. First, [{if USERMODE = WEB} we'd {else} I'd] like to collect information about your full-time attendance.]

Between [{if B22AOTST3YR01 not in (1 3)} July 1, 2020, and June 30, 2021 {else if B22AOTST2YR01 not in (1 3)} July 1, 2021, and June 30, 2022 {else} July 1, 2020, and June 30, 2022], in which months [{if before July 1, 2022 and B22AOTST3YR01 in (1 3)} have you attended, or do you plan to attend, {else} did you attend] [Other Institution] for your [Other Degree] as a full-time student? Do not include months during which [{if before July 1, 2022 and B22AOTST3YR01 in (1 3)} you are not taking classes {else} you did not take classes], such as summer break.

([If USERMODE = WEB} Selected months will be shaded dark gray.] If you attended [{if before July 1, 2022 and B22AOTST3YR01 in (1 3)}, or plan to attend,] [Other Institution] for your [Other Degree] full time for only a portion of a month, please include that month.)

2020–21

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

2021–22

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

B22AOTENRLPT01

[{If B22AOTST3YR01 = 3 or B22AOTST2YR01 = 3} Now, [{if USERMODE = WEB} we'd {else} I'd] like to collect information about your part-time attendance.]

Between [{if B22AOTST3YR01 not in (2 3)} July 1, 2020, and June 30, 2021 {else if B22AOTST2YR01 not in (2 3)} July 1, 2021, and June 30, 2022 {else} July 1, 2020, and June 30, 2022], in which months [{before July 1, 2022, and B22AOTST3YR01 in (2 3)} have you attended, or do you plan to attend, {else} did you attend] [Other Institution] for your [Other Degree] as a part-time student? Do not include months during which [{if before July 1, 2022 and B22AOTST3YR01 in (2 3)} you are not taking classes {else} you did not take classes], such as summer break.

([If USERMODE = WEB} Selected months will be shaded dark gray.] If you attended [{if before July 1, 2022 and B22AOTST3YR01 in (2 3)}, or plan to attend,] [Other Institution] for your [Other Degree] part-time for only a portion of a month, please include that month.)

2020–21

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

2021–22

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

B22AOTTNS01

Did you attempt to transfer any credits from [NPSAS Institution] to [Other Institution] for your [Other Degree]?

1 = Yes

0 = No

B22AOTCLSDG01

Between July 1, 2020, and [{if enrolled in years 2 and 3 at Other Institution} June 30, 2022 {else if enrolled in year 2 at Other Institution} June 30, 2021 {else} June 30, 2022], were you taking these [Other Degree] at [Other Institution] primarily to fulfill a degree requirement or transfer course credit to a degree or certificate program?

1 = Yes

0 = No

B22AOTCRSN01

Which of these reasons best describes why you were enrolled in classes at [Other Institution] between July 1, 2020, and [{if enrolled in year 3 at Other Institution and enrolled in year 2 at Other Institution} June 30, 2022 {else if enrolled in year 2 at Other Institution} June 30, 2021 {else} June 30, 2022]?

1 = To prepare to earn a degree later

2 = To prepare for a job certification or license

3 = To gain job or occupational skills

4 = To take courses solely for recreation, self-improvement, or personal interest

B22AOTDBLM01 *

Had you already declared a major [{if Other Institution degree is certificate or diploma or associate's degree} or field of study] for your [Other Degree] when you last attended [Other Institution]?

1 = [{If B22AOTCMPDGN01 = 1} Declared {else} Yes, declared] a single major [{if Other Institution degree is certificate or diploma or associate's degree} or field of study]

2 = [{If B22AOTCMPDGN01 = 1} Declared {else} Yes, declared] a double major [{if Other Institution degree is certificate or diploma or associate's degree} or field of study]

3 = No

B22AOTDEC01 *

Even though you did not formally declare your major, did you decide on a major for your [Other Degree] at [Other Institution]?

(Answer Yes if you have left [Other Institution] and do not plan to attend again to earn your [Other Degree] there, but you had decided what your major would be prior to leaving.)

1 = Yes

0 = No

B22AOTMAJ101 *

What [{if currently enrolled at Other Institution} is {else} was] your [{if B22AOTDBLM01 = 3} intended] major [{if Other Institution degree is certificate or diploma or associate's degree} or field of study] at [Other Institution]?

(Please provide your [{if B22AOTDBLM01 = 3} intended] major in the textbox. If you are unable to find a match in the results, please select the "Major not listed" option located at the bottom of the listed results.)

B22AOTMAJ201

What [{if currently enrolled at Other Institution} is {else} was] your second major [{if Other Institution degree is certificate or diploma or associate's degree} or field of study] at [Other Institution]?

(Please provide your second major in the textbox. If you are unable to find a match in the results, please select the "Major not listed" option located at the bottom of the listed results.)

B22AOTMJCH01

[{If on or after July 1, 2022} Before July 1, 2022, how {else} How] many times did you formally change your major [{if Other Institution degree is certificate or diploma or associate's degree} or field of study] at [Other Institution] for your [Other Degree]?

- 0 = Never
 1 = Once
 2 = More than once

B22AOTOMAJ01

What was your original declared major [{if Other Institution degree is certificate or diploma or associate's degree} or field of study] at [Other Institution]?

(Please provide your original major in the textbox. If you are unable to find a match in the results, please select the "Major not listed" option located at the bottom of the listed results.)

B22AOTDG01 *

[{If at least 1 Other Degree listed} You already provided [{if USERMODE = CATI} me {else} us] with information about your attendance at [Other Institution] for the following:

[List nonmissing base-year degree if (B22ASAMEDEG = 1 and Other Institution is NPSAS Institution in current iteration)]

[List nonmissing Other Degree from all iterations that match Other Institution]

Besides this enrollment, [{if before July 1, 2022} have you attended {else} did you attend] [Other Institution] for anything else at any time between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022]?

[{If Other Institution enrollment is coursework} (Answer **Yes** if you attended [Other Institution] for any nondegree coursework that was not part of your [Other Degree].)]

- 1 = Yes
 0 = No

B22AOTENR01 *

You've told [{if USERMODE = WEB} us {else} me] that you have attended the following schools between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022]:

[{If B22ASAMEDEG = 1} [NPSAS Institution]]

[Other Institution from iteration 1]

[Other Institution from iteration x]

[{If before July 1, 2022} Have you attended {else} Did you attend] **any other colleges, universities, or trade schools** for a degree or certificate program between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022]?

1 = Yes

0 = No

B22APRSCHLST *

You've reported attending multiple schools since July 1, 2020. Which of the following schools do you consider to be your main school?

0 = [NPSAS Institution]

1 = [Other Institution 01]

2 = [Other Institution 02]

3 = [Other Institution 03]

4 = [Other Institution 04]

5 = [Other Institution 05]

6 = [Other Institution 06]

7 = [Other Institution 07]

8 = [Other Institution 08]

9 = [Other Institution 09]

10 = [Other Institution 10]

B22ACONTENR *

[{If on or after July 1, 2022} Have you attended, or will you attend, {else} Will you continue to attend] [Primary Institution] for your [Pursued Degree] at any time between July 1, 2022, and December 31, 2022?

1 = Yes

0 = No

B22AENRPLN *

Now, we are interested in your plans for attendance at any college, university, or trade school between July 1, 2022, and December 31, 2022.

[{If on or after July 1, 2022} Have you attended, or will you attend, {else} Will you attend] any college, university, or trade school for a degree or certificate at any time between July 1, 2022, and December 31, 2022?

1 = Yes

0 = No

B22AEXPN *

In what month and year do you expect to complete the requirements for your [Pursued Degree]?

Month:

January - December

Year:

After 2029 - 2022

- ☐ Will not finish the [Pursued Degree]
- ☐ Don't know

B22AEXNCONF

On a scale from 1 to 5, with 1 being “very unlikely” and 5 being “very likely,” how unlikely or likely is it you will finish your [Pursued Degree] within five years from today?

- 1 (Very unlikely)
- 2 (Somewhat unlikely)
- 3 (Neither unlikely nor likely)
- 4 (Somewhat likely)
- 5 (Very likely)

B22AEXNCONF2

On a scale from 1 to 5, with 1 being “very unlikely” and 5 being “very likely,” how unlikely or likely is it that you will finish your [Pursued Degree]?

- 1 (Very unlikely)
- 2 (Somewhat unlikely)
- 3 (Neither unlikely nor likely)
- 4 (Somewhat likely)
- 5 (Very likely)

B22AEXPBA *

Do you plan to continue on to a bachelor's degree program within the next two years?

(A bachelor's degree is usually awarded by a 4-year college or university and usually requires at least 4 years of full-time, college-level work.)

- 1 = Yes
- 0 = No

B22AEXPEVR *

What is the highest level of education you ever expect to complete at any school?

- 1 = Undergraduate-level courses, no undergraduate degree or certificate expected
- 2 = Undergraduate certificate or diploma (usually less than 2 years), including certificates leading to a license (example: cosmetology)
- 3 = Associate's degree (usually a 2-year degree)
- 4 = Bachelor's degree (usually a 4-year degree)
- 5 = Graduate-level courses, no graduate degree or certificate expected
- 6 = Postbaccalaureate certificate
- 7 = Master's degree
- 8 = Post-master's certificate
- 10 = Professional doctoral degree (including chiropractic, dentistry, law, medicine, etc.)
- 9 = Doctoral degree, research/scholarship (including PhD, EdD, etc.)

B22APRDOCTYP

What specific degree do you expect to complete?

- 1 = Chiropractic (DC/DCM)
- 2 = Dentistry (DDS/DMD)
- 3 = Law (JD/LLB)
- 4 = Medicine (MD) or osteopathic medicine (DO)
- 5 = Ministry or divinity (MDiv/DMin/MHL/BD/ordination)
- 6 = Optometry (OD)
- 7 = Pharmacy (PharmD)
- 8 = Podiatry (DPM/DP/PodD)
- 9 = Veterinary medicine (DVM)
- 10 = Other degree not listed

B22AHIEXP LK L

On a scale from 1 to 5, with 1 being "very unlikely" and 5 being "very likely," how unlikely or likely is it that you will ever finish your [{if B22AEXPEVR = 10} [expected professional degree type] degree {else} [highest expected degree]]?

- 1 (Very unlikely)
- 2 (Somewhat unlikely)
- 3 (Neither unlikely nor likely)
- 4 (Somewhat likely)
- 5 (Very likely)

B22AMARR *

The remainder of the survey asks about your latest experiences in college and your employment and family situations.

So that [{if USERMODE = WEB we {else} I] can ask you the right set of questions in the survey, please indicate your current marital status.

- 1 = Single, never married
- 2 = Married
- 3 = Separated
- 4 = Divorced
- 5 = Widowed
- 6 = Living with partner in a marriage-like relationship

B22ASPLV

What is the highest level of education your spouse has completed?

- 1 = Did not complete high school
- 2 = High school diploma or equivalent
- 3 = Vocational/technical training
- 5 = Associate's degree (usually a 2-year degree)
- 6 = Some college but no degree
- 7 = Bachelor's degree (usually a 4-year degree)
- 8 = Master's degree or equivalent
- 9 = Professional degree (chiropractic, dentistry, law, medicine, optometry, pharmacy, podiatry, or veterinary medicine)
- 10 = Doctoral degree (PhD, EdD, etc.)
- 1 = Don't know

B22ADOB

In what month and year were you born?

Month:

January - December

Year:

1920 - 2004

B22ALT30

What is your age range? Are you...

- 1 = Under 18
- 2 = 18–23
- 3 = 24–29
- 4 = 30 or older

Education Experiences

B22BNPATND *

If you could start your education over, would you begin at [NPSAS Institution] again?

1 = Yes

0 = No

B22BREMEVER *

Remedial or developmental courses are used to strengthen your skills in math, reading, or other subjects. Students are usually assigned to these courses on the basis of a placement test taken before the academic year begins. Often, these courses do not count for credit toward graduation.

Between July 1, 2020, and [{if before July 1, 2022} today, have you taken {else} June 30, 2022, did you take] any remedial or developmental courses to improve your basic skills in English, math, reading, or writing?

1 = Yes

0 = No

B22BEDEXPINT *

[{If USERMODE = CATI} I'd {else} We'd] now like to ask a few questions about your education experiences at [Primary Institution] between July 1, 2020, and [{if currently attending Primary Institution} today {else if before July 1, 2022 and (enrolled in year 3 at Primary Institution or abbreviated survey or Spanish)} when you last attended {else if enrolled in year 3 at Primary Institution or abbreviated survey or Spanish} June 30, 2022 {else if enrolled in year 2 at Primary Institution} June 30, 2021 {else} June 30, 2022].

This information will be used to better understand the education and social experiences of college students who were enrolled after the 2019–20 academic year.

B22BGPATYP *

Was your grade point average (GPA) at [Primary Institution] measured on a 4.00 scale?

1 = Yes

0 = No

B22BGPA *

What was your cumulative GPA in all classes at [Primary Institution] through the end of your most recent term there [{if abbreviated survey or Spanish} before July 1, 2022, {else if enrolled in year 3 at Primary Institution} in the 2021–22 academic year {else if enrolled in year 2 at Primary Institution} in the 2020–21 academic year {else} before July 1, 2022]?

(If you are unsure of the exact amount, provide your best guess.)

(0.00 to 4.00)

☐ [If currently attending Primary Institution] Not applicable, have not yet earned GPA [else] Not applicable, did not earn GPA

B22BGPAEST *

Overall, which best describes your grades when you last attended [Primary Institution] [{if abbreviated survey or Spanish} before July 1, 2022, {else if enrolled in year 3 at Primary Institution} in the 2021–22 academic year {else if enrolled in year 2 at Primary Institution} in the 2020–21 academic year {else} before July 1, 2022]?

1 = Mostly A's (3.75 and above)

2 = A's and B's (3.25-3.74)

3 = Mostly B's (2.75-3.24)

4 = B's and C's (2.25-2.74)

5 = Mostly C's (1.75-2.24)

6 = C's and D's (1.25-1.74)

7 = Mostly D's or below (1.24 or below)

-1 = Don't know [{if USERMODE = CATI} your {else} my] grades

9 = [{If USERMODE = CATI} You {else} I] would describe [{If USERMODE = CATI} your {else} my] grades differently than what is listed here

B22BKNOWCLAS

On a scale from 1 to 5 with 1 being “completely disagree” and 5 being “completely agree,” please indicate how much you disagree or agree with the following statement.

I [{if completed pursued degree} knew {else} know] the requirements needed to complete my [Pursued Degree] program at [Primary Institution].

1 (Completely disagree)

2 (Somewhat disagree)

3 (Neither disagree nor agree)

4 (Somewhat agree)

5 (Completely agree)

B22BALLONLIN

Between July 1, 2020, and [{if currently attending Primary Institution} today {else if before July 1, 2022 and enrolled in year 3 at Primary Institution} when you last attended {else if enrolled in year 3 at Primary Institution} June 30, 2022 {else} June 30, 2021], [{if currently attending Primary Institution} have any of your classes at [Primary Institution] been {else} were any of your classes at [Primary Institution]] taught entirely online?

In courses taught entirely online, students access all of their instruction for class over the Internet for the entire term or semester with no in-person or face-to-face classes.

1 = Yes

0 = No

B22BONLINEP

When you last attended [Primary Institution] during the [{if enrolled in year 3 at Primary Institution} 2021–22 {else if enrolled in year 2 at Primary Institution} 2020–21 {else} 2021–22] academic year, was your entire [Pursued Degree] program online?

1 = Yes

0 = No

B22BONLINE

Would you still have attended [Primary Institution] if [{if B22BONLINEP = 1} your online [Pursued Degree] program {else} online classes] had not been available?

1 = Yes

0 = No

B22BSCALEINT *

Please use a number from 1 to 5 to answer the next few questions. One means “completely disagree” and five means “completely agree.”

B22BACDEFF

Please indicate how much you disagree or agree with the following statements.

	1 (Completely disagree)	2 (Somewhat disagree)	3 (Neither disagree nor agree)	4 (Somewhat agree)	5 (Completely agree)
Before I attended [Primary Institution], I was confident I had the ability to succeed there as a student.	☐	☐	☐	☐	☐
[If currently attending Primary Institution] After attending [Primary Institution] for a while, I am confident that I have the ability to succeed there as a student. [else] After attending [Primary Institution] for a while, I was confident that I had the ability to succeed there as a student.	☐	☐	☐	☐	☐

B22BENCOUR

Please indicate how much you disagree or agree with the following statements.

	1 (Completely disagree)	2 (Somewhat disagree)	3 (Neither disagree nor agree)	4 (Somewhat agree)	5 (Completely agree)
My parents (or guardians) [{if currently attending Primary Institution} encourage {else} encouraged] me to stay in college.	☐	☐	☐	☐	☐
My [{if B22AMARR = 2} spouse {else} partner] [{if currently attending Primary Institution} encourages {else} encouraged] me to stay in college.	☐	☐	☐	☐	☐
My friends from [Primary Institution] [{if currently attending Primary Institution} encourage {else} encouraged] me to stay in college.	☐	☐	☐	☐	☐
My friends from home [{if currently attending Primary Institution} encourage {else} encouraged] me to stay in college.	☐	☐	☐	☐	☐

B22BSENSBELNG *

I [{if currently attending Primary Institution} feel {else} felt] that I [{if currently attending Primary Institution} am {else} was] a part of [Primary Institution].

- 1 (Completely disagree)
- 2 (Somewhat disagree)
- 3 (Neither disagree nor agree)
- 4 (Somewhat agree)
- 5 (Completely agree)

B22BWBSHFAC *

Please indicate how much you disagree or agree with the following statements.

Between July 1, 2020, and [{if enrolled in year 3 at Primary Institution or abbreviated survey or Spanish} June 30, 2022 {else if enrolled in year 2 at Primary Institution} June 30, 2021 {else} June 30, 2022], most teachers at [Primary Institution] with whom I had contact...

	1 (Completely disagree)	2 (Somewhat disagree)	3 (Neither disagree nor agree)	4 (Somewhat agree)	5 (Completely agree)
[{If currently attending Primary Institution} Are {else} Were] outstanding teachers.	☐	☐	☐	☐	☐
[{If currently attending Primary Institution} Are {else} Were] genuinely interested in teaching.	☐	☐	☐	☐	☐
[{If currently attending Primary Institution} Have {else} Had] a good command of what they [{if currently attending Primary Institution} are {else} were] teaching.	☐	☐	☐	☐	☐
[{If currently attending Primary Institution} Make {else} Made] good use of examples and illustrations to explain difficult points.	☐	☐	☐	☐	☐

B22BFACULTY

My interactions with **my teachers** at [Primary Institution] [{if currently attending Primary Institution} are {else} were] positive.

	1 (Completely disagree)	2 (Somewhat disagree)	3 (Neither disagree nor agree)	4 (Somewhat agree)	5 (Completely agree)
During class	☐	☐	☐	☐	☐
Outside of class	☐	☐	☐	☐	☐

B22BPEERINT

My interactions with **other students** at [Primary Institution] [{if currently attending Primary Institution} are {else} were] positive.

	1 (Completely disagree)	2 (Somewhat disagree)	3 (Neither disagree nor agree)	4 (Somewhat agree)	5 (Completely agree)
During class	☐	☐	☐	☐	☐
Outside of class	☐	☐	☐	☐	☐

B22BFAMCOMM

[{If USERMODE = CATI} Now I'll continue to ask you questions about your experiences at [Primary Institution] that do not use a 1 to 5 scale.]

When you last attended [Primary Institution] during the [{if enrolled in year 3 at Primary Institution} 2021–22 academic year (July 1, 2021–June 30, 2022) {else if enrolled in year 2 at Primary Institution} 2020–21 academic year (July 1, 2020–June 30, 2021) {else} 2021–22 academic year (July 1, 2021–June 30, 2022)], how often did you communicate with your...

(If you communicated with individuals within a group at different frequencies, answer about the individual(s) with whom you communicated most often.)

	1 = Daily	2 = Weekly	3 = Monthly	4 = A few times a year	0 = Never	5 = Not applicable
Parents or guardians?	☐	☐	☐	☐	☐	☐
Siblings?	☐	☐	☐	☐	☐	☐
Extended family?	☐	☐	☐	☐	☐	☐
Close friends that attended other colleges?	☐	☐	☐	☐	☐	☐
Close friends that did not attend college?	☐	☐	☐	☐	☐	☐
Close friends that attended [Primary Institution]?	☐	☐	☐	☐	☐	☐

B22BACDPART *

Between July 1, 2020, and [{if currently attending Primary Institution} today {else if before July 1, 2022 and (enrolled in year 3 at Primary Institution or abbreviated survey or Spanish)} when you last attended {else if enrolled in year 3 at Primary Institution or abbreviated survey or Spanish} June 30, 2022 {else if enrolled in year 2 at Primary Institution} June 30, 2021 {else} June 30, 2022], [{if currently attending Primary Institution} have you participated {else} did you participate] in any of the following at [Primary Institution]?

	1 = Yes	0 = No
A student success course (i.e., a course that teaches the skills needed to succeed in college)	☐	☐
Course(s) with a community-based or service learning project	☐	☐
An institution-based learning community where you took two or more classes with the same group of students	☐	☐
A guided research experience (e.g., faculty research, undergraduate research, etc.)	☐	☐
An apprenticeship (not including an externship or unpaid internship)	☐	☐
An internship, co-op, field experience, student teaching, or clinical placement	☐	☐
A culminating senior experience (e.g., capstone course, senior project or thesis, comprehensive exam, portfolio, etc.)	☐	☐

B22BSOCIAL

Between July 1, 2020, and [{if currently attending Primary Institution} today {else if before July 1, 2022 and enrolled in year 3 at Primary Institution} when you last attended {else if enrolled in year 3 at Primary Institution} June 30, 2022 {else if enrolled in year 2 at Primary Institution} June 30, 2021 {else} June 20, 2022], [{if currently attending Primary Institution} have you participated {else} did you participate] in any of the following at [Primary Institution]?

	1 = Yes	0 = No
Greek letter organization (e.g., social, multicultural, service or honor fraternities and sororities, etc.)	☐	☐
Varsity athletics (e.g., NCAA, NAIA, etc.)	☐	☐
Nonvarsity sports (e.g., club sports, intramurals, etc.)	☐	☐
Institution-sponsored music, theatre, or band	☐	☐
Student affinity groups (e.g., Black Student Association, Hillel, etc.)	☐	☐
Student government	☐	☐
Other recognized or registered student organizations (e.g., chess club, photography club, etc.)	☐	☐

B22BDIVPART *

Between July 1, 2020, and [{if currently attending Primary Institution} today {else if before July 1, 2022 and (enrolled in year 3 at Primary Institution or abbreviated survey or Spanish)} when you last attended {else if enrolled in year 3 at Primary Institution or abbreviated survey or Spanish} June 30, 2022 {else if enrolled in year 2 at Primary Institution} June 30, 2021 {else} June 30, 2022], [{if currently attending Primary Institution} have you had {else} did you have] interactions outside of the classroom with students from the following groups at [Primary Institution]?

	1 = Yes	0 = No	-1 = Don't know
Student(s) of a different race or ethnicity	☐	☐	☐
Student(s) from a different economic background	☐	☐	☐
Student(s) with a different religious belief	☐	☐	☐
Student(s) with a different political belief	☐	☐	☐
Student(s) of a different sexual orientation	☐	☐	☐
Student(s) of a gender minority status (e.g., transgender, genderqueer, gender nonconforming, etc.)	☐	☐	☐

B22BDIVERSITY *

Between July 1, 2020, and [{if currently attending Primary Institution} today {else if before July 1, 2022 and (enrolled in year 3 at Primary Institution or abbreviated survey or Spanish)} when you last attended {else if enrolled in year 3 at Primary Institution or abbreviated survey or Spanish} June 30, 2022 {else if enrolled in year 2 at Primary Institution} June 30, 2021 {else} June 30, 2022], how often [{if currently attending Primary Institution} have you had {else} did you have] meaningful and honest conversations outside of the classroom with students from the following groups at [Primary Institution]?

	0 = Never	1 = Rarely	2 = Sometimes	3 = Often	4 = Always
Student(s) of a different race or ethnicity	☐	☐	☐	☐	☐
Student(s) from a different economic background	☐	☐	☐	☐	☐
Student(s) with a different religious belief	☐	☐	☐	☐	☐
Student(s) with a different political belief	☐	☐	☐	☐	☐
Student(s) of a different sexual orientation	☐	☐	☐	☐	☐
Student(s) of a gender minority status (e.g., transgender, genderqueer, gender nonconforming, etc.)	☐	☐	☐	☐	☐

B22BSRVUSE

Between July 1, 2020, and [{if currently attending Primary Institution} today {else if before July 1, 2022 and enrolled in year 3 at Primary Institution} when you last attended {else if enrolled in year 3 at Primary Institution} June 30, 2022 {else if enrolled in year 2 at Primary Institution} June 30, 2021 {else} June 30, 2022], [{if currently attending Primary Institution} have you used {else} did you use] any of the following school services at [Primary Institution]?

	1 = Yes	0 = No
Academic advising or planning	☐	☐
Career planning or job placement assistance	☐	☐
Peer or other tutoring or skills lab (e.g., writing, math, etc.)	☐	☐
Child care	☐	☐
Financial aid counseling or advice	☐	☐
Transfer advising and planning	☐	☐
Services for students with disabilities	☐	☐
Services for active military or veterans	☐	☐
Health center and services	☐	☐
Psychological or mental health services	☐	☐
Campus security or campus police	☐	☐
Diversity or multicultural student services	☐	☐

B22BSRVIMP

How important [{if currently attending Primary Institution} are {else} were] each of these services in your decision to stay in school?

	1 = Not at all important	2 = Somewhat important	3 = Important	4 = Very important
Academic advising or planning	☐	☐	☐	☐
Career planning or job placement assistance	☐	☐	☐	☐
Peer or other tutoring or skills lab (e.g., writing, math, etc.)	☐	☐	☐	☐
Child care	☐	☐	☐	☐
Financial aid counseling or advice	☐	☐	☐	☐
Transfer advising and planning	☐	☐	☐	☐
Services for students with disabilities	☐	☐	☐	☐
Services for active military or veterans	☐	☐	☐	☐
Health center and services	☐	☐	☐	☐
Psychological or mental health services	☐	☐	☐	☐
Campus security or campus police	☐	☐	☐	☐
Diversity or multicultural student services	☐	☐	☐	☐

B22BOCCINTRO

For the next few questions, think about what you intend to do when you finish all of your education.

B22BEXOCC

What do you think the job title you intend to hold will be when you finish all of your education?

(Please provide your intended job title in the textbox. If you are unable to find a match in the results, please select the “Job title not listed” option located at the bottom of the listed results.)

B22BEXJDUTY

What do you think your job duties will be [{if B22BEXJBTL ne missing} as [a(n)] [B22BEXJBTL] {else if B22BEXSTRNG ne missing} as [a(n)] [B22BEXSTRNG] {else} in your intended job]?

B22BOCCCOM

On a scale from 1 to 5 with 1 being “very unlikely” and 5 being “very likely,” how unlikely or likely do you think it is that you will hold your intended job?

- 1 (Very unlikely)
- 2 (Somewhat unlikely)
- 3 (Neither unlikely nor likely)
- 4 (Somewhat likely)
- 5 (Very likely)
- ☐ Already hold intended job

B22BFUTWAGES

When you finish [{if B22AEXPEVR ne missing} your [highest expected degree] program {else} all of your education] and begin working [{if B22BEXJBTL ne missing} as [a(n)] [B22BEXJBTL] {else if B22BEXSTRNG ne missing} as [a(n)] [B22BEXSTRNG] {else} in your intended occupation], what do you think will be your most likely beginning salary?

(If you are unsure of the exact amount, provide your best guess.)

\$.00 (per year)

Financial Aid**INTFIN ***

In the next section, we are interested in how you paid for your education after high school.

B22CSTRCVBY

Thinking back to the 2019–20 academic year (July 1, 2019–June 30, 2020), did you receive the [state aid program]?

- 1 = Yes
- 0 = No
- 1 = Don't know

B22CSTRCVY23

[{If Y_STATEAIDYR1 = 1} Thanks.] Did you [{if B22CSTRCVBY = 1} also] receive the [state aid program] in the...

	1 = Yes	0 = No	-1 = Don't know
2020–21 academic year (July 1, 2020–June 30, 2021)?	☐	☐	☐
2021–22 academic year (July 1, 2021–June 30, 2022)?	☐	☐	☐

B22COTGRTAID

Please indicate whether or not you received or used any of the following to pay for your education in the **2021–22 academic year**.

	1 = Yes	0 = No
Veteran's education benefits	☐	☐
[If respondent is younger than 24] Financial assistance from your employer or from your parents' (or guardians') employer(s) [else] Financial assistance from your employer	☐	☐
Scholarships from a private organization such as a church, PTA, fraternity/sorority, or foundation	☐	☐

B22COTGRTAMT

Again, thinking only of the 2021–22 school year, what was the total amount you received in...

(If you are unsure of the amount(s), please provide your best guess.)

Veteran's education benefits \$.00

[{If respondent is younger than 24} Financial assistance from your employer or from your parents' (or guardians') employer(s) {else} Financial assistance from your employer] \$.00

Scholarships from private organizations \$.00

B22CLOANINT

The next few questions will ask about any student loans you have taken out for your education since high school. There are two main types of loans we are interested in:

Federal student loans, such as subsidized and unsubsidized Direct Loans (also known as Stafford Loans), are from the federal government.

Private student loans are borrowed from a private lender, such as a bank or sometimes a state, usually require a cosigner, and have market interest rates based on credit history.

B22CRCVLN

Did you take out **any** student loans for the 2021–22 academic year?

(Do not include Parent PLUS Loans, grants or scholarships, or money borrowed from family or friends in your answer.)

1 = Yes

0 = No

B22CFEDLN

Thinking only about the **2021–22 academic year**, did you take out any **federal student loans**?

(Federal student loans include subsidized and unsubsidized Direct Loans [also known as Stafford Loans]. Do not include Parent PLUS Loans.)

1 = Yes

0 = No

B22CPRVLN

Thinking only about the **2021–22 academic year**, did you take out any **private loans** borrowed from a private lender?

(Private loans are borrowed from a private lender such as a bank or sometimes a state, usually require a cosigner, and have market interest rates based on credit history.)

1 = Yes

0 = No

B22CPRVAMT

For the **2021–22 academic year**, how much did you borrow in **private loans**?

Do not include any money borrowed in federal loans or any money borrowed from family or friends in your answer.

(If you are unsure of the exact amount, provide your best guess.)

\$.00

B22CPRVEST

For the **2021–22 academic year**, please indicate the range for how much you borrowed in **private loans**. Would you say it was...

- 1 = Less than \$3,000
- 2 = \$3,000–\$6,999
- 3 = \$7,000–\$9,999
- 4 = \$10,000–\$12,999
- 5 = \$13,000–\$15,999
- 6 = \$16,000–\$19,999
- 7 = \$20,000 or more
- 1 = Don't know

B22CEVRRCVLN

Have you **ever** taken out any student loans for your education?

- 1 = Yes
- 0 = No

B22CEVRFEDLN

Have you **ever** taken out any federal student loans for your education?

(Federal student loans include subsidized and unsubsidized Direct Loans [also known as Stafford Loans]. Do not include Parent PLUS Loans.)

- 1 = Yes
- 0 = No

B22CEVRPRVLN *

Have you **ever** taken out any private loans from a private lender for your education?

(Private loans are borrowed from a private lender such as a bank or sometimes a state, usually require a cosigner, and have market interest rates based on credit history.)

1 = Yes

0 = No

B22CTLPRVAMT *

What is the **total** amount that you have borrowed in **private loans** for your education?

Do not include any money borrowed in federal loans or any money borrowed from family or friends in your answer.

(If you are unsure of the exact amount, provide your best guess.)

\$.00

B22CTLPRVEST

Please indicate the range for the total amount you borrowed in **private loans** for your education. Would you say it was...

1 = Less than \$5,000

2 = \$5,000–\$9,999

3 = \$10,000–\$19,999

4 = \$20,000–\$29,999

5 = \$30,000–\$39,999

6 = \$40,000–\$59,999

7 = \$60,000 or more

-1 = Don't know

B22CPRVRYST

Are you currently repaying your **private loans**?

1 = Yes

2 = No, because they are already paid off

3 = No, because [{if USERMODE = CATI} you {else} I] have an agreement with [{if USERMODE = CATI} your {else} my] lender to delay or defer payments

0 = No

B22CPLNMOS

How much are your monthly loan payments for your **private loans**? (If you are unsure of the exact amount, provide your best guess.)

\$.00

B22CPAYSTRAT

When thinking about repaying student loans, have you heard of any...

	1 = Yes	0 = No
Income-driven repayment plans (e.g., Income-Based, Pay as You Earn, Income-Contingent Repayment Plans)?	☐	☐
Loan forgiveness programs (e.g., Teacher Loan Forgiveness, Public Service Loan Forgiveness, etc.)?	☐	☐

B22CUSEIBR

On a scale from 1 to 5 with 1 being “very unlikely” and 5 being “very likely,” how unlikely or likely is it that you will use **any income-driven repayment plans** to repay your student loans?

- 1 (Very unlikely)
- 2 (Somewhat unlikely)
- 3 (Neither unlikely nor likely)
- 4 (Somewhat likely)
- 5 (Very likely)

B22CUSELFP

On a scale from 1 to 5 with 1 being “very unlikely” and 5 being “very likely,” how unlikely or likely is it that you will use **any loan forgiveness programs** to repay your student loans?

- 1 (Very unlikely)
- 2 (Somewhat unlikely)
- 3 (Neither unlikely nor likely)
- 4 (Somewhat likely)
- 5 (Very likely)

B22CWRKSTDY

Next, [{if USERMODE = CATI} I {else} we] have a question about work-study jobs. Federal, state, and institution work-study jobs are offered to students with financial need, allowing them to work part time to earn money to help pay their education expenses.

Did you have a work-study job **between July 1, 2020, and** [{if enrolled at any institution in year 3} **June 30, 2022** {else if enrolled at any institution in year 2} **June 30, 2021** {else} **June 30, 2022**]?

(Students must file a FAFSA, Free Application for Federal Student Aid, in order to be considered for the Federal Work-Study program. Work-study jobs are often located on the campus of a student’s school and may or may not be related to a student’s course of study.)

1 = Yes

0 = No

B22CEAIDARE *

[{If currently attending Primary Institution} Does {else} Did] [Primary Institution] have an emergency aid program ?

1 = Yes

0 = No

-1 = Don't know

B22CEAIDAPPLY

Between July 1, 2020, and [{if enrolled in year 3 at Primary Institution} June 30, 2022 {else if enrolled in year 2 at Primary Institution} June 30, 2021 {else} June 30, 2022], did you **apply** for emergency aid from [Primary Institution]?

1 = Yes

0 = No

B22CEAIDRCV

Between July 1, 2020, and [{if enrolled in year 3 at Primary Institution} June 30, 2022 {else if enrolled in year 2 at Primary Institution} June 30, 2021 {else} June 30, 2022], did you **receive** emergency aid from [Primary Institution]?

1 = Yes

0 = No

2 = Applied, but have not yet received aid

Employment

B22DINTRO *

Now we are interested in collecting information about any jobs for pay you [{if before July 1, 2022} have] held between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022], even if you held them during breaks or when you weren't taking classes.

When thinking about paid employment, please include any self-employment [{if B22CWRKSTDY = 1} , work-study jobs,] and paid internships.

B22DANYJOB *

[{If before July 1, 2022} Have you worked {else} Did you work] for pay at any time between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022]?

1 = Yes

0 = No

B22DNUMEMP *

How many employers did you have between July 1, 2020, and June 30, 2022?

employer(s)

B22DEMPKLS01 *

[{If iteration = 1} Last time we contacted you, you provided us with the [{if 1 employer on picklist} name of the employer {else} names of the employers] you had during the 2019–20 academic year (July 1, 2019–June 30, 2020.)

[{If before July 1, 2022} Have you continued {else} Did you continue] to work for [{if 1 employer on picklist} this employer {else} any of these employers] at any time between **July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022]**?

1 = [base-year employer 1]

2 = [base-year employer 2]

3 = [base-year employer 3]

4 = [base-year employer 4]

5 = [base-year employer 5]

6 = [base-year employer 6]

7 = [base-year employer 7]

8 = [base-year employer 8]

9 = [base-year employer 9]

10 = [base-year employer 10]

99 = None of these employers

B22DEMPLOY01 *

What is the name of another employer you worked for [{if before July 1, 2022} between July 1, 2020, and today {else} between July 1, 2020, and June 30, 2022]?

☐ Check here to indicate self-employment

B22DWRKMON01

Between July 1, 2020, and June 30, 2022, in which months [{if before July 1, 2022} have you worked, or will you work, {else} did you work] for [{if B22DEMPNAM01 ne missing} [B22DEMPNAM01] {else if B22DEMPSLF01 = 1} yourself {else} this employer]?

([{If USERMODE = WEB} Selected months will be shaded dark gray.] If you [{if before July 1, 2022} have worked, or will work, {else} worked] for [{if B22DEMPNAM01 ne missing} [B22DEMPNAM01] {else if B22DEMPSLF01 = 1} yourself {else} this employer] only a portion of any month, please include that month.)

2020–21

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

2021–22

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

B22DEMPCUR01 *

Are you currently [{if B22DEMPNAM01 ne missing} working for [B22DEMPNAM01] {else if B22DEMPSLF01 = 1} self-employed {else} working for this employer]?

1 = Yes

0 = No

B22DEARN01

How much did you make when you [{if B22DEMPNAM01 ne missing} last worked for [B22DEMPNAM01] {else if B22DEMPSLF01 = 1} were last self-employed {else} last worked for this employer]?

Include any bonuses, tips, or commissions in your total earnings amount.
(If you are unsure of the exact amount, provide your best guess.)

\$

- 1 = Per hour
- 2 = Per month
- 3 = Per year

B22DWRKENR01

[{If before July 1, 2022} Since July 1, 2020 {else} Between July 1, 2020, and June 30, 2022], did you regularly work for [{if B22DEMPNAM01 ne missing} [B22DEMPNAM01] {else if B22DEMPSLF01 = 1} yourself {else} this employer] during weeks in which you were **also attending school**?

- 1 = Yes
- 0 = No

B22DWRKSTD01

[{If B22DEMPCUR01 = 1} Is {else} Was] your employment with [{if B22DEMPNAM01 ne missing} [B22DEMPNAM01] {else} this employer] a work-study job?

- 1 = Yes
- 0 = No

B22DONOFF01 *

[{If B22DEMPCUR01 = 1} Is {else} Was] this job on or off the campus of your school?

- 1 = On campus
- 2 = Off campus

B22DWKHREN01 *

[{If before July 1, 2022} Since July 1, 2020 {else} Between July 1, 2020, and June 30, 2022], how many hours per week did you work for [{if B22DEMPNAM01 ne missing} [B22DEMPNAM01] {else if B22DEMPSLF01 = 1} yourself {else} this employer] **while you were attending school**? (If you are unsure of the exact amount, provide your best guess.)

hours per week

B22DWRKNEN01

[{If B22DEMPCUR01 = 1} Between July 1, 2020, and today, have you regularly worked {else if before July 1, 2022} Since July 1, 2020, did you regularly work {else} Between July 1, 2020, and June 30, 2022, did you regularly work] for [{if B22DEMPNAM01 ne missing} [B22DEMPNAM01] {else if B22DEMPSLF01 = 1} yourself {else} this employer] during weeks in which you were **not attending school**?

1 = Yes

0 = No

B22DWKHRS01 *

[{If before July 1, 2022} Since July 1, 2020 {else} Between July 1, 2020, and June 30, 2022], how many hours per week did you usually work for [{if B22DEMPNAM01 ne missing} [B22DEMPNAM01] {else if B22DEMPSLF01 = 1} yourself {else} this employer] [{if B22DWRKNEN01 = 1 or abbreviated survey or Spanish)} **during weeks in which you were not attending school**]?

(If you are unsure of the exact amount, provide your best guess.)

hours per week

B22DOTHEMP01 *

[{If before July 1, 2022} Have you worked {else} Did you work] for pay for any other employers at any time [{if before July 1, 2022} Between July **1, 2020, and today?** {else} Between July **1, 2020, and June 30, 2022?**]

(Answer **Yes** for any self-employment [{if B22CWRKSTDY = 1} , work-study jobs,] or paid internships.)

1 = Yes, worked for other employers

0 = No other employers

B22DENRWORK

When you were last attending school and working, did you consider yourself to be primarily...

1 = [If currently attending any institution and currently employed]

A student who works

[else]

A student who worked

2 = An employee who decided to enroll in school

B22DREFPKLST *

The next few questions will focus on your experiences specifically with one employer. Which of the following employers do you consider to be your main employer between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022]?

- 1 = [employer from iteration 1]
- 2 = [employer from iteration 2]
- 3 = [employer from iteration 3]
- 4 = [employer from iteration 4]
- 5 = [employer from iteration 5]
- 6 = [employer from iteration 6]
- 7 = [employer from iteration 7]
- 8 = [employer from iteration 8]
- 9 = [employer from iteration 9]
- 10 = [employer from iteration 10]

B22DOCC

[{If reported 1 employer} Next, [{if USERMODE = CATI} I {else} we] have some questions that will focus on your [{if self-employed reference employer, no name} self-employment {else} employment with [Reference Employer]].]

What was your most recent job title [{if self-employed reference employer, no name} for your self-employment {else} with [Reference Employer]]?

(Please provide your job title in the textbox. If you are unable to find a match in the results, please select the “Job title not listed” option located at the bottom of the listed results.)

B22DJBDUTY

What were your most recent job duties [{if self-employed reference employer, no name} for your self-employment {else} with [Reference Employer]]?

B22DEMPBEN

[{If on or after July 1, 2022} Before July 1, 2022, when {else} When] you last worked in [job title fill], were you eligible to receive...

	1 = Yes	0 = No
Health insurance	☐	☐
Life insurance	☐	☐
Retirement or other financial benefits, such as a 401(k)/403(b)	☐	☐

B22DHLIC

Next, [{if USERMODE = CATI} I'd {else} we'd] like to ask about professional certifications or industry licenses. A professional certification or license shows you are qualified to perform a specific job and includes things like Licensed Realtor, Certified Medical Assistant, Certified Construction Manager, or an IT certification.

Do you have a currently active professional certification or a state or industry license?

(Do not include business licenses such as a liquor license or vending license.)

1 = Yes

0 = No

B22DLICREL

Was this certification or license required for [job title fill] [{if on or after July 1, 2022} between July 1, 2020, and June 30, 2022]?

1 = Yes

0 = No

B22DOCCTIMGT

[{If before July 1, 2022} Did {else} Before July 1, 2022, did] you work in [job title fill], or in any similar jobs for a year or more, even if they were not [{if self-employed reference employer, no name} self-employment {else} with [Reference Employer]]?

1 = Yes

0 = No

B22DOCCTIM

[{If before July 1, 2022} How {else} Before July 1, 2022, how] many years did you work in [job title fill], or in any similar jobs, even if they were not [{if self-employed reference employer, no name} self-employment {else} with [Reference Employer]]?

year(s)

B22DRELMAJ *

[{If currently employed at Reference Employer} Is {else} Was] [job title fill] related to what you studied in college?

1 = Yes

0 = No

B22DSTRTCR

[{If currently employed at Reference Employer} Is {else} Was] [job title fill] related to the kind of work you want to do in the future?

1 = Yes

0 = No

B22DACTLKWRK

Between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022], were there times when you were actively looking for work?

(Actively looking for work includes activities such as sending out résumés or filling out job applications, placing or responding to job advertisements, contacting employers or employment agencies, and going on job interviews.)

1 = Yes

0 = No

B22DLKWRK

Between July 1, 2020, and [{if before July 1, 2022} today {else} June 30, 2022], in which months were you not working and actively looking for a job?

([{If USERMODE = WEB} Selected months will be shaded dark gray.] If you [{if before July 1, 2022} have] looked for work for a portion of any month, please include that month.)

2020-21

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

2021-22

- ☐ July
- ☐ August
- ☐ September
- ☐ October
- ☐ November
- ☐ December
- ☐ January
- ☐ February
- ☐ March
- ☐ April
- ☐ May
- ☐ June

B22DWRKPAY *

Now [{if USERMODE = CATI} I {else} we] have some questions about what you might have done if you had not been in school at all this year.

If you had not attended college at all in the 2021–22 academic year (July 1, 2021–June 30, 2022), which of the following would you most likely have done?

- 1 = Worked for pay
- 2 = Worked, but not be paid
- 3 = Not worked (for any reason)
- 4 = Entered the military

B22DALTPAY *

How much do you think you would have earned from working if you had not attended college at all in the 2021–22 academic year?

(If you are unsure of the exact amount, provide your best guess.)

\$.00 per year

B22DEDBENFTS

When thinking about a job now or in the future, salary may be only one part of why you choose that job.

Compared to the salary, how important is each of the following to you?

	1 = Less important than salary	2 = As important as salary	3 = More important than salary
Helping others as part of your job	☐	☐	☐
Being seen as an expert in your field	☐	☐	☐
Making your own decisions about how to get your work done	☐	☐	☐
Balancing work and leisure time	☐	☐	☐
Balancing work and family	☐	☐	☐

Income and Expenses**B22EINCINTRO ***

The next set of questions will help us to better understand how the financial circumstances of students may affect their ability to attend college.

B22EINCOM *

What was your income for **calendar year 2021**, prior to taxes and deductions?

(Calendar year 2021 includes January 1, 2021, through December 31, 2021. Include all income you paid taxes on, including work, investment income, or alimony. [If B22AMARR = 2} **Do not include your spouse's income**, {else} Do not include] any grants or loans you may have used to pay for school, or any money given to you by your family.)

- 1 = No income
- 2 = Less than \$1,000
- 3 = \$1,000–\$2,499
- 4 = \$2,500–\$4,999
- 5 = \$5,000–\$9,999
- 6 = \$10,000–\$14,999
- 7 = \$15,000–\$19,999
- 8 = \$20,000–\$29,999
- 9 = \$30,000–\$49,999
- 10 = \$50,000 and above
- 1 = Don't know

B22EINCSP *

What was your spouse's income for **calendar year 2021**, prior to taxes and deductions?

(Calendar year 2021 includes January 1, 2021, through December 31, 2021. Include all income your spouse paid taxes on. Do not include any grants or loans your spouse may have used to pay for school, or any money given to your spouse by family.)

- 1 = No income
- 2 = Less than \$1,000
- 3 = \$1,000–\$2,499
- 4 = \$2,500–\$4,999
- 5 = \$5,000–\$9,999
- 6 = \$10,000–\$14,999
- 7 = \$15,000–\$19,999
- 8 = \$20,000–\$29,999
- 9 = \$30,000–\$49,999
- 10 = \$50,000 and above
- 1 = Don't know

☐ Check here if you were not married to your spouse in 2021

B22EDEPS

Do you [{if before July 1, 2022} now have, or will you have, {else} have] children who [{if before July 1, 2022} will receive {else} received] more than half of their support from you between July 1, 2021, and June 30, 2022?

- 1 = Yes
- 0 = No

B22EDEP2

How many children [{if before July 1, 2022} receive, or will receive, {else} received] more than half of their support from you between July 1, 2021, and June 30, 2022?

child(ren)

B22EOTDEPS

Do you have dependents [other than your spouse or dependent children] who live with you and [{if before July 1, 2022} receive, or will receive, {else} received] more than half of their support from you between July 1, 2021, and June 30, 2022?

- 1 = Yes
- 0 = No

B22EOTDEPS2

Other than your [spouse or dependent children], how many others [{if before July 1, 2022} live {else} lived] with you and [{if before July 1, 2022} receive, or will receive, {else} received] more than half of their support from you between July 1, 2021, and June 30, 2022?

other(s)

B22EKIDCOL

Of your [number of dependents] [{if B22EOTDEPS = 1 and B22EOTDEPS2 > 0} dependents, {else} dependent children,] how many attended a college, university, or trade school in the 2021–22 academic year (July 1, 2021–June 30, 2022)?

Only include [{if B22EOTDEPS = 1 and B22EOTDEPS2 > 0} dependents {else} dependent children] who attended at least half-time in 2021–22, a program that leads to a college degree or certificate.

dependent(s)

B22EKIDCOL1

Did your [{if B22EOTDEPS = 1 and B22EOTDEPS2 > 0} dependent {else} dependent child] attend a college, university, or trade school in the 2021–22 academic year (July 1, 2021–June 30, 2022)?

Answer **Yes** if your [{if B22EOTDEPS = 1 and B22EOTDEPS2 > 0} dependent {else} dependent child] attended at least half-time in 2021–22, a program that leads to a college degree or certificate.

1 = Yes

0 = No

B22EHHINTRO

The next few questions will help us better understand the background of students' families and households.

B22EACTDUTY

Are you currently serving on active duty in the U.S. Armed Forces for purposes other than training?

1 = Yes

0 = No

B22EVET

Are you a veteran of the U.S. Armed Forces?

1 = Yes

0 = No

B22ERESERVE

Are you currently a member of the Reserve or National Guard for any branch of the U.S. Armed Forces?

	1 = Yes	0 = No
The Reserve	☐	☐
National Guard	☐	☐

B22EHHWHO

Which of the following best describes your living situation when you turned 16 years old?

(If your parents [or guardians] lived in separate households when you turned 16, please answer this question about the parent [or guardian] with whom you lived most. If you did not live with one parent [or guardian] more than the other, answer about the parent [or guardian] who provided more financial support during this time.)

- 1 = Lived with one parent (or guardian) (including stepparent or foster parent)
- 2 = Lived with two parents (or guardians) (including stepparents or foster parents)
- 3 = Did not live with parents (or guardians)

B22EPARST

What is the current marital status of your parents (or guardians)? If your parents are divorced, please answer this question about the marital status of the parent (or guardian) whom you lived with most during the past 12 months.

(If you did not live with one parent more than the other, answer about the parent who provided more financial support during the last 12 months, or during the most recent year that you received support from a parent.)

- 1 = Married or remarried
- 2 = Single
- 3 = Divorced or separated
- 4 = Widowed
- 5 = Unmarried and both parents living together
- 6 = None of the above—Both parents (or guardians) are deceased

B22EPARNC *

In calendar year 2021 (January 1, 2021, through December 31, 2021), what was the income of the parent (or guardian) whom you lived with most in the past 12 months?

(If you did not live with one parent more than the other, answer about the parent who provided more financial support during the last 12 months, or during the most recent year that you received support from a parent.)

1 = Under \$30,000

2 = \$30,000–\$59,999

3 = \$60,000–\$89,999

4 = \$90,000–\$119,999

5 = \$120,000–\$149,999

6 = \$150,000 or more

-1 = Don't know

☐ Parents (or guardians) are deceased

B22EPRHSD

Not including yourself or the parent (or guardian) whom you lived with most in the past 12 months, how many people (e.g., brothers or sisters or grandparents) did this parent (or guardian) support financially during the most recent term you attended school in the 2021–22 academic year?

(If you did not live with one parent more than the other, answer about the parent who provided more financial support during the last 12 months, or during the most recent year that you received support from a parent.)

☐

other(s)

B22EDPNUM

Not including yourself or your parent (or guardian), how many of the people financially supported by your parent (or guardian) **attended a college, university, or trade school** during the most recent term you attended school in the 2021–22 academic year?

Only include others in your parent's household if they will attend, at least half-time in 2021–22, a program that leads to a college degree or certificate.

☐

other(s)

B22ERECSUPP

Between July 1, 2020, and June 30, 2022, did you regularly give any friends or family who did not live with you more than \$50 per month to help them out?

1 = Yes

0 = No

B22EPARHELP *

Between July 1, 2020, and [if enrolled at any institution in year 3 or abbreviated survey or Spanish) June 30, 2022 {else if enrolled at any institution in year 2} June 30, 2021 {else} June 30, 2022], did your parents (or guardians) help you pay for any of your education or living expenses while you were enrolled in school?

(Tuition and fees or school books are examples of education expenses. Rent and food are examples of living expenses.)

1 = Yes

0 = No

B22EPARRNG

Was the amount your parents (or guardians) contributed toward your education or living expenses between July 1, 2020, and [{if enrolled at any institution in year 3} June 30, 2022 {else if enrolled at any institution in year 2} June 30, 2021 {else} June 30, 2022]...

1 = \$5,000 or less

2 = More than \$5,000

B22EPARLOAMT

Okay. Was this amount...

1 = Less than \$500

2 = \$500–\$999

3 = \$1,000–\$1,499

4 = \$1,500–\$1,999

5 = \$2,000–\$5,000

B22EPARHIAMT

Okay. Was this amount...

1 = \$5,001–\$9,999

2 = \$10,000–\$14,999

3 = \$15,000–\$19,999

4 = \$20,000–\$24,999

5 = \$25,000–\$49,999

6 = \$50,000 or more

B22EFAMHELP *

Between July 1, 2020, and [{if enrolled at any institution in year 3 or abbreviated survey or Spanish} June 30, 2022 {else if enrolled at any institution in year 2} June 30, 2021 {else} June 30, 2022], did **other family members or friends** (not including [{if B22AMARR = 2} your spouse or] your parents [or guardians]) help you pay for any of your education or living expenses while you were enrolled in school?

(Tuition and fees or school books are examples of education expenses. Rent and food are examples of living expenses.)

1 = Yes

0 = No

B22EFAMRNG

Was the amount these family members contributed toward your education or living expenses between July 1, 2020, and [{if enrolled at any institution in year 3} June 30, 2022 {else if enrolled at any institution in year 2} June 30, 2021 {else} June 30, 2022]...

- 1 = \$5,000 or less
- 2 = More than \$5,000

B22EFAMLOAMT

Okay. Was this amount...

- 1 = Less than \$500
- 2 = \$500–\$999
- 3 = \$1,000–\$1,499
- 4 = \$1,500–\$1,999
- 5 = \$2,000–\$5,000

B22EFAMHIAMT

Okay. Was this amount...

- 1 = \$5,001–\$9,999
- 2 = \$10,000–\$14,999
- 3 = \$15,000–\$19,999
- 4 = \$20,000–\$24,999
- 5 = \$25,000–\$49,999
- 6 = \$50,000 or more

B22ENUMCRD *

Excluding debit or ATM cards, how many credit cards do you have in your own name that are billed to you?

(Only include credit cards for which you pay at least some of the amount owed.)

- 0 = None
- 1 = One
- 2 = More than one

B22ECARRYBAL *

Do you usually owe an amount that is carried over on your credit cards from month to month?

- 1 = Yes
- 0 = No

B22ECRDBAL

What was the **total amount you owed** on all your credit cards combined according to your last month's statements?

\$.00

B22ECCPAYMT *

What was the **total amount you paid** toward all of your credit card statements combined last month?

\$.00

B22ESCHRES

Were you living in college-owned housing, with your parents (or guardians), or off campus while you attended [Primary Institution] between July 1, 2020, and [{if enrolled in year 3 at Primary Institution} June 30, 2022 {else if enrolled in year 2 at Primary Institution} June 30, 2021 {else} June 30, 2022]?

(If you lived in more than one residence, choose the place where you lived for the longest period of time.)

- 1 = On campus or in college-owned housing (e.g., a dorm or a residence hall)
- 2 = With parents (or guardians)
- 3 = Off campus (not college-owned housing)

B22EHOME *

Do you own a home or pay a mortgage on a home?

- 1 = Yes
- 0 = No

B22ERNTAMT *

How much (on average) is your monthly [{if B22EHOME = 1} mortgage {else if B22EHOME = 0} rent {else} rent or mortgage] payment where you currently live?

(Please indicate only the amount that you [{if B22AMARR = 2} and your spouse] are responsible for paying. If someone else pays your [{if B22EHOME = 1} mortgage {else if B22EHOME = 0} rent {else} rent or mortgage] on your behalf, please indicate "0.")

\$.00

B22ECARLON

Do you [{if B22AMARR = 2} or your spouse] make loan or lease payments for a car, truck, motorcycle, or other vehicle?

(If someone other than you [{if B22AMARR = 2} or your spouse] makes the payment(s) on your behalf, please answer **No.**)

1 = Yes

0 = No

B22ECARAMT

How much do you [{if B22AMARR = 2} and your spouse] usually pay in total vehicle loan or lease payments each month?

\$.00

B22EUNTAX

At any time **during 2019 or 2020**, did you, your parents, or anyone in your parents' (or guardians') household receive benefits from any of the federal programs listed?

	1 = Yes	0 = No
Medicaid or Supplemental Security Income (SSI)	☐	☐
Supplemental Nutrition Assistance Program (SNAP)	☐	☐
Temporary Assistance for Needy Families (TANF)	☐	☐
Free or Reduced Price School Lunch	☐	☐
Special Supplemental Nutrition Program for Women, Infants, and Children (WIC)	☐	☐

B22EFIN2000 *

How confident are you that you could come up with **\$2,000**, from any available source, if an unexpected need arose within the next month? Could you...

1 = Certainly could come up with the \$2,000

2 = Probably come up with the \$2,000

3 = Probably not come up with the \$2,000

4 = Certainly could not come up with the \$2,000

B22EFIN500 *

How confident are you that you could come up with **\$500**, from any available source, if an unexpected need arose within the next month? Could you...

1 = Certainly could come up with the \$500

2 = Probably come up with the \$500

3 = Probably not come up with the \$500

4 = Certainly could not come up with the \$500

B22EFIN1YEAR

For the next few questions, we want to learn more about students' understanding of broad financial concepts. Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?

- 1 = More than today
- 2 = Exactly the same
- 3 = Less than today

B22EFIN5YEAR

Suppose you had \$100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow?

- 1 = More than \$102
- 2 = Exactly \$102
- 3 = Less than \$102

B22EFINSTOCK

Do you think that the following statement is true or false?

“Buying a single company stock usually provides a safer return than a stock mutual fund.”

- 1 = True
- 2 = False
- 1 = Don't know

B22EFEDACT

If a borrower is unable to repay their federal student loan, what steps can the government take to collect the debt? (Please check all that apply.)

- ☐ Report that the student debt is past due to the credit bureaus
- ☐ Garnish wages until the debt, plus any interest and fees, is repaid
- ☐ Retain tax refunds and Social Security payments until the debt, plus any interest and fees, is repaid
- ☐ None of the above

Background**INTBCK ***

Next, [{if USERMODE = CATI} I {else} we] have some questions that will help us better understand the unique experiences of students from all backgrounds.

B22FCITZN

Are you a U.S. citizen?

1 = Yes

2 = No, permanent resident or other eligible noncitizen (e.g., hold a temporary resident's card or other eligible noncitizen temporary resident's card)

3 = No, student visa, in the country on an F1 or F2 visa, or on a J1 or J2 exchange visitor visa

4 = No, other

B22FSX *

These next few questions will help us better understand the experiences of people of all sexual orientations and gender identities.

What sex were you assigned at birth (what the doctor put on your birth certificate)?

1 = Male

2 = Female

B22FGENDERID *

What is your gender? Your gender is how you feel inside and can be the same or different from your biological or birth sex. (Please check all that apply.)

☐ Male

☐ Female

☐ Transgender, male-to-female

☐ Transgender, female-to-male

☐ Genderqueer or gender nonconforming

Please describe:

☐ A different identity

Please describe:

B22FLGBTQ *

Now I will read a list of terms people sometimes use to describe how they think of themselves.

- Lesbian or gay, that is, homosexual
- Straight, that is, heterosexual
- Bisexual
- Another sexual orientation

As I read the list again, please say yes when you hear the option that best describes how you think of yourself.

2 = Straight, that is, heterosexual
1 = Lesbian or gay, that is, homosexual
3 = Bisexual
4 = Another sexual orientation
Please describe:

-1 = Don't know
Please describe:

B22FVTNEL *

Did you vote in the November 2020 presidential election?

1 = Yes
0 = No
☐ I am not eligible to participate in presidential elections

B22FEVRVT

Have you ever voted in a national, state, or local election?

1 = Yes
0 = No

B22FREGVT

Are you registered to vote?

1 = Yes
0 = No
-1 = Don't know

B22FACS16A

These next few questions will help us better understand the educational services available for people with disabilities.

Are you deaf or do you have serious difficulty hearing?

1 = Yes
0 = No

B22FACS16B

Are you blind or do you have serious difficulty seeing even when wearing glasses?

1 = Yes

0 = No

B22FACS17A

Because of a physical, mental, or emotional condition, do you have serious difficulty concentrating, remembering, or making decisions?

1 = Yes

0 = No

B22FACS17B

Do you have serious difficulty walking or climbing stairs?

1 = Yes

0 = No

B22FMAIN

What is the main type of condition or impairment that you have?

1 = Hearing impairment (e.g., deaf or hard of hearing)

2 = Blindness or visual impairment that cannot be corrected by wearing glasses

3 = Speech or language impairment

4 = Orthopedic or mobility impairment

5 = Specific learning disability or dyslexia

6 = Attention deficit disorder (ADD)

7 = Health impairment or problem

8 = Mental, emotional or psychiatric condition

9 = Depression

10 = Developmental disability

11 = Brain injury

12 = Other

B22FPHYSH

In general, how is your **physical** health?

1 = Excellent

2 = Very good

3 = Good

4 = Fair

5 = Poor

B22FMENTH

In general, how is your **mental** health?

- 1 = Excellent
- 2 = Very good
- 3 = Good
- 4 = Fair
- 5 = Poor

B22FHLTHMISS

In the past 30 days, how often did each of the following health concerns cause you to miss a day of school or work?

	1 = Never	2 = A few times	3 = About once a week	4 = Almost every day	5 = Every day
Physical health concern	☐	☐	☐	☐	☐
Mental health concern	☐	☐	☐	☐	☐

B22FHFINTRO

[[{If USERMODE = CATI} I'd {else} We'd] now like to ask a few questions about your food and housing experiences during college. This information will be used to better understand the basic needs of students who started college in the 2019–20 academic year.

B22FCAMPSAFE *

On a scale from 1 to 5, with 1 being “completely unsafe” and 5 being “completely safe,” how unsafe or safe did you feel while attending [Primary Institution] between July 1, 2020, and [{if enrolled in year 3 at Primary Institution or abbreviated survey or Spanish} June 30, 2022 {else if enrolled in year 2 at Primary Institution} June 30, 2021 {else} June 30, 2022]?

- 1 (Completely unsafe)
- 2 (Somewhat unsafe)
- 3 (Neither unsafe nor safe)
- 4 (Somewhat safe)
- 5 (Completely safe)

B22FHOUSEC

In the past **12 months**, did you...

	1 = Yes	0 = No
Not pay or underpay your rent or mortgage?	☐	☐
Leave your household because you felt unsafe?	☐	☐
Move three or more times ?	☐	☐
Not pay the full amount of a gas, oil, or electricity bill?	☐	☐
Move in with other people, even for a little while, because of financial problems?	☐	☐
Live with others beyond the expected capacity of the house or apartment?	☐	☐

B22FEVRHOML

In the last **30 days**, have you slept in any of the following places because you had nowhere else to go?

(Do not consider sleeping arrangements while on vacation or business travel.)

	1 = Yes	0 = No
A shelter	☐	☐
In a camper	☐	☐
Temporarily staying with a relative, friend, or couch surfing until you find other housing	☐	☐
Temporarily at a hotel or motel without a permanent home to return to	☐	☐
In transitional housing or independent living program	☐	☐
Outdoor location such as street, sidewalk or alley, bus or train stop, campground or woods, park, beach or riverbed, or under a bridge or overpass	☐	☐
In a closed area/space with a roof not meant for human habitation such as abandoned building, car or truck, van, encampment or tent, or unconverted garage, attic, or basement	☐	☐

B22FMEALPLN1

Did you purchase a school meal plan for the 2021–22 academic year?

1 = Yes

0 = No

B22FMEALPLN2

[{If before July 1, 2022} Does {else} Did] your school meal plan for the 2021–22 academic year cover 11 or more meals a week?

1 = Yes

0 = No

B22FUSDAHH

For these statements, [{if USERMODE = CATI} please tell me {else} please indicate] whether the statement was often true, sometimes true, or never true for you **in the last 30 days**.

	1 = Often true	2 = Sometimes true	3 = Never true
I worried whether my food would run out before I got money to buy more.	☐	☐	☐
The food I bought just didn't last, and I didn't have money to get more.	☐	☐	☐
I couldn't afford to eat balanced meals.	☐	☐	☐

B22FUSDAAD1

In the last 30 days, have you ever cut the size of your meals or skipped meals because there wasn't enough money for food?

1 = Yes

0 = No

B22FUSDAAD1A

In the last 30 days, how many days did this happen? (If you are unsure of the exact amount, provide your best guess.)

days

B22FUSDAAD2

In the last 30 days, did you ever eat less than you felt you should because there wasn't enough money for food?

1 = Yes

0 = No

B22FUSDAAD3

In the last 30 days, were you ever hungry but didn't eat because there wasn't enough money for food?

1 = Yes

0 = No

B22FUSDAAD4

In the last 30 days, did you lose weight because there wasn't enough money for food?

1 = Yes

0 = No

B22FUSDAAD5

In the last 30 days, did you ever not eat for a whole day because there wasn't enough money for food?

1 = Yes

0 = No

B22FUSDAAD5A

In the last 30 days, how many days did this happen? (If you are unsure of the exact amount, provide your best guess.)

days

COVID**B22FCVINTRO ***

Now we would like to collect information on how the coronavirus pandemic affected your academic and personal experiences. The next set of questions will focus on your experiences between **July 1, 2020, and June 30, 2021**.

B22FCVATND *

Did you attend any college, university, or trade school at any time **between July 1, 2020, and June 30, 2021**?

(Consider yourself as attending any institution even if all your courses were online.)

1 = Yes

0 = No

B22FCVACAD *

We're interested in knowing how your academic plans may have changed as a result of the coronavirus pandemic. Did you do any of the following between **July 1, 2020, and June 30, 2021**? (Please check all that apply.)

- ☐ Changed the institution where I enrolled
- ☐ Reduced my enrollment from full-time to part-time
- ☐ Increased my enrollment from part-time to full-time
- ☐ Did not attend for an entire term or semester
- ☐ Changed my declared or intended major
- ☐ None of the above

B22FCVNOTATND *

Please indicate how much you disagree or agree with the following statement. One means "completely disagree" and five means "completely agree."

My decision to not enroll in any college, university, or trade school **between July 1, 2020, and June 30, 2021**, was due to the coronavirus pandemic.

- 1 (Completely disagree)
- 2 (Somewhat disagree)
- 3 (Neither disagree nor agree)
- 4 (Somewhat agree)
- 5 (Completely agree)

B22FCVATNDPS *

Did you attend [Primary Institution] at any time between **July 1, 2020, and June 30, 2021**?

(Consider yourself as attending [Primary Institution] even if all your courses were online.)

- 1 = Yes
- 0 = No

B22FCVEXP *

Did you experience any of the following while attending [Primary Institution] **between July 1, 2020, and June 30, 2021**, as a result of the coronavirus pandemic? (Please check all that apply.)

- ☐ The delivery method of my classes was modified (e.g., classes started in-person and moved to hybrid or online-only instruction)
- ☐ Some or all of my classes were delivered entirely online (including labs, practicums, etc.)
- ☐ Some or all of my classes were delivered in a hybrid format (i.e., a mix of both online and in-person instruction)
- ☐ The academic calendar was modified (e.g., semester started or ended early, no fall break, etc.)
- ☐ The grading system for some or all of my classes was modified (e.g., changed from 4.00 scale to pass/fail, etc.)
- ☐ My internship, apprenticeship, student teaching, or clinical placement was modified or cancelled
- ☐ Experienced none of the above

B22FCVTHINK *

Please indicate how much you disagree or agree with the following statements. One means “completely disagree” and five means “completely agree.”

As a result of the coronavirus pandemic, while attending [Primary Institution] **between July 1, 2020, and June 30, 2021...**

	1 (Completely disagree)	2 (Somewhat disagree)	3 (Neither disagree nor agree)	4 (Somewhat agree)	5 (Completely agree)
I had fewer interactions with academic advisors	☐	☐	☐	☐	☐
I had fewer interactions with teachers (e.g., due to reduced or virtual-only office hours, etc.)	☐	☐	☐	☐	☐
I felt less connected with other students at [Primary Institution]	☐	☐	☐	☐	☐
I accessed campus learning resources less (e.g., learning labs, tutoring center, etc.)	☐	☐	☐	☐	☐
My online-only or hybrid classes were not as effective as in person classes	☐	☐	☐	☐	☐
My grades were lower than expected	☐	☐	☐	☐	☐

B22FCVPAY *

Please indicate whether or not you did any of the following to pay for your enrollment at [Primary Institution] **between July 1, 2020, and June 30, 2021**, as a result of the coronavirus pandemic.

(If the statement does not apply, please select **Not applicable**.)

	1 = Yes	0 = No	2 = Not applicable
Borrowed more money in federal student loans	☐	☐	☐
Borrowed less money in federal student loans	☐	☐	☐
Borrowed more money in private student loans	☐	☐	☐
Borrowed less money in private student loans	☐	☐	☐
Received more money from friends and/or family (e.g., gifts, loans, etc.)	☐	☐	☐
Received less money from friends and/or family (e.g., gifts, loans, etc.)	☐	☐	☐

B22FCVTSUPP *

Please indicate how much you disagree or agree with the following statements. One means “completely disagree” and five means “completely agree.”

Between July 1, 2020, and June 30, 2021, my teachers at [Primary Institution] had reasonable expectations of me during the coronavirus pandemic.

- 1 (Completely disagree)
- 2 (Somewhat disagree)
- 3 (Neither disagree nor agree)
- 4 (Somewhat agree)
- 5 (Completely agree)

B22FCVINFO *

As part of the response to the coronavirus pandemic, **between July 1, 2020, and June 30, 2021**, [Primary Institution] provided the information I needed to...

(If [Primary Institution] provided the information, but you personally did not need it, please select **Not needed**.)

	1 (Completely disagree)	2 (Somewhat disagree)	3 (Neither disagree nor agree)	4 (Somewhat agree)	5 (Completely agree)	6 = Not needed
Obtain physical health services	☐	☐	☐	☐	☐	☐
Obtain mental health services	☐	☐	☐	☐	☐	☐

B22FDISACCOM *

Earlier you reported having a physical or mental impairment or condition, and [{if USERMODE = CATI} I {else} we] would like to better understand the educational services made available to you for your online coursework. On a scale from 1 to 5 with 1 being “completely disagree” and 5 being “completely agree,” please indicate how much you disagree or agree with the following statements.

Between July 1, 2020, and June 30, 2021, [Primary Institution] provided **appropriate accommodations** for my impairment or condition so that I could...

	1 (Completely disagree)	2 (Somewhat disagree)	3 (Neither disagree nor agree)	4 (Somewhat agree)	5 (Completely agree)
Access required online course content.	☐	☐	☐	☐	☐
Complete required online coursework.	☐	☐	☐	☐	☐

B22FINTACC *

Between **July 1, 2020, and June 30, 2021**, did your residence have access to high-speed internet (e.g., such as cable, fiber optic, or DSL service, etc.)?

1 = Yes

0 = No

☐ Did not have internet access at my residence

B22FDEVHRDSFT *

Did you have the necessary hardware and software to fully participate in class and complete your coursework **between July 1, 2020, and June 30, 2021**?

	1 = Yes	0 = No
Necessary hardware (e.g., webcam, microphone, printer, other accessories, etc.)	☐	☐
Necessary software (e.g., word processor, presentation software, etc.)	☐	☐

B22FDIFFIC *

In **an average week** between July 1, 2020, and June 30, 2021, how often did you have **difficulty** with...

	0 = Never	1 = Rarely	2 = Sometimes	3 = Often	4 = Always
The internet connection at your residence when accessing your online course content (e.g., issues with speed, reliability, buffering, etc.)?	☐	☐	☐	☐	☐
Accessing a physical space free of interruption or distraction at your residence where you could complete your coursework?	☐	☐	☐	☐	☐

B22FCVEXPB *

Did you experience any of the following between **July 1, 2020, and June 30, 2021** as a result of the coronavirus pandemic? (Please check all that apply.)

- ☐ Lost a job or lost income because of reduced hours (e.g., moved away from job [{if B22FCVATND = 1} and Primary Institution not in (-3 -9)] near [Primary Institution]], nonessential job ended as result of stay-at-home orders, etc.)
- ☐ Worked more due to an additional job or increased hours (e.g., went from part-time to full-time employment, got a second job, etc.)
- ☐ Had difficulty accessing food or paying for food
- ☐ Had difficulty finding safe and stable housing arrangements
- ☐ Moved back to [{if USERMODE = CATI} your {else} my] permanent address
- ☐ Moved to another living situation
- ☐ Experienced none of the above

B22FCVPERS *

Thinking only about **July 1, 2020, through June 30, 2021**, please indicate whether or not you experienced any of the following personal impacts as a result of the coronavirus pandemic.

	1 = Yes	0 = No
Children or other dependents did not attend day care or child care (e.g., due to facility being closed, facility was open but not comfortable with health risk, etc.)	☐	☐
Children or other dependents had a non-traditional schedule for school (e.g., online learning, rotating days at school and home, etc.)	☐	☐
Had difficulty paying mortgage or rent	☐	☐
Received emergency financial assistance from [Primary Institution]	☐	☐
Received emergency financial assistance from any source [{if Primary Institution not in (-3 -9)} other than [Primary Institution]]	☐	☐
Accessed food assistance resources (e.g., food pantry, vouchers, etc.)	☐	☐

B22FCVMENTAL *

Please indicate how much you disagree or agree with the following statements. One means “completely disagree” and five means “completely agree.”

As a result of the coronavirus pandemic, between July 1, 2020, and June 30, 2021, I experienced an **increase** in...

	1 (Completely disagree)	2 (Somewhat disagree)	3 (Neither disagree nor agree)	4 (Somewhat agree)	5 (Completely agree)
Stress or anxiety	☐	☐	☐	☐	☐
Difficulty concentrating	☐	☐	☐	☐	☐
Loneliness or isolation	☐	☐	☐	☐	☐
Feeling down, depressed, or hopeless	☐	☐	☐	☐	☐

Appendix D. Contacting Materials for Survey Data Collection

Specific contact materials that have been translated into Spanish for use in Puerto Rico and upon sample member request are denoted by an asterisk (*) on the individual contact materials.

Brochure*

Who is conducting BPS?

The 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) is conducted by the National Center for Education Statistics (NCES), in the U.S. Department of Education's Institute of Education Sciences, with data collection being carried out under contract by RTI International, a U.S.-based nonprofit research organization.

NCES is authorized to conduct BPS by the Education Sciences Reform Act of 2002 (ESRA, 20 U.S.C. §9543) and to collect students' education records from educational agencies or institutions for the purposes of evaluating federally supported education programs under the Family Educational Rights and Privacy Act of 1974 (FERPA, 34 CFR §§ 99.31(a)(3)(iii) and 99.35).

How will my information be protected?

NCES is required to follow strict procedures to protect personal information in the collection, reporting, and publication of data. All of the information provided by individuals or institutions may be used only for statistical purposes and may not be disclosed, or used, in identifiable form for any other purpose except as required by law (20 U.S.C. §9573 and 6 U.S.C §151).

Data security procedures for BPS are reviewed and approved by NCES data security staff. Your answers are secured behind firewalls and are encrypted during internet transmission using Secure Sockets Layer (SSL) protocol. All data entry modules are password protected and require the user to log in before accessing the data. NCES employees and contractors are subject to large fines or imprisonment if individual responses are disclosed.

BPS Help Desk

For assistance, please contact the study Help Desk or visit the website.

800-247-6056

bps@rti.org

<https://surveys.nces.ed.gov/bps/>

If you have questions or concerns about BPS:20/22, contact:

RTI International

Michael Bryan
BPS Project Director at RTI
mbryan@rti.org
800-844-8959

National Center for Education Statistics (NCES)
David Richards, Ph.D.
BPS Study Director at NCES
david.richards@ed.gov
202-245-6202

February 2022



Conducted by:
U.S. Department of Education
Institute of Education Sciences
National Center for Education Statistics
<https://nces.ed.gov>



OMB Control Number: 1850-0631
Learn more about our confidentiality procedures at:
<https://surveys.nces.ed.gov/BPS/confidentiality>

NCES 2022-408



2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22)





What is BPS?

The 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) follows student who started postsecondary education in the 2019-20 academic year. BPS will survey about 37,000 students in 2022 to collect information about their educational progress and plans, experience in the workforce, earnings and expenses, family status, and personal and professional goals. Postsecondary education is any education beyond high school, including career and technical school, college, and graduate school.

Why am I being asked to participate?

You are being asked to participate in BPS because you were enrolled in postsecondary education for the first time during the 2019-20 academic year. Your participation is critical to help policymakers and others better understand the experiences of students who are beginning postsecondary education for the first time. Your personal circumstances are unique, and you cannot be replaced in this study. Most study members were first surveyed in 2020 as part of the National Postsecondary Student Aid Study (NPSAS).

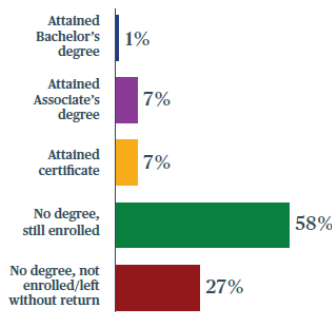
What happens to the results?

Results from the current study will be posted on the NCES website (<https://nces.ed.gov/surveys/bps/>) as soon as they are available. Your responses will be combined with those of other students and will be presented in summary form only. No individually-identifiable information will be published.

Researchers will use BPS data to explore a wide variety of topics, such as how financial aid influences persistence in and completion of postsecondary education.

Did you know?

Of all first-time beginning college students in 2011-12, 15% attained a credential by 2013-14, 58% did not attain a credential but were still enrolled, and 27% did not earn a credential and were not enrolled.



Source: U.S. Department of Education, National Center for Education Statistics, 2012-17 Beginning Postsecondary Students Longitudinal Study (BPS:12/17).

How do I participate?

Participation is easy! You can complete the BPS survey online or over the phone.

Online

Log on to the study website at <https://surveys.nces.ed.gov/bps/> using the Study ID and password provided to you in your study materials.

The survey is mobile-optimized, so it can be easily completed on a mobile device.

Phone

Call our Help Desk at 800-247-6056 to speak to one of our professional interviewers who are happy to assist you.

Additionally, during data collection you may receive reminder communications in any of the following ways:

- Mail from the National Center for Education Statistics
- E-mails from bps@rti.org
- Text messages and phone calls from area code 202 or the U.S. Department of Education



Data Collection Announcement Letter*



National Center for
Education Statistics
550 12th Street SW
Washington, DC 20202
Office: 202-403-5551

March 2022

Jane Doe
123 Main Street
Apt 2
Anywhere, Anystate 11111

Study ID: Your Study ID

Dear Jane,

Congratulations! You have been selected to participate in the 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22), an important U.S. Department of Education study that will collect information about the educational progress, experience in the workforce, earnings and expenses, family status, and personal and professional goals of current and former students who began their postsecondary education during the 2019–20 academic year. You may recall being asked to participate in the National Postsecondary Student Aid Study (NPSAS) in 2020. Data collected from the BPS survey will help researchers, educators, and policymakers gain a better understanding of enrollment in postsecondary education, school expenses and financial aid, employment during and after school, demographics, unique experiences related to the coronavirus pandemic, and living expenses. Your personal circumstances are unique, and you cannot be replaced in this study.

Help shape higher education policy by sharing your education experiences.

Here are some key pieces of information to help get you started:

- Your participation is very important!
- To complete your survey: go to our secure BPS survey website at <https://surveys.nces.ed.gov/bps/>. Log in using your Study ID: Your Study ID and Password: Your Password. Or use the camera on your phone to scan the QR code:
- If you have questions or problems completing your survey online, or prefer to complete your survey over the telephone, call the BPS Help Desk at 800-247-6056 or e-mail us at bps@rti.org.
- Because your participation is important, you may receive reminders via e-mail, mail, phone, and text message.



The survey will take approximately 30 minutes to complete and can be completed on any electronic device, including your mobile device or tablet.

We hope that you enjoy the survey and the opportunity to tell us about your postsecondary education experiences.

Para solicitar materiales de contacto en español en el futuro, por favor llama al 800-247-6056 o envía un correo electrónico a bps@rti.org.

David A. Richards
David Richards, Ph.D.
Study Director, BPS
National Center for Education Statistics
Institute of Education Sciences
U.S. Department of Education
david.richards@ed.gov | 202-245-6202

Michael Bryan
Michael Bryan
Project Director, BPS
RTI International
mbryan@rti.org | 800-844-8959

NCES is authorized to conduct the 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) by the Education Sciences Reform Act of 2002 (ESRA, 20 U.S.C. §9543), and to collect students' education records from educational agencies or institutions for the purpose of evaluating federally supported education programs under the Family Educational Rights and Privacy Act of 1974 (FERPA, 34 CFR §§ 99.31(a)(3)(iii) and 99.35). The data are being collected for NCES by RTI International, a U.S.-based nonprofit research organization.

All of the information you provide may be used only for statistical purposes and may not be disclosed, or used, in identifiable form, for any other purpose except as required by law (20 U.S.C. §9573 and 6 U.S.C. §151).

According to the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number. The valid OMB control number for this voluntary information collection is 1850-0631. The time required to complete this information collection is estimated to average approximately 30 minutes per survey response, including the time to review instructions, gather the data needed, and complete and review the information collection. If you have any comments concerning the accuracy of the time estimate, suggestions for improving this survey, or any comments or concerns regarding the status of your individual submission of this survey, please write directly to: The 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22), National Center for Education Statistics, Potomac Center Plaza, 550 12th St., SW, Room 4037, Washington, DC 20202. OMB Clearance No: 1850-0631 Expiration Date: 02/28/2025.

Data Collection Announcement E-mail*

SUBJECT: U.S. Department of Education's 2020/22 Beginning Postsecondary Students Longitudinal Study



Congratulations, Jane! You have been selected to participate in the 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22), conducted by the U.S. Department of Education's National Center for Education Statistics (NCES). The study will collect information about education progress and plans, experience in the workforce, earnings and expenses, unique experiences related to the coronavirus pandemic, family status, and personal and professional goals from you and others like you who first enrolled in postsecondary education during the 2019–20 school year. **You may recall being asked to participate in the National Postsecondary Student Aid Study (NPSAS) in 2020.** Data collected from the BPS study will help researchers and policymakers better understand the experiences of students who began their postsecondary education during the 2019–20 school year.

You were selected to represent many other students who were not chosen to participate. Therefore, your participation is critical to the success of the study.

Here are some key pieces of information to help get you started:

- Your survey will take about **30 minutes** to complete.
- To automatically start the survey, click the button below:



If you wish to go to our study website, you can log in with your Study ID and password below.

<https://surveys.nces.ed.gov/bps/>

Study ID: Your Study ID

Password: Your Password (*password is case sensitive*)

- **Once you complete the survey, we will send you \$30, payable by PayPal or check.**
- Your participation in this study will not affect any aid or other benefits you receive.
- Because your participation is important, you may receive reminders via e-mail, mail, phone, and text message.

If you have questions about the study or prefer to complete your survey over the phone, please call the BPS Help Desk at **800-247-6056** or e-mail us at bps@rti.org. You can also learn more by visiting the study website at <https://surveys.nces.ed.gov/bps/>.

We hope you enjoy the opportunity to share your experiences and help inform policy decisions.

Many thanks,

David Richards, Ph.D.
Study Director, BPS
National Center for Education Statistics
Institute of Education Sciences
U.S. Department of Education
David.Richards@ed.gov | 202-245-6202

Michael Bryan
Project Director, BPS
RTI International
mbryan@rti.org | 800-844-8959

The National Center for Education Statistics (NCES) is authorized to conduct the 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) by the Education Sciences Reform Act of 2002 (ESRA, 20 U.S.C. §9543), and to collect students' education records from educational agencies or institutions for the purpose of evaluating federally supported education programs under the Family Educational Rights and Privacy Act of 1974 (FERPA, 34 CFR §§ 99.31(a)(3)(iii) and 99.35). The data are being collected for NCES by RTI International, a U.S.-based nonprofit research organization. All of the information you provide may be used only for statistical purposes and may not be disclosed, or used, in identifiable form, for any other purpose except as required by law (20 U.S.C. §9573 and 6 U.S.C. §151).

Por favor responde a este correo electrónico para solicitar materiales en español.

Postcard 1*



You can help shape U.S. higher education policy and help generations of students!

We recently sent you a letter asking you to complete your 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) survey. If you already completed your survey, thank you very much!

If you have not yet completed your survey, please follow these instructions to complete your 30-minute survey:

Go to: <https://surveys.nces.ed.gov/bps/>
Study ID: <<CaseID>>
Password: <<Password>><<PWInd_
stage>><<PWInd_info>>

Or scan this QR code
<<QRCODE>>

Once you complete your survey, we will send you \$<<inc_amount>> to thank you for your time <<Prepaid>>.

If you have any questions, please call 800-247-6056 or e-mail us at bps@rti.org.

Please complete your survey as soon as possible!

Para solicitar materiales de contacto en español en el futuro, por favor llama al 800-247-6056 o envía un correo electrónico a bps@rti.org.

OMB Control Number: 1850-0631

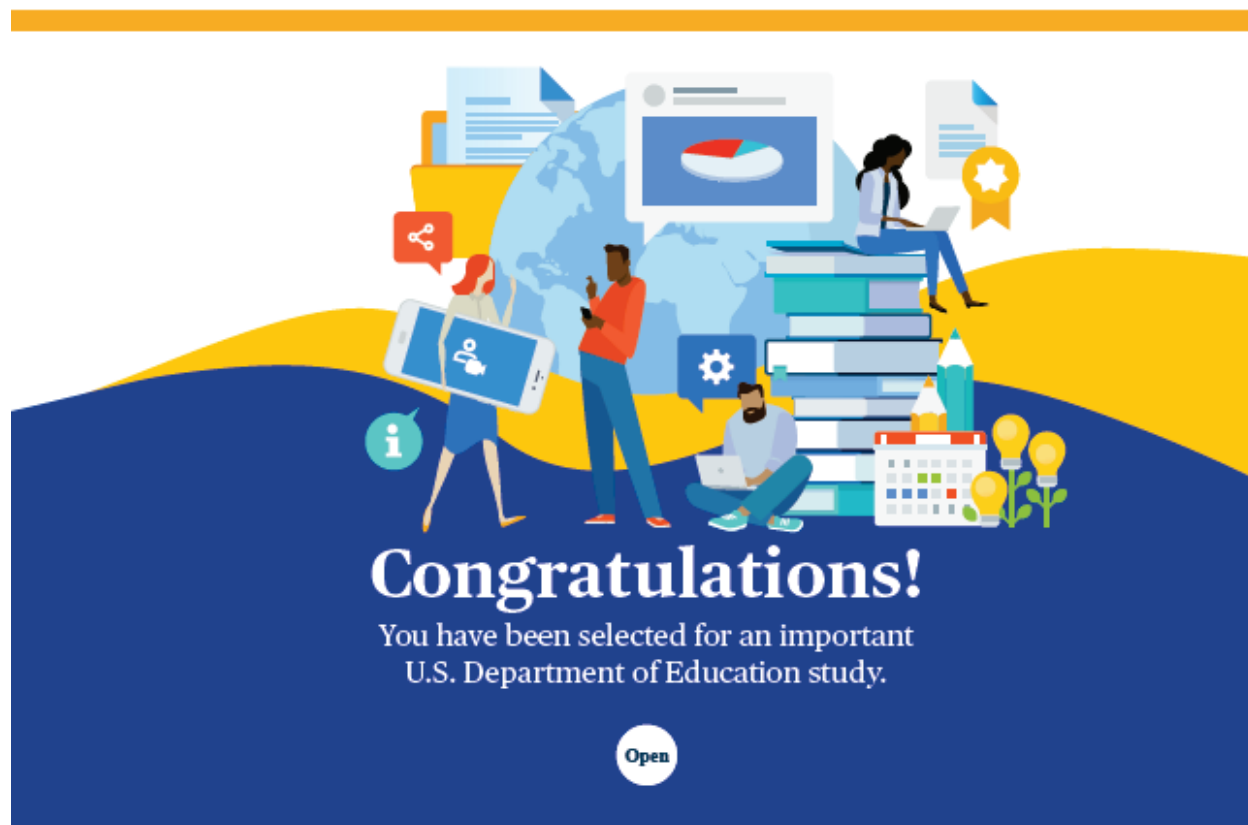
*Learn more about our confidentiality procedures at
<https://surveys.nces.ed.gov/BPS/confidentiality>*



National Center for Education Statistics
U.S. Department of Education
5265 Capital Boulevard (0216549.000.050.200.433)
Raleigh, NC 27616-2925

ADDRESS SERVICE REQUESTED

<<CaseNameNoSuffixALLCAPS>>
<<Addr1>>
<<Addr2>>
<<citystzip>>



Postcard 2*

**<<Fname>>,
help inform policymakers by sharing your education
experiences with the U.S. Department of Education.**

Make a difference for future students by sharing your experiences!

Data from previous Beginning Postsecondary Students Longitudinal Studies (BPS) help policymakers, researchers, and educators better understand the experiences of current and former students who first started their postsecondary education. Your responses to this year's BPS survey, combined with those of other students across the country, will inform education policy aimed at improving education and employment experiences of all students!

Log on to <https://surveys.nces.ed.gov/bps/> with your Study ID and password, or scan the QR code **to complete your 30-minute survey:**

Study ID: <<CaseID>>

<<QRCODE>>

Password: <<Password>><<PWInd_
stage>><<PWInd_info>>

Once you complete your survey, we will send you \$<<inc_amount>>
<<Prepaid>> to thank you for your time.

If you have any questions, please call 800-247-6056 or e-mail us at bps@rti.org.

Para solicitar materiales de contacto en español en el futuro, por favor llama
al 800-247-6056 o envía un correo electrónico a bps@rti.org



Data Collection Announcement Text*

US DEPT OF EDUC: Jane, you have been selected for the U.S. Department of Education's 2020/22 Beginning Postsecondary Students Longitudinal Study. You'll receive \$30 for participating. Click here to take your survey: [bitly link]. Reply STOP to opt out of future text messages. Responde "Español" para solicitar este mensaje en español.

Text Message—Reminder 1*

US DEPT OF EDUC: Jane, your participation in the BPS survey helps inform policy on students beginning their postsecondary education. Click here [bitly link] to begin your 30-minute survey and you'll receive \$30. Reply STOP to opt out of future text messages. Responde "Español" para solicitar este mensaje en español.

Text Message Reminder (Abbreviated)*

US DEPT OF EDUC: Great news, Jane - your BPS survey now only takes 15 minutes and you'll still receive \$40! Click here to begin: [bitly link]. Reply STOP to opt out of future text messages. Responde "Español" para solicitar este mensaje en español.

Text Message—Last Day*

US DEPT OF EDUC: Jane, today is the LAST DAY to complete your shortened 15-minute BPS survey and still receive \$40! Click here to begin: [bitly link]. Reply STOP to opt out of future text messages. Responde "Español" para solicitar este mensaje en español.

Appendix E. Training Materials for Interviewers

Data Collection Interview Training Manual Contents

1.0 BPS

- 1.1. Data Collection Interviewer Training Manual
- 1.2. What Is the 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS)?
 - 1.2.1. Overview of BPS:20/22
 - 1.2.2. How Are Sample Members Selected?
 - 1.2.3. How Will the Surveys Be Conducted?
 - 1.2.4. The Use of Incentives
- 1.3. BPS Security Procedures
- 1.4. Project Staff
- 1.5. Common Acronyms

2.0 GENERAL SURVEYING TECHNIQUES

- 2.1. Overview
- 2.2. Inbound Calls to the Help Desk
- 2.3. Types of Questions You Will Encounter
- 2.4. Responding to a Call
- 2.5. Identity Verification and Password Requests
- 2.6. Obtaining Cooperation—Outbound Calls
- 2.7. Initial Contact
 - 2.7.1. Speaking to a Contact
 - 2.7.2. Speaking to a Sample Member
 - 2.7.3. Encountering Refusals
- 2.8. Best Practices in Conducting the Survey
 - 2.8.1. Use of Active Listening and Professional Interviewing
 - 2.8.2. Conventions for Reading Questions in the BPS Survey
 - 2.8.3. Using Feedback
 - 2.8.4. Recording Responses Accurately
 - 2.8.5. Using Judgment in General Coding

3.0 CASE MANAGEMENT

- 3.1. Pre-CATI Tracing Activities
- 3.2. CATI Locating Procedures
- 3.3. Tracing Procedures
- 3.4. Scheduling a Callback
- 3.5. Voicemail and Answering Machine Message Protocol
- 3.6. Status Codes

4.0 QUALITY CONTROL

- 4.1. Ensuring Quality in the Survey
- 4.2. Training Components
 - 4.2.1. Data Collection Interviewer Training Materials

- 4.2.2. Quality Circle Meetings
- 4.2.3. Continual Training
- 4.2.4. Performance Monitoring: QUEST
- 4.3. Electronic Incident Reports
 - 4.3.1. Types of Incidents You May Encounter
 - 4.3.2. Submitting Electronic Incident Reports from the CATI-CMS
- 4.4. Conclusion

APPENDIXES

- Appendix A Frequently Asked Questions
- Appendix B Sample Members' Rights and Confidentiality
- Appendix C Refusal Aversion and Conversion Guidance
- Appendix D Data Collection Letter
- Appendix E Creating an Incident Report from the CATI-CMS
- Appendix F Event, Line-Level, and Summary Status Code
- Appendix G Survey Reference Materials

Data Collection Interviewer Training Agenda

Wednesday, April 20, 6:00-10:00pm

Welcome and Introductions, Training Objectives, and BPS Study Overview

Introduction to BPS CATI-CMS and CaseTools

Pronunciation Guide and FAQs

Break

BPS Survey Basics (part 1)

Brief break

BPS Survey Basics (part 2)

Help Desk Coverage Overview and Security Protocols

Wrap-up and Answer Questions

Thursday, April 21, 6:00-10:00pm

Training Objectives

CaseTools: Incident Reports and Sending Emails

Survey Coders: Overview

Brief break

Advanced Survey Coders and Coding Exercise

Break

Round Robin Group Survey Mock

Wrap-up and Answer Questions

Friday, April 22, 6:00-10:00pm

Training Objectives

Practice Handling Help Desk Call Scenarios and Refusal Aversion

Break

Paired Survey Mock 1

Paired Survey Mock 2

Wrap-up and Answer Questions

Appendix F. Derived Variables

Table F-1. Analysis variables

Variable name	Subject	Variable label
ID	Survey sample	Analysis ID
ACCREDREG	Institutional characteristics	Regional accreditation agency
ACCREDTYP	Institutional characteristics	Accreditation type
ACDPARTANY22	Education: Experiences	Academic activities: participated in any academic activity, AY 2020–21/2021–22
ACDPARTAPP	Education: Experiences	Academic activities: participated in an apprenticeship, AY 2019–20
ACDPARTAPP22	Education: Experiences	Academic activities: participated in an apprenticeship, AY 2020–21/2021–22
ACDPARTCOM	Education: Experiences	Academic activities: participated in community-based or service learning project, AY 2019–20
ACDPARTCOM22	Education: Experiences	Academic activities: participated in community-based or service learning project, AY 2020–21/2021–22
ACDPARTFY	Education: Experiences	Academic activities: participated in first-year experience or seminar, AY 2019–20
ACDPARTIF22	Education: Experiences	Academic activities: participated in internship or field experiment, AY 2020–21/2021–22
ACDPARTLC	Education: Experiences	Academic activities: participated in learning community, AY 2019–20
ACDPARTLC22	Education: Experiences	Academic activities: participated in learning community, AY 2020–21/2021–22
ACDPARTRSCH	Education: Experiences	Academic activities: participated in guided research experience, AY 2019–20
ACDPARTRSCH22	Education: Experiences	Academic activities: participated in guided research experience, AY 2020–21/2021–22
ACDPARTSE22	Education: Experiences	Academic activities: participated in culminating senior experience, AY 2020–21/2021–22
ACDPARTSSC	Education: Experiences	Academic activities: participated in student success course, AY 2019–20
ACDPARTSSC22	Education: Experiences	Academic activities: participated in student success course, AY 2020–21/2021–22
AGE	Student's characteristics	Age, as of 12/31/2019
AGEGROUP	Student's characteristics	Age groups, as of 12/31/2019
AIDAPP	Financial aid: Application	Applied for any aid
AIDAPP2	Financial aid: Application	Applied for any federal or non-federal aid
AIDCST	Financial aid: Ratios	Ratio of total aid to student budget
AIDSRC	Financial aid: Package	Aid package by source of aid
AIDTYPE	Financial aid: Package	Aid package by type of aid
ALTEARN	Employment	Alternative plans: annual earnings estimate if had not attended college, AY 2019–20
ALTEARN22	Employment	Alternative plans: annual earnings estimate if had not attended college, AY 2021–22
ALTSCHED	Employment	Alternative plans: would have worked full time if had not attended college, AY 2019–20
ALTWORK	Employment	Alternative plans: work status if had not attended college, AY 2019–20
ALTWORK22	Employment	Alternative plans: work status if had not attended college, AY 2021–22
ASIANTYPE	Student's characteristics	Asian type
AT1DT3Y	Education: Attainment	First degree date attained through AY 2021–22
AT1TY3Y	Education: Attainment	First degree type attained through AY 2021–22
ATAADT3Y	Education: Attainment	First associate's degree date attained through AY 2021–22

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
ATAANU3Y	Education: Attainment	Number of associate's degrees attained through AY 2021–22
ATBADT3Y	Education: Attainment	First bachelor's degree date attained through AY 2021–22
ATBANU3Y	Education: Attainment	Number of bachelor's degrees attained through AY 2021–22
ATCTDT3Y	Education: Attainment	First certificate date attained through AY 2021–22
ATCTNU3Y	Education: Attainment	Number of certificates attained through AY 2021–22
ATDEG1	Education: Attainment	Degree attained during AY 2019–20
ATDEG2	Education: Attainment	Degree attained during AY 2020–21
ATDEG3	Education: Attainment	Degree attained during AY 2021–22
ATDEG3Y	Education: Attainment	First year attained degree through AY 2021–22
ATHDT3Y	Education: Attainment	Date attained highest degree at any institution through AY 2021–22
ATHDTF3Y	Education: Attainment	Date attained highest degree at first institution through AY 2021–22
ATHTY3Y	Education: Attainment	Highest degree attained at any institution through AY 2021–22
ATHTYF3Y	Education: Attainment	Highest degree attained at first institution through AY 2021–22
ATLDT3Y	Education: Attainment	Date attained last degree through AY 2021–22
ATLTY3Y	Education: Attainment	Last degree type attained through AY 2021–22
ATNUM3Y	Education: Attainment	Degrees: number attained through AY 2021–22
ATTEND	Education: Attendance	Attendance intensity in fall 2019
ATTEND2	Education: Attendance	Attendance intensity (half-time) in fall 2019
ATTNPT	Education: Attendance	Attendance intensity (half-time), AY 2019–20
ATTNPTRN	Education: Attendance	Attendance intensity (all schools), AY 2019–20
ATTNSTAT	Education: Attendance	Attendance pattern, AY 2019–20
AWAREIDR	Finances	Aware of income-driven student loan repayment plans, AY 2019–20
AWAREIDR22	Finances	Aware of income-driven student loan repayment plans, AY 2020–21/2021–22
AWARELFP	Finances	Aware of student loan forgiveness programs, AY 2019–20
AWARELFP22	Finances	Aware of student loan forgiveness programs, AY 2020–21/2021–22
BANK1	Finances	Bank accounts: had checking or savings account
BUDGETAJ	Institution price	Student budget (attendance adjusted)
BUDNONAJ	Institution: Expenses	Non-tuition expense budget (attendance adjusted)
CAGI	Finances: Income	Adjusted Gross Income (AGI), 2017
CALSYS	Institutional characteristics	Institution calendar system
CAMPAMT	Financial aid: Federal	Federal campus-based aid (SEOG, FWS)
CAMPSAFE	Education: Experiences	Felt safe attending NPSAS, AY 2019–20
CAMPSAFE22	Education: Experiences	Felt safe attending primary institution, AY 2020–21/2021–22
CARAMT22	Finances	Monthly car loan or lease payment, AY 2021–22
CARLN22	Finances	Had a car loan or lease, AY 2021–22
CC2010B	Institution type	Carnegie Classification 2010: basic classification
CC2010C	Institution type	Carnegie Classification 2010: basic classification (collapsed)
CC2015B	Institution type	Carnegie Classification 2015: basic classification
CC2015C	Institution type	Carnegie Classification 2015: basic classification (collapsed)
CC2018B	Institution type	Carnegie Classification 2018: basic classification
CC2018C	Institution type	Carnegie Classification 2018: basic classification (collapsed)
CC2021B	Institution type	Carnegie Classification 2021: basic classification
CC2021C	Institution type	Carnegie Classification 2021: basic classification (collapsed)

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
CC2021S	Institution type	Carnegie Classification 2021: size and setting
CENSUSDIV	Institutional characteristics	Census division
CINCOME	Finances: Income	Total income (continuous), 2017
CINCOME22	Finances: Income	Total income (continuous), 2019
CITIZEN2	Student's characteristics	Citizenship, AY 2019–20
CITZN22	Student's characteristics	Citizenship, AY 2021–22
CLOCK	Institutional characteristics	Clock hour only
CLSEVR3Y	Education: Attendance	Ever took classes only (not a degree program) through AY 2021–22
CNFDIF	Education: Experiences	Academic confidence: change from before PSE to AY 2019–20
CNTLAFFI	Institution type	Institutional control of affiliation
COMPTO87	Survey sample	Comparable to 1987 NPSAS
CONTROL	Institution type	NPSAS institution control
COVATND21	Coronavirus pandemic (COVID-19)	Attended any postsecondary institution during COVID-19, AY 2020–21
COVCHNGINC21	Coronavirus pandemic (COVID-19)	Educational impacts due to COVID-19: increased enrollment from part-time to full-time, AY 2020–21
COVCHNGINS21	Coronavirus pandemic (COVID-19)	Educational impacts due to COVID-19: changed institution where enrolled, AY 2020–21
COVCHNGMAJ21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: changed declared or intended major, AY 2020–21
COVCHNGNENR21	Coronavirus pandemic (COVID-19)	Educational impacts due to COVID-19: did not attend for entire term/semester, AY 2020–21
COVCHNGRED21	Coronavirus pandemic (COVID-19)	Educational impacts due to COVID-19: reduced enrollment from full-time to part-time, AY 2020–21
COVCOMMCWRK	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: helpful communication from NPSAS regarding access to coursework, Spring 2020
COVCOMMEMP	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: helpful communication from NPSAS regarding employment at institution, Spring 2020
COVCOMMFAID	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: helpful communication from NPSAS regarding financial aid, Spring 2020
COVCOMMHOUS	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: helpful communication from NPSAS regarding housing (on campus or college owned), Spring 2020
COVCOMMPROG	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: helpful communication from NPSAS regarding degree program progress, Spring 2020
COVDEVHARD21	Coronavirus pandemic (COVID-19)	Primary institution online coursetaking during COVID-19: device had necessary hardware, AY 2020–21
COVDEVSOFT21	Coronavirus pandemic (COVID-19)	Primary institution online coursetaking during COVID-19: device had necessary software, AY 2020–21
COVDISACCES21	Coronavirus pandemic (COVID-19)	Primary institution provided COVID-19 accommodations: access online course content, AY 2020–21
COVDISCOMP21	Coronavirus pandemic (COVID-19)	Primary institution provided COVID-19 accommodations: complete online coursework, AY 2020–21
COVDISTRACT21	Coronavirus pandemic (COVID-19)	Online coursetaking during COVID-19: difficulties accessing a physical space free of interruption or distraction, AY 2020–21
COVEMRGANY21	Coronavirus pandemic (COVID-19)	Personal impacts due to COVID-19: received emergency financial assistance from any source other than primary institution, AY 2020–21
COVENRDIS	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: enrollment disruptions/changes, Spring 2020
COVEXPABS	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: took leave of absence from NPSAS, Spring 2020

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
COVEXPACTMOD21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: internship, apprenticeship, student teaching, or clinical placement modified or cancelled, AY 2020–21
COVEXPBRK	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: school break extended, Spring 2020
COVEXPCALMOD21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: academic calendar modified, AY 2020–21
COVEXPCAN	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: some or all classes canceled, Spring 2020
COVEXPCC	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: had difficulty finding safe and stable child care, Spring 2020
COVEXPFIN	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: received emergency financial assistance from NPSAS, Spring 2020
COVEXPFOOD	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: had difficulty accessing or paying for food, Spring 2020
COVEXPFOOD21	Coronavirus pandemic (COVID-19)	General experience during COVID-19: had difficulty accessing or paying for food, AY 2020–21
COVEXPGRDMOD21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: modified grading system, AY 2020–21
COVEXPHM	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: had difficulty finding safe and stable housing, Spring 2020
COVEXPHM21	Coronavirus pandemic (COVID-19)	General experience during COVID-19: had difficulty finding safe and stable housing, AY 2020–21
COVEXPHRS21	Coronavirus pandemic (COVID-19)	General experience during COVID-19: worked more due to additional job or increased hours, AY 2020–21
COVEXPHYB21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: some or all classes delivered in hybrid format, AY 2020–21
COVEXPJOB	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: lost job or lost income because of reduced hours, Spring 2020
COVEXPJOB21	Coronavirus pandemic (COVID-19)	General experience during COVID-19: lost job or lost income because of reduced hours, AY 2020–21
COVEXPMMOD21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: class delivery method modified, AY 2020–21
COVEXPMOT	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: moved to another living situation, Spring 2020
COVEXPMOT21	Coronavirus pandemic (COVID-19)	General experience during COVID-19: moved to another living situation, AY 2020–21
COVEXPONL	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: some or all classes moved to online only, Spring 2020
COVEXPONL21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: some or all classes delivered entirely online, AY 2020–21
COVEXPPAD	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: moved back to permanent address, Spring 2020
COVEXPPAD21	Coronavirus pandemic (COVID-19)	General experience during COVID-19: moved back to permanent address, AY 2020–21
COVEXPSAP	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: study-abroad program canceled or modified, Spring 2020
COVEXPWIT	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: withdrew from NPSAS, Spring 2020
COVFINDIS	Coronavirus pandemic (COVID-19)	Financial disruptions/changes due to COVID-19, Spring 2020
COVFINDIS21	Coronavirus pandemic (COVID-19)	Financial disruptions/changes due to COVID-19, AY 2020–21
COVHOUDIS	Coronavirus pandemic (COVID-19)	Housing disruptions/changes due to COVID-19, Spring 2020

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
COVHOUDIS21	Coronavirus pandemic (COVID-19)	Housing disruptions/changes due to COVID-19, AY 2020–21
COVINFOALTLV	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: information provided from NPSAS about finding alternative living arrangements, Spring 2020
COVINFOEAID	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: information provided from NPSAS about accessing emergency financial aid, Spring 2020
COVINFOFAST	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: information provided from NPSAS about access food assistance, Spring 2020
COVINFOHLTH	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: information provided from NPSAS about obtaining physical/mental health care, Spring 2020
COVINFOHLTH21	Coronavirus pandemic (COVID-19)	Primary institution COVID-19 information: obtaining physical or mental healthcare, AY 2020–21
COVINFOMHLTH21	Coronavirus pandemic (COVID-19)	Primary institution COVID-19 information: obtaining mental healthcare, AY 2020–21
COVINFOMOV	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: information provided from NPSAS about moving out of college-owned housing, Spring 2020
COVINFOPHLTH21	Coronavirus pandemic (COVID-19)	Primary institution COVID-19 information: obtaining physical healthcare, AY 2020–21
COVINTACC21	Coronavirus pandemic (COVID-19)	Residence had access to high-speed internet, AY 2020–21
COVINTPROB21	Coronavirus pandemic (COVID-19)	Online coursetaking during COVID-19: difficulties with internet connection, AY 2020–21
COVMNTDIFCON21	Coronavirus pandemic (COVID-19)	Mental impacts due to COVID-19: difficulty concentrating, AY 2020–21
COVMNTDOWN21	Coronavirus pandemic (COVID-19)	Mental impacts due to COVID-19: felt down, depressed, or hopeless, AY 2020–21
COVMNTLONLI21	Coronavirus pandemic (COVID-19)	Mental impacts due to COVID-19: experienced loneliness or isolation, AY 2020–21
COVMNTSTRESS21	Coronavirus pandemic (COVID-19)	Mental impacts due to COVID-19: experienced increased stress or anxiety, AY 2020–21
COVNOTATND21	Coronavirus pandemic (COVID-19)	Decision not to attend school due to COVID-19, AY 2020–21
COVPPSEMGR21	Coronavirus pandemic (COVID-19)	Financial impacts due to COVID-19: received emergency financial assistance from primary institution, AY 2020–21
COVPRSFOOD21	Coronavirus pandemic (COVID-19)	Personal impacts due to COVID-19: accessed food assistance resources, AY 2020–21
COVPRSKCARE21	Coronavirus pandemic (COVID-19)	Personal impacts due to COVID-19: children or other dependents did not attend day care or child care, AY 2020–21
COVPRSKSCH21	Coronavirus pandemic (COVID-19)	Personal impacts due to COVID-19: children or other dependents had a non-traditional schedule for school, AY 2020–21
COVPRSMRT21	Coronavirus pandemic (COVID-19)	Personal impacts due to COVID-19: difficulty paying mortgage or rent, AY 2020–21
COVPSATND21	Coronavirus pandemic (COVID-19)	Attended primary institution during COVID-19, AY 2020–21
COVPYADDF21	Coronavirus pandemic (COVID-19)	Financial impacts due to COVID-19: borrowed more money in federal student loans, AY 2020–21
COVPYADDP21	Coronavirus pandemic (COVID-19)	Financial impacts due to COVID-19: borrowed more money in private student loans, AY 2020–21
COVPYFEWF21	Coronavirus pandemic (COVID-19)	Financial impacts due to COVID-19: borrowed less money in federal student loans, AY 2020–21
COVPYFEWP21	Coronavirus pandemic (COVID-19)	Financial impacts due to COVID-19: borrowed less money in private student loans, AY 2020–21
COVPYLSFAM21	Coronavirus pandemic (COVID-19)	Enrollment payment impacts due to COVID-19: less money from friends and/or family, AY 2020–21

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
COVPYMORFAM21	Coronavirus pandemic (COVID-19)	Enrollment payment impacts due to COVID-19: more money from friends and/or family, AY 2020–21
COVREFTUITION	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: received tuition refund from NPSAS, Spring 2020
COVROOMREF	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: received room-and-board refund from NPSAS, Spring 2020
COVSOCEFF21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: online-only or hybrid classes not as effective, AY 2020–21
COVSOCGRD21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: grades lower than expected, AY 2020–21
COVTECHSER	Coronavirus pandemic (COVID-19)	COVID-19 pandemic: received technology or technical services from NPSAS, Spring 2020
COVTHFEWADV21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: fewer interactions with academic advisors, AY 2020–21
COVTHFEWTECH21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: fewer interactions with teachers, AY 2020–21
COVTHLSCON21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: less connected to other students, AY 2020–21
COVTHLSRES21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: accessed campus learning resources less, AY 2020–21
COVTSUPP21	Coronavirus pandemic (COVID-19)	COVID-19 impacts at primary institution: reasonable teacher expectations, AY 2020–21
CRBALCR	Finances	Credit cards: balance carried over each month, AY 2019–20
CRBALDUE	Finances	Credit cards: balance due, AY 2019–20
CRDBAL22	Finances	Credit cards: balance due, AY 2021–22
CRDNUM22	Finances	Credit cards: number in own name, AY 2021–22
CRDPAY22	Finances	Credit cards: pay off balance, AY 2021–22
CRNUMCRD	Finances	Credit cards: number in own name, AY 2019–20
CRTFNPATT3Y	Education: Transfer	Ever attempt to transfer credits from NPSAS through AY 2021–22
CRTUIT	Finances	Credit cards: used credit cards to pay tuition and fees, AY 2019–20
CRTUIT2	Finances	Credit cards: only source available to pay tuition and fees, AY 2019–20
CSTBKSDG	Finances	Cost of digital textbooks (student reported)
CSTBKSPR	Finances	Cost of print-only textbooks (student reported)
CSTOTHER	Finances	Cost of other required course materials (student reported)
CSTSUPP	Finances	Cost of required supplies (student reported)
CSTTECH	Finances	Cost of required technology (student reported)
CSTTOTAL	Finances	Total cost of required course materials (student reported)
CUMULN22	Financial aid: Borrowed cumulative	Cumulative student loans: total amount borrowed through AY 2021–22
CURCNFDIF	Education: Experiences	Academic confidence: change between AY 2019–20 to AY 2020–21/2021–22
CURCONF	Education: Experiences	Academic confidence after attending school in AY 2019–20
CURCONF22	Education: Experiences	Academic confidence, AY 2020–21/2021–22
DECMAJ	Education: Program	Formally declared major or field of study, AY 2019–20
DECMAJ22	Education: Program	Formally declared major or field of study when last enrolled through AY 2021–22
DEGENRNEXT	Education: Future	Likelihood of enrolling in next semester, AY 2019–20
DEGEVR	Education: Future	Likelihood of ever completing expected degree, AY 2019–20
DEGEXP	Education: Future	Likelihood of completing degree by expected date or within 5 years, AY 2019–20

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
DEGEXPDT	Education: Future	Date expected to complete degree requirements
DEGNUM3Y	Degree programs and goals	Number of degrees pursued through AY 2021–22
DELAYENR	Education: Attendance	Delayed enrollment into PSE: number of years
DEPANY	Student's family	Dependents: had any dependents, AY 2019–20
DEPANY22	Student's family	Had any dependents, AY 2021–22
DEPCARE	Student's family	Dependents: child in paid child care
DEPCHILD	Student's family	Dependents: had dependent children, AY 2019–20
DEPCHILD22	Student's family	Had dependent children, AY 2021–22
DEPCOL	Student's family	Independent students: number of dependents in college, AY 2019–20
DEPCOL22	Student's family	Independent students: number of dependents in college, AY 2021–22
DEPCOLCS	Finances	Amount contributed to college costs for dependents, AY 2019–20
DEPCOST	Student's family	Dependents: children in paid child care, monthly costs
DEPEND	Student's characteristics	Dependency status, AY 2019–20
DEPEND2	Student's characteristics	Dependency status (3 categories), AY 2019–20
DEPEND22	Student's characteristics	Dependency status, AY 2021–22
DEPEND4	Student's characteristics	Dependency status (4 categories), AY 2019–20
DEPEND5A	Student's characteristics	Dependency and marital status (5 categories, separated is married), AY 2019–20
DEPEND5B	Student's characteristics	Dependency and marital status (5 categories, separated is unmarried), AY 2019–20
DEPINC	Finances: Income	Dependent students: parents' income, 2017
DEPINC22	Finances: Income	Dependent students: parents' income, 2019
DEPNUM	Student's family	Dependents: had any dependents (number), AY 2019–20
DEPNUM22	Student's family	Number of dependents, AY 2021–22
DEPNUMCH	Student's family	Dependents: had dependent children (number), AY 2019–20
DEPNUMCH22	Student's family	Number of dependent children, AY 2021–22
DEPNUMOT	Student's family	Dependents: had dependent(s) other than children (number), AY 2019–20
DEPNUMOT22	Student's family	Number of dependents other than children, AY 2021–22
DEPOTCST	Student's family	Dependents: monthly cost of supporting dependent(s) other than children, AY 2019–20
DEPOTHER	Student's family	Dependents: had dependent(s) other than children, AY 2019–20
DEPOTHER22	Student's family	Had dependents other than children, AY 2021–22
DEPTYPE	Student's family	Dependents: types of dependents
DEPYNG	Student's family	Dependents: age of youngest child
DERMAJ	Education: Program	Derived major: 2020 6-digit CIP code
DGEVR22	Degree programs and goals	Highest level of education ever expected through AY 2021–22
DGEXPCHG22	Education: Experiences	Highest level of education ever expected, changes between AY 2019–20 to AY 2021–22
DIS16A	Disabilities	Disability: deaf or serious difficulty hearing, AY 2019–20
DIS16A22	Disabilities	Disability: deaf or serious difficulty hearing, AY 2021–22
DIS16B	Disabilities	Disability: blind or serious difficulty seeing, AY 2019–20
DIS16B22	Disabilities	Disability: blind or serious difficulty seeing, AY 2021–22
DIS17A	Disabilities	Disability: serious difficulty concentrating, remembering, deciding, AY 2019–20
DIS17A22	Disabilities	Disability: serious difficulty concentrating, remembering, deciding, AY 2021–22

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
DIS17B	Disabilities	Disability: serious difficulty walking or climbing stairs, AY 2019–20
DIS17B22	Disabilities	Disability: serious difficulty walking or climbing stairs, AY 2021–22
DISABL22	Disabilities	Disability: had some type of disability, AY 2021–22
DISABLE	Disabilities	Disability: had some type of disability, AY 2019–20
DISTALL	Education: Courses	Distance education: entire program at NPSAS was online, AY 2019–20
DISTALL22	Education: Courses	Distance education: entire program at primary institution was online, AY 2020–21/2021–22
DISTANCE	Institutional characteristics	Distance from student's home (in miles) to NPSAS, AY 2019–20
DISTEDUC	Education: Courses	Distance education: any classes taken at NPSAS taught entirely online, AY 2019–20
DISTEDUC22	Education: Courses	Distance education: any classes taken at primary institution taught entirely online, AY 2020–21/2021–22
DISTNCED	Institutional characteristics	Institution offers instruction exclusively online
DISTYPES	Disabilities	Disability: main type of condition or impairment, AY 2019–20
DISTYPES22	Disabilities	Disability: main type of condition or impairment, AY 2021–22
DIVECON	Education: Experiences	Interactions outside of class: different economic background, AY 2019–20
DIVECON22	Education: Experiences	Interactions outside of class: different economic background, AY 2020–21/2021–22
DIVECONFRQ	Education: Experiences	Frequency of meaningful and honest conversations: different economic background, AY 2019–20
DIVECONFRQ22	Education: Experiences	Frequency of meaningful and honest conversations: different economic background, AY 2020–21/2021–22
DIVGEN22	Education: Experiences	Interactions outside of class: gender minority, AY 2020–21/2021–22
DIVGENFRQ22	Education: Experiences	Frequency of meaningful and honest conversations: gender minority, AY 2020–21/2021–22
DIVPOL	Education: Experiences	Interactions outside of class: different political belief, AY 2019–20
DIVPOL22	Education: Experiences	Interactions outside of class: different political belief, AY 2020–21/2021–22
DIVPOLFRQ	Education: Experiences	Frequency of meaningful and honest conversations: different political belief, AY 2019–20
DIVPOLFRQ22	Education: Experiences	Frequency of meaningful and honest conversations: different political belief, AY 2020–21/2021–22
DIVRACE	Education: Experiences	Interactions outside of class: different race or ethnicity, AY 2019–20
DIVRACE22	Education: Experiences	Interactions outside of class: different race or ethnicity, AY 2020–21/2021–22
DIVRACEFRQ	Education: Experiences	Frequency of meaningful and honest conversations: different race or ethnicity, AY 2019–20
DIVRACEFRQ22	Education: Experiences	Frequency of meaningful and honest conversations: different race or ethnicity, AY 2020–21/2021–22
DIVREL	Education: Experiences	Interactions outside of class: different religious belief, AY 2019–20
DIVREL22	Education: Experiences	Interactions outside of class: different religious belief, AY 2020–21/2021–22
DIVRELFRQ	Education: Experiences	Frequency of meaningful and honest conversations: different religious belief, AY 2019–20
DIVRELFRQ22	Education: Experiences	Frequency of meaningful and honest conversations: different religious belief, AY 2020–21/2021–22
DIVSEX	Education: Experiences	Interactions outside of class: different sexual orientation, AY 2019–20

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
DIVSEX22	Education: Experiences	Interactions outside of class: different sexual orientation, AY 2020–21/2021–22
DIVSEXFRQ	Education: Experiences	Frequency of meaningful and honest conversations: different sexual orientation, AY 2019–20
DIVSEXFRQ22	Education: Experiences	Frequency of meaningful and honest conversations: different sexual orientation, AY 2020–21/2021–22
DODAMT	Financial aid: Other	Department of Defense (military) aid
DSTUINC	Finances: Income	Dependent students: income, 2017
DSTUINC22	Finances: Income	Dependent students: income, 2019
E01	Education: Attendance	Monthly enrollment status (half-time) 2019/07
E02	Education: Attendance	Monthly enrollment status (half-time) 2019/08
E03	Education: Attendance	Monthly enrollment status (half-time) 2019/09
E04	Education: Attendance	Monthly enrollment status (half-time) 2019/10
E05	Education: Attendance	Monthly enrollment status (half-time) 2019/11
E06	Education: Attendance	Monthly enrollment status (half-time) 2019/12
E07	Education: Attendance	Monthly enrollment status (half-time) 2020/01
E08	Education: Attendance	Monthly enrollment status (half-time) 2020/02
E09	Education: Attendance	Monthly enrollment status (half-time) 2020/03
E10	Education: Attendance	Monthly enrollment status (half-time) 2020/04
E11	Education: Attendance	Monthly enrollment status (half-time) 2020/05
E12	Education: Attendance	Monthly enrollment status (half-time) 2020/06
EFC	Financial aid: Need	Expected Family Contribution
EFCAID	Financial aid: Need	Aid subject to federal EFC limitation
EFFORT20	Financial aid: Ratios	Net price after grants and loans, as percent of income
EFFORT3	Financial aid: Ratios	Net price after grants, as percentage of income
EFFORT9	Financial aid: Ratios	Net tuition after all grants, as percentage of income
ELAPSE	Education: Attendance	Number of months between high school completion and PSE entry
EMPLYAM22	Financial aid: Grants	Employer aid (student and parents): amount received, AY 2021–22
EMPLYAM3	Financial aid: Grants	Employer aid (student and parents): amount received, AY 2019–20
EMPMNT3Y	Employment: History	All jobs: number of months worked through AY 2021–22
EMPMTEN3Y	Employment: History	All jobs: number of months worked since last enrolled through AY 2021–22
EMPNUM3Y	Employment: History	All jobs: number of jobs through AY 2021–22
EMRGAPPLY22	Finances	Applied for emergency aid from primary institution, AY 2021–22
EMRGAWARE22	Finances	Aware of emergency aid program at primary institution, AY 2021–22
EMRGRCV22	Finances	Received emergency aid from primary institution, AY 2021–22
ENCOEN3Y	Education: Attendance	Ever co-enrolled through AY 2021–22
ENDTFI3Y	Education: Attendance	First institution date last enrolled through AY 2021–22
ENDTFS	Education: Attendance	Date of first enrollment in AY 2019–20
ENDTLA3Y	Education: Attendance	Date of last enrollment through AY 2021–22
ENINFM	Education: Attendance	First institution: attendance intensity first month enrolled, AY 2019–20
ENINPT1	Education: Attendance	Attendance intensity pattern, AY 2019–20
ENINPT2	Education: Attendance	Attendance intensity pattern, AY 2020–21
ENINPT3	Education: Attendance	Attendance intensity pattern, AY 2021–22
ENINPT3Y	Attendance: Intensity	Attendance intensity through AY 2021–22
ENINUM1	Education: Attendance	Number of institutions attended, AY 2019–20

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
ENINUM2	Education: Attendance	Number of institutions attended, AY 2020–21
ENINUM3	Education: Attendance	Number of institutions attended, AY 2021–22
ENINUM3Y	Education: Attendance	Number of institutions attended (total) through AY 2021–22
ENLEN	Education: Attendance	First institution: months enrolled total, AY 2019–20
ENLYAT3Y	Persistence and attainment	Last year enrolled at any institution with no attainment through AY 2021–22
ENLYF3Y	Persistence and attainment	Last year enrolled at first institution with no attainment through AY 2021–22
ENMNFT1	Education: Attendance	Months enrolled full time, AY 2019–20
ENMNFT2	Education: Attendance	Months enrolled full time, AY 2020–21
ENMNFT3	Education: Attendance	Months enrolled full time, AY 2021–22
ENMNFT3Y	Education: Attendance	Months enrolled full time (total) through AY 2021–22
ENMNPT3Y	Education: Attendance	Months enrolled part time (total) through AY 2021–22
ENMNT1	Education: Attendance	Months enrolled, AY 2019–20
ENMNT2	Education: Attendance	Months enrolled, AY 2020–21
ENMNT22	Education: Attendance	Number of months since last enrolled through AY 2021–22
ENMNT3	Education: Attendance	Months enrolled, AY 2021–22
ENMNT3Y	Education: Attendance	Months enrolled (total) through AY 2021–22
ENMNTAA3Y	Persistence and attainment	Associate's degree: months enrolled before attainment through AY 2021–22
ENMNTBA3Y	Persistence and attainment	Bachelor's degree: months enrolled before attainment through AY 2021–22
ENMNTCT3Y	Persistence and attainment	Certificate: months enrolled before attainment through AY 2021–22
ENR01	Education: Attendance	Monthly enrollment status 2019/07
ENR02	Education: Attendance	Monthly enrollment status 2019/08
ENR03	Education: Attendance	Monthly enrollment status 2019/09
ENR04	Education: Attendance	Monthly enrollment status 2019/10
ENR05	Education: Attendance	Monthly enrollment status 2019/11
ENR06	Education: Attendance	Monthly enrollment status 2019/12
ENR07	Education: Attendance	Monthly enrollment status 2020/01
ENR08	Education: Attendance	Monthly enrollment status 2020/02
ENR09	Education: Attendance	Monthly enrollment status 2020/03
ENR10	Education: Attendance	Monthly enrollment status 2020/04
ENR11	Education: Attendance	Monthly enrollment status 2020/05
ENR12	Education: Attendance	Monthly enrollment status 2020/06
ENRFSIZE	Institutional characteristics	First institution: fall enrollment, AY 2019–20
ENRPLNFT	Education: Attendance	Enrollment plans: first term after AY 2021–22
ENRSPR	Education: Attendance	Enrolled any time between January and June 2020
ENRSTAT	Education: Attendance	First institution: enrollment pattern, AY 2019–20
EVERAA3Y	Education: Program	Ever enrolled in a associate's degree through AY 2021–22
EVERBA3Y	Education: Program	Ever enrolled in a bachelor's degree through AY 2021–22
EVERCT3Y	Education: Program	Ever enrolled in a certificate program through AY 2021–22
EVERDECAA	Education: Majors	Ever declared a major (associate's) through AY 2021–22
EVERDECBA	Education: Majors	Ever declared a major (bachelor's) through AY 2021–22
EVRVT22	Student's characteristics	Ever voted in a national, state, or local election through AY 2021–22
EXOCC2	Employment: Future	Expected job: type of occupation (general), AY 2019–20
EXOCC2Y22	Employment: Future	Expected job: type of occupation (general), AY 2021–22

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
EXOCC3	Employment: Future	Expected job: type of occupation (detailed), AY 2019–20
EXOCC3Y	Employment: History	Expected job: change to type of occupation, AY 2019–20 to AY 2021–22
EXOCC3Y22	Employment: Future	Expected job: type of occupation (detailed), AY 2021–22
EXOCC6	Employment: Future	Expected job: type of occupation (specific), AY 2019–20
EXOCC6Y22	Employment: Future	Expected job: type of occupation (specific), AY 2021–22
EXPBA	Education: Program	Plan to continue to a bachelor's degree program within 5 years, AY 2019–20
EXPBA22	Degree programs and goals	Plan to continue to a bachelor's degree program within 2 years, AY 2021–22
EXPFLD	Employment: Values	Job non-monetary benefits: importance of being an expert, AY 2019–20
EXPFLD22	Employment: Values	Job non-monetary benefits: importance of being an expert, AY 2021–22
EXPWAGE	Employment: Future	Expected job: annual salary expected, AY 2019–20
EXPWAGE22	Employment	Expected job: annual salary expected, AY 2021–22
FACULTYIN	Education: Experiences	Engagement: interactions with faculty during class, AY 2019–20
FACULTYIN22	Education: Experiences	Engagement: interactions with faculty during class, AY 2020–21/2021–22
FACULTYOUT	Education: Experiences	Engagement: interactions with faculty outside of class, AY 2019–20
FACULTYOUT22	Education: Experiences	Engagement: interactions with faculty outside of class, AY 2020–21/2021–22
FAMAMT22	Finances: Support	Help from family and friends: total amount received, AY 2020–21/2021–22
FAMHELP	Finances	Other financial support received
FAMHPAM	Finances	Help from family and friends: total amount received, AY 2019–20
FAMTIM	Employment: Values	Job non-monetary benefits: importance of balancing work and family, AY 2019–20
FAMTIM22	Employment: Values	Job non-monetary benefits: importance of balancing work and family, AY 2021–22
FCONTROL	Institution type	Control of first institution 2019–20
FEDAPP	Financial aid: Application	Applied for federal aid
FEDBEN	Finances: Support	Received federal benefit: any, in 2017 or 2018
FEDBEN22	Finances: Support	Received federal benefit: any, in 2019 or 2020
FEDBEN22A	Finances: Support	Received federal benefit: food stamp benefits, in 2019 or 2020
FEDBEN22B	Finances: Support	Received federal benefit: free or reduced-price lunch benefits, in 2019 or 2020
FEDBEN22C	Finances: Support	Received federal benefit: Supplemental Security Income benefits, in 2019 or 2020
FEDBEN22D	Finances: Support	Received federal benefit: TANF benefits, in 2019 or 2020
FEDBEN22E	Finances: Support	Received federal benefit: WIC benefits, in 2019 or 2020
FEDBENA	Finances: Support	Received federal benefit: food stamp benefits, in 2017 or 2018
FEDBENB	Finances: Support	Received federal benefit: free or reduced-price school lunch benefits, in 2017 or 2018
FEDBENC	Finances: Support	Received federal benefit: Supplemental Security Income benefits, in 2017 or 2018
FEDBEND	Finances: Support	Received federal benefit: TANF benefits, in 2017 or 2018
FEDBENE	Finances: Support	Received federal benefit: WIC benefits, in 2017 or 2018
FEDGRPCT	Financial aid: Ratios	Ratio of federal grants to total aid
FEDLNPAK	Financial aid: Package	Federal loan package by type of loan

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
FEDNEED	Financial aid: Federal	Federal need-based aid
FEDPACK	Financial aid: Package	Federal aid package by type of aid
FEDPCT	Financial aid: Ratios	Ratio of federal aid to total aid
FGRTLN	Financial aid: Federal	Total federal loans and grants
FHSUPP	Socioemotional support	Supportive of PSE: friends from home, AY 2019–20
FHSUPP22	Education: Experiences	Supportive of PSE: friends from home, AY 2020–21/2021–22
FIN2000	Finances	Financial security: come up with \$2,000 within the next month, AY 2019–20
FIN222000	Finances	Financial security: come up with \$2,000 within the next month, AY 2021–22
FIN22500	Finances	Financial security: come up with \$500 within the next month, AY 2021–22
FIN500	Finances	Financial security: come up with \$500 within the next month, AY 2019–20
FINLIT1	Finances	Financial literacy: effect of inflation on purchasing, AY 2019–20
FINLIT2	Finances	Financial literacy: effect of interest on savings, AY 2019–20
FINLIT3	Finances	Financial literacy: effect of diversification on risk, AY 2019–20
FINLITA22	Finances	Financial literacy: effect of inflation on purchasing, AY 2021–22
FINLITALL	Finances	Financial literacy: correctly answered all three items, AY 2019–20
FINLITB22	Finances	Financial literacy: effect of interest on savings, AY 2021–22
FINLITC22	Finances	Financial literacy: effect of diversification on risk, AY 2021–22
FINLITTOT	Finances	Financial literacy: number of correct responses, AY 2019–20
FINLITTOT22	Finances	Financial literacy: number of correct responses, AY 2021–22
FLEVEL	Institution type	Level of first institution 2019–20
FOODSEC	Student's characteristics	Food security status: previous 30 days, AY 2019–20
FOODSEC22	Student's characteristics	Food security status: previous 30 days, AY 2021–22
FRQCOMEXTFAM22	Education: Experiences	Frequency of communication: extended family, AY 2020–21/2021–22
FRQCOMNOCOL	Education: Experiences	Frequency of communication: close friends that did not attend college, AY 2019–20
FRQCOMNOCOL22	Education: Experiences	Frequency of communication: close friends that did not attend college, AY 2020–21/2021–22
FRQCOMNPSAS	Education: Experiences	Frequency of communication: close friends that attended NPSAS, AY 2019–20
FRQCOMNPSAS22	Education: Experiences	Frequency of communication: close friends that attended NPSAS, AY 2020–21/2021–22
FRQCOMOTCOL	Education: Experiences	Frequency of communication: close friends that attended other colleges, AY 2019–20
FRQCOMOTCOL22	Education: Experiences	Frequency of communication: close friends that attended other colleges, AY 2020–21/2021–22
FRQCOMPAR	Education: Experiences	Frequency of communication: parents, AY 2019–20
FRQCOMPAR22	Education: Experiences	Frequency of communication: parents, AY 2020–21/2021–22
FRQCOMSIB22	Education: Experiences	Frequency of communication: siblings, AY 2020–21/2021–22
FRQCOMSIBEXTFAM	Education: Experiences	Frequency of communication: siblings or extended family, AY 2019–20
FRSTCOL	Parents and family	First immediate family member to go to college
FSECDOC	Institution type	Doctorate-granting status of first institution, AY 2019–20
FSECTOR	Institution type	Control and level of first institution (IPEDS sector), AY 2019–20
FSSUPP	Socioemotional support	Supportive of PSE: friends from school, AY 2019–20
FSSUPP22	Education: Experiences	Supportive of PSE: friends from school, AY 2020–21/2021–22

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
GENDER22	Student's characteristics	Gender identity: gender, as of AY 2021–22
GENDER3	Student's characteristics	Gender identity: gender, as of AY 2019–20
GPA	Education: Program	Estimated grade point average, AY 2019–20
GPA22	Education: Program	Grade point average, AY 2020–21/2021–22
GPA3Y	Education: Program	Grade point average through AY 2021–22
GRTCST	Financial aid: Ratios	Ratio of grant aid to student budget
GRTLOAN	Financial aid: Ratios	Ratio of grants to total loans
GRTPCTTN	Financial aid: Ratios	Ratio of grant aid to tuition
HBCU	Institutional characteristics	Historically Black College indicator
HCMATHHI2	High school	Highest level of math completed or planned (dependent students)
HIGHLVEX	Education: Program	Highest level of education ever expected, AY 2019–20
HISPANIC	Student's characteristics	Race: Hispanic or Latino origin
HISPTYPE	Student's characteristics	Hispanic type
HLOFFER	Institutional characteristics	Highest level of offering at NPSAS institution
HLPOTH	Employment: Values	Job non-monetary benefits: importance of helping others, AY 2019–20
HLPOTH22	Employment: Values	Job non-monetary benefits: importance of helping others, AY 2021–22
HOMELESS	Student's characteristics	Homeless or at risk of homelessness, AY 2019–20
HOMELESS2	Student's characteristics	Homeless or at risk of homelessness, AY 2021–22
HOMELESS22	Student's characteristics	Experienced homelessness: previous 30 days, AY 2021–22
HOMELESS3	Student's characteristics	Experienced homelessness: previous 30 days, AY 2019–20
HOMESTUD	Finances	Owned home or paid mortgage, AY 2019–20
HOMESTUD22	Finances	Owned home or paid mortgage, AY 2021–22
HOUSESEC22	Student's characteristics	Experienced housing insecurity: previous 12 months, AY 2021–22
HRSWK20	Employment: While enrolled	Jobs while enrolled: hours worked per week, AY 2019–20
HRSWK21	Employment: While enrolled	Jobs while enrolled: hours worked per week, AY 2020–21
HRSWK22	Employment: While enrolled	Jobs while enrolled: hours worked per week, AY 2021–22
HRSWK3Y	Employment: History	Jobs while enrolled: hours worked per week when last enrolled through AY 2021–22
HSCRDAP	High school	Took AP courses while in high school
HSCRDCOL	High school	Took college-level courses while in high school
HSCRDIB	High school	Took IB courses while in high school
HSDEG	High school	High school degree type
HSGPA2	High school	Grade point average in high school (dependent students)
HSGRADYM	High school	High school completion date (year and month)
HSGRADYY	High school	High school graduation year
HSIZE	Student's family	Family size (dependent and independent), AY 2019–20
HSIZE22	Student's family	Family size (dependent and independent), AY 2021–22
HSTKANY	High school	Took any college credits in high school
HSTYPE	High school	Type of high school last attended
IMMIAGE	Student's characteristics	Age when arrived in the United States
IMMIGEN	Student's characteristics	Immigrant generational status
IMMIGRA	Student's characteristics	Immigrant status
IMPACAD	Education: Services	PSE service importance: academic advising or planning, AY 2019–20
IMPACAD22	Education: Experiences	PSE service importance: academic advising or planning, AY 2020–21/2021–22

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
IMPCAMPSEC	Education: Services	PSE service importance: campus security or campus police, AY 2019–20
IMPCAMPSEC22	Education: Experiences	PSE service importance: campus security or campus police, AY 2020–21/2021–22
IMPCHLD	Education: Services	PSE service importance: child care, AY 2019–20
IMPCHLD22	Education: Experiences	PSE service importance: child care, AY 2020–21/2021–22
IMPCPP	Education: Services	PSE service importance: career services, AY 2019–20
IMPCPP22	Education: Experiences	PSE service importance: career services, AY 2020–21/2021–22
IMPDISSRV	Education: Services	PSE service importance: services for students with disabilities, AY 2019–20
IMPDISSRV22	Education: Experiences	PSE service importance: services for students with disabilities, AY 2020–21/2021–22
IMPDIVRSE	Education: Services	PSE service importance: diversity or multicultural student services, AY 2019–20
IMPDIVRSE22	Education: Experiences	PSE service importance: diversity or multicultural student services, AY 2020–21/2021–22
IMPFINAID	Education: Services	PSE service importance: financial aid counseling or advice, AY 2019–20
IMPFINAID22	Education: Experiences	PSE service importance: financial aid services, AY 2020–21/2021–22
IMPHLTH	Education: Services	PSE service importance: health center and services, AY 2019–20
IMPHLTH22	Education: Experiences	PSE service importance: health center and services, AY 2020–21/2021–22
IMPMENSRV	Education: Services	PSE service importance: psychological or mental health services, AY 2019–20
IMPMENSRV22	Education: Experiences	PSE service importance: psychological or mental health services, AY 2020–21/2021–22
IMPMILSRV	Education: Services	PSE service importance: services for active military or veterans, AY 2019–20
IMPMILSRV22	Education: Experiences	PSE service importance: services for active military or veterans, AY 2020–21/2021–22
IMPTRPLN	Education: Services	PSE service importance: transfer advising or planning, AY 2019–20
IMPTRPLN22	Education: Experiences	PSE service importance: transfer advising or planning, AY 2020–21/2021–22
IMPTUTOR	Education: Services	PSE service importance: peer or other tutoring or skills lab, AY 2019–20
IMPTUTOR22	Education: Experiences	PSE service importance: peer or other tutoring or skills lab, AY 2020–21/2021–22
INATHAMT	Financial aid: Grants	Athletic scholarships
INCGRP	Finances: Income	Income group, AY 2019–20
INCOME	Finances: Income	Total income by dependency (categorical), 2017
INCSPS22	Finances: Income	Independent students: spouse's income, 2019
INDEPINC	Finances: Income	Independent students: student and spouse's income, 2017
INDINC22	Finances: Income	Independent students: student and spouse's income, 2019
INGRTAMT	Financial aid: Grants	Institution grants total
INJURIS	Institution: expenses	Tuition jurisdiction
INLNAMT	Financial aid: Institutional	Institutional loans
INSMERIT	Financial aid: Grants	Institution merit-only grants
INSTAMT	Financial aid: Institutional	Institution aid total
INSTCAT	Institution type	Institutional category

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
INSCAT2	Institution type	Institutional category and control
INSTNEED	Financial aid: Grants	Institutional need-based grants
INSTNOND	Financial aid: Grants	Institution non-need and merit grants
INSTSTAT	Institutional characteristics	NPSAS institution state
INSTWRK	Financial aid: Institutional	Institutional work-study
INSWAIV	Financial aid: Grants	Institution tuition and fee waivers
ISTUINC	Finances: Income	Independent students: income (continuous), 2017
ISTUINC22	Finances: Income	Independent students: income, 2019
IT2YFP3Y	Institution type	Ever attended 2-year for-profit institution through AY 2021–22
IT2YPU3Y	Institution type	Ever attended 2-year public institution through AY 2021–22
IT4Y3Y	Institution type	Ever attended 4-year institution through AY 2021–22
IT4YFP3Y	Institution type	Ever attended 4-year for-profit institution through AY 2021–22
IT4YPN3Y	Institution type	Ever attended 4-year private nonprofit institution through AY 2021–22
IT4YPU3Y	Institution type	Ever attended 4-year public institution through AY 2021–22
ITFP3Y	Institution type	Ever attended for-profit institution through AY 2021–22
ITL2FP3Y	Institution type	Ever attended less than 2-year for-profit institution through AY 2021–22
ITTYLA3Y	Institution type	Institution type last attended through AY 2021–22
JBEN22	Employment: Year 3 Job	Job: eligible for life, health, or retirement benefits, 2022
JBEN22A	Employment: Year 3 Job	Job: eligible for life insurance, 2022
JBEN22B	Employment: Year 3 Job	Job: eligible for health insurance, 2022
JBEN22C	Employment: Year 3 Job	Job: eligible for retirement benefits, 2022
JOBANY	Employment	Job: had job while enrolled (excluding work-study/fellowship), AY 2019–20
JOBEARN	Employment	Job: earnings from work while enrolled (excluding work-study/fellowship), AY 2019–20
JOBEARN2	Employment	Earnings from all work while enrolled, AY 2019–20
JOBENR	Employment	Jobs while enrolled: work intensity (excluding work-study/fellowship), AY 2019–20
JOBFT20	Employment: While enrolled	Jobs while enrolled: work intensity, AY 2019–20
JOBFT21	Employment: While enrolled	Jobs while enrolled: work intensity, AY 2020–21
JOBFT22	Employment: While enrolled	Jobs while enrolled: work intensity, AY 2021–22
JOBHOUR	Employment	Jobs while enrolled: hours worked per week (excluding work-study/fellowship), AY 2019–20
JOBHRS22	Employment: Year 3 Job	Job: hours worked per week, 2022
JOBLFP22	Employment: Not working	Labor force participation status, as of June 2022
JOBLIC22	Employment: Year 3 Job	Job: had industry certification or license, 2022
JOBMAJOR	Employment	Jobs while enrolled: related to major or field of study (excl. work-study/fellowship), AY 2019–20
JOBMNT3Y	Employment: History	All jobs: number of months worked since beginning PSE
JOBOCC22	Employment: Year 3 Job	Job: type of occupation (general), 2022
JOBONOFF	Employment	Jobs while enrolled: on or off campus (excl. work-study/fellowship), AY 2019–20
JOBRC LG22	Employment: Year 3 Job	Job: related to major or field of study, 2022
JOBRLC22	Employment: Year 3 Job	Job: job required industry certification or license, 2022
JOBRLFT22	Employment: Year 3 Job	Job: related to intended future work, 2022
JOBROLE2	Employment: While enrolled	Jobs while enrolled: primary role as student or employee (including work-study/fellowship), AY 2019–20

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
JOBSELF22	Employment: Year 3 Job	Job: self employed, 2022
JOBST22	Employment: Year 3 Job	Job: employment status, 2022
JOBTIM22	Employment: Year 3 Job	Job: years worked in same or similar job, 2022
KNOWCLAS	Education: Courses	Knew requirements needed to complete degree at first institution: AY 2019–20
KNOWCLAS22	Education: Courses	Knew requirements needed to complete pursued degree at primary institution: AY 2020–21/2021–22
LEISTIM22	Employment: Values	Job non-monetary benefits: importance of balancing work and leisure time, AY 2021–22
LEISTIME	Employment: Values	Job non-monetary benefits: importance of balancing work and leisure time, AY 2019–20
LKOCCATHD	Employment: Future	Expected job: likelihood of holding job after highest attainment, AY 2019–20
LKOCCATHD22	Employment: Future	Expected job: likelihood of holding job after highest attainment, AY 2021–22
LNLITTOT22	Finances	Loan literacy: number of correct responses, AY 2021–22
LNREPAY	Financial aid: Loans	Expected help with repaying student loans
LOANCST	Financial aid: Ratios	Ratio of loans to student budget (excludes Direct PLUS to parents)
LOANCST2	Financial aid: Ratios	Ratio of loans to student budget (includes Parent PLUS Loans)
LOANLIT1	Finances	Loan literacy: government can report unpaid debt to credit bureaus, AY 2019–20
LOANLIT2	Finances	Loan literacy: government can garnish wages for unpaid federal loan debt, AY 2019–20
LOANLIT3	Finances	Loan literacy: government can retain tax refunds or Social Security for loan debt, AY 2019–20
LOANLITA22	Finances	Loan literacy: government can report unpaid debt to credit bureaus, AY 2021–22
LOANLITALL	Finances	Loan literacy: correctly answered all three items, AY 2019–20
LOANLITB22	Finances	Loan literacy: government can garnish wages for unpaid federal loan debt, AY 2021–22
LOANLITC22	Finances	Loan literacy: government can retain tax refunds or Social Security for loan debt, AY 2021–22
LOANLITTOT	Finances	Loan literacy: number of correct responses, AY 2019–20
LOANSRC	Financial aid: Package	Loan package by source of loan
LOCALE	Institutional characteristics	Degree of urbanization of NPSAS institution address
LOCALEST	Student's characteristics	Degree of urbanization of student's permanent address
LOCALR22	Student's characteristics	Residence while attending primary institution, AY 2020–21/2021–22
LOCALRES	Student's characteristics	Residence while attending NPSAS, AY 2019–20
LOCJOB20	Employment: While enrolled	Jobs while enrolled: on or off campus, AY 2019–20
LOCJOB21	Employment: While enrolled	Jobs while enrolled: on or off campus, AY 2020–21
LOCJOB22	Employment: While enrolled	Jobs while enrolled: on or off campus, AY 2021–22
LOCJOB3Y	Employment: History	Jobs while enrolled: ever on campus through AY 2021–22
MAJ22	Education: Majors	Major or field of study: last enrolled any degree through AY 2021–22 (23 categories)
MAJ4Y22	Education: Majors	Major or field of study: last enrolled any degree through AY 2021–22 (10 categories)
MAJ4YBA22	Education: Majors	Major or field of study: last enrolled in bachelor's through AY 2021–22 (10 categories)
MAJAA22	Education: Majors	Major or field of study: last enrolled in associate's through AY 2021–22 (23 categories)

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
MAJBA22	Education: Majors	Major or field of study: last enrolled in bachelor's through AY 2021–22 (23 categories)
MAJCHG22	Education: Majors	Major changes: number anywhere through AY 2021–22
MAJCHGAA22	Education: Majors	Major changes (associate's): number anywhere through AY 2021–22
MAJCHGBA22	Education: Majors	Major changes (bachelor's): number anywhere through AY 2021–22
MAJCHGFQ	Education: Program	Major changes: frequency of formally changed through AY 2019–20
MAJCT22	Education: Program	Major or field of study: last enrolled in certificate through AY 2021–22 (23 categories)
MAJORCTE	Education: Program	Major or field of study in career and technical education (CTE) field, AY 2019–20
MAJORNSF	Education: Program	Major or field of study with a focus on NSF STEM fields, AY 2019–20
MAJORS	Education: Program	Major or field of study (detailed), AY 2019–20
MAJORS23	Education: Program	Major or field of study: undergraduate (23 categories), AY 2019–20
MAJORS2Y	Education: Program	Major or field of study: undergraduate (11 categories), AY 2019–20
MAJORS4Y	Education: Program	Major or field of study: undergraduate (10 categories), AY 2019–20
MEALPLAN	Student's characteristics	Meal plan: had meal plan, AY 2019–20
MEALPLAN22	Student's characteristics	Meal plan: had meal plan, AY 2021–22
MEALPLANFQ	Student's characteristics	Meal plan: meal plan covered 11 or more meals per week, AY 2019–20
MEALPLANFQ22	Student's characteristics	Meal plan: meal plan covered 11 or more meals per week, AY 2021–22
MENTH	Characteristics: Disability/Health	Health: general mental health, AY 2019–20
MENTH22	Student's characteristics	Health: general mental health, AY 2021–22
MENTHDIF	Student's characteristics	Health: general mental health, change between AY 2019–20 to 2021–22
MENTMISS	Characteristics: Disability/Health	Health: amount missed school/work in past 30 days for mental health: AY 2019–20
MENTMISS22	Student's characteristics	Health: amount missed school/work in past 30 days for mental health, AY 2021–22
MERITAIID	Financial aid: Total	Total merit-only grants
MILTYP22	Student's characteristics	Military type, AY 2021–22
MILTYPE	Student's characteristics	Military type, AY 2019–20
MILTYPE2	Student's characteristics	Military type (for dependency)
MJNSF22	Education: Majors	Major or field of study with a focus on NSF STEM fields, AY 2021–22
MNTJBNE20	Employment: Not enrolled	Jobs while not enrolled: number of months worked, AY 2019–20
MNTJBNE21	Employment: Not enrolled	Jobs while not enrolled: number of months worked, AY 2020–21
MNTJBNE22	Employment: Not enrolled	Jobs while not enrolled: number of months worked, AY 2021–22
MNTJOB20	Employment: While enrolled	Jobs while enrolled: number of months worked, AY 2019–20
MNTJOB21	Employment: While enrolled	Jobs while enrolled: number of months worked, AY 2020–21
MNTJOB22	Employment: While enrolled	Jobs while enrolled: number of months worked, AY 2021–22
MNTRENT	Finances	Monthly mortgage or rent amount, AY 2019–20
MSISTAT	Institutional characteristics	Minority Serving Institution indicator
MTGAMT22	Finances	Monthly mortgage or rent amount, AY 2021–22
NEEDAID	Financial aid: Total	Total need-based grant aid

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
NETCST1	Financial aid: Net price	Student budget minus all aid
NETCST10	Financial aid: Net price	Tuition and fees minus federal grants
NETCST12	Financial aid: Net price	Tuition and fees minus state grants
NETCST13	Financial aid: Net price	Tuition and fees minus institutional grants
NETCST14	Financial aid: Net price	Tuition and fees minus all non-federal grants
NETCST15	Financial aid: Net price	Tuition and fees minus state and institutional grants
NETCST16	Financial aid: Net price	Student budget minus federal and state grants
NETCST17	Financial aid: Net price	Student budget minus all grants and loans
NETCST20	Financial aid: Net price	Student budget minus all grants and loans (incl Parent PLUS Loans)
NETCST3	Financial aid: Net price	Student budget minus all grants
NETCST9	Financial aid: Net price	Tuition and fees minus all grants
NPATND	Education: Experiences	Would begin education at NPSAS again: AY 2019–20
NPATND22	Education: Experiences	Would begin education at NPSAS again: AY 2021–22
NPFRSTCH	Education: Experiences	NPSAS was first-choice college to attend for AY 2019–20
NUMAPP	Education: Experiences	Number of colleges applied to for AY 2019–20
NUMJBNE20	Employment: Not enrolled	Jobs while not enrolled: number of jobs, AY 2019–20
NUMJBNE21	Employment: Not enrolled	Jobs while not enrolled: number of jobs, AY 2020–21
NUMJBNE22	Employment: Not enrolled	Jobs while not enrolled: number of jobs, AY 2021–22
NUMJOB	Employment	Jobs while enrolled: number of jobs (excluding work-study/fellowship), AY 2019–20
NUMJOB20	Employment: While enrolled	Jobs while enrolled: number of jobs, AY 2019–20
NUMJOB21	Employment: While enrolled	Jobs while enrolled: number of jobs, AY 2020–21
NUMJOB22	Employment: While enrolled	Jobs while enrolled: number of jobs, AY 2021–22
OBereg	Institutional characteristics	NPSAS institution region
ORGMaj22	Education: Majors	Original declared major or field of study at first degree program (23 categories)
ORGMaj4Y	Education: Majors	Original declared major or field of study at first degree program (10 categories)
ORPHAN	Student's characteristics	Orphan, ward of court, emancipated minor, or in legal guardianship (under 24)
OTHGTAMT	Financial aid: Grants	Outside grants (private and employer): amount received, AY 2019–20
OTHGTAMT22	Financial aid: Grants	Outside grants (private and employer): amount received, AY 2021–22
OTHRSCR	Financial aid: Other	Outside sources total
OTHTYPE	Financial aid: Other	Total other type of aid (Direct PLUS, job training, VBA)
PAGI	Finances: Income	Dependent students: parents adjusted gross income, 2017
PARAMT22	Finances: Support	Help from parents: total amount received, AY 2020–21/2021–22
PARBORN	Student's characteristics	Parent born in U.S., PR, or U.S. territory
PAREduc	Parents and family	Parents' highest education level, AY 2019–20
PARHELP	Finances	Help from parents: housing, tuition, and other expenses
PARHELP2	Finances	Help from parents: housing, tuition, and other expenses (incl. independent students)
PARHPAMT	Finances	Help from parents: amount parents helped pay for expenses
PARHPAMT2	Finances	Help from parents: amount parents helped pay for expenses (incl. independent students)
PARSUPP	Socioemotional support	Supportive of PSE: parents, AY 2019–20
PARSUPP22	Education: Experiences	Supportive of PSE: parents, AY 2020–21/2021–22

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
PCTALL	Finances: Income	Income percentile rank
PCTDEP	Finances: Income	Income percentile (dependent respondents)
PCTENR2M	Institutional characteristics	Percent enrolled: two or more races
PCTENRAN	Institutional characteristics	Percent enrolled: American Indian/Alaskan Native
PCTENRAP	Institutional characteristics	Percent enrolled: Asian/Pacific Islander
PCTENRAS	Institutional characteristics	Percent enrolled: Asian
PCTENRBK	Institutional characteristics	Percent enrolled: Black, non-Hispanic
PCTENRHS	Institutional characteristics	Percent enrolled: Hispanic
PCTENRHH	Institutional characteristics	Percent enrolled: Native Hawaiian or other Pacific Islander
PCTENRNR	Institutional characteristics	Percent enrolled: U.S. nonresident
PCTENRUN	Institutional characteristics	Percent enrolled: race/ethnicity unknown
PCTENRWH	Institutional characteristics	Percent enrolled: White, non-Hispanic
PCTINDEP	Finances: Income	Income percentile (independent respondents)
PCTPOV	Finances: Income	Income percent of poverty level, 2017
PDADED	Parents and family	Father's highest education level
PEERINTIN	Education: Experiences	Engagement: interactions with other students during class, AY 2019–20
PEERINTIN22	Education: Experiences	Engagement: interactions with other students during class, AY 2020–21/2021–22
PEERINTOUT	Education: Experiences	Engagement: interactions with other students outside of class, AY 2019–20
PEERINTOUT22	Education: Experiences	Engagement: interactions with other students outside of class, AY 2020–21/2021–22
PELFRSTY	Financial aid: Federal grants	Federal Pell Grant: first academic year received through AY 2021–22
PELL20	Financial aid: Federal grants	Federal Pell Grant: amount received, AY 2019–20
PELL21	Financial aid: Federal grants	Federal Pell Grant: amount received, AY 2020–21
PELL22	Financial aid: Federal grants	Federal Pell Grant: amount received, AY 2021–22
PELLCST	Financial aid: Ratios	Ratio of Pell Grant to student budget
PELLCU22	Financial aid: Federal grants	Federal Pell Grant: cumulative amount received through AY 2021–22
PELLDEP	Financial aid: Federal grants	Pell Grant and dependency, AY 2019–20
PELLPACK	Financial aid: Package	Aid package with Pell grants
PELLRAT1	Financial aid: Ratios	Ratio of Pell Grant to total aid
PELLRAT2	Financial aid: Ratios	Ratio of Pell Grant to total grants
PELLST22	Financial aid: Federal grants	Federal Pell Grant: last academic year received through AY 2021–22
PELYRS22	Financial aid: Federal grants	Federal Pell Grant: number of years received through AY 2021–22
PFAMNM22	Student's family	Dependent students: family size, AY 2021–22
PFAMNUM	Parents and family	Dependent students: family size, AY 2019–20
PHYSH	Characteristics: Disability/Health	Health: general physical health, AY 2019–20
PHYSH22	Student's characteristics	Health: general physical health, AY 2021–22
PHYSHDIF	Student's characteristics	Health: general physical health, change between AY 2019–20 to 2021–22
PHYSMISS	Characteristics: Disability/Health	Health: amount missed school/work in past 30 days for physical health: 2019–20
PHYSMISS22	Student's characteristics	Health: amount missed school/work in past 30 days for physical health, AY 2021–22

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
PINCOL	Parents and family	Dependent students: number of family members in college, AY 2019–20
PINCOL22	Student's family	Dependent students: number of family members in college, AY 2021–22
PLNST22	Financial aid: Private loans	Private student loans: repayment status, AY 2021–22
PLUCUM22	Financial aid: PLUS	Direct PLUS Loans to parents: amount borrowed through AY 2021–22
PLUS20	Financial aid: PLUS	Direct PLUS Loans to parents: amount borrowed, AY 2019–20
PLUSPCT	Financial aid: Ratios	Ratio of Parent PLUS Loans to total aid
PMARIT22	Student's family	Dependent students: parents' marital status, AY 2021–22
PMARITAL	Parents and family	Dependent students: parents' marital status, AY 2019–20
PMOMED	Parents and family	Mother's highest education level
PRAT3Y	Education: Persistence	Persistence anywhere through AY 2021–22
PRATY1	Education: Persistence	Attainment and enrollment, AY 2019–20
PRATY2	Education: Persistence	Attainment and enrollment, AY 2020–21
PRATY3	Education: Persistence	Attainment and enrollment, AY 2021–22
PRECONF	Education: Experiences	Academic confidence prior to attending school in AY 2019–20
PRICUM22	Financial aid: Private loans	Private (alternative) loans: cumulative amount borrowed through AY 2021–22
PRIMLANG	Student's characteristics	First language learned to speak
PRIMLGFAQ	Student's characteristics	Frequency of speaking non-English language with primary caregiver at the start of high school
PRIVAIID	Financial aid: Grants	Private source grants: amount received, AY 2019–20
PRIVAIID22	Financial aid: Grants	Private source grants: amount received, AY 2021–22
PRIVAMT	Financial aid: Other	Total private sources grants and loans, AY 2019–20
PRIVLOAN	Financial aid: Loans	Private (alternative) loans, AY 2019–20
PRIVLRAT	Financial aid: Ratios	Ratio of private loans to total loans
PRIVPCT	Financial aid: Ratios	Ratio of private loans to total aid
PRJBAMT	Employment: Pre-Postsecondary	PSE delayers: wages from all jobs held prior to AY 2019–20
PRLVL3Y	Education: Persistence	Attainment or level of last institution enrolled, as of 2021–22 AY
PROGSTAT	Education: Attainment	Completed degree program, AY 2019–20
PROUT1_NEW	Education: Persistence	Cumulative persistence and attainment anywhere through AY 2019–20 (new version)
PROUT1_OLD	Education: Persistence	Cumulative persistence and attainment anywhere through AY 2019–20 (old version)
PROUT2_NEW	Education: Persistence	Cumulative persistence and attainment anywhere through AY 2020–21 (new version)
PROUT2_OLD	Education: Persistence	Cumulative persistence and attainment anywhere through AY 2020–21 (old version)
PROUT3_NEW	Education: Persistence	Cumulative persistence and attainment anywhere through AY 2021–22 (new version)
PROUT3_OLD	Education: Persistence	Cumulative persistence and attainment anywhere through AY 2021–22 (old version)
PROUTF1_NEW	Education: Persistence	Cumulative retention and attainment at first institution through AY 2019–20 (new version)
PROUTF1_OLD	Education: Persistence	Cumulative retention and attainment at first institution through AY 2019–20 (old version)
PROUTF2_NEW	Education: Persistence	Cumulative retention and attainment at first institution through AY 2020–21 (new version)

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
PROUTF2_OLD	Education: Persistence	Cumulative retention and attainment at first institution through AY 2020–21 (old version)
PROUTF3_NEW	Education: Persistence	Cumulative retention and attainment at first institution through AY 2021–22 (new version)
PROUTF3_OLD	Education: Persistence	Cumulative retention and attainment at first institution through AY 2021–22 (old version)
PRRPY22	Financial aid: Debt and repayment	Private student loans: monthly payment amount, AY 2021–22
PRVHRSWK	Employment: Pre-Postsecondary	PSE delayers: hours worked per week at all jobs held prior to AY 2019–20
PRVJOB	Employment: Pre-Postsecondary	PSE delayers: number of jobs for pay prior to AY 2019–20
PRVMTHWK	Employment: Pre-Postsecondary	PSE delayers: months worked in year prior to AY 2019–20
RAASIAN	Student's characteristics	Race: Asian
RABLACK	Student's characteristics	Race: Black or African American
RACE	Student's characteristics	Race/ethnicity (with multiple)
RACE2	Student's characteristics	Race/ethnicity (with multiple) without foreign students
RACECEN	Student's characteristics	Race/ethnicity Census categories
RACESEX	Student's characteristics	Race/ethnicity (with multiple) and gender
RAINDIAN	Student's characteristics	Race: American Indian or Alaska Native
RAINDTRB	Student's characteristics	Race: American Indian or Alaska Native recognized tribe
RAISLAND	Student's characteristics	Race: Native Hawaiian or other Pacific Islander
RAWHITE	Student's characteristics	Race: White
RCNTGRAD	High school	Recent (2019) high school graduate
RCVSTAD	Financial aid: State	Received financial aid from a state aid program, AY 2019–20
RCVSTAD21	Financial aid: State	Received financial aid from a state aid program, AY 2019–20
RCVSTAD22	Financial aid: State	Received financial aid from a state aid program, AY 2021–2022
RCVSTADADM	Financial aid: State	Received financial aid from a state aid program, AY 2019–20 (administrative data)
REANOAPA	Financial aid: Application	Reason for not applying for aid: did not want to take on the debt
REANOAPB	Financial aid: Application	Reason for not applying for aid: forms were too much work
REANOAPC	Financial aid: Application	Reason for not applying for aid: no information about how to apply
REANOAPD	Financial aid: Application	Reason for not applying for aid: no need
REANOAPE	Financial aid: Application	Reason for not applying for aid: thought ineligible
REANOAPF	Financial aid: Application	Reason for not applying for aid: other
REGSUPP	Finances	Provided regular financial support to family or friend, AY 2019–20
REGSUPP22	Finances	Provided regular financial support to family or friend, AY 2020–21/2021–22
REGVT22	Student's characteristics	Registered to vote in U.S. elections, AY 2021–22
REMETOOK	Education: Courses	Remedial courses: took any in AY 2019–20
REMEV3Y	Education: Courses	Ever taken remedial courses through AY 2021–22
REMEVER	Education: Courses	Remedial courses: ever taken, as of AY 2019–20
REMMATH	Education: Courses	Remedial courses: number taken in math, AY 2019–20
REMRW	Education: Courses	Remedial courses: number taken in reading/writing, AY 2019–20
REMSTAT	Education: Courses	Remedial courses: year taken and subjects taken, AY 2019–20
RISKIND2	Student's characteristics	Index of risk and nontraditional students, modified
RISKINDX	Student's characteristics	Index of risk and nontraditional students
SAGI	Finances: Income	Independent students: adjusted gross income, 2017

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
SALARY22	Employment: Year 3 Job	Job: annual salary, 2022
SAMESTAT	Student's characteristics	Attended institution in state of legal residence, AY 2019–20
SECTOR10	Institution type	NPSAS institution sector - 10 categories
SECTOR11	Institution type	NPSAS institution sector - 11 categories
SEEND3Y	Education: Attendance	First enrollment spell: date ended through AY 2021–22
SEINT3Y	Education: Attendance	First enrollment spell: intensity through AY 2021–22
SELECT3Y	Institution characteristics	Selectivity of last institution enrolled through AY 2021–22
SELECTV2	Institutional characteristics	Selectivity (4-year nonprofit institutions)
SELECTV3	Institutional characteristics	Selectivity (all 4-year institutions)
SEMON3Y	Education: Attendance	First enrollment spell: months enrolled through AY 2021–22
SENSBELNG	Education: Experiences	Engagement: felt like a part of institution, AY 2019–20
SENSBELNG22	Education: Experiences	Engagement: felt like a part of institution, AY 2020–21/2021–22
SENUM3Y	Education: Attendance	Enrollment spells: number of spells through AY 2021–22
SEOGAMT	Financial aid: Grants	Federal Supplemental Educational Opportunity Grant (SEOG)
SEROLE22	Employment: History	Jobs while enrolled: primary role as student or employee, AY 2021–22
SEXATBIRTH	Student's characteristics	Sex assigned at birth
SEXORIENTAT	Student's characteristics	Sexual orientation, as of AY 2019–20
SEXORIENTAT22	Student's characteristics	Sexual orientation, as of AY 2021–22
SFAMNM22	Student's family	Independent students: family size, AY 2021–22
SFAMNUM	Student's family	Independent students: family size, AY 2019–20
SIBCOLFT	Parents and family	First sibling to go to college
SIBINCOL	Parents and family	Dependent students: have siblings in college
SIEND3Y	Education: Attendance	First institution spells: date first enrollment spell ended through AY 2021–22
SIINT3Y	Education: Attendance	First institution spells: intensity through AY 2021–22
SINCOL	Student's family	Independent students: number of family members in college, AY 2019–20
SINCOL22	Student's family	Independent students: number of family members in college, AY 2021–22
SINGLPAR	Student's characteristics	Single parent, independent student
SJEARN	Employment	Work-study job: total earnings, AY 2019–20
SJHOURS	Employment	Work-study job: hours worked per week, AY 2019–20
SJWKST	Employment	Work-study job: had work-study job, AY 2019–20
SMAR22	Student's family	Marital status, AY 2021–22
SMARITAL	Student's characteristics	Marital status, AY 2019–20
SNEED1	Financial aid: Need	Student budget minus EFC
SNEED2	Financial aid: Need	Student budget minus EFC minus total aid
SNEED4	Financial aid: Need	Student budget minus EFC and aid subject to EFC limitation
SNEED5	Financial aid: Need	Student budget minus EFC minus all grants
SNEED8	Financial aid: Need	Student budget minus EFC and federal, state, and other grants
SNEED9	Financial aid: Need	Student budget minus EFC and all aid except private loans
SOCPARTANY22	Education: Experiences	Social activities: participated in any social activity, AY 2020–21/2021–22
SOCPARTGRK	Education: Experiences	Social activities: participated in a Greek letter organization, AY 2019–20
SOCPARTGRK22	Education: Experiences	Social activities: participated in a Greek letter organization, AY 2020–21/2021–22

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
SOCPARTMUS	Education: Experiences	Social activities: participated in institution-sponsored music, theater, or band, AY 2019–20
SOCPARTMUS22	Education: Experiences	Social activities: participated in institution-sponsored music, theater, or band, AY 2020–21/2021–22
SOCPARTNVAR	Education: Experiences	Social activities: participated in non-varsity athletics, AY 2019–20
SOCPARTNVAR22	Education: Experiences	Social activities: participated in non-varsity athletics, AY 2020–21/2021–22
SOCPARTSTGOV	Education: Experiences	Social activities: participated in student government, AY 2019–20
SOCPARTSTGOV22	Education: Experiences	Social activities: participated in student government, AY 2020–21/2021–22
SOCPARTSTGRP	Education: Experiences	Social activities: participated in student affinity groups, AY 2019–20
SOCPARTSTGRP22	Education: Experiences	Social activities: participated in student affinity groups, AY 2020–21/2021–22
SOCPARTSTORG	Education: Experiences	Social activities: participated in other recognized or registered student organizations, AY 2019–20
SOCPARTSTORG22	Education: Experiences	Social activities: participated in any other recognized or registered student organizations, AY 2020–21/2021–22
SOCPARTVAR	Education: Experiences	Social activities: participated in varsity athletics, AY 2019–20
SOCPARTVAR22	Education: Experiences	Social activities: participated in varsity athletics, AY 2020–21/2021–22
SPINCOL	Student's family	Independent students: spouse attending college
SPNM21	Student's family	Not married to spouse in 2021
SPPSUPP	Socioemotional support	Supportive of PSE: spouse or partner, AY 2019–20
SPPSUPP22	Education: Experiences	Supportive of PSE: spouse or partner, AY 2020–21/2021–22
SPSED22	Student's family	Independent students: spouse's highest education level, AY 2021–22
SPSINC	Finances: Income	Independent students: spouse's income, 2017
STABEVR	Education: Experiences	Ever studied abroad during undergraduate education through 2019–20
STAFCT1	Financial aid: Federal loans	Direct Subsidized Loan maximum
STAFCT2	Financial aid: Federal loans	Direct Subsidized and Unsubsidized Loan total maximum
STAFSUB	Financial aid: Federal loans	Direct Subsidized Loans
STATYPE	Financial aid: Federal loans	Direct Subsidized and Unsubsidized Loan types received
STAFUNSB	Financial aid: Federal loans	Direct Unsubsidized Loans, AY 2019–20
STATEAMT	Financial aid: State	State aid total
STATERP	State characteristics	State representation indicator
STDT3Y	Education: Stopout	First stopout through AY 2021–22: start date, at any institution
STDTFI3Y	Education: Stopout	First stopout through AY 2021–22: start date, at first institution
STDYABR3Y	Education: Experiences	Ever studied abroad while attending NPSAS through AY 2021–22
STEMMAJ	Education: Program	Major or field of study with a focus on STEM fields, AY 2019–20
STFCUM22	Financial aid: Direct Loans	Direct Sub. and Unsub. Loans: cumulative amt borrowed through AY 2021–22
STFIRSTY	Financial aid: Direct Loans	Direct Subsidized and Unsubsidized Loans: first academic year borrowed
STFY20	Financial aid: Direct Loans	Direct Subsidized and Unsubsidized Loans: total borrowed, AY 2019–20
STFY21	Financial aid: Direct Loans	Direct Subsidized and Unsubsidized Loans: total borrowed, AY 2020–21
STFY22	Financial aid: Direct Loans	Direct Subsidized and Unsubsidized Loans: total borrowed, AY 2021–22

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
STFYRS22	Financial aid: Direct Loans	Direct Sub. or Unsub. Loans: number of years received through AY 2021–22
STGTAMT	Financial aid: State	State grants total
STLNAMT	Financial aid: State	State loans total
STMERIT	Financial aid: State	State merit-only grants total
STNDONLY	Financial aid: State	State need-based only grants total
STNUM3Y	Education: Stopout	Stopouts through AY 2021–22: total
STSCUM22	Financial aid: Direct Loans	Direct Subsidized Loans: cumulative amount borrowed through AY 2021–22
STSTA1	Education: Stopout	Stopout status: AY 2019–20
STSTA2	Education: Stopout	Stopout status: AY 2020–21
STSTA3	Education: Stopout	Stopout status: AY 2021–22
STSUB20	Financial aid: Direct Loans	Direct Subsidized Loans: total borrowed, AY 2019–20
STSUB21	Financial aid: Direct Loans	Direct Subsidized Loans: total borrowed, AY 2020–21
STSUB22	Financial aid: Direct Loans	Direct Subsidized Loans: total borrowed, AY 2021–22
STUCUM22	Financial aid: Direct Loans	Direct Unsubsidized Loans: cumulative amount borrowed through AY 2021–22
STUDMULT	Education: Attendance	Number of institutions attended, AY 2019–20
STUSERV3	Institutional characteristics	NPSAS offered on-campus child care
STUSTATE	Student's characteristics	State of legal residence
STYB3Y	Education: Stopout	First stopout through AY 2021–22: starting academic year, at any institution
STYBF13Y	Education: Stopout	First stopout through AY 2021–22: starting academic year, at first institution
T4LNAMT2	Financial aid: Federal loans	Title IV loans (including Parent PLUS Loans): total borrowed, AY 2019–20
T4TCUM22	Financial aid: Borrowed cumulative	Title IV loans (including Parent PLUS Loans): cumulative borrowed through AY 2021–22
T4TDUE22	Financial aid: Borrowed cumulative	Title IV loans (including Parent PLUS Loans): amount owed in interest and principal through AY 2021–22
T4XDUE22	Financial aid: Federal loans	Title IV loans (excluding Parent PLUS Loans): amount owed in interest and principal through AY 2021–22
TCREFFCOM	Education: Experiences	Teacher effectiveness: had good command of subject, AY 2019–20
TCREFFCOM22	Teacher evaluation	Teacher effectiveness: had good command of subject, AY 2020–21/2021–22
TCREFFEX22	Teacher evaluation	Teacher effectiveness: made good use of examples, AY 2020–21/2021–22
TCREFFINT	Education: Experiences	Teacher effectiveness: genuinely interested in teaching, AY 2019–20
TCREFFINT22	Teacher evaluation	Teacher effectiveness: genuinely interested in teaching, AY 2020–21/2021–22
TCREFFOUT	Education: Experiences	Teacher effectiveness: outstanding teachers, AY 2019–20
TCREFFOUT22	Teacher evaluation	Teacher effectiveness: outstanding teachers, AY 2020–21/2021–22
TEACTDER	Student's characteristics	ACT derived composite score
TESATDER	Student's characteristics	SAT derived composite score
TETOOK	Student's characteristics	Took SAT or ACT exams
TFATT3Y	Education: Transfer	Transfer (first): attainment before transfer through AY 2021–22
TFDEG3Y	Education: Transfer	Transfer (first): degree program through AY 2021–22
TFEDAID	Financial aid: Federal	Total federal aid (excludes veterans/DOD)

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
TFEDAID2	Financial aid: Federal	Total federal aid (includes veterans/DOD)
TFEDAID6	Financial aid: Federal	Total federal aid (excludes Parent PLUS Loans and veterans/DOD)
TFEDGRT	Financial aid: Federal grants	Total federal grants (excludes veterans/DOD)
TFEDGRT2	Financial aid: Federal grants	Total federal grants and veterans/DOD
TFEDLN	Financial aid: Federal loans	Total federal loans (excludes Parent PLUS Loans)
TFEDLN2	Financial aid: Federal loans	Total federal loans (includes Parent PLUS Loans)
TFEDWRK	Financial aid: Federal	Federal Work-Study
TFIFTY3Y	Education: Transfer	Transfer (first): destination institution type through AY 2021–22
TFINCT3Y	Education: Transfer	Transfer direction (first): institution control through AY 2021–22
TFINLV3Y	Education: Transfer	Transfer direction (first): institution level through AY 2021–22
TFMNDI3Y	Education: Transfer	Transfer (first): months elapsed before destination institution through AY 2021–22
TFMNT3Y	Education: Transfer	Transfer (first): months enrolled before destination institution through AY 2021–22
TFNUM3Y	Education: Transfer	Transfer status through AY 2021–22
TFTYPE1	Education: Transfer	Transfer status, AY 2019–20
TFTYPE2	Education: Transfer	Transfer status, AY 2020–21
TFTYPE3	Education: Transfer	Transfer status, AY 2021–22
TFTYPE3Y	Education: Transfer	Transfer direction (first): type through AY 2021–22
TGRTLN	Financial aid: Total	Total loans and grants
TITIVAMT	Financial aid: Federal	Total federal Title IV aid
TNFEDAID	Financial aid: Total	Total non-federal aid
TNFEDGRT	Financial aid: Total	Total non-federal grants
TNFEDLN	Financial aid: Total	Total non-federal loans
TOTAID	Financial aid: Total	Total aid amount
TOTAID2	Financial aid: Total	Total federal (Title IV), state, and institutional aid
TOTAID4	Financial aid: Total	Total aid excluding Parent PLUS Loans
TOTGRT	Financial aid: Total	Total grants
TOTGRT2	Financial aid: Total	Total grants and veterans/DOD
TOTGRT4	Financial aid: Total	Total state and institutional grants
TOTLOAN	Financial aid: Total	Total loans (excluding Parent PLUS Loans)
TOTLOAN2	Financial aid: Total	Total loans (including Parent PLUS Loans)
TOTLOAN3	Financial aid: Total	Total loans excluding private loans
TOTWKST	Financial aid: Total	Total work-study
TRANSGENDER	Student's characteristics	Gender identity: transgender, as of AY 2019–20
TRANSGENDER22	Student's characteristics	Gender identity: transgender, as of AY 2021–22
TRNPLN	Education: Transfer	Plans to transfer, AY 2019–20
TRNSPEC	Education: Experiences	Plans to transfer to a specific school, AY 2019–20
TUITION2	Institution price	Tuition and fees paid
UGDEG	Education: Program	Degree program, AY 2019–20
UGDEG22	Education: Program	Degree program when last enrolled through AY 2021–22
UGLVL22	Education: Program	Highest undergraduate class level through AY 2021–22
UNEMPL22	Employment: Not working	Unemployment: currently looking for a job, 2022
UNSBLOAN	Financial aid: Total	Total unsubsidized loans (all sources)
USBORN	Student's characteristics	Born in the U.S. (student)
USEACAD	Education: Services	PSE services used: academic advising, AY 2019–20

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
USEACAD22	Education: Experiences	PSE services used: academic advising, AY 2020–21/2021–22
USEANY	Education: Services	PSE services used: any service in 2019–20
USEANY22	Education: Experiences	PSE services used: used any service, AY 2020–21/2021–22
USECAMPSEC	Education: Services	PSE services used: campus security or campus police, AY 2019–20
USECAMPSEC22	Education: Experiences	PSE services used: campus security or campus police, AY 2020–21/2021–22
USECHLD	Education: Services	PSE services used: child care, AY 2019–20
USECHLD22	Education: Experiences	PSE services used: child care, AY 2020–21/2021–22
USECPP	Education: Services	PSE services used: career services, AY 2019–20
USECPP22	Education: Experiences	PSE services used: career services, AY 2020–21/2021–22
USEDISSRV	Education: Services	PSE services used: services for students with disabilities, AY 2019–20
USEDISSRV22	Education: Experiences	PSE services used: services for students with disabilities, AY 2020–21/2021–22
USEDIVRSE	Education: Services	PSE services used: diversity or multicultural student services, AY 2019–20
USEDIVRSE22	Education: Experiences	PSE services used: diversity or multicultural student services, AY 2020–21/2021–22
USEFINAID	Education: Services	PSE services used: financial aid counseling or advice, AY 2019–20
USEFINAID22	Education: Experiences	PSE services used: financial aid counseling or advice, AY 2020–21/2021–22
USEHLTH	Education: Services	PSE services used: health center and services, AY 2019–20
USEHLTH22	Education: Experiences	PSE services used: health center and services, AY 2020–21/2021–22
USEIDR	Finances	Likelihood of using income-driven student loan repayment plans, AY 2019–20
USEIDR22	Finances	Likelihood of using income-driven student loan repayment plans, AY 2021–22
USELFP	Finances	Likelihood of using loan forgiveness programs, AY 2019–20
USELFP22	Finances	Likelihood of using loan forgiveness programs, AY 2021–22
USEMENSRV	Education: Services	PSE services used: psychological or mental health services, AY 2019–20
USEMENSRV22	Education: Experiences	PSE services used: psychological or mental health services, AY 2020–21/2021–22
USEMILSRV	Education: Services	PSE services used: services for active military or veterans, AY 2019–20
USEMILSRV22	Education: Experiences	PSE services used: services for active military or veterans, AY 2020–21/2021–22
USETRPLN	Education: Services	PSE services used: transfer advising or planning, AY 2019–20
USETRPLN22	Education: Experiences	PSE services used: transfer advising or planning, AY 2020–21/2021–22
USETUTOR	Education: Services	PSE services used: peer or other tutoring or skills lab, AY 2019–20
USETUTOR22	Education: Experiences	PSE services used: peer or other tutoring or skills lab, AY 2020–21/2021–22
VADODAMT	Financial aid: Other	Federal veterans benefits and Department of Defense, AY 2019–20
VETBEN	Financial aid: Other	Federal veterans education benefits, AY 2019–20
VETBEN22	Financial aid: Other	Veterans/DOD education benefits: amount received, AY 2021–22
VETERAN	Student's characteristics	Veteran status
VOCHELP	Financial aid: State	Vocational rehabilitation and training

See notes at end of table.

Table F-1. Analysis variables—Continued

Variable name	Subject	Variable label
VTNEL22	Student's characteristics	Voted in 2020 presidential election
WORKPCT	Financial aid: Ratios	Ratio of work-study to total aid
WRKSTDY3Y	Employment: While enrolled	Work-study job: had work-study job through AY 2021–22
WRKTSKS	Employment: Values	Job non-monetary benefits: importance of making decisions, AY 2019–20
WRKTSKS22	Employment: Values	Job non-monetary benefits: importance of making decisions, AY 2021–22
WTA000	Survey weights	BPS:20/22 response analysis weight
WTA001-WTA200	Survey weights	BPS:20/22 bootstrap replicate weight X for BPS:20/22 cross-sectional respondents

NOTE: AP = Advanced Placement; AY = academic year; CIP = Classification of Instructional Programs; DOD = Department of Defense; EFC = expected family contribution; FWS = Federal Work-Study; IB = International Baccalaureate; IPEDS = Integrated Postsecondary Education Data System; NPSAS = National Postsecondary Student Aid Study; NSF = National Science Foundation; PR = Puerto Rico; PSE = postsecondary education; SEOG = Supplemental Educational Opportunity Grant; STEM = science, technology, engineering, and mathematics; TANF = Temporary Assistance for Needy Families; VBA = Veterans Benefits Administration; WIC = Special Supplemental Nutrition Program for Women, Infants, and Children.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Appendix G. Nonresponse Bias Analysis

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Table G-1. Sample member survey nonresponse bias analysis for selected variables—All institutions: 2022

Variable	Sample counts		Before nonresponse weight adjustments				After nonresponse weight adjustment		
	Respondents, unweighted	Non-respondents, unweighted	Full-sample mean, base weighted	Respondent mean, base weighted	Estimated bias ¹	Relative bias ²	Respondent mean, nonresponse adjusted	Estimated bias ³	Relative bias ²
Control and level of institution						(Effect size = 0.09)		(Effect size = #)	
Public less-than-2-year	180	200	0.47	0.35	-0.12*	-24.72	0.47	#	#
Public 2-year	6,730	4,260	31.02	28.26	-2.76*	-8.91	31.02	#	#
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	1,530	1,000	7.80	6.69	-1.11*	-14.18	7.80	#	#
Public 4-year, non-doctorate-granting, primarily baccalaureate	1,560	650	5.50	6.06	0.55*	10.08	5.50	#	#
Public 4-year, doctorate-granting	4,930	2,290	30.25	31.96	1.71*	5.67	30.25	#	#
Private nonprofit less-than-4-year	130	90	0.23	0.23	#	-0.61	0.23	#	#
Private nonprofit 4-year, non-doctorate-granting	1,900	740	7.82	8.76	0.94*	12.01	7.82	#	#
Private nonprofit 4-year, doctorate-granting	2,440	1,040	12.68	13.58	0.89*	7.04	12.68	#	#
Private for-profit less-than-2-year	580	340	1.41	1.40	-0.01	-0.60	1.41	#	#
Private for-profit 2-year	880	550	0.92	0.91	-0.01	-1.51	0.92	#	#
Private for-profit 4-year	1,460	760	1.90	1.81	-0.09	-4.73	1.90	#	#
State of institution						(Effect size = 0.20)		(Effect size = 0.02)	
Alabama	290	180	1.24	1.10	-0.15*	-11.81	1.15	-0.09*	-7.20
Alaska	40	40	0.11	0.05	-0.06	-53.55	0.06	-0.06	-49.35
Arizona	550	320	2.28	2.16	-0.12	-5.33	2.32	0.04	1.84
Arkansas	310	180	0.86	0.83	-0.03	-2.94	0.84	-0.02	-2.51
California	1,610	810	11.60	11.60	-0.01	-0.05	11.60	#	#
Colorado	460	240	1.82	1.92	0.10	5.71	1.85	0.03	1.90
Connecticut	310	190	0.89	0.87	-0.03	-3.14	0.94	0.04	4.93
Delaware	80	40	0.34	0.38	0.04	11.09	0.32	-0.02	-7.19
Washington, DC	160	60	0.36	0.39	0.03	8.19	0.32	-0.04*	-9.95
Florida	1,150	640	5.37	5.12	-0.26	-4.81	5.37	#	#
Georgia	830	480	3.22	3.54	0.33*	10.14	3.22	#	#
Hawaii	160	110	0.28	0.25	-0.02	-8.88	0.31	0.03	10.73
Idaho	170	80	0.75	0.89	0.14	18.98	0.71	-0.04	-4.69
Illinois	650	280	3.45	3.75	0.30*	8.79	3.72	0.27*	7.85
Indiana	350	180	2.10	2.20	0.10	5.00	2.15	0.06	2.65
Iowa	290	120	1.13	1.20	0.06	5.64	1.16	0.03	2.50
Kansas	190	100	1.04	1.13	0.10	9.48	1.09	0.06	5.59
Kentucky	350	180	1.05	0.98	-0.07	-6.66	1.04	-0.01	-1.18

See notes at end of table.

Table G-1. Sample member survey nonresponse bias analysis for selected variables—All institutions: 2022—Continued

Variable	Sample counts		Before nonresponse weight adjustments				After nonresponse weight adjustment		
	Respondents, unweighted	Non-respondents, unweighted	Full-sample mean, base weighted	Respondent mean, base weighted	Estimated bias ¹	Relative bias ²	Respondent mean, nonresponse adjusted	Estimated bias ³	Relative bias ²
Louisiana	390	230	1.55	1.44	-0.11	-7.08	1.46	-0.09	-5.78
Maine	320	190	0.44	0.42	-0.03	-5.98	0.43	-0.02	-3.99
Maryland	370	130	1.72	2.09	0.38*	22.08	1.78	0.06	3.50
Massachusetts	520	280	2.88	2.66	-0.22	-7.74	2.75	-0.13	-4.47
Michigan	500	260	2.77	2.77	#	-0.11	2.79	0.02	0.71
Minnesota	450	260	1.77	1.78	0.01	0.33	1.77	#	-0.28
Mississippi	240	120	1.06	1.05	-0.01	-1.03	1.09	0.03	2.78
Missouri	320	150	1.69	1.72	0.04	2.22	1.67	-0.02	-0.96
Montana	100	40	0.25	0.34	0.09*	34.14	0.29	0.04	16.12
Nebraska	190	140	0.63	0.54	-0.08*	-12.92	0.54	-0.08*	-13.08
Nevada	160	100	0.58	0.51	-0.06*	-10.97	0.57	-0.01	-1.92
New Hampshire	160	90	0.48	0.43	-0.05	-11.34	0.51	0.03	6.09
New Jersey	640	350	2.21	2.15	-0.06	-2.59	2.21	#	#
New Mexico	140	110	0.34	0.28	-0.06*	-17.95	0.30	-0.04*	-12.33
New York	1,320	580	7.15	7.64	0.49*	6.87	7.15	#	#
North Carolina	800	460	3.28	3.25	-0.04	-1.10	3.28	#	#
North Dakota	90	60	0.34	0.38	0.04	11.81	0.37	0.03	8.09
Ohio	700	390	3.54	3.36	-0.18	-4.98	3.34	-0.19	-5.43
Oklahoma	370	230	1.14	1.03	-0.11*	-9.47	1.14	#	#
Oregon	270	130	1.18	1.09	-0.09	-7.30	1.18	#	0.32
Pennsylvania	840	410	3.99	4.24	0.25	6.28	3.99	#	#
Puerto Rico	410	190	0.90	0.93	0.04	4.23	0.90	#	#
Rhode Island	150	70	0.67	0.78	0.11	16.85	0.73	0.07	10.21
South Carolina	390	190	1.77	1.88	0.11	6.20	1.86	0.09	4.98
South Dakota	210	90	0.35	0.38	0.03	7.30	0.34	-0.01	-3.09
Tennessee	600	370	2.28	2.13	-0.15	-6.68	2.12	-0.16	-6.93
Texas	1,390	810	8.15	7.48	-0.67*	-8.20	8.15	#	#
Utah	440	280	1.26	1.31	0.05	4.33	1.24	-0.02	-1.26
Vermont	140	110	0.28	0.27	-0.01	-4.92	0.29	0.01	1.85
Virginia	510	210	2.99	3.20	0.21	7.10	3.32	0.33*	11.02
Washington	520	280	1.92	1.73	-0.19	-9.68	1.95	0.03	1.70
West Virginia	330	220	0.64	0.57	-0.06	-9.53	0.56	-0.08	-11.95
Wisconsin	280	140	1.72	1.59	-0.13	-7.46	1.57	-0.15	-8.94
Wyoming	110	70	0.18	0.17	-0.01	-3.22	0.15	-0.02*	-13.61

See notes at end of table.

Table G-1. Sample member survey nonresponse bias analysis for selected variables—All institutions: 2022—Continued

Variable	Sample counts		Before nonresponse weight adjustments				After nonresponse weight adjustment		
	Respondents, unweighted	Non-respondents, unweighted	Full-sample mean, base weighted	Respondent mean, base weighted	Estimated bias ¹	Relative bias ²	Respondent mean, nonresponse adjusted	Estimated bias ³	Relative bias ²
Selected states and Census region of institution					(Effect size = 0.04)			(Effect size = #)	
California	1,610	810	11.60	11.60	-0.01	-0.05	11.60	#	#
Florida	1,150	640	5.37	5.12	-0.26	-4.81	5.37	#	#
Georgia	830	480	3.22	3.54	0.33*	10.14	3.22	#	#
New York	1,320	580	7.15	7.64	0.49*	6.87	7.15	#	#
North Carolina	800	460	3.28	3.25	-0.04	-1.10	3.28	#	#
Pennsylvania	840	410	3.99	4.24	0.25	6.28	3.99	#	#
Texas	1,390	810	8.15	7.48	-0.67*	-8.20	8.15	#	#
Northeast & Puerto Rico	2,630	1,460	8.75	8.50	-0.25	-2.89	8.75	#	#
Midwest	4,230	2,160	20.52	20.81	0.29	1.42	20.52	#	#
South	4,390	2,330	17.00	17.09	0.08	0.49	17.00	#	#
West	3,120	1,780	10.94	10.72	-0.22	-2.02	10.94	#	#
Bureau of Economic Analysis region ⁴					(Effect size = 0.05)			(Effect size = #)	
New England	1,580	930	5.65	5.41	-0.23	-4.14	5.65	#	#
Mideast	3,410	1,560	15.77	16.90	1.13*	7.17	15.77	#	#
Great Lakes	2,490	1,250	13.58	13.68	0.10	0.74	13.58	#	#
Plains	1,740	920	6.94	7.13	0.19	2.74	6.94	#	#
Southeast	6,200	3,450	25.32	25.10	-0.22	-0.88	25.32	#	#
Southwest	2,450	1,470	11.91	10.95	-0.96*	-8.05	11.91	#	#
Rocky Mountains	1,290	700	4.26	4.64	0.38*	8.95	4.26	#	#
Far West	2,760	1,460	15.67	15.24	-0.43	-2.72	15.67	#	#
Outlying Areas	410	190	0.90	0.93	0.04	4.23	0.90	#	#
Federal aid status in 2019–20					(Effect size = 0.11)			(Effect size = #)	
Received	15,130	7,210	59.60	63.62	4.01*	6.73	59.60	#	#
Did not receive	6,000	4,260	35.00	29.92	-5.08*	-14.50	35.00	#	#
Missing	1,200	440	5.40	6.46	1.06*	19.69	5.40	#	#
Direct Loan status in 2019–20					(Effect size = 0.08)			(Effect size = #)	
Received	9,010	3,990	33.23	36.81	3.58*	10.78	33.23	#	#
Did not receive	13,320	7,930	66.77	63.19	-3.58*	-5.36	66.77	#	#

See notes at end of table.

Table G-1. Sample member survey nonresponse bias analysis for selected variables—All institutions: 2022—Continued

Variable	Sample counts		Before nonresponse weight adjustments				After nonresponse weight adjustment		
	Respondents, unweighted	Non-respondents, unweighted	Full-sample mean, base weighted	Respondent mean, base weighted	Estimated bias ¹	Relative bias ²	Respondent mean, nonresponse adjusted	Estimated bias ³	Relative bias ²
Total Direct Loan amount received in 2019–20 ⁵					(Effect size = 0.08)			(Effect size = #)	
\$0	13,320	7,930	66.77	63.19	-3.58*	-5.36	66.77	#	#
\$1–\$4,584	2,160	1,090	8.19	8.78	0.59*	7.17	8.19	#	0.04
\$4,585–\$5,500	4,830	1,990	20.05	22.57	2.53*	12.60	20.04	#	-0.02
\$5,501 or more	2,020	910	4.99	5.46	0.47*	9.40	4.99	#	#
Age as of December 31, 2019					(Effect size = ‡)			(Effect size = ‡)	
15–23	20,540	10,720	93.72	94.90	1.18*	1.26	93.72	#	#
24–29	890	590	3.07	2.59	-0.47*	-15.45	3.07	#	#
30 or older	890	610	3.19	2.49	-0.70*	-21.87	3.19	#	#
Unknown	‡	‡	‡	‡	‡	‡	‡	‡	‡
Institution aid status in 2019–20					(Effect size = 0.14)			(Effect size = #)	
Received	8,440	3,720	36.33	39.84	3.51*	9.66	36.33	#	#
Did not receive	10,840	7,200	52.17	45.47	-6.69*	-12.83	52.17	#	#
Unknown	3,040	1,000	11.50	14.69	3.18*	27.66	11.50	#	#
State aid status in 2019–20					(Effect size = 0.11)			(Effect size = #)	
Received	5,840	2,790	25.40	27.01	1.60*	6.31	25.40	#	#
Did not receive	13,280	8,020	62.36	57.81	-4.55*	-7.30	62.36	#	#
Unknown	3,200	1,110	12.24	15.18	2.95*	24.10	12.24	#	#
Received any aid excluding private loans in 2019–20					(Effect size = 0.14)			(Effect size = #)	
Yes	18,850	9,150	77.21	81.58	4.38*	5.67	77.21	#	#
No	2,560	2,420	18.29	13.03	-5.26*	-28.75	18.29	#	#
Unknown	920	350	4.51	5.39	0.88*	19.57	4.51	#	#
Veteran status					(Effect size = 0.01)			(Effect size = #)	
Yes	210	160	0.81	0.68	-0.13*	-15.47	0.81	#	#
No	22,110	11,760	99.19	99.32	0.13*	0.13	99.19	#	#

See notes at end of table.

Table G-1. Sample member survey nonresponse bias analysis for selected variables—All institutions: 2022—Continued

Variable	Sample counts		Before nonresponse weight adjustments				After nonresponse weight adjustment		
	Respondents, unweighted	Non-respondents, unweighted	Full-sample mean, base weighted	Respondent mean, base weighted	Estimated bias ¹	Relative bias ²	Respondent mean, nonresponse adjusted	Estimated bias ³	Relative bias ²
Race/ethnicity					(Effect size = 0.10)			(Effect size = #)	
American Indian or Alaska Native, not Hispanic or Latino	150	130	0.66	0.50	-0.16*	-24.24	0.66	#	#
Asian, not Hispanic or Latino	1,520	670	8.18	8.79	0.61*	7.42	8.18	#	#
Black, not Hispanic or Latino	3,100	1,560	12.16	12.32	0.15	1.27	12.16	#	#
Hispanic or Latino, of any race	5,180	2,730	22.75	22.36	-0.39	-1.73	22.75	#	#
Native Hawaiian or Other Pacific Islander, not Hispanic or Latino	80	50	0.28	0.26	-0.03	-10.25	0.28	#	#
White, not Hispanic or Latino	10,960	5,650	48.01	49.52	1.52*	3.16	48.01	#	#
Two or more races, not Hispanic or Latino	1,080	570	4.64	4.71	0.07	1.44	4.64	#	#
Unknown	270	560	3.32	1.55	-1.76*	-53.16	3.32	#	#
Gender					(Effect size = 0.07)			(Effect size = #)	
Male	9,100	5,910	46.24	42.60	-3.64*	-7.88	46.24	#	#
Female	12,910	5,900	52.59	56.02	3.43*	6.53	52.59	#	#
Unknown	320	110	1.17	1.38	0.21*	18.05	1.17	#	#
Institution total undergraduate enrollment in 2019–20 ^s					(Effect size = 0.04)			(Effect size = #)	
1–2,983	5,690	2,880	14.69	15.93	1.23*	8.40	14.69	#	#
2,984–8,499	5,470	3,100	20.81	20.23	-0.58	-2.79	20.81	#	#
8,500–19,798	5,510	3,040	29.38	28.52	-0.86*	-2.92	29.38	#	#
19,799 or more	5,650	2,910	35.12	35.33	0.20	0.58	35.12	#	#
Major					(Effect size = 0.09)			(Effect size = 0.07)	
Undecided	640	410	4.57	4.07	-0.51*	-11.06	4.24	-0.33	-7.28
Humanities	3,640	2,430	19.70	17.63	-2.09*	-10.62	17.87	-1.83*	-9.31
Social/behavioral sciences	1,630	590	7.29	8.43	1.16*	15.87	8.12	0.83*	11.38
Life sciences	2,250	940	10.56	11.49	0.94*	8.91	10.89	0.33	3.13
Physical sciences/mathematics	360	150	1.84	1.96	0.13	7.01	1.84	0.01	0.31
Computer/information science	1,090	520	4.66	4.77	0.11	2.32	5.03	0.36*	7.82
Engineering	1,500	690	7.12	7.44	0.33*	4.62	7.64	0.52*	7.32
Education	1,110	440	4.42	4.91	0.49*	11.16	4.65	0.22*	5.05
Business/management	2,540	1,420	12.55	12.19	-0.36	-2.86	12.59	0.04	0.31
Health	3,510	1,640	11.31	12.02	0.72*	6.35	11.74	0.43*	3.81
Vocational/technical	840	800	3.45	2.69	-0.77*	-22.25	2.95	-0.51*	-14.63
Other technical/professional	2,890	1,580	11.45	11.48	0.02	0.21	11.55	0.09	0.81
Unknown	160	120	1.08	0.91	-0.17	-16.15	0.91	-0.17	-15.36

See notes at end of table.

Table G-1. Sample member survey nonresponse bias analysis for selected variables—All institutions: 2022—Continued

Variable	Sample counts		Before nonresponse weight adjustments				After nonresponse weight adjustment		
	Respondents, unweighted	Non-respondents, unweighted	Full-sample mean, base weighted	Respondent mean, base weighted	Estimated bias ¹	Relative bias ²	Respondent mean, nonresponse adjusted	Estimated bias ³	Relative bias ²
Degree program					(Effect size = 0.11)			(Effect size = 0.06)	
Certificate	2,280	1,710	6.23	4.94	-1.29*	-20.75	5.40	-0.83*	-13.26
Associate's degree	8,350	5,020	36.02	33.45	-2.57*	-7.12	36.76	0.74*	2.05
Bachelor's degree	11,520	4,960	56.03	60.62	4.59*	8.19	56.70	0.67	1.19
Not in a degree program or others	160	200	1.45	0.89	-0.57*	-38.98	0.93	-0.52*	-36.11
Unknown	10	30	0.27	0.10	-0.16*	-61.78	0.21	-0.05	-20.57
Pell Grant status in 2019–20					(Effect size = 0.03)			(Effect size = #)	
Received	10,730	5,360	40.93	42.56	1.63*	3.98	40.93	#	#
Did not receive	11,590	6,560	59.07	57.44	-1.63*	-2.76	59.07	#	#
Total Pell amount received in 2019–20 ⁸					(Effect size = 0.05)			(Effect size = #)	
\$0	11,590	6,560	59.07	57.44	-1.63*	-2.76	59.07	#	#
\$1–\$3,097	2,450	1,530	10.24	9.65	-0.59*	-5.77	10.24	#	#
\$3,098–\$6,194	4,010	1,980	15.19	15.68	0.48*	3.19	15.19	#	#
\$6,195 or more	4,280	1,850	15.50	17.24	1.74*	11.21	15.50	#	#
Total income from CPS:20 ⁷					(Effect size = 0.10)			(Effect size = 0.02)	
-\$16,744–\$18,236	3,940	2,180	14.78	14.60	-0.18	-1.21	14.55	-0.23	-1.58
\$18,237–\$46,630	4,020	2,100	16.60	17.07	0.47	2.85	16.46	-0.14	-0.82
\$46,631–\$104,625	4,160	1,950	17.70	19.07	1.36*	7.70	18.00	0.30	1.68
\$104,626 or more	4,240	1,880	20.61	22.91	2.30*	11.18	20.83	0.23	1.10
Unknown	130	120	0.80	0.59	-0.21*	-26.22	0.65	-0.15*	-19.11
Not matched in CPS:20	5,840	3,690	29.51	25.76	-3.75*	-12.71	29.51	#	#
Parents' highest education from CPS:20					(Effect size = 0.09)			(Effect size = #)	
Middle school/junior high	790	440	3.31	3.28	-0.03	-0.97	3.31	#	#
High school	4,930	2,750	20.13	20.16	0.02	0.11	20.13	#	#
College or beyond	9,740	4,450	43.09	47.06	3.97*	9.22	43.09	#	#
Other/unknown	860	500	3.40	3.26	-0.14	-4.19	3.40	#	#
Missing	170	110	0.56	0.49	-0.07	-12.72	0.56	#	#
Not matched in CPS:20	5,840	3,690	29.51	25.76	-3.75*	-12.71	29.51	#	#
Has dependents from CPS:20					(Effect size = ‡)			(Effect size = ‡)	
Yes	1,350	710	3.61	3.47	-0.14	-3.86	3.61	#	#
No	15,140	7,520	66.88	70.77	3.89*	5.82	66.88	#	#
Missing	‡	‡	‡	‡	‡	‡	‡	‡	‡
Not matched in CPS:20	5,840	3,690	29.51	25.76	-3.75*	-12.71	29.51	#	#

See notes at end of table.

Table G-1. Sample member survey nonresponse bias analysis for selected variables—All institutions: 2022—Continued

Variable	Sample counts		Before nonresponse weight adjustments				After nonresponse weight adjustment		
	Respondents, unweighted	Non-respondents, unweighted	Full-sample mean, base weighted	Respondent mean, base weighted	Estimated bias ¹	Relative bias ²	Respondent mean, nonresponse adjusted	Estimated bias ³	Relative bias ²
Student's marital status from CPS:20					(Effect size = 0.08)			(Effect size = #)	
Single	15,720	7,860	68.30	72.13	3.84*	5.62	68.30	#	#
Married/remarried	580	260	1.63	1.66	0.02	1.48	1.63	#	#
Separated	80	60	0.24	0.18	-0.06*	-25.60	0.24	#	#
Divorced/widowed	100	50	0.33	0.27	-0.05	-15.67	0.33	#	#
Not matched in CPS:20	5,840	3,690	29.51	25.76	-3.75*	-12.71	29.51	#	#

Rounds to zero.

‡ Reporting standards not met (fewer than 30 unweighted nonrespondents).

* $p < 0.05$.¹ Bias in the sample mean is estimated as the difference between the weighted respondent mean and the weighted full-sample mean using the BPS:20/22 base weight.² Relative bias is calculated as 100 times the ratio of estimated bias to the weighted full-sample mean.³ Bias in the sample mean is estimated as the difference between the weighted respondent mean (using the BPS:20/22 base weight adjusted for nonresponse) and weighted full-sample mean (using the BPS:20/22 base weight).⁴ New England = Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont; Mideast = Delaware, District of Columbia, Maryland, New Jersey, New York, Pennsylvania; Great Lakes = Illinois, Indiana, Michigan, Ohio, Wisconsin; Plains = Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, South Dakota; Southeast = Alabama, Arkansas, Florida, Georgia, Kentucky, Mississippi, North Carolina, South Carolina, Tennessee, Virginia, West Virginia; Southwest = Arizona, New Mexico, Oklahoma, Texas; Rocky Mountains = Colorado, Idaho, Montana, Utah, Wyoming; Far West = Alaska, California, Hawaii, Nevada, Oregon, Washington; Outlying Areas = Puerto Rico.⁵ Categories were defined by quartiles. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of 0 or "Unknown" were excluded from quartile calculations.⁶ Pell Grant amounts include summer awards and are categorized by quartiles for students receiving less than \$6,195, with students receiving \$6,195 or more in one category. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of 0 or "Unknown" were excluded from quartile calculations.⁷ Categories were defined by quartiles. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of "Unknown" were excluded from quartile calculations. As part of the expected family contribution (EFC) formula, some calculations include allowances against income or asset amounts that can result in negative numbers.

NOTE: CPS:20 = Central Processing System data from NPSAS:20. Effect size is calculated as the square root of the sum over categories of the squared differences (respondent vs. full sample) over full-sample means. Effect sizes are not reported if any variable categories do not meet reporting standards. "Base weight" refers to the BPS:20/22 base weight. Details may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table G-2. Sample member survey analysis after poststratification for selected variables—All institutions: 2022

Variable	Before poststratification adjustment		After poststratification adjustment		
	Respondent mean, nonresponse adjusted ¹ (1)	Full sample mean, base weighted (2)	Respondent mean, after all weight adjustments ² (3)	Mean (1) – Mean (3)	Mean (2) – Mean (3)
Control and level of institution					
Public less-than-2-year	0.47	0.47	0.52	-0.05	-0.05
Public 2-year	31.02	31.02	35.62	-4.60*	-4.60*
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	7.80	7.80	8.45	-0.65*	-0.65*
Public 4-year, non-doctorate-granting, primarily baccalaureate	5.50	5.50	4.79	0.71*	0.71*
Public 4-year, doctorate-granting	30.25	30.25	26.06	4.19*	4.19*
Private nonprofit less-than-4-year	0.23	0.23	0.31	-0.08	-0.08
Private nonprofit 4-year, non-doctorate-granting	7.82	7.82	6.16	1.66*	1.66*
Private nonprofit 4-year, doctorate-granting	12.68	12.68	11.25	1.43*	1.43*
Private for-profit less-than-2-year	1.41	1.41	2.23	-0.82*	-0.82*
Private for-profit 2-year	0.92	0.92	1.60	-0.68*	-0.68*
Private for-profit 4-year	1.90	1.90	3.02	-1.12*	-1.12*
State of institution					
Alabama	1.15	1.24	1.21	-0.06	0.03
Alaska	0.06	0.11	0.09	-0.03	0.02
Arizona	2.32	2.28	2.78	-0.46*	-0.50
Arkansas	0.84	0.86	0.81	0.03	0.05
California	11.60	11.60	11.45	0.15	0.15
Colorado	1.85	1.82	1.84	0.01	-0.02
Connecticut	0.94	0.89	1.11	-0.17	-0.22
Delaware	0.32	0.34	0.31	0.01	0.03
Washington, DC	0.32	0.36	0.29	0.03	0.07*
Florida	5.37	5.37	6.19	-0.82*	-0.82*
Georgia	3.22	3.22	2.85	0.37*	0.37
Hawaii	0.31	0.28	0.48	-0.17*	-0.20*
Idaho	0.71	0.75	0.85	-0.14	-0.10
Illinois	3.72	3.45	3.56	0.16	-0.11
Indiana	2.15	2.10	2.18	-0.03	-0.08
Iowa	1.16	1.13	1.04	0.12*	0.09
Kansas	1.09	1.04	1.00	0.09	0.04
Kentucky	1.04	1.05	1.08	-0.04	-0.03

See notes at end of table.

Table G-2. Sample member survey analysis after poststratification for selected variables—All institutions: 2022—Continued

Variable	Before poststratification adjustment		After poststratification adjustment		
	Respondent mean, nonresponse adjusted ¹ (1)	Full sample mean, base weighted (2)	Respondent mean, after all weight adjustments ² (3)	Mean (1) – Mean (3)	Mean (2) – Mean (3)
Louisiana	1.46	1.55	1.67	-0.21*	-0.12
Maine	0.43	0.44	0.43	#	0.01
Maryland	1.78	1.72	1.85	-0.07	-0.13
Massachusetts	2.75	2.88	2.53	0.22	0.35*
Michigan	2.79	2.77	2.86	-0.07	-0.09
Minnesota	1.77	1.77	1.70	0.07	0.07
Mississippi	1.09	1.06	1.06	0.03	#
Missouri	1.67	1.69	1.79	-0.12	-0.10
Montana	0.29	0.25	0.29	#	-0.04*
Nebraska	0.54	0.63	0.51	0.03	0.12*
Nevada	0.57	0.58	0.66	-0.09	-0.08
New Hampshire	0.51	0.48	0.42	0.09	0.06
New Jersey	2.21	2.21	2.42	-0.21*	-0.21
New Mexico	0.30	0.34	0.32	-0.02	0.02
New York	7.15	7.15	6.39	0.76*	0.76*
North Carolina	3.28	3.28	2.95	0.33*	0.33*
North Dakota	0.37	0.34	0.31	0.06*	0.03
Ohio	3.34	3.54	3.23	0.11	0.31
Oklahoma	1.14	1.14	1.33	-0.19*	-0.19
Oregon	1.18	1.18	1.25	-0.07	-0.07
Pennsylvania	3.99	3.99	3.56	0.43	0.43
Puerto Rico	0.90	0.90	0.73	0.17*	0.17*
Rhode Island	0.73	0.67	0.74	-0.01	-0.07
South Carolina	1.86	1.77	1.82	0.04	-0.05
South Dakota	0.34	0.35	0.36	-0.02	-0.01
Tennessee	2.12	2.28	2.11	0.01	0.17
Texas	8.15	8.15	8.56	-0.41	-0.41
Utah	1.24	1.26	1.35	-0.11	-0.09
Vermont	0.29	0.28	0.25	0.04	0.03
Virginia	3.32	2.99	3.08	0.24	-0.09
Washington	1.95	1.92	2.16	-0.21*	-0.24
West Virginia	0.56	0.64	0.63	-0.07	0.01
Wisconsin	1.57	1.72	1.39	0.18	0.33*
Wyoming	0.15	0.18	0.18	-0.03	#

See notes at end of table.

Table G-2. Sample member survey analysis after poststratification for selected variables—All institutions: 2022—Continued

Variable	Before poststratification adjustment		After poststratification adjustment		
	Respondent mean, nonresponse adjusted ¹ (1)	Full sample mean, base weighted (2)	Respondent mean, after all weight adjustments ² (3)	Mean (1) – Mean (3)	Mean (2) – Mean (3)
Selected states and Census region of institution					
California	11.60	11.60	11.45	0.15	0.15
Florida	5.37	5.37	6.19	-0.82*	-0.82*
Georgia	3.22	3.22	2.85	0.37*	0.37
New York	7.15	7.15	6.39	0.76*	0.76*
North Carolina	3.28	3.28	2.95	0.33*	0.33*
Pennsylvania	3.99	3.99	3.56	0.43	0.43
Texas	8.15	8.15	8.56	-0.41	-0.41
Northeast & Puerto Rico	8.75	8.75	8.64	0.11	0.11
Midwest	20.52	20.52	19.91	0.61	0.61
South	17.00	17.00	17.25	-0.25	-0.25
West	10.94	10.94	12.24	-1.30*	-1.30*
Bureau of Economic Analysis region ³					
New England	5.65	5.65	5.49	0.16	0.16
Mideast	15.77	15.77	14.83	0.94*	0.94*
Great Lakes	13.58	13.58	13.21	0.37	0.37
Plains	6.94	6.94	6.69	0.25	0.25
Southeast	25.32	25.32	25.47	-0.15	-0.15
Southwest	11.91	11.91	12.98	-1.07*	-1.07*
Rocky Mountains	4.26	4.26	4.51	-0.25	-0.25
Far West	15.67	15.67	16.09	-0.42	-0.42
Outlying Areas	0.90	0.90	0.73	0.17*	0.17*
Federal aid status in 2019–20					
Received	59.60	59.60	61.20	-1.60*	-1.60*
Did not receive	35.00	35.00	32.81	2.19*	2.19*
Missing	5.40	5.40	5.99	-0.59*	-0.59*
Direct Loan status in 2019–20					
Received	33.23	33.23	34.17	-0.94*	-0.94*
Did not receive	66.77	66.77	65.83	0.94*	0.94*

See notes at end of table.

Table G-2. Sample member survey analysis after poststratification for selected variables—All institutions: 2022—Continued

Variable	Before poststratification adjustment		After poststratification adjustment		
	Respondent mean, nonresponse adjusted ¹ (1)	Full sample mean, base weighted (2)	Respondent mean, after all weight adjustments ² (3)	Mean (1) – Mean (3)	Mean (2) – Mean (3)
Total Direct Loan amount received in 2019–20 ⁴					
\$0	66.77	66.77	65.83	0.94*	0.94*
\$1–\$4,584	8.19	8.19	8.95	-0.76*	-0.76*
\$4,585–\$5,500	20.04	20.05	19.34	0.70*	0.71*
\$5,501 or more	4.99	4.99	5.87	-0.88*	-0.88*
Age as of December 31, 2019					
15–23	93.72	93.72	91.40	2.32*	2.32*
24–29	3.07	3.07	4.18	-1.11*	-1.11*
30 or older	3.19	3.19	4.39	-1.20*	-1.20*
Unknown	‡	‡	‡	‡	‡
Institution aid status in 2019–20					
Received	36.33	36.33	32.37	3.96*	3.96*
Did not receive	52.17	52.17	54.86	-2.69*	-2.69*
Unknown	11.50	11.50	12.77	-1.27*	-1.27*
State aid status in 2019–20					
Received	25.40	25.40	25.88	-0.48	-0.48
Did not receive	62.36	62.36	60.45	1.91*	1.91*
Unknown	12.24	12.24	13.68	-1.44*	-1.44*
Received any aid excluding private loans in 2019–20					
Yes	77.21	77.21	76.70	0.51	0.51
No	18.29	18.29	18.18	0.11	0.11
Unknown	4.51	4.51	5.12	-0.61*	-0.61*
Veteran status					
Yes	0.81	0.81	1.12	-0.31*	-0.31*
No	99.19	99.19	98.88	0.31*	0.31*

See notes at end of table.

Table G-2. Sample member survey analysis after poststratification for selected variables—All institutions: 2022—Continued

Variable	Before poststratification adjustment		After poststratification adjustment		
	Respondent mean, nonresponse adjusted ¹ (1)	Full sample mean, base weighted (2)	Respondent mean, after all weight adjustments ² (3)	Mean (1) – Mean (3)	Mean (2) – Mean (3)
Race/ethnicity					
American Indian or Alaska Native, not Hispanic or Latino	0.66	0.66	0.88	-0.22*	-0.22*
Asian, not Hispanic or Latino	8.18	8.18	10.09	-1.91*	-1.91*
Black, not Hispanic or Latino	12.16	12.16	12.47	-0.31	-0.31
Hispanic or Latino, of any race	22.75	22.75	21.35	1.40*	1.40*
Native Hawaiian or Other Pacific Islander, not Hispanic or Latino	0.28	0.28	0.52	-0.24*	-0.24*
White, not Hispanic or Latino	48.01	48.01	45.29	2.72*	2.72*
Two or more races, not Hispanic or Latino	4.64	4.64	6.79	-2.15*	-2.15*
Unknown	3.32	3.32	2.59	0.73*	0.73*
Gender					
Male	46.24	46.24	45.78	0.46	0.46
Female	52.59	52.59	52.98	-0.39	-0.39
Unknown	1.17	1.17	1.24	-0.07	-0.07
Institution total undergraduate enrollment in 2019–20 ⁴					
1–2,983	14.69	14.69	14.70	-0.01	-0.01
2,984–8,499	20.81	20.81	20.11	0.70*	0.70
8,500–19,798	29.38	29.38	29.56	-0.18	-0.18
19,799 or more	35.12	35.12	35.63	-0.51	-0.51
Major					
Undecided	4.24	4.57	3.43	0.81*	1.14*
Humanities	17.87	19.70	18.29	-0.42	1.41*
Social/behavioral sciences	8.12	7.29	7.57	0.55*	-0.28
Life sciences	10.89	10.56	9.89	1.00*	0.67*
Physical sciences/mathematics	1.84	1.84	1.69	0.15*	0.15
Computer/information science	5.03	4.66	5.37	-0.34*	-0.71*
Engineering	7.64	7.12	7.05	0.59*	0.07
Education	4.65	4.42	4.64	0.01	-0.22
Business/management	12.59	12.55	12.33	0.26	0.22
Health	11.74	11.31	13.16	-1.42*	-1.85*
Vocational/technical	2.95	3.45	3.28	-0.33*	0.17
Other technical/professional	11.55	11.45	12.22	-0.67*	-0.77*
Unknown	0.91	1.08	1.08	-0.17	#

See notes at end of table.

Table G-2. Sample member survey analysis after poststratification for selected variables—All institutions: 2022—Continued

Variable	Before poststratification adjustment		After poststratification adjustment		
	Respondent mean, nonresponse adjusted ¹ (1)	Full sample mean, base weighted (2)	Respondent mean, after all weight adjustments ² (3)	Mean (1) – Mean (3)	Mean (2) – Mean (3)
Degree program					
Certificate	5.40	6.23	7.50	-2.10*	-1.27*
Associate's degree	36.76	36.02	41.82	-5.06*	-5.80*
Bachelor's degree	56.70	56.03	49.57	7.13*	6.46*
Not in a degree program or others	0.93	1.45	1.01	-0.08	0.44*
Unknown	0.21	0.27	0.10	0.11	0.17*
Pell Grant status in 2019–20					
Received	40.93	40.93	44.83	-3.90*	-3.90*
Did not receive	59.07	59.07	55.17	3.90*	3.90*
Total Pell amount received in 2019–20 ⁵					
\$0	59.07	59.07	55.17	3.90*	3.90*
\$1–\$3,097	10.24	10.24	13.80	-3.56*	-3.56*
\$3,098–\$6,194	15.19	15.19	17.26	-2.07*	-2.07*
\$6,195 or more	15.50	15.50	13.77	1.73*	1.73*
Total income from CPS:20 ⁶					
-\$16,744–\$18,236	14.55	14.78	16.74	-2.19*	-1.96*
\$18,237–\$46,630	16.46	16.60	17.55	-1.09*	-0.95*
\$46,631–\$104,625	18.00	17.70	18.68	-0.68*	-0.98*
\$104,626 or more	20.83	20.61	17.42	3.41*	3.19*
Unknown	0.65	0.80	0.77	-0.12*	0.03
Not matched in CPS:20	29.51	29.51	28.84	0.67	0.67
Parents' highest education from CPS:20					
Middle school/junior high	3.31	3.31	3.58	-0.27*	-0.27*
High school	20.13	20.13	21.91	-1.78*	-1.78*
College or beyond	43.09	43.09	41.17	1.92*	1.92*
Other/unknown	3.40	3.40	3.76	-0.36*	-0.36*
Missing	0.56	0.56	0.73	-0.17*	-0.17*
Not matched in CPS:20	29.51	29.51	28.84	0.67	0.67
Has dependents from CPS:20					
Yes	3.61	3.61	5.38	-1.77*	-1.77*
No	66.88	66.88	65.78	1.10*	1.10*
Missing	‡	‡	‡	‡	‡
Not matched in CPS:20	29.51	29.51	28.84	0.67	0.67

See notes at end of table.

Table G-2. Sample member survey analysis after poststratification for selected variables—All institutions: 2022—Continued

Variable	Before poststratification adjustment		After poststratification adjustment		
	Respondent mean, nonresponse adjusted ¹ (1)	Full sample mean, base weighted (2)	Respondent mean, after all weight adjustments ² (3)	Mean (1) – Mean (3)	Mean (2) – Mean (3)
Student's marital status from CPS:20					
Single	68.30	68.30	67.96	0.34	0.34
Married/remarried	1.63	1.63	2.43	-0.80*	-0.80*
Separated	0.24	0.24	0.30	-0.06	-0.06
Divorced/widowed	0.33	0.33	0.47	-0.14*	-0.14*
Not matched in CPS:20	29.51	29.51	28.84	0.67	0.67

Rounds to zero.

‡ Reporting standards not met (fewer than 30 nonrespondents).

* $p < 0.05$.¹ Mean is calculated using the student base weight adjusted for nonresponse.² Mean is calculated using the student base weight adjusted for nonresponse and poststratification.³ New England = Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, Vermont; Mideast = Delaware, District of Columbia, Maryland, New Jersey, New York, Pennsylvania; Great Lakes = Illinois, Indiana, Michigan, Ohio, Wisconsin; Plains = Iowa, Kansas, Minnesota, Missouri, Nebraska, North Dakota, South Dakota; Southeast = Alabama, Arkansas, Florida, Georgia, Kentucky, Mississippi, North Carolina, South Carolina, Tennessee, Virginia, West Virginia; Southwest = Arizona, New Mexico, Oklahoma, Texas; Rocky Mountains = Colorado, Idaho, Montana, Utah, Wyoming; Far West = Alaska, California, Hawaii, Nevada, Oregon, Washington; Outlying Areas = Puerto Rico.⁴ Categories were defined by quartiles. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of 0 or "Unknown" were excluded from quartile calculations.⁵ Pell Grant amounts include summer awards and are categorized by quartiles for students receiving less than \$6,195, with students receiving \$6,195 or more in one category. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of 0 or "Unknown" were excluded from quartile calculations.⁶ Categories were defined by quartiles. Fewer than four categories will display in cases where the variable's values cause quartiles to overlap. Categories of "Unknown" or "Not matched in CPS:20" were excluded from quartile calculations. As part of the expected family contribution (EFC) formula, some calculations include allowances against income or asset amounts that can result in negative numbers.

NOTE: CPS:20 = Central Processing System data from NPSAS:20. "Base weight" refers to the BPS:20/22 base weight. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
ACDPARTANY22 (Academic activities: participated in any academic activity, AY 2020-21/2021-22)	14.39	12.17	55.88	0.11	0.31	#
ACDPARTAPP22 (Academic activities: participated in an apprenticeship, AY 2020-21/2021-22)	14.37	12.02	57.06	0.11	0.35	0.01
ACDPARTCOM22 (Academic activities: participated in community-based or service learning project, AY 2020-21/2021-22)	14.34	11.94	56.47	0.11	0.83*	0.01
ACDPARTIF22 (Academic activities: participated in internship or field experiment, AY 2020-21/2021-22)	14.31	11.88	56.47	0.11	0.14	#
ACDPARTLC22 (Academic activities: participated in learning community, AY 2020-21/2021-22)	14.34	12.07	57.06	0.11	1.19*	0.01
ACDPARTRSCH22 (Academic activities: participated in guided research experience, AY 2020-21/2021-22)	14.32	11.86	57.65	0.11	0.54	0.01
ACDPARTSE22 (Academic activities: participated in culminating senior experience, AY 2020-21/2021-22)	14.37	11.86	58.24	0.11	1.01*	0.01
ACDPARTSSC22 (Academic activities: participated in student success course, AY 2020-21/2021-22)	14.30	12.03	57.65	0.11	0.15	#
ALTEARN22 (Alternative plans: annual earning estimate if had not attended college, AY 2021-22)	18.44	12.89	59.41	0.12	0.65	0.01
ALTWORK22 (Alternative plans: work status if had not attended college, AY 2021-22)	17.41	12.75	60.00	0.12	0.62	0.01
AT1TY3Y (First degree type attained through AY 2021-22)	9.20	6.87	51.76	0.09	0.01	‡
ATAADT3Y (First associate's degree date attained through AY 2021-22)	22.00	14.75	52.35	0.16	0.57	#
ATAANU3Y (Number of associate's degrees attained through AY 2021-22)	22.00	14.75	52.35	0.16	0.21*	‡
ATBADT3Y (First bachelor's degree date attained through AY 2021-22)	21.28	14.11	61.18	0.16	0.02	‡
ATBANU3Y (Number of bachelor's degrees attained through AY 2021-22)	21.28	14.11	61.18	0.16	0.02	‡
ATCTDT3Y (First certificate date attained through AY 2021-22)	40.17	24.58	46.15	0.29	1.98*	0.02
ATCTNU3Y (Number of certificates attained through AY 2021-22)	40.17	24.58	46.15	0.29	1.68*	‡
ATDEG3Y (First year attained degree through AY 2021-22)	9.19	6.79	52.35	0.08	0.16*	‡

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
ATHDT3Y (Date attained highest degree at any institution through AY 2021-22)	9.20	6.87	51.76	0.09	0.06	#
ATLTY3Y (Last degree type attained through AY 2021-22)	9.20	6.87	51.76	0.09	0.02	‡
ATNUM3Y (Degrees: number attained through AY 2021-22)	9.20	6.87	51.76	0.09	0.04	‡
AWAREIDR22 (Aware of income-driven student loan repayment plans, AY 2020-21/2021-22)	18.56	13.13	59.17	0.11	2.12*	0.02
AWARELFP22 (Aware of student loan forgiveness programs, AY 2020-21/2021-22)	14.06	10.75	58.82	0.11	0.05	#
CAMPSAFE22 (Felt safe attending primary institution, AY 2020-21/2021-22)	14.66	11.92	57.65	0.11	1.22*	0.02
CARAMT22 (Monthly car loan or lease payment, AY 2021-22)	11.57	8.11	55.88	0.09	3.02*	0.01
CARLN22 (Had a car loan or lease, AY 2021-22)	11.51	8.17	57.65	0.09	0.47	0.01
CINCOME (Total income (continuous), 2017)	14.91	11.14	57.65	0.17	8.54*	0.06
CINCOME22 (Total income (continuous), 2019)	20.38	14.56	58.82	0.22	12.42*	0.10
CLSEVR3Y (Ever took classes only (not a degree program) through AY 2021-22)	9.20	6.87	51.76	0.09	1.47*	0.02
COVCHNGINC21 (Educational impacts due to COVID-19: increased enrollment from part-time to full-time, AY 2020-21)	15.30	13.28	59.41	0.11	#	#
COVCHNGINS21 (Educational impacts due to COVID-19: changed institution where enrolled, AY 2020-21)	15.30	13.28	59.41	0.11	1.05*	0.02
COVCHNGMAJ21 (COVID-19 impacts at primary institution: changed declared or intended major, AY 2020-21)	15.30	13.28	59.41	0.11	0.28	#
COVCHNGNENR21 (Educational impacts due to COVID-19: did not attend for entire term/semester, AY 2020-21)	15.30	13.28	59.41	0.11	3.48*	0.05
COVCHNGRED21 (Educational impacts due to COVID-19: reduced enrollment from full-time to part-time, AY 2020-21)	15.30	13.28	59.41	0.11	1.21*	0.02
COVDEVHARD21 (Primary institution online coursetaking during COVID-19: device has necessary hardware, AY 2020-21)	19.52	13.82	58.24	0.13	0.03	#
COVDEVSOFT21 (Primary institution online coursetaking during COVID-19: device has necessary software, AY 2020-21)	19.52	13.79	58.24	0.13	0.63	0.01
COVIDISACCES21 (Primary institution provided COVID-19 accommodations: access online course content, AY 2020-21)	27.54	19.81	47.65	0.22	0.94	0.01
COVIDISCOMP21 (Primary institution provided COVID-19 accommodations: complete online coursework, AY 2020-21)	27.57	19.81	47.65	0.22	0.48	0.01

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
COVDISTR21 (Online coursetaking during COVID-19: difficulties accessing a physical space free of interruption or distraction, AY 2020-21)	20.51	15.09	60.00	0.13	1.17	0.01
COVENRDIS (COVID-19 pandemic: enrollment disruptions/changes, Spring 2020)	17.05	12.62	58.58	0.11	0.65	0.01
COVEXPABS (COVID-19 pandemic: took leave of absence from NPSAS, Spring 2020)	17.05	12.62	58.58	0.11	0.25	0.01
COVEXPACTMOD21 (COVID-19 impacts at primary institution: internship, apprenticeship, student teaching, or clinical placement mod or cancelled, AY 2020-21)	17.03	12.88	60.00	0.12	0.92*	0.01
COVEXPBRK (COVID-19 pandemic: school break extended, Spring 2020)	17.05	12.62	58.58	0.11	0.27	#
COVEXPCALMOD21 (COVID-19 impacts at primary institution: academic calendar modified, AY 2020-21)	17.03	12.88	60.00	0.12	4.38*	0.04
COVEXPCAN (COVID-19 pandemic: some or all classes canceled, Spring 2020)	17.05	12.62	58.58	0.11	0.41	0.01
COVEXPFIN (COVID-19 pandemic: received emergency financial assistance from NPSAS, Spring 2020)	17.05	12.62	58.58	0.11	0.17	#
COVEXPFOOD (COVID-19 pandemic: had difficulty accessing or paying for food, Spring 2020)	15.97	11.43	54.44	0.09	0.80*	0.02
COVEXPGRDMOD21 (COVID-19 impacts at primary institution: modified grading system, AY 2020-21)	17.03	12.88	60.00	0.12	2.37*	0.03
COVEXPHM (COVID-19 pandemic: had difficulty finding safe and stable housing, Spring 2020)	15.97	11.43	54.44	0.09	0.23	0.01
COVEXPHYB21 (COVID-19 impacts at primary institution: some or all classes delivered in hybrid format, AY 2020-21)	17.03	12.88	60.00	0.12	0.31	#
COVEXPJOB (COVID-19 pandemic: lost job or lost income because of reduced hours, Spring 2020)	15.97	11.43	54.44	0.09	1.48*	0.02
COVEXPMMOD21 (COVID-19 impacts at primary institution: class delivery method modified, AY 2020-21)	17.03	12.88	60.00	0.12	0.56	0.01
COVEXPMOT (COVID-19 pandemic: moved to another living situation, Spring 2020)	15.97	11.43	54.44	0.09	0.17	#
COVEXPONL (COVID-19 pandemic: some or all classes moved to online only, Spring 2020)	11.63	8.87	52.94	0.10	0.81*	0.01

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
COVEXPONL21 (COVID-19 impacts at primary institution: some or all classes delivered entirely online, AY 2020-21)	17.03	12.88	60.00	0.12	0.52	0.01
COVEXPPAD (COVID-19 pandemic: moved back to permanent address, Spring 2020)	17.05	12.62	58.58	0.11	2.63*	0.03
COVEXPSAP (COVID-19 pandemic: study-abroad program canceled or modified, Spring 2020)	17.05	12.62	58.58	0.11	0.25	0.01
COVEXPWIT (COVID-19 pandemic: withdrew from NPSAS, Spring 2020)	17.05	12.62	58.58	0.11	0.35	0.01
COVFINDIS (Financial disruptions/changes due to COVID-19, Spring 2020)	17.05	12.62	58.58	0.11	2.69*	0.03
COVFINDIS21 (Financial disruptions/changes due to COVID-19, AY 2020-21)	10.72	8.31	55.88	0.09	3.49*	0.04
COVHOUDIS (Housing disruptions/changes due to COVID-19, Spring 2020)	17.05	12.62	58.58	0.11	2.35*	0.02
COVINFOHLTH21 (Primary institution COVID-19 information: obtaining physical or mental healthcare, AY 2020-21)	16.95	13.40	60.00	0.12	0.25	#
COVINFOMHLTH21 (Primary institution COVID-19 information: obtaining mental healthcare, AY 2020-21)	16.95	13.40	60.00	0.12	0.26	#
COVINFOPHLTH21 (Primary institution COVID-19 information: obtaining physical healthcare, AY 2020-21)	16.99	13.39	60.00	0.12	0.03	#
COVINTPROB21 (Online coursetaking during COVID-19: difficulties with internet connection, AY 2020-21)	20.54	15.04	60.00	0.13	1.29	0.02
COVNOTATND21 (Decision not to attend school due to COVID-19, AY 2020-21)	26.79	20.07	39.41	0.22	5.19*	0.06
COVPPSEMRG21 (Financial impacts due to COVID-19: received emergency financial assistance from primary institution, AY 2020-21)	15.75	12.27	57.65	0.12	0.62	0.01
COVPRSKCARE21 (Personal impacts due to COVID-19: children or other dependents did not attend day care or child care, AY 2020-21)	86.44	48.13	53.05	0.66	1.12	0.01
COVPRSKSCH21 (Personal impacts due to COVID-19: children or other dependents had a non-traditional schedule for school, AY 2020-21)	86.36	48.54	53.05	0.66	1.22	0.01
COVPYADDF21 (Financial impacts due to COVID-19: borrowed more money in federal student loans, AY 2020-21)	17.00	13.46	60.59	0.12	0.65	0.01

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
COVPYADDP21 (Financial impacts due to COVID-19: borrowed more money in private student loans, AY 2020-21)	17.00	13.29	60.59	0.12	0.21	#
COVPYFEWF21 (Financial impacts due to COVID-19: borrowed less money in federal student loans, AY 2020-21)	17.06	13.40	60.59	0.12	0.96*	0.02
COVPYFEWP21 (Financial impacts due to COVID-19: borrowed less money in private student loans, AY 2020-21)	17.00	13.24	61.76	0.12	0.86*	0.02
COVPYLSFAM21 (Enrollment payment impacts due to COVID-19: less money from friends and/or family, AY 2020-21)	17.04	13.28	61.18	0.12	0.35	#
COVPYMORFAM21 (Enrollment payment impacts due to COVID-19: more money from friends and/or family, AY 2020-21)	17.00	13.43	61.18	0.12	0.71	0.01
COVSOCEFF21 (COVID-19 impacts at primary institution: online-only or hybrid classes not as effective, AY 2020-21)	26.08	20.86	56.47	0.19	1.07	0.01
COVSOCGRD21 (COVID-19 impacts at primary institution: grades lower than expected, AY 2020-21)	16.97	13.38	60.00	0.12	1.20*	0.02
COVTHFEWADV21 (COVID-19 impacts at primary institution: fewer interactions with academic advisors, AY 2020-21)	16.99	13.34	60.00	0.12	1.45*	0.02
COVTHFEWTCH21 (COVID-19 impacts at primary institution: fewer interactions with teachers, AY 2020-21)	17.07	13.37	60.00	0.12	1.40*	0.02
COVTHLSCON21 (COVID-19 impacts at primary institution: less connected to other students, AY 2020-21)	16.97	13.41	60.00	0.12	0.96*	0.01
COVTHLSRES21 (COVID-19 impacts at primary institution: accessed campus learning resources less, AY 2020-21)	16.97	13.41	60.59	0.12	1.42*	0.02
COVTSUPP21 (COVID-19 impacts at primary institution: reasonable teacher expectations, AY 2020-21)	16.96	13.56	60.00	0.12	1.18*	0.02
CRDBAL22 (Credit cards: balance due, AY 2021-22)	13.87	11.21	56.47	0.11	8.93*	0.03
CRDPAY22 (Credit cards: pay off balance, AY 2021-22)	12.63	10.19	51.76	0.11	0.31	#
CUMULN22 (Cumulative student loans: total amount borrowed through AY 2021-22)	15.93	11.28	58.58	0.09	4.46*	0.02
CURCNFDIF (Academic confidence: change between AY 2019-20 to AY 2020-21/2021-22)	18.96	14.27	59.76	0.12	10.93*	0.11
CURCONF22 (Academic confidence, AY 2020-21/2021-22)	14.52	10.77	58.82	0.10	2.01*	0.02
DECMJ22 (Field of study: formally declared when last enrolled through AY 2021-22)	11.18	8.91	56.80	0.10	1.39*	‡

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
DEGNUM3Y (Number of degrees pursued through AY 2021-22)	9.20	6.87	51.76	0.09	2.43*	‡
DEPANY22 (Had any dependents, AY 2021-22)	10.45	7.99	56.47	0.10	0.05	#
DEPCHILD22 (Had dependent children, AY 2021-22)	10.39	7.88	55.88	0.10	0.02	#
DEPCOL22 (Independent students: number of dependents in college, AY 2021-22)	84.61	45.66	51.50	0.57	7.70*	‡
DEPEND22 (Dependency status, AY 2021-22)	18.72	13.67	55.29	0.21	5.18*	0.08
DEPINC22 (Dependent students: parents' income, 2019)	24.12	15.31	56.29	0.24	12.95*	0.11
DEPNUM22 (Number of dependents, AY 2021-22)	10.47	8.09	55.88	0.10	0.20	‡
DEPNUMCH22 (Number of dependent children, AY 2021-22)	10.49	8.01	55.88	0.10	0.28*	‡
DEPNUMOT22 (Number of dependents other than children, AY 2021-22)	10.56	8.10	55.88	0.10	0.30*	‡
DEPOTHER22 (Had dependents other than children, AY 2021-22)	10.45	7.99	56.47	0.10	0.03	‡
DGEXPCHG22 (Highest level of education ever expected, changes between AY 2019-20 to AY 2021-22)	15.08	10.68	55.62	0.09	9.75*	0.10
DIS16A22 (Disability: deaf or serious difficulty hearing, AY 2021-22)	11.54	8.07	57.06	0.09	0.03	#
DIS16B22 (Disability: blind or serious difficulty seeing, AY 2021-22)	11.53	8.10	57.06	0.09	0.12	#
DIS17A22 (Disability: serious difficulty concentrating/remembering/deciding, AY 2021-22)	11.55	8.03	58.24	0.09	0.14	#
DIS17B22 (Disability: serious difficulty walking or climbing stairs, AY 2021-22)	11.51	7.70	56.47	0.09	0.01	#
DISABL22 (Disability: had some type of disability, AY 2021-22)	11.52	7.83	56.47	0.09	0.01	#
DISTALL22 (Distance education: entire program at primary institution was online, AY 2020-21/2021-22)	15.00	11.18	58.82	0.11	1.08*	0.01
DISTEDUC22 (Distance education: any classes taken at primary institution taught entirely online, AY 2020-21/2021-22)	14.53	11.06	58.82	0.11	1.22*	0.02
DISTYPES22 (Disability: main type of condition or impairment, AY 2021-22)	19.65	13.49	44.12	0.15	0.98	‡
DIVECON22 (Interactions outside of class: different economic background, AY 2020-21/2021-22)	14.31	12.13	55.88	0.11	0.46	0.01

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
DIVECONFRQ22 (Frequency of meaningful and honest conversations: different economic background, AY 2020-21/2021-22)	17.72	14.73	60.00	0.13	1.37*	0.02
DIVGEN22 (Interactions outside of class: gender minority, AY 2020-21/2021-22)	14.28	12.15	56.47	0.11	0.36	#
DIVGENFRQ22 (Frequency of meaningful and honest conversations: gender minority, AY 2020-21/2021-22)	20.75	16.99	55.88	0.16	1.56*	0.02
DIVPOL22 (Interactions outside of class: different political belief, AY 2020-21/2021-22)	14.32	12.07	56.47	0.11	0.45	0.01
DIVPOLFRQ22 (Frequency of meaningful and honest conversations: different political belief, AY 2020-21/2021-22)	18.77	14.18	61.18	0.13	1.31*	0.01
DIVRACE22 (Interactions outside of class: different race or ethnicity, AY 2020-21/2021-22)	14.31	12.19	55.88	0.11	0.45	0.01
DIVRACEFRQ22 (Frequency of meaningful and honest conversations: different race or ethnicity, AY 2020-21/2021-22)	17.11	14.44	59.41	0.12	1.23*	0.01
DIVREL22 (Interactions outside of class: different religious belief, AY 2020-21/2021-22)	14.31	12.12	57.06	0.11	0.72*	0.01
DIVRELFQ22 (Frequency of meaningful and honest conversations: different religious belief, AY 2020-21/2021-22)	17.82	14.50	57.65	0.13	1.43*	0.02
DIVSEX22 (Interactions outside of class: different sexual orientation, AY 2020-21/2021-22)	14.29	12.31	55.88	0.11	0.62	0.01
DIVSEXFRQ22 (Frequency of meaningful and honest conversations: different sexual orientation, AY 2020-21/2021-22)	18.44	14.86	57.65	0.14	1.45*	0.02
DSTUINC22 (Dependent students: income, 2019)	24.03	15.44	55.69	0.24	18.99*	0.07
EMPLYAM22 (Employer aid (student and parents): amount received, AY 2021-22)	16.14	12.10	59.41	0.10	3.42	0.01
EMPMNT3Y (All jobs: number of months worked through AY 2021-22)	15.92	11.95	57.40	0.09	3.31*	0.05
EMPNUM3Y (All jobs: number of jobs through AY 2021-22)	10.65	7.43	54.12	0.09	3.56*	‡
EMRGAPPLY22 (Applied for emergency aid from primary institution, AY 2021-22)	24.02	16.31	48.82	0.21	1.57	0.02
EMRGAWARE22 (Aware of emergency aid program at primary institution, AY 2021-22)	14.60	11.92	57.06	0.11	0.81*	0.01

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
EMRGRCV22 (Received emergency aid from primary institution, AY 2021-22)	30.88	26.13	47.93	0.30	0.23	#
ENCOEN3Y (Ever co-enrolled through AY 2021-22)	11.25	8.60	55.88	0.09	8.54*	0.12
ENDTLA3Y (Date of last enrollment through AY 2021-22)	10.76	8.26	57.06	0.09	#	0.01
ENINPT3Y (Attendance intensity through AY 2021-22)	10.78	7.98	56.47	0.08	0.50	0.01
ENMNFT1 (Months enrolled full time, AY 2019-20)	10.20	7.75	53.53	0.09	1.23*	‡
ENMNFT3Y (Months enrolled full time (total) through AY 2021-22)	11.25	8.60	55.88	0.09	0.27	#
ENMNPT3Y (Months enrolled part time (total) through AY 2021-22)	11.25	8.60	55.88	0.09	1.43*	0.01
ENMNT1 (Months enrolled, AY 2019-20)	10.20	7.75	53.53	0.09	1.56*	‡
ENMNT22 (Number of months since last enrolled through AY 2021-22)	10.76	8.26	57.06	0.09	1.89*	0.01
ENMNT3Y (Months enrolled (total) through AY 2021-22)	11.25	8.60	55.88	0.09	#	#
ENMNTAA3Y (Associate's degree: months enrolled before attainment through AY 2021-22)	43.69	25.31	49.40	0.24	0.27	0.01
ENMNTBA3Y (Bachelor's degree: months enrolled before attainment through AY 2021-22)	63.16	38.68	34.03	0.44	‡	‡
ENMNTCT3Y (Certificate: months enrolled before attainment through AY 2021-22)	56.96	34.89	46.43	0.36	0.39	0.01
EVERDECAA (Ever declared a major (associate's) through AY 2021-22)	10.72	8.08	55.29	0.10	8.11*	0.08
EVERDECBA (Ever declared a major (bachelor's) through AY 2021-22)	9.19	6.87	52.94	0.08	2.39*	0.02
EVRVT22 (Ever voted in a national, state, or local election through AY 2021-22)	12.46	9.32	59.41	0.12	6.34*	0.07
EXOCC2Y22 (Expected job: type of occupation (general), AY 2021-22)	11.97	8.23	56.47	0.10	1.96*	‡
EXOCC3Y (Expected job: change to type of occupation, AY 2019-20 to AY 2022-23)	17.07	12.00	59.76	0.11	31.97*	0.33
EXOCC3Y22 (Expected job: type of occupation (detailed), AY 2021-22)	11.96	8.29	55.88	0.10	0.52*	0.01
EXOCC6Y22 (Expected job: type of occupation (specific), AY 2021-22)	11.97	8.57	52.94	0.10	0.52	0.01

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
EXPBA (Plan to continue to a bachelor's degree program within 5 years, AY 2019-20)	30.10	20.56	53.01	0.21	0.51	0.01
EXPBA22 (Plan to continue to a bachelor's degree program within 2 years, AY 2021-22)	24.06	19.00	50.59	0.22	2.62*	0.03
EXPFLD22 (Job non-monetary benefits: importance of being an expert, AY 2021-22)	11.50	7.94	57.65	0.09	0.36	#
EXPWAGE22 (Expected job: annual salary expected, AY 2021-22)	11.71	8.34	57.65	0.10	0.68	#
FACULTYIN22 (Engagement: interactions with faculty during class, AY 2020-21/2021-22)	14.58	11.14	58.82	0.10	0.41	0.01
FACULTYOUT22 (Engagement: interactions with faculty outside of class, AY 2020-21/2021-22)	14.62	11.13	58.82	0.10	0.61	0.01
FAMAMT22 (Help from family and friends: total amount received, AY 2020-21/2021-22)	14.86	11.16	58.24	0.11	0.36	‡
FAMTIM22 (Job non-monetary benefits: importance of balancing work and family, AY 2021-22)	11.58	7.93	57.65	0.09	0.23	#
FEDBEN22 (Received federal benefit: any, in 2019 or 2020)	15.14	11.01	56.47	0.19	1.16*	0.02
FEDBEN22A (Received federal benefit: Food Stamp Benefits, in 2019 or 2020)	13.91	10.19	50.59	0.18	1.03*	0.02
FEDBEN22B (Received federal benefit: Free or Reduced Price Lunch Benefits, in 2019 or 2020)	15.37	11.51	55.88	0.19	0.83*	0.02
FEDBEN22C (Received federal benefit: Supplemental Security Income Benefits, in 2019 or 2020)	13.93	10.29	50.59	0.18	0.74*	0.01
FEDBEN22D (Received federal benefit: TANF Benefits, in 2019 or 2020)	15.40	11.59	55.88	0.19	0.15	0.01
FEDBEN22E (Received federal benefit: WIC Benefits, in 2019 or 2020)	15.37	11.54	55.88	0.19	0.31	0.01
FHSUPP22 (Supportive of PSE: friends from home, AY 2020-21/2021-22)	14.52	10.89	58.24	0.11	2.02*	0.02
FINLITA22 (Financial literacy: effect of inflation on purchasing, AY 2021-22)	11.50	7.90	56.47	0.09	0.29	#
FINLITB22 (Financial literacy: effect of interest on savings, AY 2021-22)	11.49	7.75	57.06	0.09	0.69*	0.01
FINLITC22 (Financial literacy: effect of diversification on risk, AY 2021-22)	11.51	7.95	56.47	0.09	0.96*	0.01

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
FINLITTOT22 (Financial literacy: number of correct responses, AY 2021-22)	11.58	7.98	56.47	0.09	3.45*	0.04
FRQCOMEXTFAM22 (Frequency of communication: extended family, AY 2020-21/2021-22)	14.65	11.07	59.41	0.10	0.59	0.01
FRQCOMNOCOL22 (Frequency of communication: close friends that did not attend college, AY 2020-21/2021-22)	14.64	11.16	58.82	0.11	1.42*	0.02
FRQCOMNPSAS22 (Frequency of communication: close friends that attended NPSAS institution, AY 2020-21/2021-22)	14.63	11.16	58.24	0.11	0.69	0.01
FRQCOMOTCOL22 (Frequency of communication: close friends that attended other colleges, AY 2020-21/2021-22)	14.62	11.10	59.41	0.10	0.97	0.01
FRQCOMPAR22 (Frequency of communication: parents, AY 2020-21/2021-22)	17.65	12.86	60.12	0.12	1.19*	‡
FRQCOMSIB22 (Frequency of communication: siblings, AY 2020-21/2021-22)	14.62	11.13	58.82	0.11	0.59*	0.01
FSSUPP22 (Supportive of PSE: friends from school, AY 2020-21/2021-22)	14.53	10.91	58.82	0.11	2.01*	0.02
GENDER3 (Gender, as of AY 2019-20)	14.90	10.39	53.25	0.09	2.00*	‡
GPA22 (Grade point average, AY 2020-21/2021-22)	17.42	13.11	60.00	0.11	1.24*	0.07
GPA3Y (Grade point average through AY 2021-22)	15.18	12.16	61.18	0.10	1.10*	0.05
HIGHLVEX (Highest level of education ever expected, AY 2019-20)	15.00	10.43	55.03	0.09	2.71*	‡
HLPOTH22 (Job non-monetary benefits: importance of helping others, AY 2021-22)	11.52	7.93	57.65	0.09	0.58	0.01
HOMELESS22 (Experienced homelessness: previous 30 days, AY 2021-22)	11.68	8.19	56.47	0.09	1.19*	0.02
HOMESTUD22 (Owned home or paid mortgage, AY 2021-22)	14.67	11.69	53.53	0.12	0.29	0.01
HOUSESEC22 (Experienced housing insecurity: previous 12 months, AY 2021-22)	11.56	7.95	56.47	0.09	3.00*	0.04
HRSWK21 (Jobs while enrolled: hours worked, AY 2020-21)	10.29	7.65	47.06	0.08	11.61*	0.09
HRSWK22 (Jobs while enrolled: hours worked, AY 2021-22)	10.33	7.25	43.53	0.08	9.40*	0.08
HRSWK3Y (Jobs while enrolled: hours worked when last enrolled through AY 2021-22)	11.20	7.88	53.53	0.09	0.30	#
HSIZE22 (Family size (dependent and independent), AY 2021-22)	18.84	13.90	56.47	0.22	6.50*	‡

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
IMPACAD22 (PSE service importance: academic advising or planning, AY 2020-21/2021-22)	17.42	12.69	57.06	0.12	0.83	0.01
IMPCAMPSEC22 (PSE service importance: campus security or campus police, AY 2020-21/2021-22)	34.72	26.26	57.99	0.27	1.95	0.02
IMPCHLD22 (PSE service importance: child care, AY 2020-21/2021-22)	43.65	34.73	36.53	0.31	‡	‡
IMPCPP22 (PSE service importance: career services, AY 2020-21/2021-22)	22.50	17.30	53.53	0.16	1.07	0.02
IMPDISSRV22 (PSE service importance: services for students with disabilities, AY 2020-21/2021-22)	30.51	24.63	42.94	0.20	2.29	0.02
IMPDIVRSE22 (PSE service importance: diversity or multicultural student services, AY 2020-21/2021-22)	35.01	24.62	51.48	0.25	1.76	‡
IMPFINAID22 (PSE service importance: financial aid services, AY 2020-21/2021-22)	20.26	16.34	51.76	0.23	1.71*	0.02
IMPHLTH22 (PSE service importance: student health services, AY 2020-21/2021-22)	32.24	23.92	57.65	0.22	1.25	0.01
IMPMENSRV22 (PSE service importance: psychological or mental health services, AY 2020-21/2021-22)	34.84	26.43	55.88	0.25	2.02*	0.02
IMPMILSRV22 (PSE service importance: services for active military or veterans, AY 2020-21/2021-22)	32.64	20.22	28.99	0.20	0.56	‡
IMPTRPLN22 (PSE service importance: transfer advising or planning, AY 2020-21/2021-22)	23.44	17.79	49.41	0.16	1.91*	0.03
IMPTUTOR22 (PSE service importance: peer or other tutoring or skills lab, AY 2020-21/2021-22)	21.06	14.91	47.65	0.18	1.75*	0.02
INCSPS22 (Independent students: spouse's income, 2019)	85.42	39.41	40.24	0.47	3.91	0.03
INDINC22 (Independent students: student and spouse's income, 2019)	66.21	33.30	44.38	0.48	4.81	0.03
ISTUINC22 (Independent students: income, 2019)	66.21	33.30	44.38	0.48	6.70	0.03
JBEN22 (Job: eligible for life, health, and retirement benefits, 2022)	29.19	21.59	44.71	0.22	5.74*	0.06
JBEN22A (Job: eligible for life insurance, 2022)	30.11	21.00	45.29	0.24	1.27	0.01
JBEN22B (Job: eligible for health insurance, 2022)	30.11	21.59	45.88	0.24	1.62	0.02
JBEN22C (Job: eligible for retirement benefits, 2022)	29.83	21.53	45.88	0.23	0.94	0.01
JOBFT21 (Jobs while enrolled: work intensity, AY 2020-21)	10.29	7.65	47.06	0.08	10.92*	0.11

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
JOBFT22 (Jobs while enrolled: work intensity, AY 2021-22)	10.33	7.25	43.53	0.08	9.07*	0.09
JOBHRS22 (Job: hours worked weekly, 2022)	25.99	19.43	41.18	0.21	0.26	0.01
JOBLFP22 (Labor force participation status in June 2022)	27.51	20.35	40.59	0.20	8.88*	0.09
JOBLIC22 (Job: has industry certification or license, 2022)	28.58	20.63	45.88	0.22	0.91	0.01
JOBMNT3Y (All jobs: number of months worked since beginning PSE)	13.79	9.29	52.94	0.07	5.45*	0.08
JOBOCC22 (Job: occupation, 2022)	29.11	20.61	45.29	0.23	4.58*	‡
JOBRLG22 (Job: related to college studies, 2022)	27.88	20.63	43.53	0.24	1.28	0.01
JOBRLC22 (Job: job required industry certification or license, 2022)	28.56	20.64	45.88	0.22	3.34*	0.05
JOBRLFT22 (Job: related to intended future work, 2022)	28.53	20.80	45.29	0.22	0.60	0.01
JOBSELF22 (Job: self employed, 2022)	25.99	18.75	43.53	0.21	0.59	0.01
JOBST22 (Job: employment status, 2022)	23.71	17.25	42.94	0.20	2.93*	0.03
JOBTIM22 (Job: years worked in same or similar job, 2022)	28.62	20.02	44.12	0.23	0.26	0.05
KNOWCLAS22 (Knew requirements needed to complete pursued degree at primary institution)	15.29	11.51	59.41	0.11	0.79	0.01
LEISTIM22 (Job non-monetary benefits: importance of balancing work and leisure time, AY 2021-22)	11.54	7.90	57.65	0.09	0.12	#
LKOCCATHD22 (Expected job: likelihood of holding job after highest attainment, AY 2021-22)	11.65	8.24	58.82	0.10	1.02*	0.02
LNLITTOT22 (Loan literacy: number of correct responses, AY 2021-22)	11.55	7.90	56.47	0.09	1.02*	0.01
LOANLITA22 (Loan literacy: government can report unpaid debt to credit bureaus, AY 2021-22)	11.55	7.90	56.47	0.09	1.05*	0.01
LOANLITB22 (Loan literacy: government can garnish wages for unpaid federal loan debt, AY 2021-22)	11.55	7.90	56.47	0.09	0.78	0.01
LOANLITC22 (Loan literacy: government can retain tax refunds or Social Security for loan debt, AY 2021-22)	11.55	7.90	56.47	0.09	0.76	0.01
LOCALR22 (Residence while attending primary institution, AY 2020-21/2021-22)	15.04	11.64	58.24	0.11	0.78	0.01
LOCJOB21 (Jobs while enrolled: on or off campus, AY 2020-21)	19.32	14.99	61.76	0.14	0.92*	0.02
LOCJOB22 (Jobs while enrolled: on or off campus, AY 2021-22)	19.92	13.26	60.59	0.14	1.93*	0.02

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
LOCJOB3Y (Jobs while enrolled: ever on campus through AY 2021-22)	18.04	12.58	53.85	0.11	2.34*	0.03
MAJ22 (Field of study: last enrolled any degree through AY 2021-22 (23 categories))	11.27	8.99	59.17	0.10	0.12	0.04
MAJ4Y22 (Field of study: last enrolled any degree through AY 2021-22 (10 categories))	11.27	8.99	59.17	0.10	1.74*	0.03
MAJ4YBA22 (Field of study: last enrolled in bachelor's through AY 2021-22 (10 categories))	21.94	14.17	61.18	0.16	2.74*	‡
MAJAA22 (Field of study: last enrolled in associate's through AY 2021-22 (23 categories))	24.36	17.62	48.82	0.19	1.05*	0.23
MAJBA22 (Field of study: last enrolled in bachelor's through AY 2021-22 (23 categories))	21.94	14.17	61.18	0.16	0.78	0.07
MAJCHG22 (Major changes: number anywhere through AY 2021-22)	13.30	9.87	61.54	0.11	1.64*	0.02
MAJCHGAA22 (Major changes (associate's): number anywhere through AY 2021-22)	29.23	20.60	54.71	0.22	3.45*	0.04
MAJCHGBA22 (Major changes (bachelor's): number anywhere through AY 2021-22)	23.05	15.52	61.76	0.16	1.55*	0.02
MAJCT22 (Field of study: last enrolled in certificate through AY 2021-22 (23 categories))	38.64	23.71	44.05	0.28	1.93*	0.06
MEALPLAN22 (Meal plan: had meal plan, AY 2021-22)	17.34	12.65	60.00	0.12	0.14	#
MEALPLANFQ22 (Meal plan: meal plan covered 11 or more meals per week, AY 2021-22)	43.90	40.11	68.67	0.35	1.55	0.02
MENTH22 (Health: mental, AY 2021-22)	11.54	7.86	56.47	0.09	0.79	0.01
MENTHDIF (Health: mental from 2020 to 2022)	17.08	13.04	60.36	0.10	14.56*	0.16
MENTMISS22 (Health: amount missed school/work in past 30 days for mental health, AY 2021-22)	11.59	8.02	55.88	0.09	0.52*	0.01
MJNSF22 (Major field of study with a focus on NSF STEM fields, AY 2019-20)	11.27	8.99	59.17	0.10	1.40*	0.03
MNTJBNE21 (Jobs while not enrolled: number of months worked, AY 2020-21)	10.84	7.94	56.47	0.09	4.79*	0.06
MNTJBNE22 (Jobs while not enrolled: number of months worked, AY 2021-22)	10.82	7.98	57.06	0.09	5.13*	0.07

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
MNTJOB21 (Jobs while enrolled: number of months worked, AY 2020-21)	10.84	7.95	56.47	0.09	2.59*	0.03
MNTJOB22 (Jobs while enrolled: number of months worked, AY 2021-22)	10.83	7.98	57.06	0.09	3.41*	0.05
MTGAMT22 (Monthly mortgage or rent amount, AY 2021-22)	11.73	8.16	48.82	0.10	0.26	#
NUMJOB21 (Jobs while enrolled: number of jobs, AY 2020-21)	10.84	7.95	56.47	0.09	1.15*	‡
NUMJOB22 (Jobs while enrolled: number of jobs, AY 2021-22)	10.83	7.97	57.06	0.09	1.40*	‡
OTHGTAMT22 (Outside grants (private and employer): amount received, AY 2021-22)	17.12	12.50	60.00	0.11	0.19	#
PARAMT22 (Help from parents: total amount received, AY 2020-21/2021-22)	14.96	11.35	59.41	0.11	2.60*	0.03
PAREduc (Parents' highest education level, AY 2019-20)	14.83	10.86	51.48	0.09	4.90*	0.07
PARSUPP22 (Supportive of PSE: parents, AY 2020-21/2021-22)	17.55	13.00	60.12	0.12	1.25*	0.02
PEERINTIN22 (Engagement: interactions with other students during class, AY 2020-21/2021-22)	14.71	11.13	57.65	0.11	0.63	0.01
PEERINTOUT22 (Engagement: interactions with other students outside of class, AY 2020-21/2021-22)	14.69	11.11	57.65	0.11	0.68	0.01
PFAMNM22 (Dependent students: family size, AY 2021-22)	24.08	15.37	55.69	0.24	5.78*	‡
PHYSH22 (Health: physical, AY 2021-22)	11.56	8.09	58.24	0.09	0.84	0.01
PHYSHDIF (Health: physical from 2020 to 2022)	17.01	13.24	61.54	0.10	16.28*	0.17
PHYSMISS22 (Health: amount missed school/work in past 30 days for physical health, AY 2021-22)	11.60	8.03	55.88	0.09	0.24	0.01
PINCOL22 (Dependent students: number family members in college, AY 2021-22)	24.23	15.81	57.49	0.25	6.74*	‡
PLNST22 (Private student loans: repayment status, AY 2021-22)	16.14	11.37	59.17	0.09	1.10*	0.02
PMARIT22 (Dependent students: parents' marital status, AY 2021-22)	24.06	15.38	55.69	0.24	7.88*	0.09
PRAT3Y (Persistence anywhere through AY 2021-22)	10.75	8.23	57.06	0.09	1.32*	0.02
PRICUM22 (Private (alternative) loans: cumulative amount borrowed through AY 2021-22)	15.93	11.28	58.58	0.09	17.85*	0.04
PRIVAI22 (Private source grants: amount received, AY 2021-22)	17.16	12.38	60.59	0.11	2.37	#
PRLVL3Y (Attainment or level of last institution enrolled as of 2021-22 AY)	10.32	8.09	55.88	0.08	3.42*	‡

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
PROUT1_NEW (Cumulative persistence and attainment anywhere through 2019-20 AY (new version))	9.98	7.65	52.35	0.09	0.11*	‡
PROUT1_OLD (Cumulative persistence and attainment anywhere through 2019-20 AY (old version))	10.89	8.88	54.71	0.09	3.27*	‡
PROUT2_NEW (Cumulative persistence and attainment anywhere through 2020-21 AY (new version))	10.18	7.74	55.88	0.08	2.94*	‡
PROUT2_OLD (Cumulative persistence and attainment anywhere through 2020-21 AY (old version))	10.38	8.09	55.88	0.08	2.02*	‡
PROUT3_NEW (Cumulative persistence and attainment anywhere through 2021-22 AY (new version))	10.32	8.08	55.88	0.08	3.50*	‡
PROUT3_OLD (Cumulative persistence and attainment anywhere through 2021-22 AY (old version))	10.32	8.08	55.88	0.08	2.55*	‡
PROUTF1_NEW (Cumulative retention and attainment at first institution through 2019-20 (new version))	10.04	7.66	53.53	0.09	0.29*	‡
PROUTF1_OLD (Cumulative retention and attainment at first institution through 2019-20 (old version))	10.04	7.66	53.53	0.09	1.80*	‡
PRRPY22 (Private student loans: monthly payment amount, AY 2021-22)	46.69	36.30	48.50	0.38	2.36	0.02
RCVSTAD21 (Received financial aid from a state aid program, AY 2019-20)	22.00	14.97	66.09	0.15	0.64	0.01
RCVSTAD22 (Received financial aid from a state aid program, AY 2021-2022)	22.01	14.50	66.96	0.15	1.34	0.01
REGSUPP22 (Provided regular financial support to family or friend, AY 2020-21/2021-22)	11.46	7.99	58.24	0.09	0.09	#
REGVT22 (Registered to vote in U.S. elections, AY 2021-22)	12.81	10.15	61.18	0.12	5.20*	0.08
REMEV3Y (Ever taken remedial courses through AY 2021-22)	13.58	9.08	48.82	0.08	6.44*	0.07
SALARY22 (Job: annual salary, 2022)	27.84	19.41	44.71	0.21	1.19	0.02
SEEND3Y (First enrollment spell: date ended through AY 2021-22)	11.25	8.60	55.88	0.09	#	0.04
SEINT3Y (First enrollment spell: intensity through AY 2021-22)	11.25	8.60	55.88	0.09	2.55*	0.03
SEMON3Y (First enrollment spell: months enrolled through AY 2021-22)	11.25	8.60	55.88	0.09	1.58*	0.03
SENSBELNG22 (Engagement: felt like a part of institution, AY 2020-21/2021-22)	14.21	11.81	57.06	0.11	1.33*	0.02

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
SENUM3Y (Enrollment spells: number of spells through AY 2021-22)	11.25	8.60	55.88	0.09	6.63*	‡
SEROLE22 (Jobs while enrolled: primary role as student or employee, AY 2021-22)	18.62	11.59	52.94	0.13	0.80	0.01
SFAMNM22 (Independent students: family size, AY 2021-22)	59.57	33.05	46.15	0.43	3.84*	‡
SIEND3Y (First institution spells: date first enrollment spell ended through AY 2021-22)	11.02	8.07	56.47	0.09	#	0.03
SIINT3Y (First institution spells: intensity through AY 2021-22)	11.02	8.07	56.47	0.09	1.69*	0.02
SINCOL22 (Independent students: number of family members in college 2021-22)	60.46	29.74	43.79	0.44	0.86	‡
SOCPARTANY22 (Social activities: participated in any social activity, AY 2020-21/2021-22)	14.82	11.21	58.24	0.11	0.31	#
SOCPARTGRK22 (Social activities: participated in a Greek letter organization, AY 2020-21/2021-22)	14.64	11.04	58.82	0.11	0.03	#
SOCPARTMUS22 (Social activities: participated in institution sponsored music, theatre, or band, AY 2020-21/2021-22)	14.67	11.24	60.00	0.11	0.19	#
SOCPARTNVAR22 (Social activities: participated in non-varsity athletics, AY 2020-21/2021-22)	14.69	11.39	60.00	0.11	0.25	#
SOCPARTSTGOV22 (Social activities: participated in student government, AY 2020-21/2021-22)	14.61	11.30	60.59	0.11	0.38	0.01
SOCPARTSTGRP22 (Social activities: participated in student affinity groups, AY 2020-21/2021-22)	14.65	11.11	60.00	0.11	0.78*	0.01
SOCPARTSTORG22 (Social activities: participated in any other recognized or registered student organizations, AY 2020-21/2021-22)	14.71	11.43	59.41	0.11	2.26*	0.03
SOCPARTVAR22 (Social activities: participated in varsity athletics, AY 2020-21/2021-22)	14.70	11.44	59.41	0.11	0.47	0.01
SPNM21 (Not married to spouse in 2021)	68.31	38.12	47.34	0.42	0.01	‡
SPPSUPP22 (Supportive of PSE: spouse/partner, AY 2020-21/2021-22)	55.78	29.34	44.31	0.33	1.30	‡
SPSED22 (Independent students: spouse's highest education level, AY 2021-22)	78.80	34.23	39.13	0.48	1.20	‡
STDT3Y (First stopout through AY 2021-22: start date)	11.25	8.60	55.88	0.09	18.64*	0.09

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
STDTFI3Y (First stopout at first institution through AY 2021-22: start date)	11.02	8.07	56.47	0.09	18.39*	0.08
STDYABR3Y (Ever studied abroad while attending NPSAS institution through AY 2021-22)	16.75	12.37	57.99	0.11	1.60*	0.03
STNUM3Y (Stopouts through AY 2021-22: total)	11.25	8.60	55.88	0.09	6.63*	‡
STSTA1 (Stopout status: AY 2019-20)	11.25	8.60	55.88	0.09	6.70*	0.09
STSTA2 (Stopout status: 2020-21)	11.25	8.60	55.88	0.09	6.71*	0.09
STSTA3 (Stopout status: AY 2021-22)	11.25	8.60	55.88	0.09	6.64*	0.09
STYB3Y (First stopout through AY 2021-22: starting academic year)	11.25	8.60	55.88	0.09	6.68*	0.09
STYBFI3Y (First stopout at first institution through AY 2021-22: starting academic year)	11.02	8.07	56.47	0.09	5.03*	0.08
TCREFFCOM22 (Teacher effectiveness: had good command of subject, AY 2020-21/2021-22)	14.30	11.88	57.06	0.11	1.08	0.01
TCREFFEX22 (Teacher effectiveness: make good use of examples, AY 2020-21/2021-22)	14.30	11.80	57.06	0.11	1.38*	0.02
TCREFFINT22 (Teacher effectiveness: genuinely interested in teaching, AY 2020-21/2021-22)	14.29	11.82	57.06	0.11	0.66	0.01
TCREFFOUT22 (Teacher effectiveness: outstanding teachers, AY 2020-21/2021-22)	14.29	11.90	57.06	0.11	1.36*	0.02
TFATT3Y (Transfer (first): attainment before transfer through AY 2021-22)	11.15	8.46	55.88	0.09	1.05*	‡
TFDEG3Y (Transfer (first): degree program through AY 2021-22)	11.24	8.58	56.47	0.09	1.27*	0.03
TFIFTY3Y (Transfer (first): destination institution type through AY 2021-22)	11.25	8.60	55.88	0.09	0.67*	‡
TFINCT3Y (Transfer direction (first): institution control through AY 2021-22)	11.25	8.60	55.88	0.09	0.41*	‡
TFINLV3Y (Transfer direction (first): institution level through AY 2021-22)	11.25	8.60	55.88	0.09	0.71*	‡
TFMNDI3Y (Transfer (first): months elapsed before destination inst through AY 2021-22)	28.61	22.05	48.52	0.18	2.20*	0.06
TFMNT3Y (Transfer (first): months enrolled before destination inst through AY 2021-22)	28.61	22.05	48.52	0.18	4.18*	0.08
TFNUM3Y (Transfer status through AY 2021-22)	11.25	8.60	55.88	0.09	0.28	‡

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
TFTYPE1 (Transfer status during 2019-20 AY)	11.25	8.60	55.88	0.09	1.04*	0.03
TFTYPE2 (Transfer status during 2020-21 AY)	11.25	8.60	55.88	0.09	0.82*	0.02
TFTYPE3 (Transfer status during 2021-22 AY)	11.25	8.60	55.88	0.09	0.82*	‡
TFTYPE3Y (Transfer direction (first): type through AY 2021-22)	11.25	8.60	55.88	0.09	1.00*	0.03
TRNPLN (Plans to transfer, AY 2019-20)	15.82	11.14	56.80	0.09	4.17*	0.04
UGDEG22 (Degree program when last enrolled through AY 2021-22)	13.16	10.17	58.24	0.11	2.70*	0.05
UGLVL22 (Highest undergraduate class level through AY 2021-22)	15.40	11.36	58.82	0.15	4.15*	‡
UNEMPL22 (Unemployment: currently looking for a job, 2022)	37.88	25.27	35.50	0.30	0.41	#
USEACAD22 (PSE services used: academic advising, AY 2020-21/2021-22)	14.70	11.64	58.82	0.11	1.93*	0.02
USEANY22 (PSE services used: used any service, AY 2020-21/2021-22)	14.77	11.71	58.82	0.11	0.50	0.01
USECAMPSEC22 (PSE services used: campus security or campus police, AY 2020-21/2021-22)	14.72	11.49	58.82	0.11	0.35	#
USECHLD22 (PSE services used: child care, AY 2020-21/2021-22)	14.75	11.74	59.41	0.11	0.11	#
USECPP22 (PSE services used: career services, AY 2020-21/2021-22)	14.72	11.69	59.41	0.11	0.13	#
USEDISSRV22 (PSE services used: services for students with disabilities, AY 2020-21/2021-22)	14.76	11.69	60.59	0.11	0.39	0.01
USEDIVRSE22 (PSE services used: diversity or multicultural student services, AY 2020-21/2021-22)	14.74	11.58	60.00	0.11	0.37	0.01
USEFINAID22 (PSE services used: financial aid services, AY 2020-21/2021-22)	14.70	11.57	58.82	0.11	0.65	0.01
USEHLTH22 (PSE services used: student health services, AY 2020-21/2021-22)	14.82	11.63	60.00	0.11	1.60*	0.02
USEIDR22 (Likelihood of using income-driven repayment plan, AY 2020-21/2021-22)	28.23	21.09	51.48	0.23	2.75	0.03
USELFP22 (Likelihood of using loan forgiveness programs, AY 2020-21/2021-22)	23.70	18.75	60.59	0.20	0.84	0.01
USEMENSrv22 (PSE services used: psychological or mental health services, AY 2020-21/2021-22)	14.71	11.60	58.82	0.11	0.96*	0.01

See notes at end of table.

Table G-3. Summary of survey item nonresponse bias, by selected variables: 2022—Continued

Variable	Before imputation ¹				After imputation	
	Mean percent relative bias across characteristics ²	Median percent relative bias across characteristics ²	Percentage of characteristics with significant bias	Median effect size ³	Difference between pre- and post-imputation means ⁴	Effect size for difference ⁵
USEMILSRV22 (PSE services used: services for active military or veterans, AY 2020-21/2021-22)	14.71	11.53	58.82	0.11	0.19	#
USETRPLN22 (PSE services used: transfer advising or planning, AY 2020-21/2021-22)	14.70	11.63	58.82	0.10	1.40*	0.02
USETUTOR22 (PSE services used: peer or other tutoring or skills lab, AY 2020-21/2021-22)	14.74	11.56	59.41	0.11	0.02	#
VTNEL22 (Voted in 2020 presidential election)	11.61	8.51	56.47	0.12	0.99*	0.03
WRKSTDY3Y (Work-study job: had work-study job through AY 2021-22)	10.97	8.07	55.88	0.09	0.47	0.01
WRKTSKS22 (Job non-monetary benefits: importance of making decisions, AY 2021-22)	11.51	7.99	57.65	0.09	0.54	0.01

Rounds to zero.

‡ Reporting standards not met (fewer than 30 unweighted nonrespondents).

* The difference between the pre- and postimputation means is significant at the 0.05 level. For categorical variables, at least one category difference is significant.

¹ *Before imputation* calculations use the weighted differences between respondent and eligible-sample mean, using the BPS:20/22 base weight, times 100.² *Percent relative bias* is calculated as the ratio of estimated bias to the weighted eligible-sample mean, using the BPS:20/22 base weight, times 100.³ *Effect size* for categorical variables is calculated as the square root of the weighted sum over categories of the squared differences over eligible-sample means.⁴ For categorical variables, the *Difference between pre- and post-imputation means* is the size-weighted average percentage difference across categories before and after imputation. "Size" refers to the unweighted count of respondents in a category.⁵ For categorical variables, the *Effect size for difference* is calculated as the square root of the sum over categories of the squared differences over weighted postimputation means using the BPS:20/22 analysis weight. For continuous variables, the *Effect size for difference* is calculated as the weighted difference over the postimputation standard deviation using the BPS:20/22 analysis weight.

NOTE: AY = academic year; NPSAS = National Postsecondary Student Aid Study; PSE = postsecondary education; TANF = Temporary Assistance for Needy Families; WIC = Special Supplemental Nutrition Program for Women, Infants, and Children.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

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Table H-1. Design effects for selected variables for all eligible BPS:20/22 sample members: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	65.31	0.50	0.32	1.56	2.45
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	12.54	0.31	0.22	1.41	1.98
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	87.46	0.31	0.22	1.41	1.98
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.67	0.07	0.05	1.27	1.62
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	6.78	0.24	0.17	1.44	2.08
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	5.08	0.23	0.15	1.54	2.36
Attended more than 1 institution in 2019-20	STUDMULT > 1	6.19	0.33	0.16	2.03	4.13
Ever transferred through June 2022	TFNUM3Y > 0	9.82	0.27	0.20	1.33	1.78
Ever stopped out through June 2022	STNUM3Y > 0	18.53	0.41	0.26	1.59	2.54
Received any aid in 2019-20	TOTAID > 0	78.17	0.37	0.28	1.33	1.78
Received any loans in 2019-20	TOTLOAN2 > 0	36.02	0.39	0.32	1.22	1.50
Ever received Pell Grant through June 2022	PELLCU22 > 0	50.51	0.39	0.33	1.16	1.34
Attended full time in fall 2019	ATTEND = 1	67.35	0.48	0.31	1.53	2.34
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	41.18	0.44	0.33	1.35	1.81
Employed in 2022	JOBST22 = 1	17.65	0.43	0.26	1.68	2.84
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	3.32	0.22	0.12	1.82	3.33
Single, never married as of 2021-22	SMAR22 = 1	89.29	0.31	0.21	1.50	2.24
Had any dependent children as of 2021-22	DEPANY22 = 1	9.40	0.35	0.20	1.77	3.14
Had a disability in 2021-22	DISABL22 = 1	24.11	0.47	0.29	1.63	2.66
Ever borrowed student loans through June 2022	CUMULN22 > 0	43.94	0.46	0.33	1.38	1.89
Formally declared a major when last enrolled through 2021-22	DECMAJ22 = 1	88.81	0.34	0.21	1.59	2.53
Summary statistics						
Minimum		†	†	†	1.16	1.34
25th percentile		†	†	†	1.35	1.81
Median		†	†	†	1.50	2.24
75th percentile		†	†	†	1.59	2.54
Maximum		†	†	†	2.03	4.13

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-2. Design effects for selected variables for all eligible BPS:20/22 sample members at public less-than-2-year base-year institutions: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	24.68	7.20	3.22	2.24	5.00
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	63.41	9.96	3.60	2.77	7.66
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	36.59	9.96	3.60	2.77	7.66
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	†	†	†	†	†
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	2.52	2.08	1.17	1.77	3.14
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	60.89	10.31	3.65	2.83	7.98
Attended more than 1 institution in 2019-20	STUDMULT > 1	2.60	2.32	1.19	1.95	3.81
Ever transferred through June 2022	TFNUM3Y > 0	5.21	4.72	1.66	2.84	8.07
Ever stopped out through June 2022	STNUM3Y > 0	24.63	6.52	3.22	2.03	4.11
Received any aid in 2019-20	TOTAID > 0	71.28	11.94	3.38	3.53	12.46
Received any loans in 2019-20	TOTLOAN2 > 0	14.80	6.23	2.65	2.35	5.52
Ever received Pell Grant through June 2022	PELLCU22 > 0	49.44	8.04	3.74	2.15	4.63
Attended full time in fall 2019	ATTEND = 1	53.95	8.54	3.73	2.29	5.25
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	22.95	7.29	3.14	2.32	5.38
Employed in 2022	JOBST22 = 1	49.50	6.25	3.74	1.67	2.79
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	6.33	3.21	1.82	1.76	3.10
Single, never married as of 2021-22	SMAR22 = 1	77.52	8.10	3.12	2.60	6.74
Had any dependent children as of 2021-22	DEPANY22 = 1	19.90	8.01	2.98	2.68	7.21
Had a disability in 2021-22	DISABL22 = 1	29.57	6.20	3.41	1.82	3.31
Ever borrowed student loans through June 2022	CUMULN22 > 0	27.17	7.57	3.32	2.28	5.19
Formally declared a major when last enrolled through 2021-22	DECMAJ22 = 1	89.07	5.48	2.33	2.35	5.53
Summary statistics						
Minimum		†	†	†	1.67	2.79
25th percentile		†	†	†	1.99	3.96
Median		†	†	†	2.31	5.32
75th percentile		†	†	†	2.73	7.43
Maximum		†	†	†	3.53	12.46

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-3. Design effects for selected variables for all eligible BPS:20/22 sample members at public 2-year base-year institutions: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	51.16	0.96	0.61	1.57	2.48
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	18.92	0.63	0.48	1.33	1.77
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	81.08	0.63	0.48	1.33	1.77
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.13	0.06	0.04	1.34	1.79
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	13.03	0.54	0.41	1.32	1.73
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	5.76	0.44	0.28	1.56	2.42
Attended more than 1 institution in 2019-20	STUDMULT > 1	2.44	0.20	0.19	1.08	1.16
Ever transferred through June 2022	TFNUM3Y > 0	12.47	0.49	0.40	1.23	1.50
Ever stopped out through June 2022	STNUM3Y > 0	23.19	0.79	0.51	1.53	2.35
Received any aid in 2019-20	TOTAID > 0	66.28	0.76	0.58	1.33	1.76
Received any loans in 2019-20	TOTLOAN2 > 0	12.34	0.43	0.40	1.08	1.17
Ever received Pell Grant through June 2022	PELLCU22 > 0	52.59	0.67	0.61	1.10	1.21
Attended full time in fall 2019	ATTEND = 1	50.64	0.74	0.61	1.21	1.46
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	20.67	0.57	0.49	1.16	1.35
Employed in 2022	JOBST22 = 1	25.93	0.79	0.53	1.47	2.17
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	4.65	0.40	0.26	1.57	2.47
Single, never married as of 2021-22	SMAR22 = 1	85.33	0.55	0.43	1.27	1.60
Had any dependent children as of 2021-22	DEPANY22 = 1	12.71	0.62	0.41	1.54	2.37
Had a disability in 2021-22	DISABL22 = 1	26.56	0.79	0.54	1.46	2.14
Ever borrowed student loans through June 2022	CUMULN22 > 0	22.81	0.62	0.51	1.22	1.48
Formally declared a major when last enrolled through 2021-22	DECMAY22 = 1	84.48	0.68	0.44	1.55	2.39
Summary statistics						
Minimum		†	†	†	1.08	1.16
25th percentile		†	†	†	1.22	1.48
Median		†	†	†	1.33	1.77
75th percentile		†	†	†	1.53	2.35
Maximum		†	†	†	1.57	2.48

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-4. Design effects for selected variables for all eligible BPS:20/22 sample members at public 4-year, non-doctorate-granting, primarily subbaccalaureate base-year institutions: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	59.68	1.64	1.25	1.31	1.72
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	19.89	1.42	1.02	1.39	1.93
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	80.11	1.42	1.02	1.39	1.93
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.11	0.04	0.09	0.48	0.23
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	14.45	1.33	0.90	1.48	2.20
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	5.33	0.78	0.57	1.37	1.87
Attended more than 1 institution in 2019-20	STUDMULT > 1	3.28	0.42	0.46	0.93	0.87
Ever transferred through June 2022	TFNUM3Y > 0	14.86	1.25	0.91	1.38	1.89
Ever stopped out through June 2022	STNUM3Y > 0	25.04	1.46	1.11	1.32	1.74
Received any aid in 2019-20	TOTAID > 0	65.20	1.31	1.22	1.08	1.16
Received any loans in 2019-20	TOTLOAN2 > 0	13.94	0.84	0.89	0.95	0.90
Ever received Pell Grant through June 2022	PELLCU22 > 0	51.00	1.13	1.28	0.89	0.79
Attended full time in fall 2019	ATTEND = 1	51.94	1.41	1.28	1.11	1.22
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	21.34	1.13	1.05	1.08	1.16
Employed in 2022	JOBST22 = 1	19.37	1.43	1.01	1.41	2.00
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	4.00	0.63	0.50	1.25	1.56
Single, never married as of 2021-22	SMAR22 = 1	86.11	1.19	0.88	1.34	1.81
Had any dependent children as of 2021-22	DEPANY22 = 1	10.54	0.96	0.78	1.23	1.50
Had a disability in 2021-22	DISABL22 = 1	25.59	1.72	1.12	1.54	2.38
Ever borrowed student loans through June 2022	CUMULN22 > 0	24.87	1.31	1.10	1.19	1.40
Formally declared a major when last enrolled through 2021-22	DECMAY22 = 1	84.33	1.40	0.93	1.51	2.28
Summary statistics						
Minimum		†	†	†	0.48	0.23
25th percentile		†	†	†	1.08	1.16
Median		†	†	†	1.31	1.72
75th percentile		†	†	†	1.39	1.93
Maximum		†	†	†	1.54	2.38

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-5. Design effects for selected variables for all eligible BPS:20/22 sample members at public 4-year, non-doctorate-granting, primarily baccalaureate base-year institutions: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	75.22	1.26	1.09	1.15	1.33
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	4.36	0.61	0.52	1.19	1.41
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	95.64	0.61	0.52	1.19	1.41
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.85	0.26	0.23	1.13	1.28
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	2.17	0.45	0.37	1.21	1.47
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	1.34	0.32	0.29	1.09	1.18
Attended more than 1 institution in 2019-20	STUDMULT > 1	7.33	0.72	0.66	1.09	1.19
Ever transferred through June 2022	TFNUM3Y > 0	10.22	0.86	0.77	1.12	1.26
Ever stopped out through June 2022	STNUM3Y > 0	13.81	1.24	0.87	1.42	2.01
Received any aid in 2019-20	TOTAID > 0	88.23	0.90	0.82	1.11	1.23
Received any loans in 2019-20	TOTLOAN2 > 0	45.69	1.26	1.26	1.00	1.00
Ever received Pell Grant through June 2022	PELLCU22 > 0	51.74	1.35	1.26	1.07	1.14
Attended full time in fall 2019	ATTEND = 1	82.49	1.05	0.96	1.09	1.19
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	51.66	1.44	1.26	1.14	1.30
Employed in 2022	JOBST22 = 1	11.80	1.04	0.82	1.28	1.63
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	1.99	0.47	0.35	1.32	1.76
Single, never married as of 2021-22	SMAR22 = 1	94.29	0.72	0.59	1.23	1.52
Had any dependent children as of 2021-22	DEPANY22 = 1	4.09	0.85	0.50	1.70	2.88
Had a disability in 2021-22	DISABL22 = 1	24.12	1.44	1.08	1.33	1.76
Ever borrowed student loans through June 2022	CUMULN22 > 0	54.22	1.43	1.26	1.13	1.28
Formally declared a major when last enrolled through 2021-22	DECMJ22 = 1	90.35	1.03	0.75	1.38	1.91
Summary statistics						
Minimum		†	†	†	1.00	1.00
25th percentile		†	†	†	1.11	1.23
Median		†	†	†	1.15	1.33
75th percentile		†	†	†	1.28	1.63
Maximum		†	†	†	1.70	2.88

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-6. Design effects for selected variables for all eligible BPS:20/22 sample members at public 4-year, doctorate-granting, base-year institutions: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	83.01	0.66	0.54	1.23	1.51
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	3.46	0.36	0.26	1.38	1.91
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	96.54	0.36	0.26	1.38	1.91
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	1.38	0.19	0.17	1.15	1.32
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	1.07	0.16	0.15	1.10	1.21
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	1.00	0.26	0.14	1.82	3.31
Attended more than 1 institution in 2019-20	STUDMULT > 1	8.21	0.51	0.39	1.30	1.68
Ever transferred through June 2022	TFNUM3Y > 0	6.44	0.46	0.35	1.30	1.70
Ever stopped out through June 2022	STNUM3Y > 0	12.72	0.87	0.47	1.84	3.39
Received any aid in 2019-20	TOTAID > 0	85.55	0.59	0.50	1.18	1.39
Received any loans in 2019-20	TOTLOAN2 > 0	49.15	0.66	0.71	0.93	0.86
Ever received Pell Grant through June 2022	PELLCU22 > 0	43.75	0.73	0.71	1.03	1.05
Attended full time in fall 2019	ATTEND = 1	87.26	0.54	0.47	1.15	1.31
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	52.38	0.75	0.71	1.05	1.10
Employed in 2022	JOBST22 = 1	8.66	0.59	0.40	1.46	2.13
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	0.81	0.17	0.13	1.30	1.70
Single, never married as of 2021-22	SMAR22 = 1	95.94	0.51	0.28	1.80	3.25
Had any dependent children as of 2021-22	DEPANY22 = 1	2.42	0.41	0.22	1.86	3.45
Had a disability in 2021-22	DISABL22 = 1	21.37	0.86	0.58	1.47	2.16
Ever borrowed student loans through June 2022	CUMULN22 > 0	55.49	0.72	0.71	1.02	1.04
Formally declared a major when last enrolled through 2021-22	DECMAJ22 = 1	93.00	0.49	0.36	1.34	1.80
Summary statistics						
Minimum		†	†	†	0.93	0.86
25th percentile		†	†	†	1.15	1.31
Median		†	†	†	1.30	1.70
75th percentile		†	†	†	1.46	2.13
Maximum		†	†	†	1.86	3.45

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-7. Design effects for selected variables for all eligible BPS:20/22 sample members at private nonprofit 2-year or less base-year institutions: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	50.98	13.19	4.33	3.04	9.26
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	27.75	8.40	3.88	2.16	4.68
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	72.25	8.40	3.88	2.16	4.68
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	†	†	†	†	†
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	15.30	7.17	3.12	2.30	5.28
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	12.45	7.23	2.86	2.53	6.38
Attended more than 1 institution in 2019-20	STUDMULT > 1	39.55	21.04	4.24	4.96	24.62
Ever transferred through June 2022	TFNUM3Y > 0	2.09	2.64	1.24	2.12	4.51
Ever stopped out through June 2022	STNUM3Y > 0	14.71	10.85	3.07	3.53	12.48
Received any aid in 2019-20	TOTAID > 0	95.74	8.26	1.75	4.72	22.26
Received any loans in 2019-20	TOTLOAN2 > 0	78.15	21.00	3.58	5.86	34.34
Ever received Pell Grant through June 2022	PELLCU22 > 0	88.71	20.87	2.74	7.60	57.79
Attended full time in fall 2019	ATTEND = 1	43.18	17.63	4.30	4.11	16.86
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	78.68	23.17	3.55	6.52	42.55
Employed in 2022	JOBST22 = 1	30.25	11.69	3.98	2.94	8.62
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	3.45	1.83	1.58	1.16	1.34
Single, never married as of 2021-22	SMAR22 = 1	71.84	11.55	3.90	2.96	8.78
Had any dependent children as of 2021-22	DEPANY22 = 1	27.15	9.36	3.86	2.43	5.89
Had a disability in 2021-22	DISABL22 = 1	17.51	9.97	3.30	3.03	9.15
Ever borrowed student loans through June 2022	CUMULN22 > 0	80.88	20.09	3.41	5.89	34.71
Formally declared a major when last enrolled through 2021-22	DECMJ22 = 1	85.55	8.18	3.05	2.68	7.19
Summary statistics						
Minimum		†	†	†	1.16	1.34
25th percentile		†	†	†	2.36	5.58
Median		†	†	†	2.99	8.97
75th percentile		†	†	†	4.84	23.44
Maximum		†	†	†	7.60	57.79

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-8. Design effects for selected variables for all eligible BPS:20/22 sample members at private nonprofit 4-year, non-doctorate-granting, base-year institutions: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	77.40	1.69	0.96	1.76	3.09
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	2.88	0.85	0.38	2.21	4.86
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	97.12	0.85	0.38	2.21	4.86
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.32	0.13	0.13	0.97	0.93
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	1.12	0.42	0.24	1.74	3.04
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	1.44	0.62	0.27	2.25	5.08
Attended more than 1 institution in 2019-20	STUDMULT > 1	5.72	0.64	0.53	1.21	1.45
Ever transferred through June 2022	TFNUM3Y > 0	8.07	0.86	0.63	1.37	1.89
Ever stopped out through June 2022	STNUM3Y > 0	16.97	1.24	0.86	1.44	2.08
Received any aid in 2019-20	TOTAID > 0	89.29	1.33	0.71	1.88	3.53
Received any loans in 2019-20	TOTLOAN2 > 0	58.28	2.19	1.13	1.94	3.76
Ever received Pell Grant through June 2022	PELLCU22 > 0	47.08	2.41	1.15	2.10	4.43
Attended full time in fall 2019	ATTEND = 1	84.55	1.59	0.83	1.92	3.67
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	59.42	2.11	1.13	1.87	3.51
Employed in 2022	JOBST22 = 1	10.69	1.51	0.71	2.13	4.53
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	2.05	0.56	0.33	1.72	2.95
Single, never married as of 2021-22	SMAR22 = 1	94.06	1.01	0.54	1.85	3.44
Had any dependent children as of 2021-22	DEPANY22 = 1	3.65	0.89	0.43	2.06	4.24
Had a disability in 2021-22	DISABL22 = 1	23.66	1.53	0.98	1.57	2.45
Ever borrowed student loans through June 2022	CUMULN22 > 0	63.85	2.15	1.10	1.95	3.81
Formally declared a major when last enrolled through 2021-22	DECMAJ22 = 1	91.60	1.72	0.64	2.69	7.26
Summary statistics						
Minimum		†	†	†	0.97	0.93
25th percentile		†	†	†	1.72	2.95
Median		†	†	†	1.88	3.53
75th percentile		†	†	†	2.10	4.43
Maximum		†	†	†	2.69	7.26

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-9. Design effects for selected variables for all eligible BPS:20/22 sample members at private nonprofit 4-year, doctorate-granting, base-year institutions: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	85.74	1.43	0.71	2.01	4.06
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	2.08	0.56	0.29	1.95	3.79
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	97.92	0.56	0.29	1.95	3.79
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.67	0.19	0.17	1.17	1.37
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	1.01	0.46	0.20	2.27	5.18
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	0.40	0.22	0.13	1.72	2.94
Attended more than 1 institution in 2019-20	STUDMULT > 1	5.47	0.59	0.46	1.27	1.62
Ever transferred through June 2022	TFNUM3Y > 0	8.49	0.97	0.56	1.71	2.94
Ever stopped out through June 2022	STNUM3Y > 0	14.08	1.55	0.70	2.20	4.84
Received any aid in 2019-20	TOTAID > 0	87.21	1.56	0.68	2.30	5.29
Received any loans in 2019-20	TOTLOAN2 > 0	55.33	2.99	1.01	2.97	8.82
Ever received Pell Grant through June 2022	PELLCU22 > 0	38.63	2.03	0.99	2.05	4.22
Attended full time in fall 2019	ATTEND = 1	85.31	1.59	0.72	2.21	4.90
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	56.87	2.87	1.00	2.86	8.17
Employed in 2022	JOBST22 = 1	5.41	0.99	0.46	2.16	4.66
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	2.11	1.04	0.29	3.57	12.75
Single, never married as of 2021-22	SMAR22 = 1	95.97	1.00	0.40	2.50	6.25
Had any dependent children as of 2021-22	DEPANY22 = 1	3.18	1.04	0.36	2.93	8.60
Had a disability in 2021-22	DISABL22 = 1	19.57	1.29	0.80	1.61	2.58
Ever borrowed student loans through June 2022	CUMULN22 > 0	59.59	2.73	0.99	2.75	7.56
Formally declared a major when last enrolled through 2021-22	DECMJ22 = 1	94.27	0.74	0.47	1.58	2.50
Summary statistics						
Minimum		†	†	†	1.17	1.37
25th percentile		†	†	†	1.72	2.94
Median		†	†	†	2.16	4.66
75th percentile		†	†	†	2.50	6.25
Maximum		†	†	†	3.57	12.75

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-10. Design effects for selected variables for all eligible BPS:20/22 sample members at private for-profit less-than-2-year base-year institutions: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	16.91	2.08	1.55	1.34	1.80
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	44.60	3.11	2.06	1.51	2.29
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	55.40	3.11	2.06	1.51	2.29
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	†	†	†	†	†
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	†	†	†	†	†
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	44.60	3.11	2.06	1.51	2.29
Attended more than 1 institution in 2019-20	STUDMULT > 1	26.77	7.41	1.83	4.04	16.35
Ever transferred through June 2022	TFNUM3Y > 0	6.14	1.62	0.99	1.63	2.65
Ever stopped out through June 2022	STNUM3Y > 0	23.41	3.11	1.75	1.78	3.15
Received any aid in 2019-20	TOTAID > 0	94.21	1.25	0.97	1.30	1.69
Received any loans in 2019-20	TOTLOAN2 > 0	74.92	3.27	1.79	1.82	3.32
Ever received Pell Grant through June 2022	PELLCU22 > 0	87.19	1.84	1.38	1.33	1.77
Attended full time in fall 2019	ATTEND = 1	45.71	3.13	2.06	1.52	2.31
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	79.00	3.81	1.69	2.26	5.11
Employed in 2022	JOBST22 = 1	41.08	3.14	2.04	1.54	2.37
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	12.48	2.77	1.37	2.03	4.11
Single, never married as of 2021-22	SMAR22 = 1	69.70	3.09	1.90	1.63	2.65
Had any dependent children as of 2021-22	DEPANY22 = 1	39.55	3.67	2.02	1.82	3.30
Had a disability in 2021-22	DISABL22 = 1	26.03	3.54	1.82	1.95	3.79
Ever borrowed student loans through June 2022	CUMULN22 > 0	80.94	3.79	1.63	2.33	5.45
Formally declared a major when last enrolled through 2021-22	DECMAJ22 = 1	86.48	1.80	1.42	1.27	1.61
Summary statistics						
Minimum		†	†	†	1.27	1.61
25th percentile		†	†	†	1.51	2.29
Median		†	†	†	1.63	2.65
75th percentile		†	†	†	1.95	3.79
Maximum		†	†	†	4.04	16.35

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-11. Design effects for selected variables for all eligible BPS:20/22 sample members at private for-profit 2-year base-year institutions: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	17.88	2.79	1.29	2.17	4.70
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	49.01	3.12	1.68	1.86	3.45
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	50.99	3.12	1.68	1.86	3.45
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.21	0.34	0.15	2.23	4.96
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	7.44	1.31	0.88	1.48	2.20
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	41.36	2.81	1.66	1.70	2.89
Attended more than 1 institution in 2019-20	STUDMULT > 1	41.87	9.56	1.66	5.76	33.21
Ever transferred through June 2022	TFNUM3Y > 0	6.38	1.52	0.82	1.86	3.44
Ever stopped out through June 2022	STNUM3Y > 0	20.41	2.46	1.36	1.82	3.30
Received any aid in 2019-20	TOTAID > 0	97.80	0.72	0.49	1.45	2.10
Received any loans in 2019-20	TOTLOAN2 > 0	85.40	2.10	1.19	1.77	3.12
Ever received Pell Grant through June 2022	PELLCU22 > 0	92.59	1.40	0.88	1.59	2.53
Attended full time in fall 2019	ATTEND = 1	48.18	3.78	1.68	2.25	5.06
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	85.44	2.18	1.19	1.83	3.37
Employed in 2022	JOBST22 = 1	43.01	3.20	1.67	1.92	3.69
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	11.99	1.90	1.09	1.73	3.01
Single, never married as of 2021-22	SMAR22 = 1	72.02	3.15	1.51	2.09	4.35
Had any dependent children as of 2021-22	DEPANY22 = 1	33.70	3.35	1.59	2.11	4.43
Had a disability in 2021-22	DISABL22 = 1	21.98	1.82	1.39	1.30	1.70
Ever borrowed student loans through June 2022	CUMULN22 > 0	87.25	2.06	1.12	1.83	3.36
Formally declared a major when last enrolled through 2021-22	DECMAJ22 = 1	85.92	2.54	1.17	2.17	4.73
Summary statistics						
Minimum		†	†	†	1.30	1.70
25th percentile		†	†	†	1.73	3.01
Median		†	†	†	1.86	3.44
75th percentile		†	†	†	2.11	4.43
Maximum		†	†	†	5.76	33.21

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-12. Design effects for selected variables for all eligible BPS:20/22 sample members at private for-profit 4-year base-year institutions: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	48.24	3.34	1.31	2.55	6.51
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	13.38	2.34	0.89	2.62	6.89
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	86.62	2.34	0.89	2.62	6.89
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	3.90	1.16	0.51	2.28	5.20
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	5.86	1.96	0.61	3.19	10.17
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	3.62	0.91	0.49	1.86	3.47
Attended more than 1 institution in 2019-20	STUDMULT > 1	6.07	2.21	0.63	3.53	12.48
Ever transferred through June 2022	TFNUM3Y > 0	7.54	1.74	0.69	2.51	6.32
Ever stopped out through June 2022	STNUM3Y > 0	17.38	2.05	0.99	2.06	4.26
Received any aid in 2019-20	TOTAID > 0	95.89	1.60	0.52	3.09	9.52
Received any loans in 2019-20	TOTLOAN2 > 0	75.67	6.90	1.12	6.14	37.70
Ever received Pell Grant through June 2022	PELLCU22 > 0	79.02	5.63	1.07	5.28	27.90
Attended full time in fall 2019	ATTEND = 1	40.57	5.10	1.29	3.97	15.72
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	77.68	6.65	1.09	6.10	37.24
Employed in 2022	JOBST22 = 1	24.33	3.23	1.12	2.87	8.26
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	4.77	1.13	0.56	2.03	4.13
Single, never married as of 2021-22	SMAR22 = 1	72.39	3.39	1.17	2.90	8.41
Had any dependent children as of 2021-22	DEPANY22 = 1	31.96	3.89	1.22	3.18	10.14
Had a disability in 2021-22	DISABL22 = 1	31.77	3.39	1.22	2.78	7.75
Ever borrowed student loans through June 2022	CUMULN22 > 0	80.64	5.60	1.03	5.41	29.31
Formally declared a major when last enrolled through 2021-22	DECMJ22 = 1	91.49	1.61	0.73	2.20	4.84
Summary statistics						
Minimum		†	†	†	1.86	3.47
25th percentile		†	†	†	2.51	6.32
Median		†	†	†	2.87	8.26
75th percentile		†	†	†	3.53	12.48
Maximum		†	†	†	6.14	37.70

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-13. Design effects for selected variables for Asian, not Hispanic or Latino BPS:20/22 sample members: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	80.30	1.38	1.01	1.36	1.84
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	9.24	0.98	0.74	1.33	1.76
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	90.76	0.98	0.74	1.33	1.76
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.81	0.24	0.23	1.06	1.12
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	5.16	0.72	0.56	1.27	1.62
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	3.27	0.62	0.45	1.37	1.88
Attended more than 1 institution in 2019-20	STUDMULT > 1	6.71	0.78	0.64	1.22	1.50
Ever transferred through June 2022	TFNUM3Y > 0	8.09	0.81	0.70	1.16	1.35
Ever stopped out through June 2022	STNUM3Y > 0	14.53	1.33	0.90	1.48	2.18
Received any aid in 2019-20	TOTAID > 0	69.15	1.48	1.18	1.26	1.59
Received any loans in 2019-20	TOTLOAN2 > 0	23.32	1.46	1.08	1.35	1.82
Ever received Pell Grant through June 2022	PELLCU22 > 0	43.80	1.54	1.27	1.22	1.48
Attended full time in fall 2019	ATTEND = 1	74.22	1.57	1.12	1.41	1.99
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	27.06	1.52	1.13	1.35	1.81
Employed in 2022	JOBST22 = 1	7.46	1.00	0.67	1.49	2.23
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	2.44	0.51	0.39	1.30	1.68
Single, never married as of 2021-22	SMAR22 = 1	94.44	0.89	0.58	1.52	2.30
Had any dependent children as of 2021-22	DEPANY22 = 1	4.06	0.76	0.50	1.50	2.26
Had a disability in 2021-22	DISABL22 = 1	17.43	1.46	0.97	1.51	2.28
Ever borrowed student loans through June 2022	CUMULN22 > 0	29.73	1.62	1.17	1.39	1.92
Formally declared a major when last enrolled through 2021-22	DECMJ22 = 1	91.10	1.02	0.73	1.40	1.97
Summary statistics						
Minimum		†	†	†	1.06	1.12
25th percentile		†	†	†	1.27	1.62
Median		†	†	†	1.35	1.82
75th percentile		†	†	†	1.41	1.99
Maximum		†	†	†	1.52	2.30

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-14. Design effects for selected variables for Black, not Hispanic or Latino BPS:20/22 sample members: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	56.66	1.35	0.89	1.53	2.33
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	10.67	0.93	0.55	1.69	2.84
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	89.33	0.93	0.55	1.69	2.84
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.26	0.08	0.09	0.85	0.72
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	6.21	0.75	0.43	1.73	2.99
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	4.20	0.48	0.36	1.35	1.82
Attended more than 1 institution in 2019-20	STUDMULT > 1	6.98	0.76	0.46	1.67	2.79
Ever transferred through June 2022	TFNUM3Y > 0	10.73	0.77	0.55	1.38	1.92
Ever stopped out through June 2022	STNUM3Y > 0	21.51	1.08	0.74	1.46	2.15
Received any aid in 2019-20	TOTAID > 0	86.78	0.97	0.61	1.60	2.56
Received any loans in 2019-20	TOTLOAN2 > 0	47.98	1.19	0.89	1.33	1.78
Ever received Pell Grant through June 2022	PELLCU22 > 0	75.44	1.12	0.77	1.46	2.13
Attended full time in fall 2019	ATTEND = 1	58.95	1.53	0.88	1.74	3.03
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	55.32	1.17	0.89	1.32	1.74
Employed in 2022	JOBST22 = 1	18.58	1.12	0.70	1.62	2.61
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	5.25	0.63	0.40	1.57	2.47
Single, never married as of 2021-22	SMAR22 = 1	90.64	0.85	0.52	1.63	2.67
Had any dependent children as of 2021-22	DEPANY22 = 1	16.17	1.16	0.66	1.77	3.12
Had a disability in 2021-22	DISABL22 = 1	21.84	1.13	0.74	1.53	2.34
Ever borrowed student loans through June 2022	CUMULN22 > 0	56.77	1.19	0.89	1.35	1.81
Formally declared a major when last enrolled through 2021-22	DECMAJ22 = 1	89.14	0.86	0.56	1.55	2.40
Summary statistics						
Minimum		†	†	†	0.85	0.72
25th percentile		†	†	†	1.38	1.92
Median		†	†	†	1.55	2.40
75th percentile		†	†	†	1.67	2.79
Maximum		†	†	†	1.77	3.12

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-15. Design effects for selected variables for Hispanic, of any race BPS:20/22 sample members: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	60.15	0.94	0.68	1.39	1.94
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	13.37	0.65	0.47	1.37	1.89
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	86.63	0.65	0.47	1.37	1.89
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.44	0.10	0.09	1.12	1.25
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	6.82	0.44	0.35	1.26	1.60
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	6.12	0.49	0.33	1.47	2.17
Attended more than 1 institution in 2019-20	STUDMULT > 1	5.92	0.62	0.33	1.89	3.58
Ever transferred through June 2022	TFNUM3Y > 0	8.49	0.50	0.39	1.30	1.70
Ever stopped out through June 2022	STNUM3Y > 0	20.07	0.95	0.55	1.72	2.95
Received any aid in 2019-20	TOTAID > 0	79.53	0.83	0.56	1.49	2.23
Received any loans in 2019-20	TOTLOAN2 > 0	26.40	0.80	0.61	1.31	1.71
Ever received Pell Grant through June 2022	PELLCU22 > 0	62.52	0.94	0.67	1.41	1.99
Attended full time in fall 2019	ATTEND = 1	62.31	1.03	0.67	1.54	2.39
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	30.94	0.89	0.64	1.39	1.92
Employed in 2022	JOBST22 = 1	18.72	0.91	0.54	1.69	2.86
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	4.31	0.48	0.28	1.70	2.88
Single, never married as of 2021-22	SMAR22 = 1	89.35	0.58	0.43	1.37	1.87
Had any dependent children as of 2021-22	DEPANY22 = 1	11.25	0.72	0.44	1.65	2.72
Had a disability in 2021-22	DISABL22 = 1	26.17	0.91	0.61	1.50	2.25
Ever borrowed student loans through June 2022	CUMULN22 > 0	33.05	0.88	0.65	1.36	1.84
Formally declared a major when last enrolled through 2021-22	DECMJ22 = 1	86.15	0.84	0.48	1.76	3.09
Summary statistics						
Minimum		†	†	†	1.12	1.25
25th percentile		†	†	†	1.37	1.87
Median		†	†	†	1.41	1.99
75th percentile		†	†	†	1.65	2.72
Maximum		†	†	†	1.89	3.58

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-16. Design effects for selected variables for White, not Hispanic or Latino BPS:20/22 sample members: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	67.80	0.67	0.44	1.51	2.28
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	13.80	0.46	0.33	1.41	2.00
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	86.20	0.46	0.33	1.41	2.00
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.73	0.09	0.08	1.13	1.28
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	7.74	0.36	0.25	1.41	1.99
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	5.33	0.33	0.21	1.54	2.36
Attended more than 1 institution in 2019-20	STUDMULT > 1	6.08	0.37	0.23	1.64	2.69
Ever transferred through June 2022	TFNUM3Y > 0	10.70	0.40	0.29	1.36	1.84
Ever stopped out through June 2022	STNUM3Y > 0	17.99	0.64	0.36	1.75	3.06
Received any aid in 2019-20	TOTAID > 0	77.29	0.57	0.40	1.45	2.09
Received any loans in 2019-20	TOTLOAN2 > 0	39.48	0.76	0.46	1.65	2.71
Ever received Pell Grant through June 2022	PELLCU22 > 0	39.13	0.67	0.46	1.44	2.08
Attended full time in fall 2019	ATTEND = 1	70.79	0.73	0.43	1.69	2.86
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	44.74	0.79	0.47	1.68	2.83
Employed in 2022	JOBST22 = 1	18.11	0.63	0.37	1.71	2.94
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	2.71	0.30	0.15	1.92	3.70
Single, never married as of 2021-22	SMAR22 = 1	87.92	0.51	0.31	1.66	2.77
Had any dependent children as of 2021-22	DEPANY22 = 1	8.03	0.44	0.26	1.70	2.88
Had a disability in 2021-22	DISABL22 = 1	24.22	0.64	0.41	1.57	2.47
Ever borrowed student loans through June 2022	CUMULN22 > 0	48.08	0.82	0.47	1.74	3.02
Formally declared a major when last enrolled through 2021-22	DECMAJ22 = 1	89.52	0.52	0.29	1.79	3.20
Summary statistics						
Minimum		†	†	†	1.13	1.28
25th percentile		†	†	†	1.44	2.08
Median		†	†	†	1.64	2.69
75th percentile		†	†	†	1.70	2.88
Maximum		†	†	†	1.92	3.70

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-17. Design effects for selected variables for BPS:20/22 sample members of another race, not Hispanic or Latino: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	59.46	2.02	1.35	1.50	2.24
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	10.32	1.17	0.84	1.40	1.96
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	89.68	1.17	0.84	1.40	1.96
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	1.38	0.45	0.32	1.40	1.96
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	4.32	0.65	0.56	1.17	1.36
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	4.61	0.88	0.58	1.53	2.35
Attended more than 1 institution in 2019-20	STUDMULT > 1	5.72	0.88	0.64	1.38	1.89
Ever transferred through June 2022	TFNUM3Y > 0	9.15	0.97	0.79	1.22	1.48
Ever stopped out through June 2022	STNUM3Y > 0	17.99	1.60	1.06	1.52	2.30
Received any aid in 2019-20	TOTAID > 0	77.76	1.68	1.14	1.47	2.16
Received any loans in 2019-20	TOTLOAN2 > 0	39.34	2.07	1.34	1.54	2.38
Ever received Pell Grant through June 2022	PELLCU22 > 0	53.42	1.86	1.37	1.36	1.84
Attended full time in fall 2019	ATTEND = 1	65.42	2.20	1.31	1.68	2.83
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	44.09	2.00	1.37	1.46	2.14
Employed in 2022	JOBST22 = 1	23.41	1.80	1.16	1.54	2.38
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	2.35	0.57	0.42	1.37	1.86
Single, never married as of 2021-22	SMAR22 = 1	88.39	1.31	0.88	1.49	2.22
Had any dependent children as of 2021-22	DEPANY22 = 1	8.68	1.14	0.77	1.48	2.18
Had a disability in 2021-22	DISABL22 = 1	29.65	2.25	1.26	1.79	3.22
Ever borrowed student loans through June 2022	CUMULN22 > 0	47.38	2.09	1.37	1.52	2.31
Formally declared a major when last enrolled through 2021-22	DECMAJ22 = 1	88.50	1.36	0.88	1.54	2.39
Summary statistics						
Minimum		†	†	†	1.17	1.36
25th percentile		†	†	†	1.40	1.96
Median		†	†	†	1.48	2.18
75th percentile		†	†	†	1.53	2.35
Maximum		†	†	†	1.79	3.22

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-18. Design effects for selected variables for male BPS:20/22 sample members: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	63.81	0.81	0.50	1.60	2.57
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	11.45	0.51	0.33	1.54	2.36
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	88.55	0.51	0.33	1.54	2.36
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.51	0.11	0.07	1.50	2.26
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	6.58	0.35	0.26	1.34	1.80
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	4.35	0.37	0.21	1.75	3.06
Attended more than 1 institution in 2019-20	STUDMULT > 1	5.10	0.39	0.23	1.69	2.86
Ever transferred through June 2022	TFNUM3Y > 0	9.30	0.38	0.30	1.24	1.53
Ever stopped out through June 2022	STNUM3Y > 0	18.86	0.61	0.41	1.50	2.24
Received any aid in 2019-20	TOTAID > 0	74.66	0.58	0.46	1.28	1.63
Received any loans in 2019-20	TOTLOAN2 > 0	32.91	0.65	0.49	1.32	1.75
Ever received Pell Grant through June 2022	PELLCU22 > 0	44.81	0.67	0.52	1.29	1.67
Attended full time in fall 2019	ATTEND = 1	66.95	0.68	0.49	1.38	1.90
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	36.92	0.65	0.51	1.28	1.64
Employed in 2022	JOBST22 = 1	19.26	0.67	0.41	1.63	2.66
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	3.65	0.32	0.20	1.65	2.71
Single, never married as of 2021-22	SMAR22 = 1	91.36	0.51	0.29	1.72	2.97
Had any dependent children as of 2021-22	DEPANY22 = 1	6.43	0.47	0.26	1.81	3.28
Had a disability in 2021-22	DISABL22 = 1	19.20	0.66	0.41	1.60	2.57
Ever borrowed student loans through June 2022	CUMULN22 > 0	40.07	0.67	0.51	1.30	1.69
Formally declared a major when last enrolled through 2021-22	DECMAJ22 = 1	88.29	0.50	0.34	1.50	2.24
Summary statistics						
Minimum		†	†	†	1.24	1.53
25th percentile		†	†	†	1.32	1.75
Median		†	†	†	1.50	2.26
75th percentile		†	†	†	1.63	2.66
Maximum		†	†	†	1.81	3.28

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table H-19. Design effects for selected variables for female BPS:20/22 sample members: 2022

Variable	Defined as	Percent estimate	Design standard error	Simple random sample standard error	DEFT ¹	DEFF ²
Still enrolled at any institution through June 2022	PRAT3Y = 1, 3	66.59	0.64	0.42	1.53	2.34
Attained a degree or certificate at any institution through June 2022	PRAT3Y = 1, 2	13.53	0.40	0.30	1.33	1.77
No degree or certificate at any institution through June 2022	PRAT3Y = 3, 4	86.47	0.40	0.30	1.33	1.77
Attained a bachelor's degree anywhere through June 2022	PROUT3_OLD = 1	0.80	0.09	0.08	1.12	1.25
Attained an associate's degree anywhere through June 2022	PROUT3_OLD = 2	6.99	0.32	0.22	1.43	2.04
Attained a certificate degree anywhere through June 2022	PROUT3_OLD = 3	5.73	0.29	0.20	1.42	2.01
Attended more than 1 institution in 2019-20	STUDMULT > 1	7.19	0.48	0.23	2.12	4.48
Ever transferred through June 2022	TFNUM3Y > 0	10.31	0.36	0.27	1.35	1.82
Ever stopped out through June 2022	STNUM3Y > 0	18.19	0.58	0.34	1.71	2.92
Received any aid in 2019-20	TOTAID > 0	81.28	0.49	0.34	1.41	2.00
Received any loans in 2019-20	TOTLOAN2 > 0	38.85	0.61	0.43	1.43	2.04
Ever received Pell Grant through June 2022	PELLCU22 > 0	55.51	0.56	0.44	1.28	1.63
Attended full time in fall 2019	ATTEND = 1	67.69	0.75	0.41	1.83	3.35
Received any loans (Subsidized, Unsubsidized, or PLUS) through June 2022	T4TCUM22 > 0	45.03	0.64	0.44	1.45	2.11
Employed in 2022	JOBST22 = 1	16.24	0.52	0.32	1.61	2.59
For unemployed, currently looking for job between January and June 2022	UNEMPL22 = 1	2.99	0.26	0.15	1.71	2.94
Single, never married as of 2021-22	SMAR22 = 1	87.55	0.48	0.29	1.64	2.70
Had any dependent children as of 2021-22	DEPANY22 = 1	12.10	0.50	0.29	1.76	3.09
Had a disability in 2021-22	DISABL22 = 1	27.53	0.62	0.39	1.58	2.51
Ever borrowed student loans through June 2022	CUMULN22 > 0	47.49	0.64	0.44	1.45	2.10
Formally declared a major when last enrolled through 2021-22	DECMAJ22 = 1	89.34	0.42	0.27	1.53	2.35
Summary statistics						
Minimum		†	†	†	1.12	1.25
25th percentile		†	†	†	1.41	2.00
Median		†	†	†	1.45	2.11
75th percentile		†	†	†	1.64	2.70
Maximum		†	†	†	2.12	4.48

† Not applicable.

Rounds to zero.

¹ DEFT is the square root of DEFF and can also be defined as the ratio of the design-based standard error over the standard error that would have been obtained from a simple random sample of the same size (if that were practical).² DEFF is the survey design effect for a statistic and is defined as the ratio of the design-based variance estimate over the variance estimate that would have been obtained from a simple random sample of the same size (if that were practical).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Appendix I. Imputations

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Table I-1. Weighted item response rates, by control and level of institution and variable: 2022

			Public						Private nonprofit			Private for-profit		
					4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year	
Variable	Sample Size	All sample members	Less-than-2-year	2-year										
ACDPARTANY22 (Academic activities: participated in any academic activity, AY 2020-21/2021-22)	19,120	84.4	68.8	81.45	80.38	85.48	87.99	77.51	85.66	89.11	72.06	65.36	84.2	
ACDPARTAPP22 (Academic activities: participated in an apprenticeship, AY 2020-21/2021-22)	19,120	84.71	64.19	81.89	80.84	86.05	88.22	77.51	85.71	89.35	72.06	65.38	84.7	
ACDPARTCOM22 (Academic activities: participated in community-based or service learning project, AY 2020-21/2021-22)	19,120	84.71	68.8	81.75	81.07	85.91	88.37	77.51	85.63	89.26	71.36	65.41	84.7	
ACDPARTIF22 (Academic activities: participated in internship or field experiment, AY 2020-21/2021-22)	19,120	84.75	68.8	81.99	80.76	85.76	88.4	77.51	85.71	89.26	70.25	64.88	84.69	
ACDPARTLC22 (Academic activities: participated in learning community, AY 2020-21/2021-22)	19,120	84.74	68.8	81.91	80.91	85.89	88.27	77.51	85.7	89.31	72.06	65.41	84.65	
ACDPARTRSCH22 (Academic activities: participated in guided research experience, AY 2020-21/2021-22)	19,120	84.63	68.8	81.76	81.01	85.86	88.23	77.51	85.75	89.08	71.82	65.38	83.47	
ACDPARTSE22 (Academic activities: participated in culminating senior experience, AY 2020-21/2021-22)	19,120	84.74	68.8	81.96	80.67	86.1	88.19	77.51	85.8	89.4	72.06	65.4	84.55	
ACDPARTSSC22 (Academic activities: participated in student success course, AY 2020-21/2021-22)	19,120	84.77	68.8	81.99	80.8	86.07	88.29	77.51	85.8	89.24	72.06	65.41	84.57	
AGE (Age, as of 12/31/2019)	22,320	99.98	100	100	100	100	100	100	100	99.83	100	100	100	
ALTEARN22 (Alternative plans: annual earning estimate if had not attended college, AY 2021-22)	15,700	74.85	49.92	70.48	68.83	74.1	79.56	72.1	77.01	80.83	45.45	43.99	70	
ALTWORK22 (Alternative plans: work status if had not attended college, AY 2021-22)	16,600	78.47	55.37	74.66	74.37	77.97	82.47	76.67	80.39	83.33	49.97	50.32	76.38	
AT1DT3Y (First degree date attained through AY 2021-22)	22,320	99.9	100	99.81	99.91	100	99.98	99.93	100	99.94	99.73	99.69	100	

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
AT1TY3Y (First degree type attained through AY 2021-22)	22,320	98.19	99.96	98.53	98.64	97.4	97.71	74.64	98.61	98.15	98.96	97.46	99.29
ATAADT3Y (First associate's degree date attained through AY 2021-22)	10,150	96.27	‡	98.44	98.48	85.83	81.79	67.5	88.41	80.71	‡	90.55	97.64
ATAANU3Y (Number of associate's degrees attained through AY 2021-22)	10,150	96.27	‡	98.44	98.48	85.83	81.79	67.5	88.41	80.71	‡	90.55	97.64
ATBADT3Y (First bachelor's degree date attained through AY 2021-22)	13,280	96.81	‡	90.39	95.57	97.26	97.64	‡	98.49	98.1	‡	‡	99.03
ATBANU3Y (Number of bachelor's degrees attained through AY 2021-22)	13,280	96.81	‡	90.39	95.57	97.26	97.64	‡	98.49	98.1	‡	‡	99.03
ATCTDT3Y (First certificate date attained through AY 2021-22)	4,520	88.61	99.96	92.46	91.52	67.67	66.48	55.24	75.7	57.49	98.95	96.96	94.49
ATCTNU3Y (Number of certificates attained through AY 2021-22)	4,520	88.61	99.96	92.46	91.52	67.67	66.48	55.24	75.7	57.49	98.95	96.96	94.49
ATDEG1 (Degree attained during AY 2019-20)	22,320	100	100	99.99	100	100	100	100	100	100	100	100	100
ATDEG2 (Degree attained during AY 2020-21)	22,320	99.94	100	99.88	99.97	100	99.98	100	100	99.96	99.73	99.9	100
ATDEG3 (Degree attained during AY 2021-22)	22,320	99.97	100	99.94	99.94	100	100	99.93	100	99.98	100	99.79	100
ATDEG3Y (First year attained degree through AY 2021-22)	22,320	98.35	99.96	98.76	99.03	97.53	97.77	76.45	98.72	98.25	98.96	97.87	99.3
ATHDT3Y (Date attained highest degree at any institution through AY 2021-22)	22,320	98.19	99.96	98.53	98.64	97.4	97.71	74.64	98.61	98.15	98.96	97.46	99.29
ATHDTF3Y (Date attained highest degree at first institution through AY 2021-22)	22,320	99.69	100	99.7	99.83	99.25	99.76	98.34	99.97	99.79	99.27	97.91	99.74
ATHTY3Y (Highest degree attained at any institution through AY 2021-22)	22,320	99.9	100	99.81	99.91	100	99.98	99.93	100	99.94	99.73	99.69	100
ATHTYF3Y (Highest degree attained at first institution through AY 2021-22)	22,320	99.94	100	99.85	100	100	100	100	100	100	99.73	99.69	100
ATLDT3Y (Date attained last degree through AY 2021-22)	22,320	99.9	100	99.81	99.91	100	99.98	99.93	100	99.94	99.73	99.69	100

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
ATLTY3Y (Last degree type attained through AY 2021-22)	22,320	98.19	99.96	98.53	98.64	97.4	97.71	74.64	98.61	98.15	98.96	97.46	99.29
ATNUM3Y (Degrees: number attained through AY 2021-22)	22,320	98.19	99.96	98.53	98.64	97.4	97.71	74.64	98.61	98.15	98.96	97.46	99.29
AWAREIDR22 (Aware of income-driven student loan repayment plans, AY 2020-21/2021-22)	19,120	61.09	49.28	64.88	68.45	66.03	51.27	68.15	63.6	63.33	56.8	58.91	75.28
AWARELFP22 (Aware of student loan forgiveness programs, AY 2020-21/2021-22)	19,120	82.65	68.68	82.19	80.91	80.82	84.05	78.46	82	84.89	73.18	70.33	85.45
CAMPSAFE22 (Felt safe attending primary institution, AY 2020-21/2021-22)	19,120	81.62	62.1	78.06	77.84	83.45	85.38	76.78	82.98	87.4	68.87	62.85	80.06
CARAMT22 (Monthly car loan or lease payment, AY 2021-22)	22,320	83.04	70.1	82.33	81.03	82.09	84.56	77.83	84.09	85.47	78.91	75.39	84.08
CARLN22 (Had a car loan or lease, AY 2021-22)	22,320	83.47	70.6	82.8	81.56	82.71	84.83	80.62	84.46	85.53	80.92	76.2	84.68
CINCOME (Total income (continuous), 2017)	22,320	82.47	57.33	78.64	78.05	89.54	86.09	93.65	82.48	81.23	92.99	94.41	91.3
CINCOME22 (Total income (continuous), 2019)	16,600	65.49	33.06	63.82	63.27	70.8	66.85	69.85	65.28	65.74	49.92	49.53	76.19
CITZN22 (Citizenship, AY 2021-22)	22,320	96.44	90.09	94.73	94.85	97.93	97.67	97.37	97.99	98.65	96.04	96.49	98.17
CLSEVR3Y (Ever took classes only (not a degree program) through AY 2021-22)	22,320	98.19	99.96	98.53	98.64	97.4	97.71	74.64	98.61	98.15	98.96	97.46	99.29
COVATND21 (Attended any postsecondary institution during COVID-19, AY 2020-21)	22,320	91.26	85.73	92.22	90.81	90.42	90.8	86.91	89.19	91.97	88.78	86.62	93.77
COVCHNGINC21 (Educational impacts due to COVID-19: increased enrollment from part-time to full-time, AY 2020-21)	18,460	78.71	56.47	74.78	75.71	80.84	82.9	75.49	79.71	84.88	59.37	58.92	74.28
COVCHNGINS21 (Educational impacts due to COVID-19: changed institution where enrolled, AY 2020-21)	18,460	78.71	56.47	74.78	75.71	80.84	82.9	75.49	79.71	84.88	59.37	58.92	74.28

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

			Public						Private nonprofit			Private for-profit		
					4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	4-year, non-doctorate-granting		4-year, doctorate-granting				
Variable	Sample Size	All sample members	Less-than-2-year	2-year					Less-than-4-year			Less-than-2-year	2-year	4-year
COVCHNGMAJ21 (COVID-19 impacts at primary institution: changed declared or intended major, AY 2020-21)	18,460	78.71	56.47	74.78		75.71	80.84	82.9	75.49	79.71	84.88	59.37	58.92	74.28
COVCHNGNENR21 (Educational impacts due to COVID-19: did not attend for entire term/semester, AY 2020-21)	18,460	78.71	56.47	74.78		75.71	80.84	82.9	75.49	79.71	84.88	59.37	58.92	74.28
COVCHNGRED21 (Educational impacts due to COVID-19: reduced enrollment from full-time to part-time, AY 2020-21)	18,460	78.71	56.47	74.78		75.71	80.84	82.9	75.49	79.71	84.88	59.37	58.92	74.28
COVDEVHARD21 (Primary institution online coursetaking during COVID-19: device has necessary hardware, AY 2020-21)	15,190	71.06	37.89	59.88		61.62	75.74	80.12	45.76	77.06	81.98	45.4	44.77	56.54
COVDEVSOFT21 (Primary institution online coursetaking during COVID-19: device has necessary software, AY 2020-21)	15,190	71.07	37.89	59.89		61.62	75.74	80.12	45.76	77.06	81.98	45.4	44.77	56.54
COVDISACCES21 (Primary institution provided COVID-19 accommodations: access online course content, AY 2020-21)	5,460	38.03	15.8	33.6		33.95	39.54	42.69	25.15	44.23	44.92	25.02	14.35	33.05
COVDISCOMP21 (Primary institution provided COVID-19 accommodations: complete online coursework, AY 2020-21)	5,460	38.01	15.8	33.6		33.95	39.74	42.69	25.15	43.84	44.92	25.02	14.35	33.05
COVDISTRACT21 (Online coursetaking during COVID-19: difficulties accessing a physical space free of interruption or distraction, AY 2020-21)	14,940	65.9	29.44	54.55		56.53	69.05	74.88	35.72	71.85	77.47	38.07	38.05	52.54
COVEMRGANY21 (Personal impacts due to COVID-19: received emergency financial assistance from any source other than primary institution, AY 2020-21)	22,320	95.12	90.46	94.23		95.06	95.76	95.81	98.57	96.81	95.87	93.18	92.7	95.69

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
COVENRDIS (COVID-19 pandemic: enrollment disruptions/changes, Spring 2020)	20,870	68.97	76.65	72.71	77.24	77.33	57.25	72.61	72.95	72.22	69.7	70.98	71.72
COVEXPABS (COVID-19 pandemic: took leave of absence from NPSAS, Spring 2020)	20,870	68.97	76.65	72.71	77.24	77.33	57.25	72.61	72.95	72.22	69.7	70.98	71.72
COVEXPACTMOD21 (COVID-19 impacts at primary institution: internship, apprenticeship, student teaching, or clinical placement mod or cancelled, AY 2020-21)	17,040	74.27	51.61	65.86	66.76	78.18	81.4	58.26	78.75	83.48	55	52.69	70.37
COVEXPBRK (COVID-19 pandemic: school break extended, Spring 2020)	20,870	68.97	76.65	72.71	77.24	77.33	57.25	72.61	72.95	72.22	69.7	70.98	71.72
COVEXPCALMOD21 (COVID-19 impacts at primary institution: academic calendar modified, AY 2020-21)	17,040	74.27	51.61	65.86	66.76	78.18	81.4	58.26	78.75	83.48	55	52.69	70.37
COVEXPCAN (COVID-19 pandemic: some or all classes canceled, Spring 2020)	20,870	68.97	76.65	72.71	77.24	77.33	57.25	72.61	72.95	72.22	69.7	70.98	71.72
COVEXPFIN (COVID-19 pandemic: received emergency financial assistance from NPSAS, Spring 2020)	20,870	68.97	76.65	72.71	77.24	77.33	57.25	72.61	72.95	72.22	69.7	70.98	71.72
COVEXPFOOD (COVID-19 pandemic: had difficulty accessing or paying for food, Spring 2020)	22,320	73.65	81.5	78.31	84.11	80.57	58.77	80.02	74.81	75.92	82.77	83.04	82.51
COVEXPFOOD21 (General experience during COVID-19: had difficulty accessing or paying for food, AY 2020-21)	22,320	95.95	90.46	95.07	95.63	97.27	96.49	98.78	97.37	96.91	94.1	93.74	97.17
COVEXPGRDMOD21 (COVID-19 impacts at primary institution: modified grading system, AY 2020-21)	17,040	74.27	51.61	65.86	66.76	78.18	81.4	58.26	78.75	83.48	55	52.69	70.37
COVEXPHM (COVID-19 pandemic: had difficulty finding safe and stable housing, Spring 2020)	22,320	73.65	81.5	78.31	84.11	80.57	58.77	80.02	74.81	75.92	82.77	83.04	82.51

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
COVEXPHM21 (General experience during COVID-19: had difficulty finding safe and stable housing, AY 2020-21)	22,320	95.95	90.46	95.07	95.63	97.27	96.49	98.78	97.37	96.91	94.1	93.74	97.17
COVEXPHRS21 (General experience during COVID-19: worked more due to additional job or increased hours, AY 2020-21)	22,320	95.95	90.46	95.07	95.63	97.27	96.49	98.78	97.37	96.91	94.1	93.74	97.17
COVEXPHYB21 (COVID-19 impacts at primary institution: some or all classes delivered in hybrid format, AY 2020-21)	17,040	74.27	51.61	65.86	66.76	78.18	81.4	58.26	78.75	83.48	55	52.69	70.37
COVEXPJOB (COVID-19 pandemic: lost job or lost income because of reduced hours, Spring 2020)	22,320	73.65	81.5	78.31	84.11	80.57	58.77	80.02	74.81	75.92	82.77	83.04	82.51
COVEXPJOB21 (General experience during COVID-19: lost job or lost income because of reduced hours, AY 2020-21)	22,320	95.95	90.46	95.07	95.63	97.27	96.49	98.78	97.37	96.91	94.1	93.74	97.17
COVEXPMMOD21 (COVID-19 impacts at primary institution: class delivery method modified, AY 2020-21)	17,040	74.27	51.61	65.86	66.76	78.18	81.4	58.26	78.75	83.48	55	52.69	70.37
COVEXPMOT (COVID-19 pandemic: moved to another living situation, Spring 2020)	22,320	73.65	81.5	78.31	84.11	80.57	58.77	80.02	74.81	75.92	82.77	83.04	82.51
COVEXPMOT21 (General experience during COVID-19: moved to another living situation, AY 2020-21)	22,320	95.95	90.46	95.07	95.63	97.27	96.49	98.78	97.37	96.91	94.1	93.74	97.17
COVEXPONL (COVID-19 pandemic: some or all classes moved to online only, Spring 2020)	20,720	90.77	86.42	88.79	89.26	90.95	94.02	78.82	94.06	94.68	80.04	80.25	78.91
COVEXPONL21 (COVID-19 impacts at primary institution: some or all classes delivered entirely online, AY 2020-21)	17,040	74.27	51.61	65.86	66.76	78.18	81.4	58.26	78.75	83.48	55	52.69	70.37
COVEXPPAD (COVID-19 pandemic: moved back to permanent address, Spring 2020)	20,870	68.97	76.65	72.71	77.24	77.33	57.25	72.61	72.95	72.22	69.7	70.98	71.72

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
COVEXPPAD21 (General experience during COVID-19: moved back to permanent address, AY 2020-21)	22,320	95.95	90.46	95.07	95.63	97.27	96.49	98.78	97.37	96.91	94.1	93.74	97.17
COVEXPSAP (COVID-19 pandemic: study-abroad program canceled or modified, Spring 2020)	20,870	68.97	76.65	72.71	77.24	77.33	57.25	72.61	72.95	72.22	69.7	70.98	71.72
COVEXPWIT (COVID-19 pandemic: withdrew from NPSAS, Spring 2020)	20,870	68.97	76.65	72.71	77.24	77.33	57.25	72.61	72.95	72.22	69.7	70.98	71.72
COVFINDIS (Financial disruptions/changes due to COVID-19, Spring 2020)	20,870	68.97	76.65	72.71	77.24	77.33	57.25	72.61	72.95	72.22	69.7	70.98	71.72
COVFINDIS21 (Financial disruptions/changes due to COVID-19, AY 2020-21)	22,320	88.39	82.08	86.47	86.42	90.15	90.22	88.87	89.13	91.53	88.07	83.23	88.91
COVHOUDIS (Housing disruptions/changes due to COVID-19, Spring 2020)	20,870	68.97	76.65	72.71	77.24	77.33	57.25	72.61	72.95	72.22	69.7	70.98	71.72
COVHOUDIS21 (Housing disruptions/changes due to COVID-19, AY 2020-21)	22,320	95.95	90.46	95.07	95.63	97.27	96.49	98.78	97.37	96.91	94.1	93.74	97.17
COVINFOHLTH21 (Primary institution COVID-19 information: obtaining physical or mental healthcare, AY 2020-21)	17,040	74.78	51.87	66.56	67.31	78.64	81.63	57.69	79.21	84.07	55.17	54.93	71.06
COVINFOMHLTH21 (Primary institution COVID-19 information: obtaining mental healthcare, AY 2020-21)	17,040	74.78	51.87	66.56	67.31	78.64	81.63	57.69	79.21	84.07	55.17	54.93	71.06
COVINFOPHLTH21 (Primary institution COVID-19 information: obtaining physical healthcare, AY 2020-21)	17,040	74.73	51.87	66.46	67.36	78.65	81.57	57.69	79.21	84.06	55.17	54.93	71.06
COVINTACC21 (Residence had access to high-speed internet, AY 2020-21)	22,320	95.48	90.1	94.48	95.39	96.65	96.14	98.51	97	96.83	92.88	92.34	95.97
COVINTPROB21 (Online coursetaking during COVID-19: difficulties with internet connection, AY 2020-21)	14,940	65.8	25.55	54.45	56.21	68.93	74.77	35.72	71.85	77.54	38.07	38.05	52.54

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

			Public						Private nonprofit			Private for-profit		
					4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year	
Variable	Sample Size	All sample members	Less-than-2-year	2-year										
COVMNTDIFCON21 (Mental impacts due to COVID-19: difficulty concentrating, AY 2020-21)	22,320	95.19	88.88	94.28		94.9	96.07	95.93	98.57	96.94	96.06	92.51	92.74	96.42
COVMNTDOWN21 (Mental impacts due to COVID-19: felt down, depressed, or hopeless, AY 2020-21)	22,320	95.13	88.88	94.29		94.97	95.96	95.67	98.53	96.89	96.13	92.39	92.81	96.32
COVMNTLONLI21 (Mental impacts due to COVID-19: experienced loneliness or isolation, AY 2020-21)	22,320	95.18	88.66	94.35		94.99	96.01	95.9	98.57	96.94	95.94	92.07	92.68	96.05
COVMNTSTRESS21 (Mental impacts due to COVID-19: experienced increased stress or anxiety, AY 2020-21)	22,320	95.21	88.88	94.27		94.99	96.12	95.93	98.57	96.96	96.12	92.53	92.71	96.36
COVNOTATND21 (Decision not to attend school due to COVID-19, AY 2020-21)	5,830	54.66	59.53	63.56		58.98	44.83	36.79	36.1	39.35	37.67	62.47	55.83	67.68
COVPPSEMRG21 (Financial impacts due to COVID-19: received emergency financial assistance from primary institution, AY 2020-21)	17,040	81.82	66.35	76.5		76.6	83.69	86.57	68.89	84.05	88.68	66.75	62.87	79.62
COVPRSFOOD21 (Personal impacts due to COVID-19: accessed food assistance resources, AY 2020-21)	22,320	95	90.46	94.08		94.69	95.71	95.72	98.57	96.71	95.85	92.95	92.83	95.87
COVPRSKCARE21 (Personal impacts due to COVID-19: children or other dependents did not attend day care or child care, AY 2020-21)	2,280	49.07	44.06	53.97		47.04	53.37	15.44	63.84	34.37	19.84	76.66	67.21	82.29
COVPRSKSCH21 (Personal impacts due to COVID-19: children or other dependents had a non-traditional schedule for school, AY 2020-21)	2,280	49.13	44.06	54.06		47.25	53.37	15.44	64.75	34.37	19.84	76.56	67.35	82.29
COVPRSMRT21 (Personal impacts due to COVID-19: difficulty paying mortgage or rent, AY 2020-21)	22,320	94.97	90.46	94.08		94.42	95.81	95.59	98.3	96.59	96.14	93.16	92.57	95.83
COVPYADDF21 (Financial impacts due to COVID-19: borrowed more money in federal student loans, AY 2020-21)	17,040	74.71	51.87	66.46		67.49	78.39	81.73	57.69	78.75	83.91	55.07	53.33	70.93

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
COVPYADDP21 (Financial impacts due to COVID-19: borrowed more money in private student loans, AY 2020-21)	17,040	74.55	51.87	66.35	66.98	78.16	81.54	57.58	78.92	83.77	54.69	53.52	70.82
COVPYFEWF21 (Financial impacts due to COVID-19: borrowed less money in federal student loans, AY 2020-21)	17,040	74.59	51.87	66.36	67.13	78.33	81.62	57.69	78.83	83.81	54.64	52.99	70.76
COVPYFEWP21 (Financial impacts due to COVID-19: borrowed less money in private student loans, AY 2020-21)	17,040	74.45	51.87	66.31	66.91	78.16	81.47	57.69	78.27	83.74	54.69	53.27	70.6
COVPYLSFAM21 (Enrollment payment impacts due to COVID-19: less money from friends and/or family, AY 2020-21)	17,040	74.53	51.87	66.35	67.18	78.02	81.5	57.69	78.81	83.76	54.69	53.52	70.71
COVPYMORFAM21 (Enrollment payment impacts due to COVID-19: more money from friends and/or family, AY 2020-21)	17,040	74.52	51.87	66.25	67.31	77.94	81.42	57.69	79.05	83.8	55.17	53.76	70.81
COVSOCEFF21 (COVID-19 impacts at primary institution: online-only or hybrid classes not as effective, AY 2020-21)	15,190	40.57	13.63	23.13	24.96	48.25	52.66	14.6	53.23	56.09	17.91	24.21	21.22
COVSOCGRD21 (COVID-19 impacts at primary institution: grades lower than expected, AY 2020-21)	17,040	74.77	51.87	66.53	67.58	78.46	81.75	57.69	78.92	83.93	55.17	54.34	70.89
COVTHFEWADV21 (COVID-19 impacts at primary institution: fewer interactions with academic advisors, AY 2020-21)	17,040	74.72	51.87	66.49	67.58	78.49	81.6	57.69	79.04	83.93	55.17	54.34	70.8
COVTHFEWTCH21 (COVID-19 impacts at primary institution: fewer interactions with teachers, AY 2020-21)	17,040	74.64	51.87	66.32	67.58	78.56	81.72	57.69	79.14	83.4	54.75	54.34	70.89
COVTHLSCON21 (COVID-19 impacts at primary institution: less connected to other students, AY 2020-21)	17,040	74.8	51.87	66.55	67.58	78.58	81.74	57.69	79.16	83.97	55.17	54.34	70.89
COVTHLSRES21 (COVID-19 impacts at primary institution: accessed campus learning resources less, AY 2020-21)	17,040	74.73	51.87	66.5	67.58	78.2	81.74	57.69	79.14	83.77	54.76	54.34	70.89

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
COVTSUPP21 (COVID-19 impacts at primary institution: reasonable teacher expectations, AY 2020-21)	17,040	74.76	51.87	66.58	67.55	78.65	81.57	57.69	79.3	83.84	55.17	54.93	71.06
CRDBAL22 (Credit cards: balance due, AY 2021-22)	13,110	87.25	73.85	85.59	87.18	89.51	89.11	85.07	90.16	90.07	84.46	75.39	83.77
CRDNUM22 (Credit cards: number in own name, AY 2021-22)	22,320	96.18	87.1	95.52	96.07	96.9	96.87	99.08	97.2	96.83	94.13	93.97	96.52
CRDPAY22 (Credit cards: pay off balance, AY 2021-22)	13,110	93.32	78.03	92.41	93.86	94.36	94.41	98.76	94.34	94.12	90.05	88.86	93.77
CRTFNPATT3Y (Ever attempt to transfer credits from NPSAS institution through AY 2021-22)	22,320	94.21	86.85	93.58	94.08	93.2	94.9	92.46	94.6	95.32	93.42	92.95	95.5
CUMULN22 (Cumulative student loans: total amount borrowed through AY 2021-22)	22,320	66.82	64.73	70.63	74.15	72.29	55.34	72.3	68.69	68.62	73.21	67.06	76.13
CURCNFDIF (Academic confidence: change between AY 2019-20 to AY 2020-21/2021-22)	19,120	58.32	44.96	60.33	62.63	64.32	50.4	65.43	61.46	62.55	53.63	53.27	71.97
CURCONF22 (Academic confidence, AY 2020-21/2021-22)	19,120	81.3	65.55	78.91	77.5	81	84.51	74.41	82.65	85.4	72.29	64.78	80.18
DECMJ22 (Field of study: formally declared when last enrolled through AY 2021-22)	21,720	87.87	76.64	87.14	86.6	87.72	89.18	81.58	88.18	90.1	78.42	83.23	91.69
DEGNUM3Y (Number of degrees pursued through AY 2021-22)	22,320	98.19	99.96	98.53	98.64	97.4	97.71	74.64	98.61	98.15	98.96	97.46	99.29
DEPANY22 (Had any dependents, AY 2021-22)	22,320	94.87	84.13	94.06	92.41	96.31	95.75	94.78	97.28	96.55	95.24	90.62	94.02
DEPCHILD22 (Had dependent children, AY 2021-22)	22,320	95.15	84.13	94.34	92.7	96.43	96.12	94.78	97.4	96.81	95.3	90.87	94.29
DEPCOL22 (Independent students: number of dependents in college, AY 2021-22)	2,930	23.5	20.13	30.86	28.94	14.63	6.5	44.55	4.48	16.65	33.69	28.91	63.38
DEPEND22 (Dependency status, AY 2021-22)	16,600	68.09	38.52	68.8	66.34	72.12	67.62	74.59	66.13	67.63	52.02	55.27	81.62

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
DEPEND5B (Dependency and marital status (5 categories, separated is unmarried), AY 2019-20)	22,320	95.1	88.51	94.83	94.59	98.48	94.29	95.45	95.64	95.23	98.14	96.51	97.77
DEPIN22 (Dependent students: parents' income, 2019)	14,880	63.62	23.95	60.79	58.72	69.91	66.35	63.69	63.91	65.81	25.67	27.42	64.54
DEPNUM22 (Number of dependents, AY 2021-22)	22,320	94.75	83.48	93.89	92.32	96.25	95.73	94.71	97.1	96.54	94.6	90.2	93.93
DEPNUMCH22 (Number of dependent children, AY 2021-22)	22,320	94.97	83.48	94.14	92.59	96.33	96.01	94.48	97.22	96.8	94.85	90.27	93.58
DEPNUMOT22 (Number of dependents other than children, AY 2021-22)	22,320	94.68	83.79	93.86	92.22	96.21	95.63	94.55	97.28	96.55	94.15	89.61	93.22
DEPOTHER22 (Had dependents other than children, AY 2021-22)	22,320	94.87	84.13	94.06	92.41	96.31	95.75	94.78	97.28	96.55	95.24	90.62	94.02
DGEVR22 (Highest level of education ever expected through AY 2021-22)	22,320	98.77	98.24	98.67	98.59	98.96	98.75	99.12	99.23	99.18	97.45	97.76	99.45
DGEXPCHG22 (Highest level of education ever expected, changes between AY 2019-20 to AY 2021-22)	22,320	75.89	81.6	79.75	86.21	84.24	61.54	93.14	77.64	77.05	87.11	85.64	87.91
DIS16A22 (Disability: deaf or serious difficulty hearing, AY 2021-22)	22,320	83.84	73.04	83.34	82.46	83.01	84.93	80.69	84.54	85.72	81.59	76.31	85.03
DIS16B22 (Disability: blind or serious difficulty seeing, AY 2021-22)	22,320	83.82	73.04	83.23	82.46	82.96	84.83	80.69	84.65	85.97	81.46	76.73	85.14
DIS17A22 (Disability: serious difficulty concentrating/remembering/deciding, AY 2021-22)	22,320	83.71	72.19	83.19	82.07	82.92	84.84	80.69	84.5	85.55	81.67	76.48	85.18
DIS17B22 (Disability: serious difficulty walking or climbing stairs, AY 2021-22)	22,320	83.82	71.46	83.22	82.34	83.02	84.87	80.69	84.87	86	81.85	76.73	84.66
DISABL22 (Disability: had some type of disability, AY 2021-22)	22,320	83.45	70.6	82.89	82	82.43	84.73	80.69	84.22	85.35	80.99	76.06	84.25
DISTALL22 (Distance education: entire program at primary institution was online, AY 2020-21/2021-22)	18,590	80.28	63.88	77.37	76.83	79.23	83.78	72.25	81.28	85.19	69.42	62.41	79.99

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public						Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year	
DISTEDUC22 (Distance education: any classes taken at primary institution taught entirely online, AY 2020-21/2021-22)	19,120	83.51	66.89	80.44	80.3	84.5	87.25	75.22	84.2	88.17	72.08	65.47	81.87	
DISTYPES22 (Disability: main type of condition or impairment, AY 2021-22)	8,110	52.17	39.35	53.06	51.09	52.05	51.96	33.59	54.61	50.62	50.22	36.97	59.82	
DIVECON22 (Interactions outside of class: different economic background, AY 2020-21/2021-22)	19,120	84.56	66.27	81.71	81.04	85.96	88.27	76.23	85.23	88.84	71.7	65.41	83.83	
DIVECONFRQ22 (Frequency of meaningful and honest conversations: different economic background, AY 2020-21/2021-22)	16,070	81.08	59.33	75.37	75	83.46	86.69	71.13	83.56	87.19	64.96	57.53	73.39	
DIVGEN22 (Interactions outside of class: gender minority, AY 2020-21/2021-22)	19,120	84.61	66.27	81.75	81.04	86.04	88.19	77.51	85.13	89.33	71.7	65.41	83.69	
DIVGENFRQ22 (Frequency of meaningful and honest conversations: gender minority, AY 2020-21/2021-22)	12,190	75.09	42.66	66.69	67.66	79.11	82.3	40.79	78.65	83.89	58.49	43.47	67.31	
DIVPOL22 (Interactions outside of class: different political belief, AY 2020-21/2021-22)	19,120	84.56	66.27	81.62	80.94	86.17	88.2	77.51	85.25	89.33	71.7	65.03	83.48	
DIVPOLFRQ22 (Frequency of meaningful and honest conversations: different political belief, AY 2020-21/2021-22)	15,020	79.5	58.28	72.81	73.25	82.77	85.51	72.19	81.74	86.71	63.13	56.38	72.23	
DIVRACE22 (Interactions outside of class: different race or ethnicity, AY 2020-21/2021-22)	19,120	84.66	66.27	81.8	81.04	86.21	88.27	77.51	85.23	89.33	71.7	65.41	83.67	
DIVRACEFRQ22 (Frequency of meaningful and honest conversations: different race or ethnicity, AY 2020-21/2021-22)	16,630	81.93	57.36	76.87	76.23	84.33	87.05	74.04	84.07	87.95	64.68	58.9	75.61	
DIVREL22 (Interactions outside of class: different religious belief, AY 2020-21/2021-22)	19,120	84.59	66.27	81.66	81.01	85.95	88.21	77.51	85.27	89.33	71.7	65.12	83.83	
DIVRELFQ22 (Frequency of meaningful and honest conversations: different religious belief, AY 2020-21/2021-22)	15,600	80.43	58.54	74.22	74.6	83.14	86.17	73.08	82.62	86.94	64.07	58.51	74.58	

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

			Public						Private nonprofit			Private for-profit		
					4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year	
Variable	Sample Size	All sample members	Less-than-2-year	2-year										
DIVSEX22 (Interactions outside of class: different sexual orientation, AY 2020-21/2021-22)	19,120	84.51	66.27	81.56	80.91	86.21	88.17	77.51	84.95	89.22	71.35	65.01	83.83	
DIVSEFRQ22 (Frequency of meaningful and honest conversations: different sexual orientation, AY 2020-21/2021-22)	15,240	79.89	54.96	72.98	73.05	82.97	86.01	68.96	82.09	87.13	61.98	53.84	73.37	
DSTUINC22 (Dependent students: income, 2019)	14,880	63.8	25.09	61.33	58.87	70.1	66.35	63.69	63.91	65.85	25.04	27.45	64.71	
EMPLYAM22 (Employer aid (student and parents): amount received, AY 2021-22)	16,600	81.2	58.95	80.4	80.08	79.26	82.91	76.74	80.55	83.77	54.96	55.16	82.93	
EMPMNT3Y (All jobs: number of months worked through AY 2021-22)	22,320	61.95	58.32	66.43	69.71	68.66	52.35	65.91	58.1	60.77	64.76	57.49	72.24	
EMPMTEN3Y (All jobs: number of months worked since last enrolled through AY 2021-22)	22,320	92.92	88.64	90.09	92.8	94.65	95.31	89.92	95.23	97.2	88.62	85.62	90.84	
EMPNUM3Y (All jobs: number of jobs through AY 2021-22)	22,320	86.06	75.88	86.04	84.91	84.94	87.18	83.12	86.19	86.89	85.16	80.08	84.11	
EMRGAPPLY22 (Applied for emergency aid from primary institution, AY 2021-22)	7,260	51.5	14.99	45.41	49.14	54.26	60.83	31.35	52.65	60.15	19.98	21.04	26.98	
EMRGAWARE22 (Aware of emergency aid program at primary institution, AY 2021-22)	19,120	82.67	62.14	79.22	78.7	84.05	86.75	77.45	83.51	87.95	69.7	63.75	81.74	
EMRGRCV22 (Received emergency aid from primary institution, AY 2021-22)	5,180	30.15	2.53	26.09	30.83	29.84	39.21	16.94	27.55	35.49	11.56	5.38	10.08	
ENCOEN3Y (Ever co-enrolled through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37	
ENDTFI3Y (First institution date last enrolled through AY 2021-22)	22,320	94.94	89.67	95.71	95.56	94.56	94.15	86.88	94.78	96.02	91.1	88.55	95.95	
ENDTFS (Date of first enrollment)	22,320	94.03	90.29	93.86	95.38	95.96	91.94	92.78	97.27	96.22	91.43	92.06	96.29	
ENDTLA3Y (Date of last enrollment through AY 2021-22)	22,320	90.53	80.32	91.16	90.02	89.94	90.67	86.92	89.25	91.9	85.13	81.77	92.4	

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
ENINFM (First institution: attendance intensity first month enrolled, AY 2019-20)	22,320	94.03	90.29	93.86	95.38	95.96	91.94	92.78	97.27	96.22	91.43	92.06	96.29
ENINPT1 (Attendance intensity pattern, AY 2019-20)	22,320	94.68	91.29	94.58	96.26	96.74	92.19	97.1	97.4	96.72	93.6	95.98	96.87
ENINPT2 (Attendance intensity pattern, AY 2020-21)	22,320	91.26	85.73	92.22	90.81	90.42	90.8	86.91	89.19	91.97	88.78	86.62	93.77
ENINPT3 (Attendance intensity pattern, AY 2021-22)	22,320	91.47	86.37	92.47	90.9	90.34	90.97	86.92	89.41	92.12	89.6	87.13	94.33
ENINPT3Y (Attendance intensity through AY 2021-22)	22,320	89.94	79.72	90.79	91.81	90.44	88.56	84.95	89.81	90.86	84.47	84.1	92.03
ENINUM1 (Number of institutions attended, AY 2019-20)	22,320	94.17	90.91	94.17	95.84	96.5	91.67	95.57	96.68	96.36	92.09	94.13	96.2
ENINUM2 (Number of institutions attended, AY 2020-21)	22,320	91.26	85.73	92.22	90.81	90.42	90.8	86.91	89.19	91.97	88.78	86.62	93.77
ENINUM3 (Number of institutions attended, AY 2021-22)	22,320	91.47	86.37	92.47	90.9	90.34	90.97	86.92	89.41	92.12	89.6	87.13	94.33
ENINUM3Y (Number of institutions attended (total) through AY 2021-22)	22,320	94.79	91.87	94	95	94.28	95.48	92.46	95.45	95.76	93.79	93.46	95.79
ENLYAT3Y (Last year enrolled at any institution with no attainment through AY 2021-22)	22,320	91.2	90.82	91.84	91.33	90.31	90.88	90.05	89.41	91.91	88.15	86.91	93.22
ENLYF3Y (Last year enrolled at first institution with no attainment through AY 2021-22)	22,320	95.22	95.74	95.91	95.97	94.56	94.23	89.99	94.78	96.02	92.34	92.48	96.58
ENMNFT1 (Months enrolled full time, AY 2019-20)	22,320	92.89	87.67	93.04	94.71	95.43	90.8	90.05	95.57	95.51	87.41	84.24	94.5
ENMNFT2 (Months enrolled full time, AY 2020-21)	22,320	91.26	85.73	92.22	90.81	90.42	90.8	86.91	89.19	91.97	88.78	86.62	93.77
ENMNFT3 (Months enrolled full time, AY 2021-22)	22,320	91.47	86.37	92.47	90.9	90.34	90.97	86.92	89.41	92.12	89.6	87.13	94.33
ENMNFT3Y (Months enrolled full time (total) through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
ENMNPT3Y (Months enrolled part time (total) through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
ENMNT1 (Months enrolled, AY 2019-20)	22,320	92.89	87.67	93.04	94.71	95.43	90.8	90.05	95.57	95.51	87.41	84.24	94.5
ENMNT2 (Months enrolled, AY 2020-21)	22,320	91.26	85.73	92.22	90.81	90.42	90.8	86.91	89.19	91.97	88.78	86.62	93.77
ENMNT22 (Number of months since last enrolled through AY 2021-22)	22,320	90.53	80.32	91.16	90.02	89.94	90.67	86.92	89.25	91.9	85.13	81.77	92.4
ENMNT3 (Months enrolled, AY 2021-22)	22,320	91.47	86.37	92.47	90.9	90.34	90.97	86.92	89.41	92.12	89.6	87.13	94.33
ENMNT3Y (Months enrolled (total) through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
ENMNTAA3Y (Associate's degree: months enrolled before attainment through AY 2021-22)	2,500	70.05	‡	81.42	81.46	40.79	24.93	28.25	36.95	23.66	‡	68.47	70.03
ENMNTBA3Y (Bachelor's degree: months enrolled before attainment through AY 2021-22)	920	24.81	‡	6.65	3.12	19.41	36.77	‡	18.77	18.02	‡	‡	84.65
ENMNTCT3Y (Certificate: months enrolled before attainment through AY 2021-22)	2,490	68.26	94.26	76.63	75.51	33.75	19.73	‡	46.97	15.31	87.89	80.9	75.57
ENRPLNFT (Enrollment plans: first term after AY 2021-22)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
EVERDECAA (Ever declared a major (associate's) through AY 2021-22)	22,320	91.64	99.65	84.82	88.11	95.79	96.33	69.08	97.84	96.99	98.54	95.41	96.14
EVERDECBA (Ever declared a major (bachelor's) through AY 2021-22)	22,320	96.47	99.91	98.43	97.53	94.14	94.12	74.64	96.29	96	98.96	97.36	95.8
EVRVT22 (Ever voted in a national, state, or local election through AY 2021-22)	21,360	87.96	77.79	85.14	85.49	89.99	90.79	87.2	90.46	92.21	84.14	80.33	87.49
EXOCC2Y22 (Expected job: type of occupation (general), AY 2021-22)	21,770	76.46	74.52	74.42	72.17	74.66	79.41	69.98	76.68	81.19	71.72	70.25	79.09
EXOCC3Y (Expected job: change to type of occupation, AY 2019-20 to AY 2022-23)	21,900	55.01	55.83	56.37	59.5	60.83	46.76	61.33	56.62	60.06	54.48	55.74	65.72
EXOCC3Y22 (Expected job: type of occupation (detailed), AY 2021-22)	21,770	76.31	74.52	74.26	72.11	74.27	79.25	69.98	76.54	81.09	71.58	69.87	79.09

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
EXOCC6Y22 (Expected job: type of occupation (specific), AY 2021-22)	21,770	76.16	74.52	74.04	72.11	73.93	79.19	69.98	76.29	80.89	71.58	69.7	79.09
EXPBA (Plan to continue to a bachelor's degree program within 5 years, AY 2019-20)	8,530	81.28	‡	82.55	81.38	72.59	39.78	90.93	50.44	85.17	‡	86.06	82.04
EXPBA22 (Plan to continue to a bachelor's degree program within 2 years, AY 2021-22)	6,450	73.86	70.26	85.12	80.39	53.36	47.56	77.58	50.75	43.67	79.23	79.89	80.66
EXPFLD22 (Job non-monetary benefits: importance of being an expert, AY 2021-22)	22,320	84.21	73.04	83.73	82.66	83.55	85.54	80.62	84.82	86.02	81.15	76.38	84.6
EXPWAGE22 (Expected job: annual salary expected, AY 2021-22)	21,770	79.83	76.33	78.3	77.64	77.18	81.7	74.21	80.36	83.82	77.12	73.94	82.06
FACULTYIN22 (Engagement: interactions with faculty during class, AY 2020-21/2021-22)	19,120	81.2	65.48	78.66	77.56	80.97	84.36	74.41	82.53	85.69	71.89	63.39	80.84
FACULTYOUT22 (Engagement: interactions with faculty outside of class, AY 2020-21/2021-22)	19,120	81.12	65.48	78.66	77.14	81	84.35	74.41	82.51	85.39	71.44	63.2	80.78
FAMAMT22 (Help from family and friends: total amount received, AY 2020-21/2021-22)	19,120	77.98	57.31	74.83	73.92	77.88	81.49	73.87	80.23	83.16	69.47	61.45	76.1
FAMTIM22 (Job non-monetary benefits: importance of balancing work and family, AY 2021-22)	22,320	84.17	73.04	83.75	82.35	83.25	85.6	80.62	84.71	85.71	81.91	76.46	84.54
FEDBEN22 (Received federal benefit: any, in 2019 or 2020)	22,320	65.68	56.73	65.15	62.99	68.92	66.54	73.29	65.42	65.87	66.18	58.17	71.12
FEDBEN22A (Received federal benefit: Food Stamp Benefits, in 2019 or 2020)	22,320	68.91	61.48	70.09	67.08	70.42	67.09	74.79	66.9	67.08	74.8	70.39	79.78
FEDBEN22B (Received federal benefit: Free or Reduced Price Lunch Benefits, in 2019 or 2020)	22,320	65.24	54.07	64.68	62.61	68.92	66.48	69.68	65.47	65.52	62.71	55.91	69.53
FEDBEN22C (Received federal benefit: Supplemental Security Income Benefits, in 2019 or 2020)	22,320	68.89	61.48	70.01	67.13	70.46	67.1	74.79	66.9	67.08	75.23	70.16	79.69

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
FEDBEN22D (Received federal benefit: TANF Benefits, in 2019 or 2020)	22,320	65.18	54.07	64.56	62.6	68.92	66.47	68.9	65.47	65.47	62.45	55.56	69.55
FEDBEN22E (Received federal benefit: WIC Benefits, in 2019 or 2020)	22,320	65.22	54.07	64.67	62.6	68.92	66.49	68.9	65.22	65.52	62.84	55.49	69.51
FHSUPP22 (Supportive of PSE: friends from home, AY 2020-21/2021-22)	19,120	81.18	65.48	78.76	77.25	81.08	84.29	74.41	82.65	85.39	72.46	63.7	81.01
FIN222000 (Financial security: come up with \$2,000 in the next month, AY 2021-22)	22,320	95.76	85.67	94.75	96.03	96.51	96.55	99.15	97.24	96.87	93.36	92.28	96.84
FIN22500 (Financial security: come up with \$500 within the next month, AY 2021-22)	22,320	95.5	87.04	94.45	95.81	96.91	96.4	98.95	97.1	96.23	91.32	92.33	96.83
FINLITA22 (Financial literacy: effect of inflation on purchasing, AY 2021-22)	22,320	83.19	71.35	82.5	81.72	82.28	84.48	80.2	84.03	85.7	79.63	74.57	84.3
FINLITB22 (Financial literacy: effect of interest on savings, AY 2021-22)	22,320	83.19	71.35	82.69	81.75	81.74	84.36	78.75	83.89	85.53	80.28	75.82	83.88
FINLITC22 (Financial literacy: effect of diversification on risk, AY 2021-22)	22,320	83.69	71.23	83.11	82.2	82.47	84.76	80.62	84.81	86.06	80.93	76.24	84.6
FINLITTOT22 (Financial literacy: number of correct responses, AY 2021-22)	22,320	82.59	71.12	81.85	80.95	81.45	83.96	78.33	83.58	85.39	78.73	73.58	83.51
FOODSEC22 (Food security status: previous 30 days, AY 2021-22)	22,320	95.56	89.51	94.74	95.08	96.11	96.32	99.08	96.89	96.66	94.86	93.07	94.94
FRQCOMEXTFAM22 (Frequency of communication: extended family, AY 2020-21/2021-22)	19,120	80.9	62.95	78.28	77.4	81.05	84.23	73.92	82.12	84.96	72.05	63.19	80.69
FRQCOMNOCOL22 (Frequency of communication: close friends that did not attend college, AY 2020-21/2021-22)	19,120	80.98	62.95	78.37	77.44	81.04	84.32	73.92	82.11	85.1	72.05	63.25	81
FRQCOMNPSAS22 (Frequency of communication: close friends that attended NPSAS institution, AY 2020-21/2021-22)	19,120	81.04	62.95	78.46	77.44	81	84.37	73.92	82.4	85.1	71.94	63.39	80.77

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public						Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year	
FRQCOMOTCOL22 (Frequency of communication: close friends that attended other colleges, AY 2020-21/2021-22)	19,120	80.89	62.95	78.36	77.17	80.81	84.27	73.92	81.81	85.06	72	63.25	80.28	
FRQCOMPAR22 (Frequency of communication: parents, AY 2020-21/2021-22)	17,580	80.68	63.2	77.7	76.4	80.79	84.25	71.87	82.43	84.84	60.97	53.95	77.99	
FRQCOMSIB22 (Frequency of communication: siblings, AY 2020-21/2021-22)	19,120	80.97	62.95	78.35	77.34	81.06	84.3	73.92	82.38	84.98	72.05	63.29	81	
FSECTOR (Control and level of first institution (IPEDS sector), AY 2019-20)	22,320	100	100	100	100	100	100	100	100	100	100	100	100	
FSSUPP22 (Supportive of PSE: friends from school, AY 2020-21/2021-22)	19,120	81.12	65.48	78.73	76.7	81.08	84.32	72.81	82.65	85.33	72.46	63.63	80.9	
GENDER22 (Gender identity: gender, as of AY 2021-22)	22,320	95.94	88.02	95.04	96.02	97.31	96.32	99.15	97.15	97.17	94	93.97	97.35	
GENDER3 (Gender, as of AY 2019-20)	22,320	77.68	84.28	82.53	87.59	86.22	62.59	92.9	78.2	77.69	89.92	89.7	90.39	
GPA22 (Grade point average, AY 2020-21/2021-22)	19,120	72.58	44.47	62.65	67.62	77.03	83.15	56.11	76.97	82.81	36.02	39.3	66.34	
GPA3Y (Grade point average through AY 2021-22)	22,320	69.74	44.21	59.83	66.5	75.6	82.17	51.47	75.01	81.31	39.17	47.52	65.93	
HIGHLVEX (Highest level of education ever expected, AY 2019-20)	22,320	76.79	82.82	80.87	87.21	85.22	62.34	93.14	77.95	77.48	88.94	87.87	88.44	
HLPOTH22 (Job non-monetary benefits: importance of helping others, AY 2021-22)	22,320	84.33	73.04	83.79	82.79	83.62	85.66	80.62	84.89	86.27	82.04	76.5	84.6	
HOMELESS22 (Experienced homelessness: previous 30 days, AY 2021-22)	22,320	82.44	71.14	81.85	80.19	81.5	83.74	79.44	83.44	84.57	79.89	75.05	83.93	
HOMESTUD22 (Owned home or paid mortgage, AY 2021-22)	15,480	73.34	38.94	73.32	72.83	72.41	77.39	64.9	58.8	72.9	68.29	58.4	73.06	
HOUSESEC22 (Experienced housing insecurity: previous 12 months, AY 2021-22)	22,320	82.93	70.74	82.2	81.39	81.61	84.16	80.69	83.79	85.24	81.05	76.2	84.27	

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
HRSWK21 (Jobs while enrolled: hours worked, AY 2020-21)	22,320	74.89	67.15	78.37	77.57	70.11	71.57	76.97	70.65	72.54	77.77	73.48	79.61
HRSWK22 (Jobs while enrolled: hours worked, AY 2021-22)	22,320	75.67	70.46	79.38	77.45	70.53	72.35	76.37	71.4	73.11	81.43	76.09	78.46
HRSWK3Y (Jobs while enrolled: hours worked when last enrolled through AY 2021-22)	22,320	79.78	68.76	79.82	81.33	79.12	79.69	78.28	76.22	82.12	75.42	73.23	84.05
HSIZE22 (Family size (dependent and independent), AY 2021-22)	16,600	67.82	37.59	68.25	65.93	71.92	67.59	74.59	65.86	67.62	51.91	54.74	80.21
IMPACAD22 (PSE service importance: academic advising or planning, AY 2020-21/2021-22)	16,440	77.12	44.25	72.39	71.25	78.08	82.38	68.58	79.28	83.18	55.73	44.44	73.22
IMPCAMPSEC22 (PSE service importance: campus security or campus police, AY 2020-21/2021-22)	6,480	38.44	17.53	24.12	27.36	40.09	36.43	51.75	61.72	63.24	20.24	18.32	32.05
IMPCHLD22 (PSE service importance: child care, AY 2020-21/2021-22)	4,130	8.07	2.5	9.12	7.2	7.61	5.11	16.46	4.06	11.44	12.86	9.89	13.55
IMPCPP22 (PSE service importance: career services, AY 2020-21/2021-22)	8,930	57.13	48	46.8	47.38	55	63.24	67.71	66.25	70.56	55.71	44.8	51.93
IMPDISSRV22 (PSE service importance: services for students with disabilities, AY 2020-21/2021-22)	5,440	29.47	28.99	25.48	21.41	32.6	31.95	14.13	35.81	39.49	20.97	14.21	38.61
IMPDIRSE22 (PSE service importance: diversity or multicultural student services, AY 2020-21/2021-22)	5,340	29.11	10.58	17.19	19.55	32.09	34.36	21.74	44.5	48.32	18.89	15.73	24.18
IMPFINAID22 (PSE service importance: financial aid services, AY 2020-21/2021-22)	11,670	66.71	47.92	64.29	63.54	67.45	68.19	54.58	68.89	73.55	63.77	53.68	70.98
IMPHLTH22 (PSE service importance: student health services, AY 2020-21/2021-22)	8,660	56.1	37.7	29.26	26.66	60.48	69.73	47.83	69.13	75.03	20.9	15.7	34.95
IMPMENSRV22 (PSE service importance: psychological or mental health services, AY 2020-21/2021-22)	6,530	40.36	31.14	22.52	21.11	46.8	49.46	23.02	57.38	61.39	10.81	17.05	35.4
IMPMILSRV22 (PSE service importance: services for active military or veterans, AY 2020-21/2021-22)	4,640	18.88	4.83	18.41	16.03	21.18	18.87	16.74	17.26	20.05	14.69	19.73	33.45

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
IMPTRPLN22 (PSE service importance: transfer advising or planning, AY 2020-21/2021-22)	7,410	49.34	12.71	57.24	52.33	40.63	42.03	31.31	41.79	43.37	29.44	24.74	42.53
IMPTUTOR22 (PSE service importance: peer or other tutoring or skills lab, AY 2020-21/2021-22)	9,660	60.95	43.56	52.23	54.37	61.64	67.35	62.67	66.8	72.42	40.24	33.34	61.88
INCSPS22 (Independent students: spouse's income, 2019)	2,370	17.31	‡	24	19.73	8.05	8.18	36.57	12.08	11.98	17.52	8.42	45.72
INDINC22 (Independent students: student and spouse's income, 2019)	6,610	21.44	16.88	29.17	29.23	16.19	8.27	45.6	13.24	8.94	39.98	39.28	64.21
ISTUINC22 (Independent students: income, 2019)	6,610	21.44	16.88	29.17	29.23	16.19	8.27	45.6	13.24	8.94	39.98	39.28	64.21
IT2YFP3Y (Ever attended 2-year for-profit institution through AY 2021-22)	22,320	99.89	100	99.94	99.92	99.96	99.94	100	99.97	99.96	99.7	96.98	99.95
IT2YPU3Y (Ever attended 2-year public institution through AY 2021-22)	22,320	98.49	99.45	97.87	99.63	98.39	98.42	100	98.49	98.9	99.91	99.82	99.79
IT4Y3Y (Ever attended 4-year institution through AY 2021-22)	22,320	97.64	97.83	97.4	96.79	97.41	97.99	99.07	98.09	97.77	98.81	98.48	97.27
IT4YFP3Y (Ever attended 4-year for-profit institution through AY 2021-22)	22,320	99.85	100	99.89	99.95	99.83	99.96	100	99.96	99.89	99.29	99.77	98.29
IT4YPN3Y (Ever attended 4-year private nonprofit institution through AY 2021-22)	22,320	99.59	100	99.59	99.35	99.64	99.92	99.07	99.05	99.23	100	99.04	99.69
IT4YPU3Y (Ever attended 4-year public institution through AY 2021-22)	22,320	98.17	97.83	97.88	97.41	97.94	98.08	100	99.08	98.6	99.53	99.66	99.28
ITFP3Y (Ever attended for-profit institution through AY 2021-22)	22,320	99.59	100	99.74	99.82	99.72	99.79	100	99.79	99.85	95.53	96.85	98.23
ITL2FP3Y (Ever attended less than 2-year for-profit institution through AY 2021-22)	22,320	99.83	100	99.9	99.88	99.93	99.87	100	99.85	100	96.42	99.95	99.98
JBEN22 (Job: eligible for life, health, and retirement benefits, 2022)	4,020	76.15	78.84	76.89	74.03	70.73	76.3	59.32	77.98	78.77	75.59	73.24	76.55
JBEN22A (Job: eligible for life insurance, 2022)	3,680	79.2	77.54	79.65	75.73	74.82	74.79	70.35	81.1	82.18	86.4	82.32	83.2

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
JBEN22B (Job: eligible for health insurance, 2022)	3,680	79.42	77.54	79.88	76.6	75.32	74.8	70.35	81	82.18	86.4	82.32	83.24
JBEN22C (Job: eligible for retirement benefits, 2022)	3,680	79.25	77.54	79.75	76.6	75.32	74.49	70.35	81.36	82.18	86.4	82.32	81.5
JOBFT21 (Jobs while enrolled: work intensity, AY 2020-21)	22,320	74.89	67.15	78.37	77.57	70.11	71.57	76.97	70.65	72.54	77.77	73.48	79.61
JOBFT22 (Jobs while enrolled: work intensity, AY 2021-22)	22,320	75.67	70.46	79.38	77.45	70.53	72.35	76.37	71.4	73.11	81.43	76.09	78.46
JOBHRS22 (Job: hours worked weekly, 2022)	4,330	80.19	78.25	80.76	79.21	82.29	81.71	63.11	82.69	88.06	74.98	73.85	76.84
JOBLFP22 (Labor force participation status in June 2022)	6,080	57.13	60.44	58.72	52.42	62.45	59.48	37.55	63.21	52.02	53.79	49.24	52.36
JOBLIC22 (Job: has industry certification or license, 2022)	4,330	73.52	75.12	74.32	70.82	67.22	72.8	54.51	78.54	77.31	73.96	68.91	75.02
JOBMNT3Y (All jobs: number of months worked since beginning PSE)	22,320	69.35	63.06	72.57	75.15	72.86	63.77	75.11	63.27	66.44	75.39	68.68	77.29
JOBOCC22 (Job: occupation, 2022)	4,330	70.87	71.97	72.13	69.52	63.01	70.21	54.61	72.27	67.93	72.66	66.03	71.69
JOBRLG22 (Job: related to college studies, 2022)	3,980	79.17	78.14	79.07	78.95	81.7	80.34	66.45	85.66	87.93	75.45	73.74	76.38
JOBRLC22 (Job: job required industry certification or license, 2022)	4,330	73.37	74.13	74.21	70.82	67.22	72.8	54.51	78.54	77.31	73.96	68.91	73.29
JOBRLFT22 (Job: related to intended future work, 2022)	4,330	73.15	75.32	74.21	69.77	68.42	71.31	54.51	78.26	77.23	73.62	67.25	75.7
JOBSELF22 (Job: self employed, 2022)	4,330	82.66	84.19	83.13	82.37	84.47	83.28	66.78	85.54	88.89	78.49	77.15	79.39
JOBST22 (Job: employment status, 2022)	6,080	90.71	90.41	91.27	92.24	90.57	89.03	78.58	93.62	95.08	88.7	86.16	87.82
JOBTIM22 (Job: years worked in same or similar job, 2022)	4,330	71.21	72.01	72.32	66.78	65.28	70.73	54.51	74.53	74.05	71.96	68.52	71.56
KNOWCLAS22 (Knew requirements needed to complete pursued degree at primary institution)	18,570	78.39	62.47	74.55	74.15	77.9	82.25	70.67	80.55	84.34	66.63	61.44	78.75
LEISTIM22 (Job non-monetary benefits: importance of balancing work and leisure time, AY 2021-22)	22,320	84.22	73.04	83.66	82.72	83.55	85.61	80.62	84.8	85.93	82.04	76.4	84.58

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
LKOCCATHD22 (Expected job: likelihood of holding job after highest attainment, AY 2021-22)	21,770	80.15	76.33	79.02	76.89	77.59	81.8	74.94	81.09	83.77	78.18	75.18	81.87
LNLITTOT22 (Loan literacy: number of correct responses, AY 2021-22)	22,320	82.42	71.19	81.83	80.56	81.33	83.65	80.49	82.93	84.39	80.54	75.42	84.51
LOANLITA22 (Loan literacy: government can report unpaid debt to credit bureaus, AY 2021-22)	22,320	82.42	71.19	81.83	80.56	81.33	83.65	80.49	82.93	84.39	80.54	75.42	84.51
LOANLITB22 (Loan literacy: government can garnish wages for unpaid federal loan debt, AY 2021-22)	22,320	82.42	71.19	81.83	80.56	81.33	83.65	80.49	82.93	84.39	80.54	75.42	84.51
LOANLITC22 (Loan literacy: government can retain tax refunds or Social Security for loan debt, AY 2021-22)	22,320	82.42	71.19	81.83	80.56	81.33	83.65	80.49	82.93	84.39	80.54	75.42	84.51
LOCALR22 (Residence while attending primary institution, AY 2020-21/2021-22)	19,120	77.99	56.26	74.74	74.04	78.04	81.74	73.65	79.53	83.47	68.55	60.62	75.34
LOCJOB21 (Jobs while enrolled: on or off campus, AY 2020-21)	13,190	71.12	46.09	68.42	66.37	71.69	75.47	67.72	73.79	77.1	42.88	46.32	63.36
LOCJOB22 (Jobs while enrolled: on or off campus, AY 2021-22)	13,790	73.27	39.6	68.56	68.99	73.7	78.78	66.26	76.32	80.22	29.52	30.17	64.54
LOCJOB3Y (Jobs while enrolled: ever on campus through AY 2021-22)	19,160	52.17	48.94	55.49	59.96	60.32	44.83	59.65	48.47	52.72	46.31	42.38	61.12
MAJ22 (Field of study: last enrolled any degree through AY 2021-22 (23 categories))	21,720	86.98	76.54	86.16	85.43	87.28	88.13	80.95	87.97	89.12	78.25	82.14	91.47
MAJ4Y22 (Field of study: last enrolled any degree through AY 2021-22 (10 categories))	21,720	86.98	76.54	86.16	85.43	87.28	88.13	80.95	87.97	89.12	78.25	82.14	91.47
MAJ4YBA22 (Field of study: last enrolled in bachelor's through AY 2021-22 (10 categories))	13,280	91.17	‡	89.45	88.02	91.06	91.2	‡	93.55	92.61	‡	‡	91.41
MAJAA22 (Field of study: last enrolled in associate's through AY 2021-22 (23 categories))	10,150	74.97	‡	75.77	77.76	71.92	65.52	52.73	79.33	66.87	‡	69.88	77.18

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
MAJBA22 (Field of study: last enrolled in bachelor's through AY 2021-22 (23 categories))	13,280	91.17	‡	89.45	88.02	91.06	91.2	‡	93.55	92.61	‡	‡	91.41
MAJCHG22 (Major changes: number anywhere through AY 2021-22)	20,530	83.3	32.53	83.54	82.95	84.18	86.16	73.59	85.52	87.57	21.56	80.36	90.29
MAJCHGAA22 (Major changes (associate's): number anywhere through AY 2021-22)	9,360	66.16	‡	71.1	67.9	31.48	36.77	53.6	49.6	39.69	‡	64	73.57
MAJCHGBA22 (Major changes (bachelor's): number anywhere through AY 2021-22)	12,990	82.37	‡	75.63	83.34	84.98	81.37	‡	84.82	86.91	‡	‡	80.9
MAJCT22 (Field of study: last enrolled in certificate through AY 2021-22 (23 categories))	4,520	71.34	81.43	75.93	74.55	60.18	59.78	38.13	68.11	51.84	74.89	68.86	59.1
MEALPLAN22 (Meal plan: had meal plan, AY 2021-22)	16,600	78.04	55.37	74.32	74.15	77.6	81.55	76.67	80.15	82.88	49.25	49.77	79.78
MEALPLANFQ22 (Meal plan: meal plan covered 11 or more meals per week, AY 2021-22)	6,670	41.73	#	15.33	13.15	46.26	45.76	‡	68.02	67.4	0.76	3.16	33.01
MENTH22 (Health: mental, AY 2021-22)	22,320	83.4	71.46	82.76	82.04	82.8	84.52	80.65	83.84	85.81	80.64	76.16	84.61
MENTHDIF (Health: mental from 2020 to 2022)	22,320	58.55	53.64	60.56	64.85	63.71	49	66.36	61.68	62.08	62.71	55.83	70.2
MENTMISS22 (Health: amount missed school/work in past 30 days for mental health, AY 2021-22)	22,320	83.62	71.46	83.05	82.15	82.81	84.7	80.69	84.7	85.65	81.22	76.3	84.61
MILTYP22 (Military type, AY 2021-22)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
MJNSF22 (Major field of study with a focus on NSF STEM fields, AY 2019-20)	21,720	86.98	76.54	86.16	85.43	87.28	88.13	80.95	87.97	89.12	78.25	82.14	91.47
MNTJBNE21 (Jobs while not enrolled: number of months worked, AY 2020-21)	22,320	83.56	75.19	83.39	82.16	82.07	84.6	79.92	83.84	85.3	81.39	77.04	82.78
MNTJBNE22 (Jobs while not enrolled: number of months worked, AY 2021-22)	22,320	83.73	75.83	83.62	82.18	81.99	84.78	79.93	84.05	85.38	82.21	77.2	82.61

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
MNTJOB21 (Jobs while enrolled: number of months worked, AY 2020-21)	22,320	83.55	75.19	83.38	82.16	82.07	84.58	79.92	83.84	85.3	81.39	77.04	82.78
MNTJOB22 (Jobs while enrolled: number of months worked, AY 2021-22)	22,320	83.73	75.83	83.63	82.18	81.99	84.75	79.93	84.06	85.38	82.21	77.2	82.61
MTGAMT22 (Monthly mortgage or rent amount, AY 2021-22)	18,680	73.65	59.15	73.77	71.82	72.68	76.34	64.09	61.86	73.28	74.75	67.88	78.31
NPATND22 (Would begin education at NPSAS institution again)	22,320	99.83	100	99.78	99.88	99.72	99.88	100	99.97	99.82	100	99.42	99.83
NUMJBNE21 (Jobs while not enrolled: number of jobs, AY 2020-21)	22,320	93.42	89.47	93.75	93.73	92.44	93.6	92.9	92.47	93.5	93.09	89.75	93.34
NUMJBNE22 (Jobs while not enrolled: number of jobs, AY 2021-22)	22,320	95.24	95.41	96.15	94.72	94.1	94.89	94.88	94.1	95	94.28	93.63	95.66
NUMJOB21 (Jobs while enrolled: number of jobs, AY 2020-21)	22,320	83.55	75.19	83.38	82.16	82.07	84.58	79.92	83.84	85.3	81.39	77.04	82.78
NUMJOB22 (Jobs while enrolled: number of jobs, AY 2021-22)	22,320	83.72	75.83	83.62	82.18	81.99	84.75	79.93	84.05	85.38	82.21	77.2	82.61
ORGMAJ22 (Original declared field of study at first degree program (23 categories))	22,160	97.97	100	98.22	97.17	99.6	97.86	99.54	98.19	97.04	95.64	98.05	99.98
ORGMAJ4Y (Original declared field of study at first degree program (10 categories))	22,160	97.97	100	98.22	97.17	99.6	97.86	99.54	98.19	97.04	95.64	98.05	99.98
OTHGTAMT22 (Outside grants (private and employer): amount received, AY 2021-22)	16,600	76.83	55.06	73.3	73.19	75.98	80.44	76.05	77.5	81.42	49.06	50.68	79.53
PARAMT22 (Help from parents: total amount received, AY 2020-21/2021-22)	19,120	77.34	56.63	73.89	73.78	77.3	81.09	73.81	79.82	82.12	68.99	61.1	76.18
PAREduc (Parents' highest education level, AY 2019-20)	22,320	72.95	74.51	76.54	81.68	81.42	59.81	87.07	75.46	73.94	83.83	81.86	82.92
PARSUPP22 (Supportive of PSE: parents, AY 2020-21/2021-22)	17,580	80.74	66.13	77.96	75.99	80.75	84.16	71.87	82.7	84.87	60.97	54.14	78.1
PEERINTIN22 (Engagement: interactions with other students during class, AY 2020-21/2021-22)	19,120	80	65.44	77.66	76.17	80.56	82.99	71.96	80.41	84.11	72.46	63.17	80.57

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

			Public						Private nonprofit			Private for-profit		
					4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year	
Variable	Sample Size	All sample members	Less-than-2-year	2-year										
PEERINTOUT22 (Engagement: interactions with other students outside of class, AY 2020-21/2021-22)	19,120	80	65.44	77.63		76.62	80.47	82.94	71.96	80.48	84.05	72.46	63.17	80.52
PELFRSTY (Federal Pell Grant: first academic year received through AY 2021-22)	11,860	100	100	100		100	100	100	100	100	100	100	100	100
PELL20 (Federal Pell Grant: amount received, AY 2019-20)	22,320	100	100	100		100	100	100	100	100	100	100	100	100
PELL21 (Federal Pell Grant: amount received, AY 2020-21)	22,320	100	100	100		100	100	100	100	100	100	100	100	100
PELL22 (Federal Pell Grant: amount received, AY 2021-22)	22,320	100	100	100		100	100	100	100	100	100	100	100	100
PELLCU22 (Federal Pell Grant: cumulative amount received through AY 2021-22)	22,320	100	100	100		100	100	100	100	100	100	100	100	100
PELLDEP (Pell grant and dependency, AY 2019-20)	22,320	99.76	99.55	99.8		99.69	100	99.81	100	99.88	99.96	98.51	97.29	99.94
PELLST22 (Federal Pell Grant: last academic year received through AY 2021-22)	11,860	100	100	100		100	100	100	100	100	100	100	100	100
PELYRS22 (Federal Pell Grant: number of years received through AY 2021-22)	22,320	100	100	100		100	100	100	100	100	100	100	100	100
PFAMNM22 (Dependent students: family size, AY 2021-22)	14,880	63.75	23.95	61.21		58.66	70.1	66.34	63.69	63.85	65.85	25.67	27.45	64.71
PHYSH22 (Health: physical, AY 2021-22)	22,320	83.79	70.89	83.25		82.16	83.09	84.84	79.33	84.62	85.96	81.57	76.86	85.05
PHYSHDIF (Health: physical from 2020 to 2022)	22,320	59.2	53.93	61.33		65.08	64.36	49.34	64.96	62.69	62.7	64.66	58	71.29
PHYSMISS22 (Health: amount missed school/work in past 30 days for physical health, AY 2021-22)	22,320	83.62	71.46	83.03		82.15	82.91	84.68	80.69	84.68	85.71	81.41	76.11	84.62
PINCOL22 (Dependent students: number family members in college, AY 2021-22)	14,880	63.37	23.19	60.6		57.53	69.9	66.15	63.69	63.74	65.77	22.44	27.45	64.42

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
PLNST22 (Private student loans: repayment status, AY 2021-22)	22,320	66.13	64.73	70.28	73.31	71.23	54.59	73.04	67.56	67.61	72.32	65.55	75.63
PLUCUM22 (Direct PLUS Loans to parents: amount borrowed through AY 2021-22)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
PLUS20 (Direct PLUS Loans to parents: amount borrowed, AY 2019-20)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
PMARIT22 (Dependent students: parents' marital status, AY 2021-22)	14,880	63.79	23.95	61.32	58.73	70.1	66.35	63.69	63.85	65.85	25.67	27.45	64.71
PRAT3Y (Persistence anywhere through AY 2021-22)	22,320	90.45	80.32	90.98	89.95	90.12	90.65	86.85	89.25	91.91	84.87	81.46	92.4
PRATY1 (Attainment and enrollment during the 2019-20 AY)	22,320	93.05	90.21	93.25	94.78	95.49	90.84	90.42	95.57	95.51	88.2	86.32	94.55
PRATY2 (Attainment and enrollment during the 2020-21 AY)	22,320	91.49	87.03	92.49	91.35	90.46	90.78	89.67	89.23	91.97	89.8	89.46	94.03
PRATY3 (Attainment and enrollment during the 2021-22 AY)	22,320	91.73	87.88	92.73	91.57	90.62	91.2	86.94	89.54	92.17	89.6	87.23	94.8
PRICUM22 (Private (alternative) loans: cumulative amount borrowed through AY 2021-22)	22,320	66.82	64.73	70.63	74.15	72.29	55.34	72.3	68.69	68.62	73.21	67.06	76.13
PRIVAI22 (Private source grants: amount received, AY 2021-22)	16,600	78.2	55.06	74.65	74.08	78.07	81.75	76.67	79.34	82.8	49.97	51.01	81.08
PRLVL3Y (Attainment or level of last institution enrolled as of 2021-22 AY)	22,320	89.74	90.78	90.74	90.25	88.42	88.87	64.76	88.22	90.35	87.55	86.17	92.67
PROUT1_NEW (Cumulative persistence and attainment anywhere through 2019-20 AY (new version))	22,320	91.53	90.21	91.93	93.62	93.23	89.12	66.61	94.62	93.95	87.16	84.01	93.87
PROUT1_OLD (Cumulative persistence and attainment anywhere through 2019-20 AY (old version))	22,320	84.73	86	85.73	85.82	85.27	82.44	56.76	85.15	87.19	79.66	77	89.12
PROUT2_NEW (Cumulative persistence and attainment anywhere through 2020-21 AY (new version))	22,320	90.13	94.87	91.41	90.32	88.54	88.78	64.76	88.02	90.48	90.14	90.06	93.5
PROUT2_OLD (Cumulative persistence and attainment anywhere through 2020-21 AY (old version))	22,320	89.34	90.78	90.3	89.46	88.1	88.53	64.76	87.7	90.35	87	85.91	92.13

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
PROUT3_NEW (Cumulative persistence and attainment anywhere through 2021-22 AY (new version))	22,320	89.74	90.78	90.75	90.25	88.42	88.87	64.76	88.22	90.35	87.55	86.17	92.67
PROUT3_OLD (Cumulative persistence and attainment anywhere through 2021-22 AY (old version))	22,320	89.74	90.78	90.75	90.25	88.42	88.87	64.76	88.22	90.35	87.55	86.17	92.67
PROUTF1_NEW (Cumulative retention and attainment at first institution through 2019-20 (new version))	22,320	92.76	90.21	92.95	94.68	94.68	90.65	88.76	95.53	95.33	87.47	84.1	94.29
PROUTF1_OLD (Cumulative retention and attainment at first institution through 2019-20 (old version))	22,320	92.76	90.21	92.95	94.68	94.68	90.65	88.76	95.53	95.33	87.47	84.1	94.29
PROUTF2_NEW (Cumulative retention and attainment at first institution through 2020-21 (new version))	22,320	91.44	93.68	92.43	91.23	89.96	90.62	88.37	89.16	91.93	90.27	90.4	93.82
PROUTF2_OLD (Cumulative retention and attainment at first institution through 2020-21 (old version))	22,320	91.44	93.68	92.43	91.23	89.96	90.62	88.37	89.16	91.93	90.27	90.4	93.82
PROUTF3_NEW (Cumulative retention and attainment at first institution through AY 2021-22 (new version))	22,320	91.83	94.32	92.95	91.82	90.02	90.88	88.46	89.38	92.09	91.1	90.68	94.83
PROUTF3_OLD (Cumulative retention and attainment at first institution through AY 2021-22 (old version))	22,320	95.21	95.74	95.91	95.97	94.56	94.23	88.33	94.78	96.02	92.34	92.17	96.58
PRRPY22 (Private student loans: monthly payment amount, AY 2021-22)	6,960	5.64	9.3	3.18	4.69	8.06	3.7	9.58	12.78	9.42	14.85	13.04	11.93
RACE (Race/ethnicity (with multiple), AY 2019-20)	22,320	96.79	96.68	96.91	97.63	98.68	97.44	95.82	94.87	96.8	91.27	91.23	95.32
RCVSTAIID (Received financial aid from a state aid program, AY 2019-20)	7,950	100	‡	100	100	100	100	100	100	100	100	100	100
RCVSTAIID21 (Received financial aid from a state aid program, AY 2019-20)	7,950	66.15	‡	63.42	61.79	74.09	73.42	67.54	63.13	68	52.93	55.94	62.72
RCVSTAIID22 (Received financial aid from a state aid program, AY 2021-2022)	7,950	66.14	‡	63.28	62.35	72.76	74.07	67.54	63.22	67.75	53.18	52.1	60.94

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
REGSUPP22 (Provided regular financial support to family or friend, AY 2020-21/2021-22)	22,320	84.11	71.46	83.66	82.44	83.48	85.35	79.02	84.62	85.97	81.66	76.8	84.91
REGVT22 (Registered to vote in U.S. elections, AY 2021-22)	21,360	84.79	76.17	81.31	81.36	87.57	88.44	78.03	88.85	90.37	76.31	74.29	83.61
REMEV3Y (Ever taken remedial courses through AY 2021-22)	22,320	81.85	88.33	86.02	90.19	88.4	69.34	94.08	84.15	80.61	91.01	89.99	93.33
SALARY22 (Job: annual salary, 2022)	5,860	52.63	57.18	56.51	50.5	47.9	48.11	42.55	49.59	38.97	52.45	49.87	52.23
SEEND3Y (First enrollment spell: date ended through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
SEINT3Y (First enrollment spell: intensity through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
SEMON3Y (First enrollment spell: months enrolled through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
SENSBELNG22 (Engagement: felt like a part of institution, AY 2020-21/2021-22)	19,120	85.34	68.8	82.71	81.98	86.55	88.68	78	86.21	89.77	72.31	65.48	84.72
SENUM3Y (Enrollment spells: number of spells through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
SEROLE22 (Jobs while enrolled: primary role as student or employee, AY 2021-22)	13,790	61.83	26.88	61.87	62.73	58.99	63.5	63.84	59.93	65.5	27.75	27.93	57.2
SEXATBIRTH (Sex assigned at birth)	22,320	98.29	94.98	97.95	98.61	99.30	98.21	99.93	98.31	98.83	97.28	98.17	99.74
SEXORIENTAT22 (Sexual orientation, as of AY 2021-22)	22,320	95.68	88.02	94.76	95.83	96.57	96.36	99.15	96.69	96.68	93.65	92.56	97.32
SFAMNM22 (Independent students: family size, AY 2021-22)	6,610	26.52	22.59	37.29	34.48	18.89	10.35	54.16	14.85	13.85	42.36	45.52	70.15
SIEND3Y (First institution spells: date first enrollment spell ended through AY 2021-22)	22,320	89.71	79.86	90.72	90.74	91.39	89.08	78.76	89.74	91.41	79.35	72.67	90.94
SIINT3Y (First institution spells: intensity through AY 2021-22)	22,320	89.71	79.86	90.72	90.74	91.39	89.08	78.76	89.74	91.41	79.35	72.67	90.94

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
SINCOL22 (Independent students: number of family members in college 2021-22)	6,610	23.9	17	33.67	29.88	17.01	9.57	52.57	12.76	12.73	38.53	42.48	63.71
SMAR22 (Marital status, AY 2021-22)	22,320	99.8	98.41	99.77	99.95	99.81	99.91	100	99.62	99.64	100	99.57	99.98
SOCPARTANY22 (Social activities: participated in any social activity, AY 2020-21/2021-22)	19,120	79.83	64.19	76.38	75.81	79.84	83.61	72.57	81.89	84.88	71.27	60.97	80.27
SOCPARTGRK22 (Social activities: participated in a Greek letter organization, AY 2020-21/2021-22)	19,120	80.51	65.48	77.74	76.46	80.14	83.95	73.85	81.9	85.04	71.47	62.72	80.75
SOCPARTMUS22 (Social activities: participated in institution sponsored music, theatre, or band, AY 2020-21/2021-22)	19,120	80.41	65.48	77.56	76.36	80.55	83.94	73.85	81.34	84.99	71.69	61.68	80.87
SOCPARTNVAR22 (Social activities: participated in non-varsity athletics, AY 2020-21/2021-22)	19,120	80.45	65.48	77.52	76.49	80.52	83.86	73.85	81.91	85.08	71.46	62.86	80.96
SOCPARTSTGOV22 (Social activities: participated in student government, AY 2020-21/2021-22)	19,120	80.43	65.48	77.73	76.41	80.23	83.72	73.85	81.72	84.96	71.62	62.96	80.88
SOCPARTSTGRP22 (Social activities: participated in student affinity groups, AY 2020-21/2021-22)	19,120	80.53	64.19	77.68	76.56	80.44	84	72.57	81.95	85.11	71.69	63.14	80.63
SOCPARTSTORG22 (Social activities: participated in any other recognized or registered student organizations, AY 2020-21/2021-22)	19,120	80.49	65.48	77.72	75.97	80.54	83.89	73.85	81.94	85.19	71.6	62.98	80.62
SOCPARTVAR22 (Social activities: participated in varsity athletics, AY 2020-21/2021-22)	19,120	80.46	65.48	77.52	76.53	80.54	83.9	73.85	81.98	85.09	71.61	62.74	80.69
SPEED (Measured internet download speed)	21,390	100	100	100	100	100	100	100	100	100	100	100	100
SPNM21 (Not married to spouse in 2021)	1,190	89.52	‡	89.13	87.56	94.09	90.98	‡	83.56	85.35	96.91	92.43	92.03
SPPSUPP22 (Supportive of PSE: spouse/partner, AY 2020-21/2021-22)	1,400	69.37	‡	66.85	70.27	73.57	76.88	61.45	64.85	75.3	75.55	55.72	76.68

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

			Public					Private nonprofit			Private for-profit		
					4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	4-year, non-doctorate-granting		4-year, doctorate-granting			
Variable	Sample Size	All sample members	Less-than-2-year	2-year				Less-than-4-year			Less-than-2-year	2-year	4-year
SPSED22 (Independent students: spouse's highest education level, AY 2021-22)	610	68.23	‡	67.28	81.58	‡	68.25	‡	‡	‡	‡	‡	76.98
STDT3Y (First stopout through AY 2021-22: start date)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
STDTFI3Y (First stopout at first institution through AY 2021-22: start date)	22,320	89.71	79.86	90.72	90.74	91.39	89.08	78.76	89.74	91.41	79.35	72.67	90.94
STDYABR3Y (Ever studied abroad while attending NPSAS institution through AY 2021-22)	22,320	56.94	54.64	57.58	59.39	62.72	49.38	63.27	59.55	61.42	65.94	59.8	68.16
STEMMAJ (Major field of study with a focus on STEM fields - 2020 CIP)	22,320	97.99	100	98.26	97.21	99.6	97.86	99.55	98.2	97.04	95.7	98.09	99.98
STFCUM22 (Direct Sub. and Unsub. Loans: cumulative amt borrowed through AY 2021-22)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
STFIRSTY (Direct Subsidized and Unsubsidized Loans: first academic year borrowed)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
STFY20 (Direct Subsidized and Unsubsidized Loans: total borrowed, AY 2019-20)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
STFY21 (Direct Subsidized and Unsubsidized Loans: total borrowed, AY 2020-21)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
STFY22 (Direct Subsidized and Unsubsidized Loans: total borrowed 2021-22)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
STFYRS22 (Direct Sub. or Unsub. Loans: number of years received through AY 2021-22)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
STNUM3Y (Stopouts through AY 2021-22: total)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
STSCUM22 (Direct Subsidized Loans: cumulative amount borrowed through AY 2021-22)	22,320	100	100	100	100	100	100	100	100	100	100	100	100

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
STSTA1 (Stopout status: AY 2019-20)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
STSTA2 (Stopout status: 2020-21)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
STSTA3 (Stopout status: AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
STSUB20 (Direct Subsidized Loans: total borrowed, AY 2019-20)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
STSUB21 (Direct Subsidized Loans: total borrowed, AY 2020-21)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
STSUB22 (Direct Subsidized Loans: total borrowed, AY 2021-22)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
STUCUM22 (Direct Unsubsidized Loans: cumulative amount borrowed through AY 2021-22)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
STYB3Y (First stopout through AY 2021-22: starting academic year)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37
STYBF13Y (First stopout at first institution through AY 2021-22: starting academic year)	22,320	89.71	79.86	90.72	90.74	91.39	89.08	78.76	89.74	91.41	79.35	72.67	90.94
T4TCUM22 (Title IV loans (incl Parent PLUS loans): cumulative borrowed through AY 2021-22)	22,320	100	100	100	100	100	100	100	100	100	100	100	100
T4TDUE22 (Title IV loans (incl Parent PLUS): amount owed in interest and principal through AY 2021-22)	10,560	100	100	100	100	100	100	100	100	100	100	100	100
T4XDUE22 (Title IV loans (excl Parent PLUS): amount owed in interest and principal through AY 2021-22)	10,480	100	100	100	100	100	100	100	100	100	100	100	100
TCREFFCOM22 (Teacher effectiveness: had good command of subject, AY 2020-21/2021-22)	19,120	84.98	68.8	82.12	81.69	86.01	88.55	77.99	85.41	89.73	71.91	65.26	84.31
TCREFFEX22 (Teacher effectiveness: make good use of examples, AY 2020-21/2021-22)	19,120	84.97	68.8	82.2	81.53	85.92	88.55	77.99	85.45	89.74	71.36	65.28	83.8
TCREFFINT22 (Teacher effectiveness: genuinely interested in teaching, AY 2020-21/2021-22)	19,120	84.97	68.8	82.17	81.7	86.05	88.47	76.43	85.45	89.7	71.91	65.23	84.31

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public						Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year	
TCREFFOUT22 (Teacher effectiveness: outstanding teachers, AY 2020-21/2021-22)	19,120	85.06	68.8	82.22	81.78	86.02	88.57	77.99	85.39	89.97	71.91	65.28	84.31	
TFATT3Y (Transfer (first): attainment before transfer through AY 2021-22)	22,320	84.2	76.91	85.22	85.47	85.13	82.3	55.61	84.68	86.89	76.96	71.15	88.84	
TFDEG3Y (Transfer (first): degree program through AY 2021-22)	22,320	85.27	76.91	86.14	86.25	86.59	83.65	77.66	85.26	87.94	77.77	71.57	89.32	
TFIFTY3Y (Transfer (first): destination institution type through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37	
TFINCT3Y (Transfer direction (first): institution control through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37	
TFINLV3Y (Transfer direction (first): institution level through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37	
TFMNDI3Y (Transfer (first): months elapsed before destination inst through AY 2021-22)	5,270	36.25	7.04	44.23	49.95	38.63	23.82	8.57	32.09	36.72	13.36	12.8	38.11	
TFMNT3Y (Transfer (first): months enrolled before destination inst through AY 2021-22)	5,270	36.25	7.04	44.23	49.95	38.63	23.82	8.57	32.09	36.72	13.36	12.8	38.11	
TFNUM3Y (Transfer status through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37	
TFTYPE1 (Transfer status during 2019-20 AY)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37	
TFTYPE2 (Transfer status during 2020-21 AY)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37	
TFTYPE3 (Transfer status during 2021-22 AY)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37	
TFTYPE3Y (Transfer direction (first): type through AY 2021-22)	22,320	85.44	76.91	86.29	86.48	86.82	83.82	77.67	85.47	88.21	77.82	71.57	89.37	
TRANSGENDER22 (Gender identity: transgender, as of AY 2021-22)	22,320	95.94	88.02	95.04	96.02	97.31	96.32	99.15	97.15	97.17	94	93.97	97.35	
TRNPLN (Plans to transfer, AY 2019-20)	22,320	68.05	71.62	70.76	76.39	76.18	55.31	82.48	70.58	70.07	78.24	77.91	82.39	
UGDEG (Undergraduate degree program, AY 2019-20)	22,320	99.9	100	99.88	99.62	100	99.91	100	100	100	100	100	100	

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

Variable	Sample Size	All sample members	Public					Private nonprofit			Private for-profit		
			Less-than-2-year	2-year	4-year, non-doctorate-granting, primarily subbaccalaureate	4-year, non-doctorate-granting, primarily baccalaureate	4-year, doctorate-granting	Less-than-4-year	4-year, non-doctorate-granting	4-year, doctorate-granting	Less-than-2-year	2-year	4-year
UGDEG22 (Degree program when last enrolled through AY 2021-22)	19,120	89.28	77.3	89.55	88.54	88.55	89.95	82.81	87.8	91.24	75.42	77.38	90.97
UGLVL22 (Highest undergraduate class level through AY 2021-22)	19,120	80.63	43.67	76.09	74.72	83.67	85.95	77.69	80.71	84.44	72.52	71.81	86.8
UNEMPL22 (Unemployment: currently looking for a job, 2022)	2,350	51.81	49.71	52.17	48.02	55.79	46.74	‡	58.39	68.33	55.79	43.36	48.1
USEACAD22 (PSE services used: academic advising, AY 2020-21/2021-22)	19,120	80.6	65.48	77.98	76.18	80.73	83.99	73.92	81.72	85.01	71.69	62.85	80.96
USEANY22 (PSE services used: used any service, AY 2020-21/2021-22)	19,120	80.41	65.48	77.55	75.77	80.73	83.94	73.92	81.74	84.99	71.69	63.09	80.8
USECAMPSEC22 (PSE services used: campus security or campus police, AY 2020-21/2021-22)	19,120	80.39	65.23	77.71	75.89	80.71	83.81	73.92	81.43	84.82	71.69	62.99	80.87
USECHLD22 (PSE services used: child care, AY 2020-21/2021-22)	19,120	80.33	60.87	77.77	75.93	80.59	83.72	73.92	81.51	84.56	70.79	63.15	80.92
USECPP22 (PSE services used: career services, AY 2020-21/2021-22)	19,120	80.47	65.48	77.81	75.99	80.75	83.92	73.92	81.58	84.91	71.69	62.88	80.01
USEDISSRV22 (PSE services used: services for students with disabilities, AY 2020-21/2021-22)	19,120	80.4	65.48	77.71	75.7	80.81	83.88	72.54	81.68	84.84	70.55	62.8	80.96
USEDIVRSE22 (PSE services used: diversity or multicultural student services, AY 2020-21/2021-22)	19,120	80.52	65.48	77.86	76.17	80.78	83.94	73.92	81.74	84.76	71.44	63.08	80.96
USEFINAID22 (PSE services used: financial aid services, AY 2020-21/2021-22)	19,120	80.44	65.48	77.64	76.03	80.59	83.88	73.92	81.71	85	71.69	63.15	80.95
USEHLTH22 (PSE services used: student health services, AY 2020-21/2021-22)	19,120	80.43	65.48	77.66	76.17	80.8	83.92	69.34	81.41	84.96	71.69	62.7	80.54
USEIDR22 (Likelihood of using income-driven repayment plan, AY 2020-21/2021-22)	11,620	24.63	21.97	23.93	26.75	25.34	19.16	54.42	29.61	30.84	27.58	27.64	40.36
USELFP22 (Likelihood of using loan forgiveness programs, AY 2020-21/2021-22)	11,550	49.14	43.68	44.44	42.44	45.18	50.75	66.17	53.87	57.47	42.8	42.34	58.32

See notes at end of table.

Table I-1. Weighted item response rates, by control and level of institution and variable: 2022—Continued

			Public						Private nonprofit			Private for-profit		
			Less- than- 2-year	2-year	4-year, non- doctorate- granting, primarily subbacca- laureate	4-year, non- doctorate- granting, primarily bacca- laureate	4-year, doctorate- granting	Less- than- 4-year	4-year, non- doctorate- granting	4-year, doctorate- granting	Less- than- 2-year	2-year	4-year	
Variable	Sample Size	All sample members												
USEMNSRV22 (PSE services used: psychological or mental health services, AY 2020-21/2021-22)	19,120	80.44	64.76	77.79	75.83	80.8	83.84	73.92	81.5	84.99	70.77	63.03	80.78	
USEMILSRV22 (PSE services used: services for active military or veterans, AY 2020-21/2021-22)	19,120	80.42	65.48	77.77	75.86	80.76	83.83	73.92	81.39	84.85	71.69	62.9	80.84	
USETRPLN22 (PSE services used: transfer advising or planning, AY 2020-21/2021-22)	19,120	80.48	65.48	77.82	76.06	80.65	83.83	73.92	81.71	84.95	71.45	62.82	80.94	
USETUTOR22 (PSE services used: peer or other tutoring or skills lab, AY 2020-21/2021-22)	19,120	80.54	65.48	77.91	76.19	80.81	83.89	72.38	81.66	84.95	71.69	62.87	80.95	
VETBEN22 (Veterans'/DOD education benefits: amount received, AY 2021-22)	22,320	100	100	100	100	100	100	100	100	100	100	100	100	
VTNEL22 (Voted in 2020 presidential election)	21,040	92.27	82.13	89.84	90.26	94.96	93.95	95.78	94.88	95.32	89.86	89.57	95.02	
WRKSTDY3Y (Work-study job: had work- study job through AY 2021-22)	22,320	85.05	77.18	84.01	83.44	84.05	86.56	78.05	85.68	88.27	79.74	78.45	86.43	
WRKTSKS22 (Job non-monetary benefits: importance of making decisions, AY 2021-22)	22,320	84.16	71.46	83.59	82.74	83.53	85.56	80.62	84.81	85.86	82	76.08	84.54	

Rounds to zero.

‡ Reporting standards not met (fewer than 30 unweighted nonrespondents).

NOTE: AY = academic year; CIP = Classification of Instructional Programs; DOD = Department of Defense; IPEDS = Integrated Postsecondary Education Data System; NPSAS = National Postsecondary Student Aid Study; NSF = National Science Foundation; PSE = postsecondary education; STEM = science, technology, engineering, and mathematics; TANF = Temporary Assistance for Needy Families; WIC = Special Supplemental Nutrition Program for Women, Infants, and Children. Response rates are weighted using the final BPS:20/22 analysis weight.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table I-2. Mean values of continuous variables before and after imputation: 2022

Variable	Description	Mean before imputation	Mean after imputation	Difference	Percent relative difference
AGE	Age, as of 12/31/2019	19.78	19.79	-0.01	-0.04
ALTEARN22	Alternative plans: annual earning estimate if had not attended college, AY 2021-22	27,211.36	27,036.95	174.41	0.65
CARAMT22	Monthly car loan or lease payment, AY 2021-22	75.52	77.88	-2.36*	-3.02
CINCOME	Total income (continuous), 2017	79,632.84	87,067.69	-7,434.84*	-8.54
CINCOME22	Total income (continuous), 2019	83,292.47	95,104.88	-11,812.40*	-12.42
CRDBAL22	Credit cards: balance due, AY 2021-22	711.62	781.41	-69.79*	-8.93
CUMULN22	Cumulative student loans: total amount borrowed through AY 2021-22	9,040.81	9,462.57	-421.76*	-4.46
DEPINC22	Dependent students: parents' income, 2019	93,259.33	107,138.80	-13,879.50*	-12.95
DSTUINC22	Dependent students: income, 2019	3,821.12	4,717.09	-895.97*	-18.99
EMPLYAM22	Employer aid (student and parents): amount received, AY 2021-22	3,034.49	3,141.95	-107.46	-3.42
EMPMNT3Y	All jobs: number of months worked through AY 2021-22	16.47	17.04	-0.56*	-3.31
EMPMTEN3Y	All jobs: number of months worked since last enrolled through AY 2021-22	2.87	3.61	-0.74*	-20.46
ENMNFT3Y	Months enrolled full time (total) through AY 2021-22	19.28	19.23	0.05	0.27
ENMNP3Y	Months enrolled part time (total) through AY 2021-22	3.57	3.62	-0.05*	-1.43
ENMNT22	Number of months since last enrolled through AY 2021-22	6.88	6.75	0.13*	1.89
ENMNT3Y	Months enrolled (total) through AY 2021-22	22.85	22.85	0.00	0.00
ENMNTAA3Y	Associate's degree: months enrolled before attainment through AY 2021-22	20.67	20.62	0.05	0.27
ENMNTCT3Y	Certificate: months enrolled before attainment through AY 2021-22	15.32	15.38	-0.06	-0.39
EXPWAGE22	Expected job: annual salary expected, AY 2021-22	76,767.66	76,249.55	518.11	0.68
GPA22	Grade point average, AY 2020-21/2021-22	334.09	329.98	4.10*	1.24
GPA3Y	Grade point average through AY 2021-22	323.98	320.46	3.52*	1.10
HRSWK21	Jobs while enrolled: hours worked, AY 2020-21	10.64	12.03	-1.40*	-11.61
HRSWK22	Jobs while enrolled: hours worked, AY 2021-22	11.37	12.55	-1.18*	-9.40
HRSWK3Y	Jobs while enrolled: hours worked when last enrolled through AY 2021-22	11.37	11.34	0.03	0.30
INCSPS22	Independent students: spouse's income, 2019	26,841.00	27,934.12	-1,093.13	-3.91
INDINC22	Independent students: student and spouse's income, 2019	23,603.25	24,795.25	-1,192.00	-4.81
ISTUINC22	Independent students: income, 2019	16,353.94	17,527.43	-1,173.49	-6.70
JOBHRS22	Job: hours worked weekly, 2022	36.27	36.17	0.09	0.26
JOBMNT3Y	All jobs: number of months worked since beginning PSE	15.19	16.06	-0.88*	-5.45
MTGAMT22	Monthly mortgage or rent amount, AY 2021-22	488.85	490.11	-1.27	-0.26
OTHGTAMT22	Outside grants (private and employer): amount received, AY 2021-22	4,088.78	4,096.55	-7.78	-0.19
PRICUM22	Private (alternative) loans: cumulative amount borrowed through AY 2021-22	1,839.23	2,238.98	-399.75*	-17.85
PRIVAI22	Private source grants: amount received, AY 2021-22	932.02	954.60	-22.58	-2.37
PRRPY22	Private student loans: monthly payment amount, AY 2021-22	228.84	223.56	5.28	2.36
SALARY22	Job: annual salary, 2022	30,745.42	30,384.35	361.07	1.19
TFMNDI3Y	Transfer (first): months elapsed before destination inst through AY 2021-22	19.23	18.82	0.41*	2.20
TFMNT3Y	Transfer (first): months enrolled before destination inst through AY 2021-22	14.08	13.52	0.57*	4.18

* $p < .05$.

NOTE: AY = academic year; PSE = postsecondary education. Means were weighted using the final BPS:20/22 analysis weight. Cases with legitimate skips for the item are not included in the estimated means. The difference is computed as the mean before imputation minus the mean after imputation.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Table I-3. Distributions of categorical variables before and after imputation: 2022

Variable	Description	Value	Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
ACDPARTANY22	Academic activities: participated in any academic activity, AY 2020–21/2021–22	0	No	37.92	38.07	-0.16	-0.41
		1	Yes	62.08	61.93	0.16	0.25
ACDPARTAPP22	Academic activities: participated in an apprenticeship, AY 2020–21/2021–22	0	No	93.49	93.32	0.17	0.18
		1	Yes	6.51	6.68	-0.17	-2.51
ACDPARTCOM22	Academic activities: participated in community-based or service learning project, AY 2020–21/2021–22	0	No	78.82	78.41	0.40*	0.51
		1	Yes	21.18	21.59	-0.40*	-1.86
ACDPARTIF22	Academic activities: participated in internship or field experiment, AY 2020–21/2021–22	0	No	74.30	74.37	-0.07	-0.09
		1	Yes	25.70	25.63	0.07	0.26
ACDPARTLC22	Academic activities: participated in learning community, AY 2020–21/2021–22	0	No	75.43	74.86	0.57*	0.77
		1	Yes	24.57	25.14	-0.57*	-2.28
ACDPARTRSCH22	Academic activities: participated in guided research experience, AY 2020–21/2021–22	0	No	80.11	79.84	0.27	0.33
		1	Yes	19.89	20.16	-0.27	-1.32
ACDPARTSE22	Academic activities: participated in culminating senior experience, AY 2020–21/2021–22	0	No	87.15	86.67	0.48*	0.55
		1	Yes	12.85	13.33	-0.48*	-3.60
ACDPARTSSC22	Academic activities: participated in student success course, AY 2020–21/2021–22	0	No	69.64	69.57	0.07	0.10
		1	Yes	30.36	30.43	-0.07	-0.24
ALTWORK22	Alternative plans: work status if had not attended college, AY 2021–22	1	Worked for pay	89.65	89.97	-0.32	-0.35
		2	Worked, but not be paid	1.59	1.59	0.00	0.10
		3	Not worked (for any reason)	6.33	6.11	0.21	3.50
		4	Entered the military	2.44	2.33	0.10	4.36

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
AT1TY3Y	First degree type attained through AY 2021–22	0 No degree attained	87.47	87.46	0.01	0.01
		1 Certificate	5.88	5.88	0.00	0.05
		2 Associate's degree	6.03	6.03	0.00	0.00
		3 Bachelor's degree	0.62	0.63	-0.01	-1.59
ATAANU3Y	Number of associate's degrees attained through AY 2021–22	0 No degree attained	85.72	85.80	-0.08	-0.09
		1 One	14.11	14.00	0.10*	0.75
		2 Two	0.16	0.19	-0.02	-12.98
		3 Three	0.00	0.00	0.00	2.61
ATBADT3Y	First bachelor's degree date attained through AY 2021–22	0 No degree attained	98.84	98.81	0.02	0.02
		201911 November 2019	0.01	0.01	0.00	2.61
		202001 January 2020	0.02	0.02	0.00	2.61
		202005 May 2020	0.00	0.00	0.00	2.61
		202012 December 2020	0.01	0.01	0.00	2.61
		202104 April 2021	0.01	0.01	0.00	2.61
		202105 May 2021	0.04	0.04	0.00	-11.03
		202106 June 2021	0.05	0.05	0.00	-6.73
		202107 July 2021	0.02	0.03	-0.01	-28.81
		202108 August 2021	0.02	0.02	0.00	2.61
		202110 October 2021	0.01	0.01	0.00	2.61
		202111 November 2021	0.02	0.02	0.00	2.61
		202112 December 2021	0.16	0.18	-0.02	-12.98
		202201 January 2022	0.01	0.01	0.00	2.61
		202202 February 2022	0.00	0.00	0.00	2.61
		202203 March 2022	0.07	0.07	0.00	2.61
		202204 April 2022	0.07	0.07	0.00*	2.61
		202205 May 2022	0.58	0.57	0.01	1.48
		202206 June 2022	0.07	0.08	0.00	-0.40
ATBANU3Y	Number of bachelor's degrees attained through AY 2021–22	0 No degree attained	98.84	98.81	0.02	0.02
		1 One	1.16	1.19	-0.02	-2.06
ATCTNU3Y	Number of certificates attained through AY 2021–22	0 No degree attained	57.93	58.75	-0.82*	-1.39
		1 One	41.39	40.55	0.83*	2.05
		2 Two	0.68	0.69	-0.01	-2.02

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
ATDEG1	Degree attained during AY 2019–20	0 No degree attained	98.50	98.49	0.00	0.00
		1 One or more degrees attained	1.50	1.51	0.00	-0.30
ATDEG2	Degree attained during AY 2020–21	0 No degree attained	94.34	94.30	0.03*	0.04
		1 One or more degrees attained	5.66	5.70	-0.03*	-0.60
ATDEG3	Degree attained during AY 2021–22	0 No degree attained	94.36	94.33	0.03*	0.03
		1 One or more degrees attained	5.64	5.67	-0.03*	-0.54
ATDEG3Y	First year attained degree through AY 2021–22	0 No degree attained	87.32	87.46	-0.14*	-0.16
		1 Academic year 2019–20	1.53	1.51	0.02*	1.37
		2 Academic year 2020–21	5.70	5.64	0.06*	1.04
		3 Academic year 2021–22	5.45	5.39	0.06*	1.12
ATHTY3Y	Highest degree attained at any institution through AY 2021–22	0 No degree attained	87.52	87.46	0.06*	0.07
		1 Certificate	5.07	5.08	-0.01	-0.24
		2 Associate's degree	6.75	6.78	-0.04*	-0.56
		3 Bachelor's degree	0.66	0.67	-0.01	-1.81
ATHTYF3Y	Highest degree attained at first institution through AY 2021–22	0 No degree attained	88.58	88.55	0.03*	0.04
		1 Certificate	4.44	4.46	-0.01	-0.25
		2 Associate's degree	6.38	6.40	-0.01	-0.23
		3 Bachelor's degree	0.60	0.60	-0.01	-1.01
ATLTY3Y	Last degree type attained through AY 2021–22	0 No degree attained	87.47	87.46	0.01	0.01
		1 Certificate	5.26	5.26	0.00	0.09
		2 Associate's degree	6.62	6.62	0.00	0.05
		3 Bachelor's degree	0.65	0.66	-0.02	-2.28
ATNUM3Y	Degrees: number attained through AY 2021–22	0 No degree attained	87.47	87.46	0.01	0.01
		1 One	11.39	11.37	0.03	0.24
		2 Two	1.12	1.14	-0.02	-1.85
		3 Three	0.02	0.03	-0.01	-39.82
AWAREIDR22	Aware of income-driven student loan repayment plans, AY 2020–21/2021–22	0 No	60.39	59.33	1.06*	1.78
		1 Yes	39.61	40.67	-1.06*	-2.60

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
AWARELFP22	Aware of student loan forgiveness programs, AY 2020–21/2021–22	0 No	49.02	48.99	0.03	0.05
		1 Yes	50.98	51.01	-0.03	-0.05
CAMPSAFE22	Felt safe attending primary institution, AY 2020–21/2021–22	1 Completely unsafe	1.49	1.66	-0.17*	-10.12
		2 Somewhat unsafe	3.71	3.68	0.03	0.91
		3 Neither unsafe nor safe	9.90	9.90	0.00	-0.04
		4 Somewhat safe	31.43	30.85	0.58*	1.87
		5 Completely safe	53.48	53.92	-0.44	-0.81
CARLN22	Had a car loan or lease, AY 2021–22	0 No	80.23	79.99	0.24	0.30
		1 Yes	19.77	20.01	-0.24	-1.18
CITZN22	Citizenship, AY 2021–22	1 U.S. citizen	93.49	92.26	1.23*	1.33
		2 Permanent resident or other eligible non-citizen	4.25	4.90	-0.64*	-13.13
		3 Foreign or international student	2.26	2.84	-0.58*	-20.50
CLSEVR3Y	Ever took classes only (not a degree program) through AY 2021–22	0 No	87.72	86.99	0.74*	0.85
		1 Yes	12.28	13.01	-0.74*	-5.67
COVATND21	Attended any postsecondary institution during COVID-19, AY 2020–21	0 No	20.31	19.27	1.04*	5.37
		1 Yes	79.69	80.73	-1.04*	-1.28
COVCHNGINC21	Educational impacts due to COVID-19: increased enrollment from part-time to full-time, AY 2020–21	0 No	97.49	97.49	0.00	0.00
		1 Yes	2.51	2.51	0.00	0.04
COVCHNGINS21	Educational impacts due to COVID-19: changed institution where enrolled, AY 2020–21	0 No	92.19	91.68	0.51*	0.56
		1 Yes	7.81	8.32	-0.51*	-6.18
COVCHNGMAJ21	COVID-19 impacts at primary institution: changed declared or intended major, AY 2020–21	0 No	87.49	87.34	0.14	0.16
		1 Yes	12.51	12.66	-0.14	-1.12
COVCHNGNENR21	Educational impacts due to COVID-19: did not attend for entire term/semester, AY 2020–21	0 No	89.41	87.64	1.77*	2.01
		1 Yes	10.59	12.36	-1.77*	-14.29

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value	Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
COVCHNGRED21	Educational impacts due to COVID-19: reduced enrollment from full-time to part-time, AY 2020–21	0	No	89.97	89.33	0.64*	0.71
		1	Yes	10.03	10.67	-0.64*	-5.98
COVDEVHARD21	Primary institution online coursetaking during COVID-19: device had necessary hardware, AY 2020–21	0	No	9.00	8.98	0.01	0.15
		1	Yes	91.00	91.02	-0.01	-0.01
COVDEVSOFT21	Primary institution online coursetaking during COVID-19: device had necessary software, AY 2020–21	0	No	7.87	8.19	-0.32	-3.86
		1	Yes	92.13	91.81	0.32	0.34
COVDISACCES21	Primary institution provided COVID-19 accommodations: access online course content, AY 2020–21	0	No	86.04	86.49	-0.46	-0.53
		1	Yes	13.96	13.51	0.46	3.38
COVDISCOMP21	Primary institution provided COVID-19 accommodations: complete online coursework, AY 2020–21	0	No	83.15	82.91	0.24	0.29
		1	Yes	16.85	17.09	-0.24	-1.42
COVDISTRRACT21	Online coursetaking during COVID-19: difficulties accessing a physical space free of interruption or distraction, AY 2020–21	0	Never	15.04	14.65	0.38	2.60
		1	Rarely	33.86	34.21	-0.35	-1.02
		2	Sometimes	34.64	34.56	0.08	0.23
		3	Often	11.96	12.20	-0.23	-1.92
		4	Always	4.50	4.38	0.12	2.79
COVEMRGANY21	Personal impacts due to COVID-19: received emergency financial assistance from any source other than primary institution, AY 2020–21	0	No	87.09	87.01	0.08	0.09
		1	Yes	12.91	12.99	-0.08	-0.62
COVENRDIS	COVID-19 pandemic: enrollment disruptions/changes, spring 2020	0	No	5.89	5.57	0.32	5.74
		1	Yes	94.11	94.43	-0.32	-0.34

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
COVEXPABS	COVID-19 pandemic: took leave of absence from NPSAS, spring 2020	0 No	95.71	95.58	0.13	0.14
		1 Yes	4.29	4.42	-0.13	-2.94
COVEXPACTMOD21	COVID-19 impacts at primary institution: internship, apprenticeship, student teaching, or clinical placement mod or cancelled, AY 2020–21	0 No	87.78	88.22	-0.44*	-0.50
		1 Yes	12.22	11.78	0.44*	3.76
COVEXPBRK	COVID-19 pandemic: school break extended, spring 2020	0 No	56.70	56.56	0.13	0.24
		1 Yes	43.30	43.44	-0.13	-0.31
COVEXPCALMOD21	COVID-19 impacts at primary institution: academic calendar modified, AY 2020–21	0 No	55.60	57.78	-2.18*	-3.77
		1 Yes	44.40	42.22	2.18*	5.16
COVEXPCAN	COVID-19 pandemic: some or all classes canceled, spring 2020	0 No	88.57	88.77	-0.20	-0.23
		1 Yes	11.43	11.23	0.20	1.81
COVEXPCC	COVID-19 pandemic: had difficulty finding safe and stable childcare, spring 2020	0 No	65.42	63.71	1.71	2.69
		1 Yes	34.58	36.29	-1.71	-4.71
COVEXPFIN	COVID-19 pandemic: received emergency financial assistance from NPSAS, spring 2020	0 No	84.90	84.98	-0.08	-0.10
		1 Yes	15.10	15.02	0.08	0.54
COVEXPFOOD	COVID-19 pandemic: had difficulty accessing or paying for food, spring 2020	0 No	92.45	92.85	-0.40*	-0.43
		1 Yes	7.55	7.15	0.40*	5.54
COVEXPFOOD21	General experience during COVID-19: had difficulty accessing or paying for food, AY 2020–21	0 No	88.37	88.36	0.00	0.00
		1 Yes	11.63	11.64	0.00	-0.03
COVEXPGRDMOD21	COVID-19 impacts at primary institution: modified grading system, AY 2020–21	0 No	64.67	65.86	-1.19*	-1.81
		1 Yes	35.33	34.14	1.19*	3.49

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
COVEXPHM	COVID-19 pandemic: had difficulty finding safe and stable housing, spring 2020	0 No	96.96	97.08	-0.12	-0.12
		1 Yes	3.04	2.92	0.12	3.98
COVEXPHM21	General experience during COVID-19: had difficulty finding safe and stable housing, AY 2020–21	0 No	95.27	95.27	-0.01	-0.01
		1 Yes	4.73	4.73	0.01	0.13
COVEXPHRS21	General experience during COVID-19: worked more due to additional job or increased hours, AY 2020–21	0 No	84.46	84.38	0.08	0.09
		1 Yes	15.54	15.62	-0.08	-0.51
COVEXPHYB21	COVID-19 impacts at primary institution: some or all classes delivered in hybrid format, AY 2020–21	0 No	45.41	45.26	0.15	0.34
		1 Yes	54.59	54.74	-0.15	-0.28
COVEXPJOB	COVID-19 pandemic: lost job or lost income because of reduced hours, spring 2020	0 No	76.18	75.45	0.73*	0.97
		1 Yes	23.82	24.55	-0.73*	-2.99
COVEXPJOB21	General experience during COVID-19: lost job or lost income because of reduced hours, AY 2020–21	0 No	80.41	80.38	0.03	0.04
		1 Yes	19.59	19.62	-0.03	-0.16
COVEXPMMOD21	COVID-19 impacts at primary institution: class delivery method modified, AY 2020–21	0 No	26.59	26.31	0.28	1.07
		1 Yes	73.41	73.69	-0.28	-0.38
COVEXPMOT	COVID-19 pandemic: moved to another living situation, spring 2020	0 No	94.68	94.60	0.08	0.09
		1 Yes	5.32	5.40	-0.08	-1.55
COVEXPMOT21	General experience during COVID-19: moved to another living situation, AY 2020–21	0 No	87.69	87.69	0.00	0.00
		1 Yes	12.31	12.31	0.00	0.00
COVEXPONL	COVID-19 pandemic: some or all classes moved to online only, spring 2020	0 No	8.79	9.20	-0.40*	-4.40
		1 Yes	91.21	90.80	0.40*	0.45

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
COVEXPONL21	COVID-19 impacts at primary institution: some or all classes delivered entirely online, AY 2020–21	0 No	20.45	20.71	-0.26	-1.24
		1 Yes	79.55	79.29	0.26	0.32
COVEXPPAD	COVID-19 pandemic: moved back to permanent address, spring 2020	0 No	68.49	67.19	1.30*	1.93
		1 Yes	31.51	32.81	-1.30*	-3.95
COVEXPPAD21	General experience during COVID-19: moved back to permanent address, AY 2020–21	0 No	80.96	81.15	-0.19*	-0.23
		1 Yes	19.04	18.85	0.19*	0.99
COVEXPSAP	COVID-19 pandemic: study-abroad program canceled or modified, spring 2020	0 No	98.08	97.96	0.12	0.12
		1 Yes	1.92	2.04	-0.12	-5.91
COVEXPWIT	COVID-19 pandemic: withdrew from NPSAS, spring 2020	0 No	95.34	95.51	-0.18	-0.19
		1 Yes	4.66	4.49	0.18	3.96
COVFINDIS	Financial disruptions/changes due to COVID-19, spring 2020	0 No	65.23	63.90	1.33*	2.09
		1 Yes	34.77	36.10	-1.33*	-3.69
COVFINDIS21	Financial disruptions/changes due to COVID-19, AY 2020–21	0 No	56.46	58.20	-1.74*	-2.99
		1 Yes	43.54	41.80	1.74*	4.17
COVHOUDIS	Housing disruptions/changes due to COVID-19, spring 2020	0 No	63.58	62.42	1.16*	1.86
		1 Yes	36.42	37.58	-1.16*	-3.09
COVHOUDIS21	Housing disruptions/changes due to COVID-19, AY 2020–21	0 No	70.47	70.65	-0.18*	-0.26
		1 Yes	29.53	29.35	0.18*	0.62
COVINFOHLTH21	Primary institution COVID-19 information: obtaining physical or mental healthcare, AY 2020–21	0 No	80.52	80.40	0.12	0.15
		1 Yes	19.48	19.60	-0.12	-0.62
COVINFOMHLTH21	Primary institution COVID-19 information: obtaining mental healthcare, AY 2020–21	0 No	51.40	51.41	-0.01	-0.02
		1 Yes	16.19	16.31	-0.12	-0.75
		2 Not needed	32.41	32.28	0.13	0.40

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
COVINFOPHLTH21	Primary institution COVID-19 information: obtaining physical healthcare, AY 2020–21	0 No	51.13	51.12	0.01	0.03
		1 Yes	14.85	14.85	-0.01	-0.06
		2 Not needed	34.02	34.03	0.00	-0.01
COVINTACC21	Residence had access to high-speed internet, AY 2020–21	0 Residence in 2020–21 did not have internet access	2.28	2.28	-0.01	-0.37
		1 Residence in 2020–21 had access to high-speed internet	86.52	86.57	-0.05	-0.05
		2 Residence in 2020–21 had access to non-high-speed internet	11.21	11.15	0.06	0.50
COVINTPROB21	Online coursetaking during COVID-19: difficulties with internet connection, AY 2020–21	0 Never	21.41	21.20	0.21	1.00
		1 Rarely	24.40	23.96	0.43	1.80
		2 Sometimes	28.01	28.09	-0.08	-0.28
		3 Often	16.73	17.16	-0.43	-2.51
		4 Always	9.46	9.59	-0.14	-1.41
COVMNTDIFCON21	Mental impacts due to COVID-19: difficulty concentrating, AY 2020–21	1 Completely disagree	11.72	11.71	0.01	0.12
		2 Somewhat disagree	6.42	6.41	0.01	0.09
		3 Neither disagree nor agree	18.02	17.91	0.11	0.63
		4 Somewhat agree	31.81	31.88	-0.07	-0.23
		5 Completely agree	32.03	32.09	-0.06	-0.19
COVMNTDOWN21	Mental impacts due to COVID-19: felt down, depressed, or hopeless, AY 2020–21	1 Completely disagree	14.69	14.71	-0.03	-0.18
		2 Somewhat disagree	8.44	8.42	0.02	0.26
		3 Neither disagree nor agree	20.30	20.20	0.10	0.48
		4 Somewhat agree	26.95	26.98	-0.03	-0.12
		5 Completely agree	29.63	29.69	-0.06	-0.20
COVMNTLONLI21	Mental impacts due to COVID-19: experienced loneliness or isolation, AY 2020–21	1 Completely disagree	12.44	12.47	-0.03	-0.28
		2 Somewhat disagree	7.15	7.15	-0.01	-0.08
		3 Neither disagree nor agree	18.01	17.92	0.09	0.51
		4 Somewhat agree	28.21	28.16	0.06	0.20
		5 Completely agree	34.19	34.30	-0.11	-0.31

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
COVMNTSTRESS21	Mental impacts due to COVID-19: experienced increased stress or anxiety, AY 2020–21	1 Completely disagree	9.24	9.18	0.06	0.63
		2 Somewhat disagree	4.22	4.27	-0.04	-1.02
		3 Neither disagree nor agree	13.52	13.46	0.06	0.48
		4 Somewhat agree	34.34	34.36	-0.02	-0.06
		5 Completely agree	38.68	38.74	-0.06	-0.15
COVNOTATND21	Decision not to attend school due to COVID-19, AY 2020–21	1 Completely disagree	28.55	26.94	1.61*	5.97
		2 Somewhat disagree	5.53	6.09	-0.57	-9.29
		3 Neither disagree nor agree	17.15	16.30	0.85*	5.20
		4 Somewhat agree	21.93	21.81	0.12	0.56
		5 Completely agree	26.85	28.86	-2.01*	-6.97
COVPPSEMGR21	Financial impacts due to COVID-19: received emergency financial assistance from primary institution, AY 2020–21	0 No	69.17	69.48	-0.31	-0.44
		1 Yes	30.83	30.52	0.31	1.01
COVPRSFOOD21	Personal impacts due to COVID-19: accessed food assistance resources, AY 2020–21	0 No	89.58	89.56	0.02	0.03
		1 Yes	10.42	10.44	-0.02	-0.23
COVPRSKCARE21	Personal impacts due to COVID-19: children or other dependents did not attend day care or child care, AY 2020–21	0 No	61.28	60.73	0.56	0.92
		1 Yes	38.72	39.27	-0.56	-1.42
COVPRSKSCH21	Personal impacts due to COVID-19: children or other dependents had a non-traditional schedule for school, AY 2020–21	0 No	54.38	53.77	0.61	1.14
		1 Yes	45.62	46.23	-0.61	-1.32
COVPRSMRT21	Personal impacts due to COVID-19: difficulty paying mortgage or rent, AY 2020–21	0 No	85.31	85.21	0.10	0.11
		1 Yes	14.69	14.79	-0.10	-0.64
COVPSATND21	Attended primary institution during COVID-19, AY 2020–21	0 No	26.97	25.73	1.25*	4.84
		1 Yes	73.03	74.27	-1.25*	-1.68

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
COVPYADDF21	Financial impacts due to COVID-19: borrowed more money in federal student loans, AY 2020–21	0 No	48.65	48.45	0.19	0.40
		1 Yes	12.13	12.43	-0.31	-2.49
		2 Not applicable	39.23	39.11	0.12	0.30
COVPYADDP21	Financial impacts due to COVID-19: borrowed more money in private student loans, AY 2020–21	0 No	49.88	49.78	0.10	0.19
		1 Yes	3.09	3.17	-0.08	-2.64
		2 Not applicable	47.03	47.05	-0.01	-0.03
COVPYFEWF21	Financial impacts due to COVID-19: borrowed less money in federal student loans, AY 2020–21	0 No	53.69	53.22	0.46	0.87
		1 Yes	4.70	5.17	-0.47*	-9.01
		2 Not applicable	41.61	41.61	0.00	0.01
COVPYFEWP21	Financial impacts due to COVID-19: borrowed less money in private student loans, AY 2020–21	0 No	50.24	49.84	0.40	0.81
		1 Yes	1.61	1.94	-0.33*	-17.18
		2 Not applicable	48.15	48.22	-0.07	-0.14
COVPYLSFAM21	Financial impacts due to COVID-19: less money from friends and/or family, AY 2020–21	0 No	58.45	58.57	-0.12	-0.20
		1 Yes	6.07	6.12	-0.06	-0.90
		2 Not applicable	35.48	35.31	0.17	0.49
COVPYMORFAM21	Financial impacts due to COVID-19: more money from friends and/or family, AY 2020–21	0 No	54.79	54.75	0.05	0.08
		1 Yes	12.35	12.70	-0.35	-2.76
		2 Not applicable	32.86	32.55	0.30	0.94
COVREFTUITION	COVID-19 pandemic: received tuition refund from NPSAS, spring 2020	0 No	73.02	73.44	-0.42	-0.57
		1 Yes	26.98	26.56	0.42	1.59
COVROOMREF	COVID-19 pandemic: received room-and-board refund from NPSAS, spring 2020	0 No	27.05	29.55	-2.50*	-8.46
		1 Yes	72.95	70.45	2.50*	3.55

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
COVSOCEFF21	COVID-19 impacts at primary institution: online-only or hybrid classes not as effective, AY 2020–21	1 Completely disagree	5.30	5.32	-0.03	-0.50
		2 Somewhat disagree	7.51	7.54	-0.04	-0.49
		3 Neither agree nor disagree	13.86	14.13	-0.28	-1.95
		4 Somewhat agree	29.84	30.03	-0.20	-0.65
		5 Completely agree	43.50	42.97	0.54	1.25
COVSOCGRD21	COVID-19 impacts at primary institution: grades lower than expected, AY 2020–21	1 Completely disagree	22.85	22.96	-0.11	-0.49
		2 Somewhat disagree	14.62	14.84	-0.22	-1.47
		3 Neither disagree nor agree	22.05	22.32	-0.27	-1.21
		4 Somewhat agree	20.81	20.80	0.01	0.03
		5 Completely agree	19.67	19.08	0.59*	3.11
COVTECHSER	COVID-19 pandemic: received technology or technical services from NPSAS, spring 2020	0 No	61.54	61.95	-0.41*	-0.66
		1 Yes	38.46	38.05	0.41*	1.07
COVTHFEWADV21	COVID-19 impacts at primary institution: fewer interactions with academic advisors, AY 2020–21	1 Completely disagree	7.42	7.33	0.08	1.16
		2 Somewhat disagree	6.87	7.00	-0.13	-1.88
		3 Neither agree nor disagree	20.60	20.23	0.37	1.84
		4 Somewhat agree	29.40	29.13	0.27	0.94
		5 Completely agree	35.70	36.30	-0.60*	-1.65
COVTHFEWTCH21	COVID-19 impacts at primary institution: fewer interactions with teachers, AY 2020–21	1 Completely disagree	5.29	5.17	0.12	2.33
		2 Somewhat disagree	4.92	4.94	-0.01	-0.23
		3 Neither agree nor disagree	14.95	14.41	0.55*	3.78
		4 Somewhat agree	31.14	31.10	0.04	0.13
		5 Completely agree	43.69	44.39	-0.70*	-1.57
COVTHLSCON21	COVID-19 impacts at primary institution: less connected to other students, AY 2020–21	1 Completely disagree	6.31	6.02	0.29*	4.86
		2 Somewhat disagree	4.82	4.82	0.01	0.13
		3 Neither agree nor disagree	16.63	16.46	0.17	1.05
		4 Somewhat agree	26.97	26.99	-0.03	-0.09
		5 Completely agree	45.26	45.71	-0.45	-0.98

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
COVTHLSRES21	COVID-19 impacts at primary institution: accessed campus learning resources less, AY 2020–21	1 Completely disagree	17.62	17.03	0.59*	3.47
		2 Somewhat disagree	10.23	10.50	-0.27	-2.57
		3 Neither agree nor disagree	24.55	24.43	0.12	0.50
		4 Somewhat agree	21.12	21.13	-0.01	-0.05
		5 Completely agree	26.48	26.91	-0.43	-1.60
COVTSUPP21	COVID-19 impacts at primary institution: reasonable teacher expectations, AY 2020–21	1 Completely disagree	3.62	3.85	-0.23	-6.01
		2 Somewhat disagree	10.78	10.39	0.40*	3.82
		3 Neither agree nor disagree	21.61	21.42	0.19	0.88
		4 Somewhat agree	37.58	37.70	-0.13	-0.33
		5 Completely agree	26.41	26.64	-0.23	-0.86
CRDNUM22	Credit cards: number in own name, AY 2021–22	0 None	41.64	41.74	-0.10	-0.23
		1 One	37.28	37.30	-0.02	-0.05
		2 Two or more	21.08	20.96	0.12	0.55
CRDPAY22	Credit cards: pay off balance, AY 2021–22	0 No	36.71	36.56	0.16	0.43
		1 Yes	63.29	63.44	-0.16	-0.25
CRTFNPATT3Y	Ever attempt to transfer credits from NPSAS through AY 2021–22	0 Never enrolled at another institution	80.66	78.11	2.55*	3.26
		1 Attempted to transfer credits	11.07	12.18	-1.11*	-9.12
		2 Did not attempt to transfer credits	8.28	9.71	-1.44*	-14.78
CURCNFDIF	Academic confidence: change between AY 2019–20 to AY 2020–21/2021–22	1 Confidence declined	21.25	23.12	-1.87*	-8.09
		2 Confidence remained constant	55.81	50.33	5.48*	10.88
		3 Confidence improved	22.94	26.54	-3.61*	-13.59
CURCONF22	Academic confidence, AY 2020–21/2021–22	1 Completely disagree	2.07	2.25	-0.18	-8.13
		2 Somewhat disagree	4.21	4.35	-0.14	-3.15
		3 Neither disagree nor agree	8.55	8.92	-0.38*	-4.21
		4 Somewhat agree	24.37	24.69	-0.32	-1.29
		5 Completely agree	60.80	59.78	1.02*	1.70

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
DECMJ22	Formally declared major or field of study when last enrolled through AY 2021–22	1 Formally declared	93.07	92.40	0.67*	0.72
		2 Decided but not formally declared	1.50	1.37	0.13*	9.13
		3 Undecided	5.44	6.23	-0.79*	-12.73
DEGNUM3Y	Number of degrees pursued through AY 2021–22	0 Zero	0.53	0.52	0.01*	1.85
		1 One	76.41	75.27	1.14*	1.51
		2 Two	19.96	20.43	-0.47*	-2.31
		3 Three	2.88	3.33	-0.45*	-13.46
		4 Four	0.21	0.36	-0.15*	-41.59
		5 Five	0.01	0.09	-0.08	-84.72
		6 Six	0.00	0.00	0.00	-87.62
DEPANY22	Had any dependents, AY 2021–22	0 No	90.57	90.60	-0.03	-0.03
		1 Yes	9.43	9.40	0.03	0.27
DEPCHILD22	Had dependent children, AY 2021–22	0 No	91.79	91.79	-0.01	-0.01
		1 Yes	8.21	8.21	0.01	0.09
DEPCOL22	Independent students: number of dependents in college, AY 2021–22	0 Zero	93.88	88.63	5.25*	5.93
		1 One	6.04	8.79	-2.76*	-31.33
		2 Two	0.08	1.30	-1.22	-93.97
		3 Three	0.00	0.73	-0.73*	-100.00
		4 Four	0.00	0.55	-0.55	-100.00
DEPEND22	Dependency status, AY 2021–22	1 Dependent student	82.68	85.39	-2.70*	-3.17
		2 Independent student	17.32	14.61	2.70*	18.50
DEPEND5B	Dependency and marital status (5 categories, separated is unmarried), AY 2019–20	1 Dependent	84.74	85.06	-0.32*	-0.38
		2 Independent, no dependents, unmarried/separated	6.61	6.68	-0.06	-0.93
		3 Independent, no dependents, married	1.35	1.30	0.05*	3.85
		4 Independent, with dependents, unmarried/separated	4.80	4.59	0.21*	4.65
		5 Independent, with dependents, married	2.50	2.38	0.12*	5.01

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
DEPNUM22	Number of dependents, AY 2021–22	0 No dependents	90.69	90.60	0.09	0.09
		1 One	4.58	4.66	-0.08	-1.81
		2 Two	2.78	2.76	0.02	0.77
		3 Three	1.30	1.29	0.01	0.41
		4 Four	0.45	0.44	0.00	0.35
		5 Five	0.12	0.12	0.00	1.50
		6 Six	0.03	0.04	-0.01	-26.43
		7 Seven	0.03	0.05	-0.02	-44.99
		8 Eight	0.02	0.02	0.00	5.54
		10 Ten	0.01	0.01	0.00	5.54
DEPNUMCH22	Number of dependent children, AY 2021–22	0 No dependent children	91.96	91.79	0.16*	0.18
		1 One	4.32	4.38	-0.06	-1.37
		2 Two	2.31	2.35	-0.04	-1.89
		3 Three	0.99	1.03	-0.04	-4.22
		4 Four	0.28	0.29	-0.01	-2.74
		5 Five	0.12	0.11	0.01*	5.06
		6 Six	0.01	0.02	-0.01	-51.30
		7 Seven	0.02	0.02	0.00	5.29
DEPNUMOT22	Number of dependents other than children, AY 2021–22	8 Eight	0.00	0.00	0.00	5.29
		0 No other dependents	98.17	98.01	0.16*	0.16
		1 One	1.22	1.37	-0.15*	-10.64
		2 Two	0.33	0.35	-0.02	-7.00
		3 Three	0.15	0.14	0.00	2.16
		4 Four	0.08	0.08	0.00*	5.62
		5 Five	0.03	0.03	0.00	2.69
		6 Six	0.00	0.00	0.00	5.62
		7 Seven	0.00	0.00	0.00	5.62
		8 Eight	0.02	0.02	0.00	5.62
DEPOTHER22	Had dependents other than children, AY 2021–22	0 No	97.98	98.01	-0.03	-0.03
		1 Yes	2.02	1.99	0.03	1.63

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
DGEVR22	Highest level of education ever expected through AY 2021–22	1 No degree or certificate expected	6.91	6.90	0.00	0.04
		2 Undergraduate certificate or diploma	5.84	5.86	-0.02	-0.41
		3 Associate's degree	12.77	12.66	0.10*	0.83
		4 Bachelor's degree	39.99	40.17	-0.18*	-0.44
		5 Post-bachelor's or post-master's certificate	1.00	1.00	0.01*	0.87
		6 Master's degree	21.43	21.38	0.06	0.26
		7 Doctoral degree - professional practice	5.78	5.77	0.00	0.02
		8 Doctoral degree - research/scholarship or other	6.29	6.26	0.03	0.43
DGEXPCHG22	Highest level of education ever expected, changes between AY 2019–20 to AY 2021–22	1 Highest level of education ever expected decreased	34.84	37.72	-2.89*	-7.66
		2 Highest level of education ever expected remained constant	51.19	46.34	4.84*	10.45
		3 Highest level of education ever expected increased	13.98	15.93	-1.96*	-12.27
DIS16A22	Disability: deaf or serious difficulty hearing, AY 2021–22	0 No	98.67	98.65	0.01	0.01
		1 Yes	1.33	1.35	-0.01	-1.04
DIS16B22	Disability: blind or serious difficulty seeing, AY 2021–22	0 No	97.33	97.27	0.06	0.06
		1 Yes	2.67	2.73	-0.06	-2.17
DIS17A22	Disability: serious difficulty concentrating, remembering, deciding, AY 2021–22	0 No	78.47	78.40	0.07	0.09
		1 Yes	21.53	21.60	-0.07	-0.32
DIS17B22	Disability: serious difficulty walking or climbing stairs, AY 2021–22	0 No	98.42	98.43	-0.01	-0.01
		1 Yes	1.58	1.57	0.01	0.41
DISABL22	Disability: had some type of disability, AY 2021–22	0 No	75.90	75.89	0.01	0.01
		1 Yes	24.10	24.11	-0.01	-0.03
DISTALL22	Distance education: entire program at primary institution was online, AY 2020–21/2021–22	0 No	82.98	82.43	0.55*	0.67
		1 Yes	17.02	17.57	-0.55*	-3.14
DISTEDUC22	Distance education: any classes taken at primary institution taught entirely online, AY 2020–21/2021–22	0 No	7.74	7.14	0.61*	8.51
		1 Yes	92.26	92.86	-0.61*	-0.65

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
DISTYPES22	Disability: main type of condition or impairment, AY 2021–22	1 Hearing impairment (for example, deaf or hard of hearing)	3.45	3.15	0.30*	9.52
		2 Blindness or visual impairment that cannot be corrected by wearing glasses	2.52	2.36	0.16	6.80
		3 Speech or language impairment	0.80	0.72	0.08	10.57
		4 Orthopedic or mobility impairment	1.62	1.52	0.10	6.40
		5 Specific learning disability or dyslexia	2.60	2.53	0.06	2.45
		6 Attention deficit disorder (ADD)	22.13	22.59	-0.46	-2.03
		7 Health impairment or problem	1.66	1.74	-0.07	-4.28
		8 Mental, emotional, or psychiatric condition	22.95	23.14	-0.19	-0.82
		9 Depression	29.21	29.32	-0.11	-0.36
		10 Developmental disability	1.11	1.05	0.06	5.67
		11 Brain injury	0.61	0.64	-0.03	-4.54
		12 Other	11.35	11.24	0.10	0.90
DIVECON22	Interactions outside of class: different economic background, AY 2020–21/2021–22	0 No	7.96	8.09	-0.13	-1.55
		1 Yes	79.60	79.72	-0.12	-0.15
		2 Don't Know	12.44	12.19	0.24	1.98
DIVECONFRQ22	Frequency of meaningful and honest conversations: different economic background, AY 2020–21/2021–22	0 Never	3.93	4.11	-0.18	-4.28
		1 Rarely	7.75	7.53	0.23*	3.00
		2 Sometimes	28.33	27.94	0.39	1.41
		3 Often	37.09	37.61	-0.51	-1.36
		4 Always	22.89	22.82	0.07	0.31
DIVGEN22	Interactions outside of class: gender minority, AY 2020–21/2021–22	0 No	16.47	16.53	-0.06	-0.38
		1 Yes	55.79	55.91	-0.12	-0.21
		2 Don't Know	27.74	27.56	0.18	0.66
DIVGENFRQ22	Frequency of meaningful and honest conversations: gender minority, AY 2020–21/2021–22	0 Never	5.22	5.51	-0.29	-5.20
		1 Rarely	15.44	15.04	0.40*	2.67
		2 Sometimes	30.78	30.39	0.38	1.25
		3 Often	27.95	28.36	-0.41	-1.43
		4 Always	20.60	20.69	-0.09	-0.44

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
DIVPOL22	Interactions outside of class: different political belief, AY 2020–21/2021–22	0 No	8.90	8.90	0.00	0.00
		1 Yes	72.76	72.99	-0.23	-0.32
		2 Don't Know	18.34	18.11	0.23	1.28
DIVPOLFRQ22	Frequency of meaningful and honest conversations: different political belief, AY 2020–21/2021–22	0 Never	4.96	5.00	-0.04	-0.87
		1 Rarely	12.21	12.22	-0.01	-0.10
		2 Sometimes	32.10	31.45	0.66*	2.08
		3 Often	31.48	31.93	-0.45	-1.40
		4 Always	19.25	19.40	-0.15	-0.79
DIVRACE22	Interactions outside of class: different race or ethnicity, AY 2020–21/2021–22	0 No	8.50	8.72	-0.22	-2.51
		1 Yes	83.35	83.37	-0.02	-0.02
		2 Don't Know	8.15	7.92	0.24	3.01
DIVRACEFRQ22	Frequency of meaningful and honest conversations: different race or ethnicity, AY 2020–21/2021–22	0 Never	4.59	4.72	-0.12	-2.61
		1 Rarely	8.60	8.54	0.06	0.69
		2 Sometimes	28.82	28.26	0.56*	1.97
		3 Often	34.71	35.19	-0.47*	-1.34
		4 Always				
DIVREL22	Interactions outside of class: different religious belief, AY 2020–21/2021–22	0 No	8.06	8.13	-0.07	-0.85
		1 Yes	76.80	77.10	-0.30	-0.39
		2 Don't Know	15.14	14.77	0.37*	2.50
DIVRELFQ22	Frequency of meaningful and honest conversations: different religious belief, AY 2020–21/2021–22	0 Never	4.45	4.50	-0.05	-1.15
		1 Rarely	9.99	9.91	0.08	0.85
		2 Sometimes	31.27	30.64	0.62*	2.04
		3 Often	33.26	33.92	-0.66*	-1.95
		4 Always	21.03	21.02	0.00	0.02
DIVSEX22	Interactions outside of class: different sexual orientation, AY 2020–21/2021–22	0 No	9.50	9.67	-0.17	-1.79
		1 Yes	74.22	74.37	-0.15	-0.20
		2 Don't Know	16.28	15.96	0.32	2.02

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
DIVSEFRQ22	Frequency of meaningful and honest conversations: different sexual orientation, AY 2020–21/2021–22	0 Never	5.37	5.49	-0.12	-2.17
		1 Rarely	11.75	11.40	0.34*	3.01
		2 Sometimes	29.48	29.13	0.35	1.20
		3 Often	30.66	31.26	-0.60*	-1.93
		4 Always	22.74	22.71	0.03	0.14
EMPNUM3Y	All jobs: number of jobs through AY 2021–22	0 Zero	12.37	10.65	1.72*	16.20
		1 One	28.55	27.82	0.73*	2.62
		2 Two	30.50	31.38	-0.88*	-2.82
		3 Three	18.12	18.95	-0.83*	-4.39
		4 Four	6.93	7.49	-0.56*	-7.54
		5 Five	2.45	2.56	-0.11	-4.31
		6 Six	0.77	0.78	-0.01	-1.78
		7 Seven	0.20	0.24	-0.05	-19.58
		8 Eight	0.07	0.07	0.00	1.34
		9 Nine	0.01	0.02	-0.01	-29.72
		10 Ten	0.02	0.02	0.00	16.20
		11 Eleven	0.02	0.02	0.00	3.33
EMRGAPPLY22	Applied for emergency aid from primary institution, AY 2021–22	0 No	56.50	57.28	-0.78	-1.36
		1 Yes	43.50	42.72	0.78	1.83
EMRGAWARE22	Aware of emergency aid program at primary institution, AY 2021–22	0 No	4.95	5.20	-0.25*	-4.78
		1 Yes	25.84	26.00	-0.16	-0.60
		2 Don't know	69.21	68.80	0.41	0.59
EMRGRCV22	Received emergency aid from primary institution, AY 2021–22	0 No	7.76	7.63	0.12	1.62
		1 Yes	92.24	92.37	-0.12	-0.13
ENCOEN3Y	Ever co-enrolled through AY 2021–22	0 No	91.24	87.16	4.08*	4.68
		1 Yes	8.76	12.84	-4.08*	-31.77

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
ENDTFS	Date of first enrollment in AY 2019–20	201907 July 2019	2.75	2.83	-0.08	-2.88
		201908 August 2019	59.83	59.73	0.10	0.16
		201909 September 2019	14.82	14.79	0.02	0.17
		201910 October 2019	1.25	1.27	-0.02	-1.74
		201911 November 2019	0.80	0.80	0.00	-0.35
		201912 December 2019	0.92	0.92	0.00	0.00
		202001 January 2020	15.56	15.57	-0.01	-0.04
		202002 February 2020	1.47	1.48	-0.01	-0.46
		202003 March 2020	1.14	1.19	-0.05	-4.40
		202004 April 2020	0.77	0.74	0.03*	3.68
		202005 May 2020	0.37	0.37	0.01	2.17
		202006 June 2020	0.32	0.31	0.01	4.42
ENINFM	First institution: attendance intensity first month enrolled, AY 2019–20	1 Full-time	83.51	82.85	0.66*	0.80
		2 Part-time	16.49	17.15	-0.66*	-3.87
ENINPT1	Attendance intensity pattern, AY 2019–20	1 Full-time	68.93	68.50	0.43*	0.63
		2 Part-time	12.65	13.15	-0.49*	-3.76
		3 Mixed	18.42	18.36	0.06	0.34
ENINPT2	Attendance intensity pattern, AY 2020–21	0 Not enrolled	20.31	19.27	1.04*	5.37
		1 Full-time	52.44	52.84	-0.40*	-0.76
		2 Part-time	10.40	10.06	0.34*	3.36
		3 Mixed	16.85	17.82	-0.97*	-5.44
ENINPT3	Attendance intensity pattern, AY 2021–22	0 Not enrolled	27.63	26.22	1.40*	5.35
		1 Full-time	47.53	47.99	-0.45*	-0.94
		2 Part-time	9.67	9.59	0.08	0.81
		3 Mixed	15.17	16.20	-1.03*	-6.34
ENINPT3Y	Attendance intensity through AY 2021–22	1 Always full-time	47.27	47.14	0.13	0.27
		2 Always part-time	7.46	7.31	0.15	2.05
		3 Mixed	45.27	45.55	-0.28	-0.60

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
ENINUM1	Number of institutions attended, AY 2019–20	1 One	95.12	92.77	2.36*	2.54
		2 Two	4.58	6.70	-2.12*	-31.67
		3 Three	0.23	0.46	-0.23*	-49.34
		4 Four	0.07	0.08	-0.01	-13.12
ENINUM2	Number of institutions attended, AY 2020–21	0 Zero	20.31	19.27	1.04*	5.37
		1 One	73.38	72.80	0.57*	0.79
		2 Two	5.94	7.42	-1.48*	-19.96
		3 Three	0.35	0.45	-0.10*	-22.33
		4 Four	0.02	0.05	-0.03	-53.26
ENINUM3	Number of institutions attended, AY 2021–22	0 Zero	27.63	26.22	1.40*	5.35
		1 One	66.14	66.32	-0.18	-0.27
		2 Two	5.90	7.02	-1.12*	-15.97
		3 Three	0.31	0.40	-0.08*	-21.14
		4 Four	0.02	0.04	-0.02	-53.08
		5 Five	0.01	0.01	0.00	9.32
ENINUM3Y	Number of institutions attended (total) through AY 2021–22	1 One	80.16	77.69	2.47*	3.17
		2 Two	17.88	19.79	-1.91*	-9.67
		3 Three	1.76	2.25	-0.49*	-21.75
		4 Four	0.19	0.25	-0.06*	-23.28
		5 Five	0.01	0.02	0.00	-26.13
		6 Six	0.00	0.00	0.00	5.49
ENLYAT3Y	Last year enrolled at any institution with no attainment through AY 2021–22	1 No degree, last enrolled in 2019–20 AY	14.23	13.81	0.42*	3.03
		2 No degree, last enrolled in 2020–21 AY	8.64	8.81	-0.17	-1.92
		3 No degree, last enrolled Fall 2021	4.60	5.70	-1.10*	-19.24
		4 No degree, last enrolled in Spring 2022	58.86	59.14	-0.28	-0.48
		5 Attained degree as of June 2022	13.67	12.54	1.13*	9.00

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
ENLYF3Y	Last year enrolled at first institution with no attainment through AY 2021–22	1 No degree, last enrolled in 2019–20 AY	21.48	20.84	0.63*	3.05
		2 No degree, last enrolled in 2020–21 AY	12.26	12.46	-0.19*	-1.55
		3 No degree, last enrolled Fall 2021	4.88	5.98	-1.11*	-18.50
		4 No degree, last enrolled in Spring 2022	49.39	49.26	0.13	0.27
		5 Attained degree as of June 2022	11.99	11.45	0.53*	4.65
ENMNFT1	Months enrolled full time, AY 2019–20	0 Zero	12.59	13.15	-0.56*	-4.24
		1 One	1.37	1.45	-0.08	-5.40
		2 Two	2.10	2.09	0.02	0.79
		3 Three	2.17	2.18	-0.01	-0.46
		4 Four	3.99	4.03	-0.04	-1.01
		5 Five	13.32	13.33	-0.01	-0.09
		6 Six	4.65	4.65	0.00	-0.10
		7 Seven	1.97	1.93	0.05	2.35
		8 Eight	3.97	3.97	0.00	-0.03
		9 Nine	10.81	10.67	0.15	1.36
		10 Ten	36.25	35.79	0.46*	1.28
		11 Eleven	6.23	6.22	0.01	0.15
		12 Twelve	0.58	0.56	0.03*	5.35
ENMNFT2	Months enrolled full time, AY 2020–21	0 Zero	30.71	29.33	1.37*	4.68
		1 One	0.98	1.02	-0.04	-3.91
		2 Two	1.04	1.01	0.04	3.57
		3 Three	0.73	0.72	0.01	1.66
		4 Four	1.54	1.83	-0.29*	-15.60
		5 Five	4.71	5.86	-1.15*	-19.64
		6 Six	1.78	2.49	-0.70*	-28.31
		7 Seven	0.81	0.81	0.00	0.40
		8 Eight	2.35	2.26	0.08*	3.71
		9 Nine	7.62	7.38	0.24*	3.30
		10 Ten	29.75	29.05	0.70*	2.42
		11 Eleven	6.67	7.33	-0.66*	-8.98
		12 Twelve	11.30	10.92	0.38*	3.52

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
ENMNFT3	Months enrolled full time, AY 2021–22	0 Zero	37.29	35.81	1.48*	4.14
		1 One	1.81	1.77	0.04	2.10
		2 Two	1.00	0.97	0.03	3.53
		3 Three	0.52	0.56	-0.04	-7.74
		4 Four	1.24	1.52	-0.27*	-18.10
		5 Five	4.33	5.87	-1.54*	-26.31
		6 Six	2.09	2.77	-0.68*	-24.68
		7 Seven	0.83	0.92	-0.09*	-10.18
		8 Eight	1.78	1.69	0.09*	5.13
		9 Nine	6.50	6.27	0.23*	3.63
		10 Ten	26.41	25.57	0.84*	3.27
		11 Eleven	7.53	7.88	-0.35*	-4.44
		12 Twelve	8.69	8.40	0.29*	3.42
ENMNT1	Months enrolled, AY 2019–20	1 One	0.89	0.99	-0.10	-10.22
		2 Two	1.43	1.56	-0.13*	-8.28
		3 Three	2.08	2.10	-0.03	-1.33
		4 Four	3.74	3.80	-0.06	-1.49
		5 Five	14.44	14.24	0.20	1.43
		6 Six	6.74	6.87	-0.13	-1.85
		7 Seven	1.94	1.98	-0.03	-1.76
		8 Eight	4.13	4.14	-0.01	-0.21
		9 Nine	10.80	10.65	0.15	1.43
		10 Ten	39.43	38.95	0.48*	1.23
		11 Eleven	13.48	13.78	-0.30*	-2.20
		12 Twelve	0.89	0.94	-0.04	-4.72
ENMNT2	Months enrolled, AY 2020–21	0 Zero	20.31	19.27	1.04*	5.37
		1 One	0.98	1.00	-0.02	-2.02
		2 Two	1.40	1.38	0.03	1.92
		3 Three	0.94	0.93	0.01	1.61
		4 Four	1.55	1.73	-0.18*	-10.49
		5 Five	4.51	5.27	-0.76*	-14.41
		6 Six	2.49	3.21	-0.73*	-22.56
		7 Seven	0.99	1.06	-0.07	-6.16
		8 Eight	2.46	2.38	0.08*	3.51
		9 Nine	6.90	6.76	0.15*	2.19
		10 Ten	26.51	25.75	0.76*	2.96
		11 Eleven	13.37	13.88	-0.51*	-3.66
		12 Twelve	17.58	17.39	0.19	1.08

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
ENMNT3	Months enrolled, AY 2021–22	0 Zero	27.63	26.22	1.40*	5.35
		1 One	1.18	1.22	-0.04	-3.24
		2 Two	1.60	1.60	0.00	-0.03
		3 Three	0.79	0.79	0.00	-0.13
		4 Four	1.34	1.56	-0.22*	-13.88
		5 Five	3.92	4.97	-1.05*	-21.07
		6 Six	2.77	3.60	-0.83*	-23.05
		7 Seven	1.29	1.40	-0.11*	-8.09
		8 Eight	2.01	1.94	0.06	3.25
		9 Nine	5.72	5.53	0.19*	3.43
		10 Ten	24.34	23.57	0.77*	3.28
		11 Eleven	14.04	14.44	-0.40*	-2.75
		12 Twelve	13.36	13.15	0.22*	1.65
ENMNTBA3Y	Bachelor's degree; months enrolled before attainment through AY 2021–22	0 Zero	1.88	1.71	0.17	10.25
		1 One	0.75	0.68	0.07	10.25
		3 Three	0.11	0.10	0.01	10.25
		15 Fifteen	2.14	1.94	0.20	10.25
		18 Eighteen	0.00	0.87	-0.87	-100.00
		19 Nineteen	0.83	0.75	0.08	10.25
		20 Twenty	2.88	3.08	-0.20	-6.58
		21 Twenty-one	6.89	6.24	0.64*	10.25
		22 Twenty-two	3.23	2.93	0.30	10.25
		23 Twenty-three	1.74	2.25	-0.51	-22.73
		24 Twenty-four	1.96	1.78	0.18	10.25
		25 Twenty-five	0.84	0.76	0.08	10.25
		26 Twenty-six	8.83	9.29	-0.46	-4.94
		27 Twenty-seven	5.38	5.73	-0.36	-6.22
		28 Twenty-eight	8.09	8.68	-0.58	-6.74
		29 Twenty-nine	17.02	15.44	1.58*	10.25
		30 Thirty	11.52	10.69	0.83	7.75
		31 Thirty-one	13.45	14.12	-0.67	-4.76
		32 Thirty-two	4.26	4.04	0.22	5.39
		33 Thirty-three	8.22	8.92	-0.71	-7.93

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
EVERDECAA	Ever declared a major (associate's) through AY 2021–22	0 Never enrolled in associate's	56.02	51.93	4.09*	7.88
		1 Declared a major in associate's	39.39	42.67	-3.28*	-7.70
		2 Never declared a major in associate's	4.60	5.40	-0.81*	-14.94
EVERDECBA	Ever declared a major (bachelor's) through AY 2021–22	0 Never enrolled in bachelor's	44.66	43.45	1.20*	2.76
		1 Declared a major in bachelor's	54.10	55.22	-1.12*	-2.03
		2 Never declared a major in bachelor's	1.25	1.33	-0.08*	-6.06
EVRVT22	Ever voted in a national, state, or local election through AY 2021–22	0 No	25.19	28.38	-3.19*	-11.24
		1 Yes	74.81	71.62	3.19*	4.45
EXOCC2Y22	Expected job: type of occupation (general), AY 2021–22	11 Management	10.85	10.92	-0.07	-0.60
		13 Business and Financial Operations	8.24	8.21	0.03	0.32
		15 Computer and Mathematical	8.53	8.04	0.50*	6.17
		17 Architecture and Engineering	4.91	4.70	0.20	4.27
		19 Life, Physical, Social Science	6.98	6.81	0.16	2.42
		21 Community and Social Service	4.10	4.16	-0.07	-1.56
		23 Legal	2.70	2.71	0.00	-0.14
		25 Educational Instruction and Library	8.40	8.41	-0.01	-0.12
		27 Arts, Design, Entertainment, Sports, and Media	9.58	9.86	-0.28	-2.86
		29 Healthcare Practitioners and Technical	17.93	18.06	-0.13	-0.73
		31 Healthcare Support	2.30	2.41	-0.11	-4.70
		33 Protective Service	2.41	2.44	-0.02	-0.97
		35 Food Preparation and Serving Related	0.82	0.78	0.04	4.68
		37 Building and Grounds Cleaning and Maintenance	0.20	0.23	-0.02	-9.94
		39 Personal Care and Service	2.63	2.73	-0.10	-3.52
		41 Sales and Related	1.83	1.77	0.06	3.48
		43 Office and Administrative Support	1.63	1.65	-0.03	-1.72
		45 Farming, Fishing, and Forestry	0.41	0.45	-0.04	-8.90
		47 Construction and Extraction	1.15	1.17	-0.02	-1.38
		49 Installation, Maintenance, and Repair	1.57	1.67	-0.10	-5.95
		51 Production	1.39	1.37	0.02	1.52
		53 Transportation and Material Moving	1.07	1.08	-0.01	-0.66
		55 Military Specific	0.36	0.37	0.00	-0.55

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
EXOCC3Y	Expected job: change to type of occupation, AY 2019–20 to AY 2021–22	1 Expects job in a different field	46.78	62.57	-15.79*	-25.24
		2 Expects job in the same field	53.22	37.43	15.79*	42.19
EXPBA	Plan to continue to a bachelor's degree program within 5 years, AY 2019–20	0 Did not plan to continue to bachelor's degree	24.42	24.17	0.25	1.04
		1 Yes, planned to continue to bachelor's degree	75.58	75.83	-0.25	-0.33
EXPBA22	Plan to continue to a bachelor's degree program within 2 years, AY 2021–22	0 No	21.69	22.97	-1.28*	-5.59
		1 Yes	78.31	77.03	1.28*	1.67
EXPFLD22	Job non-monetary benefits: importance of being an expert, AY 2021–22	1 Less important than salary	23.18	23.23	-0.05	-0.21
		2 As important as salary	51.74	51.87	-0.13	-0.25
		3 More important than salary	25.08	24.90	0.18	0.71
FACULTYIN22	Engagement: interactions with faculty during class, AY 2020–21/2021–22	1 Completely disagree	1.31	1.27	0.04	3.43
		2 Somewhat disagree	2.27	2.20	0.07	3.02
		3 Neither disagree nor agree	10.37	10.31	0.06	0.57
		4 Somewhat agree	35.67	35.63	0.04	0.10
		5 Completely agree	50.39	50.59	-0.20	-0.40
FACULTYOUT22	Engagement: interactions with faculty outside of class, AY 2020–21/2021–22	1 Completely disagree	2.26	2.29	-0.03	-1.18
		2 Somewhat disagree	3.86	3.91	-0.05	-1.17
		3 Neither disagree nor agree	20.39	20.25	0.13	0.66
		4 Somewhat agree	28.89	28.72	0.17	0.59
		5 Completely agree	44.60	44.83	-0.23	-0.52

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
FAMAMT22	Help from family and friends: total amount received, AY 2020–21/2021–22	0 Zero	83.17	83.26	-0.08	-0.10
		1 Less than \$500	3.37	3.32	0.05	1.46
		2 \$500 - \$999	2.76	2.87	-0.11	-3.81
		3 \$1,000 - \$1,499	1.74	1.67	0.07	3.89
		4 \$1,500 - \$1,999	1.10	1.09	0.02	1.59
		5 \$2,000 - \$5,000	2.61	2.61	0.00	-0.06
		6 \$5,001 - \$9,999	1.63	1.63	0.00	0.26
		7 \$10,000 - \$14,999	1.34	1.35	-0.01	-0.94
		8 \$15,000 - \$19,999	0.63	0.58	0.05	8.72
		9 \$20,000 - \$24,999	0.55	0.52	0.03	5.48
		10 \$25,000 - \$49,999	0.58	0.61	-0.02	-4.08
		11 \$50,000 or more	0.52	0.50	0.02	3.08
FAMTIM22	Job non-monetary benefits: importance of balancing work and family, AY 2021–22	1 Less important than salary	37.73	37.61	0.12	0.31
		2 As important as salary	56.88	56.96	-0.09	-0.15
		3 More important than salary	0.00	0.00	0.00*	.
FEDBEN22	Received federal benefit: any, in 2019 or 2020	0 No	84.98	85.55	-0.57*	-0.67
		1 Yes	15.02	14.45	0.57*	3.95
FEDBEN22A	Received federal benefit: Food Stamp Benefits, in 2019 or 2020	0 No	93.70	94.19	-0.49*	-0.52
		1 Yes	6.30	5.81	0.49*	8.44
FEDBEN22B	Received federal benefit: Free or Reduced Price Lunch Benefits, in 2019 or 2020	0 No	93.02	92.60	0.41*	0.45
		1 Yes	6.98	7.40	-0.41*	-5.59
FEDBEN22C	Received federal benefit: Supplemental Security Income Benefits, in 2019 or 2020	0 No	92.31	92.68	-0.36*	-0.39
		1 Yes	7.69	7.32	0.36*	4.97
FEDBEN22D	Received federal benefit: TANF Benefits, in 2019 or 2020	0 No	99.37	99.44	-0.07	-0.07
		1 Yes	0.63	0.56	0.07	12.11
FEDBEN22E	Received federal benefit: WIC Benefits, in 2019 or 2020	0 No	97.50	97.65	-0.14	-0.15
		1 Yes	2.50	2.35	0.14	6.12

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
FHSUPP22	Supportive of PSE: friends from home, AY 2020–21/2021–22	1 Completely disagree	2.71	2.82	-0.11	-3.98
		2 Somewhat disagree	2.58	2.69	-0.10	-3.89
		3 Neither disagree nor agree	20.48	21.27	-0.78*	-3.68
		4 Somewhat agree	17.62	17.65	-0.03	-0.15
		5 Completely agree	56.60	55.58	1.03*	1.85
FIN222000	Financial security: come up with \$2,000 within the next month, AY 2021–22	1 Certainly come up with the \$2,000	27.96	27.88	0.07	0.26
		2 Probably come up with the \$2,000	30.15	30.24	-0.09	-0.31
		3 Probably not come up with the \$2,000	23.95	23.92	0.03	0.14
		4 Certainly not come up with the \$2,000	17.94	17.95	-0.01	-0.08
FIN22500	Financial security: come up with \$500 within the next month, AY 2021–22	1 Certainly come up with the \$500	55.81	55.78	0.03	0.05
		2 Probably come up with the \$500	28.12	28.06	0.06	0.22
		3 Probably not come up with the \$500	9.66	9.76	-0.10	-1.01
		4 Certainly not come up with the \$500	6.40	6.40	0.01	0.08
FINLITA22	Financial literacy: effect of inflation on purchasing, AY 2021–22	1 More than today	17.91	17.82	0.10	0.54
		2 Exactly the same	23.01	22.96	0.05	0.22
		3 Less than today	59.08	59.22	-0.15	-0.25
FINLITB22	Financial literacy: effect of interest on savings, AY 2021–22	1 More than \$102	79.85	80.19	-0.34*	-0.43
		2 Exactly \$102	11.72	11.58	0.14	1.19
		3 Less than \$102	8.42	8.22	0.20*	2.48
FINLITC22	Financial literacy: effect of diversification on risk, AY 2021–22	1 True	8.30	8.14	0.16	2.00
		2 False	37.00	36.68	0.32	0.88
		3 Don't know	54.70	55.18	-0.48*	-0.88
FINLITTOT22	Financial literacy: number of correct responses, AY 2021–22	0 Zero	7.61	7.18	0.43*	6.03
		1 One	33.35	33.24	0.11	0.33
		2 Two	34.15	35.89	-1.74*	-4.84
		3 Three	24.89	23.69	1.20*	5.05

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
FOODSEC22	Food security status: previous 30 days, AY 2021–22	1 High food security	67.57	67.17	0.39*	0.59
		2 Marginal food security	9.75	9.93	-0.18*	-1.81
		3 Low food security	8.44	8.61	-0.18*	-2.06
		4 Very low food security	14.25	14.28	-0.04	-0.26
FRQCOMEXTFAM22	Frequency of communication: extended family, AY 2020–21/2021–22	0 Never	6.59	6.64	-0.06	-0.84
		1 Daily	5.73	5.80	-0.07	-1.23
		2 Weekly	20.68	20.83	-0.15	-0.71
		3 Monthly	31.98	31.82	0.15	0.47
		4 A few times a year	30.38	30.23	0.14	0.48
		5 Not Applicable	4.65	4.67	-0.02	-0.47
FRQCOMNOCOL22	Frequency of communication: close friends that did not attend college, AY 2020–21/2021–22	0 Never	4.71	4.87	-0.16	-3.19
		1 Daily	19.87	20.04	-0.17	-0.86
		2 Weekly	26.74	27.04	-0.30	-1.10
		3 Monthly	17.60	17.36	0.24	1.37
		4 A few times a year	15.12	15.21	-0.09	-0.59
		5 Not Applicable	15.96	15.48	0.48*	3.09
FRQCOMNPSAS22	Frequency of communication: close friends that attended primary institution, AY 2020–21/2021–22	0 Never	4.14	4.25	-0.11	-2.63
		1 Daily	43.63	43.44	0.19	0.43
		2 Weekly	23.23	23.08	0.15	0.67
		3 Monthly	6.26	6.38	-0.12	-1.90
		4 A few times a year	6.19	6.19	0.00	0.08
		5 Not Applicable	16.54	16.65	-0.11	-0.68
FRQCOMOTCOL22	Frequency of communication: close friends that attended other colleges, AY 2020–21/2021–22	0 Never	3.96	4.16	-0.20	-4.90
		1 Daily	22.18	22.11	0.07	0.31
		2 Weekly	31.67	31.38	0.29	0.92
		3 Monthly	21.12	21.29	-0.17	-0.81
		4 A few times a year	13.93	13.80	0.13	0.95
		5 Not Applicable	7.16	7.27	-0.11	-1.52

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
FRQCOMPAR22	Frequency of communication: parents, AY 2020–21/2021–22	0 Never	0.78	0.77	0.01	1.00
		1 Daily	59.09	59.68	-0.59*	-0.99
		2 Weekly	30.79	30.32	0.47*	1.55
		3 Monthly	5.53	5.55	-0.02	-0.35
		4 A few times a year	2.95	2.85	0.10	3.48
		5 Not Applicable	0.87	0.83	0.04	4.38
FRQCOMSIB22	Frequency of communication: siblings, AY 2020–21/2021–22	0 Never	1.89	1.92	-0.04	-1.85
		1 Daily	39.27	39.48	-0.22	-0.55
		2 Weekly	29.72	29.76	-0.04	-0.13
		3 Monthly	14.20	14.21	-0.01	-0.09
		4 A few times a year	6.09	6.07	0.01	0.23
		5 Not Applicable	8.85	8.56	0.29*	3.38
FSSUPP22	Supportive of PSE: friends from primary institution, AY 2020–21/2021–22	1 Completely disagree	3.35	3.48	-0.13	-3.73
		2 Somewhat disagree	1.98	2.03	-0.05	-2.50
		3 Neither disagree nor agree	21.54	22.14	-0.60*	-2.72
		4 Somewhat agree	15.96	16.19	-0.23	-1.42
		5 Completely agree	57.17	56.16	1.01*	1.81
GENDER3	Gender identity: gender, as of AY 2019–20	1 Male	44.67	45.84	-1.17*	-2.56
		2 Female	53.87	53.02	0.85*	1.60
		3 Genderqueer, gender nonconforming, or a different identity	1.46	1.13	0.33*	28.73
HIGHLVEX	Highest level of education ever expected, AY 2019–20	1 No degree or certificate expected	0.13	0.11	0.02*	21.01
		2 Undergraduate certificate or diploma	3.89	3.60	0.29*	8.03
		3 Associate's degree	10.68	10.08	0.60*	5.97
		4 Bachelor's degree	41.18	40.77	0.40	0.99
		5 Post-baccalaureate or post-master's certificate	1.30	1.37	-0.08	-5.58
		6 Master's degree	25.66	26.39	-0.73*	-2.78
		7 Doctoral degree - professional practice	10.68	11.11	-0.43*	-3.83
		8 Doctoral degree - research/scholarship and other	6.48	6.56	-0.08	-1.21

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
HLPOTH22	Job non-monetary benefits: importance of helping others, AY 2021–22	1 Less important than salary	18.91	19.12	-0.21	-1.11
		2 As important as salary	54.70	54.41	0.29	0.54
		3 More important than salary	26.40	26.47	-0.08	-0.30
HOMELESS22	Experienced homelessness: previous 30 days, AY 2021–22	0 No	92.11	92.70	-0.59*	-0.64
		1 Yes	7.89	7.30	0.59*	8.08
HOMESTUD22	Owned home or paid mortgage, AY 2021–22	0 No	94.39	94.24	0.15	0.16
		1 Yes	5.61	5.76	-0.15	-2.61
HOUSESEC22	Experienced housing insecurity: previous 12 months, AY 2021–22	0 No	76.30	77.79	-1.49*	-1.91
		1 Yes	23.70	22.21	1.49*	6.70
HSIZE22	Family size (dependent and independent), AY 2021–22	1 One	7.03	6.07	0.96*	15.80
		2 Two	16.12	14.44	1.68*	11.64
		3 Three	25.54	24.81	0.74*	2.96
		4 Four	26.49	27.74	-1.25*	-4.51
		5 Five	15.32	15.93	-0.61*	-3.82
		6 Six	5.78	6.67	-0.89*	-13.38
		7 Seven	2.30	2.54	-0.24*	-9.57
		8 Eight	0.75	1.03	-0.28*	-26.83
		9 Nine	0.34	0.35	0.00	-1.36
		10 Ten	0.14	0.22	-0.08	-35.66
		11 Eleven	0.11	0.13	-0.02	-14.71
		12 Twelve	0.02	0.02	0.01	40.87
		13 Thirteen	0.02	0.03	-0.01	-26.03
		14 Fourteen	0.04	0.03	0.01	21.92
		18 Eighteen	0.00	0.01	-0.01	-100.00
IMPACAD22	PSE service importance: academic advising or planning, AY 2020–21/2021–22	1 Not at all important	7.34	7.19	0.15	2.07
		2 Somewhat important	15.66	15.72	-0.06	-0.37
		3 Important	30.87	30.60	0.27	0.87
		4 Very important	46.13	46.49	-0.36	-0.77

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
IMPCAMPSEC22	PSE service importance: campus security or campus police, AY 2020–21/2021–22	1 Not at all important	12.81	12.74	0.06	0.48
		2 Somewhat important	21.57	21.29	0.28	1.31
		3 Important	27.95	28.93	-0.98	-3.37
		4 Very important	37.67	37.04	0.64	1.72
IMPCHLD22	PSE service importance: child care, AY 2020–21/2021–22	1 Not at all important	11.87	11.06	0.81	7.30
		2 Somewhat important	21.52	21.09	0.43	2.03
		3 Important	26.36	28.34	-1.98	-6.99
		4 Very important	40.24	39.50	0.74	1.89
IMPCPP22	PSE service importance: career services, AY 2020–21/2021–22	1 Not at all important	6.38	6.76	-0.38	-5.66
		2 Somewhat important	15.03	15.19	-0.17	-1.09
		3 Important	31.05	30.54	0.51	1.68
		4 Very important	47.55	47.51	0.04	0.07
IMPDISSRV22	PSE service importance: services for students with disabilities, AY 2020–21/2021–22	1 Not at all important	6.85	6.83	0.02	0.34
		2 Somewhat important	14.39	14.63	-0.23	-1.59
		3 Important	24.61	25.52	-0.91	-3.55
		4 Very important	54.14	53.03	1.12	2.11
IMPDIVRSE22	PSE service importance: diversity or multicultural student services, AY 2020–21/2021–22	1 Not at all important	6.50	6.85	-0.35	-5.13
		2 Somewhat important	16.89	17.27	-0.38	-2.18
		3 Important	31.17	31.45	-0.28	-0.88
		4 Very important	45.44	44.43	1.00	2.26
IMPFINAID22	PSE service importance: financial aid services, AY 2020–21/2021–22	1 Not at all important	2.84	2.77	0.07	2.51
		2 Somewhat important	8.72	8.47	0.25	2.98
		3 Important	26.53	26.00	0.53	2.04
		4 Very important	61.91	62.77	-0.85*	-1.36
IMPHLTH22	PSE service importance: health center and services, AY 2020–21/2021–22	1 Not at all important	13.50	13.29	0.21	1.57
		2 Somewhat important	20.49	20.85	-0.36	-1.73
		3 Important	28.83	29.10	-0.27	-0.93
		4 Very important	37.18	36.76	0.42	1.15

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
IMPMENSRV22	PSE service importance: psychological or mental health services, AY 2020–21/2021–22	1 Not at all important	8.42	8.18	0.24	2.92
		2 Somewhat important	16.69	15.91	0.79*	4.94
		3 Important	24.04	24.49	-0.45	-1.86
		4 Very important	50.85	51.42	-0.57	-1.11
IMPMLSRV22	PSE service importance: services for active military or veterans, AY 2020–21/2021–22	1 Not at all important	8.43	8.48	-0.05	-0.53
		2 Somewhat important	13.05	13.36	-0.31	-2.34
		3 Important	28.46	28.51	-0.04	-0.14
		4 Very important	50.05	49.65	0.40	0.80
IMPTRPLN22	PSE service importance: transfer advising or planning, AY 2020–21/2021–22	1 Not at all important	4.94	5.61	-0.67*	-12.00
		2 Somewhat important	13.21	13.01	0.20	1.54
		3 Important	32.58	32.84	-0.26	-0.79
		4 Very important	49.28	48.54	0.73	1.51
IMPTUTOR22	PSE service importance: peer or other tutoring or skills lab, AY 2020–21/2021–22	1 Not at all important	8.03	8.43	-0.40	-4.74
		2 Somewhat important	18.92	18.25	0.67*	3.69
		3 Important	32.21	32.69	-0.47	-1.44
		4 Very important	40.83	40.63	0.20	0.49
IT2YFP3Y	Ever attended 2-year for-profit institution through AY 2021–22	0 No	97.92	97.81	0.11*	0.11
		1 Yes	2.08	2.19	-0.11*	-4.81
IT2YPU3Y	Ever attended 2-year public institution through AY 2021–22	0 No	59.68	58.78	0.90*	1.53
		1 Yes	40.32	41.22	-0.90*	-2.19
IT4Y3Y	Ever attended 4-year institution through AY 2021–22	0 No	33.89	33.09	0.80*	2.42
		1 Yes	66.11	66.91	-0.80*	-1.20
IT4YFP3Y	Ever attended 4-year for-profit institution through AY 2021–22	0 No	96.28	96.14	0.14*	0.15
		1 Yes	3.72	3.86	-0.14*	-3.67
IT4YPN3Y	Ever attended 4-year private nonprofit institution through AY 2021–22	0 No	80.93	80.60	0.33*	0.41
		1 Yes	19.07	19.40	-0.33*	-1.71

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
IT4YPU3Y	Ever attended 4-year public institution through AY 2021–22	0 No	54.41	53.41	1.00*	1.87
		1 Yes	45.59	46.59	-1.00*	-2.14
ITFP3Y	Ever attended for-profit institution through AY 2021–22	0 No	91.96	91.59	0.38*	0.41
		1 Yes	8.04	8.41	-0.38*	-4.49
ITL2FP3Y	Ever attended less than 2-year for-profit institution through AY 2021–22	0 No	97.37	97.20	0.17*	0.18
		1 Yes	2.63	2.80	-0.17*	-6.08
JBEN22	Job: eligible for life, health, or retirement benefits, 2022	0 No	34.59	31.70	2.89*	9.11
		1 Yes	65.41	68.30	-2.89*	-4.23
JBEN22A	Job: eligible for life insurance, 2022	0 No	55.62	56.25	-0.63	-1.13
		1 Yes	44.38	43.75	0.63	1.45
JBEN22B	Job: eligible for health insurance, 2022	0 No	40.87	41.68	-0.81	-1.95
		1 Yes	59.13	58.32	0.81	1.39
JBEN22C	Job: eligible for retirement benefits, 2022	0 No	46.59	46.11	0.47	1.02
		1 Yes	53.41	53.89	-0.47	-0.88
JOBFT21	Jobs while enrolled: work intensity, AY 2020–21	1 No job	56.48	51.03	5.46*	10.70
		2 Part-time	27.65	30.98	-3.33*	-10.76
		3 Full-time	15.87	17.99	-2.12*	-11.81
JOBFT22	Jobs while enrolled: work intensity, AY 2021–22	1 No job	50.79	46.25	4.54*	9.82
		2 Part-time	33.45	36.32	-2.86*	-7.88
		3 Full-time	15.75	17.43	-1.68*	-9.63
JOBLFP22	Labor force participation status, as of June 2022	0 Employed	61.38	56.91	4.47*	7.85
		1 Unemployed	4.72	5.67	-0.95*	-16.70
		2 Not in the labor force	33.90	37.42	-3.52*	-9.41
JOBLIC22	Job: had industry certification or license, 2022	0 No	80.17	79.74	0.43	0.54
		1 Yes	19.83	20.26	-0.43	-2.14

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
JOBOCC22	Job: type of occupation (general), 2022	11 Management	5.94	5.96	-0.02	-0.35
		13 Business/Financial Operation	2.83	2.77	0.06	2.22
		15 Computer and Mathematical	2.12	2.00	0.12	5.75
		17 Architecture and Engineering	0.87	0.82	0.05	6.61
		19 Life, Physical, Social Science	0.46	0.65	-0.20	-29.93
		21 Community/Social Services	0.76	0.77	-0.02	-2.01
		23 Legal	0.29	0.29	0.01	1.97
		25 Educational Instruction and Library	2.44	2.38	0.06	2.43
		27 Arts, Design, Entertainment, Sports, Media	3.04	2.78	0.27	9.61
		29 Healthcare Practitioners/Technical	3.14	4.07	-0.93*	-22.92
		31 Healthcare Support	5.56	5.48	0.08	1.46
		33 Protective Service	1.69	1.64	0.06	3.49
		35 Food Prep/Serving Related	11.02	11.39	-0.36	-3.18
		37 Building/Grounds Cleaning/Maintenance	2.68	2.38	0.30	12.69
		39 Personal Care and Service	4.22	4.41	-0.19	-4.33
		41 Sales and Related	14.02	13.37	0.64	4.80
		43 Office/Administrative Support	10.58	11.25	-0.68	-6.03
		45 Farming, Fishing, Forestry	0.97	0.92	0.05	5.61
		47 Construction and Extraction	3.29	3.26	0.03	0.96
		49 Installation, Maintenance, Repair	4.97	4.62	0.34	7.45
JOBRCLG22	Job: related to major or field of study, 2022	0 No	78.49	77.90	0.59	0.76
		1 Yes	21.51	22.10	-0.59	-2.68
JOBRLC22	Job: job required industry certification or license, 2022	0 No	87.52	89.03	-1.51*	-1.70
		1 Yes	12.48	10.97	1.51*	13.76
JOBRFLT22	Job: related to intended future work, 2022	0 No	60.49	60.19	0.30	0.49
		1 Yes	39.51	39.81	-0.30	-0.75
JOBSELF22	Job: self employed, 2022	0 No	90.48	90.78	-0.30	-0.34
		1 Yes	9.52	9.22	0.30	3.30
JOBST22	Job: employment status, 2022	0 No	32.30	33.76	-1.47*	-4.34
		1 Yes	67.70	66.24	1.47*	2.21

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
JOBTIM22	Job: years worked in same or similar job, 2022	0 Zero	64.89	64.69	0.19	0.30
		1 One	12.34	12.25	0.08	0.69
		2 Two	9.77	9.54	0.23	2.41
		3 Three	5.09	5.03	0.06	1.12
		4 Four	3.68	4.06	-0.38	-9.31
		5 Five	1.51	1.49	0.02	1.02
		6 Six	0.49	0.51	-0.02	-3.23
		7 Seven	0.17	0.15	0.02	14.96
		8 Eight	0.35	0.31	0.03	10.98
		9 Nine	0.11	0.09	0.01	13.34
		10 Ten	0.37	0.40	-0.03	-7.75
		12 Twelve	0.06	0.05	0.01	13.31
		13 Thirteen	0.07	0.07	0.00	-1.19
		14 Fourteen	0.15	0.19	-0.05	-23.28
		15 Fifteen	0.17	0.20	-0.03	-14.25
		16 Sixteen	0.05	0.04	0.01	31.07
		17 Seventeen	0.03	0.02	0.01	31.07
		18 Eighteen	0.05	0.04	0.01	31.07
		20 Twenty	0.14	0.13	0.00	1.16
		22 Twenty-two	0.09	0.15	-0.06	-40.99
KNOWCLAS22	Knew requirements needed to complete pursued degree at primary institution, AY 2020–21/2021–22	1 Completely disagree	1.68	1.73	-0.04	-2.59
		2 Somewhat disagree	2.85	2.99	-0.14	-4.72
		3 Neither disagree nor agree	5.51	5.74	-0.23	-4.06
		4 Somewhat agree	26.03	25.69	0.34	1.34
		5 Completely agree	63.93	63.86	0.08	0.12
LEISTIM22	Job non-monetary benefits: importance of balancing work and leisure time, AY 2021–22	1 Less important than salary	6.65	6.66	-0.01	-0.20
		2 As important as salary	48.52	48.46	0.06	0.12
		3 More important than salary	44.84	44.88	-0.04	-0.10

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
LKOCCATHD22	Expected job: likelihood of holding job after highest attainment, AY 2021–22	1 Very unlikely	3.65	3.97	-0.33*	-8.22
		2 Somewhat unlikely	4.78	4.90	-0.12	-2.53
		3 Neither unlikely nor likely	17.18	17.24	-0.07	-0.39
		4 Somewhat likely	35.79	35.41	0.38	1.08
		5 Very likely	38.60	38.47	0.13	0.35
LNLITTOT22	Loan literacy: number of correct responses, AY 2021–22	0 Zero	17.94	17.64	0.30	1.69
		1 One	30.48	30.27	0.21	0.69
		2 Two	21.34	21.36	-0.02	-0.07
		3 Three	30.24	30.73	-0.49*	-1.60
LOANLITA22	Loan literacy: government can report unpaid debt to credit bureaus, AY 2021–22	0 No, cannot report debt past due to credit bureaus	36.21	35.69	0.53*	1.48
		1 Yes, can report debt past due to credit bureaus	63.79	64.31	-0.53*	-0.82
LOANLITB22	Loan literacy: government can garnish wages for unpaid federal loan debt, AY 2021–22	0 No, cannot have employer withhold money	53.64	53.25	0.39	0.73
		1 Yes, can have employer withhold money	46.36	46.75	-0.39	-0.83
LOANLITC22	Loan literacy: government can retain tax refunds or Social Security for loan debt, AY 2021–22	0 No, cannot retain tax refunds until debt is repaid	46.26	45.88	0.38	0.83
		1 Yes, can retain tax refunds until the debt is repaid	53.74	54.12	-0.38	-0.70
LOCALR22	Residence while attending primary institution, AY 2020–21/2021–22	1 On campus or in college-owned housing	21.46	21.09	0.37	1.76
		2 Off campus	39.97	39.98	-0.01	-0.02
		3 Living with parents	38.57	38.93	-0.36	-0.94
LOCJOB21	Jobs while enrolled: on or off campus, AY 2020–21	1 On campus	8.43	7.99	0.44*	5.51
		2 Off campus	84.94	85.10	-0.16	-0.19
		3 Both on and off campus	6.63	6.91	-0.28	-4.05
LOCJOB22	Jobs while enrolled: on or off campus, AY 2021–22	1 On campus	10.62	10.02	0.60*	5.98
		2 Off campus	79.46	80.40	-0.94*	-1.17
		3 Both on and off campus	9.92	9.57	0.34*	3.58
LOCJOB3Y	Jobs while enrolled: ever on campus through AY 2021–22	0 No	79.90	78.75	1.15*	1.46
		1 Yes	20.10	21.25	-1.15*	-5.39

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
MAJ22	Major or field of study: last enrolled any degree through AY 2021–22 (23 categories)	0 Undecided or Undeclared	5.49	6.23	-0.74*	-11.83
		1 Computer and information sciences	6.03	5.82	0.21*	3.53
		2 Engineering and engineering technology	6.45	6.55	-0.09	-1.41
		3 Biological and physical sciences, science tech	7.11	7.04	0.07	0.99
		4 Mathematics	0.86	0.82	0.03	3.81
		5 Agriculture and natural resources	2.29	2.39	-0.10	-4.20
		6 General studies and other	4.99	5.08	-0.09	-1.70
		7 Social sciences	4.71	4.56	0.15*	3.25
		8 Psychology	5.45	5.41	0.04	0.70
		9 Humanities	5.63	5.55	0.08	1.50
		10 History	0.60	0.58	0.03	4.48
		11 Personal and consumer services	3.69	3.77	-0.08	-2.18
		12 Manufacturing, construction, repair and transportation	2.78	2.79	-0.01	-0.41
		13 Military technology and protective services	3.55	3.60	-0.05	-1.40
		14 Health care fields	13.29	13.19	0.10	0.78
		15 Business	13.73	13.71	0.01	0.08
		16 Education	5.11	4.88	0.23*	4.62
		17 Architecture	0.71	0.68	0.03	4.12
		18 Communications	3.45	3.32	0.13*	3.83
		19 Public administration and human services	1.39	1.42	-0.03	-1.89
		20 Design and applied arts	1.98	1.94	0.05	2.34
		21 Law and legal studies	0.48	0.45	0.03*	6.36
		22 Library sciences	0.03	0.02	0.00*	13.82
		23 Theology and religious vocations	0.20	0.19	0.01	6.71
MAJ4Y22	Major or field of study: last enrolled any degree through AY 2021–22 (10 categories)	0 Undecided or undeclared	5.49	6.23	-0.74*	-11.83
		1 Computer and information sciences	6.03	5.82	0.21*	3.53
		2 Engineering and engineering technologies/technicians	6.45	6.55	-0.09	-1.41
		3 Bio and phys science, sci tech, math, agriculture	10.25	10.25	0.00	0.01
		4 General studies and other	4.99	5.08	-0.09	-1.70
		5 Social sciences	10.16	9.97	0.19	1.87
		6 Humanities	6.24	6.13	0.11	1.78
		7 Health care fields	13.29	13.19	0.10	0.78
		8 Business	13.73	13.71	0.01	0.08
		9 Education	5.11	4.88	0.23*	4.62
		10 Other applied	18.26	18.19	0.07	0.41

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
MAJ4YBA22	Major or field of study: last enrolled in bachelor's through AY 2021–22 (10 categories)	0 Undecided or undeclared	1.36	2.35	-0.98*	-41.96
		1 Computer and information sciences	6.01	5.87	0.14	2.32
		2 Engineering and engineering technologies/technicians	7.70	7.45	0.25*	3.36
		3 Bio and phys science, sci tech, math, agriculture	14.00	13.76	0.24*	1.74
		4 General studies and other	1.31	1.32	-0.02	-1.15
		5 Social sciences	14.32	14.65	-0.33*	-2.24
		6 Humanities	8.10	8.14	-0.04	-0.46
		7 Health care fields	9.09	9.08	0.02	0.17
		8 Business	16.07	15.60	0.48*	3.05
		9 Education	5.89	5.76	0.13	2.29
		10 Other applied	16.15	16.03	0.12	0.74
MAJAA22	Major or field of study: last enrolled in associate's through AY 2021–22 (23 categories)	0 Undecided or undeclared	4.16	11.24	-7.08*	-62.99
		1 Computer and information sciences	6.67	6.26	0.41*	6.63
		2 Engineering and engineering technology	5.61	5.17	0.44*	8.61
		3 Biological and physical sciences, science tech	5.21	4.81	0.40*	8.40
		4 Mathematics	0.38	0.39	-0.01	-2.39
		5 Agriculture and natural resources	2.52	2.68	-0.16	-5.82
		6 General studies and other	10.19	10.65	-0.46	-4.31
		7 Social sciences	2.34	2.14	0.20*	9.44
		8 Psychology	4.49	4.50	0.00	-0.09
		9 Humanities	4.07	3.77	0.30*	7.96
		10 History	0.28	0.23	0.06*	24.34
		11 Personal and consumer services	2.39	2.01	0.38*	18.76
		12 Manufacturing, construction, repair and transportation	3.17	2.85	0.32*	11.27
		13 Military technology and protective services	5.44	4.86	0.58*	11.90
		14 Health care fields	16.28	14.67	1.61*	10.98
		15 Business	14.62	13.02	1.60*	12.26
		16 Education	5.33	4.67	0.66*	14.11
		17 Architecture	0.35	0.38	-0.04	-9.59
		18 Communications	2.43	2.25	0.19	8.30
		19 Public administration and human services	1.39	1.28	0.12	9.15
		20 Design and applied arts	1.70	1.43	0.28*	19.38
		21 Law and legal studies	0.73	0.59	0.14*	23.90
		22 Library sciences	0.04	0.03	0.01	31.76
		23 Theology and religious vocations	0.21	0.16	0.05	31.76

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
MAJBA22	Major or field of study: last enrolled in bachelor's through AY 2021–22 (23 categories)	0 Undecided or undeclared	1.36	2.35	-0.98*	-41.96
		1 Computer and information sciences	6.01	5.87	0.14	2.32
		2 Engineering and engineering technology	7.70	7.45	0.25*	3.36
		3 Biological and physical sciences, science tech	10.15	10.05	0.10	0.99
		4 Mathematics	1.30	1.26	0.04	3.48
		5 Agriculture and natural resources	2.55	2.46	0.10*	3.93
		6 General studies and other	1.31	1.32	-0.02	-1.15
		7 Social sciences	7.05	7.23	-0.18	-2.52
		8 Psychology	7.28	7.42	-0.15	-1.97
		9 Humanities	7.19	7.25	-0.06	-0.85
		10 History	0.91	0.89	0.02	2.72
		11 Personal and consumer services	2.69	2.66	0.04	1.44
		12 Manufacturing, construction, repair and transportation	0.40	0.46	-0.06	-12.33
		13 Military technology and protective services	3.08	3.08	0.00	-0.15
		14 Health care fields	9.09	9.08	0.02	0.17
		15 Business	16.07	15.60	0.48*	3.05
		16 Education	5.89	5.76	0.13	2.29
		17 Architecture	0.96	0.97	0.00	-0.42
		18 Communications	4.64	4.49	0.15*	3.32
		19 Public administration and human services	1.48	1.44	0.04	2.60
		20 Design and applied arts	2.30	2.30	0.00	0.13
		21 Law and legal studies	0.36	0.37	-0.01	-3.48
		22 Library sciences	0.03	0.02	0.00	8.96
		23 Theology and religious vocations	0.22	0.25	-0.03	-13.45
MAJCHG22	Major changes: number anywhere through AY 2021–22	0 Never	68.66	69.48	-0.83*	-1.19
		1 One time	25.25	24.68	0.58*	2.33
		2 More than one time	6.09	5.84	0.25*	4.28
MAJCHGAA22	Major changes (associate's): number anywhere through AY 2021–22	0 Never	70.15	71.89	-1.74*	-2.42
		1 One time	23.84	22.20	1.64*	7.38
		2 More than one time	6.00	5.90	0.10	1.71
MAJCHGBA22	Major changes (bachelor's): number anywhere through AY 2021–22	0 Never	69.19	69.96	-0.77*	-1.11
		1 One time	25.07	24.56	0.52	2.10
		2 More than one time	5.74	5.48	0.26*	4.70

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
MAJCT22	Major or field of study: last enrolled in certificate through AY 2021–22 (23 categories)	1 Computer and information sciences	3.83	3.79	0.04	1.14
		2 Engineering and engineering technology	4.92	4.78	0.14	2.98
		3 Biological and physical sciences, science tech	2.38	2.37	0.01	0.53
		4 Mathematics	0.07	0.11	-0.04	-33.77
		5 Agriculture and natural resources	1.47	1.48	-0.01	-0.40
		6 General studies and other	8.60	8.86	-0.26	-2.94
		7 Social sciences	1.16	1.07	0.09	7.96
		8 Psychology	0.71	0.90	-0.18	-20.53
		9 Humanities	3.51	3.96	-0.45	-11.42
		10 History	0.19	0.16	0.03	17.53
		11 Personal and consumer services	12.93	12.26	0.67	5.44
		12 Manufacturing, construction, repair and transportation	13.23	13.29	-0.06	-0.42
		13 Military technology and protective services	2.47	2.39	0.09	3.71
		14 Health care fields	29.22	27.88	1.34	4.80
		15 Business	8.81	9.87	-1.05*	-10.69
		16 Education	2.56	2.59	-0.03	-1.17
		17 Architecture	0.23	0.33	-0.10	-30.13
		18 Communications	1.46	1.58	-0.12	-7.65
		19 Public administration and human services	0.96	0.89	0.07	8.39
		20 Design and applied arts	0.73	0.82	-0.09	-11.22
		21 Law and legal studies	0.55	0.63	-0.09	-13.83
MEALPLAN22	Meal plan: had meal plan, AY 2021–22	0 No	79.69	79.62	0.07	0.08
		1 Yes	20.31	20.38	-0.07	-0.32
MEALPLANFQ22	Meal plan: meal plan covered 11 or more meals per week, AY 2021–22	0 No	28.83	28.06	0.77	2.75
		1 Yes	71.17	71.94	-0.77	-1.07
MENTH22	Health: general mental health, AY 2021–22	1 Excellent	12.76	12.80	-0.04	-0.28
		2 Very good	20.05	19.87	0.18	0.90
		3 Good	31.50	31.28	0.21	0.68
		4 Fair	25.87	26.14	-0.27	-1.03
		5 Poor	9.82	9.91	-0.09	-0.87

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
MENTHDIF	Health: general mental health, change between AY 2019–20 to 2021–22	1 Mental health rating improved	38.11	40.87	-2.76*	-6.76
		2 Mental health remained constant	41.54	34.31	7.23*	21.07
		3 Mental health rating declined	20.34	24.81	-4.47*	-18.01
MENTMISS22	Health: amount missed school/work in past 30 days for mental health, AY 2021–22	1 Never	59.51	59.58	-0.06	-0.11
		2 A few times	25.31	25.33	-0.02	-0.08
		3 About once a week	8.60	8.34	0.26*	3.12
		4 Almost every day	4.11	4.28	-0.17	-4.05
		5 Every day	2.47	2.47	0.00	-0.10
MJNSF22	Major or field of study with a focus on NSF STEM fields, AY 2021–22	0 Undecided	5.49	6.23	-0.74*	-11.83
		1 Science, engineering, and math	23.22	23.15	0.07	0.32
		2 Psychology and other social science	14.24	14.12	0.12	0.85
		3 Health and medicine	13.37	13.26	0.11	0.79
		4 Other field of study not supported by NSF	43.68	43.24	0.44*	1.02
MNTJBNE21	Jobs while not enrolled: number of months worked, AY 2020–21	0 Zero	50.83	48.64	2.19*	4.51
		1 One	16.09	16.63	-0.54*	-3.23
		2 Two	12.17	11.93	0.24	2.00
		3 Three	3.55	3.73	-0.18*	-4.87
		4 Four	1.83	1.96	-0.14*	-6.97
		5 Five	1.32	1.52	-0.20*	-13.36
		6 Six	2.37	2.65	-0.28*	-10.56
		7 Seven	1.99	2.31	-0.32*	-13.77
		8 Eight	1.09	1.18	-0.08	-7.15
		9 Nine	0.96	1.00	-0.04	-3.87
		10 Ten	1.24	1.26	-0.02	-1.49
		11 Eleven	0.99	1.08	-0.09	-8.36
		12 Twelve	5.57	6.11	-0.54*	-8.91

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
MNTJBNE22	Jobs while not enrolled: number of months worked, AY 2021–22	0 Zero	41.73	39.26	2.47*	6.30
		1 One	17.31	17.19	0.12	0.68
		2 Two	12.27	12.28	-0.01	-0.07
		3 Three	3.83	3.91	-0.08	-2.15
		4 Four	2.49	2.75	-0.26*	-9.47
		5 Five	2.02	2.52	-0.50*	-19.74
		6 Six	2.77	3.50	-0.73*	-20.79
		7 Seven	2.21	2.63	-0.43*	-16.26
		8 Eight	1.31	1.45	-0.14*	-9.63
		9 Nine	1.56	1.54	0.03	1.65
		10 Ten	2.87	2.93	-0.06	-2.05
		11 Eleven	1.79	1.81	-0.02	-1.00
		12 Twelve	7.83	8.22	-0.39*	-4.75
MNTJOB21	Jobs while enrolled: number of months worked, AY 2020–21	0 Zero	50.52	49.96	0.56*	1.13
		1 One	5.59	5.65	-0.05	-0.97
		2 Two	4.09	4.27	-0.17	-4.07
		3 Three	3.45	3.54	-0.08	-2.37
		4 Four	3.08	3.22	-0.14	-4.22
		5 Five	4.01	4.41	-0.40*	-9.05
		6 Six	2.96	3.17	-0.21*	-6.49
		7 Seven	1.84	1.86	-0.02	-1.22
		8 Eight	2.75	2.82	-0.07	-2.49
		9 Nine	3.64	3.59	0.05	1.51
		10 Ten	9.22	8.83	0.39*	4.46
		11 Eleven	3.95	4.07	-0.12	-3.02
		12 Twelve	4.88	4.62	0.26*	5.55

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
MNTJOB22	Jobs while enrolled: number of months worked, AY 2021–22	0 Zero	45.80	45.10	0.70*	1.56
		1 One	5.16	5.30	-0.14	-2.65
		2 Two	4.34	4.54	-0.20	-4.38
		3 Three	3.36	3.38	-0.02	-0.45
		4 Four	3.31	3.57	-0.26*	-7.20
		5 Five	4.47	5.05	-0.57*	-11.33
		6 Six	3.27	3.71	-0.44*	-11.82
		7 Seven	2.48	2.55	-0.07	-2.69
		8 Eight	3.42	3.22	0.20*	6.24
		9 Nine	5.11	4.93	0.18*	3.59
		10 Ten	10.14	9.78	0.36*	3.69
		11 Eleven	5.21	5.18	0.03	0.57
		12 Twelve	3.91	3.69	0.22*	5.92
NPATND22	Would begin education at NPSAS again, AY 2021–22	0 No	29.00	29.02	-0.01	-0.05
		1 Yes	71.00	70.98	0.01	0.02
NUMJBNE21	Jobs while not enrolled: number of jobs, AY 2020–21	0 Zero	45.47	48.64	-3.17*	-6.52
		1 One	43.12	40.38	2.74*	6.79
		2 Two	9.88	9.41	0.47*	5.00
		3 Three	1.39	1.41	-0.02	-1.35
		4 Four	0.11	0.13	-0.02	-16.62
		5 Five	0.01	0.01	0.00	-5.68
		6 Six	0.00	0.00	0.00	7.04
		7 Seven	0.00	0.00	0.00	7.04
		10 Ten	0.02	0.02	0.00	7.04
NUMJBNE22	Jobs while not enrolled: number of jobs, AY 2021–22	0 Zero	36.63	39.26	-2.63*	-6.70
		1 One	47.80	45.63	2.17*	4.75
		2 Two	13.35	12.89	0.45*	3.50
		3 Three	2.00	1.96	0.03	1.78
		4 Four	0.18	0.20	-0.03	-12.58
		5 Five	0.03	0.03	0.00	-4.49
		6 Six	0.01	0.01	0.00	4.99
		10 Ten	0.01	0.01	0.00	4.99

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
NUMJOB21	Jobs while enrolled: number of jobs, AY 2020–21	0 Zero	50.52	49.96	0.56*	1.13
		1 One	37.83	37.83	0.00	0.00
		2 Two	10.11	10.54	-0.44*	-4.16
		3 Three	1.34	1.46	-0.11*	-7.87
		4 Four	0.14	0.15	-0.01	-6.30
		5 Five	0.02	0.03	-0.01	-22.03
		8 Eight	0.01	0.01	0.00	19.68
		10 Ten	0.02	0.01	0.00	19.68
NUMJOB22	Jobs while enrolled: number of jobs, AY 2021–22	0 Zero	45.80	45.10	0.71*	1.56
		1 One	37.83	38.12	-0.28	-0.75
		2 Two	13.82	14.11	-0.29*	-2.04
		3 Three	2.27	2.37	-0.10	-4.24
		4 Four	0.21	0.22	-0.01	-5.35
		5 Five	0.03	0.05	-0.03	-50.69
		6 Six	0.01	0.01	0.00	12.66
		10 Ten	0.03	0.02	0.00	19.44
ORGMJ22	Original declared major or field of study at first degree program (23 categories)	0 Undecided or undeclared	2.28	3.03	-0.75*	-24.81
		1 Computer and information sciences	5.68	5.60	0.08*	1.51
		2 Engineering and engineering technology	7.25	7.15	0.10*	1.35
		3 Biological and physical sciences, science tech	7.97	7.88	0.09*	1.10
		4 Mathematics	0.63	0.62	0.01*	2.07
		5 Agriculture and natural resources	2.21	2.18	0.03*	1.33
		6 General studies and other	11.48	11.55	-0.07	-0.60
		7 Social sciences	3.95	3.91	0.04	0.90
		8 Psychology	4.48	4.43	0.04*	0.93
		9 Humanities	5.15	5.10	0.05	0.99
		10 History	0.52	0.53	-0.01	-1.12
		11 Personal and consumer services	3.84	3.85	-0.01	-0.35
		12 Manufacturing, construction, repair and transportation	2.85	2.82	0.03*	1.09
		13 Military technology and protective services	3.62	3.59	0.04	1.00
		14 Health care fields	13.63	13.57	0.06	0.47

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
ORGMJ22 (continued)	Original declared major or field of study at first degree program (23 categories) (continued)	15 Business	12.45	12.34	0.12*	0.95
		16 Education	4.80	4.73	0.07*	1.46
		17 Architecture	0.60	0.59	0.01*	2.07
		18 Communications	2.83	2.79	0.05*	1.67
		19 Public administration and human services	1.29	1.28	0.02	1.19
		20 Design and applied arts	1.82	1.80	0.02	1.19
		21 Law and legal studies	0.47	0.48	-0.01	-2.32
		22 Library sciences	0.01	0.01	0.00	2.07
		23 Theology and religious vocations	0.20	0.19	0.00*	2.07
ORGMJ4Y	Original declared major or field of study at first degree program (10 categories)	0 Undecided	2.28	3.03	-0.75*	-24.81
		1 Computer and information sciences	5.68	5.60	0.08*	1.51
		2 Engineering and engineering-related technologies/technicians	7.25	7.15	0.10*	1.35
		3 Bio and phys science, sci tech, math, agriculture	10.80	10.67	0.13*	1.20
		4 General studies and other	11.48	11.55	-0.07	-0.60
		5 Social sciences	8.42	8.35	0.08*	0.92
		6 Humanities	5.67	5.63	0.04	0.79
		7 Health care fields	13.63	13.57	0.06	0.47
		8 Business	12.45	12.34	0.12*	0.95
		9 Education	4.80	4.73	0.07*	1.46
PARAMT22	Help from parents: total amount received, AY 2020–21/2021–22	10 Other applied	17.52	17.38	0.14*	0.81
		0 Zero	30.31	31.49	-1.19*	-3.77
		1 Less than \$500	5.83	5.83	0.00	-0.07
		2 \$500 - \$999	6.29	6.25	0.04	0.63
		3 \$1,000 - \$1,499	4.76	4.68	0.08	1.81
		4 \$1,500 - \$1,999	3.43	3.30	0.13	3.95
		5 \$2,000 - \$5,000	10.08	10.10	-0.02	-0.19
		6 \$5,001 - \$9,999	11.14	11.24	-0.10	-0.89
		7 \$10,000 - \$14,999	7.73	7.71	0.02	0.27
		8 \$15,000 - \$19,999	5.14	4.97	0.17	3.41
		9 \$20,000 - \$24,999	4.04	3.93	0.11	2.77
		10 \$25,000 - \$49,999	5.43	5.05	0.38*	7.49
		11 \$50,000 or more	5.83	5.45	0.38*	6.94

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
PAREduc	Parents' highest education level, AY 2019–20	1 Did not complete high school	8.95	7.66	1.29*	16.81
		2 High school diploma or equivalent	17.20	16.55	0.65*	3.95
		3 Vocational/technical training	3.66	3.49	0.17	4.92
		4 Associate's degree	7.57	7.22	0.35*	4.87
		5 Some college but no degree	14.83	15.42	-0.59*	-3.84
		6 Bachelor's degree	24.96	25.40	-0.44	-1.73
		7 Master's degree or equivalent	16.82	17.44	-0.62*	-3.55
		8 Doctoral degree - professional practice	3.04	3.67	-0.63*	-17.12
		9 Doctoral degree - research/scholarship	2.96	3.15	-0.19	-5.93
PARSUPP22	Supportive of PSE: parents, AY 2020–21/2021–22	1 Completely disagree	1.11	1.30	-0.18*	-14.22
		2 Somewhat disagree	0.98	1.08	-0.11	-9.88
		3 Neither disagree nor agree	6.96	7.19	-0.23	-3.19
		4 Somewhat agree	10.87	10.98	-0.11	-1.00
		5 Completely agree	80.09	79.46	0.63*	0.79
PEERINTIN22	Engagement: interactions with other students during class, AY 2020–21/2021–22	1 Completely disagree	1.34	1.31	0.02	1.69
		2 Somewhat disagree	2.48	2.49	-0.01	-0.47
		3 Neither disagree nor agree	15.95	15.93	0.02	0.15
		4 Somewhat agree	34.04	34.34	-0.30	-0.88
		5 Completely agree	46.20	45.93	0.27	0.58
PEERINTOUT22	Engagement: interactions with other students outside of class, AY 2020–21/2021–22	1 Completely disagree	2.46	2.45	0.00	0.08
		2 Somewhat disagree	3.31	3.44	-0.13	-3.65
		3 Neither disagree nor agree	20.01	20.22	-0.21	-1.06
		4 Somewhat agree	31.49	31.42	0.07	0.23
		5 Completely agree	42.73	42.47	0.27	0.63
PELLDEP	Pell grant and dependency, AY 2019–20	0 Did not receive Pell grant	54.97	54.83	0.13*	0.24
		1 Received Pell grant, dependent	35.53	35.68	-0.15*	-0.42
		2 Received Pell grant, independent	9.51	9.49	0.02*	0.20

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
PFAMNM22	Dependent students: family size, AY 2021–22	2 Two	13.78	11.90	1.88*	15.83
		3 Three	27.64	26.57	1.07*	4.01
		4 Four	30.28	31.26	-0.98*	-3.12
		5 Five	17.26	17.77	-0.51	-2.88
		6 Six	6.67	7.58	-0.91*	-11.96
		7 Seven	2.69	2.90	-0.21	-7.31
		8 Eight	0.90	1.17	-0.27*	-23.18
		9 Nine	0.38	0.35	0.03	8.70
		10 Ten	0.17	0.26	-0.09	-33.74
		11 Eleven	0.13	0.15	-0.02	-12.16
		12 Twelve	0.03	0.02	0.01	45.07
		13 Thirteen	0.02	0.03	-0.01	-23.83
		14 Fourteen	0.04	0.04	0.01	25.55
		18 Eighteen	0.00	0.01	-0.01	-100.00
PHYSH22	Health: general physical health, AY 2021–22	1 Excellent	20.85	21.03	-0.18	-0.84
		2 Very good	31.56	31.19	0.36	1.17
		3 Good	33.41	33.52	-0.11	-0.33
		4 Fair	12.27	12.40	-0.13	-1.07
		5 Poor	1.91	1.86	0.05	2.96
PHYSHDIF	Health: general physical health, change between AY 2019–20 to 2021–22	1 Physical health rating improved	32.68	36.22	-3.54*	-9.78
		2 Physical health remained constant	48.21	40.12	8.09*	20.16
		3 Physical health rating declined	19.11	23.66	-4.54*	-19.21
PHYSMISS22	Health: amount missed school/work in past 30 days for physical health, AY 2021–22	1 Never	65.41	65.39	0.02	0.03
		2 A few times	27.07	27.01	0.07	0.26
		3 About once a week	4.50	4.57	-0.07	-1.59
		4 Almost every day	1.57	1.62	-0.05	-2.79
		5 Every day	1.44	1.41	0.03	2.28

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
PINC0L22	Dependent students: number family members in college, AY 2021–22	1 One	70.77	67.27	3.50*	5.21
		2 Two	25.29	27.17	-1.88*	-6.92
		3 Three	3.35	4.35	-1.01*	-23.10
		4 Four	0.47	0.86	-0.39*	-45.15
		5 Five	0.07	0.21	-0.14*	-68.09
		6 Six	0.00	0.04	-0.04*	-100.00
		7 Seven	0.00	0.01	-0.01	-100.00
		8 Eight	0.05	0.05	0.00	9.02
		9 Nine	0.00	0.04	-0.04	-100.00
		11 Eleven	0.00	0.01	-0.01	-100.00
PLNST22	Private student loans: repayment status, AY 2021–22	0 No private student loans	90.38	89.87	0.51*	0.57
		1 In repayment	3.19	3.44	-0.25*	-7.20
		2 Not in repayment	2.51	2.60	-0.09	-3.58
		3 Paid in full	0.64	0.67	-0.03	-4.37
		4 In deferment	3.28	3.42	-0.14	-4.15
PMARIT22	Dependent students: parents' marital status, AY 2021–22	1 Married/remarried	62.21	66.00	-3.79*	-5.75
		2 Single	11.59	10.70	0.88*	8.26
		3 Divorced/Separated	22.82	19.87	2.96*	14.90
		4 Widowed	2.64	2.56	0.08	3.16
		5 Unmarried and both parents living together	0.74	0.87	-0.13	-15.08
PRAT3Y	Persistence anywhere through AY 2021–22	1 Attained, still enrolled	6.60	6.46	0.14*	2.17
		2 Attained, not enrolled	6.38	6.08	0.30*	4.89
		3 No degree, still enrolled	59.03	58.85	0.18	0.31
		4 No degree, not enrolled	27.99	28.61	-0.62*	-2.16
PRATY1	Attainment and enrollment, AY 2019–20	1 Attained degree	1.62	1.51	0.11*	7.15
		2 No degree, enrolled 8 months or more	67.73	67.54	0.20	0.29
		3 No degree, enrolled less than 8 months	30.65	30.95	-0.30	-0.98
PRATY2	Attainment and enrollment, AY 2020–21	1 Attained degree	6.19	5.70	0.49*	8.58
		2 No degree, enrolled 8 months or more	62.40	62.08	0.31	0.51
		3 No degree, enrolled less than 8 months	11.85	13.59	-1.74*	-12.83
		4 No degree, not enrolled	19.57	18.63	0.94*	5.05

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
PRATY3	Attainment and enrollment, AY 2021–22	1 Attained degree	6.14	5.67	0.48*	8.38
		2 No degree, enrolled 8 months or more	55.65	55.10	0.55*	1.00
		3 No degree, enrolled less than 8 months	10.87	13.21	-2.34*	-17.71
		4 No degree, not enrolled	27.34	26.03	1.31*	5.04
PRLVL3Y	Attainment or level of last institution enrolled, as of 2021–22 AY	1 Attained bachelor's degree	0.71	0.67	0.04*	6.36
		2 Attained associate's degree	7.42	6.78	0.63*	9.32
		3 Attained certificate	5.58	5.08	0.50*	9.79
		4 No degree, enrolled at 4-year institution	47.16	47.92	-0.76*	-1.58
		5 No degree, enrolled at less-than-4-year institution	16.02	16.92	-0.90*	-5.31
		6 No degree, not enrolled	23.10	22.62	0.48*	2.14
PROUT1_NEW	Cumulative persistence and attainment anywhere through AY 2019–20 (new version)	1 Attained bachelor's degree	0.02	0.02	0.00	9.25
		2 Attained associate's degree	0.36	0.34	0.03*	7.77
		3 Attained certificate	1.24	1.15	0.08*	7.36
		4 No degree, enrolled in 2019–20 AY	98.38	98.49	-0.11*	-0.11
PROUT1_OLD	Cumulative persistence and attainment anywhere through AY 2019–20 (old version)	1 Attained bachelor's degree	0.02	0.02	0.00	18.02
		2 Attained associate's degree	0.39	0.34	0.06*	16.43
		3 Attained certificate	1.34	1.15	0.18*	15.98
		4 No degree, still enrolled in 2019–20 AY	82.29	84.06	-1.77*	-2.11
		6 No degree, left without return as of June 2022	15.96	14.43	1.53*	10.59
PROUT2_NEW	Cumulative persistence and attainment anywhere through AY 2020–21 (new version)	1 Attained bachelor's degree	0.08	0.08	0.00	2.75
		2 Attained associate's degree	3.90	3.57	0.32*	9.00
		3 Attained certificate	3.82	3.50	0.32*	9.09
		4 No degree, enrolled in 2020–21 AY	73.40	75.12	-1.72*	-2.28
		5 No degree, not enrolled in 2020–21 AY	18.80	17.73	1.07*	6.05

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
PROUT2_OLD	Cumulative persistence and attainment anywhere through AY 2020–21 (old version)	1 Attained bachelor's degree	0.08	0.08	0.00	3.65
		2 Attained associate's degree	3.93	3.57	0.36*	9.96
		3 Attained certificate	3.86	3.50	0.35*	10.05
		4 No degree, still enrolled in 2020–21 AY	65.11	66.18	-1.07*	-1.62
		5 No degree, not enrolled in 2020–21 AY	3.62	3.85	-0.24*	-6.10
		6 No degree, left without return as of June 2022	23.41	22.82	0.60*	2.61
PROUT3_NEW	Cumulative persistence and attainment anywhere through AY 2021–22 (new version)	1 Attained bachelor's degree	0.71	0.67	0.04*	6.36
		2 Attained associate's degree	7.42	6.78	0.63*	9.32
		3 Attained certificate	5.58	5.08	0.50*	9.79
		4 No degree, enrolled in 2021–22 AY	63.18	64.84	-1.66*	-2.55
		5 No degree, not enrolled in 2021–22 AY	23.10	22.62	0.48*	2.14
PROUT3_OLD	Cumulative persistence and attainment anywhere through AY 2021–22 (old version)	1 Attained bachelor's degree	0.71	0.67	0.04*	6.36
		2 Attained associate's degree	7.42	6.78	0.63*	9.32
		3 Attained certificate	5.58	5.08	0.50*	9.79
		4 No degree, still enrolled in 2021–22 AY	58.26	58.85	-0.59*	-1.00
		6 No degree, left without return as of June 2022	28.03	28.61	-0.58*	-2.04
PROUTF1_NEW	Cumulative retention and attainment at first institution through AY 2019–20 (new version)	1 Attained bachelor's degree	0.01	0.01	0.00	7.81
		2 Attained associate's degree	0.36	0.33	0.02*	6.34
		3 Attained certificate	1.19	1.10	0.09*	7.81
		4 No degree, enrolled in 2019–20 AY	94.27	94.19	0.07	0.08
		6 No degree, transferred as of 2019–20 AY	4.18	4.36	-0.18*	-4.14
PROUTF1_OLD	Cumulative retention and attainment at first institution through AY 2019–20 (old version)	1 Attained bachelor's degree	0.01	0.01	0.00	7.81
		2 Attained associate's degree	0.36	0.33	0.02*	6.34
		3 Attained certificate	1.19	1.10	0.09*	7.81
		4 No degree, still enrolled in 2019–20 AY	77.85	77.02	0.83*	1.08
		6 No degree, transferred as of 2019–20 AY	4.18	4.36	-0.18*	-4.14
		7 No degree, left without return as of June 2022	16.41	17.17	-0.76*	-4.43

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
PROUTF2_NEW	Cumulative retention and attainment at first institution through AY 2020–21 (new version)	1 Attained bachelor's degree	0.07	0.06	0.00	3.85
		2 Attained associate's degree	3.79	3.47	0.31*	8.98
		3 Attained certificate	3.48	3.20	0.28*	8.63
		4 No degree, enrolled in 2020–21 AY	66.58	67.48	-0.89*	-1.33
		5 No degree, not enrolled in 2020–21 AY	19.63	19.19	0.44*	2.29
		6 No degree, transferred as of 2020–21 AY	6.46	6.59	-0.14	-2.05
PROUTF2_OLD	Cumulative retention and attainment at first institution through AY 2020–21 (old version)	1 Attained bachelor's degree	0.07	0.06	0.00	3.85
		2 Attained associate's degree	3.79	3.47	0.31*	8.98
		3 Attained certificate	3.48	3.20	0.28*	8.63
		4 No degree, still enrolled in 2020–21 AY	56.72	57.15	-0.43*	-0.75
		5 No degree, not enrolled in 2020–21 AY	2.13	2.59	-0.47*	-17.94
		6 No degree, transferred as of 2020–21 AY	6.46	6.59	-0.14	-2.05
		7 No degree, left without return as of June 2022	27.37	26.93	0.44*	1.63
PROUTF3_NEW	Cumulative retention and attainment at first institution through AY 2021–22 (new version)	1 Attained bachelor's degree	0.65	0.60	0.05*	7.73
		2 Attained associate's degree	6.94	6.40	0.55*	8.57
		3 Attained certificate	4.82	4.46	0.37*	8.27
		4 No degree, enrolled in 2021–22 AY	53.33	54.69	-1.36*	-2.49
		5 No degree, not enrolled in 2021–22 AY	27.25	26.75	0.51*	1.89
		6 No degree, transferred as of 2021–22 AY	7.00	7.11	-0.11	-1.53
PROUTF3_OLD	Cumulative retention and attainment at first institution through AY 2021–22 (old version)	1 Attained bachelor's degree	0.63	0.60	0.02*	3.90
		2 Attained associate's degree	6.70	6.40	0.30*	4.72
		3 Attained certificate	4.65	4.46	0.20*	4.43
		4 No degree, still enrolled in 2021–22 AY	49.13	48.98	0.15	0.30
		5 No degree, not enrolled in 2021–22 AY	4.58	5.71	-1.13*	-19.78
		6 No degree, transferred as of 2021–22 AY	7.36	7.11	0.25*	3.56
		7 No degree, left without return as of June 2022	26.95	26.75	0.21*	0.77

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
RACE	Race/ethnicity (with multiple)	1 White	46.36	46.68	-0.32*	-0.69
		2 Black or African American	12.78	12.60	0.19*	1.47
		3 Hispanic or Latino	22.06	21.76	0.30*	1.39
		4 Asian	10.39	10.37	0.02	0.18
		5 American Indian or Alaska Native	0.88	0.89	-0.01	-0.81
		6 Native Hawaiian/other Pacific Islander	0.53	0.56	-0.03	-5.08
		7 More than one race	6.99	7.14	-0.15	-2.10
RCVSTAI21	Received financial aid from a state aid program, AY 2019–20	0 No	60.15	59.83	0.32	0.53
		1 Yes	39.85	40.17	-0.32	-0.80
RCVSTAI22	Received financial aid from a state aid program, AY 2021–2022	0 No	67.98	67.32	0.66	0.98
		1 Yes	32.02	32.68	-0.66	-2.02
REGSUPP22	Provided regular financial support to family or friend, AY 2020–21/2021–22	0 No	81.93	81.98	-0.05	-0.06
		1 Yes	18.07	18.02	0.05	0.26
REGVT22	Registered to vote in U.S. elections, AY 2021–22	0 No	10.55	13.18	-2.63*	-19.97
		1 Yes	89.45	86.82	2.63*	3.03
REMEV3Y	Ever taken remedial courses through AY 2021–22	0 No	61.30	64.55	-3.26*	-5.04
		1 Yes	38.70	35.45	3.26*	9.19
SEINT3Y	First enrollment spell: intensity through AY 2021–22	1 Full-time	52.96	51.72	1.24*	2.40
		2 Part-time	9.09	9.12	-0.04	-0.42
		3 Mixed	37.95	39.16	-1.20*	-3.07
SENSBELNG22	Engagement: felt like a part of institution, AY 2020–21/2021–22	1 Completely disagree	4.84	5.08	-0.25*	-4.85
		2 Somewhat disagree	7.88	8.10	-0.22	-2.75
		3 Neither disagree nor agree	21.01	21.19	-0.19	-0.88
		4 Somewhat agree	30.98	30.87	0.12	0.38
		5 Completely agree	35.30	34.76	0.54*	1.55
SENUM3Y	Enrollment spells: number of spells through AY 2021–22	1 One	84.93	81.47	3.45*	4.24
		2 Two	14.03	16.95	-2.92*	-17.25
		3 Three	1.03	1.55	-0.52*	-33.62
		4 Four	0.02	0.03	-0.01	-30.94

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
SEROLE22	Jobs while enrolled: primary role as student or employee, AY 2021–22	1 A student working to meet expenses	90.15	90.57	-0.41	-0.46
		2 An employee who decided to enroll in school	9.85	9.43	0.41	4.40
SEXATBIRTH	Sex assigned at birth	1 Male	45.78	45.86	-0.08	-0.18
		2 Female	54.22	54.14	0.08	0.15
SEXORIENTAT22	Sexual orientation, as of AY 2021–22	1 Lesbian or gay, that is, homosexual	3.77	3.76	0.01	0.14
		2 Straight, that is, heterosexual	80.02	80.04	-0.02	-0.03
		3 Bisexual	11.37	11.35	0.03	0.24
		4 Another sexual orientation	2.55	2.56	-0.01	-0.38
		5 Don't know	2.29	2.28	0.00	0.06
SFAMNM22	Independent students: family size, AY 2021–22	1 One	41.12	41.52	-0.40	-0.96
		2 Two	27.43	29.25	-1.82*	-6.22
		3 Three	15.39	14.52	0.87	6.00
		4 Four	8.06	7.15	0.91*	12.72
		5 Five	5.95	5.20	0.74*	14.30
		6 Six	1.42	1.34	0.08	5.86
		7 Seven	0.39	0.43	-0.03	-8.22
		8 Eight	0.06	0.24	-0.17	-72.61
		9 Nine	0.18	0.36	-0.18	-49.32
SIINT3Y	First institution spells: intensity through AY 2021–22	1 Full-time	55.93	55.06	0.88*	1.59
		2 Part-time	9.26	9.62	-0.36*	-3.76
		3 Mixed	34.81	35.32	-0.52*	-1.46
SINCOL22	Independent students: number of family members in college, AY 2021–22	1 One	91.25	90.62	0.62	0.69
		2 Two	8.15	7.91	0.24	2.97
		3 Three	0.61	0.94	-0.33	-35.23
		4 Four	0.00	0.30	-0.30*	-100.00
		5 Five	0.00	0.23	-0.23	-100.00
SMAR22	Marital status, AY 2021–22	1 Single, never married	89.27	89.29	-0.02*	-0.02
		2 Married	5.32	5.31	0.01*	0.20
		3 Separated	0.59	0.59	0.00*	0.20
		4 Divorced	0.90	0.90	0.00*	0.20
		5 Widowed	0.13	0.13	0.00*	0.20
		6 With partner in marriage-like relationship	3.79	3.78	0.00	0.07

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
SOCPARTANY22	Social activities: participated in any social activity, AY 2020–21/2021–22	0 No	53.22	53.37	-0.15	-0.29
		1 Yes	46.78	46.63	0.15	0.33
SOCPARTGRK22	Social activities: participated in a Greek letter organization, AY 2020–21/2021–22	0 No	87.48	87.46	0.01	0.02
		1 Yes	12.52	12.54	-0.01	-0.12
SOCPARTMUS22	Social activities: participated in institution sponsored music, theatre, or band, AY 2020–21/2021–22	0 No	92.50	92.40	0.09	0.10
		1 Yes	7.50	7.60	-0.09	-1.25
SOCPARTNVAR22	Social activities: participated in non-varsity athletics, AY 2020–21/2021–22	0 No	84.42	84.55	-0.12	-0.14
		1 Yes	15.58	15.45	0.12	0.79
SOCPARTSTGOV22	Social activities: participated in student government, AY 2020–21/2021–22	0 No	95.23	95.41	-0.18	-0.19
		1 Yes	4.77	4.59	0.18	3.98
SOCPARTSTGRP22	Social activities: participated in student affinity groups, AY 2020–21/2021–22	0 No	88.59	88.98	-0.39*	-0.43
		1 Yes	11.41	11.02	0.39*	3.50
SOCPARTSTORG22	Social activities: participated in any other recognized or registered student organizations, AY 2020–21/2021–22	0 No	71.65	72.77	-1.12*	-1.53
		1 Yes	28.35	27.23	1.12*	4.09
SOCPARTVAR22	Social activities: participated in varsity athletics, AY 2020–21/2021–22	0 No	93.16	93.38	-0.21	-0.23
		1 Yes	6.84	6.62	0.21	3.21
SPNM21	Not married to spouse in 2021	0 Married to spouse in 2021	94.98	94.99	-0.01	-0.01
		1 Not married to spouse in 2021	5.02	5.01	0.01	0.11
SPPSUPP22	Supportive of PSE: spouse or partner, AY 2020–21/2021–22	1 Completely disagree	3.02	2.88	0.13	4.58
		2 Somewhat disagree	0.98	1.70	-0.72	-42.29
		3 Neither disagree nor agree	13.02	12.41	0.61	4.91
		4 Somewhat agree	13.80	14.15	-0.34	-2.42
		5 Completely agree	69.18	68.86	0.32	0.46

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
SPSED22	Independent students: spouse's highest education level, AY 2021–22	1 Did not complete high school	4.21	4.21	0.00	-0.08
		2 High school diploma or equivalent	34.21	33.81	0.41	1.20
		3 Vocational/technical training	9.56	8.26	1.30*	15.77
		5 Associate's degree	10.86	11.31	-0.44	-3.92
		6 Some college but no degree	15.39	14.71	0.68	4.64
		7 Bachelor's degree	17.70	17.92	-0.22	-1.23
		8 Master's degree or equivalent	7.47	9.25	-1.77	-19.18
		9 Professional degree	0.08	0.06	0.02	31.87
		10 Doctoral degree	0.50	0.47	0.03	6.51
STDYABR3Y	Ever studied abroad while attending NPSAS through AY 2021–22	0 No	93.18	93.97	-0.78*	-0.83
		1 Yes	6.82	6.03	0.78*	12.98
STEMMAJ	Major or field of study with a focus on STEM fields, AY 2019–20	1 Math/computer/sciences/engineering/technologies	23.02	22.69	0.32*	1.42
		2 Social/behavioral sciences	7.73	7.63	0.10*	1.31
		4 Non-STEM field	65.65	65.25	0.40*	0.61
		5 Undecided or not in a degree program	3.61	4.42	-0.82*	-18.52
STNUM3Y	Stopouts through AY 2021–22: total	0 Never stopped out	84.93	81.47	3.45*	4.24
		1 One	14.03	16.95	-2.92*	-17.25
		2 Two	1.03	1.55	-0.52*	-33.62
		3 Three	0.02	0.03	-0.01	-30.94
STSTA1	Stopout status: AY 2019–20	0 Never stopped out	84.93	81.47	3.45*	4.24
		1 Stopout began	6.85	8.27	-1.42*	-17.21
		3 Stopout beginning and ending during same year	1.53	1.66	-0.14*	-8.20
		5 Did not stopout during academic year	6.70	8.59	-1.89*	-22.03
STSTA2	Stopout status: AY 2020–21	0 Never stopped out	84.93	81.47	3.45*	4.24
		1 Stopout began	4.52	5.43	-0.91*	-16.70
		2 Stopout ended	3.93	4.81	-0.88*	-18.32
		3 Stopout beginning and ending during same year	1.91	2.93	-1.02*	-34.65
		4 During a stopout, no enrollment spell	2.45	2.80	-0.34*	-12.30
		5 Did not stopout during academic year	2.26	2.57	-0.31*	-11.98
STSTA3	Stopout status: AY 2021–22	0 Never stopped out	84.93	81.47	3.45*	4.24
		2 Stopout ended	7.49	8.94	-1.45*	-16.27
		3 Stopout beginning and ending during same year	1.21	1.69	-0.48*	-28.21
		5 Did not stopout during academic year	6.38	7.90	-1.52*	-19.28

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
STYB3Y	First stopout through AY 2021–22: starting academic year, at any institution	0 Never stopped out	84.93	81.47	3.45*	4.24
		1 Academic year 2019–20	8.37	9.93	-1.56*	-15.71
		2 Academic year 2020–21	5.84	7.49	-1.65*	-21.98
		3 Academic year 2021–22	0.86	1.10	-0.25*	-22.36
STYBF13Y	First stopout through AY 2021–22: starting academic year, at first institution	0 Never stopped out at first institution	88.32	85.69	2.63*	3.07
		1 Academic year 2019–20	6.57	7.64	-1.07*	-14.04
		2 Academic year 2020–21	4.48	5.92	-1.44*	-24.40
		3 Academic year 2021–22	0.64	0.75	-0.12*	-15.33
TCREFFCOM22	Teacher effectiveness: had good command of subject, AY 2020–21/2021–22	1 Completely disagree	1.53	1.63	-0.10	-6.30
		2 Somewhat disagree	3.99	4.13	-0.15	-3.52
		3 Neither disagree nor agree	13.68	13.99	-0.31	-2.20
		4 Somewhat agree	39.40	39.13	0.28	0.70
		5 Completely agree	41.40	41.12	0.28	0.68
TCREFFEX22	Teacher effectiveness: make good use of examples, AY 2020–21/2021–22	1 Completely disagree	2.20	2.30	-0.10	-4.27
		2 Somewhat disagree	6.41	6.74	-0.34*	-5.02
		3 Neither disagree nor agree	16.69	16.95	-0.27	-1.59
		4 Somewhat agree	40.20	39.76	0.44*	1.11
		5 Completely agree	34.51	34.24	0.26	0.77
TCREFFINT22	Teacher effectiveness: genuinely interested in teaching, AY 2020–21/2021–22	1 Completely disagree	1.99	2.11	-0.12	-5.68
		2 Somewhat disagree	4.83	4.96	-0.12	-2.45
		3 Neither disagree nor agree	14.27	14.36	-0.09	-0.63
		4 Somewhat agree	39.07	38.98	0.09	0.23
		5 Completely agree	39.83	39.58	0.24	0.61
TCREFFOUT22	Teacher effectiveness: outstanding teachers, AY 2020–21/2021–22	1 Completely disagree	1.96	2.07	-0.11	-5.13
		2 Somewhat disagree	6.35	6.62	-0.27*	-4.01
		3 Neither disagree nor agree	15.78	16.10	-0.32*	-2.01
		4 Somewhat agree	44.23	43.70	0.54*	1.23
		5 Completely agree	31.68	31.52	0.16	0.50

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
TFATT3Y	Transfer (first): attainment before transfer through AY 2021–22	0 Never transferred	90.56	90.18	0.37*	0.42
		1 No attainment before transfer	6.97	7.56	-0.59*	-7.78
		2 Attained a certificate before transfer	0.25	0.27	-0.02	-6.47
		3 Attained an associate's degree before transfer	2.22	1.99	0.23*	11.79
		4 Attained a bachelor's degree before transfer	0.00	0.00	0.00	-100.00
TFDEG3Y	Transfer (first): degree program through AY 2021–22	0 Never transferred	90.49	90.18	0.31*	0.35
		1 Not in a degree program	0.81	0.99	-0.18*	-18.08
		2 Certificate program	0.78	0.96	-0.18*	-18.67
		3 Associate's degree program	1.85	2.06	-0.22*	-10.56
		4 Bachelor's degree program	6.07	5.81	0.26*	4.53
TFIFTY3Y	Transfer (first): destination institution type through AY 2021–22	0 Never transferred	90.31	90.18	0.13	0.14
		1 Public 4-year	5.56	5.31	0.25*	4.65
		2 Private not-for-profit 4-year	1.51	1.48	0.02	1.62
		3 Private for-profit 4-year	0.31	0.38	-0.08*	-19.91
		4 Public 2-year	1.80	2.00	-0.20*	-9.92
		5 Private not-for-profit 2-year	0.04	0.05	-0.01	-21.55
		6 Private for-profit 2-year	0.18	0.25	-0.06*	-24.92
		7 Public less than 2-year	0.05	0.06	-0.01	-18.50
		9 Private for-profit less than 2-year	0.24	0.29	-0.04	-15.19
TFINCT3Y	Transfer direction (first): institution control through AY 2021–22	0 Never transferred	90.31	90.18	0.13	0.14
		1 Public to public	6.33	6.25	0.09	1.41
		2 Public to private not-for-profit	0.99	0.98	0.02	1.66
		3 Public to private for-profit	0.54	0.65	-0.10*	-15.86
		4 Private not-for-profit to public	0.81	0.86	-0.04	-5.11
		5 Private not-for-profit to private not-for-profit	0.48	0.48	0.00	0.57
		6 Private not-for-profit to private for-profit	0.11	0.12	-0.01	-6.99
		7 Private for-profit to public	0.26	0.27	-0.01	-2.78
		8 Private for-profit to private not-for-profit	0.07	0.07	0.00	-6.87
		9 Private for-profit to private for-profit	0.08	0.15	-0.07*	-46.12

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
TFINLV3Y	Transfer direction (first): institution level through AY 2021–22	0 Never transferred	90.31	90.18	0.13	0.14
		1 4-year to 4-year	3.60	3.48	0.12	3.38
		2 4-year to 2-year	1.26	1.45	-0.18*	-12.79
		3 4-year to less than 2-year	0.15	0.20	-0.05	-24.28
		4 2-year to 4-year	3.74	3.61	0.13	3.52
		5 2-year to 2-year	0.69	0.77	-0.07	-9.35
		6 2-year to less than 2-year	0.14	0.15	-0.01	-3.42
		7 Less than 2-year to 4-year	0.03	0.08	-0.05	-59.16
		8 Less than 2-year to 2-year	0.06	0.08	-0.01	-16.30
		9 Less than 2-year to less than 2-year	0.00	0.00	0.00	-100.00
TFNUM3Y	Transfer status through AY 2021–22	0 Zero	90.31	90.18	0.13	0.14
		1 One	9.41	9.55	-0.14	-1.44
		2 Two	0.27	0.27	0.01	3.02
		3 Three	0.00	0.00	0.00	-26.67
TFTYPE1	Transfer status, AY 2019–20	0 Never transferred	90.31	90.18	0.13	0.14
		1 Left origin institution	3.87	4.28	-0.41*	-9.59
		3 Left origin institution and began at destination institution	0.39	0.44	-0.05*	-11.49
		4 Did not transfer in 2019–20 AY	5.43	5.10	0.33*	6.50
TFTYPE2	Transfer status, AY 2020–21	0 Never transferred	90.31	90.18	0.13	0.14
		1 Left origin institution	3.04	2.83	0.21*	7.54
		2 Began at destination institution	2.75	3.12	-0.37*	-11.84
		3 Left origin institution and began at destination institution	1.05	1.02	0.04	3.71
		4 Did not transfer in 2020–21 AY	2.84	2.86	-0.01	-0.40
TFTYPE3	Transfer status, AY 2021–22	0 Never transferred	90.31	90.18	0.13	0.14
		1 Left origin institution	0.01	0.01	0.00	17.04
		2 Began at destination institution	4.15	3.97	0.18*	4.57
		3 Left origin institution and began at destination institution	1.56	1.47	0.08*	5.75
		4 Did not transfer in 2021–22 AY	3.97	4.37	-0.40*	-9.12

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
TFTYPE3Y	Transfer direction (first): type through AY 2021–22	0 Never transferred	90.31	90.18	0.13	0.14
		1 Immediate, upward transfer	3.00	2.80	0.20*	7.08
		2 Immediate, lateral transfer	3.04	2.91	0.13*	4.50
		3 Immediate, downward transfer	0.87	0.96	-0.09*	-9.32
		4 Delayed, upward transfer	0.84	0.98	-0.13*	-13.68
		5 Delayed, lateral transfer	1.26	1.34	-0.09	-6.46
		6 Delayed, downward transfer	0.69	0.84	-0.15*	-17.84
TRNPLN	Plans to transfer, AY 2019–20	0 Did not plan to transfer	60.21	62.32	-2.12*	-3.39
		1 Yes, planned to transfer	39.79	37.68	2.12*	5.62
UGDEG	Degree program, AY 2019–20	1 Certificate	7.50	7.50	0.01*	0.10
		2 Associate's degree	41.87	41.90	-0.03	-0.08
		3 Bachelor's degree	49.62	49.59	0.03	0.05
		4 Not in a degree program or others	1.02	1.01	0.00	0.10
UGDEG22	Degree program when last enrolled through AY 2021–22	1 Certificate	6.16	6.69	-0.54*	-8.02
		2 Associate's degree	32.80	32.23	0.57*	1.76
		3 Bachelor's degree	57.65	56.89	0.76*	1.33
		4 Not in a degree program or other	3.40	4.18	-0.79*	-18.81
UGLVL22	Highest undergraduate class level through AY 2021–22	1 1st year undergraduate	25.21	26.74	-1.54*	-5.75
		2 2nd year undergraduate	31.08	31.63	-0.55*	-1.74
		3 3rd year undergraduate	36.25	34.63	1.62*	4.67
		4 4th year undergraduate	7.11	6.66	0.45*	6.80
		5 5th year undergraduate	0.08	0.06	0.01*	23.82
		6 Unclassified undergraduate	0.28	0.28	0.00	0.13
UNEMPL22	Unemployment: currently looking for a job, 2022	0 No	62.89	63.10	-0.20	-0.32
		1 Yes	37.11	36.90	0.20	0.55
USEACAD22	PSE services used: academic advising, AY 2020–21/2021–22	0 No	18.33	19.30	-0.98*	-5.05
		1 Yes	81.67	80.70	0.98*	1.21
USEANY22	PSE services used: used any service, AY 2020–21/2021–22	0 No	8.54	8.28	0.26	3.14
		1 Yes	91.46	91.72	-0.26	-0.28

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
USECAMPSEC22	PSE services used: campus security or campus police, AY 2020–21/2021–22	0 No 1 Yes	84.71 15.29	84.88 15.12	-0.16 0.16	-0.19 1.09
USECHLD22	PSE services used: child care, AY 2020–21/2021–22	0 No 1 Yes	97.85 2.15	97.90 2.10	-0.06 0.06	-0.06 2.63
USECPP22	PSE services used: career services, AY 2020–21/2021–22	0 No 1 Yes	67.42 32.58	67.49 32.51	-0.07 0.07	-0.10 0.20
USEDISSRV22	PSE services used: services for students with disabilities, AY 2020–21/2021–22	0 No 1 Yes	89.80 10.20	89.61 10.39	0.19 -0.19	0.21 -1.84
USEDIVRSE22	PSE services used: diversity or multicultural student services, AY 2020–21/2021–22	0 No 1 Yes	90.03 9.97	90.22 9.78	-0.18 0.18	-0.20 1.88
USEFINAID22	PSE services used: financial aid counseling or advice, AY 2020–21/2021–22	0 No 1 Yes	50.81 49.19	50.48 49.52	0.32 -0.32	0.64 -0.65
USEHLTH22	PSE services used: health center and services, AY 2020–21/2021–22	0 No 1 Yes	68.83 31.17	69.63 30.37	-0.79* 0.79*	-1.14 2.61
USEIDR22	Likelihood of using income-driven student loan repayment plans, AY 2021–22	1 Very unlikely 2 Somewhat unlikely 3 Neither unlikely nor likely 4 Somewhat likely 5 Very likely	21.65 6.59 25.22 26.58 19.96	20.65 6.49 26.45 26.30 20.11	1.01 0.09 -1.23 0.28 -0.15	4.88 1.46 -4.65 1.05 -0.74
USELFP22	Likelihood of using loan forgiveness programs, AY 2021–22	1 Very unlikely 2 Somewhat unlikely 3 Neither unlikely nor likely 4 Somewhat likely 5 Very likely	26.48 8.92 27.43 21.85 15.31	26.06 9.09 27.44 22.00 15.41	0.42 -0.17 -0.01 -0.15 -0.10	1.61 -1.85 -0.02 -0.67 -0.64

See notes at end of table.

Table I-3. Distributions of categorical variables before and after imputation: 2022—Continued

Variable	Description	Value Label	Percent distribution before imputation	Percent distribution after imputation	Difference	Percent relative difference
USEMENSrv22	PSE services used: psychological or mental health services, AY 2020–21/2021–22	0 No	83.49	83.95	-0.46*	-0.55
		1 Yes	16.51	16.05	0.46*	2.89
USEMILSRV22	PSE services used: services for active military or veterans, AY 2020–21/2021–22	0 No	94.31	94.40	-0.09	-0.10
		1 Yes	5.69	5.60	0.09	1.69
USETRPLN22	PSE services used: transfer advising or planning, AY 2020–21/2021–22	0 No	76.15	75.45	0.70*	0.93
		1 Yes	23.85	24.55	-0.70*	-2.87
USETUTOR22	PSE services used: peer or other tutoring or skills lab, AY 2020–21/2021–22	0 No	62.11	62.12	-0.01	-0.02
		1 Yes	37.89	37.88	0.01	0.03
VTNEL22	Voted in 2020 presidential election	0 No	31.96	32.06	-0.10	-0.33
		1 Yes	66.87	66.37	0.50*	0.75
		2 Ineligible to vote in 2020 national election	1.17	1.57	-0.39*	-25.06
WRKSTDY3Y	Work-study job: had work-study job through AY 2021–22	0 No	82.00	82.23	-0.23	-0.28
		1 Yes	18.00	17.77	0.23	1.30
WRKTSKS22	Job non-monetary benefits: importance of making decisions, AY 2021–22	1 Less important than salary	14.65	14.80	-0.14	-0.97
		2 As important as salary	57.11	57.24	-0.13	-0.22
		3 More important than salary	28.23	27.96	0.27	0.97

* $p < .05$.

NOTE: AY = academic year; NPSAS = National Postsecondary Student Aid Study; NSF = National Science Foundation; PSE = postsecondary education; STEM = science, technology, engineering, and mathematics; TANF = Temporary Assistance for Needy Families; WIC = Special Supplemental Nutrition Program for Women, Infants, and Children. Distributions were computed using the BPS:20/22 final analysis weight. Cases with legitimate skips for the item are not included in the distributions. The difference is computed as the percentage before imputation minus the percentage after imputation.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22).

Appendix J. 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test Data File Documentation

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Section 1. Overview of BPS:20/22 Field Test

The 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) field test, conducted by the U.S. Department of Education's National Center for Education Statistics (NCES), provides information on student persistence and attainment in postsecondary education. The BPS:20/22 field test is the first follow-up of a panel of students who began their postsecondary education for the first time during the 2018–19 academic year, identified in the 2019–20 National Postsecondary Student Aid Study (NPSAS:20).

The respondent universe for the BPS:20/22 field test consisted of students enrolled in postsecondary education for the first time between July 1, 2018, and June 30, 2019, at any Title IV eligible postsecondary institution in the United States and Puerto Rico. The field test sample included a total of 3,700 sample members. BPS:20/22 field test surveys were conducted between March 2, 2021, and June 30, 2021.

This appendix describes procedures and results of the BPS:20/22 field test. The field test was designed to evaluate data collection methodologies, the student survey, and systems proposed for use in the full-scale study. These procedures and systems were based on established protocols from previous cycles of BPS, NPSAS, and other NCES postsecondary education studies. Included as an appendix to the full-scale data file documentation, the following sections provide information that is unique to the field test. Specific field test evaluation goals for the student survey included identifying problematic data elements in the BPS:20/22 student survey, determining optimal survey design for collecting month-level enrollment intensity, evaluating the functionality of collecting address information using a new predictive search database, identifying the most analytically useful data elements concerning the impacts of the coronavirus pandemic, and testing two data collection protocols aimed at reducing nonresponse error.

Section 2 details the sampling design for institutions and students in NPSAS:20 and outlines the process for identifying BPS-eligible sample members. Section 3 presents information on data collection procedures and results, including

evaluations of data quality. Section 3 also presents information on the design and results of the data collection experiments. Lastly, section 4 details recommendations for the BPS:20/22 full-scale student survey and data collection procedures based on field test results.

Tables and figures throughout this report present relevant analyses from the field test. Unless otherwise indicated, a probability level of 0.05 was used for all tests of significance conducted for BPS:20/22 field test evaluations. Unlike the full-scale sample, the field test sample is not a random sample and is not weighted; therefore, statistically significant results are not representative of the population. Rounding is used to protect the confidentiality of respondents. Due to rounding, row and column entries in tables may not sum to their respective totals and reported percentages may differ somewhat from those that would result from the rounded numbers. Rounding is done to the nearest 10.

Section 2. Sampling Design

This section describes how the BPS:20/22 field test sample was selected and details the different categories that made up the field test sample. In prior BPS data collections, a NPSAS field test, conducted a year before the full-scale study, had been used to identify a field test sample for the longitudinal follow-up. However, because NPSAS:20 did not include a field test, the BPS:20/22 field test sample was drawn from the NPSAS:20 full-scale sample. BPS cohorts are typically surveyed in their first year of postsecondary education (i.e., base-year NPSAS) and followed up at the end of their third (first BPS follow-up) and sixth years (second BPS follow-up) after entry into postsecondary education. For the BPS:20/22 field test sample, the NPSAS:20 full-scale survey collected relevant BPS information about their 2019–20 (second) academic year, while the BPS:20/22 field test survey collected information about their 2020–21 (third) academic year to replicate the temporal components associated with the first follow-up survey for a typical BPS cohort, which collects information about enrollment and educational experiences during the 2 academic years since they first enrolled. Thus, the BPS:20/22 field test cohort followed students whose first postsecondary enrollment after high school occurred during the 2018–19 academic year.

The BPS:20/22 field test sample design had three stages. The first two stages occurred within the NPSAS:20 full-scale sample: first, the selection of a sample of NPSAS-eligible institutions and second, the selection of a sample of students within institutions. The third stage was specific to the BPS:20/22 field test and identified confirmed and potential first-time beginning students (FTBs) during the second stage of the sample design.

Students who self-identified as FTBs in the 2018–19 academic year (i.e., prior-year FTBs) in the NPSAS:20 student survey, and those identified as potential prior-year FTBs using student records, enrollment lists, or other administrative data, were eligible for selection for the BPS:20/22 field test. The field test sample of approximately 3,700 students was identified through enrollment lists, administrative data, and survey responses collected during the NPSAS:20 full-scale study. The BPS:20/22 field test sample consisted of four distinct groups based on the NPSAS:20 full-scale data collection outcomes:

1. NPSAS:20-eligible survey respondents who self-identified as FTBs between July 1, 2018, and April 30, 2019 (approximately 2,440 students): This group covered students who were confirmed FTBs based on being NPSAS base-year survey respondents.
2. NPSAS:20-ineligible survey respondents who self-identified as FTBs between July 1, 2018, and April 30, 2019 (approximately 180 students): This group covered students who completed a short-term credential or stopped or dropped out of the NPSAS institution within a year of initial enrollment.
3. NPSAS:20 survey nonrespondents who were potential 2018–19 academic year FTBs based on administrative data (approximately 540 students): This group covered students identified as potential FTBs, based on administrative data, who were survey nonrespondents normally subsampled into the full-scale BPS cohort.
4. NPSAS:20 sample members who were not invited to participate in the survey (administrative-only students) and were potential 2018–19 academic year FTBs based on administrative data (approximately 540 students): Students within this group were new for the BPS:20/22 cohort and also covered those who were potential FTBs based on administrative data as in group 3 above. The difference is that students in this group were never invited to participate in the NPSAS survey, and including them in the field test allowed for determining whether they had different participation patterns to a BPS follow-up survey.

For the first group, NPSAS:20-eligible respondents, rather than relying on random sampling from 2018–19 academic year FTBs in the NPSAS:20 sample, the BPS:20/22 field test used a quota sampling approach to reduce burden on NPSAS:20 respondents. Potentially eligible BPS:20/22 field test sample members received an additional question module targeted at 2018–19 FTB status in the NPSAS:20 student survey. Quotas were defined across the 11 institution sectors and three NPSAS:20 data collection phases (the latter is associated with a change in data collection protocols to convert nonrespondents and improve sample representativeness).

Once specific subgroup quotas were met, further respondents in said subgroup were no longer eligible for the BPS:20/22 field test sample and, as a result, obtained the standard NPSAS:20 survey without the additional FTB module. In addition to reducing burden, this sampling approach allowed a longer “cooling off” period before sample members were contacted again and asked to participate in the BPS:20/22 field test. NPSAS:20 data collection continued until

February 1, 2021, while the BPS:20/22 field test began in March 2021. Quotas were filled for the BPS:20/22 field test by November 13, 2020, which allowed for an additional 2 months before some sample members were contacted again, compared with if sampling had continued through the end of NPSAS:20 data collection.

The second group, the NPSAS:20-ineligible survey respondents, was unusual because students were not enrolled during the 2019–20 academic year but were enrolled during the prior 2018–19 academic year, during which they either completed a short-term credential or stopped or dropped out of the NPSAS institution. Lack of enrollment during the 2019–20 academic year made them ineligible for NPSAS:20. However, because full-scale BPS first follow-up data collection includes students who were not enrolled 1 year after they first began postsecondary education, this group was useful in field testing BPS:20/22 methods and procedures across different eligible populations. While these students were not eligible for NPSAS:20, they were asked to complete a subset of survey questions to confirm FTB status in the 2018–19 academic year. If eligible, the respondent was notified about the potential for a follow-up and administered the locating section of the NPSAS:20 survey so that they could be contacted and asked to participate in the BPS:20/22 field test.

NPSAS:20 consisted of an administrative sample of 380,100 (undergraduate and graduate) students. The NPSAS:20 survey sample was a stratified random sample of approximately 173,360 students from the total administrative student sample. The remaining students who were part of the administrative sample but were not selected for the survey sample are referred to as administrative-only students, which also included potential prior-year FTBs as identified using administrative data sources. The third and fourth groups for the field test were subsamples of potential prior-year FTBs from the NPSAS:20 survey nonrespondents or administrative-only students. The BPS:20/22 field test subsample size and distribution across 11 institution sectors were identical for survey nonrespondents and administrative-only students so that their response behaviors could be evaluated for the full-scale sample.

Table J-1 displays the BPS:20/22 field test sample distributions, by sampling subgroup and control (classification of whether an institution is operated by publicly or privately elected or appointed officials or by privately elected or appointed officials and derives its major source of funds from private sources) and level (whether an institution's programs are 4-year or higher [4-year], 2-but-less-than-4-year [2-year], or less than 2-year) of institution.

Table J-1. BPS:20/22 field test sample distributions, by sampling subgroup and control and level of institution: 2021

Control and level of institution	Total sample		NPSAS:20-eligible survey respondents		NPSAS:20-ineligible survey respondents		NPSAS:20 survey nonrespondents		NPSAS:20 administrative-only students	
	Number	Percent	Number	Percent	Number	Percent	Number	Percent	Number	Percent
All institutions	3,700	100.0	2,440	100.0	180	100.0	540	100.0	540	100.0
Control of institution										
Public	2,190	59.0	1,400	57.4	130	71.1	330	60.6	330	60.6
Private nonprofit	850	23.0	700	28.5	20	10.0	70	12.9	70	12.9
Private for-profit	660	17.9	340	14.0	30	18.9	140	26.5	140	26.5
Level of institution										
Less-than-2-year	190	5.2	90	3.7	20	9.4	40	7.9	40	8.1
2-year	1,270	34.4	560	22.8	120	64.4	300	55.6	300	55.4
4-year, non-doctorate-granting	1,170	31.6	970	39.9	30	16.7	90	17.3	80	13.8
4-year, doctorate-granting	1,070	28.8	820	33.7	20	9.4	100	19.2	120	22.7
Control and level of institution										
Public less-than-2-year	50	1.4	30	1.4	#	1.7	10	1.3	10	1.3
Public 2-year	940	25.3	420	17.1	100	55.0	210	38.9	210	38.9
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	300	8.1	230	9.3	10	6.7	30	5.5	30	5.5
Public 4-year, non-doctorate-granting, primarily baccalaureate	240	6.5	200	8.2	10	3.3	20	3.1	20	3.1
Public 4-year, doctorate-granting	660	17.9	530	21.5	10	4.4	60	11.8	60	11.8
Private nonprofit 2-year or less	40	1.0	10	0.5	#	0.6	10	2.2	10	2.2
Private nonprofit 4-year, non-doctorate-granting	460	12.5	410	16.9	10	5.0	20	3.9	20	3.9
Private nonprofit 4-year, doctorate-granting	350	9.5	270	11.1	10	4.4	40	6.8	40	6.8
Private for-profit less-than-2-year	140	3.8	60	2.3	10	7.8	40	6.6	40	6.6
Private for-profit 2-year	300	8.1	130	5.2	20	8.9	80	14.5	80	14.5
Private for-profit 4-year	220	6.0	160	6.5	#	2.2	30	5.3	30	5.3

Rounds to zero.

NOTE: Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

Section 3. Survey Data Collection Activities, Outcomes, and Evaluation

The BPS:20/22 field test survey instrument was a multimodal survey designed for web and telephone administration. This section describes select aspects of the BPS:20/22 field test survey, beginning with training and monitoring of telephone interviewers, who administered the survey by interacting with the web instrument and reading questions to respondents. The section then details the process of locating, tracing (identifying sample members' most recent contact information), and contacting sample members, including refusal conversion. It also includes details on the various phases of data collection and response rates, presenting the results of efforts to locate and gain participation of sample members. It describes data collection experiments and results. Finally, it presents evaluations of several aspects of the survey's functionality, including the survey design and tests of survey design, cognitive interview results, time required to complete the survey by section, the completion rates of predictive coding forms, and nonresponse rates for specific items.

3.1 Survey Data Collection Activities and Outcomes

This section describes the data collection procedures and results of the BPS:20/22 field test. It also presents data quality evaluations conducted to inform the BPS:20/22 full-scale data collection.

3.1.1 *Training and Monitoring of Telephone Interviewers*

Training for data collection staff. The field test data collection team included five quality control supervisors, four quality experts primarily responsible for monitoring interviews, five telephone interviewers, and two intensive tracing staff. As a result of the coronavirus pandemic, training sessions were modified to accommodate a virtual setting utilizing videoconferencing software. Electronic versions of training materials were provided for virtual trainings, with hard copy training manuals available upon request.

Telephone interviewers acted as the primary point of contact for sample members, conducting telephone interviews and employing strategies to avert or convert refusals. Telephone interviewers also served as help desk agents to respond to sample member concerns, e-mail log-in instructions, and address incentive receipt inquiries and issues. Training programs for these staff members were critical to maximizing response rates and collecting accurate and reliable data. All data collection staff, regardless of role, completed a project training program before beginning work on the study. Additionally, all data collection interviewing staff completed a general training program on topics such as proper interviewing techniques, confidentiality procedures, and sample member rights. All BPS:20/22 field test telephone interviewers had previous experience on either NPSAS:20 or the 2016/20 Baccalaureate and Beyond Longitudinal Study (B&B:16/20), therefore, an abbreviated project training was completed in one 4-hour session that included an overview of the study and a thorough review of the instrument. The intensive tracing staff members had a training on project-specific tracing procedures.

After training, all data collection interviewing staff met certification requirements by administering a full-length interview to project staff, obtaining a security certification, and passing an oral evaluation of both the study's frequently asked questions and pronunciation of the data collection's key words.

Monitoring of data collection staff. Quality experts regularly monitored interviewers to ensure that they administered telephone interviews accurately and professionally. They monitored interviews and recorded observations on interviewer professionalism, question administration, conversational interviewing, and familiarity with the survey instrument. Quality experts and quality control supervisors used recorded interviews during feedback sessions with each interviewer to point out areas for improvement. Segments from recorded interviews were also used during project trainings and quality meetings.

3.1.2 Locating and Tracing

To achieve a high rate of response, data collection staff implemented several procedures to identify sample members' updated contact information (e.g., tracing) and to confirm the accuracy of contact information (e.g., locating). BPS:20/22 field test tracing procedures are provided, followed by locating procedures and results.

Batch tracing. Contact information obtained during NPSAS:20 full-scale data collection was confirmed or updated through matching with various sources of

locating data once the BPS:20/22 field test sample was identified. Before the data collection announcement mailing, the sample was matched with the National Change of Address database, LexisNexis's Single Best Address search, and PhoneAppend to update locating information. Any new information received was loaded into the BPS:20/22 field test database and available for the start of data collection. In addition, during the field test data collection, Premium Phone and Single Best Phone batch searches were run for cases with no good telephone information after the start of telephone interviewing to collect additional telephone locating information. During BPS:20/22 field test data collection, 99 percent of cases were sent to at least one batch tracing source.

Intensive tracing. Intensive tracing was initiated when all telephone locating methods were exhausted and no other telephone number was available. Tracing staff used an approach that included public domain and proprietary databases to identify updated contact information. Tracers identified sample members in consumer databases (e.g., LexisNexis, Experian, and Accurant) using Social Security numbers. If this search resulted in an updated telephone number, tracers sent the case back to the case management system for interviewer follow-up. If the search resulted only in a new address, tracers used directory assistance searches to locate a telephone number. This approach minimized the effort required to locate cases through intensive tracing and reduced the time that cases were unavailable to telephone interviewers. Overall, five percent of the 3,700 BPS:20/22 field test sample members required intensive tracing.

Locating results. A sample member was defined as located if at any point during data collection, contact information was confirmed to be accurate for the individual. For example, if an answering machine confirmed the sample member's name, or a member of the sample member's household confirmed the contacting information, then the sample member was considered located. Similarly, when intensive tracing efforts successfully confirmed contacting information for a sample member, then the case was considered located. Thus, a sample member was considered not located if no contact information was ever verified as accurate for the student. Approximately 3,090 (83 percent) of BPS:20/22 field test sample members were located, and 2,240 (73 percent) of located sample members responded to the survey.

By prior-round (NPSAS:20) response, the field test sample members who were NPSAS:20 survey respondents had a locate rate of 89 percent, compared with 63 percent of NPSAS:20 nonrespondents ($p < .001$). Overall, regardless of located status, 87 percent of the field test sample members who were NPSAS:20 survey

respondents completed the BPS:20/22 field test survey, compared with only 16 percent of NPSAS:20 survey nonrespondents ($p < .001$).

Table J-2 shows the located and field test survey response rates based on prior-response status as well as control and level of institution.

Table J-2. BPS:20/22 field test located and respondent status, by prior-round response status and control and level of institution: 2021

Prior-round response status and control and level of institution	Total sample	Eligible sample ¹	Located ²		Respondents ³		
			Number	Percent of total	Number	Percent of located	Percent of eligible
Total	3,700	3,520	3,090	83.4	2,240	72.5	63.7
NPSAS:20 (prior-round) response status							
Eligible survey respondents	2,440	2,440	2,160	88.6	1,890	87.3	77.4
Ineligible survey respondents ⁴	180	180	160	86.1	130	80.6	69.4
Survey nonrespondents	540	480	340	63.2	50	15.7	11.2
Administrative-only sample	540	420	430	79.0	170	40.3	41.5
Control of institution							
Public	2,190	2,080	1,850	84.5	1,320	71.3	63.4
Private nonprofit	850	830	750	87.5	620	83.0	74.3
Private for-profit	660	610	490	74.4	300	61.3	49.9
Level of institution							
Less-than-2-year	190	180	130	68.9	80	57.9	43.0
2-year	1,270	1,140	1,010	79.2	580	57.5	50.7
4-year, non-doctorate-granting	1,170	1,150	1,010	86.5	830	81.7	72.2
4-year, doctorate-granting	1,070	1,050	930	87.5	750	80.8	72.1
Control and level of institution							
Public less-than-2-year	50	50	40	78.0	30	69.2	56.3
Public 2-year	940	850	760	81.3	450	58.8	52.5
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	300	290	250	83.6	190	74.8	64.9
Public 4-year, non-doctorate-granting, primarily baccalaureate	240	240	220	90.8	180	84.3	77.2
Public 4-year, doctorate-granting	660	650	580	87.6	470	81.3	72.6
Private nonprofit 2-year or less	40	30	30	75.7	10	50.0	46.7
Private nonprofit 4-year, non-doctorate-granting	460	460	410	87.5	350	86.2	76.6
Private nonprofit 4-year, doctorate-granting	350	350	310	88.6	260	81.7	73.7
Private for-profit less-than-2-year	140	130	90	65.5	50	52.7	37.7
Private for-profit 2-year	300	260	220	73.1	120	54.1	45.4
Private for-profit 4-year	220	220	180	81.9	140	74.6	62.8

¹ Excludes 180 cases determined to be ineligible for the study, using data from one or more sources.

² Sample members were considered located if they were ever located at some point during data collection.

³ Respondents count includes sample members who met the criteria for qualification as a survey respondent, which required completing at least a partial survey.

⁴ Ineligible survey respondents are those ineligible for the 2019–20 National Postsecondary Student Aid Study (NPSAS:20) who self-identified as first-time beginning students between July 1, 2018, and April 30, 2019.

NOTE: Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

3.1.3 Contacting Efforts

Data collection announcement. At the start of data collection on March 2, 2021, sample members were sent a data collection announcement mailing followed by an e-mail on March 4, 2021, and a text message on March 9, 2021. The mailing included a study brochure and a letter that announced the start of data collection. The letter informed sample members of any incentive being offered for completing the survey, provided unique log-in information for the web survey, and included the study toll-free help desk number and e-mail address (see sections 3.1.6.1 and 4.1.2 for further information on incentives). The brochure provided information about the purpose of the study, confidentiality, security concerns, and contact information. The e-mail included equivalent information, along with a case-specific link that logged sample members directly into their survey when clicked. Additional reminders were sent to nonrespondents periodically throughout data collection, via e-mail, postcard and other mailings, and text message (see section 3.1.6 for details on reminder mode experiments conducted in the field test).

Contacting methods. The web survey was available 24 hours a day, 7 days a week throughout the entire data collection, which began on March 2, 2021, and ended on June 30, 2021. Although outbound telephone interviewing continued throughout the entire data collection, the e-mail, mailings, and text messages encouraged sample members to complete the web survey, particularly during the early response phase (see section 3.1.4 for further information on data collection phases).

Once outbound calling began, interviewers attempted to locate sample members, answered questions about the study, gained sample members' cooperation, and conducted interviews. When they successfully reached sample members, interviewers encouraged them to complete the interview over the telephone. Alternatively, interviewers could e-mail secure survey log-in credentials to sample members who preferred to participate on the Web. Interviewers followed up with sample members by telephone roughly 1 week after they selected the web option if they had not yet completed the survey.

The automated call scheduler assigned cases to interviewers by type and priority (e.g., best day and time to call, scheduled appointments, and based on the timing of prior calls). Scheduled appointments were specific dates and times, or a window of time, suggested by sample members when previously contacted. The scheduler organized cases into queues based on a variety of factors, including refusal status, prior number of calls, and appointments set during a prior contact

attempt. The scheduler also automatically ordered numbers to call by prioritizing those most likely to reach the sample member. New numbers were added continuously, based on telephone contacts and tracing efforts as well as updates received through mailings, e-mails, or telephone calls to the help desk. The call scheduler reprioritized telephone numbers based on new information as it came in.

3.1.4 Response Rates

The BPS:20/22 field test sample was divided into two groups that received different treatments based on prior-round (i.e., NPSAS:20 full-scale) response status. As described below, the protocols differed in terms of the amounts of incentives offered, how long certain incentives were available, and when prospective respondents were first approached by telephone. The data collection groups were defined as follows:

- Default protocol group: All NPSAS:20-eligible survey respondents and NPSAS:20-ineligible survey respondents received the default data collection protocol.
- Aggressive protocol group: All NPSAS:20 survey nonrespondents and NPSAS:20 survey administrative-only students received the aggressive data collection protocol.

3.1.4.1 Data collection phase and response rates

The BPS:20/22 field test included three distinct phases of data collection: early response phase, production phase I, and production phase II.

In the early response phase, there was no telephone prompting for the default protocol group, but telephone prompting began after the first 2 weeks for the aggressive protocol group. Light outbound telephone prompting began for the default protocol group in production phase I and regular telephone prompting continued for the aggressive protocol group. During production phase I all cases were part of a 3-week experiment in which a case either received text message reminders with no phone calls or received phone calls and no text message reminders. Finally, during production phase II the aggressive protocol group received a \$10 prepaid incentive.

Of the 2,150 completed surveys, over two-thirds of the BPS:20/22 field test respondents (68 percent) completed their survey during the early response phase. The remaining respondents completed their survey during production phase I or

production phase II (32 percent).¹ Approximately 70 percent of sample members who were NPSAS:20-eligible survey respondents completed the survey during the early response phase, as did 38 percent of sample members who were NPSAS:20 survey nonrespondents.

Table J-3 displays survey response rates by data collection phase, prior-response (i.e., NPSAS:20 full-scale study) status, and control and level of institution.

¹ Only the aggressive protocol group was included in production phase II.

Table J-3. BPS:20/22 field test completion rates, by data collection phase, prior-round response status, and control and level of institution: 2021

Prior-round response status and control and level of institution	Total eligible ¹	Total completes ²		Early response		Data collection phase Production phase I		Production phase II ³	
		Number	Percent of eligible	Number	Percent of completes	Number	Percent of completes	Number	Percent of completes
Total	3,520	2,150	61.1	1,460	68.0	650	30.3	40	1.7
NPSAS:20 (prior-round) response status									
Eligible survey respondents	2,440	1,820	74.8	1,270	69.5	560	30.5	†	†
Ineligible survey respondents ⁴	180	110	63.3	80	65.8	40	34.2	†	†
Survey nonrespondents	480	50	10.4	20	38.0	10	28.0	20	34.0
Administrative-only sample	420	160	38.6	100	62.1	40	26.1	20	11.8
Control of institution									
Public	2,080	1,260	60.5	850	67.4	390	30.8	20	1.8
Private nonprofit	830	610	72.6	430	70.2	170	28.8	10	1.0
Private for-profit	610	290	47.6	190	66.1	90	31.5	10	2.4
Level of institution									
Less-than-2-year	180	70	40.2	40	58.3	30	40.3	#	1.4
2-year	1,140	540	47.4	360	66.8	160	30.1	20	3.1
4-year, non-doctorate-granting	1,150	810	70.2	560	69.7	240	30.1	#	0.2
4-year, doctorate-granting	1,050	730	69.9	500	68.1	220	29.7	20	2.2
Control and level of institution									
Public less-than-2-year	50	30	52.1	10	56.0	10	40.0	#	4.0
Public 2-year	850	420	49.0	280	66.0	130	30.6	10	3.3
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	290	180	62.2	110	62.6	70	36.9	#	0.6
Public 4-year, non-doctorate-granting, primarily baccalaureate	240	180	74.7	130	72.3	50	27.7	†	†
Public 4-year, doctorate-granting	650	460	70.3	320	69.3	130	29.2	10	1.5
Private nonprofit 2-year or less	30	10	43.3	10	53.8	10	46.2	†	†
Private nonprofit 4-year, non-doctorate-granting	460	350	75.5	250	73.0	90	27.0	†	†
Private nonprofit 4-year, doctorate-granting	350	250	71.4	170	67.2	80	30.4	10	2.4
Private for-profit less-than-2-year	130	50	35.4	30	60.9	20	39.1	†	†
Private for-profit 2-year	260	110	42.7	80	70.5	30	26.8	#	2.7
Private for-profit 4-year	220	130	60.9	80	64.1	40	32.8	#	3.1

† Not applicable.

Rounds to zero.

¹ Excludes 180 cases determined to be ineligible for the study, using data from one or more sources.² Total completes count excludes 90 respondents who met the criteria as a student survey respondent but were partial completes. Partial survey respondents are defined as sample members who completed at least the enrollment section of the survey but did not complete the entire survey.³ Only the aggressive protocol group cases were included in production phase II.⁴ Ineligible survey respondents are those ineligible for the 2019–20 National Postsecondary Student Aid Study (NPSAS:20) who self-identified as first-time beginning students between July 1, 2018, and April 30, 2019.

NOTE: Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

3.1.4.2 Survey completeness

Overall, of the 2,240 field test survey respondents, 96 percent completed the entire survey, and 4 percent were classified as partial survey respondents. Partial survey completers were considered respondents if they at least completed the enrollment section of the survey.

Table J-4 provides BPS:20/22 field test survey completion rates, by prior-round response status and control and level of institution.

Table J-4. BPS:20/22 field test survey completion, by prior-round response status and control and level of institution: 2021

Prior-round response status and control and level of institution	Total respondents ¹	Survey completeness			
		Full survey		Partial survey ²	
		Number	Percent of total	Number	Percent of total
Total	2,240	2,150	96.0	90	4.0
NPSAS:20 (prior-round) response status					
Eligible survey respondents	1,890	1,820	96.7	60	3.3
Ineligible survey respondents ³	130	110	91.2	10	8.8
Survey nonrespondents	50	50	92.6	#	7.4
Administrative-only students	170	160	93.1	10	6.9
Control of institution					
Public	1,320	1,260	95.4	60	4.6
Private nonprofit	620	610	97.7	10	2.3
Private for-profit	300	290	95.4	10	4.6
Level of institution					
Less-than-2-year	80	70	93.5	10	6.5
2-year	580	540	93.4	40	6.6
4-year, non-doctorate-granting	830	810	97.2	20	2.8
4-year, doctorate-granting	750	730	96.9	20	3.1
Control and level of institution					
Public less-than-2-year	30	30	92.6	#	7.4
Public 2-year	450	420	93.3	30	6.7
Public 4-year, non-doctorate-granting, primarily subbaccalaureate	190	180	95.7	10	4.3
Public 4-year, non-doctorate-granting, primarily baccalaureate	180	180	96.7	10	3.3
Public 4-year, doctorate-granting	470	460	96.8	20	3.2
Private nonprofit 2-year or less	10	10	92.9	#	7.1
Private nonprofit 4-year, non-doctorate-granting	350	350	98.6	10	1.4
Private nonprofit 4-year, doctorate-granting	260	250	96.9	10	3.1
Private for-profit less-than-2-year	50	50	93.9	#	6.1
Private for-profit 2-year	120	110	94.1	10	5.9
Private for-profit 4-year	140	130	97.0	#	3.0

Rounds to zero.

¹ Total respondents count includes sample members who met the criteria for qualification as a survey respondent, which required completing at least a partial survey.

² Partial survey respondents are defined as sample members who completed at least the enrollment section of the survey but did not complete the entire survey.

³ Ineligible survey respondents are those ineligible for the 2019–20 National Postsecondary Student Aid Study (NPSAS:20) who self-identified as first-time beginning students between July 1, 2018, and April 30, 2019.

NOTE: Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

3.1.4.3 Mode of completion

The BPS:20/22 field test survey was available to complete by web and telephone. For analysis purposes, the self-administered web completion mode was automatically identified, differentiating surveys completed on nonmobile (e.g., desktop or laptop) or mobile (e.g., smartphone or tablet) devices.² By the end of data collection, 1,030 respondents (48 percent) completed the survey on a nonmobile device, and 1,040 respondents (48.3 percent) completed the survey on a mobile device. Eighty respondents (3.7 percent) completed the survey by telephone.

3.1.5 Refusal Conversion

For the purposes of data collection, a “refusal” occurred when a sample member refused to participate or when someone else refused on a sample member’s behalf (i.e., a gatekeeper). BPS:20/22 field test staff integrated refusal conversion techniques into interviewer training and reinforced these strategies throughout data collection in quality circle meetings. Interviewers shared their experiences avoiding refusals and sought guidance from the group about particularly difficult cases. In order to encourage conversion, sample members who ever refused, or had a gatekeeper refuse on their behalf, were only called by a subset of interviewers who were trained in specialized refusal conversion techniques.

Overall, 2 percent of eligible cases³ ever refused or had someone refuse on their behalf. Of those who ever had a refusal, 8 percent subsequently responded to the survey. One percent of field test sample members who were NPSAS:20 survey respondents were refusals in the BPS:20/22 field test, statistically significantly less than the 6 percent of refusals by sample members who were NPSAS:20 survey nonrespondents in the BPS:20/22 field test ($p < .001$). Fifty-nine BPS:20/22 field test sample members refused during NPSAS:20, and 12 (20.3 percent) responded to the BPS:20/22 field test survey.

3.1.6 Data Collection Experiments

Two data collection experiments were designed for the BPS:20/22 field test to investigate the effects of (1) an “early bird” incentive experiment in which respondents received an additional \$5 incentive if they completed the survey

² Two elements of instrument paradata identified web mobile mode: the rendering of the instrument (e.g., how the instrument displayed on a browser for a respondent) and parsing the browser agent string to obtain information about device type, browser, and touch-screen capability.

³ Excludes 180 cases determined to be ineligible for the study using data obtained from one or more sources.

within the first 3 weeks of data collection and (2) a survey reminder mode experiment in which text messages, instead of telephone calls, were exclusively used as reminders for a limited time during data collection.

The experiments were evaluated on three criteria: survey response, sample representativeness, and data collection efficiency. Survey response was evaluated for both experiments using response rates. Pearson chi-squared tests assessed whether the experimental treatments statistically significantly increased survey response. See section 3.1.4 for details on survey response rates.

Next, using administrative frame data, sample representativeness was assessed across age, sex, ethnicity, race, and control of institution (i.e., public institution, private nonprofit institution, and private for-profit institution). Two-sided t -tests were used to assess continuous respondent characteristics (i.e., age), and Pearson chi-squared tests were used to assess categorical respondent characteristics (i.e., percentages for sex, ethnicity, race, and institutional control). Nonresponse bias was also calculated (Groves et al. 2009) for each representativeness metric within each group (i.e., within control and experimental groups) as

$$\frac{m}{n}(\bar{y}_r - \bar{y}_m),$$

where m is the number of nonrespondents in a group, n is the total sample size for that group, $\bar{y}_r$ is the respondent mean for that group, and $\bar{y}_m$ is the nonrespondent mean for that group. This provides a measure of the magnitude of bias. To compare the magnitude of biases across the control and treatment group, the difference of absolute biases was calculated as

$$|Bias_T| - |Bias_C|,$$

where $|Bias_T|$ is the absolute value of nonresponse bias for the treatment group and $|Bias_C|$ is the absolute value of the control group.

Finally, data collection efficiency is operationalized as the number of days between the start of the experiment and survey completion. One-sided t -tests were used to explore whether the experimental treatments statistically significantly reduced the number of days from the start of the experiment it took respondents to complete the survey.⁴

⁴ Results use Satterthwaite (1946) approximation in difference-of-means tests with unequal variances.

Table J-5 summarizes these evaluation indicators, their operationalization, and the analytic approaches.

Table J-5. BPS:20/22 field test: Overview of indicators, operationalization, and analytic approaches for data collection experiments

Indicator	Operationalization	Analytic approach
Survey response	Response rates (eligible sample members only, including partial completions)	χ^2 test
Sample representativeness	Compare estimates for respondents with estimates for nonrespondents	t -test - continuous variables χ^2 test – categorical variables
Data collection efficiency	Number of days from start of experiment to survey completion	t -test

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

3.1.6.1 Experiment #1: Early bird incentives

Offering time-limited incentives for early responders (i.e., early bird incentives) can lead to faster responses and increased participation rates within a specified incentive period (e.g., Coopersmith et al. 2016; LeClere et al. 2012; Ward et al. 2014). To investigate the effects of an early bird incentive, BPS:20/22 field test sample members ($n = 3,520$) were randomly assigned to one of two groups: a control group ($n = 1,760$) that received the baseline incentive (\$30 for sample members receiving the default protocol; \$40 for those receiving the aggressive protocol) or an experimental group ($n = 1,760$) that received a \$5 early bird incentive in addition to the baseline incentive if they completed the survey within the first 3 weeks of data collection.

Results.

Survey response. Response rates were observed at two time points during data collection: at the end of the 3-week early bird period (3 weeks into data collection) and at the end of data collection (17 weeks into data collection).

Table J-6 summarizes the response rates for the control and the experimental early bird groups at these two time points.

Table J-6. BPS:20/22 field test response rates, by early bird experimental condition and evaluation period: 2021

Evaluation period	Control group ¹	Experimental group ²	χ^2	p-value
End of early bird period	37.2	42.6	10.53	< 0.01
End of data collection	63.4	63.9	0.08	0.78

¹ Received baseline incentive.

² Received baseline incentive plus early bird incentive.

NOTE: Results exclude cases determined to be ineligible for the study, using data from one or more sources.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

At the end of the 3-week early bird period, the response rate for the experimental early bird group (42.6 percent) was statistically significantly higher than that of the control group (37.2 percent; $p < .01$). However, at the end of data collection, response rates for the control group (63.4 percent) and the experimental group (63.9 percent) did not statistically significantly differ ($\chi^2 = 0.08$, $p = .78$).⁵

The results from this experiment therefore indicate that a \$5 early bird incentive may be effective at increasing response at the beginning of data collection, but these differences disappear as time in the field increases. This limits the utility of an early bird incentive for surveys with a long field period, like the BPS:20/22 full-scale survey.

Representativeness. Sample representativeness of respondents was investigated at the end of data collection.

Table J-7 displays the sample composition for each respondent characteristic by experimental condition.

⁵ Notably, these results diverge from past research that finds increased survey participation with early bird incentives (e.g., Coopersmith et al. 2016; LeClere et al. 2012). This difference may have occurred because studies from this past research offered larger early bird incentives (\$20 or more) than those offered by the BPS:20/22 field test (\$5).

Table J-7. BPS:20/22 field test sample composition, by early bird experimental condition at end of data collection: 2021

Sample member characteristic	Control group ¹	Experimental group ²	Difference in Bias Experimental - control
Age (mean)			
Respondents	21.6	21.5	
Nonrespondents	23.4	23.6	
Respondents – nonrespondents	-1.8***	-2.1***	
Nonresponse bias	-0.65	-0.76	0.11
Overall sample (<i>n</i> = 3,510) ³	22.3	22.2	
Female (in percent)			
Respondents	58	57.4	
Nonrespondents	56.1	56	
Respondents – nonrespondents	1.9	1.4	
Nonresponse bias	0.64	0.47	-0.18
Overall sample (<i>n</i> = 3,360) ³	57.3	56.9	
White⁴ (in percent)			
Respondents	70.2	71.2	
Nonrespondents	62.3	65.6	
Respondents – nonrespondents	7.9**	5.6*	
Nonresponse bias	2.61	1.81	-0.80
Overall sample (<i>n</i> = 3,180) ³	67.6	69.4	
Hispanic or Latino (in percent)			
Respondents	21.7	21.9	
Nonrespondents	23.3	23.1	
Respondents – nonrespondents	-1.6	-1.2	
Nonresponse bias	-0.53	-0.39	-0.14
Overall sample (<i>n</i> = 3,190) ³	22.2	22.2	
Control of institution (in percent)			
Public			
Respondents	57.9	59.7	
Nonrespondents	59.9	59	
Respondents – nonrespondents	-2	0.7	
Nonresponse bias	-0.73	0.25	-0.48
Overall sample (<i>n</i> = 3,520)	58.6	59.5	
Private nonprofit			
Respondents	28	27.3	
Nonrespondents	17	16.6	
Respondents – nonrespondents	11.0***	10.7***	
Nonresponse bias	4.02	3.86	-0.16
Overall sample (<i>n</i> = 3,520)	24	23.4	
Private for-profit			
Respondents	14.1	13	
Nonrespondents	23.2	24.4	
Respondents – nonrespondents	-9.1***	-11.4***	
Nonresponse bias	-3.33	-4.12	0.79
Overall sample (<i>n</i> = 3,520)	17.4	17.1	

* $p < .05$, ** $p < .01$, *** $p < .001$.¹ Received baseline incentive.² Received baseline incentive plus early bird incentive.³ Sample sizes for the overall sample differ due to missing data.⁴ "White" includes those who are and are not of Hispanic or Latino background. Hispanic or Latino is considered an ethnicity rather than a race. People of Hispanic or Latino origin may be of any race.

NOTE: Results exclude cases determined to be ineligible for the study, using data from one or more sources. Nonresponse bias is calculated as (respondent mean - nonrespondent mean) * (nonresponse rate).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

BPS:20/22 field test respondents were statistically significantly younger than nonrespondents in both the experimental early bird group ($p < 0.001$) and the control group ($p < 0.001$). This indicates the age of respondents and nonrespondents was similar in both the experimental early bird and control groups. The sex composition of respondents and nonrespondents was not statistically significantly different in either the early bird group ($\chi^2 = 0.32$, $p = 0.57$) or the control group ($\chi^2 = 0.57$, $p = 0.45$). These results indicate that the proportion of female respondents and nonrespondents in both groups was similar.

There were more White respondents than nonrespondents in both the early bird group ($p < .05$) and the control group ($p < 0.01$), but there was no statistically significant difference between the percentage of Hispanic or Latino respondents and nonrespondents in the early bird group ($\chi^2 = 0.31$, $p = 0.58$) or the control group ($\chi^2 = 0.54$, $p = 0.46$). This indicates the proportion of respondents and nonrespondents by race was consistent between the experimental and control groups.

There was no statistically significant difference between the percentage of respondents and nonrespondents from public institutions in either the early bird group ($\chi^2 = 0.09$, $p = 0.78$) or the control group ($\chi^2 = 0.67$, $p = 0.41$). There were more respondents than nonrespondents from private nonprofit institutions in both the early bird group ($p < .001$) and the control group ($p < 0.001$). In turn, there were fewer respondents than nonrespondents from private for-profit institutions in the early bird group ($p < .001$) and the control group ($p < 0.001$).

Nonresponse bias tended to be lower in the early bird group (last column of table J-7). Five of seven indicators had smaller biases with the early bird experimental group (ranging from 0.14 to 0.80). However, the magnitude of these biases were similar across the control and experimental groups for all indicators (average difference across the five indicators = 0.35).

Taken together, these results indicate that offering an early bird incentive encouraged response from the same types of sample members as did the baseline incentive alone.

Efficiency. Data collection efficiency was examined (i.e., average number of days it took respondents to complete the survey) at two time points during data collection: (1) at the end of the 3-week early bird period (3 weeks into data collection) and (2) at the end of data collection (17 weeks in the field).

Table J-8 summarizes the average number of days to survey completion for the control and early bird groups at these two time points.

Table J-8. BPS:20/22 field test average number of days to complete survey, by early bird experimental condition and evaluation period: 2021

Evaluation period	Control group ¹	Experimental group ²	<i>t</i>	<i>p</i> -value
End of early bird period	8.8	10.2	-4.02	1.00
End of data collection	30.9	28.1	2.09	0.02

¹ Received baseline incentive.² Received baseline incentive plus early bird incentive.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

At the end of the early bird period, the average number of days that it took respondents in the experimental early bird group to complete the survey (10.2 days) was not statistically significantly fewer than that of the control group (8.8 days; $t(1,400.7) = -4.02, p = 1.00$). At the end of data collection, respondents in the early bird group took statistically significantly fewer days (28.1 days) than respondents in the control group (30.9 days) to complete the survey ($p < 0.05$). However, this difference was small (2.8 days) and not long enough to allow for any cost savings in the data collection process (e.g., via fewer reminder calls, texts, or mailings).

Recommendations for full-scale data collection.

Offering sample members an early bird incentive did not statistically significantly improve response rates or representativeness by the end of data collection. Further, while offering an early bird incentive did bring in responses a few days sooner than offering the baseline incentive alone, this small gain in efficiency did not justify the cost of the early bird incentive. Therefore, the use of an early bird incentive in the BPS:20/22 full-scale data collection was not recommended. See section 4.1 for complete full-scale data collection design recommendations.

3.1.6.2 Experiment #2: Reminder modes

Text message advance notifications and reminders have been shown to statistically significantly increase response rates (e.g., Callegaro et al. 2011; Schober et al. 2015). Further, NPSAS:20 focus group ($n = 50$ overall; including $n = 20$ FTBs) results suggest that text messaging is the preferred mode of communication for most focus group participants (21 mentions). E-mails were mentioned as the preferred mode 11 times, and telephone six times. Text messaging was also used in NPSAS:20 to prompt sample members to complete the student survey. When compared with prompts by telephone, text message reminders have comparable rates of student survey completes and a higher absolute number of completes and are more cost efficient.

After 2 months of data collection, all nonresponding BPS:20/22 field test sample members ($n = 1,840$)⁶ were randomly assigned to one of two groups: a control group that received only telephone reminders ($n = 930$) and an experimental group that received only text message reminders ($n = 910$). Sample members received reminders in their assigned mode for 3 weeks. During this 3-week reminder period, all other data collection activities (e.g., reminder e-mails, hard copy mailings) continued for both groups.

Results.

Survey response. Response rates were examined at two time points during data collection: (1) at the end of the 3-week reminder period (i.e., 10 weeks in the field) and (2) at the end of data collection (i.e., 17 weeks in the field).

Table J-9 summarizes the response rates for the telephone reminder control group and the text message reminder experimental group at these two time points.

Table J-9. BPS:20/22 field test response rates, by reminder mode experimental condition and evaluation period: 2021

Evaluation period	Control group ¹	Experimental group ²	χ^2	<i>p</i> -value
End of reminder period	14.2	13.6	0.12	0.73
End of data collection	31.5	29.6	0.83	0.36

¹ Received telephone reminders.

² Received text message reminders.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

At the end of the 3-week reminder period, the response rate for the experimental text message group (13.6 percent) did not statistically significantly differ from that of the control group, which received only telephone reminders (14.2 percent; $\chi^2 = 0.12$, $p = 0.73$). Similarly, at the end of data collection, response rates for the control group (31.5 percent) and the experimental group (29.6 percent) did not statistically significantly differ ($\chi^2 = 0.83$, $p = 0.36$). These results indicate that the less expensive text messaging method of sending reminders was just as effective as telephone reminders at prompting survey response.

Representativeness. Respondent sample representativeness was investigated at the end of data collection.

Table J-10 displays the sample composition for each respondent characteristic, by experimental condition.

⁶ Excluding ineligible sample members.

Table J-10. BPS:20/22 field test sample composition, by reminder mode experimental condition at end of data collection: 2021

Sample member characteristic	Control group ¹	Experimental group ²	Difference in Bias Experimental - control
Age (mean)			
Respondents	21.4	21.8	
Nonrespondents	23.8	23.2	
Respondents – nonrespondents	-2.4***	-1.4***	
Nonresponse bias	-1.64	-0.98	-0.65
Overall sample ($n = 1,830$) ³	23	22.8	
Female (in percent)			
Respondents	55.2	53	
Nonrespondents	55.5	56.5	
Respondents – nonrespondents	-0.3	-3.5	
Nonresponse bias	-0.20	-2.38	2.18
Overall sample ($n = 1,690$) ³	55.4	55.4	
White⁴ (in percent)			
Respondents	74.1	65.3	
Nonrespondents	64.5	63.3	
Respondents – nonrespondents	9.6**	2	
Nonresponse bias	6.26	1.34	-4.92
Overall sample ($n = 1,570$) ³	67.8	64	
Hispanic or Latino (in percent)			
Respondents	21.3	23.6	
Nonrespondents	25.8	20.6	
Respondents – nonrespondents	-4.5	3	
Nonresponse bias	-2.97	2.01	-0.96
Overall sample ($n = 1,570$) ³	24.3	21.6	
Control of institution (in percent)			
Public			
Respondents	58.6	63.3	
Nonrespondents	59.9	58.9	
Respondents – nonrespondents	-1.3	4.4	
Nonresponse bias	-0.89	3.10	2.21
Overall sample ($n = 1,840$)	59.5	60.2	
Private nonprofit			
Respondents	26	23.3	
Nonrespondents	15.9	17.6	
Respondents – nonrespondents	10.1***	5.7*	
Nonresponse bias	6.92	4.01	-2.90
Overall sample ($n = 1,840$)	19.1	19.3	
Private for-profit			
Respondents	15.4	13.3	
Nonrespondents	24.1	23.5	
Respondents – nonrespondents	-8.7**	-10.2**	
Nonresponse bias	-5.96	-7.18	1.23
Overall sample ($n = 1,840$)	21.4	20.5	

* $p < .05$, ** $p < .01$, *** $p < .001$.¹ Received telephone reminders.² Received text message reminders.³ Sample sizes for the overall sample differ due to missing data.⁴ "White" includes those who are and are not of Hispanic or Latino background. Hispanic or Latino is considered an ethnicity rather than a race. People of Hispanic or Latino origin may be of any race.

NOTE: Results exclude cases determined to be ineligible for the study, using data from one or more sources. Bias is calculated as (respondent mean - nonrespondent mean) * (nonresponse rate).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

BPS:20/22 field test respondents were statistically significantly younger than nonrespondents in both the experimental text message reminder group ($p < 0.001$) and the control telephone reminder group ($p < 0.001$). This indicates that the age of respondents was similar in both the experimental and control groups. The sex composition of respondents and nonrespondents was not statistically significantly different in either the text message reminder group ($\chi^2 = 0.91, p = 0.34$) or the telephone reminder group ($\chi^2 = 0.01, p = 0.92$). These results indicate that the proportion of female respondents and nonrespondents in both groups was similar.

There were more White respondents than nonrespondents in the control group ($p < 0.01$). However, there was not a statistically significant difference between the percentage of White respondents and nonrespondents in the text message group ($\chi^2 = 0.28, p = 0.60$). Notably, the magnitude of nonresponse bias for the control condition (6.26) was much larger than the text message group (1.34; difference = 4.92). This indicates that the text message reminders were more effective than telephone reminders at encouraging participation from respondents of races other than White. There was not, however, a statistically significant difference between the percentage of Hispanic or Latino respondents and nonrespondents in the text group ($\chi^2 = 0.87, p = 0.35$) or the telephone group ($\chi^2 = 1.92, p = 0.17$).

There was no statistically significant difference between the percentage of respondents and nonrespondents from public institutions in either the text group ($\chi^2 = 1.53, p = 0.22$) or the telephone group ($\chi^2 = 0.16, p = 0.69$). There were more respondents than nonrespondents from private nonprofit institutions in both the text message reminder group ($p < .05$) and the telephone reminder group ($p < 0.001$). Conversely, there were fewer respondents than nonrespondents from private for-profit institutions in the text message group ($p < .01$) and the telephone group ($p < 0.01$).

Overall, four of the seven indicators examined here had lower nonresponse bias in the text messaging group versus the control group (last column of table J-10; average difference across the four indicators = 2.36). As noted above, for race, nonresponse bias was much higher for the control condition than the text message condition (difference = 4.92). Differences for other indicators were smaller (ranging from 0.65 to 2.90).

Taken together, these results indicate that generally, text message reminders encouraged response from the same types of sample members as did telephone reminders. However, text message reminders have the added benefit of recruiting more respondents of races other than White compared with telephone reminders.

Efficiency. Data collection efficiency was examined (i.e., average number of days it took respondents to complete the survey) at two time points during data collection: (1) at the end of the 3-week reminder period (i.e., 10 weeks in the field) and (2) at the end of data collection (i.e., 17 weeks in the field).

Table J-11 summarizes the average number of days to survey completion for the control and experimental groups at these two time points.

Table J-11. BPS:20/22 field test average number of days to complete survey, by reminder mode experimental condition and evaluation period: 2021

Evaluation period	Control group ¹	Experimental group ²	<i>t</i>	<i>p</i> -value
End of reminder period	52.7	48.4	1.99	0.02
End of data collection	77.0	75.5	0.62	0.27

¹ Received telephone reminders.

² Received text message reminders.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

At the end of the 3-week reminder period, respondents who received text message reminders completed the survey in 48.4 days on average, which was 4.3 days sooner than respondents who received telephone reminders (52.7 days; a statistically significant difference, $p < .05$). However, at the end of data collection, this difference disappeared. The number of days it took for respondents in the text message reminder group to complete the survey (75.5 days) was not statistically significantly different from that of the telephone reminder group (77.0 days; $t(542.8) = 0.62$, $p = 0.27$). These results demonstrate that text message reminders are just as efficient as telephone reminders across the entire field period and are more efficient early in data collection.

Recommendations for full-scale data collection.

Text message reminders achieved response rates, representativeness, and efficiency that was comparable to more expensive telephone reminders. Based on these results, the use of text reminders as a part of the BPS:20/22 full-scale data collection was recommended. However, there are still situations where telephone reminders would be useful (e.g., sample members who do not have a cell phone number, those who opt out of text message reminders). Therefore, a mix of telephone and text message reminders was recommended for the BPS:20/22 full-scale data collection. See section 4.1 for complete full-scale data collection design recommendations.

3.2 Evaluation of Survey

This section discusses the development of survey content and design of the BPS:20/22 field test survey. This section also presents various metrics related to the experience of taking the survey, such as the timing burden on respondents, instrument coders, item-level nonresponse, and reliability reinterviews. In addition, the results of two field test survey instrument experiments and two randomly assigned modules that collected information about the impacts of the coronavirus pandemic are covered.

3.2.1 Student Survey Design

The data elements included in the BPS:20/22 field test survey were determined with consideration of the base-year survey (NPSAS:20) and past BPS studies, while also incorporating new and modified data elements. Common BPS items related to enrollment history, education experiences, financial aid, employment history, income and expenses, and demographic characteristics were retained. New data elements were identified and developed with Technical Review Panel (TRP) member input and results from pretesting methods including cognitive interviewing and usability testing. For a description of the development process and results of pretesting methods, see section 3.2.2.

As with the NPSAS:20 survey, data elements pertaining to food security and an indicator of homelessness were added to the BPS:20/22 field test survey. These items were identified and developed during NPSAS:20 survey design with input from the U.S. Department of Agriculture and the Hope Center.⁷ Additionally, per TRP recommendation, items measuring aspects of housing security were added to the BPS:20/22 field test survey. These items were also identified and developed with input from the Hope Center.

The coronavirus pandemic created unprecedented disruptions for postsecondary education. To maximize opportunities to test questions about impacts of the coronavirus pandemic, BPS:20/22 field test respondents were randomly assigned into two groups that received one of two modules of questions. The first module included a series of revised questions from NPSAS:20 and the other included new items of interest determined during the field test TRP.

⁷ The Hope Center for College, Community, and Justice is an action research center focused on the basic needs of college students through applied scientific research, technical assistance to colleges and universities, policy advising with state and federal governments, and strategic communications. For more information, see: <https://hope4college.com>.

The survey consisted of seven key sections focused on specific content areas: enrollment, education experiences, financial aid, employment, income and expenses, background, and COVID. A brief overview of each area follows. A locating and incentives section collected information for future contacting purposes but no substantive data for public use.

Enrollment confirmed eligibility to participate in the BPS:20/22 field test for NPSAS:20 survey nonrespondents. The survey collected additional information on continued enrollment at the sampled institution (referred to hereafter as the NPSAS institution) as well as any additional postsecondary institutions attended during the 2020–21 academic year. This section also determined the primary institution of interest for each respondent. The primary institution was the current or most recent institution the respondent attended between July 1, 2020, and June 30, 2021. If respondents attended more than one current or most recent institution at the same time, they selected their primary institution. Questions throughout the remainder of the survey were anchored to experiences at respondents' primary institution.

Education experiences collected information on remedial education as well as grade point average and online-coursetaking at the primary institution. This section also captured information regarding academic and social experiences of college students. These items included interactions with family and friends, faculty, and other students; social and academic groups and clubs participated in; services used and the importance of those services; and expected occupation and wages.

Financial aid captured information about sources of aid used in the 2020–21 academic year, such as federal and private loans, grants or scholarships, employer assistance, and veterans' benefits. Information about cumulative private loan amounts and repayment plans were also collected in this section. Additionally, awareness and use of income-driven repayment plans and loan forgiveness programs as well as information about emergency aid were collected.

Finally, for the first time in BPS, this section collected information about receipt of state financial aid programs, for 11 select states,⁸ to take advantage of the state-representativeness of NPSAS:20 and answer questions such as how state aid affects retention, persistence, and attainment; how loss of state aid can contribute to academic hardships; the efficiency of merit- versus need-based aid; and the interaction between state and federal financial aid. Respondents whose NPSAS

⁸ The 11 states identified for state aid questions were Arizona, California, Florida, Georgia, Illinois, Michigan, New York, North Carolina, Ohio, Pennsylvania, and Texas.

institution was in one of the 11 states identified with a major financial aid program (e.g., HOPE scholarship in Georgia), were administered questions about receipt of the applicable state financial aid program in the 2019–20 and 2020–21 academic years. Respondents who indicated not receiving the state aid program in the 2020–21 academic year were then asked a follow-up question about reasons why they did not receive the state aid.

Employment items collected information about all work for pay during the 2020–2021 academic year. This included dates employed, hours worked while enrolled and while not enrolled, amount earned, and whether the job was on or off campus, for all full- and part-time employment, paid internships, and self-employment. The employment section also collected more detailed information about the current or most recent employer such as job title, if the job was related to respondents' major, and whether respondents had looked for work while not employed.

The **income and expenses** section of the BPS:20/22 field test survey collected information including respondent income, military status, information about dependent children and other dependents, parent information including parental income, marital status, living situation as of age 16, financial support for education and living expenses from family and friends, number of credit cards and balances, residence while attending school, mortgage and rent amount for those who did not live on campus, broad financial concept knowledge, and untaxed benefits received. For respondents with a spouse, information on income and education level were collected.

Background items obtained information about student demographic characteristics such as U.S. citizenship, immigration status, race and ethnicity, sex, sexual orientation, gender identity, and voting behavior. Measures of housing security, homelessness, and items from the U.S. Department of Agriculture 10-item food security module were also included in the background section.

Finally, the **COVID** section of the BPS:20/22 field test survey included items that pertained to student experiences during the coronavirus pandemic. This section collected information in two modules. The first module included questions administered in the base year and captured information about attendance, general experiences, refunds received, and institutional communication and information provided to students. Module two collected a new set of constructs, such as changes in enrollment and borrowing, changes in academic engagement, and access to support resources. Additionally, this section included items that collected information about respondents' access to the Internet.

3.2.2 Cognitive Interview Results

To evaluate quality, performance, and reliability of BPS:20/22 survey data, EurekaFacts, on behalf of RTI International and NCES, conducted cognitive and usability testing of a portion of the survey questions that are either new to this BPS cohort or have been revised from existing questions before their inclusion in the study.⁹ Insights from testing helped refine survey questions, maximize data quality, and provide information on issues with important implications for the overall survey design. Full details of the cognitive interview results were included in the approved field test Office of Management and Budget (OMB) package in December 2019, OMB# 1850-0631 v. 18.¹⁰

A sample of 30 students whose first postsecondary enrollment occurred between January 1, 2017, and June 30, 2019, participated in the cognitive testing. Further, students had to be attending their first postsecondary institution and working on their first postsecondary degree or certificate since completing high school during this time frame. The time frame included an additional academic year to increase the eligible participant pool for this round of cognitive testing. Table J-12 provides a breakdown of participant demographics.

⁹ Full details of the cognitive interview components were originally approved in December 2019, OMB# 1850-0803 v. 260 (revised and approved in April 2020, OMB# 1850-0803 v. 267). Refer to OMB# 1850-0803 v. 267 Attachment V for the BPS:20/22 cognitive survey instrument:

<https://www.reginfo.gov/public/do/DownloadDocument?objectID=100368601>.

¹⁰ For complete cognitive testing results, refer to OMB# 1850-0631 v.18 Appendix D:

<https://www.reginfo.gov/public/do/DownloadDocument?objectID=104083901>.

Table J-12. Percentage of BPS cognitive participants, by demographics: 2021

Demographic	Percent
Total participants	100.0
Gender	
Female	71.4
Male	28.6
Age	
18–24	89.3
25–29	7.1
30–49	3.6
Race	
Asian	7.1
Asian/White	3.6
Black or African American	28.6
White	53.6
Prefer not to answer	7.1
Hispanic/Latino origin	
Yes	14.3
No	82.1
Prefer not to answer	3.6
Income	
Less than \$20,000	53.6
\$20,000 to \$49,999	10.7
\$50,000 to \$99,999	7.1
\$100,000 or more	17.9
Prefer not to answer	10.7
Current education level	
Associate's degree	17.9
Bachelor's degree	82.1

NOTE: Percentages are based on unrounded numbers.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Cognitive Interviews.

Overall, cognitive interview participants considered the survey easy to answer and understand. A few participants reported on the burden involved in that the survey included a large amount of text that respondents were required to read, while others were concerned about the content of the survey and how data collected would be used. When asked to provide feedback about specific items in the survey, the most common topics discussed included the collection of enrollment intensity, main institution and main employer, education experiences, emergency aid, and housing security. The following sections provide a summary of cognitive feedback and any subsequent field test changes for each of these topics.

3.2.2.1 Enrollment intensity

Participants were asked to provide their enrollment intensity (i.e., full time or part time) in two ways: a single forced-choice question with options of (1) full time, (2) part time, (3) mix of both full- and part-time enrollment, and (4) not enrolled (B22ASTST2, B22AOTSTST201) or two yes/no radio button gate questions that

asked about full- and part-time enrollment separately (B22ANENRLFT, B22ANENRLPT, B22AOTENRLFT01, B22AOTENRLPT01). Many students mentioned they were generally able to comprehend and provide enrollment information across both versions of enrollment intensity collection, and there was no clear consensus on a preferred approach. Due to this feedback, both versions (i.e., the single forced-choice question and the two yes/no radio button questions) were included in the field test survey. Analysis of this experiment can be found in section 3.2.8.

3.2.2.2 Main institution and main employer

BPS surveys collect additional data elements about a student's primary school enrollment ("main institution") and reference employment ("main employer"). Historically, the BPS survey has calculated and selected the main institution and employer for respondents. Cognitive interview participants with multiple postsecondary school enrollments and/or multiple employers were asked to identify their main school and/or main employer, to gather feedback on a survey respondent's process of identifying this information and compare the respondent selection to a calculated selection. However, an insufficient number of cognitive interview participants fell into either of these categories; thus, the field test survey included two debriefing forms after both the respondent selection of primary school (B22APRSCHDB1, B22APRSCHDB2) and reference employer (B22DREFPKLTDB, B22DREFPKLTDB2) to collect further information about the primary institution and employer identification.

3.2.2.3 Education experiences

Cognitive feedback from the education experiences section resulted in clarification of language for two field test survey questions. First, participants experienced accuracy issues when reporting how often they communicate with five different groups of people (B22BFAMCOMM), primarily due to "siblings and extended family" as a single response option and unclear instructions about the frequency scale. Given this feedback, the field test survey separated "siblings" and "extended family" into two response options, and additional language was added to clarify how to report frequency of communication with the question's scale. Additionally, cognitive testing results indicated that participants were unfamiliar with, and not able to accurately define, a "learning community" when asked to report on their participation in academic activities (B22BACDPART). This feedback illustrated the importance of providing a clearer description of a "learning community," and this response option was modified for the field test survey to provide further clarification.

3.2.2.4 Emergency aid

Next, cognitive interview participants were asked to report on whether they were aware of emergency aid programs (B22CEAIDAWARE). Given the low rate of awareness of these programs, the field test survey included additional emergency aid questions to observe respondent experiences with emergency aid (B22CEAIDAPPLY, B22CEAIDRCV, B22CEAIDFINASST, B22CEAIDLN, B22CEAIDPAY).

3.2.2.5 Housing security

Most cognitive interview participants had no difficulty answering the new set of housing security questions that were included the field test survey, except for the question that required respondents to report on the number of times they moved (B22FNUMMOVE). Some participants had difficulty understanding what should be included in a “move.” Given the importance of this data point in the identification of housing-insecure students, the field test question wording was modified to provide additional clarification on types of “moves” to include. Additionally, the field test included a follow-up question for those who reported three or more moves in the year (B22FNMOVEFU01). The data resulting from the field test was used to inform the full-scale collection of this information.

3.2.3 Timing Burden

To calculate the time required to complete the survey, the survey instrument recorded the elapsed time respondents took to complete each form or survey screen. The summed form-level timing values resulted in section times and total survey times. Respondents could complete the survey in multiple sessions. When beginning a new session to continue the survey, respondents began on the form they last saw in their prior session. Among cases that completed the survey in multiple sessions, the timing for the last form the respondent saw before the breakoff could not be measured, therefore, it was imputed to the median time other respondents spent on the same form. All significance testing reported in this section is statistically significant at the $p < .0001$ level, unless otherwise indicated.

Of the 2,240 field test surveys completed, 1,470 (66 percent) are included in the timing report (Table J-13). About 770 cases (34 percent) were excluded from analyses, including partial cases, cases that completed in multiple sessions

(i.e., cases with more than two forms requiring mean time imputation¹¹), and total time outliers.¹²

Table J-13. Number and percentage of BPS:20/22 field test surveys included and excluded from timing report: 2021

Survey type	Number of cases	Percent
Total surveys (including partials)	2,240	100.0
Total completed surveys included in timing report	1,470	65.8
Surveys excluded from timing report	770	34.2
Completed in multiple sessions ¹	640	28.4
Total time outliers ²	40	1.9
Partial surveys	90	4.0

¹ Interviews completed in more than one session were included in the timing analysis when possible. However, if a respondent completed the survey in more than two sessions, typically more than two forms would be missing timing information; thus, the case was excluded from analysis.

² Outliers were calculated by normalizing the data and excluding extreme values identified by Tukey's formula (1977) which is not sensitive to distributional assumptions. Timing cases outside of the interquartile range (interquartile range = 3rd quartile - 1st quartile) multiplied by 1.5 were identified as total time outliers.

NOTE: Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Student Longitudinal Study (BPS:20/22) Field Test.

BPS:20/22 field test survey respondents had the opportunity to complete the survey by computer-assisted telephone survey (CATI) and self-administered web survey. Self-administered web completion was separated by surveys completed on a mobile device (e.g., smartphone or tablet), referred to hereafter as “web mobile” mode, and surveys completed on a nonmobile device, referred to hereafter as “web nonmobile” mode. Overall, the field test survey took 33.7 minutes on average to complete. Web nonmobile surveys were completed in an average of 33.6 minutes. Web mobile surveys took statistically significantly less time with an average of 31 minutes.¹³ Telephone surveys took an average of 60.9 minutes to complete, statistically significantly the highest timing burden of all modes of administration, compared with both web nonmobile and web mobile surveys. Data collection interviewers read survey questions aloud to respondents for telephone mode, therefore, surveys administered by telephone often yielded longer completion times.

¹¹ To avoid introducing excessive imputation and uncertainty into the timing estimates, cases that required more than two form imputations were excluded.

¹² To detect total time outliers, the distribution of all survey times (highly right-skewed) was first normalized using a Box-Cox power transformation (Box and Cox 1964). Cases were then removed from the overall total using an interquartile range formula adopted from Tukey (1977) with a multiplier of 1.5. Cases were excluded as outlier if total time > 75th percentile + (1.5 * interquartile range), or if total time < 25th percentile – (1.5 * interquartile range).

¹³ Results use Satterthwaite (1946) approximation in difference-of-means tests with unequal variances.

The BPS:20/22 field test is the first follow-up of a cohort of beginning postsecondary students who entered postsecondary education for the first time in the 2018–19 academic year. The first attempt to contact these respondents occurred in the 2019–20 academic year with the NPSAS:20 student survey (i.e., base year). The BPS:20/22 field test sample consisted of two groups: NPSAS:20 nonrespondents (survey nonrespondents and administrative-only students) and NPSAS:20 respondents. In the BPS:20/22 field test survey, NPSAS:20 nonrespondents received more questions about their enrollment during the 2018–19 academic year to determine eligibility for the survey, while NPSAS:20 respondents received questions pertaining only to continued or additional enrollment in the 2020–21 academic year (i.e., year 3).

Table J-14 provides a breakdown of the average time to complete the BPS:20/22 field test survey by prior-response status and mode of administration.

Table J-14. Average time, in minutes, to complete BPS:20/22 field test, by mode of administration and prior-response status: 2021

Prior-response status	All respondents			Mode of administration ¹								
				Web nonmobile			Web mobile			Telephone		
	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time
Overall	1,470	33.73	30.58	670	33.61	30.91	730	31.01	28.57	80	60.89	63.14
NPSAS:20 nonrespondent	170	39.69	35.91	60	41.06	38.33	90	35.43	32.30	20	58.32	51.18
NPSAS:20 respondent	1,310	32.98	29.98	600	32.82	30.64	640	30.42	27.88	60	61.53	63.86

¹ Mode of administration is the mode in which the respondent completed the student survey; this mode may be different than starting mode for respondents who completed the survey in more than one session.

NOTE: NPSAS:20 = 2019–20 National Postsecondary Student Aid Study. Analysis includes only completed cases that have two or fewer forms of imputed timing data. Respondents who completed the survey in multiple sessions, total time outliers, and respondents who did not complete the entire survey (partial surveys) were excluded. Sample sizes rounded to the nearest 10. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

As anticipated, NPSAS:20 nonrespondents took an average of 39.7 minutes to complete the BPS:20/22 field test survey, statistically significantly longer than NPSAS:20 respondents, who averaged 33 minutes. NPSAS:20 respondents took statistically significantly longer in telephone mode (61.5 minutes) than in both web nonmobile mode (32.8 minutes) and web mobile mode (30.4 minutes). The survey was also statistically significantly longer in web nonmobile mode than web mobile mode for NPSAS:20 respondents ($p < .001$). Significance testing by mode of administration was not conducted for NPSAS:20 nonrespondents due to insufficient cell sizes.

The BPS:20/22 field test survey contained eight sections: (1) enrollment, (2) education experiences, (3) financial aid, (4) employment, (5) income and expenses, (6) background, (7) COVID, and (8) locating and incentives. Table J-15 provides the average timing for the survey, by survey section and mode of administration.

Table J-15. Average time, in minutes, to complete BPS:20/22 field test, by mode of administration and survey section: 2021

Survey section	All respondents			Mode of administration ¹								
				Web nonmobile			Web mobile			Telephone		
	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time	Number of cases	Mean time	Median time
Total	1,470	33.73	30.58	670	33.61	30.91	730	31.01	28.57	80	60.89	63.14
Enrollment	1,470	4.95	4.11	670	5.01	4.14	730	4.59	3.86	80	7.83	7.34
Education experiences	1,470	6.87	6.25	670	7.31	6.69	730	6.09	5.80	80	10.57	12.59
Financial aid	1,470	1.29	1.29	670	1.72	1.35	730	1.48	1.15	80	3.12	2.98
Employment	1,470	3.73	3.01	670	3.38	2.91	730	3.67	2.97	80	7.30	7.52
Income and expenses	1,470	4.90	4.09	670	4.92	4.11	730	4.51	3.87	80	8.40	8.37
Background	1,470	3.51	2.90	670	3.21	2.83	730	3.30	2.81	80	8.05	7.78
COVID ²	1,310	3.92	3.53	630	3.99	3.66	610	3.45	3.28	60	7.68	9.59
Locating and incentives	1,470	4.03	3.29	670	3.69	3.02	730	3.92	3.33	80	8.01	7.76

¹ Mode of administration is the mode in which the respondent completed the student survey; this mode may be different than starting mode for respondents who completed the survey in more than one session.

² Respondents without a primary institution were not administered the COVID module.

NOTE: Analysis includes only completed cases that have two or fewer forms of imputed timing data. Respondents who completed the survey in multiple sessions, total time outliers, and respondents who did not complete the entire survey (partial surveys) were excluded. Sample sizes rounded to the nearest 10. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

Overall, the education experiences section, which collected social and academic experiences at the main institution in which respondents were enrolled during year 3, took respondents the longest time to complete: 6.9 minutes on average. Telephone respondents took an average of 10.6 minutes to complete the education experiences section, statistically significantly longer than both web nonmobile respondents (7.3 minutes) and web mobile respondents (6.1 minutes). Similarly, web nonmobile respondents took statistically significantly longer than web mobile respondents to complete the survey.

Coder forms were associated with an underlying database that provided standardized responses to an open-ended response from the respondent. The BPS:20/22 field test survey used predictive coders. The predictive coder generated and updated a list of potential results as the respondent entered a search word. The BPS:20/22 field test used predictive search coders for certain standardized data elements such as major or field of study, occupation, postsecondary institutions, and addresses. Coders represented some of the longest

overall form times in the survey. Coders required respondents or the telephone interviewer to interact with the predictive text search functionality of the form, which made it a more involved process than simply typing in a response or selecting a response option on noncoder forms (i.e., forms not linked to an underlying database of standardized responses). The following analyses detail the average time to administer each individual substantive form within the survey—excluding the front end, locating, and incentive sections. The form-level timing analysis separated coders from other form types.

Table J-16 provides the average form-level timing for all coders and the top 10 longest noncoder forms.

Table J-16. Mean and median timing of forms with highest average administration time, by form and item type: 2021

Form name	Form label	Type	Number	Mean (seconds)	Median (seconds)
Coder form					
B22BEXOCC	Expected occupation coder	Coder	1,400	68.40	48.39
B22DOCC	Reference employer: occupation coder	Coder	270	61.11	42.26
B22AOTMAJ101 ¹	Other attendance 1: major 1 coder	Coder	250	41.70	24.45
B22AMAJ1 ¹	Major 1 coder for base-year NPSAS degree/certificate	Coder	280	39.41	26.56
B22AMAJ2 ¹	Major 2 coder for base-year NPSAS degree/certificate	Coder	60	31.77	21.61
B22AOMJ1A	Original major coder for base-year NPSAS degree/certificate	Coder	330	30.29	18.79
B22AOTSCH01 ¹	Other attendance 1: school coder	Coder	180	28.58	18.59
B22AOTOMAJ01 ¹	Other attendance 1: original major coder	Coder	40	26.89	19.05
Noncoder form					
B22FCVTHINK	COVID module 2: social or academic experiences at primary institution due to COVID	Grid	490	42.31	33.54
B22FCOVCOMM	COVID module 1: communication received from primary institution due to COVID	Grid	550	41.55	33.25
B22BACDPART	Participation in academic activities at primary institution	Grid	1,120	40.66	30.85
B22FNMOVEFU01 ¹	Four or more moves follow-up	Radio	100	37.86	28.81
B22BFAMCOMM	Frequency of communication with family and friends	Grid	1,120	35.63	27.56
B22ANENRLFT2	Full-time months of attendance at NPSAS in year 3	Calendar	860	34.85	26.57
B22FCVEXP2	COVID module 2: general experiences at primary institution due to COVID	Checkbox	490	34.20	29.19
B22BSRVUSE	Used school services at primary institution	Grid	1,120	34.06	26.70
B22EINCSP	Spouse's income from all sources in previous calendar year	Radio	50	34.02	14.87
B22FCOVEXPA	COVID module 1: experiences or disruptions while attending primary institution due to COVID	Checkbox	550	33.62	26.05

¹ This form may be administered multiple times to each respondent, so each respondent's time value is the average of all administrations. That is, it is the number of seconds to complete all iterations of the form divided by the number of iterations.

NOTE: Analysis includes only completed cases that have two or fewer forms of imputed timing data. Respondents who completed the survey in multiple sessions, total time outliers, and respondents who did not complete the entire survey (partial surveys) were excluded. For individual form time calculations, forms from the front end, locating and incentive sections, and forms administered to less than 25 respondents were excluded from the analysis. Sample sizes rounded to the nearest 10.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

The *Expected occupation coder* (B22BEXOCC) had the longest average form time with an average of 68.4 seconds. This form asked respondents to code their own expected occupation using the Occupational Information Network Online database (O*NET-SOC v 24.0). It may be cognitively burdensome and therefore take time for some respondents to think ahead to completing their education.

Reference employer: occupation coder (B22DOCC), which also utilized the O*NET-SOC database, had the next longest average coder time with an average of 61.1 seconds. The shorter time to report the reference employer occupation is likely due to the reporting of actual events; for example, providing a current or most recent occupation is less burdensome than providing an expected future occupation.

The noncoder forms that took the longest to answer represented some of the most cognitively burdensome questions in the survey. These forms either relied on respondents' recall of detailed information or took longer to read due to the form type; five of these longest noncoder forms were in a grid format, which displays several items and scaled response options in a grid. The noncoder form with the highest timing burden was *COVID module 2: social and academic experiences at primary institution due to COVID* (B22FCVTHINK). It took respondents an average of 42.3 seconds to provide a response to a grid of seven academic experiences they had between July 1, 2020, and December 31, 2020, which is a departure from the majority of BPS survey questions with a time frame of the entire 2020–21 academic year. Additionally, for *COVID module 1: communication received from primary institution due to COVID-19* (B22FCOVCOMM), respondents took an average of 41.6 seconds to respond to a grid of five communication topics received from their primary institution during their enrollment from July 1, 2020, through December 31, 2020. This subset time period likely introduced additional burden on respondents, reflected in the fact that four of the ten longest noncoder forms were from the COVID section that collected information from this subset time frame. The third highest noncoder form time was for *Participation in academic activities at primary institution* (B22ACDPART). Respondents took an average of 40.7 seconds to select all academic activities they participated in during their enrollment at their primary institution between July 1, 2020, and June 30, 2021, from a grid of seven activities.

3.2.4 Coder Usability

Coder usability rates (e.g., successful rate of coding by respondents in survey) were reported by coder type. Success rates for each coding system were calculated by dividing the number of responses where a complete code was

selected, by either the respondent or a telephone interviewer in the survey, by the total number of times the coder was administered. Analysis of successful coding rates in the survey was limited to cases with a final complete or final partial complete status ($n = 2,240$). Overall, respondents successfully coded their response 86 percent of the time when administered a coder (Table J-17).

All coder systems in the survey, except for the current/most recent occupation coder, had a success rate of 77 percent or above. By coding system, coding success rates ranged from 98 percent overall success with the major coder to 68 percent overall success with the current/most recent occupation coder. Overall coding success rates by mode of administration ranged from 93 percent for telephone mode, 90 percent for web nonmobile mode, and 82 percent for web mobile mode.

Significance tests were conducted for each of the four coders to determine statistically significant differences in coding success rates between modes of administration. Results are statistically significant at the $p < .05$ level unless otherwise noted. Rates of successful coding were statistically significantly higher in web nonmobile mode than in web mobile mode for every coder system: postsecondary institution, major, expected occupation, and current/most recent occupation. These differences could be attributed to coder forms being more difficult to navigate on small mobile screens. Additionally, coding success rates were statistically significantly higher in web nonmobile mode than in telephone mode for the major coder. However, coding success rates were statistically significantly higher in telephone mode than in web mobile mode for two out of the four coding systems: expected occupation and current/most recent occupation. Although experienced telephone interviewers administered the survey, respondents may have provided text strings that did not produce desired results when coding, making it difficult to choose a final code.

Table J-17 shows a summary of coding success rate, by mode of administration and coding system.

Table J-17. Percentage of responses coded in survey, by mode of administration and coding system: 2021

Coding system	Mode of administration ¹			
	Overall	Web nonmobile	Web mobile	Telephone
Total	85.5	90.3	81.8	92.8
Major ²	98.3	99.1	97.4	95.7
Postsecondary institution ²	97.8	99.1	96.6	100.0
Occupation: expected occupation	77.9	84.0	71.6	83.0
Occupation: current/most recent occupation	68.1	79.1	61.5	92.6

¹ Mode of administration is the mode in which the respondent completed the student survey, this mode may be different than starting mode for respondents who completed the survey in more than one session.

² When a form is administered multiple times to a respondent (e.g., the postsecondary institution coder is administered for every additional attendance in the 2020–21 academic year) the coder system success rate is calculated using the total number of administrations across respondents (i.e., the number of successful coding attempts of the form divided by the number of administrations).

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

3.2.5 Help Text

Respondents and telephone interviewers could click a help button provided on each form to obtain form-specific help text. Additionally, some questions provided embedded hyperlinks within question wording or response option text to prompt respondents to access the help text on the form. Whether accessed through the help button or through an embedded hyperlink, the help text provided definitions of key terms and phrases used in question wording and response options as well as any explanations to help clarify and standardize interpretation for respondents.

Help text access rates were calculated by dividing the number of times help text was accessed by the number of times forms were administered. Overall, respondents and telephone interviewers accessed help text less than 1 percent of the time. The rate of help text access was also analyzed at the form level, overall, and by mode of administration to identify specific forms that may have been difficult for users. Form-level help text access rates were calculated by dividing the number of times help text was accessed on each form by the number of times that form was administered. Table J-18 shows the 10 forms administered to at least 10 percent of respondents for which help text was accessed at a rate of 1 percent or greater, by mode of administration.

Aware of emergency aid program at primary institution (B22CEAIDAWARE) was administered to 1,670 respondents and had the highest rate of help text access of 7 percent. This form was included in the NPSAS:20 student survey in which it also had the highest help text access rate. Results from BPS:20/22 cognitive testing indicated that students typically have little awareness regarding emergency aid programs (see section 3.2.2). The term “emergency aid program” was

displayed with an embedded help text hyperlink that defined these programs. The lower familiarity with these programs and the embedded hyperlink likely contributed to the higher help text access rate.

Table J-18. Forms with at least 1 percent help text access rate, by mode of administration: 2021

Form	Form label	Overall		Web nonmobile		Web mobile		Telephone	
		Number administered to	Percent of help text access	Number administered to	Percent of help text access	Number administered to	Percent of help text access	Number administered to	Percent of help text access
B22CEAIDAWARE	Aware of emergency aid program at primary institution	1,670	7.0	910	9.8	700	2.6	50	15.4
B22DWRKSTD01	Employer 1: work-study job	290	5.9	180	9.1	110	0.9	10	0.0
B22ECARRYBAL	Credit card amount carried over each month	1,130	4.1	540	5.6	550	2.5	40	7.0
B22AOTCLSDG01	Other attendance 1: courses to fulfill degree requirements or transfer credit to degree or certificate program	220	3.6	120	5.0	100	2.0	#	0.0
B22ASTDABR	Studied abroad while at NPSAS for base-year degree or certificate	1,360	2.1	780	1.7	540	2.6	40	5.0
B22ADBLMAJ	Declared major (or double major) for base-year degree or certificate at NPSAS	1,390	1.9	790	2.0	560	1.6	40	2.4
B22DEDBENFTS	Importance of benefits	2,150	1.7	1,030	1.5	1,040	1.9	80	1.3
B22CPRVLN	Took out private loans in 2020–21 academic year	700	1.1	380	0.8	300	1.0	20	11.8
B22DEMPLOY01	Employer 1: employer name	2,210	1.0	1,090	0.8	1,030	0.9	90	5.6
B22CEVRRCVLN	Ever took out any student loans	1,200	1.0	560	1.6	580	0.3	60	1.8

Rounds to zero.

NOTE: NPSAS = National Postsecondary Student Aid Study. Mode of administration is the mode in which the respondent completed the student survey; this mode may be different than starting mode for respondents who completed the survey in more than one session. This table only includes those items that were administered to at least 10 percent of respondents. Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals due to rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

3.2.6 Survey Item Nonresponse Rates

An item in the survey is a single response option (e.g., check box) or set of response options (e.g., radio button list) that will become a variable in survey data files. A form in the survey can include more than one item. The rate of nonresponse for individual items in the survey is used to identify potentially burdensome survey questions and to better understand the experiences of respondents completing the survey. Item-level nonresponse rates in the BPS:20/22 field test student survey were calculated for all items administered to at least 25 respondents; items with a nonresponse rate of 10 percent or more are reported here. Of the 500 items administered to more than 25 respondents, only 3 percent (13 items) had a nonresponse rate of 10 percent or more. Table J-19 shows item nonresponse rates for these 13 items with 10 percent or more of data missing, overall and by mode of administration, ordered by item nonresponse.

Table J-19. Item nonresponse for items with more than 10 percent of data missing: 2021

Item	Item label	Overall		Web nonmobile		Web mobile		Telephone	
		Number administered to	Percent missing	Number administered to	Percent missing	Number administered to	Percent missing	Number administered to	Percent missing
B22FLGBTQ_DK	Sexual orientation: don't know: please describe	50	60.4	20	40.9	30	74.2	†	†
B22DREFPKLTDB2	Main reason selected reference employer	90	44.6	20	47.8	60	46.9	10	0.0
B22FGENDERQR_OTHER	Gender identity: genderqueer or gender nonconforming: please describe	60	19.3	30	14.8	30	20.7	#	100.0
B22EKIDCOL	Number of dependents in college in 2020–21 academic year	100	10.2	20	8.3	70	11.8	10	0.0
B22FCOVEXPJOB	General experiences during COVID-19: lost a job or lost income because of reduced hours	870	10.1	460	10.3	380	10.6	30	0.0
B22FCOVEXPHRS	General experiences during COVID-19: worked more due to additional job/increased hours	870	10.1	460	10.3	380	10.6	30	0.0
B22FCOVEXPFOOD	General experiences during COVID-19: had difficulty accessing food or paying for food	870	10.1	460	10.3	380	10.6	30	0.0
B22FCOVEXPCC	General experiences during COVID-19: had difficulty finding safe and stable child care	870	10.1	460	10.3	380	10.6	30	0.0
B22FCOVEXPHM	General experiences during COVID-19: had difficulty finding safe and stable housing	870	10.1	460	10.3	380	10.6	30	0.0
B22FCOVEXPPAD	General experiences during COVID-19: moved back to my permanent address	870	10.1	460	10.3	380	10.6	30	0.0
B22FCOVEXPMOT	General experiences during COVID-19: moved to another living situation	870	10.1	460	10.3	380	10.6	30	0.0
B22FCOVEXPOTH	General experiences during COVID-19: another experience not listed	870	10.1	460	10.3	380	10.6	30	0.0
B22FCOVEXPNONEB	General experiences during COVID-19: experienced none of the above	870	10.1	460	10.3	380	10.6	30	0.0

Rounds to zero.

† Not applicable.

NOTE: Mode of administration is the mode in which the respondent completed the student survey; this mode may be different than starting mode for respondents who completed the survey in more than one session. This table only includes those items that were administered to at least 25 respondents. Sample sizes rounded to the nearest 10. Percentages are based on unrounded numbers. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

The item with the highest overall nonresponse rate was *Sexual orientation: don't know: please describe* (B22FLGBTQ_DK). This item was administered to 50 respondents and had an overall nonresponse rate of 60 percent. This item was administered to respondents who selected “don't know” when asked about their sexual orientation. When “don't know” was selected, the textbox item appeared underneath the response option to allow respondents to provide further explanation. The high rate of nonresponse may be due to the sensitive nature of the question.

Main reason selected reference employer (B22DREFPKLTDB2) had the next highest overall nonresponse rate. In the employment section, respondents who worked for pay in the 2020–21 academic year and reported more than one employer were first asked to indicate all the reasons (e.g., currently employed, longest duration, related to degree) they selected their reference employer as their main employer. If respondents indicated more than one reason, they were administered this item, which displayed all the reasons previously selected in a radio button list for respondents to choose one main reason for their selection. This item was administered to 90 respondents and had an overall nonresponse rate of 45 percent. If no displayed options captured why a respondent selected their main employer, they had the option to select the seventh item, “some other reason.” Self-administered web respondents who felt no displayed reasons applied but did not select the seventh item contributed to the 45 percent nonresponse rate. Telephone interviewers were trained to select “some other reason” if a respondent indicated that none of the reasons listed applied to them.

In the COVID section, *General experiences during COVID-19* (B22FCOVEXP) asked respondents to report if they experienced any of the listed experiences between 2020 and 2021 due to the coronavirus pandemic. These experiences included hardships such as losing a job or reducing hours, difficulty accessing food, and whether the respondent moved or had any difficulty finding housing. This question had eight experience items that were administered to 870 respondents and had a nonresponse rate of 10 percent. If a respondent did not experience any of the listed items, they had the option to select the ninth item, “experienced none of the above” (B22FCOVEXPNONEB), which is also considered a valid response. The 10 percent nonresponse rate may be due to respondents for whom none of the eight experiences applied but who did not select B22FCOVEXPNONEB. This is evidenced by the fact that the nonresponse rate was driven solely by self-administered web respondents, as telephone interviewers are trained to select B22FCOVEXPNONEB if a respondent indicated not experiencing any of the mentioned items.

Item-level nonresponse rates were also examined by mode of administration for the 13 items with 10 percent or more missing data.¹⁴ *Sexual orientation: don't know please describe* (B22FLGBTQ_DK) had statistically significant higher nonresponse rates in web mobile mode (74 percent) than web nonmobile mode (41 percent). The higher nonresponse rate in web mobile mode could be a result of difficulty navigating this form on a small screen when the textbox entry appeared. Ten items had statistically significant higher nonresponse rates in web nonmobile mode than telephone mode: *Main reason selected reference employer* (B22DREFPKLTDB2, $p < .001$) and all nine *General experiences during COVID-19* (B22FCOVEXP) items ($p < .0001$). Finally, those same 10 items had statistically significant higher nonresponse rates in web mobile mode than telephone mode ($p < .0001$), along with *Number of dependents in college in 2020-21 academic year* (B22EKIDCOL) ($p < .01$). Telephone interviewers were trained to utilize survey tools, such as help text and conversion text, which likely led to the lower nonresponse rates for these items in telephone mode.

3.2.7 Reinterview Analysis

To assess the reliability of self-reported survey responses gathered during the BPS:20/22 field test, all respondents who completed the survey were presented with an opportunity to participate in a reinterview survey. The reinterview consisted of 30 items selected from 19 questions from the following BPS:20/22 field test survey sections: enrollment, education experiences, financial aid, employment, income and expenses, and background. The items selected for the reinterview had a high degree of temporal stability, meaning that responses from respondents should remain constant over time.

Each respondent who expressed interest in participating in the reinterview was contacted approximately 3 weeks after completing the survey and asked to participate in a brief reliability reinterview. The follow-up contacting letter for potential reinterview respondents prompted them to complete the reinterview in the same mode in which they completed the survey, either via self-administered web (desktop, laptop, tablet, or smartphone), or by telephone with an interviewer. Twenty-eight percent of reinterview respondents completed the reinterview in a different mode than their main survey. With the exception of one respondent who completed the main survey by telephone and the reinterview by self-administered web, all mode changes were between the self-administered modes.

¹⁴ Results use Satterthwaite (1946) approximation in difference-of-means tests with unequal variances.

Of the 2,150 reinterview-eligible respondents who completed the survey, 1,740 agreed to participate (81 percent), and of those, 520 (30 percent) completed the reinterview. On average, the reinterview took 4.5 minutes to complete. As shown in Table J-20, most respondents (60 percent) completed the reinterview using a desktop or laptop (i.e., web nonmobile mode).

Table J-20. BPS:20/22 field test reinterview responses, by mode of administration: 2021

Mode of administration	Number of cases	Percent
Total	520	100.0
Web nonmobile	310	59.9
Web mobile	200	38.7
Telephone	10	1.3

NOTE: Sample sizes rounded to the nearest 10. Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

Administration of reinterview survey questions aligned with the main survey administration. If the respondent was administered the question in the main survey, it was also administered in the reinterview.¹⁵ For items with categorical response options, such as “yes” or “no,” agreement was defined as when the reinterview response was identical to the main survey response. For items with a continuous value or ordinal categories response options (i.e., dollar amounts or degree type), agreement was defined as the reinterview value falling within one standard deviation (plus or minus) of the survey value. When applicable, outliers were removed for continuous response items before calculating agreement rates.

In addition to percent agreement, a relational statistic quantifies the strength of association between the pairs of items with a value ranging between 0 and 1. The closer the statistic is to 1, the stronger the association is between the survey response and the reinterview response (i.e., a relational statistic of 1.00 indicates perfect association with no difference in responses). Cramer’s V was used as the relational statistic for items with discrete, unordered response categories. The Pearson product-moment correlation coefficient (r) was used for items yielding continuous responses or ordinal responses.

¹⁵ For example, respondents who reported living on campus (B22ESCHRES) were not administered the question about owning a home or paying a mortgage in the main survey (B22EHOME). The reinterview did not include B22ESCHRES; however, that response was preserved to incorporate appropriate routing to R22EHOME in the reinterview. If a respondent reported living on campus in the main survey (B22ESCHRES = 1), they were not administered R22EHOME in the reinterview.

Table J-21 provides the survey items, percent agreement, and relational statistic for all items administered in the reinterview, sorted by percent agreement.

Table J-21. BPS:20/22 field test reliability estimates: 2021

Item	Item label	Number of cases	Percent agreement	Relational statistic
B22FVTINELG	Voted in 2020 presidential election: not eligible	480	99.4	!
B22FVTNEL	Voted in 2020 presidential election	480	97.9	0.94
B22EHOME	Owned home or paid mortgage on a home	430	97.7	0.54
B22BGPATYP	Grade point average at primary school measured on 4.00 scale	450	97.5	0.66
B22CRCVLN	Took out any student loans in 2020–21 academic year	450	96.0	0.92
B22ASAMESCH	Continued attending NPSAS after 2018–19 academic year	480	95.8	0.86
B22EDEPS	Any children financially supported in 2020–21 academic year	190	95.7	0.84
B22CSTRCVBY	Received state aid program in 2019–20 academic year	200	95.4	0.90
B22DANYJOB	Worked for pay in 2020–21 academic year	520	94.1	0.85
B22BPARTAPP	Academic activities participated in primary school: apprenticeship	440	92.6	!
B22EINCOM	Income from all sources in 2020	520	92.1	0.82 ¹
B22EREGSUPP	Regularly gave friends/family more than \$50/month	520	90.2	0.52
B22BNPATND	Would begin education at NPSAS if started over	520	89.7	0.71
B22EHHWHO	Living situation at age 16	520	89.7	0.69
B22EDEP2	Number of children financially supported in 2020–21 academic year	30	88.5	0.68 ¹
B22CWRKSTDY	Held a work-study job in 2020–21 academic year	450	86.3	0.55
B22BPARTINTRN	Academic activities participated in primary school: internship/co-op/field experience	450	83.8	0.62
B22ENUMCRD	Number of credit cards in own name	520	83.5	0.78 ¹
B22BFUTWAGES ²	Most likely beginning salary	480	82.3	0.60 ¹
B22AEXPEVR	Highest level of education expected at any postsecondary institution	520	78.6	0.64 ¹
B22BPARTCOMM	Academic activities participated in primary school: community/service	450	78.3	0.43
B22BPARTCRDT	Academic activities participated in primary school: student success	450	75.8	0.40
B22BPARTPEER	Academic activities participated in primary school: learning community	440	75.4	0.35
B22DFAMTIM	Importance of benefits: balancing work and family	520	72.2	0.44
B22DHLPTH	Importance of benefits: helping others as part of job	520	71.8	0.56
B22DLEISTIME	Importance of benefits: balancing work and leisure	520	69.0	0.41
B22DEXPFLD	Importance of benefits: seen as an expert in field	520	65.3	0.48
B22DWRKTSKS	Importance of benefits: making own decisions about how to get work done	520	61.4	0.38

! The relational statistic for this item cannot be calculated; 75 percent of cells in the response contingency table have less than five respondents (e.g., majority of respondents indicated "no" in both main survey and reinterview).

¹ The relational statistic presented is Pearson's product-moment correlation coefficient r .

² This item had a non-normal distribution of response values in the main and reinterview survey, so values were excluded as outliers before calculating agreement and relational statistics. Outliers were calculated by normalizing the data and excluding extreme values identified by Tukey's formula (1977) which is not sensitive to distributional assumptions (when the response was $> 75\text{th percentile} + (1.5 * \text{interquartile range})$, or when the response was $< 25\text{th percentile} - (1.5 * \text{interquartile range})$).

NOTE: NPSAS = National Postsecondary Student Aid Study. The relational statistic presented is Cramer's V unless otherwise noted. Sample sizes rounded to the nearest 10.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

The reinterview items tested showed strong reliability across all sections included in the reinterview analysis, indicating that the survey yielded data of high quality with consistently reliable results. All reinterview items in which a relational statistic could be calculated had a range of 0.38 to 0.94, indicating these items had a medium to high level of association across the main survey and reinterview. Of the 30 individual items examined, 21 produced agreement rates at or above 80 percent, and 13 produced agreement rates at or above 90 percent. Only nine items examined produced an agreement rate below 80 percent (ranging from 61 to 78 percent). The mean agreement rate across all items included in the reinterview was 86 percent. Two of the three questions new to the BPS:20/22 survey that were included in the reinterview produced high reliability: *Received state aid program in 2019–20 academic year* (B22CSTRCVBY) (95 percent) and *Living situation at age 16* (B22EHHWHO) (90 percent). The other new question, *Academic activities participated in primary school* (B22BPART*) had a mean agreement rate of 81 percent across all items in the set.

3.2.8 Tests of Survey Design

Two field test survey instrument experiments tested different methods of collecting month-level enrollment intensity and address information to identify which method could reduce burden and result in higher quality data. In addition, two randomly assigned modules that collected information about the impacts of the coronavirus pandemic were fielded. The results of this testing informed which survey questions were best suited for inclusion in the full-scale study. All analyses in this section use Satterthwaite (1946) approximation of the degree of freedom used in difference-of-means tests with unequal variances.

3.2.8.1 Collection of month-level enrollment intensity

Enrollment intensity (i.e., full- or part-time enrollment) is an important measure for BPS:20/22 due to its role in financial aid eligibility and other analyses of student persistence and credential attainment. The BPS:20/22 field test survey randomly assigned respondents into one of two methods of collecting enrollment information to determine which method increased the reliability of self-reported enrollment information. The first method, comparable to past BPS surveys (control group), used a forced-choice grid question that collected enrollment intensity for each academic year (B22ASTST2) on a single form, with options of (1) full time, (2) part time, (3) mix of full and part time, and (4) no enrollment. The treatment group received the second method, in which respondents were administered separate questions for full- and part-time enrollment intensity for specified academic years (B22ANENRLFT, B22ANENRLPT).

Following the control or experimental gates, both groups were administered the custom calendar survey questions, which displayed individual month buttons by academic year. Respondents could choose a “select all” option to select all months in the academic year or select months individually. This burden concern is particularly important given the BPS:20/22 full-scale survey collects information about 2 years of academic enrollment (24 months) rather than 1 academic year (12 months) as in the field test. Following the administration of each version, specific months of enrollment were collected for the applicable enrollment intensity.

Of the 2,240 BPS:20/22 field test respondents, 1,320 were included in this analysis (59 percent).¹⁶ Of the analysis cases, 680 respondents were in the control group (51 percent), and 640 respondents were in the treatment group (49 percent).

There was no statistically significant difference in missingness between the treatment and control group. No respondents assigned to the control group left the enrollment intensity gate missing, and less than 1 percent of respondents left either of the yes/no radio gates missing in the treatment group ($t(636) = 1.42$, $p = 0.1575$). Across both treatment and control groups, answering the enrollment intensity gates took 13.7 seconds, on average. The control group took an average of 10.5 seconds to answer the single forced-choice grid question. The treatment group took statistically significantly longer, at an average of 17.2 seconds to answer both the separate yes/no radio gates by enrollment intensity ($p < .0001$). For the full-scale survey, respondents were asked to report their enrollment for 2 academic years, which increased the size of the forced-choice grid (i.e., the grid had 2 years instead of 1). However, the anticipated burden increase of the forced-choice grid was unlikely to exceed that of the separate yes/no radio gates by enrollment intensity.

To determine whether the enrollment intensity gate questions impacted the concordance of enrollment data collected, field test respondents’ NPSAS base-year degree enrollment reported in the survey data were compared with

¹⁶ Field test respondents could be excluded for multiple reasons; therefore, the exclusion categorization was prioritized according to the following order: (1) if respondents indicated not continuing enrollment at NPSAS and were not administered enrollment intensity questions for their NPSAS base-year degree ($n = 790$), (2) respondents whose NPSAS institution did not report to the National Student Clearinghouse ($n = 90$), and (3) respondents identified as timing outliers for the NPSAS enrollment intensity gates ($n = 50$). Outliers were calculated by normalizing the data and excluding extreme values identified by Tukey’s formula (1977), which is not sensitive to distributional assumptions. Respondents with timing burden for enrollment intensity gates outside of the interquartile range (interquartile range = 3rd quartile - 1st quartile) multiplied by 1.5 were identified as timing outliers.

administrative enrollment data from the National Student Clearinghouse (NSC). Enrollment was compared for the months of July 2020 through December 2020, a retrospective period before BPS:20/22 field test data collection. The survey response was considered to be in concordance with administrative data from NSC if the respondent either (1) reported no enrollment for that intensity during the 6 months of interest and there were no enrollment records in NSC for the respondent attending the NPSAS institution during that same time frame or (2) reported enrollment for that intensity during the 6 months of interest and NSC had a record of the respondent attending the NPSAS institution for that same intensity at any point in those same 6 months of interest.

Differences in concordance between the treatment and control group were not statistically significant. For both experimental groups, 70 percent of respondents were in concordance with NSC enrollment records during the time frame of interest regardless of intensity ($t(1311.1) = 0.24, p = 0.8119$). Full-time enrollment had the highest concordance, the control group had an 89 percent concordance rate, and the treatment group an 87 percent concordance rate ($t(1292.8) = 1.04, p = 0.2991$). See Table J-22 for concordance rates overall and by enrollment intensity.

Table J-22. BPS:20/22 field test month-level enrollment intensity concordance with National Student Clearinghouse, overall and by intensity: 2021

Group	Number of cases	Concordance rates		
		Overall ¹	Full time	Part time
Overall agreement	1,320	70.0	88.1	77.9
Control: single forced-choice grid	680	70.1	89.0	77.6
Treatment: yes/no radio gates by intensity	640	70.7	87.1	78.4

¹ Overall concordance between field test response and National Student Clearinghouse administrative records is defined as the percentage of respondents for which full- and part-time enrollment agreed across both sources. Because a respondent could have agreement for one enrollment intensity and not another, totaling full- and part-time agreement rates will not sum to total of overall agreement rate.

NOTE: Concordance rates were determined by comparing the same 6-month time frame between field test response and National Student Clearinghouse enrollment records for which enrollment could be determined (e.g., enrollment that had occurred prior to the field test survey data collection period), from July 2020 to December 2020. Sample sizes rounded to nearest 10.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test.

Overall, the method of collecting enrollment intensity did not impact missingness rates or reporting agreement between sources, as determined by administrative concordance. However, administering the separate intensity yes/no radio questions did statistically significantly increase respondent burden by an average of 6.7 seconds (64 percent increase), though the timing difference may be reduced in the full-scale survey when the grid form includes 2 academic years of enrollment instead of 1. Given the field test results, data will continue to be

collected using the single forced-choice grid in the BPS:20/22 full-scale survey instrument.

3.2.8.2 Collection of address information using predictive search database

Predictive search forms have become commonplace in web-based data collections. The collection of respondent addresses is critical for survey incentive payments and future locating efforts. Therefore, the BPS:20/22 field test implemented a predictive search method of obtaining address information to increase data quality and lower respondent burden typically associated with manual entries. The BPS:20/22 field test collected addresses for the entire responding field test sample using a predictive search format, with each address entry as a single textbox linked to an underlying Experian database of U.S. addresses standardized to U.S. Postal Service guidelines. The address textbox entry allowed results of matching addresses to populate while the respondent entered an address. The respondent then selected the best match from the list of results provided, or if no matching result was found, the respondent could manually enter in each address field. To assess the feasibility and benefits of the predictive search address coder, timing and missingness of the BPS:20/22 predictive search method was compared with the traditional collection method. Data from respondents who provided address data in both the BPS:20/22 field test (predictive method) and the NPSAS:20 full-scale survey (traditional method) was compared.

Of the 2,240 BPS:20/22 field test respondents, 1,620 respondents were included in this analysis (72 percent).¹⁷ The results described below are from the first address collection form in the survey, respondent permanent address (N20G1ADR, B22G1ADR), which was administered to all 1,620 respondents included in the analysis.

Completion rates were compared across each method of address collection to determine feasibility and benefits of utilizing a predictive search database.

¹⁷ Field test respondents could be excluded for multiple reasons; therefore, the exclusion categorization was prioritized according to the following order: if respondents (1) did not respond to the NPSAS:20 student survey ($n = 340$), (2) were permanent address timing outliers in either study ($n = 250$), (3) completed NPSAS:20 in Spanish ($n = 20$), (4) were partial survey respondents in either study ($n = 10$), and (5) reported a foreign address in either study ($n = 10$). Respondents were considered time outliers for the permanent address collection from either study (N20G1ADR, B22G1ADR) or respondents for which timing data were unavailable. Outliers were calculated by normalizing the data and excluding extreme values identified by Tukey's formula (1977) which is not sensitive to distributional assumptions. Respondents with timing burden for enrollment intensity gates outside of the interquartile range (interquartile range = 3rd quartile - 1st quartile) multiplied by 1.5 were identified as timing outliers.

Overall, 98 percent of respondents provided a complete permanent address in NPSAS:20 compared with 85 percent in the BPS:20/22 field test ($p < .0001$). NPSAS:20 consistently yielded a higher rate of complete addresses across all modes of administration (see Table J-23) and when controlling for mode change across studies. Telephone mode had the highest completion rate for both studies.

Table J-23. Permanent address completion rate by study, overall and by mode of administration: 2019–21

Study	Overall		Mode of administration					
			Web nonmobile		Web mobile		Telephone	
	N	Percent	N	Percent	N	Percent	N	Percent
NPSAS:20 full-scale	1,620	98.4	800	98.5	780	98.2	50	100.0
BPS:20/22 field test	1,620	85.5	830	89.1	740	81.1	50	91.7

NOTE: Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test and 2019–20 National Postsecondary Student Aid Study (NPSAS:20).

Table J-24 provides results of timing burden overall and by mode of administration across studies. Overall, it took respondents an average of 29.5 seconds to respond to the permanent address form in the NPSAS:20 full-scale study, statistically significantly higher than the 27.1 seconds it took respondents in the BPS:20/22 field test ($p < .0001$). The most impactful burden saving was for those who completed the survey over the telephone. Interviewers spent 18 seconds longer, on average, administering the permanent address in the NPSAS:20 full-scale study compared with the BPS:20/22 field test ($p < .0001$). It is important to note that the cumulative reduction of burden resulting from the Experian coder was dependent on the number of addresses collected. While all respondents were administered the permanent address form, respondents could be administered up to six address forms to collect information for parents and guardians and another friend or family member for future study contacting purposes.

Table J-24. Permanent address timing burden by study, overall and by mode of administration: 2019–21

Study	Overall		Mode of administration					
			Web nonmobile		Web mobile		Telephone	
	Mean time N (in seconds)		Mean time N (in seconds)		Mean time N (in seconds)		Mean time N (in seconds)	
NPSAS:20 full-scale	1,620	29.5	800	23.4	780	33.4	50	72.3
BPS:20/22 field test	1,620	27.1	830	24.2	740	28.5	50	54.1

NOTE: Detail may not sum to totals because of rounding.

SOURCE: U.S. Department of Education, National Center for Education Statistics, 2020/22 Beginning Postsecondary Students Longitudinal Study (BPS:20/22) Field Test and 2019–20 National Postsecondary Student Aid Study (NPSAS:20).

As a result of implementing the Experian coder to collect respondents' permanent address, collection of this address required less time for BPS:20/22 field test survey respondents compared with NPSAS:20 full-scale survey respondents. However, with the latter, a higher rate of complete addresses (i.e., all elements were provided, including street address, city, state, and zip code) was obtained. While more complete addresses were obtained without the use of the Experian database in NPSAS:20, this does not indicate all entries were error free and valid. However, by definition, all addresses obtained from the BPS:20/22 field test Experian database were complete and accurate (i.e., a true verifiable address), reducing the labor costs associated with data collection staff resolving "undeliverable" check addresses (i.e., check sent back as not a real address). For NPSAS:20, around 1.2 percent of FTB check addresses were considered "undeliverable," and the BPS:20/22 field test did not collect any "undeliverable" addresses.

Survey instrument functionalities can improve the completion rate for the BPS:20/22 full-scale study by implementing conversion text and soft check validations when address fields are left incomplete. These functionalities were not incorporated in the BPS:20/22 field test in order to analyze initial respondent behavior with this new coder; survey instrumentation results from past studies indicate these are effective methods of improving nonresponse or partial response. Given the implementation of improved instrument functionality to reduce incomplete address entries, higher quality data resulting from Experian verifiable addresses, and the statistically significant reduction in burden using Experian compared with manual entry, the continuation of use of the Experian database is proposed for the BPS:20/22 full-scale study.

3.2.8.3 Evaluation of coronavirus pandemic items

The coronavirus pandemic has created unprecedented disruptions for postsecondary education. BPS:20/22 full-scale sample members were FTBs throughout the 2019–20 academic year, when the pandemic began. Researchers, including members of the BPS:20/22 TRP, expressed interest in using BPS:20/22 data to examine impacts of the coronavirus pandemic on postsecondary students. As described in section 3.2.1, BPS:20/22 field test respondents were randomly assigned into two groups that received one of two modules asking about the coronavirus pandemic. The questions in module 1 included revised questions from the NPSAS:20 full-scale survey¹⁸ and collected information about attendance, general experiences, refunds received, and institutional

¹⁸ See Appendix C of NPSAS:20 Data File Documentation:
<https://nces.ed.gov/pubs2024/2024482.pdf>.

communication and information provided to students. Module 2 collected data on a new set of constructs identified during the BPS:20/22 field test TRP that may be of analytic value to researchers and policymakers, such as changes in enrollment and borrowing, changes in academic engagement, and access to support resources.¹⁹

This randomized assignment of modules allowed a greater number of survey questions to be fielded, providing more data to inform full-scale decisions while reducing burden to individual field test respondents. Questions were assessed using common measures such as response timing, item missingness, and response patterns (e.g., contradictory responses or responses lacking variation).

Of the cases eligible for analysis, respondents were equally split between the random module assignment: 50.4 percent in the module 1 group and 49.7 percent in the module 2 group.²⁰

Regardless of module assignment, the coronavirus pandemic questions took respondents an average of 2.7 minutes to complete, just under the allotted goal of 3 minutes. Respondents administered the coronavirus module by a telephone interviewer took an average of 6.2 minutes, statistically significantly longer than web nonmobile and web mobile modes (2.7 minutes and 2.5 minutes ($p < .0001$), respectively). The most burdensome questions administered were large Likert scale grids, questions that contained multiple response options within a grid with scaled response options (e.g., “strongly agree” to “strongly disagree”). For those respondents assigned to module 1, the six-item Likert grid *Helpful communication from primary school* (B22FCOVCOMM) took respondents an average of 42 seconds to answer. Likewise, the seven-item Likert grid for respondents assigned to module two was *Social/academic experiences at primary school during*

¹⁹ The wording for these questions can be found in appendix E of the BPS:20/22 field test package (OMB #1850-0631 v.18).

²⁰ Analyses of coronavirus pandemic questions were limited to 1,180 respondents (53 percent of all field test respondents). Field test respondents could be excluded for multiple reasons; therefore, the exclusion categorization was prioritized according to the following order: respondents who (1) did not complete the entire survey ($n = 90$), (2) did not report attending their primary institution in the 2020–21 academic year in the enrollment section of the survey ($n = 420$) (this is because most coronavirus pandemic questions were administered to respondents who attended their primary institution between July 2020 and December 2020), (3) had no timing data available due to completing the survey in more than one session ($n = 530$), and (4) identified as total timing outliers ($n = 20$). To detect total time outliers, the distribution of all survey times (highly right-skewed) was first normalized using a Box-Cox power transformation (Box and Cox 1964). Cases were then removed from the overall total using an interquartile range formula adopted from Tukey (1977) with a multiplier of 1.5. Cases were excluded as outlier if total time > 75 th percentile + (1.5 * interquartile range) or if total time < 25 th percentile – (1.5 * interquartile range).

COVID-19 (B22FCVTHINK) and also took respondents an average of 42 seconds to answer.

Module 1 had an average nonresponse rate of 3 percent, statistically significantly higher than the 0.6 percent nonresponse rate of module 2 ($p < .0001$). For module 1, the question that increased overall nonresponse rate was *General experiences during COVID-19* (B22FCOVEXPB), for which 10 percent of respondents left all items on this checkbox list missing, likely due to the design of the form. If a respondent did not select any of the individual experiences, nor opted to select “none of the above,” it was considered a nonresponse. Comparatively, for module 2, no single item on any of the questions administered had a higher nonresponse rate than 1 percent.

To analyze the response patterns of the coronavirus pandemic questions, the rate of contradictory responses and straightlining (i.e., selecting the same scaled response for the entire set of response options on a grid) was examined.

To determine contradictory responses, the rate at which field test respondents disagreed regarding their primary institution attendance during the coronavirus pandemic was analyzed. Disagreement is defined as respondents indicating they attended their primary institution between July 2020 and December 2020 in the enrollment section of the main survey but indicating not having attended the primary institution in the coronavirus pandemic module during this same time frame. Respondents administered module 2 had a statistically significant higher rate of disagreement at 9 percent, compared with just 2 percent for module 1 ($p < .0001$). Much of this difference in disagreement can be attributed to design differences between the modules because module 1 only had one question related to respondents’ enrollment at primary institution (*Attended primary school during COVID-19* [B22FCOVATND]), whereas module 2 had two opportunities to report enrollment information (*Attended any postsecondary institution during COVID-19* [B22FCVATND2] and *Attended primary school during COVID-19* [B22FCVATNDPS]). For module 2, if respondents indicated not attending any postsecondary institution during the time frame of interest, they were not administered the follow-up specific to their primary institution (and would thus be considered in disagreement had they reported attendance at the primary institution in the enrollment section of the survey).

The rate of straightlining was calculated for the largest grid-format question administered to each module group. Straightlining in and of itself does not necessarily indicate poor question performance, as some lack of variation in response is likely valid (e.g., selecting “strongly disagree” for all if respondents

had negative experiences during the coronavirus pandemic). However, comparison across similarly constructed questions can indicate respondent fatigue or lack of comprehension. For module 1, the response patterns of the six-item Likert grid *Helpful communication from primary school* (B22FCOVCOMM) were analyzed. For module 2, the response patterns of the seven-item Likert grid *Social/academic experiences at primary school during COVID-19* (B22FCVTHINK) were analyzed. The rate of straightlining for respondents in module 1 was 20 percent, statistically significantly higher than the 12 percent for module 2 ($p < .001$).

Based upon field test results and TRP recommendations, a subset of items from both modules will be administered in the BPS:20/22 full-scale survey as a single module, with a burden goal of 3 minutes. In general, response timing, item missingness, and response patterns indicate that both modules performed well in the field test. The most burdensome grid with the highest straightlining observed in the field test, *Helpful communication from primary school* (B22FCOVCOMM), is not recommended for the full-scale survey. The two-step enrollment questions from module 2 will be included in the full-scale survey to collect information about whether the coronavirus pandemic impacted respondents' decision to not enroll at all or not enroll at their primary institution. The contradictory response rate was higher for this version in the field test, though the contradictory response rate is anticipated to decline, as the full-scale survey will ask about enrollment for an entire academic year rather than a single term. Additionally, the full-scale survey will include a validation to compare the response from the enrollment section to improve accuracy for these coronavirus pandemic gate questions.

Section 4. Recommendations for BPS:20/22 Full-Scale Study

The BPS:20/22 field test was conducted to test and evaluate study methods and results before implementing the full-scale study. This section includes recommendations for the full-scale study based on field test results.

4.1 Recommendations for Data Collection Design

The data collection design proposed for the BPS:20/22 full-scale study builds on the designs implemented in past studies such as the 2012/14 Beginning Postsecondary Students Longitudinal Study (BPS:12/14), NPSAS:20, and Baccalaureate and Beyond Longitudinal Study (B&B). Additionally, results from the BPS:20/22 field test data collection experiments (see section 3.1.6) inform the BPS:20/22 full-scale data collection design.

A primary goal of the full-scale design is to minimize the potential for nonresponse bias that could be introduced into BPS:20/22, especially bias that could be due to lower response rates among NPSAS:20 nonrespondents. Another important goal is to reduce the amount of time for and cost of data collection efforts.

4.1.1 *Define Protocols by Prior-Round Response Status*

The BPS:20/22 full-scale data collection is planned to achieve at least a 70 percent response rate. Doing so will minimize potential nonresponse bias, optimize statistical power, and enable subgroup analyses. BPS:20/22 full-scale staff plan to divide the sample into two groups and implement differential data collection treatments based on prior-round response status. A similar approach was successfully implemented in the BPS:20/22 field test and the latest B&B studies in which more reluctant sample members received a more aggressive protocol.²¹

²¹ For an experimental comparison see the 2016/17 Baccalaureate and Beyond Longitudinal Study field test: <https://nces.ed.gov/pubsearch/pubsinfo.asp?pubid=2020441>.

For the BPS:20/22 full-scale design, the following groupings will be used:

- **NPSAS:20 survey respondents:** Sample members who responded to NPSAS:20 and self-identified that they began their postsecondary education between July 1, 2019, and April 30, 2020, will receive a default data collection protocol ($n = 26,500$).
- **NPSAS:20 survey nonrespondents and administrative-only students:** NPSAS:20 administrative respondents who failed to respond to NPSAS:20 and are potential 2019–20 academic year FTBs based on administrative data (i.e., NPSAS:20 survey nonrespondents; $n = 5,500$) will receive an aggressive data collection protocol. For NPSAS:20 survey nonrespondents, the goal of this treatment is to convert reluctant sample members as early in data collection as possible, based on prior evidence that the default data collection protocol is not as successful for this group. The aggressive protocol will also apply to NPSAS:20 administrative respondent students who were part of the NPSAS:20 administrative-only sample and are potential 2019–20 academic year FTBs based on administrative data and who were never invited to complete the NPSAS:20 survey (referred to as administrative-only students; $n = 5,350$). As administrative-only students likely have no prior experience with NPAS-related studies, the goal of this treatment is to establish trust and legitimacy, two key drivers of survey participation (Dillman, Smyth, and Christian 2014).

Table J-25 presents the type and timing of interventions to be applied in data collection by group assignment and data collection phase. The details of these interventions are described in the following sections.

Table J-25. BPS:20/22 full-scale data collection protocols, by group assignment and data collection phase

	Data collection group assignments	
	Default protocol	Aggressive protocol
Sample	NPSAS:20 survey respondents (26,500)	- NPSAS:20 survey nonrespondents (5,500) - NPSAS:20 administrative-only students(5,350)
Data collection protocol		
Prior to data collection	- Panel maintenance - Prenotification/greeting (from NCES)	- Panel maintenance - Prenotification/greeting (from NCES)
Early completion phase	- Data collection announcement mail, text, and e-mail - Mode tailoring with NPSAS:20 completion mode	- Data collection announcement mail, text, and e-mail, \$2 cash prepaid - Targeted CATI begins 2 weeks after mail outs – continued throughout
Production phase I	- Targeted light CATI outbound begins - Postcard, e-mail, text message reminders – continued throughout	Postcard, e-mail, text message reminders – continued throughout
Production phase II	\$10 boost postpaid	\$20 boost postpaid - Abbreviated survey offered
Nonresponse conversion phase	Abbreviated survey offered	
Total incentives	\$30 baseline \$10 boost - Maximum = \$40	\$2 prepaid \$45 baseline \$20 boost - Maximum = \$67

NOTE: BPS:20/22 = 2020/22 Beginning Postsecondary Students Longitudinal Study; CATI = computer-assisted telephone interviewing; NCES = National Center for Education Statistics; NPSAS:20 = 2019–20 National Postsecondary Student Aid Study.

4.1.2 Data Collection Protocol Design Elements

4.1.2.1 Incentives

The baseline incentive for the default protocol will be \$30 with a \$10 incentive boost for nonresponse conversion (see section 4.1.2.6) leading to a maximum possible total incentive of \$40. The baseline incentive for the aggressive protocol will be \$45. An experiment conducted in BPS:12/14 demonstrated that a \$45 baseline incentive yielded the highest response rates (Hill et al. 2016). However, this experiment was underpowered to detect differences from \$30 in the lower propensity response groups (Wilson, Shepherd, and Wine 2015). Nonetheless, implementing a higher baseline incentive is recommended given the \$30 baseline incentive and the \$10 incentive boost from NPSAS:20 was not enough to encourage prior-year nonrespondents to participate. Further, the \$40 BPS:20/22 field test incentive yielded a response rate of only 25.3 percent among these “aggressive protocol” sample members. The baseline incentive will be paid in addition to a \$2 prepaid incentive (see section 4.1.2.3) and a \$20 incentive boost (see section 4.1.2.6). The maximum possible total incentive is \$67 in the aggressive data collection protocol.

Results from the BPS:20/22 field test showed that offering sample members an early bird incentive did not statistically significantly improve response rates or representativeness by the end of data collection, nor did it practically improve data collection efficiency (see section 3.1.6.1 for details). Therefore, early bird incentives will not be used in the full-scale study for either the default or aggressive protocols.

Beyond the baseline incentives, both data collection protocols employ similar interventions, although the timing and intensity of these interventions differs across groups. Interventions occur sooner in the aggressive protocol and are more intense.

4.1.2.2 Prenotification

Sample members in both the default and aggressive data collection protocols will receive a greeting card as the first mailing. This mailing is aimed to increase the perceived legitimacy of the upcoming survey request (e.g., Groves, Cialdini, and Couper 1992) in both data collection groups and announce the incentive amounts. Greeting cards, in particular, have been shown to statistically significantly increase response rates in longitudinal studies (Griggs et al. 2019), and this method will be used as a precursor to the invitation letter. The greeting card will be mailed a few weeks in advance of data collection.

4.1.2.3 \$2 prepaid incentive

Cash prepaid incentives have been shown to statistically significantly increase response rates in both interviewer-administered and self-administered surveys. These prepaid incentives increase the perceived legitimacy of the survey request and therefore reduce the potential for nonresponse bias (e.g., Cantor, O'Hare, and Connor 2008; Church 1993; Goeritz 2006; Medway and Tourangeau 2015; Messer and Dillman 2011; Parsons and Manierre 2014; Singer 2002). During the early completion phase in the 2016/17 Baccalaureate and Beyond Longitudinal Study (B&B:16/17) field test, prepaid incentives (\$10 via check or PayPal) in combination with telephone prompting statistically significantly increased response rates by 4.4 percentage points in the aggressive protocol group. Given these positive findings combined with general recommendations in the literature (e.g., DeBell et al. 2019; Singer and Ye 2013), a small \$2 cash prepaid “visible” incentive or, where necessary due to low address quality, a \$2 prepaid PayPal incentive announced on a separate index card will be sent to all cases in the aggressive protocol for the BPS:20/22 full-scale survey.²² Sample members will

²² See results from the B&B:16/20 calibration experiment (Kirchner, Peytcheva, and Tate 2021).

be notified of this prepaid incentive in the data collection prenotification, and it will be included in the data collection announcement letter.

4.1.2.4 Mode tailoring

The leverage-saliency theory suggests that respondents have different hooks that drive their likelihood of survey participation (Groves, Singer, and Corning 2000); thus, offering a person the survey mode (e.g., web, mail, telephone) that they prefer may increase their likelihood of responding. This is further supported by empirical evidence that shows offering people their preferred mode speeds up their response and is associated with higher participation rates (e.g., Olson, Smyth, and Wood 2012). Using the NPSAS:20 survey completion mode as a proxy for mode preference, the BPS:20/22 full-scale early completion phase will approach sample members with their mode of completion for NPSAS:20 (i.e., the default protocol). Specifically, while all sample members in the default protocol will receive identical data collection announcement letters and e-mails, those who completed the NPSAS:20 survey by telephone (4.3 percent) will be approached by telephone from the start of data collection. Likewise, those who completed the NPSAS:20 student survey by web will not be contacted by telephone before a preassigned outbound telephone data collection date.

4.1.2.5 (Light) outbound CATI calling and text messaging

Sample members in the default group who qualify for telephone calls will receive a light CATI protocol. Light CATI involves a minimal number of phone calls, used mainly to prompt web response (as opposed to regular CATI efforts that involve more frequent phone efforts, with the goal to locate sample members and encourage participation via CATI). In the B&B:16/17 field test, introduction of light CATI appeared to increase production phase response rates in the default protocol. Although one should use caution when interpreting these results—group assignment in B&B:16/17 field test was not random but instead compared 2015–16 National Postsecondary Student Aid Study (NPSAS:16) “early” and “late” respondents—the findings are consistent with the literature which has shown that web surveys tend to have lower response rates compared with interviewer-administered surveys (e.g., Lozar Manfreda et al. 2008).

Attempting to survey sample members by telephone also increases the likelihood of initiating locating efforts sooner. B&B:16/17 field test results showed higher locate rates in the default protocol (93.7 percent), which had light CATI, compared with a more relaxed protocol without light CATI (77.8 percent; $p < 0.001$). For BPS:20/22 full-scale data collection, light CATI will be used in the default protocol once CATI begins in production phase I (see Table J-25).

Additionally, all students in the aggressive protocol will receive earlier and more intense telephone prompting than eligible students in the default group.

The results from the BPS:20/22 field test showed that there were no statistically significant differences in response rates for the text message reminder and the telephone-only group at the end of the experimental period (see section 3.1.6.2 for details). As a result, both data collection groups will receive early text message reminders combined with prioritized telephone calls. Telephone calls will be prioritized to students for whom no cell phone number exists, those who opt out of the text message reminders, and sample members who should be prioritized based on other criteria (e.g., from lower performing institutional sectors). The BPS:20/22 full-scale survey will also allow for two-way text messaging whereby interviewers can respond to text message questions sent by sample members, enabling a direct exchange.

4.1.2.6 Incentive boosts

Researchers have commonly used incentive boosts as a nonresponse conversion strategy for sample members who have implicitly or explicitly refused to complete the survey (e.g., Groves and Heeringa 2006; Singer and Ye 2013). These boosts are especially common in large federal surveys during the nonresponse follow-up phase and have been implemented successfully in other postsecondary education surveys.²³ In NPSAS:20, a \$10 incentive boost increased the overall response rate by about 3.2 percentage points above the projected response rate. Therefore, a \$10 incentive boost increase to the BPS:20/22 full-scale baseline incentive is planned for all remaining nonrespondents in the default data collection protocol, before the abbreviated survey is offered. Remaining nonrespondents in the aggressive data collection protocol will be offered a \$20 incentive boost increase to the baseline incentive before the abbreviated survey. This is because the \$10 incentive boost in NPSAS:20 did not show any effect among sample members in the aggressive protocol. If necessary, incentive boosts may be targeted only at certain groups of nonrespondents to achieve response goals (e.g., targeting nonrespondents from certain states to ensure representativeness, targeting aggressive group nonrespondents to reduce the potential for nonresponse bias).

²³ For example, the following studies have utilized incentive boosts during the nonresponse follow-up phase: The National Survey of Family Growth (NSFG), High School Longitudinal Study of 2009 (HSL:09) Second Follow-Up, 2012/17 Beginning Postsecondary Students Longitudinal Study (BPS:12/17), and NPSAS:20.

4.1.2.7 Abbreviated survey

Obtaining responses from all sample members is an important assumption of the inferential paradigm. The leverage-saliency theory (Groves, Singer, and Corning 2000) and the social exchange theory (Dillman, Smyth, and Christian 2014) suggest that the participation decision of an individual is driven by different survey design factors or perceived cost of participating. As such, reducing the perceived burden of participating by reducing the survey length may motivate sample members to participate.

During the B&B:16/17 field test, sample members in the aggressive protocol (i.e., prior-round nonrespondents) who were offered the abbreviated survey during the production phase responded at higher rates (22.7 percent) than those who were not offered the abbreviated survey at the same time (12.1 percent; $t(2,097) = 3.67, p < 0.001$). The 2008/12 Baccalaureate and Beyond Longitudinal Study (B&B:08/12) and 2008/18 Baccalaureate and Beyond Longitudinal Study (B&B:08/18) full-scale studies showed similar results; offering the abbreviated survey to prior-round nonrespondents at a later point in data collection increased overall response rates of that group by 18.2 and 8.8 percentage points respectively (Cominole, Shepherd, and Siegel 2015). In NPSAS:20, 14.4 percent of those offered the abbreviated survey completed it. An abbreviated survey option will be offered to all sample members in BPS:20/22 full-scale data collection. For the aggressive protocol, the abbreviated survey will be offered during production phase II, which is the latter half of the production phase of data collection (see Table J-25). For the default protocol, the abbreviated survey will be offered as the last step in nonresponse conversion.

4.1.2.8 Other interventions

RTI recommends sending e-mail contacts to sample members from both RTI and NCES email domains. While all BPS studies are conducted by NCES, the data collection contractor, RTI, has typically used the study-specific e-mail “bps@rti.org” to contact and support sample members. Changing the e-mail sender to the BPS study director’s NCES e-mail address or the BPS project director’s RTI e-mail address may increase the perceived importance of the survey and help personalize the contact materials, thereby potentially increasing relevance. Periodically switching the sender throughout data collection also increases the chance that the survey invitation is delivered to the sample member rather than to a spam folder.

4.2 Recommendations for Survey Design

The survey design proposed for the BPS:20/22 full-scale study builds on the designs implemented in past studies such as BPS:12/14 and NPSAS:20. Additionally, results from the BPS:20/22 field test experiments inform recommendations for the BPS:20/22 full-scale survey design (see section 3.2.8 for details):

- **Collection of month-level enrollment intensity.** The BPS:20/22 full-scale survey instrument will continue to use the single forced-choice grid, as used in prior studies. This method collects enrollment intensity for each academic year on a single form (B22ASTST) and results in lower timing burden and accurate self-reported enrollment information as determined by concordance with NSC administrative data.
- **Collection of address information using predictive search database.** Given the implementation of improved instrument functionality to reduce incomplete address entries, higher quality data resulting from Experian verifiable addresses, and the statistically significant reduction in burden using Experian compared with manual entry, the continuation of the Experian predictive search database is proposed for the BPS:20/22 full-scale survey.
- **Coronavirus pandemic questions.** The BPS:20/22 full-scale survey instrument will administer a subset of the questions from both field test coronavirus pandemic modules, based upon field test performance and TRP feedback. The coronavirus pandemic module for the full-scale survey maintains the burden goal of 3 minutes.

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